2025 Mathematics, Rebuilt Month by Month

351 developed treatments from twelve months of recovered mathematical evidence.

Mathematics-only retrospective · prepared 21 October 2026

This volume rebuilds mathematics recovered from conversations dated in 2025, month by month. Each developed treatment contains a newly revised question, definitions and hypotheses, a precise result, a proof or derivation, a worked example, limits and a further mathematical problem. Related source formulas may be grouped or developed separately where they ask distinct questions.

All twelve months have dated mathematical evidence. Recovery is bounded and fragmentary, rather than an exhaustive export of every historical message. Dates are retained in UTC when available. U identifies a retrieved user-attributed topic or passage; A identifies assistant-only evidence; T identifies a topic or secondary-source fragment. U does not by itself establish that an original prompt was recovered verbatim.

Historical evidence and the new mathematics are separate. The revised questions, proofs and stipulated examples are authored for this volume. Exact source excerpts and receipt locators are retained in the searchable edition and source bundle. Missing original operators, tables, data and proofs are not reconstructed as if their historical bytes were available.

The scope is mathematics. Classical results, new expositions, exact finite certificates, numerical examples and open conjectures are distinguished. Claims about physical devices, model-training performance, or major open problems require their own evidence; the mathematical developments here do not certify such claims. The supplied check programs test their stated finite cases and do not replace written proofs.

The searchable HTML edition provides complete question tables, month filtering and the full source summaries. The accompanying ZIP contains machine-readable entries and offline replay programs.

Month-by-month coverage

Month	Treatments	U / A / T	Source coverage
January	25	8 / 17 / 0	Dated fragments recovered; complete transcript unavailable
February	25	6 / 18 / 1	Dated fragments recovered; complete transcript unavailable
March	35	3 / 32 / 0	Dated fragments recovered; complete transcript unavailable
April	28	13 / 15 / 0	Dated fragments recovered; complete transcript unavailable
May	24	9 / 13 / 2	Dated fragments recovered; complete transcript unavailable
June	24	11 / 13 / 0	Dated fragments recovered; complete transcript unavailable
July	35	17 / 18 / 0	Dated fragments recovered; complete transcript unavailable
August	40	20 / 19 / 1	Dated fragments recovered; complete transcript unavailable
September	21	7 / 14 / 0	Dated fragments recovered; complete transcript unavailable
October	36	16 / 19 / 1	Dated fragments recovered; complete transcript unavailable
November	31	17 / 13 / 1	Dated fragments recovered; complete transcript unavailable
December	27	18 / 9 / 0	Dated fragments recovered; complete transcript unavailable

The counts measure developed treatments, not conversations. Related source formulas can be grouped in one treatment. Different treatments may develop distinct aspects of the same conversation. Source dates are not newly assigned publication dates.

How to read the edition

Each month begins with a revised-question table. The complete developments follow it in source-date order. The PDF outline links months and treatment identifiers. A is assistant-only evidence; U is user-attributed recovered content rather than a promise of exact original wording. The new mathematical question is always separate from the recovered starting point.

January 2025

Dated fragments recovered; complete transcript unavailable. 25 developed treatments.

The recovered material ranges from oscillator quantization and limiting Hodge structures to moonshine, group actions, intersection cohomology and deformations. The developments construct explicit mathematical models with checked identities and retain the missing bridges as precise further problems.

ID · date	Concept and revised question	Result and mathematical core
Y25-01-025 2025-01-01 · U	The6.02dB identity refers to an amplitude ratio When does doubling produce6.02dB, and how do coherent amplitudes differ from independent power additions?	Exact doubling and interference laws Doubling amplitude at fixed conversionc quadruples power and adds$20\log_{10}2\approx6.02060$dB. Doubling power adds$10\log_{10}2\approx3.01030$dB. Equal coherent amplitudes give$P_{total}/P_{single}=2+2\cos\varphi$.
Y25-01-024 2025-01-02 · U	A controlled gate is unitary but protection needs an error model Which algebraic facts certify a controlled gate, and what quantitative assumption is needed to call a circuit protected?	Exact entangling gate and circuit error bound $CNOT^2=I$ andCNOT is unitary. $CNOT(H\otimes I)\lvert00\rangle=(\lvert00\rangle+\lvert11\rangle)/\sqrt{2}$, whose one-qubit marginal is$I/2$. The two circuit products differ by at most$\Sigma_j\varepsilon_j$.
Y25-01-021 2025-01-10 · U	The oscillator connects a connection, a star product and an eigenstate Which explicit constructions connect the classical oscillator to prequantization and a quantum ground state?	Exact curvature and quantization checks The connection has curvature$-i\omega/\hbar$. The star commutator$q\star p-p\star q=i\hbar$, and$H\star H=H^2-\hbar^2/4$. The Gaussian$\psi_0=(\pi\hbar)^{-1/4}\exp(-q^2/(2\hbar))$ is normalized and has energy$\hbar/2$.
Y25-01-022 2025-01-10 · A	Equivariant cohomology retains the stabilizer of a fixed point What is$H^*_{S^1}(R^2)$ for the oscillator rotation action, and why does contractibility not make it ordinary point cohomology?	Complete equivariant calculation for the plane $H^*_{S^1}(C;k)\cong k[u]$, with one generator in every nonnegative even degree and none in odd degree. $H^*(C;k)=H^*(LC;k)=k$ in degree zero only.
Y25-01-001 2025-01-11 · A	A nilpotent monodromy has a computable weight filtration What algebraic data can actually be extracted from a specified unipotent monodromy before adding a quantum correction?	Explicit monodromy weight calculation The filtration is$W_k=0$ for$k<0$,$W_0=W_1=Qe_1$,$W_k=V$ for$k\geq2$. The map$N:Gr^2_W\to Gr^0_W$ is an isomorphism. The nontriviality of this filtration does not alone define a limiting mixed Hodge structure.
Y25-01-002 2025-01-11 · A	A kernel transform is an equivalence only when it has an inverse Which explicit kernel operation is an autoequivalence, and what distinguishes that theorem from merely adding a correction term?	Classification in a finite derived model Everyk-linear triangulated equivalence of$D$ permutesS_1,S_2 and assigns an integer shift to each. In particular$F(V,W)=(W[1],V[-1])$ is an equivalence with$F^2\cong\mathrm{Id}$.

ID · date	Concept and revised question	Result and mathematical core
Y25-01-003 2025-01-15 · U	Noncommutative observables still have identities and symmetries Which algebraic property separates jointly measurable finite observables, without discarding identity operators or symmetry?	Joint diagonalization and a sharp uncertainty example A finite commuting family of self-adjoint matrices is simultaneously unitarily diagonalizable. Forψ=(1,i)/√2,ΔX=ΔZ=1 and|⟨[X,Z]⟩|/2=1, attaining the Robertson lower bound.
Y25-01-004 2025-01-15 · U	Closing a curve is distinct from choosing a polygon or a winding What does returning to the starting point establish, and when can a smooth closed path be replaced by a polygon without changing its winding?	A quantitative polygonal replacement theorem The interpolationH(s,t)=(1−s)γ(t)+sp(t) is a homotopy of closed curves avoiding zero, soγ andp have equal winding. Every continuous nonvanishing loop admits such a polygonal approximation.
Y25-01-005 2025-01-15 · A	Boundary conditions are part of a spectral problem Which operator domain makes an eigenvalue condition a theorem, and how does changing that domain change the spectrum?	Two domains, two exact spectra The Dirichlet eigenvalues aren² forn≥1, with eigenfunctionssin(nx). The periodic eigenvalues are4n² forn∈Z, including zero, with eigenfunctionse^{2inx}.
Y25-01-006 2025-01-15 · A	A positive metric can repair a matrix inner product When does a non-Hermitian matrix become self-adjoint in a positive inner product, and which obstruction rules this out?	Explicit quasi-Hermitian metric and its boundary η_s=[[1,−s],[−s,1+s²]] satisfiesA_s*η_s=η_sA_s and is positive definite. A matrix with a nontrivial Jordan block and real eigenvalue admits no positive metric making it self-adjoint.
Y25-01-023 2025-01-15 · A	A modular Hamiltonian is defined by a state What mathematical relation does a modular Hamiltonian have to entropy, and which extra data are needed before inserting it into a geometric action?	Entropy first variation and strict concavity S′(0)=Tr(δρK)=−Σv_i logp_i, andS″(0)=−Σv_i²/p_i. The second derivative is strictly negative for every nonzero trace-zero diagonal perturbation.
Y25-01-007 2025-01-17 · U	Moonshine coefficients are graded dimensions and traces What information is contained in a graded character, and how can character arithmetic be verified before attaching a physical interpretation?	An exact tensor-character decomposition For the two-dimensional standard representationW ofS₃, W⊗W≅1⊕sgn⊕W. At the dimension level the first moonshine coefficient satisfies196884=1+196883; the second satisfies21493760=1+196883 +21296876.
Y25-01-008 2025-01-17 · U	Modular invariance has a concrete weight and coefficient test How do theE₄ and discriminant expansions producej, and what modular-invariance statement follows from their weights?	Weight-zero quotient and exact initial Fourier coefficients WhereΔ≠0, j transforms invariantly underSL₂(Z). Its expansion beginsq⁻¹+744+196884q+21493760q². ConsequentlyJ has zero constant term and the same positive-power coefficients.
Y25-01-009 2025-01-18 · U	Projective phases require a genuine two-cocycle Which phase factors define an associative projective representation, and how can one prove that they cannot be removed by rephasing?	A nontrivial projective representation with an exact certificate U_gU_h=σ(g,h)U_(g+h), andσ is a two-cocycle that is not a coboundary. Thus the projective action cannot be made an ordinary representation by choosing new scalar phases.

ID · date	Concept and revised question	Result and mathematical core
Y25-01-010 2025-01-21 · A	Six pentads give an exhaustive outer automorphism certificate Can the exceptional symmetry ofS_6 be exhibited by a finite construction rather than inferred from an analogy with particle generations?	Explicit exceptional permutation action There are fifteen synthemes and six pentads. The action$S_6 \to \mathrm{Sym}(\text{pentads})$ is bijective. A vertex transposition acts on pentads as three disjoint transpositions, so this automorphism is outer.
Y25-01-012 2025-01-24 · A	Platonic boundaries share homology but have different symmetry groups Which topology remains unchanged between a sphere and a convex polyhedral boundary, and which symmetry data can distinguish the Platonic cases?	Topological invariance and finite symmetry counts ∂P is homeomorphic toS^2 and has Betti numbers$(1,0,1)$. The tetrahedron,cube,octahedron,dodecahedron,icosahedron have(V,E,F) respectively$(4,6,4),(8,12,6),(6,12,8),(20,30,12),(12,30,20)$, with rotational group orders$12,24,24,60,60$.
Y25-01-013 2025-01-24 · A	A scalar wave potential has critical points, not a square Jacobian Which derivative detects a singular level of a scalar wave field, and how is that different from the Jacobian determinant of a vector map?	Complete critical-point calculation and fold model The potential has eight critical points, each coordinate zero orπ. Its critical values are$3,1,-1,-3$ and all eight points are Morse. The vector map$F(x,y,z)=(x^2,y,z)$ has singular plane$\{x=0\}$ and critical-value plane$\{u=0\}$.
Y25-01-014 2025-01-24 · A	Intersection homology retains cycles that an ordinary cone contracts What does a perversity allow near a singular point, and can it change the homology of a contractible cone?	Explicit cone-formula instance $IH_0(X)=k, IH_1(X)=k^2$ and$IH_i(X)=0$ for$i\geq 2$ under the stated perversity. Ordinary$H_1(X)=0$ because the whole cone contracts.
Y25-01-015 2025-01-26 · A	The derived tangent complex comes from the cotangent complex Which complex captures infinitesimal deformations and obstructions at a singular point?	Double-point deformation and obstruction At$x=0$ the tangent-complex differential is zero, so$H^0\cong k$ and$H^1\cong k$. Every first-order displacement$x=at$ satisfies$x^2=0$ modulot^2, but it extends to$x=at+bt^2$ modulot^3 only if$a^2=0$.
Y25-01-016 2025-01-26 · A	Perverse degree is fixed by support and cosupport Which degree shift makes a local system perverse on a smooth complex curve, and what changes for a point-supported object?	Curve and point perverse objects A finite-rank local system$L[1]$ is perverse on$C$. A skyscraperk_p is perverse. The unshifted nonzero local system$L$ is not perverse.
Y25-01-017 2025-01-26 · A	A persistence module retains finite bars that its colimit forgets Which object records birth and death times, and why is a terminal colimit not an adequate definition of persistence?	Complete barcode of the triangle filtration H_0 consists of one interval$[0,\infty)$ and two intervals$[0,1)$. H_1 consists of$[2,3)$. Higher homology vanishes. The colimitH_1 is zero, while the rank of$H_1(X_2)\to H_1(X_2)$ is one.
Y25-01-018 2025-01-26 · A	A weight filtration need not split canonically What is the difference between a filtration, its associated graded object and a chosen direct-sum splitting?	No functorial splitting for this filtered object Every complement ofW_0 has the form$k(e_2+ce_1)$. The filtered automorphisms$T_a(e_1)=e_1, T_a(e_2)=e_2+ae_1$ act transitively on these complements. No complement is fixed by allT_a.

ID · date	Concept and revised question	Result and mathematical core
Y25-01-019 2025-01-26 · A	Localization gives a valid motivic and cohomological subtraction Which additivity law underlies a motivic expression, and how does a concrete open/closed decomposition act on cohomology?	Exact class and compact-support cohomology ofC* [C*]=L−1. Over a fieldk,H_c¹(C*;k)=k,H_c²(C*;k)=k and the other compact-support groups vanish. Its compact-support Euler characteristic is zero.
Y25-01-011 2025-01-28 · A	An E_8 projection has exact fibers and changes root lengths What does a specifiedE_8-to-four-coordinate projection actually produce, including multiplicities and zero fibers?	Complete projected-root fiber distribution The24 D_4vectors each have one preimage. Each of the24 shortF_4vectors has eight preimages. Zero has24 preimages. Thus24+24·8+24=240, and squared projected lengths are2,1 and0.
Y25-01-020 2025-01-28 · A	A Jacobiator detects nonassociativity only after multiplication is defined Can a claimed nonzero flux Jacobiator be distinguished from the identically zero Jacobiator of associative operator commutators?	An exact nonzero Jacobiator For(a,b,c)=(e_1,e_2,e_4), the associator is2e_7 andJ=−12e_7. In every associative algebra the same commutator formula hasJ=0.

Y25-01-025 The6.02dB identity refers to an amplitude ratio

2025-01-01 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

When does doubling produce6.02dB, and how do coherent amplitudes differ from independent power additions?

The same physical ratio has different numerical logarithmic factors depending on whether one starts with power or amplitude. Coherent addition and independent addition are different hypotheses, with exact phase-dependent formulas.

Definitions and assumptions

For positive powers define$L=10\log_{10}(P/P_0)$. If$P=cA^2$ with fixed positivec, the same level difference is$20\log_{10}(A/A_0)$.

Two equal complex amplitudesA andAe^{iφ} sum coherently. Independent zero-mean signals with no cross covariance add their powers instead.

All logarithm arguments must be dimensionless ratios with the same reference convention.

Exact doubling and interference laws

Doubling amplitude at fixed conversionc quadruples power and adds$20\log_{10}2\approx6.02060$dB. Doubling power adds$10\log_{10}2\approx3.01030$dB. Equal coherent amplitudes giveP_total/P_single=2+2cosφ.

Derivation or proof

The amplitude-to-power law gives$P_2/P_1=(A_2/A_1)^2$, so ten times the logarithm of power becomes twenty times the amplitude logarithm. For coherent addition, $|A+Ae^{i\varphi}|^2=|A|^2(2+e^{i\varphi}+e^{-i\varphi})=2|A|^2(1+\cos\varphi)$. At$\varphi=0$ power is four times one signal; at$\varphi=\pi$ it is zero. For independent or uncorrelated zero-mean signals, expectation of the cross term vanishes, leaving twice the single-signal power and the3.01dB result. These calculations identify exactly which statistical or phase assumption changes the reported gain.

Worked example

Compare phases zero,$\pi/2$ andπ for equal-amplitude inputs.

The power ratios are four,two and zero. The first gives6.02060dB, the second3.01030dB, and exact cancellation has no finite logarithmic level relative to positiveP_0. Rounding6.02 does not convert an amplitude statement into a universal gain for two devices or two noisy sources.

The original identity is preserved with its mathematical domain and operational meaning made explicit.

Scope

Changing impedance or another amplitude-to-power conversion factor changes the simple20log formula. The result does not validate any proposed phonon, curvature or material-protection mechanism.

Further development

For unequal amplitudesA,B, derive$P=A^2+B^2+2AB\cos\varphi$ and propagate a bounded phase error through it. The derivative$-2AB\sin\varphi$ shows where a phase-sensitive operating point is locally robust and where it is sensitive. Near destructive interference write $\varphi=\pi+\delta$. Then $P_{total}=2A^2(1-\cos\delta)\le A^2\delta^2$. The inequality $1-\cos\delta\le\delta^2/2$ gives a global quadratic upper bound on leakage from a phase error, a stronger and more useful robustness statement than merely reporting the zero at perfect cancellation.

Source receipt: months/01/root_coverage.json record 0. Mathematical references: R001.

Y25-01-024 A controlled gate is unitary but protection needs an error model

2025-01-02 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which algebraic facts certify a controlled gate, and what quantitative assumption is needed to call a circuit protected?

CNOT is an exact permutation unitary and creates an entangled state from a separable input. A telescoping estimate bounds accumulated implementation error, but unitarity alone does not prove fault tolerance.

Definitions and assumptions

DefineCNOT|a,b⟩=|a,b⊕a⟩ for binary labels. The Hadamard gate sends|0⟩ to(|0⟩+|1⟩)/√2.

For unitary gatesU₁,...,U_m and unitary approximationsV₁,...,V_m assume||V_j−U_j||≤ε_j in operator norm.

Protection is a statement about specified errors, encoding, correction and a performance bound, not merely a name for the intended gate.

Exact entangling gate and circuit error bound

CNOT²=I andCNOT is unitary. CNOT(H⊗I)|00⟩=(|00⟩+|11⟩)/√2, whose one-qubit marginal isI/2. The two circuit products differ by at mostΣ_jε_j.

Derivation or proof

CNOT permutes the four orthonormal basis states and is its own inverse, proving unitarity. Applying it to(|00⟩+|10⟩)/√2 gives the displayed Bell state. Taking a partial trace eliminates the two off-diagonal cross terms and leaves equal populations, so the pure joint state is not a product. For the error bound, subtractV_m⋯V₁−U_m⋯U₁ and insert mixed products one gate at a time. The resulting sum has termsV_m⋯V_{j+1}(V_j−U_j)U_{j−1}⋯U₁. Unitary factors have norm one, so submultiplicativity and the triangle inequality giveΣ ε_j. This is a worst-case coherent-error estimate and requires no stochastic independence.

Worked example

Take ten gates each with norm error at most10^{-3}.

The final unitary error is at most10^{-2}. Acting on any normalized state gives state-vector error at most10^{-2}; an expectation of an observable of norm one changes by at most$2\cdot10^{-2}$. This bound does not imply a threshold theorem or a percentage of operational coherence.

A correct gate and a protected implementation are separate mathematical claims with a calculable bridge only after errors are specified.

Scope

The calculation supplies no material implementation or hardware error rate. Fault-tolerant thresholds involve coding, locality, noise and repeated recovery assumptions absent from the recovered formulas.

Further development

Choose an explicit encoding and noise channel, verify the Knill-Laflamme conditions and compute a logical error bound after recovery. Compare it with the coherent telescoping bound before claiming protection.

Source receipt: months/01/root_coverage.json record 0. Mathematical references: R002.

Y25-01-021 The oscillator connects a connection, a star product and an eigenstate

2025-01-10 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which explicit constructions connect the classical oscillator to prequantization and a quantum ground state?

On the symplectic plane one can write the prequantum connection and verify its curvature directly. The constant-Poisson Moyal product and a normalized Gaussian then give exact, separately checkable quantization statements.

Definitions and assumptions

Use coordinates(q,p),$\omega=dq\wedge dp$,$H=(q^2+p^2)/2$ and$\hbar>0$. DefineX_H by$\iota_X\omega=dH$, giving$X_H=p\partial q-q\partial p$.

On the trivial complex line bundle set$\nabla=d-(i/\hbar)q\,dp$. The formal Moyal product is$f\star g=f\exp[(i\hbar/2)(\leftarrow\partial q\rightarrow\partial p-\leftarrow\partial p\rightarrow\partial q)]g$.

The Schrödinger operator is$\hat{H}=-(\hbar^2/2)d^2/dq^2+q^2/2$ on an appropriateL^2 domain.

Exact curvature and quantization checks

The connection has curvature$-i\omega/\hbar$. The star commutator$q\star p-p\star q=i\hbar$, and$H\star H=H^2-\hbar^2/4$. The Gaussian$\psi_0=(\pi\hbar)\hat{}-1/4\exp(-q^2/(2\hbar))$ is normalized and has energy$\hbar/2$.

Derivation or proof

Because the connection one-form is scalar, its wedge square vanishes and$\nabla^2=d[-iq\,dp/\hbar]=-i\,dq\wedge dp/\hbar$. For$q,p$ the bidifferential expansion stops after first derivatives, yielding$qp\pm i\hbar/2$. For$H\star H$ the first Poisson term is zero; the second derivative combination is$H_{qq}H_{pp}+H_{pp}H_{qq}=2$, so its coefficient is$-\hbar^2/8$ times two. Higher derivatives vanish. Differentiating the Gaussian gives$\psi_0''=(q^2/\hbar^2-1/\hbar)\psi_0$, so substitution cancelsq^2 and leaves$\hbar\psi_0/2$. Its squared norm is one by the elementary Gaussian integral. Formal associativity of the Moyal product follows because the constant-coefficient operators on three tensor factors commute, making both parenthesizations the same exponential of the sum of pair operators.

Worked example

Set$\hbar=1$ and compare the classical fixed point with the Gaussian.

The classical point$(q,p)=(0,0)$ has$H=0$. The normalized quantum state has energy$1/2$ and variances$\langle q^2\rangle=\langle p^2\rangle=1/2$, giving$\Delta q\Delta p=1/2$. The difference is a quantified statement about a specified representation, not a change in the classical curvature formula.

Each bridge has an explicit map or calculation and can be checked independently.

Scope

Prequantization alone does not choose a polarization or prove the full oscillator spectrum. The formal star product is not automatically a convergent product on every function space.

Further development

Construct the ladder operators, prove their commutator on Schwartz functions and derive all Gaussian-Hermite eigenstates. Then justify completeness and the self-adjoint operator domain rather than relying only on formal algebra. Since H is quadratic, every third derivative vanishes and its Moyal commutator with a polynomial equals iħ times the Poisson bracket exactly. In particular $(q\star H-H\star q)/(i\hbar)=p$ and $(p\star H-H\star p)/(i\hbar)=-q$, reproducing the classical linear oscillator equations with no higher-order correction.

Source receipt: months/01/root_followups.json record 0. Mathematical references: R003.

Y25-01-022 Equivariant cohomology retains the stabilizer of a fixed point

2025-01-10 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What is$H^*_{S^1}(R^2)$ for the oscillator rotation action, and why does contractibility not make it ordinary point cohomology?

The plane contracts to its origin equivariantly, so its equivariant cohomology is$H^*(BS^1)=k[u]$. Ordinary plane and free-loop-space cohomology are instead just$k$. These are different constructions with different retained symmetry data.

Definitions and assumptions

LetS^1 rotate$C\cong R^2$. The Borel construction is$ES^1\times_{S^1}C$, and equivariant cohomology is its ordinary cohomology.

Use characteristic-zero coefficients and$u\in H^2(BS^1;k)$, with$BS^1\simeq CP^\infty$. Its cohomology ring is$k[u]$.

The ordinary free-loop spaceLC consists of continuous maps$S^1\to C$; it does not include the Borel quotient unless explicitly added.

Complete equivariant calculation for the plane

> $H^*_{S^1}(C;k)\cong k[u]$, with one generator in every nonnegative even degree and none in odd degree. $H^*(C;k)=H^*(LC;k)=k$ in degree zero only.

Derivation or proof

The homotopy$r_s(z)=(1-s)z$ commutes with rotations and ends at the fixed origin. It descends to a deformation retraction of$ES^1\times_{S^1}C$ onto$ES^1\times_{S^1}\{0\}=BS^1$. The standard cell decomposition ofCP^∞ has one cell in every even dimension and no odd cells; its ring generator$u$ has nonzero powers, yielding$k[u]$. For the ordinary loop space, contract every loop pointwise byr_s. This is a continuous contraction to the zero loop, giving the ordinary cohomology claim. Thus an exterior factor from the two-dimensional tangent space does not belong automatically to Borel cohomology; such a factor would require a separately specified complex or theory.

Worked example

Compare the first four cohomological degrees.

The equivariant dimensions in degrees0,1,2,3 are1,0,1,0. OrdinaryC andLC have dimensions1,0,0,0. The origin’s stabilizer is the whole circle, which explains the surviving degree-two class in the quotient construction despite the underlying plane being contractible.

The calculation repairs a conflation of loop cohomology, tangent exterior algebra and equivariant cohomology.

Scope

A claimed spectral-sequence degeneration needs its actual fibration or filtered complex and differential bidegrees. No such general degeneration follows merely from the oscillator notation.

Further development

Replace the weight-one rotation byz↦g^m z and compute the equivariant Euler classmu of the normal complex line. Check fixed-point localization only after specifying the coefficient localization and an equivariant integration map. Remove the origin and the rotation action becomes free. The Borel construction of C* is homotopy equivalent to its radial quotient (0,∞), so its equivariant cohomology is k. The restriction map from k[u] therefore kills u. This example shows exactly where a fixed stabilizer contributes additional classes.

Source receipt: months/01/root_followups.json record 0. Mathematical references: R004.

Y25-01-001 A nilpotent monodromy has a computable weight filtration

2025-01-11 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What algebraic data can actually be extracted from a specified unipotent monodromy before adding a quantum correction?

A two-dimensional unipotent matrix supplies an exact logarithm, a centered weight filtration and a compatible sl_2 representation. This concrete calculation isolates the linear algebra used in degeneration theory from an unprovided quantum enrichment.

Definitions and assumptions

Let $V=Qe_1⊕Qe_2$ and $M=[[1,1],[0,1]]$. Its nilpotent logarithm is $N=\log M=M-I$ because$(M-I)^2=0$.

A monodromy weight filtration centered at one is increasing, satisfies$N(W_k)⊂W_{k-2}$, and makes$N^j:Gr^W_{1+j}→Gr^W_{1-j}$ an isomorphism for every $j≥0$.

Explicit monodromy weight calculation

The filtration is$W_k=0$ for$k<0$,$W_0=W_1=Qe_1$,$W_k=V$ for$k\geq 2$. The map$N:Gr^2_W\to Gr^0_W$ is an isomorphism. The nontriviality of this filtration does not alone define a limiting mixed Hodge structure.

Derivation or proof

Since$Ne_2=e_1$ and$Ne_1=0$, image and kernel of$N$ are bothQe_1. The proposed filtration is stable under$N$ with the required two-step decrease. Both nonzero graded pieces are one-dimensional and the induced$N$ maps$[e_2]$ to$[e_1]$, proving the required isomorphism. Higher powers are zero and relate zero graded pieces. For ansl_2 triple set$E=N$,$F=[[0,0],[1,0]]$,$H=\mathrm{diag}(1,-1)$. Multiplication gives$[H,E]=2E$,$[H,F]=-2F$,$[E,F]=H$. Weight index$1-H$ assigns index zero toe_1 and two toe_2, agreeing with the filtration. This is the exact finite representation-theoretic content; a Hodge filtration, polarization and geometric degeneration are additional data. Over the integer lattice, replacing M by $M_n=I+nN$ changes the coinvariants to$\mathbb{Z}\oplus\mathbb{Z}/n\mathbb{Z}$ for$n>0$: the image is$n\mathbb{Z}e_1$. Tensoring with $\mathbb{Q}$ removes this torsion and leaves the same rational rank. This gives an explicit arithmetic distinction invisible to the real matrix rank alone.

Worked example

Apply$M$ repeatedly and calculate the invariant subspace and coinvariants.

$M^n=I+nN=[[1,n],[0,1]]$ for every integer$n$. Invariants areQe_1; coinvariantsV/NV are generated by$[e_2]$. They each have dimension one, yet one is a subspace and the other a quotient. Confusing them would erase the nonzero connecting map$N$.

The example supplies explicit degeneration linear algebra instead of an undefined list of corrected filtrations.

Scope

The computation is not a proof of the geometric monodromy theorem or the Clemens-Schmid sequence. A genuine limiting mixed Hodge structure must satisfy its Hodge and polarization requirements; attaching the word quantum does not provide them.

Further development

For a supplied nilpotent matrix, compute its Jordan strings. A length$r$ string receives centered weights$1-(r-1),1-(r-3),\ldots,1+(r-1)$. Prove the graded$N^j$ isomorphisms string by string and then test compatibility with an independently specified Hodge filtration.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R005.

Y25-01-002 A kernel transform is an equivalence only when it has an inverse

2025-01-11 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which explicit kernel operation is an autoequivalence, and what distinguishes that theorem from merely adding a correction term?

On a two-point space the bounded derived category can be classified completely. Equivalences permute the two simple objects and shift them; a shift-and-swap gives an exact involution.

Definitions and assumptions

Letk be a field andD=D^b(Vec_k×Vec_k), represented by pairs of bounded finite-dimensional complexes.

Write[1] for the cohomological shift. Each complex over a field is isomorphic in the derived category to the direct sum of its cohomology spaces placed in their degrees.

The simple objects areS_1=(k,0),S_2=(0,k). Their shifts are the indecomposable objects, and different simples have no morphisms in any degree.

Classification in a finite derived model

> Everyk-linear triangulated equivalence ofD permutesS_1,S_2 and assigns an integer shift to each. In particularF(V,W)=(W[1],V[−1]) is an equivalence withF^2≅Id.

Derivation or proof

Split every differential into its image, a complement in its kernel and a remaining complement; the image/complement pairs form contractible two-term summands. This leaves precisely the graded cohomology, proving the decomposition. An equivalence preserves indecomposability and all graded morphism spaces. It therefore sends eachS_i to a shifted simple. The two images cannot be shifts of the same simple, since then some shifted morphism between them would be nonzero although every$\mathrm{Hom}(S_1,S_2[n])$ vanishes. Essential surjectivity forces both simple types to occur. Conversely a permutation with prescribed shifts has an inverse permutation with the opposite shifts and acts componentwise on morphisms, so it is an equivalence. For the displayedF, applying it twice cancels both shifts.

Worked example

TakeV=k in degree zero and$W=k^2$ in degree zero.

F(V,W) has its two-dimensional first component shifted by one and its one-dimensional second component shifted by minus one. ApplyingF again restores both original complexes exactly up to the standard shift isomorphisms. The operationG(V,W)=(V⊕W,0), by contrast, loses the distinction betweenS_1 andS_2 and has no inverse.

An equivalence is supported by a functor, an inverse and their composition identities, not by the existence of a suggestive kernel notation.

Scope

This finite classification is not a Fourier-Mukai theorem for arbitrary varieties. Coherent sheaves on positive-dimensional spaces have extensions and support geometry absent here. The historical quantum-derived categories were not defined sufficiently to reconstruct them.

Further development

For a genuine integral transformΦ_K, specify the source/target categories and convolution of kernels, then proveK*L andL*K are diagonal identity kernels. A computable finite model can test these identities before any geometric generalization is claimed.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R006.

Y25-01-003 Noncommutative observables still have identities and symmetries

2025-01-15 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which algebraic property separates jointly measurable finite observables, without discarding identity operators or symmetry?

Commuting self-adjoint matrices admit a common orthonormal eigenbasis. Noncommuting examples retain an identity and can have exact unitary symmetries; their uncertainty is a quantitative statement about a state.

Definitions and assumptions

An observable in this finite model is a self-adjoint complex matrix. Joint diagonalization means one unitary basis makes every matrix diagonal.

Use PauliX=[[0,1],[1,0]],Z=diag(1,−1),Y=[[0,−i],[i,0]], and identityI.

For normalizedψ, variance is$(\Delta A)^2=||(A-\langle A\rangle)\psi||^2$. A symmetry may act by unitary conjugation.

Joint diagonalization and a sharp uncertainty example

> A finite commuting family of self-adjoint matrices is simultaneously unitarily diagonalizable. Forψ=(1,i)/√2,ΔX=ΔZ=1 and$|\langle[X,Z]\rangle|/2=1$, attaining the Robertson lower bound.

Derivation or proof

Diagonalize one self-adjoint matrix. Every matrix commuting with it preserves each eigenspace, becauseA(Bv)=B(Av)=λBv. Restrict the remaining self-adjoint matrices to those invariant eigenspaces and iterate; their orthonormal eigenbases combine into the required basis. Conversely diagonal matrices commute. For the inequality putu=(A−⟨A⟩)ψ andv=(B−⟨B⟩)ψ. Cauchy-Schwarz gives$||u||||v||\geq|\mathrm{Im}\langle u,v\rangle|=|\langle[A,B]\rangle|/2$. With the statedψ,⟨X⟩=⟨Z⟩=0,$X^2=Z^2=I$ and[X,Z]=−2iY whileYψ=ψ. All claimed numbers follow directly. The same noncommutative algebra still containsI, andXZX=−Z is an exact conjugation symmetry.

Worked example

CompareI,Z withX,Z and evaluate their products on the chosen state.

I andZ share the standard basis. In contrastXZ=−ZX, so no common eigenbasis exists. NeverthelessI commutes with everything, each Pauli matrix is unitary, and conjugation byX exchanges the twoZ

eigenstates. Noncommutativity therefore does not imply absence of identity, symmetry or algebraic consistency.

The original contrast becomes a theorem about simultaneous diagonalization and state-dependent variance.

Scope

The finite result does not settle interpretations of quantum measurement or a Platonic ontology. Unbounded observables need common domains; zero commutator expectation in one state does not mean the operators commute.

Further development

For an explicit matrix algebra, compute its center and commutant. These identify classical sectors and exact symmetries without assuming that every noncommutative system has the same observable structure. The commutator expectation can vanish in a particular state even for noncommuting operators. For the maximally mixed qubit state I/2, both Pauli variances equal one but the commutator has trace zero. The displayed inequality is then valid but not sharp; operator commutation is stronger than one state’s expectation test.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R002.

Y25-01-004 Closing a curve is distinct from choosing a polygon or a winding

2025-01-15 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What does returning to the starting point establish, and when can a smooth closed path be replaced by a polygon without changing its winding?

A loop is a continuous map with equal endpoints; it need not be polygonal or noncontractible. A uniformly close polygonal approximation preserves winding whenever the straight-line interpolation avoids the reference point.

Definitions and assumptions

Letγ:[0,1]→C\{0} be a closed piecewise smooth curve andd=min_t|γ(t)|>0.

Letp be a closed polygonal path withsup_t|p(t)−γ(t)|<d. Winding about zero is the integer degree ofγ/|γ|, or its logarithmic-derivative integral when smooth enough.

A polygon additionally consists of finitely many straight segments. Closure alone supplies neither this regularity nor a nonzero degree.

A quantitative polygonal replacement theorem

The interpolationH(s,t)=(1−s)γ(t)+sp(t) is a homotopy of closed curves avoiding zero, soγ andp have equal winding. Every continuous nonvanishing loop admits such a polygonal approximation.

Derivation or proof

The reverse triangle inequality gives|H(s,t)|≥|γ(t)|−s|p(t)−γ(t)|>d−d=0. Endpoints agree for every intermediate curve becauseγ andp are both closed. Homotopy invariance of degree gives equal winding. For existence, uniform continuity provides a partition fine enough thatγ changes by less thanε<d/2 on each subinterval. Linear interpolation between successive sample values then stays within2ε ofγ there; choosing a smallerε gives the required strict bound. This proof explains why polygonal approximation preserves a topological invariant only with a separation margin from the excluded point.

Worked example

Compare the unit circle, the constant loopγ≡1 and the triangle joining the cube roots of unity.

The circle and triangle each wind once around zero; the constant loop has winding zero although it returns to its start. Refining equal-angle inscribed polygons of the circle makes the uniform interpolation error tend to zero. The strict distance condition prevents an approximating edge from cutting through zero.

Strings, loops and polygons are now distinguished by precise properties rather than by an assumed progression from closure to a particular shape.

Scope

Winding is relative to a chosen excluded point. Self-intersections are permitted, and a loop of winding zero need not have a simple image. A path through zero is outside this invariant’s domain.

Further development

For a sampled curve with a known Lipschitz constantL and meshh, bound interpolation error byLh and compare it with a certified distance lower boundd. This produces a finite sampling condition for preserving winding under approximation.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R007.

Y25-01-005 Boundary conditions are part of a spectral problem

2025-01-15 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which operator domain makes an eigenvalue condition a theorem, and how does changing that domain change the spectrum?

A differential expression alone does not specify an operator. The same second derivative has different spectra under Dirichlet and periodic domains, illustrating the domain gap in a proposed zeta Hamiltonian without pretending to resolve it.

Definitions and assumptions

OnL²(0,π), use the expression−u″. The Dirichlet domain isH²∩H₀¹; the periodic domain requiresu(0)=u(π),u′(0)=u′(π).

An eigenpair consists of a nonzero vector in the chosen domain satisfyingAu=λu. Self-adjointness includes equality with the adjoint domain, not merely a symmetric formal expression.

Two domains, two exact spectra

The Dirichlet eigenvalues aren² forn≥1, with eigenfunctionssin(nx). The periodic eigenvalues are4n² forn∈Z, including zero, with eigenfunctionse^{2inx}.

Derivation or proof

Integration by parts gives⟨u,−v″⟩−⟨−u″,v⟩=[u′ conjugate(v)−u conjugate(v′)]₀^π, with the precise conjugation placement adjusted to the inner-product convention. Either stated domain kills this boundary form. Solving−u″=λu gives trigonometric solutions forλ>0; Dirichlet endpoints force√λ to be a positive integer. Periodicity forces e^{i√λπ}=1, hence√λ=2|n|. The constant solves the periodic zero eigenvalue but is excluded by Dirichlet endpoints. Completeness follows from the classical sine and Fourier bases. These regular self-adjoint realizations demonstrate how domain information enters both the eigenvalue equation and its spectral interpretation.

Worked example

Compare the lowest spectral values and boundary behavior.

Dirichlet has lowest eigenvalue one and no constant mode. Periodic has zero followed by four, with the nonzero frequencies occurring in sine/cosine pairs. The same expression therefore gives a different gap and multiplicity structure. A solution of a formal equation that violates an endpoint condition is not an eigenvector of the specified realization.

The repaired question asks for the domain, boundary form and completeness argument before interpreting a spectral list.

Scope

The Bender-Brody-Müller proposal is a specific research construction, not a generic synonym for pseudo-Hermitian matrices. Its primary paper explicitly makes the self-adjointness argument conditional. This example proves neither that construction nor the Riemann hypothesis.

Further development

For a proposed nonlocal Hamiltonian, state a dense invariant domain, prove closability and characterize the adjoint. Then prove that its boundary map both produces the desired zeros and excludes unwanted spectral points; these are separate implications.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R008, R009.

Y25-01-006 A positive metric can repair a matrix inner product

2025-01-15 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

When does a non-Hermitian matrix become self-adjoint in a positive inner product, and which obstruction rules this out?

An explicit similarity gives a positive metric and real spectrum for a triangular matrix. A nontrivial Jordan block cannot have such a metric, so pseudo-Hermitian notation alone is insufficient.

Definitions and assumptions

LetA_s=[[1,s],[0,2]] for reals andS=[[1,s],[0,1]], soA_s=Sdiag(1,2)S^{-1}.

A positive metric is a Hermitian positive-definite matrixη defining⟨u,v⟩_η=u*ηv. Self-adjointness in this inner product isA*η=ηA.

Explicit quasi-Hermitian metric and its boundary

η_s=[[1,−s],[−s,1+s^2]] satisfiesA_s*η_s=η_sA_s and is positive definite. A matrix with a nontrivial Jordan block and real eigenvalue admits no positive metric making it self-adjoint.

Derivation or proof

Setη=(S^{-1})*S^{-1}. Thenv*ηv=||S^{-1}v||²>0 for nonzerov. SubstitutingA=SDS^{-1}, withD real diagonal, provesA*η=ηA. For the obstruction, a positiveη has a positive square root. The transformed matrixB=η^{1/2}Aη^{−1/2} satisfiesB*=B, hence is unitarily diagonalizable and has real eigenvalues. Similarity then makesA diagonalizable too. Thus a Jordan defect cannot be hidden in a positive change of inner product. ForJ=[[1,1],[0,1]], writing a real symmetricη=[[a,b],[b,c]] givesJ^Tη−ηJ=[[0,−a],[a,0]], forcinga=0 and contradicting positivity. The general similarity argument also covers complex Hermitian metrics.

Worked example

Takes=1 and compare the two eigenvectors with their ordinary and metric inner products.

Eigenvectorsv_1=(1,0),v_2=(1,1) have ordinary inner product one. Underη=[[1,−1],[−1,2]], their inner product is zero and both have metric norm one. The eigenvalues remain one and two because a change of inner product changes orthogonality, not the algebraic characteristic polynomial. In ordinary coordinates e^{itA_s} has off-diagonal entry s(e^{2it}−e^{it}). At t=π and s=1 its matrix is [[−1,2],[0,1]], whose Euclidean operator norm is 1+√2, although the evolution has norm one in the η metric. Thus real eigenvalues do not alone control Euclidean transient amplification.

This is a full finite positive-metric construction with an explicit failure case, rather than a conjectural appeal to spectral reality.

Scope

In infinite dimensions, a formal similarity may be unbounded, fail to preserve domains or lack a bounded inverse. The matrix theorem cannot silently be transferred to a proposed zeta Hamiltonian.

Further development

Quantify stability through the condition number ofS ornη. Even real spectra can have large transient Euclidean amplification in nonnormal coordinates; compare||e^{itA_s}|| with its unit norm in thenη metric.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R008, R010.

Y25-01-023 A modular Hamiltonian is defined by a state

2025-01-15 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What mathematical relation does a modular Hamiltonian have to entropy, and which extra data are needed before inserting it into a geometric action?

For a positive density matrix the modular Hamiltonian is−logρ. In a commuting finite model its entropy variation is exact to first order, and the negative second variation quantifies concavity. This is distinct from the weight transformation of a modular form.

Definitions and assumptions

Letρ=diag(p_1,…,p_n) withp_i>0 andΣp_i=1. SetK=−logρ andS(ρ)=−Tr(ρlogρ).

A commuting perturbation isρ(ε)=diag(p_i+εv_i), withΣv_i=0 and positivity for sufficiently smallε.

The word modular in this operator definition does not by itself mean a function transforms under$SL_2(Z)$.

Entropy first variation and strict concavity

S′(0)=Tr(δρK)=−Σv_i logp_i, andS″(0)=−Σv_i²/p_i. The second derivative is strictly negative for every nonzero trace-zero diagonal perturbation.

Derivation or proof

Differentiate−(p_i+εv_i)log(p_i+εv_i). The derivative is−v_i(log(p_i+εv_i)+1). At zero the terms−Σv_i cancel by normalization, giving the first-law identity. Differentiating again gives the negative sum. It is strictly negative unless everyv_i is zero. For a Gibbs stateρ=e^{−βH}/Z, functional calculus givesK=βH+logZ·I, so the trace-zero perturbation law becomesδS=βδ⟨H⟩ whenβ,H are held fixed in that variation. These are consequences of the stated state and normalization, not a prescription for an arbitrary objectH_mod(R) in a gravitational action.

Worked example

Usep=(3/4,1/4) andv=(1,−1).

The entropy derivative is$-\log 3$, while the second derivative is$-(4/3+4)=-16/3$. A perturbation toward the more uneven distribution lowers entropy. The formula requires$p_i>0$; at an eigenvalue zero,$\log p$ is unbounded on that eigenspace and the differentiability argument changes.

The construction provides a real operator and variation theorem while keeping two different uses of modular terminology separate.

Scope

This finite commuting derivation does not define a modular gravitational action, renormalized black-hole entropy or a modular Hamiltonian for a quantum field region. Matrix positivity and trace normalization are essential.

Further development

For noncommuting perturbations, use the Fréchet derivative of$\log p$ or its resolvent integral to derive the second variation. State a positive lower spectral bound to obtain quantitative continuity and concavity estimates. For the same diagonal family, relative entropy $D(p+\varepsilon v \,||\, p)$ has zero first derivative and second derivative $\Sigma v_i^2/p_i$. This positive quadratic form is the Fisher metric on trace-zero directions. It is the negative of the entropy Hessian here because the reference distribution is fixed in the relative-entropy expression.

Source receipt: months/01/root_followups.json record 0. Mathematical references: R011.

Y25-01-007 Moonshine coefficients are graded dimensions and traces

2025-01-17 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What information is contained in a graded character, and how can character arithmetic be verified before attaching a physical interpretation?

A graded trace separates dimension from the action of a symmetry. A smallS_3 representation gives a complete exact decomposition, while the familiar moonshine dimensions are retained as representation-theoretic data.

Definitions and assumptions

For a graded representation$V=\oplus V_n$ of a finite group, its character series is$\mathrm{Tr}_V(gq^{L_0})=\Sigma \mathrm{Tr}(g|V_n)q^n$, formally when every graded piece is finite.

At$g=1$ the coefficients are dimensions. Away from the identity they are traces, which can differ even for representations of equal dimension.

TheS_3 classes are identity, transpositions and three-cycles, with sizes1,3,2. Its standard character is$(2,0,-1)$.

An exact tensor-character decomposition

For the two-dimensional standard representationW ofS_3, $W \otimes W \cong 1 \oplus \mathrm{sgn} \oplus W$. At the dimension level the first moonshine coefficient satisfies$196884 = 1 + 196883$; the second satisfies$21493760 = 1 + 196883 + 21296876$.

Derivation or proof

The permutation representation on three coordinates splits into the constant line and the plane with coordinate sum zero. Counting fixed coordinates gives its trace(3,1,0), so subtracting the constant line givesW character(2,0,−1). Tensor traces multiply, makingW⊗W character(4,0,1). Summing the characters of1,sgn andW gives(1,1,1)+(1,−1,1)+(2,0,−1)=(4,0,1). Finite-group character orthogonality, proved by averaging matrix coefficients, makes equality of characters equivalent to isomorphism overC. This is a genuine representation calculation; the moonshine dimension sums alone do not prove the existence of a vertex operator algebra or identify physical particle states.

Worked example

Inspect the tensor trace on a transposition and a three-cycle.

The tensor traces are zero and one, despite the representation having dimension four. The decomposition reproduces both:1−1+0=0 and1+1−1=1. A count of conjugacy classes describes how many irreducible characters exist, not a count of energy levels, measured transitions or spatial directions.

Graded characters provide exact constraints, with their mathematical and observational meanings kept separate.

Scope

The Monster moonshine theorem and construction ofV^natural are deep established results, not proved by two integer equalities. The old links to biology, gravity and character-ratio constants are not inherited.

Further development

Given a finite graded trace table, compute each irreducible multiplicity by the class-weighted character inner product. Nonintegral or negative multiplicities reject a proposed representation immediately; passing these tests still does not supply its multiplication or VOA axioms.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R012.

Y25-01-008 Modular invariance has a concrete weight and coefficient test

2025-01-17 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

How do theE_4 and discriminant expansions producej, and what modular-invariance statement follows from their weights?

The classical quotientj=E_4^3/Δ has weight zero. Formal division recovers its first coefficients exactly, providing arithmetic content without deriving a conformal field theory from a numerical coincidence.

Definitions and assumptions

Useq=e^{2πiτ}, $E_4=1+240q+2160q^2+6720q^3+\ldots$, and$\Delta=q-24q^2+252q^3-1472q^4+\ldots$.

A weightk modular form transforms byf((aτ+b)/(cτ+d))=(cτ+d)^k f(τ), for matrices in$SL_2(Z)$. E_4 has weight four andΔ weight twelve.

The normalized moonshine series is$J=j-744$, so a constant shift distinguishesj fromJ.

Weight-zero quotient and exact initial Fourier coefficients

Where$\Delta\neq 0$, j transforms invariantly under$SL_2(Z)$. Its expansion begins$q^{-1}+744+196884q+21493760q^2$. ConsequentlyJ has zero constant term and the same positive-power coefficients.

Derivation or proof

The transformation factors ofE_4^3 andΔ both have exponent twelve, so their quotient factors cancel. This gives the invariance directly from the stated modular-form theorem. For the arithmetic calculation, $E_4^3=1+720q+179280q^2+16954560q^3+\ldots$. Writej=$q^{-1}+c_0+c_1q+c_2q^2+\ldots$ and multiply byΔ. Matchingq terms gives$c_0-24=720$, so$c_0=744$. Matchingq^2 gives$c_1-24\cdot 744+252=179280$, yielding196884. Matchingq^3 gives$c_2-24\cdot 196884+252\cdot 744-1472=16954560$, yielding21493760. Every step uses finite exact coefficient arithmetic; no floating-point fitting is involved.

Worked example

Comparej andJ and test the basic translation$\tau\mapsto\tau+1$.

Translation leavesq unchanged, so each displayed Fourier series is invariant under that generator. Inversion$\tau\mapsto-1/\tau$ follows from the weight cancellation, not from a finite truncation. Subtracting744 changes only the graded constant coefficient, which matters when matching a proposed state-space character.

The result explains exactly which modular statement is proved and which coefficients are certified.

Scope

A finiteq-series match is not a proof of a global modular identity unless a theorem such as a dimension or Sturm bound supplies the upgrade. Modular invariance alone does not construct a positive Hilbert space, OPE or physical measurement model.

Further development

For a proposed character vector, state its representation of$SL_2(Z)$, transformation law and pole orders. Then use an appropriate finite-dimensional modular-form bound to determine whether a certified finite coefficient comparison proves an identity. A stronger uniqueness argument is possible after compactification: if two modular functions on a compact modular curve have the same principal parts at every pole, their difference is holomorphic and hence constant by the maximum principle. Fixing one constant coefficient then determines the difference. The pole hypotheses are essential to this upgrade.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R013.

Y25-01-009 Projective phases require a genuine two-cocycle

2025-01-18 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which phase factors define an associative projective representation, and how can one prove that they cannot be removed by rephasing?

A two-cocycle on$C_2 \times C_2$ has an explicit Pauli realization and a nontrivial commutator phase. This supplies a rigorous model of projective symmetry instead of assuming a claimed Monster cohomology value.

Definitions and assumptions

Let$G = F_2^2$ under addition and$\sigma((a,b),(c,d)) = (-1)^{bc}$. Define$U_{(a,b)} = X^a Z^b$ using the Pauli matrices.

A normalized two-cocycle satisfies$\sigma(g,h)\sigma(g+h,k) = \sigma(h,k)\sigma(g,h+k)$ and$\sigma(0,g) = \sigma(g,0) = 1$.

RephasingU_g by a nonzero scalar changesσ by a coboundary. On an abelian group every scalar coboundary has symmetric values in its two arguments.

A nontrivial projective representation with an exact certificate

> $U_g U_h = \sigma(g,h) U_{g+h}$, and$\sigma$ is a two-cocycle that is not a coboundary. Thus the projective action cannot be made an ordinary representation by choosing new scalar phases.

Derivation or proof

The relation$ZX = -XZ$ gives the displayed multiplier when movingZ^b pastX^c. For triple elements$g=(a,b), h=(c,d), k=(e,f)$, the cocycle exponents are$bc + (b+d)e$ and$de + b(c+e)$, equal modulo two. This proves associativity of the central-extension multiplication. For the obstruction choose$g=(1,0), h=(0,1)$. Their multiplier ratio$\sigma(g,h)/\sigma(h,g) = -1$. Replacing$\sigma$ by a coboundary cannot change that ratio because the group is abelian and a coboundary$c(g)c(h)/c(g+h)$ is symmetric. Hence the class is nonzero. The eight matrices$\{\pm I, \pm X, \pm Z, \pm XZ\}$ form the corresponding noncommutative central extension. The extension is non-split: a splitting would give commuting lifts of the two quotient generators, whereas every scalar rephasing leaves their commutator equal to$-I$. This demonstrates a group-level obstruction separately from the already checked cocycle identity.

Worked example

Compute the commutator of the lifts of the two order-two generators.

Both$X$ and$Z$ square to$I$, but$XZX^{-1}Z^{-1} = -I$ and$(XZ)^2 = -I$. Their images commute after quotienting by$\{\pm I\}$, exactly recovering$G$. The lift group has order eight, while its quotient has four elements. Orbit-stabilizer likewise requires a specified action; the order of an ambient group alone does not determine orbit sizes.

A central phase now has a checked algebraic origin and a concrete obstruction to removal.

Scope

This example concerns$G=F_2^2$. It does not establish the historical assertion$H^2(M,C^*)=Z/2$ for the Monster; a group-specific Schur-multiplier calculation must be cited separately. No continuous spacetime symmetry follows from this finite extension.

Further development

Given a proposed finite multiplier table, test the cocycle identity exhaustively, compute its alternating commutator form and solve the rephasing equations. This separates a removable convention from a nontrivial projective class.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R014.

Y25-01-010 Six pentads give an exhaustive outer automorphism certificate

2025-01-21 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Can the exceptional symmetry ofS_6 be exhibited by a finite construction rather than inferred from an analogy with particle generations?

Perfect matchings ofK_6 form fifteen synthemes and six complete factorizations called pentads. Permuting the six original vertices acts on these six pentads and gives an outer automorphism, with all720 images independently enumerable.

Definitions and assumptions

A duad is an unordered pair of six labels. A syntheme is a partition of those labels into three duads, equivalently a perfect matching ofK_6.

A pentad is a set of five synthemes whose fifteen edges are pairwise distinct, hence cover all edges ofK_6 exactly once.

An inner automorphism ofS_6 is conjugation by a permutation and therefore preserves cycle type.

Explicit exceptional permutation action

There are fifteen synthemes and six pentads. The action$S_6 \to \mathrm{Sym}(\text{pentads})$ is bijective. A vertex transposition acts on pentads as three disjoint transpositions, so this automorphism is outer.

Derivation or proof

Construct a matching by pairing the least unused vertex with each remaining vertex and recursing; there are$5\cdot3\cdot1=15$ outcomes. Enumerate all five-subsets of these fifteen matchings and retain exactly those whose edge union has size fifteen. This finite exhaustive procedure yields six pentads, retained in the verification certificate. Relabeling vertices preserves disjointness and coverage, so it permutes the six retained objects and defines a homomorphism. Enumerating all720 vertex permutations yields720 distinct permutations of the pentads, proving injectivity and, since the target also has order720, surjectivity. The image of a transposition has no fixed pentad and order two, so its cycle type is2^3. Conjugation cannot take cycle type$2\cdot1^4$ to2^3, proving outerness. The construction and exhaustive checks together are a finite proof certificate; the source supplies no physical identification for this automorphism.

Worked example

Apply the label swap$(0\ 1)$ to each of the six certified pentads.

The pentads pair into three exchanged pairs. Applying the swap twice restores every matching, but one application fixes none of the six pentads. This contrasts with its four fixed original labels and makes the change of cycle type directly visible.

The exceptional automorphism is now fully specified by finite combinatorial data and a complete enumeration.

Scope

The numbering of the six pentads chooses a particular automorphism; different numberings differ by an inner automorphism. This does not identify particle families, Monster representations or a continuous gauge group.

Further development

Construct the induced permutation on conjugacy classes and check that transpositions exchange with triple transpositions. Then restrict toA_6 and distinguish inner automorphisms ofA_6 from those induced by odd permutations ofS_6.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R015.

Y25-01-012 Platonic boundaries share homology but have different symmetry groups

2025-01-24 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which topology remains unchanged between a sphere and a convex polyhedral boundary, and which symmetry data can distinguish the Platonic cases?

Radial projection identifies every convex polyhedral boundary surrounding the origin withS^2. Its homology is therefore the same across the Platonic solids, while face counts and rotational group orders distinguish their geometry.

Definitions and assumptions

LetP⊂R³ be a convex polytope containing zero in its interior. Every ray meets∂P in exactly one point.

Use homology over a field. The boundary is a two-dimensional cell complex; including the filled interior changes the space to a contractible ball.

For a regular solid withF regularp-gon faces, rotations act transitively on faces and a face stabilizer has orderp.

Topological invariance and finite symmetry counts

∂P is homeomorphic toS² and has Betti numbers(1,0,1). The tetrahedron,cube,octahedron,dodecahedron,icosahedron have(V,E,F) respectively(4,6,4),(8,12,6),(6,12,8),(20,30,12),(12,30,20), with rotational group orders12,24,24,60,60.

Derivation or proof

Map a boundary pointx tox/||x||. Convexity and the interior-origin condition give a unique positive intersection on every ray, so this map is bijective. It is continuous from a compact space to the Hausdorff sphere, hence a homeomorphism. The classical homology ofS² gives the Betti numbers. For the listed regular solids, each face hasp edges and every edge bounds two faces, soFp=2E. Their Euler countsV−E+F=2 agree with the topological result. Orbit-stabilizer gives rotation orderFp: a face can be sent to any ofF faces, withp rotations preserving its center and cyclically permuting its vertices. In the tetrahedral case, the induced even vertex permutations realizeA_4; cube/octahedron rotations realizeS_4, and the remaining pair realizesA_5.

Worked example

Build the tetrahedron boundary chain complex and compare it with the filled tetrahedron.

The boundary has chain dimensions4,6,4 and boundary ranks3,3, giving Betti numbers1,0,1. Adding the single3-cell has nonzero boundary equal to the oriented sum of the four faces, killingH_2 and leaving onlyH_0. This is an exact certificate of the difference between boundary and filling.

Homology alone cannot distinguish the five regular boundaries, but the underlying cell and group structures can.

Scope

The historical22-equation list is truncated; its missing members are not reconstructed. The argument requires convexity and an interior reference point; an arbitrary singular wave level set need not be a sphere.

Further development

For an explicitly supplied nonconvex or singular mesh, verify its boundary maps and manifold links before using Euler characteristic as a topology classifier. Add the actual symmetry action as separate data when homology is insufficient.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R007.

Y25-01-013 A scalar wave potential has critical points, not a square Jacobian

2025-01-24 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which derivative detects a singular level of a scalar wave field, and how is that different from the Jacobian determinant of a vector map?

A scalar field onR^3 has a gradient and Hessian. A determinant belongs to a square derivative of a declared three-component map. Two explicit examples distinguish critical values, Morse indices and fold singularities.

Definitions and assumptions

For$\Phi:R^3\to R$, a level$\Phi=c$ is regular when$\nabla\Phi\neq0$ at every point on that level.

For$F:R^3\to R^3$, its derivative is3×3 and$\det DF$ detects rank loss. These are different maps and cannot be denoted by an untyped determinant of one scalar.

OnT^3 use$\Phi(x,y,z)=\cos x+\cos y+\cos z$, with coordinates modulo2π.

Complete critical-point calculation and fold model

The potential has eight critical points, each coordinate zero orπ. Its critical values are$3,1,-1,-3$ and all eight points are Morse. The vector map$F(x,y,z)=(x^2,y,z)$ has singular plane$\{x=0\}$ and critical-value plane$\{u=0\}$.

Derivation or proof

The gradient is$(-\sin x,-\sin y,-\sin z)$, so it vanishes precisely at the eight listed points. The Hessian is$\operatorname{diag}(-\cos x,-\cos y,-\cos z)$, with every diagonal entry±1 there, so its determinant is nonzero. A zero coordinate contributes a negative eigenvalue; the index counts the number of zero coordinates. There are$\binom{3}{k}$ points of index$k$. Their alternating count$1-3+3-1=0$ matches$\chi(T^3)$. For the vector fold, $DF=\operatorname{diag}(2x,1,1)$, so rank drops exactly at$x=0$. An image point(u,v,w) has two preimages for$u>0$, one for$u=0$ and none for$u<0$. This is a complete local multiplicity change, obtained from a correctly typed determinant.

Worked example

Compare the maximum ofΦ at$(0,0,0)$ with a point of the fold plane.

The potential maximum has value three and Hessian$-I$, so its scalar level becomes singular there. The fold map instead has derivative rank two at$(0,v,w)$, with a one-dimensional kernel in the$x$ direction. The two tests answer different questions even though both involve a loss of regular behavior.

Wave geometry is now expressed through explicit functions and regularity hypotheses rather than an undefined singular-locus symbol.

Scope

The chosen cosine field is a new tractable example, not a recovered experimental reconstruction. Morse counts do not identify a derived stack, intersection complex or a sound-driven geometry.

Further development

For a supplied finite wave sum, compute its gradient and Hessian symbolically, isolate all critical points and certify the Hessian signs. For a vector-valued measurement map, instead study its rank strata and the stability of inverse branches.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R016.

Y25-01-014 Intersection homology retains cycles that an ordinary cone contracts

2025-01-24 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What does a perversity allow near a singular point, and can it change the homology of a contractible cone?

The open cone on a torus is contractible in ordinary topology, yet lower-middle intersection homology retains two first-homology classes. The discrepancy comes from which bounding chains may touch the cone point.

Definitions and assumptions

Let$L=T^2$ and$X=cL$ be its open real three-dimensional cone, with the apex as a codimension-three singular stratum.

Use field coefficients and lower-middle perversity $p(3)=0$. An allowable$i$-chain may meet the apex only in dimension at most$i-3$; its boundary must satisfy the analogous condition.

This is a fixed stratified pseudomanifold and perversity, not an unspecified object$IC(X)$.

Explicit cone-formula instance

$IH_0(X)=k$,$IH_1(X)=k^2$ and$IH_i(X)=0$ for$i\geq 2$ under the stated perversity. Ordinary$H_1(X)=0$ because the whole cone contracts.

Derivation or proof

Allowable chains in dimensions zero,one and two cannot meet the apex: their permitted intersection dimensions are negative. Thus first cycles and their allowable bounding two-chains lie in$X\setminus\{\text{apex}\}=T^2\times(0,\infty)$, which retracts onto$T^2$. Its two independent one-cycles therefore survive inIH_1. A two-cycle in the link can be coned to an allowable three-chain, because a three-chain may meet the apex at a zero-dimensional set. This kills the link's fundamental two-cycle. More generally the standard cone contraction applies above the cutoff; the cone formula states$IH_i(cL)=IH_i(L)$ for$i<\dim L-p(3)=2$ and zero above. The chain explanation proves the low-degree distinction directly and identifies why an ordinary apex-crossing disk is inadmissible.

Worked example

Take the two coordinate circles ofT^2 at a fixed cone radius.

They generate a two-dimensionalIH_1. In ordinary homology each bounds its cone disk, but that two-chain meets the apex and violates$2-3<0$. The link's two-dimensional fundamental cycle does bound its three-dimensional cone, where$3-3=0$ allows the vertex. The verifier checks the torus chain ranks and ordinary cone contraction as finite supporting calculations.

Perversity controls allowable fillings, so contractibility does not force this refined homology to vanish.

Scope

Changing to upper-middle perversity $p(3)=1$ changes the cutoff and removes the retainedH_1. Shifts and indexing of an intersection complex must also be specified before equating its hypercohomology to anIH group.

Further development

Construct an explicit triangulated link and its cone, label singular strata, and implement allowable-chain restrictions degree by degree. Compare the resulting homology with the stated cone formula before using it in a larger singular space.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R017.

Y25-01-015 The derived tangent complex comes from the cotangent complex

2025-01-26 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which complex captures infinitesimal deformations and obstructions at a singular point?

For a hypersurface the cotangent complex is an explicit two-term derivative map. At a double point it detects both a tangent direction and an obstruction, information not supplied by$RHom(O_X,O_X)$.

Definitions and assumptions

Over a characteristic-zero field$k$ let$A=k[x]/(x^2)$, with point$x=0$.

For a hypersurface the cotangent complex is$[I/I^2\to\Omega_{k[x]/k}\otimes A]$ in degrees$-1,0$. Here it is$[A\to A]$ with differential multiplication by$2x$.

The tangent complex at the point is$RHom_A(L_{A/k},k)$, with cohomology in degrees zero and one.

Double-point deformation and obstruction

At$x=0$ the tangent-complex differential is zero, so$H^0\cong k$ and$H^1\cong k$. Every first-order displacement$x=at$ satisfies$x^2=0$ modulot^2, but it extends to$x=at+bt^2$ modulot^3 only if$a^2=0$.

Derivation or proof

The relation generatorx² differentiates to2x dx, giving the two-term cotangent map. Tensoring with the residue field killsx, and dualizing givesk→k with zero differential in degrees0,1. For an explicit deformation, substituteat into the defining relation overk[t]/t²: its square vanishes automatically. Overk[t]/t³, the square ofat+bt² isa²t², so extension requiresa²=0. For a field this forcesa=0, exhibiting the nonlinear obstruction to a nonzero first-order tangent direction. The complex predicts a possible obstruction space; the equation computes the actual obstruction mapa↦a². By contrastRHom_A(A,A)=A in degree zero becauseA is a free module over itself, so it misses this deformation structure.

Worked example

Compare with the smooth hypersurfacek[x]/(x), whose derivative map is multiplication by one.

The smooth one-point scheme has zero tangent complex: the mapk→k is an isomorphism. The double point has the same underlying topological point but nontrivial tangent and obstruction spaces. Thus topology of the underlying set cannot recover its derived infinitesimal geometry.

The correction supplies a concrete deformation problem and an actual failed lift, not only a replacement symbol.

Scope

The displayed hypersurface model uses a regular-sequence presentation. General derived schemes require the full cotangent complex and derived base change. A nonzero obstruction space does not mean every deformation is obstructed.

Further development

Forf₁,…,f_r forming a regular sequence, compute the Jacobian mapkⁿ→kʳ at a point. Its kernel and cokernel give tangent and first obstruction spaces; then compute higher-order lift equations to distinguish a potential obstruction from a nonzero one.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R018.

Y25-01-016 Perverse degree is fixed by support and cosupport

2025-01-26 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which degree shift makes a local system perverse on a smooth complex curve, and what changes for a point-supported object?

Middle perversity is a pair of support conditions, not a freely adjustable label. On a smooth complex curve a local system shifted by one is perverse, while a skyscraper is perverse in degree zero.

Definitions and assumptions

Work with bounded constructible complexes of finite-dimensionalk-vector spaces on a smooth complex algebraic curve C.

Use the middle-perverse criteriondim_C supp H^i(K)≤−i and the same inequalities for Verdier dualD K.

For a smooth oriented real two-manifoldD(k_C)=k_C[2]; for a point-supported finite-dimensional sheaf, Verdier duality takes its vector-space dual without a shift.

Curve and point perverse objects

A finite-rank local systemL[1] is perverse onC. A skyscraperk_p is perverse. The unshifted nonzero local systemL is not perverse.

Derivation or proof

The only nonzero ordinary cohomology sheaf ofL[1] lies in degree−1 and has one-dimensional support, exactly satisfying1≤−(−1). Its Verdier dual isL^v[1], which satisfies the same inequality. A skyscraper has degree-zero cohomology supported in dimension zero and is self-dual up to vector-space dual, so both inequalities hold. The unshiftedL instead has degree-zero support of dimension one, violating1≤0. This computation checks both halves of the definition and demonstrates why ordinary cohomological degree and perverse degree differ. Direct sums of these two types remain perverse because support dimensions take maxima and duality distributes over finite sums.

Worked example

TakeK=k_C[1]⊕k_p and compute its ordinary cohomology sheaves.

$H^{-1}(K)=k_C$ and $H^{0}(K)=k_p$, with support dimensions one and zero. Both are allowed simultaneously. Moving the skyscraper to degree one would give a nonzero degree-one sheaf with support dimension zero, violating0≤−1. Thus even a tiny complex has a sharply constrained perverse grading.

The repaired formulation gives a testable definition and two explicit objects rather than assuming a quantum perverse category exists.

Scope

The general theory requires constructibility, duality and a chosen perversity. An extra parameterq must come with a defined category or functor and a proof of compatibility; the recovered symbolP_q(T) does not supply those data.

Further development

For an open inclusionj:C\{p}→C, comparej_!L[1],Rj_*L[1] and the intermediate extensionj_!*L[1]. Compute local monodromy invariants and coinvariants to identify which point-supported subobjects or quotients are removed.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R019.

Y25-01-017 A persistence module retains finite bars that its colimit forgets

2025-01-26 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which object records birth and death times, and why is a terminal colimit not an adequate definition of persistence?

A finite triangle filtration has a one-cycle born at time two and killed at time three. ItsH_1 colimit is zero despite a nontrivial persistent interval.

Definitions and assumptions

LetX_0 contain three vertices. At time one add edgesAB andBC; at time two addAC; at time three fill the triangleABC.

Persistent homology is the vector-space diagram$t \mapsto H_i(X_t;k)$ together with maps induced by inclusions.

An interval module$k[b,d)$ has a one-dimensional space for $b \le t < d$, zero otherwise, and identity maps within its lifetime.

Complete barcode of the triangle filtration

H_0 consists of one interval$[0,\infty)$ and two intervals$[0,1)$. H_1 consists of$[2,3)$. Higher homology vanishes. The colimitH_1 is zero, while the rank of$H_1(X_2) \to H_1(X_2)$ is one.

Derivation or proof

At time zero all three vertices are separate, soH_0 has dimension three. The two added edges form a tree, whose incidence matrix has rank two, merging all components without creatingH_1. The third edge raises the edge count to three while the incidence rank stays two, producing one cycle. Adding the face gives a nonzero boundary equal to that cycle, so it dies at time three. All inclusion maps follow directly from these chain generators. The interval decomposition is therefore explicit rather than merely asserted. The directed colimit identifies every vector with its later image; the cycle maps to zero after the filling and disappears. It records the eventual space but not how long a transient feature existed.

Worked example

Write the three edge boundaries in the vertex basis and the face boundary in the edge basis.

Use$\partial AB = B - A, \partial BC = C - B, \partial AC = C - A$, so the cycle$AB + BC - AC$ has zero boundary. The face boundary is exactly$AB + BC - AC$. Exact rank calculation gives Betti sequencesH_0:$(3,1,1,1)$,H_1:$(0,0,1,0)$.

The reconstructed definition preserves the actual diagram and its maps; replacing it by$\operatorname{colim}_t H^i$ loses the feature this example is designed to detect.

Scope

Interval decomposition here uses a one-parameter finite filtration over a field. Multi-parameter persistence has no general barcode classification by intervals. The motivic adjective requires additional coefficient and functorial data.

Further development

For any supplied filtration, retain the rank invariant rank(H_i(X_s)→H_i(X_t)) as a certificate. Test whether a proposed simplification preserves these ranks, rather than comparing only the final Betti numbers.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R020.

Y25-01-018 A weight filtration need not split canonically

2025-01-26 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What is the difference between a filtration, its associated graded object and a chosen direct-sum splitting?

A two-step filtered vector space admits many splittings, none invariant under all filtration-preserving automorphisms. This elementary example identifies the extra choice suppressed by writing a weight filtration as a direct sum.

Definitions and assumptions

Let$V=ke_1\oplus ke_2$,$W_0=ke_1$,$W_1=W_0$,$W_2=V$, with the evident zero and full extensions outside this range.

The associated graded is$Gr^0_W=ke_1$ and$Gr^2_W=V/ke_1$. A splitting selects a lift of each quotient vector and identifies$V$ with their direct sum.

A canonical splitting would have to be preserved by every automorphism of the given filtered object.

No functorial splitting for this filtered object

Every complement ofW_0 has the form$k(e_2+ce_1)$. The filtered automorphisms$T_a(e_1)=e_1$,$T_a(e_2)=e_2+ae_1$ act transitively on these complements. No complement is fixed by allT_a.

Derivation or proof

A line complementary toke₁ has a vector with nonzeroe₂ coordinate; rescaling it gives e₂+ce₁, proving the classification. ApplyingT_a sends this line to the line with parameterc+a. Choosing any nonzeroa changes it, so no complement can be invariant under the whole automorphism group. EachT_a induces the identity on both graded quotients, showing that the associated graded forgets genuine extension automorphisms. The equality of graded dimensions withdimV therefore does not provide a distinguished decomposition. In mixed Hodge theory additional structures constrain the extensions, and certain splittings may exist after changing scalars or adding hypotheses; those theorems must be stated rather than replacingW_k by an unexplained Hodge-index sum.

Worked example

Compare the complementske₂ andk(e₂+3e₁).

Both project isomorphically toGr²_W and yield identical graded dimensions(1,1). T₃ exchanges the first with the second while acting trivially onGr_W. A numerical routine that returns one complement has selected coordinates, not found an invariant hidden direct-sum decomposition.

The finite calculation explains exactly why filtrations and decompositions encode different information.

Scope

This is a filtered-vector-space theorem, not a classification of mixed Hodge extensions. A pure Hodge decomposition is a different structure and can be canonical under its own hypotheses. The historical formulaW_k=⊕_{p+q=k}H_M^{p,q} conflates indices unless its objects are redefined.

Further development

Add a specified nondegenerate inner product and take the orthogonal complement. Then identify the smaller symmetry group preserving that metric and show the splitting is natural only for that smaller category. This exposes the cost of the additional choice.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R005.

Y25-01-019 Localization gives a valid motivic and cohomological subtraction

2025-01-26 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which additivity law underlies a motivic expression, and how does a concrete open/closed decomposition act on cohomology?

The projective line minus two points gives a fully specified localization example. It has classL−1 and compactly supported cohomology in degrees one and two; no undefined infinite motivic series is needed.

Definitions and assumptions

In the Grothendieck ring of varieties,[X]=[Z]+[X\Z] for a closed subvarietyZ. SetL=[A¹].

LetX=P¹_C,Z={0,∞},U=C*. The open/closed trianglej_!k_U→k_X→i_*k_Z produces a long exact compact-support cohomology sequence.

The existence of this sheaf-theoretic localization sequence is standard background; its maps and dimensions are computed below.

Exact class and compact-support cohomology ofC*

[C*]=L−1. Over a fieldk,H_c¹(C*;k)=k,H_c²(C*;k)=k and the other compact-support groups vanish. Its compact-support Euler characteristic is zero.

Derivation or proof

DecomposeP¹ intoA¹ and the point at infinity to obtain[P¹]=L+1. Removing two points givesL−1. In the long exact sequence, H⁰(P¹)=k maps toH⁰(Z)=k² by the diagonal mapc↦(c,c). Its kernel is zero and cokernel is one-dimensional, soH_c⁰(U)=0 andH_c¹(U)=k. SinceH¹(P¹)=0,H¹(Z)=0,H²(P¹)=k andH²(Z)=0, exactness givesH_c²(U)=k. Hence the Euler characteristic is−1+1=0. The cohomological subtraction includes connecting maps and degrees; it is not literal subtraction of ordinary vector spaces. Its numerical Euler realization agrees with evaluatingL at one in the classL−1.

Worked example

Write the finite matrix for the only nontrivial map in low degrees.

The column(1,1)ᵀ has rank one. The quotientk²/diag(k) is detected by(a,b)↦b−a. Thus two removed points create one compact-support degree-one class, even thoughC* itself has no compact connected component.

This is a constructive interpretation of additivity, six-operation notation and a motivic class in one exact example.

Scope

A motivic integral over arcs additionally needs a measure, cylinder normalization and a completed coefficient ring. The recovered sumΣ[X_n]L^−n does not automatically converge or define that measure. No quantum volume is inferred.

Further development

ForP¹ withr punctures, repeat the same diagonal-map calculation to obtainH_c¹ dimensionr−1 andH_c² dimension one, and verify the classL+1−r. Then state what additional completion would be needed for an actual arc-space sum. Over field coefficients, the compact-support Künneth theorem gives dimensions 1,2,1 in degrees two, three and four for (C*)². Its class is (L−1)² and its Euler characteristic is again zero. These matching additivity and product rules provide a second consistency check without identifying all motivic information with Euler characteristic.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R021, R019.

Y25-01-011 An E_8 projection has exact fibers and changes root lengths

2025-01-28 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What does a specifiedE_8-to-four-coordinate projection actually produce, including multiplicities and zero fibers?

Projecting the standard240 E_8roots onto their first four coordinates gives the48 F_4root vectors with unequal multiplicities, together with zero. The24 long projected roots formD_4, but the entire image is larger.

Definitions and assumptions

TheE_8roots are112 vectors$\pm e_i \pm e_j$ for$i<j$ and128 half-integral vectors$(\pm 1/2,\dots,\pm 1/2)$ with an even number of minus signs.

Let$\pi:\mathbb{R}^8\to\mathbb{R}^4$ retain the first four coordinates. Root length is measured with the standard Euclidean inner product.

TheD_4roots are the24 vectors$\pm e_i \pm e_j$ in four coordinates. F_4 adds the eight vectors$\pm e_i$ and sixteen half-sign vectors.

Complete projected-root fiber distribution

The24 D_4vectors each have one preimage. Each of the24 shortF_4vectors has eight preimages. Zero has24 preimages. Thus$24+24\cdot 8+24=240$, and squared projected lengths are2,1 and0.

Derivation or proof

Integral roots supported among the first four coordinates give$6\cdot 4=24$ D_4vectors. Those supported among the last four give24 zero images. Mixed support selects one of four visible and one of four hidden coordinates and two signs, giving64 roots: for each visible$\pm e_i$ there are four hidden-coordinate choices and two hidden signs, hence eight preimages. For a fixed visible half-sign pattern, the last four signs must have a prescribed parity. Exactly eight of their sixteen patterns satisfy it, producing128 half-root images with multiplicity eight. This exhausts both original root types. The length assertions follow by summing coordinate squares and explicitly show why this projection is not an isometry or a bijection of root systems.

Worked example

Track one long root, one short image and the zero fiber.

e_1+e_2 maps to aD_4root of squared length two. The eight vectors $e_1\pm e_j$ for$j=5,\dots,8$ all map toe_1 of squared length one. Every$\pm e_i\pm e_j$ with$5\le i<j\le 8$ maps to zero. The verifier constructs all240 roots as integer doubled coordinates and records the complete fiber histogram without rounding.

An$E_8\to D_4$ slogan becomes a precise linear map with a complete accounting of what it preserves and collapses.

Scope

The projection does not by itself define a Lie-algebra homomorphism$E_8 \to D_4$ or a branching rule for arbitrary representations. Root-image multiplicities are not particle masses or coupling constants.

Further development

Compare this root projection with restriction of the adjoint representation to an explicitly embeddedD_4subalgebra. Compute its actual weight multiplicities, including Cartan zero weights, rather than treating projected roots as a complete representation decomposition.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R022.

Y25-01-020 A Jacobiator detects nonassociativity only after multiplication is defined

2025-01-28 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Can a claimed nonzero flux Jacobiator be distinguished from the identically zero Jacobiator of associative operator commutators?

Associative commutators satisfy Jacobi by cancellation. An explicit octonion triple has a nonzero Jacobiator, furnishing a concrete nonassociative example without identifying it with a physicalR-flux.

Definitions and assumptions

For a bilinear multiplication define$[a,b]=ab-ba$ and$J(a,b,c)=[a,[b,c]]+[b,[c,a]]+[c,[a,b]]$.

Use octonion units$e_1,\dots,e_7$ with squares-1 and oriented Fano triples$(123),(145),(176),(246),(257),(347),(365)$; reversing an ordered pair changes sign.

The associator is$(a,b,c)=(ab)c-a(bc)$. Parentheses cannot be omitted in a nonassociative product.

An exact nonzero Jacobiator

> For$(a,b,c)=(e_1,e_2,e_4)$, the associator is$2e_7$ and$J=-12e_7$. In every associative algebra the same commutator formula has$J=0$.

Derivation or proof

In an associative algebra expand the three nested commutators. Each of the six ordered triple products occurs twice with opposite signs, giving zero. In the specified octonion table,$e_2e_4=e_6$,$e_1e_6=-e_7$, so$[e_1,[e_2,e_4]]=-4e_7$. Also$e_4e_1=-e_5$,$e_2e_5=e_7$, giving the second term$-4e_7$. Finally$e_1e_2=e_3$,$e_4e_3=-e_7$ gives the third$-4e_7$. Their sum is$-12e_7$. Separately$(e_1e_2)e_4=e_3e_4=e_7$, whereas$e_1(e_2e_4)=e_1e_6=-e_7$, giving associator$2e_7$. This verifies the sign convention rather than appealing to a generic nonassociative slogan. Left multiplication still represents every octonion as a linear map, but it is not a product-preserving representation: $L_{e_1}L_{e_2}e_4=-e_7$ whereas $L_{e_1e_2}e_4=e_7$. Commutators of the linear maps satisfy Jacobi, while the octonion commutators need not. This prevents confusing an operator embedding with an associative algebra embedding.

Worked example

Compare this triple with the quaternionic subalgebra spanned by$1, e_1, e_2, e_3$.

Within that four-dimensional subalgebra multiplication is associative and every commutator Jacobiator vanishes. The chosen triple involvinge_4 leaves that subalgebra and gives the nonzero value above. Thus a large algebra can contain associative sectors without being associative globally.

A nonzero Jacobiator is now attached to a declared multiplication table and a checked triple.

Scope

Ordinary bounded matrices cannot realize this multiplication associatively while preserving all products, since their commutators always satisfy Jacobi. A physicalR-flux interpretation needs a separately defined bracket, observables and consistency rules; it is not derived from the table.

Further development

Test alternativity$(a,a,b)=(b,a,a)=0$ on the full basis and extend by polarization. Then compare the resulting commutator algebra with the Malcev identity, which replaces Lie Jacobi in this setting.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R023.

February 2025

Dated fragments recovered; complete transcript unavailable. 25 developed treatments.

The recovered clusters include weighted Hardy-space and BMO methods, numerical-relativity discretization, braided structures, categorical representation theory and quantitative resource models. Each treatment specifies its own assumptions, normalization and mathematical scope.

ID · date	Concept and revised question	Result and mathematical core
Y25-02-016 2025-02-01 · U	Berry curvature and a Chern number require a state bundle Which explicitly defined connection produces a nonzero topological invariant, and what normalization makes the integral an integer?	Explicit Chern-one line bundle A=sin²(θ/2)dφ,F=(1/2)sinθ dθ∧dφ, and(2π)⁻¹∫_{S²}F=1. No globally smooth one-formA onS² can have this curvature.
Y25-02-002 2025-02-05 · A	A network multiplier is an inverse matrix with a convergence condition Which matrix and convergence condition produce a claimed multiplier, and how does an input shock propagate through its network?	Exact two-sector propagation ForA=[[0,1/4],[1/2,0]],(I−A)⁻¹=[[8/7,2/7],[4/7,8/7]]. A unit shock in sector one gives total output12/7; a unit shock in sector two gives10/7.
Y25-02-003 2025-02-05 · A	Weighted scores and capacity quotas need different constraints How can a capacity-based allocation be formulated so that its units, feasible set and fairness objective are explicit?	Unique concave allocation The log-utility problem has a unique optimumx_i=max(a_i/λ−1,0), withλ>0 chosen soΣx_i=C. For(a₁,a₂)=(2,1),C=2, the allocation is(5/3,1/3).
Y25-02-004 2025-02-05 · A	Debt-to-output dynamics include the primary flow When does a debt ratio fall, and what is missing from comparing growth with the interest rate alone?	Exact ratio recurrence and fixed point b_{t+1}=[(1+r)b_t+p]/(1+g). It decreases exactly whenp<(g−r)b_t. Ifg>r and1+r≥0, the fixed pointb*=p/(g−r) is attracting.
Y25-02-001 2025-02-07 · U	A cumulative wealth equation is a stock-flow identity What does the recovered integral equation imply, and which independent assumptions turn it into a predictive dynamical model?	Exact affine feedback solution Forb≠0,W(t)=(W₀+a/b)e^{bt}−a/b. Forb=0,W=W₀+at. Ifb<0 the equilibrium−a/b attracts exponentially; ifb>0 it is unstable.
Y25-02-006 2025-02-16 · A	A simplified metric RHS has an exact solution but no relativity certificate What can the simplifiedg′=−2A,A′=−κA prototype establish before full geometric evolution terms are supplied?	Exact damped-source evolution A(t)=A₀e^{−κt} andg(t)=g₀−(2A₀/κ)(1−e^{−κt}). ForA₀>0,g stays strictly positive for all finite times ifg₀≥2A₀/κ, with a positive uniform lower bound only when the inequality is strict.
Y25-02-007 2025-02-16 · A	CFL stability and halo exchange have separate exact tests What does a CFL bound prove for one actual discretization, and how can domain decomposition be tested without inventing scaling benchmarks?	Stability interval and decomposition equivalence For0≤ν≤1, the update is anℓ∞ contraction. Forν>1, the alternating Fourier mode has amplification1−2ν with modulus greater than one. Two subdomains with correct old-time halos give the identical update to one serial domain.

ID · date	Concept and revised question	Result and mathematical core
Y25-02-008 2025-02-16 · A	A spline element has exact partition and stiffness certificates Which finite identities should a basis satisfy before its derivative arrays are used in a geometric solver?	Exact local element identities $\Sigma B_i=1, x=(1/2)B_1+B_2$ and $x^2=B_2$. The stiffness matrix has diagonal4/3 and off-diagonal$-2/3$, hence eigenvalues0,2,2.
Y25-02-009 2025-02-17 · A	Weighted atoms must cancel in the same measure as their pairing Which normalization and cancellation make a weighted atomic pairing provably bounded?	Compatible-measure atomic bound $\lvert\int_a^b d\nu\rvert \le \lVert b\rVert_{BMO_2(\nu)}$ and $\lvert\int f b\, d\nu\rvert \le \lVert f\rVert_{H^1_{at}(\nu)} \lVert b\rVert_{BMO_2(\nu)}$ for finite atomic sums.
Y25-02-010 2025-02-17 · U	Hyperbolic volume growth changes global analysis Which Euclidean assumptions fail on the full hyperbolic space, and what spectral statement can be proved in a radial model?	Ball growth and exact radial spectral identity $V(R)=\pi(\sinh 2R-2R)$, so$V(2R)/V(R)$ is unbounded. For radialu, its Rayleigh quotient is$1+\int\lvert v'\rvert^2 dr/\int\lvert v\rvert^2 dr$, whose infimum is one and is not attained by a nonzero radialL^2 eigenfunction at the bottom.
Y25-02-011 2025-02-17 · A	Frame bounds, coordinate changes and sampling errors are separate What finite certificate makes a family a frame, and which correction preserves its norm under a change of coordinates?	Exact reconstruction and norm-preserving pullback The f_j are orthonormal, so$f=\Sigma\langle f,h_j\rangle h_j$ and the frame bounds are$A=B=1$. Two copies of everyh_j scaled by$1/\sqrt{2}$ give a redundant tight frame with the same bounds. T_g is unitary onL^2.
Y25-02-005 2025-02-18 · U	A Gini calculation and a growth decomposition measure different objects Which distribution statistic and production identity are actually being compared, and what assumptions determine each?	Scale invariance and exact logarithmic decomposition G is unchanged by multiplying every income by a positive constant. For(0,0,2,2),$G=1/2$; for(1,1,1,1),$G=0$. The production formula differentiates exactly to$d\log Y=d\log A+\alpha d\log K+\beta d\log L$.
Y25-02-025 2025-02-18 · A	A hyperbolic quotient needs an action and can have a noncompact cusp Which explicit group action defines a hyperbolic quotient, and why do finite volume and compactness remain different properties?	Exact cusp area and noncompactness The cusp area is$1/Y$ and its horizontal section at$y$ has length$1/y$. It is noncompact despite finite area. The entire translation quotient has infinite area.
Y25-02-012 2025-02-19 · U	Searching many strategies needs an independent validation contract How does optimizing a large parameter grid change the uncertainty of the selected score?	Uniform finite-family validation bound With probability at least$1-\delta$, every candidate satisfies$\lvert\hat{R}_k-R_k\rvert \le \varepsilon=\sqrt{[\log(2K/\delta)/(2n)]}$. The empirically best candidate then has true risk at most2ε above the best candidate's true risk.
Y25-02-022 2025-02-19 · A	Creation operators and Boolean cumulants have distinct finite tests Which exact identities distinguish a creation algebra from a finite matrix approximation, and what does a Boolean Gaussian actually mean?	Vacuum defect and exact Boolean cumulants $L_i^* L_j=\delta_{ij} I$ and$\Sigma_i L_i L_i^*=I-P_{vac}$. For a symmetric variable$X=\pm 1, m_{2n}=1, m_{2n+1}=0$ and its only nonzero Boolean cumulant is$b_2=1$.

ID · date	Concept and revised question	Result and mathematical core
Y25-02-013 2025-02-22 · U	A categorical equivalence has explicit units and counits What finite example demonstrates the actual data required by a categorical equivalence and induction compatibility?	Explicit Morita inverse and induction adjunction F,G are inverse equivalences. AlsoHom_G(Ind_H^G V,W)≅Hom_H(V,Res W), naturally. ForG=S_3,H=A_3,V=1, the induced representation is1⊕sgn.
Y25-02-014 2025-02-22 · A	Monodromy-free means identity, not eigenvalues equal to one Which monodromy condition can be checked for a concrete second-order equation, and why is unipotent monodromy still nontrivial?	Complete constant-potential monodromy test Forλ=$-4\pi^2n^2$ with nonzero integern,M_λ=I. Forλ=0,M_0=[[1,1],[0,1]]≠I although both eigenvalues are one.
Y25-02-015 2025-02-24 · A	A reflection deformation has an explicit polynomial representation Can a Weyl algebra crossed with a reflection group have a real deformation parameter, contrary to a blanket rigidity claim?	Exact deformed Weyl relation [D_κ,X]=I+κs,s^2=I,sX=−Xs,sD_κ=−D_κs. For the Weyl algebraA_1 crossed with{±I}, the standard Hochschild theorem givesHH^2≅C, not zero.
Y25-02-017 2025-02-25 · A	Heat dimension and summability are definitions with calculable limits What does a heat trace measure in a finite model, and which analytic conditions are needed before a proposed operator is a spectral triple?	Exact heat dimension and a summability distinction For the cycle,P(t)=(1+e^{−2t})²/4 andd_s(t)=8t/(e^{2t}+1), tending to zero at both ends. The diagonalD has compact resolvent andTr(e^{−tD})<∞ for everyt>0, butTr(D^{−p})=∞ for every finitep>0.
Y25-02-018 2025-02-25 · A	A logarithmic L-derivative has a pole at a zero What data make anL-value constant well defined, and what happens toL′/L when the proposed evaluation point is a zero?	Exact local logarithmic-derivative decomposition L′(s)/L(s)=m/(s−s_0)+g′(s)/g(s). Thus a positive-order zero produces a simple pole with residuem, and multiplyingL by a nonzero constant does not change its logarithmic derivative.
Y25-02-019 2025-02-25 · A	A repetition code corrects specified flips but has quantum distance one Which error family is corrected by an explicit code, and why must code distance be an integer rather than a fitted real constant?	Exact correction and exact uncorrectable error For the four bit-flip errors,PE_a†E_bP=δ_abP, so they are jointly correctable. ButPZ_1P acts as logicaldiag(1,−1), not a scalar. Hence the code has parameters[[3,1,1]], despite its classical bit-flip distance three.
Y25-02-020 2025-02-27 · A	Descent and factorization need the actual diagram What finite gluing diagram and algebraic quotient illustrate the limit and colimit constructions being invoked?	Explicit gluing and trace quotient Compatible functions onU_1,U_2 glue uniquely. ForA=M_2(k) in characteristic zero,HH_0(A) is one-dimensional and is detected by trace.
Y25-02-021 2025-02-27 · A	A CPTP update gives a quantitative mixing law What does a specified completely positive update actually preserve, and which coherence decays under repetition?	Exact depolarizing contraction Kraus operators are√(1−3p/4)I and√(p/4)X,Y,Z. The channel sendsr to(1−p)r and satisfiesD(E_p^nρ,E_p^nσ)=(1−p)^nD(ρ,σ).

ID · date	Concept and revised question	Result and mathematical core
Y25-02-024 2025-02-27 · A	Pentagon consistency is a path identity with a norm error bound What finite identity is tested by an associativity pentagon, and what can a tolerance of0.07 actually imply?	Exact pentagon and quantitative perturbation The two directed associator paths from(((AB)C)D) toA(B(CD)) are equal. If their edge counts are three and two, their approximate products differ by at most the sum of the five edge errors.
Y25-02-023 date unava · T	Parity observations alone do not certify quantum coherence Which distinct states have identical parity observations, and what additional observable can distinguish their coherence?	An exact observational collision ρ_B andρ_M have identical computational-basis probabilities, identical one-qubit marginalsI/2 andZZ expectation one. ButTr(ρ_B XX)=1 andTr(ρ_M XX)=0, and their trace distance is1/2.

Y25-02-016 Berry curvature and a Chern number require a state bundle

2025-02-01 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which explicitly defined connection produces a nonzero topological invariant, and what normalization makes the integral an integer?

A two-component state over a sphere gives a unit Chern number through a direct patch calculation. The example separates a geometric curvature theorem from unprovided links to phonemes, gravity or algebraic rigidity.

Definitions and assumptions

On a northern chart ofS^2 use$u(\theta,\varphi)=(\cos(\theta/2),e^{i\varphi}\sin(\theta/2))$, with$u^\dagger u=1$.

Use Berry connection$A=-iu^\dagger du$ and curvature$F=dA$. The first Chern number in this convention is$(2\pi)^{-1}\int F$.

Changing the local state by a phase changesA by an exact local differential and leavesF unchanged. A single chart need not extend over both poles.

Explicit Chern-one line bundle

$A=\sin^2(\theta/2)d\varphi$,$F=(1/2)\sin\theta\, d\theta\wedge d\varphi$, and$(2\pi)^{-1}\int_{S^2}F=1$. No globally smooth one-formA onS^2 can have this curvature.

Derivation or proof

Differentiateu and take its inner product withu. Theθ contributions cancel and theφ contribution isi sin²(θ/2)dφ, givingA. Differentiation givesF. Integratingφ from zero to2π andθ from zero toπ yields2π. IfF=dA for a single smooth globalA, Stokes' theorem on the closed sphere would force the integral to zero, a contradiction. The state therefore needs at least two phase charts. Multiplyingu bye^{−iφ} gives a southern choice whose potential differs by−dφ, encoding a winding-one transition. This is the local-to-global mechanism behind the integer, not an assumed physical protection law.

Worked example

Compare the complete sphere with a contractible small patch.

On a patch the sameF is exact by the displayed potential, so no local calculation alone detects the global obstruction. The full integral distinguishes the bundle. For an abelian four-dimensional potential, d(A∧dA)=F∧F; that differential identity is separate from whether a normalized integral is an integer on a closed manifold.

Connection, gauge changes, curvature and global topology are all explicit.

Scope

An arbitrary integral ofTr exp(iF/2π) needs degree selection and bundle/normalization hypotheses before it is a Chern number. No claimed vanishing follows from a separate algebra'sHH², and no material or neural effect is derived.

Further development

For a supplied smooth family of gapped Hermitian matrices, compute its eigenprojectorP and Berry curvature fromP dP∧dP. Track gap closing separately, since the bundle can cease to be defined there.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R024.

Y25-02-002 A network multiplier is an inverse matrix with a convergence condition

2025-02-05 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which matrix and convergence condition produce a claimed multiplier, and how does an input shock propagate through its network?

A finite nonnegative input network has a convergent response series when its spectral radius is below one. A two-sector example yields an exact multiplier, showing why a universal scalar such as2.68 needs a declared model.

Definitions and assumptions

Letx=Ax+d, whereA has nonnegative entries andd is an external input. All coordinates use compatible units.

Assume spectral radius$\rho(A)<1$. Then$(I-A)^{-1}=\Sigma_{n\geq 0}A^n$ and the response to a shockδd is$\delta x=(I-A)^{-1}\delta d$.

A scalar multiplier also needs an aggregation roww^T; it is$w^T\delta x$ divided by a stated input normalization.

Exact two-sector propagation

For$A=[[0,1/4],[1/2,0]]$,$(I-A)^{-1}=[[8/7,2/7],[4/7,8/7]]$. A unit shock in sector one gives total output12/7; a unit shock in sector two gives10/7.

Derivation or proof

$A^2=I/8$, so even and odd powers form two geometric series and$\rho(A)=1/\sqrt{8}<1$. Their sum is$(I+A)/(1-1/8)$, giving the inverse directly. Multiplication by$I-A$ independently verifies the identity. The two different column sums demonstrate that even one fixed network need not have a single shock-independent scalar multiplier. For a general finite matrix the Neumann series follows by Jordan form or an equivalent norm in which high powers decay; the spectral-radius hypothesis is essential to that series interpretation.

Worked example

Use$\delta d=(1,0)$ and compare the first three response rounds.

The direct response is(1,0), the first spillover(0,1/2), and the next(1/8,0). Summing all rounds gives(8/7,4/7). Counting the direct input once and then adding the full inverse response again would double count the zeroth round.

The mathematical multiplier is computable, directional and conditional on a network that must be independently justified.

Scope

This is a linear propagation model, not evidence for any particular sectoral multiplier or monetary return. An inverse may exist even when$\rho(A)\geq 1$, but its Neumann-series and nonnegative-round interpretation can then fail.

Further development

For perturbations$A+E$, use the resolvent identity$B'-B=B'EB$. If$||B||||E||<1$, derive$||B'-B||\leq||B||^2||E||/(1-||B||||E||)$, quantifying how uncertain links can amplify multiplier uncertainty. For the concrete network, the sum of all response rounds after round m is bounded by a geometric tail because $A^2=I/8$. Keeping only even powers through A^{2m} leaves their scalar geometric remainder (1/8)^{m+1}/(1−1/8). Thus a finite-round approximation can carry a rigorous truncation error rather than an arbitrary stopping choice.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R025.

Y25-02-003 Weighted scores and capacity quotas need different constraints

2025-02-05 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

How can a capacity-based allocation be formulated so that its units, feasible set and fairness objective are explicit?

A quotient of capacity by demand is not itself an allocation theorem. A concave two-recipient model yields a unique feasible optimum and an exact water-filling rule.

Definitions and assumptions

Letx_i≥0 be allocated units withΣx_i≤C,C>0. Let benefit beΣa_i log(1+x_i), witha_i>0.

A weighted index such as0.3A+0.3TC+0.2LP+0.2CE lies in[0,1] only after all four inputs are normalized to that interval.

A capacity ratioK/[θ(P+U)] requiresθ>0 andP+U>0. Its interpretation and rounding rule are additional modeling choices.

Unique concave allocation

The log-utility problem has a unique optimumx_i=max(a_i/λ−1,0), withλ>0 chosen soΣx_i=C. For$(a_1,a_2)=(2,1)$,C=2, the allocation is(5/3,1/3).

Derivation or proof

The compact feasible simplex gives existence. The Hessian is diagonal with strictly negative entries$-a_i/(1+x_i)^2$, so the objective is strictly concave and has at most one maximizer. Each benefit increases, so capacity is exhausted. Lagrange optimality gives$a_i/(1+x_i)=\lambda$ on positive coordinates and $a_i\le\lambda$ on zero coordinates. These conditions yield the formula and are sufficient by concavity. For the example both coordinates are positive, so$2/\lambda-1+1/\lambda-1=2$ gives$\lambda=3/4$ and the displayed allocation.

Worked example

Compare the optimum with equal allocation(1,1) and concentrated allocation(2,0).

At the optimum both marginal benefits equal3/4. Equal allocation has marginal benefits one and one half, so moving a small unit from the second to the first improves the objective. Concentration at(2,0) has first marginal2/3 and second marginal one, so it is also improvable. The certificate uses exact marginal comparisons rather than promotional ranking.

A feasible allocation and an aggregate index have explicit but different mathematical contracts.

Scope

No choice of utility or normalization is scientifically or socially compelled by this calculation. Zero demand makes the quoted ratio undefined, and unbounded capacity claims cannot be repaired by optimization alone.

Further development

Add lower guarantees, upper capacity bounds and multiple resource constraints. Derive the KKT multipliers and use complementary slackness to identify which resource actually limits expansion. The solution also has a precise activation threshold: recipient i receives a positive allocation exactly when a_i>λ. As capacity grows, λ decreases and additional recipients become active. This is a mathematical consequence of the specified concave utility, not a universal conclusion about participation or social value.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R025.

Y25-02-004 Debt-to-output dynamics include the primary flow

2025-02-05 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

When does a debt ratio fall, and what is missing from comparing growth with the interest rate alone?

An exact recurrence separates inherited debt, interest, output growth and the primary flow. Its stable fixed point can be calculated, giving a constructive alternative to an unconditional growth claim.

Definitions and assumptions

LetD_{t+1}=(1+r)D_t+P_t andY_{t+1}=(1+g)Y_t withg>−1. Writeb_t=D_t/Y_t andp_t=P_t/Y_t.

PositiveP is a primary deficit under this sign convention; negativeP is a primary surplus.

For a fixed-flow model letp_t=p andr,g be constant.

Exact ratio recurrence and fixed point

b_{t+1}=[(1+r)b_t+p]/(1+g). It decreases exactly whenp<(g−r)b_t. Ifg>r and1+r≥0, the fixed pointb*=p/(g−r) is attracting.

Derivation or proof

Divide the debt equation by the new output level to obtain the recurrence. Subtractb_t and combine denominators to get[b_{t+1}−b_t]=[p−(g−r)b_t]/(1+g), proving the strict condition. Solveb*=((1+r)b*+p)/(1+g) to obtain the fixed point. Subtracting it givesb_{t+1}−b*=a(b_t−b*) witha=(1+r)/(1+g). Under the stated inequalities0≤a<1, so the difference decays geometrically. Thus a positive primary deficit can make the ratio increase despiteg>r, and the initial ratio’s position relative tob* determines the direction.

Worked example

Taker=1/50,g=1/25,b_0=1 and comparep=1/100 withp=1/20.

The first case givesb_1=103/104<1 and fixed point1/2. The second givesb_1=107/104>1 and fixed point5/2. The interest and growth rates are identical; only the primary flow changes. Repeating the first case givesb_t=1/2+(1/2)(51/52)^t exactly.

The recurrence gives conditions and trajectories rather than an unsupported prediction.

Scope

These are accounting dynamics under fixed rates, not financial advice or a forecast. Inflation, valuation changes, maturity structure and changing primary flows require additional terms.

Further development

For bounded time-varyingr_t,g_t,p_t, derive product-sum bounds for the nonautonomous recurrence. Identify conditions giving a uniform contraction and an invariant interval for all admissible rate paths. With variable primary flows but fixed rates, b_t=a^t b₀+(1+g)⁻¹Σ_{j=0}^{t−1}a^{t−1−j}p_j. If 0≤a<1 and 0≤p_j≤p_max, the limiting upper bound is p_max/(g−r). This follows by summing a geometric series and makes the required bound on future flows explicit.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R026.

Y25-02-001 A cumulative wealth equation is a stock-flow identity

2025-02-07 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What does the recovered integral equation imply, and which independent assumptions turn it into a predictive dynamical model?

Integrating a flow defines a stock but does not determine the flow. An explicitly specified linear feedback gives a solvable growth law, a stable regime and a transparent accounting of units.

Definitions and assumptions

LetW(t)=W₀+∫₀ᵗ[αV(s)+βM(s)+γI(s)]ds. Each summand must have units of wealth per time.

For a new test model set the total flow equal toa+bW, wherea is constant wealth per time andb has inverse-time units.

The initial condition already fixes the integration constant; adding a second arbitrary constant would duplicate it.

Exact affine feedback solution

Forb≠0,W(t)=(W₀+a/b)e^{bt}−a/b. Forb=0,W=W₀+at. Ifb<0 the equilibrium−a/b attracts exponentially; ifb>0 it is unstable.

Derivation or proof

The fundamental theorem of calculus givesW' equal to the integrand when that flow is continuous. Substituting the declared feedback gives$W'-bW=a$. Multiplication bye^{-bt} makes the left side the derivative of$e^{-bt}W$; integrate from zero to$t$ to obtain the formula. Perturbing the equilibrium byδ gives$\delta'=b\delta$, proving the stability statement. Thus the stock-flow identity and a causal feedback assumption are mathematically different premises. The integral by itself can represent every differentiable$W$ after defining its flow asW' and therefore cannot certify projected asset targets.

Worked example

Take$W_0=0, a=10, b=-1/5$ in consistent units.

The solution is$W(t)=50(1-e^{-t/5})$, with initial flow ten and limiting stock fifty. In contrast$b=1/5$ gives$50(e^{t/5}-1)$, with accelerating growth. Both start with the same value and derivative but give radically different long-run paths. A numerical projection must therefore report its feedback coefficient and not only its starting flow.

The revised equation becomes a testable dynamical model once its flow is independently specified.

Scope

No historic asset-growth forecast or multiplier is validated. IfV, M, I already contain overlapping receipts, summing them can double count; a common unit does not establish disjoint economic contributions.

Further development

Add a capacity constraint or saturation term and establish an invariant nonnegative interval. Estimate parameters from independent flow observations and report whether the proposed long-run regime is identifiable from those data. If the coefficients are time dependent but$b(t)$ is prescribed, the integrating factor gives$W(t)=e^{B(t)}[W_0+\int_0^t e^{-B(s)}a(s)ds]$, where$B(t)=\int_0^t b(s)ds$. This formula separates observable input flows from amplification and exposes exactly which missing functions a numerical forecast would need.

Source receipt: months/02/root_coverage.json record 0. Mathematical references: R027.

Y25-02-006 A simplified metric RHS has an exact solution but no relativity certificate

2025-02-16 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What can the simplified$g'=-2A, A'=-\kappa A$ prototype establish before full geometric evolution terms are supplied?

The stated prototype is exactly solvable and gives a parameter-dependent positivity condition. This provides a useful code reference while exposing why damping one variable does not automatically preserve a metric.

Definitions and assumptions

For scalar components take$A' = -\kappa A, g' = -2A$ with$\kappa > 0$ and initial valuesA_0, g_0.

A positive scalar metric component requires$g(t) > 0$; positive definiteness of a full matrix is a stronger joint condition.

Curvature, constraints, gauge equations and boundary conditions are not included in this two-variable model.

Exact damped-source evolution

$A(t) = A_0 e^{-\kappa t}$ and$g(t) = g_0 - (2A_0/\kappa)(1 - e^{-\kappa t})$. For$A_0 > 0$,$g$ stays strictly positive for all finite times if$g_0 \geq 2A_0/\kappa$, with a positive uniform lower bound only when the inequality is strict.

Derivation or proof

The first equation solves by separation. Integratingg' using that solution gives the displayed expression. If$A_0 > 0$,g is decreasing and has limit$g_\infty = g_0 - 2A_0/\kappa$. A strictly positive limit gives uniform positivity. At equality$g(t) = g_0 e^{-\kappa t} > 0$ for finite time but tends to zero, so uniform coercivity fails. If the limit is negative, continuity forces a finite zero crossing. Solving$g = 0$ gives$e^{-\kappa t} = 1 - \kappa g_0/(2A_0)$ when$0 < g_0 < 2A_0/\kappa$. This is a complete exact diagnostic for the simplified system, independent of any GPU implementation.

Worked example

Use$\kappa = 1/10, A_0 = 1/10, g_0 = 1$.

Then$A = (1/10)e^{-t/10}$ and$g = 2e^{-t/10} - 1$. The component reaches zero at$t = 10\log 2$ despite the source$A$ decaying. Using$g_0 = 3$ instead gives$g = 1 + 2e^{-t/10}$, uniformly bounded below by one.

An analytic reference trajectory can test a numerical RHS while keeping its geometric interpretation conditional.

Scope

The source explicitly labels the CUDA prototype simplified. Passing this ODE check proves neither the BSSN constraints nor stable Einstein evolution; omitted Christoffel/Ricci terms cannot be treated as validated placeholders.

Further development

For a full symmetric matrix$g$, derive an eigenvalue lower bound from an operator-norm integral estimate on$A$. Then test constraint propagation and convergence for a declared complete system and exact geometric benchmark. At$\kappa = 0$ the solution is$A = A_0$ and$g = g_0 - 2A_0 t$, obtained either directly or as the continuous$\kappa \to 0$ limit on every bounded time interval. Substituting$\kappa = 0$ into a formula containing$1/\kappa$ without taking that limit would create a spurious singularity. Zero damping is a distinct parameter case.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R028.

Y25-02-007 CFL stability and halo exchange have separate exact tests

2025-02-16 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What does a CFL bound prove for one actual discretization, and how can domain decomposition be tested without inventing scaling benchmarks?

For periodic upwind advection, the CFL interval is an exact maximum-norm stability condition. Correct halo exchange reproduces the serial update algebraically, but says nothing by itself about speedup.

Definitions and assumptions

Foru_t+cu_x=0,c>0, useu_j^{n+1}=(1−ν)u_j^n+νu_{j−1}^n withν=cΔt/Δx and periodic indices.

A one-cell halo stores the neighboring old-time value required at a subdomain boundary.

Strong-scaling speedup is measured wall time relative to a specified baseline; it cannot be inferred from a stable stencil.

Stability interval and decomposition equivalence

For0≤ν≤1, the update is anℓ^∞ contraction. Forν>1, the alternating Fourier mode has amplification1−2ν with modulus greater than one. Two subdomains with correct old-time halos give the identical update to one serial domain.

Derivation or proof

In the stable interval each new value is a convex combination of two old values, so|u_j^{n+1}|≤max_k|u_k^n|. For the alternating modeu_j=(−1)^j, substitution gives(1−2ν)(−1)^j, proving the instability assertion. On a decomposed mesh, interior points use the same stencil. At the first point of each block, the needed predecessor is supplied by the halo; if it is the old-time predecessor from the adjacent block, the expression is exactly the serial one. Updating that halo from mixed time levels instead changes the method. Local refinement must therefore respect both a stability restriction and consistent interlevel synchronization.

Worked example

Use four periodic cellsu=(0,1,2,3),ν=1/2 and split after the second cell.

The serial result is(3/2,1/2,3/2,5/2). Supplying left halos three and one to the two blocks reproduces it exactly. The mean remains3/2 because the periodic shift has the same sum as the original vector.

The example supplies meaningful stability and communication certificates without claiming an unmeasured MPI/GPU efficiency.

Scope

Higher-order, multidimensional and relativistic systems have different characteristic speeds and stability regions. A CFL inequality is generally necessary but may not suffice for a chosen time integrator and boundary treatment.

Further development

For an AMR interface, derive a conservative flux synchronization rule and verify equal-and-opposite interface fluxes. Then measure scaling with fixed problem, hardware, precision and synchronization policy.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R029.

Y25-02-008 A spline element has exact partition and stiffness certificates

2025-02-16 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which finite identities should a basis satisfy before its derivative arrays are used in a geometric solver?

A quadratic Bernstein element gives exact partition of unity, polynomial reproduction and a positive-semidefinite stiffness matrix. These are local approximation facts that can be checked independently of a full relativity code.

Definitions and assumptions

On$[0,1]$, use$B_0=(1-x)^2, B_1=2x(1-x), B_2=x^2$, the degree-two Bernstein basis, also a single-spanB-spline basis.

The stiffness matrix is$K_{ij}=\int_0^1 B_i' B_j' dx$. Its quadratic form is the derivative energy of the represented polynomial.

The mass matrix$M_{ij}=\int B_i B_j dx$ is a Gram matrix and is positive definite because the basis functions are linearly independent.

Exact local element identities

$\Sigma B_i=1, x=(1/2)B_1+B_2$ and$x^2=B_2$. The stiffness matrix has diagonal$4/3$ and off-diagonal$-2/3$, hence eigenvalues$0,2,2$.

Derivation or proof

Expand the three polynomials to prove the reproduction identities coefficient by coefficient. Their derivatives are$-2+2x, 2-4x, 2x$. Integrating each derivative product gives the stated matrix, equivalently$K=2I-(2/3)11^T$. The constant coefficient vector$(1,1,1)$ is in its kernel, reflecting zero derivative of a constant. Every vector orthogonal to it has eigenvalue two, proving positivity on nonconstant modes. Independently,$c^TKc=\int|(\Sigma c_i B_i)'|^2 \geq 0$; equality means the polynomial is constant, giving the same kernel without spectral algebra.

Worked example

Use coefficients$c=(0,0,1)$ forx^2 and$c=(0,1/2,1)$ for$x$.

The derivative energies are$\int(2x)^2dx=4/3$ and$\int 1dx=1$. Direct multiplicationc^TKc gives the same values exactly. The constant vector gives zero energy and unit mass integral, supplying a small patch test for assembled operators.

These finite identities certify basis differentiation and an energy form, not a complete physical simulation.

Scope

Multi-span splines need knot multiplicities and continuity specified. Curved geometry mappings add Jacobians, and boundary conditions determine whether the assembled stiffness becomes strictly positive.

Further development

On an explicitly mapped element, verify a positive Jacobian and transform the stiffness integrand by the inverse metric. Assemble adjacent spans and test global constant reproduction, flux consistency and the expected null space. The exact null vector must be retained before boundary conditions are imposed. Artificially inverting the local stiffness matrix would fail because its determinant is zero. Removing a Dirichlet degree of freedom or passing to a quotient by constants changes the operator domain and can produce a positive-definite reduced system with a meaningful inverse.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R030.

Y25-02-009 Weighted atoms must cancel in the same measure as their pairing

2025-02-17 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which normalization and cancellation make a weighted atomic pairing provably bounded?

A single consistent measure gives an elementaryH^1-BMO_2 estimate, including an exact sharp weighted two-point example. Mixing unweighted cancellation with a weighted pairing generally breaks the constant-removal step.

Definitions and assumptions

Let$\nu=w\mu$ be a positive measure and$B$ have$0<\nu(B)<\infty$. Aν-atom$a$ is supported in$B$, has$\int a\,d\nu=0$ and$||a||_{L^2(\nu)}\le\nu(B)^{-1/2}$.

Set$b_B=\nu(B)^{-1}\int_B b\,d\nu$ and$||b||_{BMO_2(\nu)}=\sup_B[\nu(B)^{-1}\int_B|b-b_B|^2d\nu]^{1/2}$.

For finite atomic sums$f=\sum\lambda_ja_j$, define the atomic norm by the infimum of$\sum|\lambda_j|$ over such representations.

Compatible-measure atomic bound

$|\int ab\,d\nu|\le||b||_{BMO_2(\nu)}$ and$|\int fb\,d\nu|\le||f||_{H^1_{at}(\nu)}||b||_{BMO_2(\nu)}$ for finite atomic sums.

Derivation or proof

Cancellation gives$\int ab = \int a(b - b_B)$. Cauchy-Schwarz bounds this by$\nu(B)^{-1/2}\|b - b_B\|_2$, proving the single-atom estimate. Sum the bounds and take the infimum over representations to get the finite-sum inequality. Completion needs a separately justified extension of the pairing. If instead$\int a\, d\mu = 0$ but$\int a\, d\nu \neq 0$, subtractingb_B changes the pairing by$b_B \int a\, d\nu$ and the proof fails. Thus a weighted norm is not repaired by inserting$w$ into one integral while keeping the rest of the Euclidean definition unchanged.

Worked example

Use a two-point ball withν masses$(1,3)$,$b = (0,2)$, and$a = (3c, -c)$,$c = 1/(4\sqrt{3})$.

The weighted mean of$b$ is$3/2$ and its normalized squared oscillation is$3/4$. The atom has zeroν mean and squaredν norm$12c^2 = 1/4$, exactly the required bound. Its pairing is$-6c = -\sqrt{3}/2$, attaining the oscillation bound. All weighted constants and factors are visible.

The result gives a valid constructive fragment of weighted duality under declared definitions.

Scope

It does not prove full weightedH^1-BMO duality on hyperbolic space. RelatingBMO_2 toBMO_1 requires geometric/weight hypotheses such as an appropriate doubling and John-Nirenberg theorem.

Further development

State an admissible ball family and weight class, prove its measure estimates and the required oscillation comparison, then establish both directions of the completed duality. Treat that as a theorem to prove, not a Monte Carlo conclusion. In the weighted two-point example, the normalized atom is proportional to $b - b_B$ with the opposite sign, which is precisely the equality condition in Cauchy-Schwarz. The sharp constant therefore has an explicit optimizer. If cancellation were imposed with unweighted masses instead, a constant b could already produce a nonzero weighted pairing.

Source receipt: months/02/root_followups.json record 0. Mathematical references: R031.

Y25-02-010 Hyperbolic volume growth changes global analysis

2025-02-17 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which Euclidean assumptions fail on the full hyperbolic space, and what spectral statement can be proved in a radial model?

Hyperbolic three-space has exponential ball growth and is not globally doubling. A radial energy identity proves the bottom spectral value one and explains why a global discrete eigenbasis cannot simply be assumed.

Definitions and assumptions

For curvature-1 hyperbolic three-space, radial volume is$d\mu = 4\pi \sinh^2 r\, dr$ after angular integration.

For a radial compactly supported functionu, writeu(r)=v(r)/sinh r, withv(0)=0 and the endpoint behavior needed for integration by parts.

Use the nonnegative Laplacian convention−Δ and its Rayleigh quotient∫|∇u|²/∫|u|².

Ball growth and exact radial spectral identity

V(R)=π(sinh2R−2R), soV(2R)/V(R) is unbounded. For radialu, its Rayleigh quotient is1+∫|v′|²dr/∫|v|²dr, whose infimum is one and is not attained by a nonzero radialL² eigenfunction at the bottom.

Derivation or proof

Integrate4πsinh²r using sinh²r=(cosh2r−1)/2 to getV. Its leading exponential is(π/2)e^{2R}, proving the ratio divergence. For the energy,sinh²r·|u′|²=|v′−v coth r|². Integrate the cross term−(v²)′coth r and use(coth r)′=−csch²r; the remaining coefficient is coth²r−csch²r=1. Boundary terms vanish under the stated conditions. For general smooth compactly supportedu, apply the radial identity to each angular direction and add the nonnegative angular-gradient energy. This proves the lower bound one for all such functions, not only radial ones. Takingv(r)=sin(πr/R) on[0,R], with standard smooth approximation at the endpoint, gives quotient1+π²/R²→1. Equality would forcev′=0 almost everywhere; its zero endpoint andL² condition then forcev=0.

Worked example

Compare radiiR and2R and long radial test functions.

The volume-doubling ratio grows asymptotically ase^{2R}. Increasing a radial support length fromR to2R divides the excess Rayleigh quotient above one by four. These are geometric and spectral effects with exact formulas, not evidence of a finite discrete mode list.

The example identifies which global hypotheses must be rebuilt for hyperbolic Hardy spaces and spectral algorithms.

Scope

The radial calculation does not prove the entire spectral resolution or a weighted duality theorem. A bounded domain or compact quotient can have a discrete spectrum, unlike the complete noncompact model; boundary and quotient assumptions must be stated.

Further development

Separate local balls of bounded radius from global geometry. Establish local doubling constants and compare local atomic spaces with global spaces before transferring a Euclidean proof or wavelet construction.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R032.

Y25-02-011 Frame bounds, coordinate changes and sampling errors are separate

2025-02-17 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What finite certificate makes a family a frame, and which correction preserves its norm under a change of coordinates?

A four-point Haar basis gives exact Parseval reconstruction. A redundant duplication is still a tight frame, and a Jacobian factor-not a name such as Möbius-scaled-controls unitarity of coordinate pullback.

Definitions and assumptions

InR^4 use$h_0=(1,1,1,1)/2, h_1=(1,1,-1,-1)/2, h_2=(1,-1,0,0)/\sqrt{2}, h_3=(0,0,1,-1)/\sqrt{2}$.

A frame has$A||f||^2 \le \Sigma|\langle f,\psi_j\rangle|^2 \le B||f||^2$ with$A>0$. Its frame operator is$Sf=\Sigma\psi_j\langle\psi_j,f\rangle$.

For a smooth bijectiong in Euclidean coordinates, define$T_gf(x)=|detDg^{-1}(x)|^{1/2}f(g^{-1}x)$.

Exact reconstruction and norm-preserving pullback

The fourh_j are orthonormal, so$f=\Sigma\langle f,h_j\rangle h_j$ and the frame bounds are$A=B=1$. Two copies of everyh_j scaled by$1/\sqrt{2}$ give a redundant tight frame with the same bounds. T_g is unitary onL^2.

Derivation or proof

Direct dot products give$h_i^T h_j=\delta_{ij}$, and four orthonormal vectors spanR^4. Parseval and reconstruction follow. In the duplicate family each squared coefficient is halved but occurs twice, so their total remains$||f||^2$ and$S=I$; redundancy does not imply unique coefficients. For the pullback, substitute$x=g(y)$ in$\int|T_gf(x)|^2dx$. The inverse Jacobian cancels the change-of-variables Jacobian, leaving$\int|f(y)|^2dy$. A hyperbolic isometry preserves its invariant volume measure and needs no such Euclidean-coordinate correction once that measure is used consistently.

Worked example

Apply the Haar transform to$f=(0,1,2,3)$.

The coefficients are$3,-2,-1/\sqrt{2},-1/\sqrt{2}$; their squared sum is$9+4+1/2+1/2=14$, equal to$||f||^2$. Omitting the scaling on duplicate vectors doubles the frame energy to28 rather than preserving the stated normalization.

A proposed algorithm now has independent algebraic and geometric checks before performance claims are considered.

Scope

A finite Haar example does not construct a hyperbolic wavelet frame. Monte Carlo agreement with one signal cannot prove a uniform lower frame bound; a missing lower bound permits unrecoverable modes.

Further development

For a finite discretization form the analysis matrixΨ and certify eigenvalue bounds forΨ*Ψ. Report discretization and quadrature errors separately from sampling variance and from GPU runtime. For uniform Monte Carlo sampling of this four-point signal, the mean is 3/2 and the population variance is 5/4. An average of m independent samples therefore has variance 5/(4m). This sampling uncertainty is distinct from the exact Parseval identity and does not decrease a deterministic missing-mode reconstruction error.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R033.

Y25-02-005 A Gini calculation and a growth decomposition measure different objects

2025-02-18 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which distribution statistic and production identity are actually being compared, and what assumptions determine each?

The finite-sample Gini is a normalized pairwise-difference sum. A Cobb-Douglas log differential is instead a production identity with a residual. Both can be computed exactly without validating a national comparison.

Definitions and assumptions

For nonnegative incomesx_i with positive sum, useG=Σ_{i,j}|x_i−x_j|/[2nΣ_i x_i].

For positive outputY=AK^αL^β, take logarithms. A growth-accounting residual isdlogA=dlogY−αdlogK−βdlogL.

A dependency ratio also needs an explicit numerator population and denominator population; changing either changes the statistic.

Scale invariance and exact logarithmic decomposition

G is unchanged by multiplying every income by a positive constant. For(0,0,2,2),G=1/2; for(1,1,1,1),G=0. The production formula differentiates exactly todlogY=dlogA+αdlogK+βdlogL.

Derivation or proof

Both numerator and denominator ofG scale linearly, proving invariance. In the unequal example eight ordered pairs cross between zero and two, each contributing two, so the numerator is sixteen and the denominator2·4·4=32. Equal values have zero pairwise differences. For production, logY=logA+αlogK+βlogL, so ordinary differentiation gives the identity whenα,β are fixed. A measured residual is therefore whatever remains under the chosen functional form and coefficients; the algebra does not prove that it measures a unique causal technology effect.

Worked example

Setα=β=1/2, holdA fixed, doubleK and halveL.

The output ratio is$\sqrt{2}\cdot\sqrt{(1/2)}=1$, while the exact log contributions cancel. A first-order percentage approximation should not replace this finite-change calculation for large changes. Separately,341,000,000 divided by10,000 is exactly34,100; a capacity rule requiring whole representatives uses a ceiling if the division leaves a remainder.

The revised comparison distinguishes inequality, growth accounting and representation arithmetic instead of treating them as interchangeable evidence.

Scope

Zero total income makesG undefined. A finite Gini does not recover the whole distribution, and a growth residual depends on measurement and model choices. No country-specific data claim is established here.

Further development

For an inequality-reducing transfer preserving total income, derive the change in the pairwise sum while accounting for rank crossings. For production data, compare alternative elasticities and report the resulting range of residual growth. Dividing each income by its mean leaves the Gini unchanged, but adding the same positive constant to every income generally lowers it: the pairwise numerator stays fixed while the total-income denominator increases. This exact distinction separates scale changes from a uniform transfer and shows why both should not be called normalization.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R034, R035.

Y25-02-025 A hyperbolic quotient needs an action and can have a noncompact cusp

2025-02-18 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which explicit group action defines a hyperbolic quotient, and why do finite volume and compactness remain different properties?

The translation quotient of the upper half-plane gives a concrete cusp geometry with an exact area and shrinking circular sections. This supplies real quotient mathematics without deriving an automorphic economic model from notation.

Definitions and assumptions

Let$H^2=\{x+iy:y>0\}$ with$ds^2=(dx^2+dy^2)/y^2$ and area form$dxdy/y^2$.

The group$Z$ acts freely and properly discontinuously by$z\mapsto z+n$. Its quotient is a smooth hyperbolic cylinder, with$x$ periodic modulo one.

The cusp region$y\geq Y>0$ in that quotient has a boundary circle at$y=Y$ and remains unbounded in the logarithmic radial direction.

Exact cusp area and noncompactness

The cusp area is1/Y and its horizontal section aty has length1/y. It is noncompact despite finite area. The entire translation quotient has infinite area.

Derivation or proof

Translation preservesy anddx, so it preserves the metric. For any compact subset ofH^2 only finitely many integer translates meet it, proving proper discontinuity; no nonzero translation fixes a point. A strip$0 \le x < 1$ is a fundamental domain. Integrating its cusp area gives$\int_0^1 dx \int_Y^\infty y^{-2} dy = 1/Y$. The circle at heighty has length$\int_0^1 dx/y = 1/y$. Along a vertical path the distance fromY toy is$\log(y/Y)$, which tends to infinity, so no compact set contains the whole cusp. Extending the area integral down toy=0 diverges, showing that the full quotient and its cusp subregion must not be confused.

Worked example

SetY=2 and compare sections aty=2 andy=8.

The cusp area is1/2. Its two section lengths are1/2 and1/8, while the vertical separation is$\log 4$. The cross sections shrink but recede to infinite distance, explaining finite area without compactness.

Quotient, metric, measure and global geometry are all specified independently of an application label.

Scope

A quotient$[H^3/\Gamma]$ requires an actual group, action and a choice between an orbifold/stack and a smooth manifold. Automorphic functions and Hecke operators need additional arithmetic structure; arbitrary urban data do not supply it.

Further development

For a supplied discrete arithmetic group, determine stabilizers, fundamental-domain volume and cusp structure. Then formulate the exact function space and Laplacian domain before invoking a discrete spectral expansion or trace formula.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R036.

Y25-02-012 Searching many strategies needs an independent validation contract

2025-02-19 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

How does optimizing a large parameter grid change the uncertainty of the selected score?

A finite-family concentration bound quantifies selection error under explicit independent bounded-data assumptions. It shows why a best synthetic Sharpe value is not a validated predictive result.

Definitions and assumptions

LetK fixed candidate models have losses$\ell_k(Z)\in[0,1]$ on independent identically distributed validation observations$Z_1,\dots,Z_n$.

Write$R_k=E\ell_k$ and$\hat{R}_k=n^{-1}\Sigma\ell_k(Z_i)$. The candidates must be fixed independently of this validation sample for the elementary bound.

The recovered grid$j_{min},j_{max},\lambda_1,\lambda_2$,Q and500-ms sampling describe a search procedure, not an independence or boundedness theorem for market returns.

Uniform finite-family validation bound

With probability at least$1-\delta$, every candidate satisfies$|\hat{R}_k-R_k|\le\varepsilon=\sqrt{[\log(2K/\delta)/(2n)]}$. The empirically best candidate then has true risk at most2ε above the best candidate’s true risk.

Derivation or proof

Hoeffding’s inequality gives$P(|\hat{R}_k-R_k|>\varepsilon)\le 2e^{-2n\varepsilon^2}$ for each fixed candidate. A union bound overK candidates gives$2Ke^{-2n\varepsilon^2}$; setting this equal toδ gives the radius. If$\hat{k}$ minimizes empirical risk andk^* minimizes true risk, then$R_{\hat{k}}\le\hat{R}_{\hat{k}}+\varepsilon\le\hat{R}_{k^*}+\varepsilon\le R_{k^*}+2\varepsilon$. This is a complete selection argument under the assumptions, not an assertion that arbitrary rolling returns satisfy them. Fitting candidates on the same validation sample violates the fixed-family premise unless the entire possible family was already included inK.

Worked example

Take$K=1000,n=1000,\delta=0.05$.

The radius is approximately0.07279 and the excess-risk bound approximately0.14558. Increasing the number of searched combinations without more independent observations enlarges the uniform uncertainty. A Sharpe ratio also needs a nonzero variance denominator; a constant positive synthetic return makes its naive ratio undefined or infinite rather than exceptionally certified.

The revised optimization specifies an out-of-sample inferential guarantee and its limitations.

Scope

Actual time series can be dependent, nonstationary and heavy-tailed. This result is not financial guidance and does not certify a strategy. Synthetic signals test implementation behavior, not historical market performance.

Further development

Use a chronological training/validation/test split and an explicit dependence model. Derive a concentration or bootstrap procedure justified for that model and report the complete search family, costs and uncertainty of the final score. Reusing the validation sample adaptively changes the inference problem. A final untouched test set can evaluate the selected model after selection, but its confidence statement then concerns that fixed selected model, not a fresh search over the test data. Repeatedly checking and selecting on the final set destroys this separation.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R037.

Y25-02-022 Creation operators and Boolean cumulants have distinct finite tests

2025-02-19 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which exact identities distinguish a creation algebra from a finite matrix approximation, and what does a Boolean Gaussian actually mean?

Prefix creation on word space satisfies Cuntz-Toeplitz relations with a vacuum defect. Separately, the Boolean moment-cumulant recursion makes a symmetric two-point law Gaussian in the Boolean sense, unlike the classical Gaussian.

Definitions and assumptions

Let the full word Hilbert space have orthonormal basis$|w\rangle$ over finite words in two letters, including the empty vacuum. Define$L_i|w\rangle=|iw\rangle$.

Boolean cumulantsb_n satisfy $m_0=1$ and$m_n=\Sigma_{j=1}^{n}b_j m_{n-j}$, equivalent to summing products over interval partitions.

These constructions require stated notions of independence; a word such as Itô does not transfer classical stochastic rules to a different probability theory.

Vacuum defect and exact Boolean cumulants

$L_i^*L_j=\delta_{ij}I$ and$\Sigma_i L_i L_i^*=I-P_{vac}$. For a symmetric variable$X=\pm1$,$m_{2n}=1$,$m_{2n+1}=0$ and its only nonzero Boolean cumulant is$b_2=1$.

Derivation or proof

Removing the first letter undoes prefixing when the letters agree and kills it otherwise, proving the first relation. Every nonempty word begins with exactly one letter, while the vacuum lies in neither range, proving the defect formula. In a finite-dimensional space two exact isometries with orthogonal ranges cannot both exist: their ranges would each have the full ambient dimension. Finite truncations therefore add boundary defects. For cumulants, the moment series is$M(z)=1/(1-z^2)$. The recursion implies$M=1+B\,M$, so$B=1-1/M=z^2$. Thus onlyb_2 survives. This does not give the classical Gaussian, whose fourth moment is three; the two-point law has fourth moment one.

Worked example

Inspect words of length at most two and compute the first six moments.

There are$1+2+4=7$ basis words. Prefixing and truncating annihilates length-two inputs, so its adjoint-product is not$I$ on that top layer. The moment sequence is$1,0,1,0,1,0,1$; interval-pairing recursion reproduces it exactly.

The result supplies concrete algebra and probability instead of treating advanced operator names as interchangeable.

Scope

It does not construct a Cuntz-Pimsner market model or a Boolean stochastic differential equation. A stochastic integral and its Itô rule need their own Fock-space and adaptedness definitions.

Further development

State a correspondence and coefficient algebra for a genuine Cuntz-Pimsner construction. For Boolean dynamics, build a process with specified increment independence and derive its integral isometry before claiming a stochastic calculus.

Source receipt: months/02/root_followups.json record 0. Mathematical references: R038.

Y25-02-013 A categorical equivalence has explicit units and counits

2025-02-22 · historical evidence U · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What finite example demonstrates the actual data required by a categorical equivalence and induction compatibility?

Matrix Morita equivalence and a finite Frobenius-reciprocity calculation supply complete constructive models. They illustrate categorical language without asserting the recovered broad FLE formula under missing level and support hypotheses.

Definitions and assumptions

LetA=$M_2(k)$,e=E_{11}. For a leftA-moduleM defineF(M)=eM; for a vector spaceV defineG(V)=$k^2\otimes V$ withA acting onk^2.

For finite groupsH⊂G, induction isk[G]⊗_{k[H]}V and restriction forgets part of the group action.

A claimed equivalence needs functors and natural isomorphismsFG≅Id,GF≅Id, not only a pair of category names.

Explicit Morita inverse and induction adjunction

F,G are inverse equivalences. AlsoHom_G(Ind_H^G V,W)≅Hom_H(V,Res W), naturally. ForG=S_3,H=A_3,V=1, the induced representation is1⊕sgn.

Derivation or proof

The map$k^2\otimes eM\to M$ sendsbasis vector e_i⊗m toE_{i1}m. Its inverse sendsm toΣ_i e_i⊗E_{1i}m. The matrix-unit identities show both composites are identities and that the maps respectA actions. On the other side,e($k^2\otimes V$) identifies directly withV. For Frobenius reciprocity, send aG-linear mapT to theH-linear mapv↦T(1⊗v). Conversely anH-linear mapf definesT(g⊗v)=gf(v); the tensor relation makes this well defined. For the exampleG/H has two cosets, and its permutation representation splits into the constant and alternating lines, giving1⊕sgn.

Worked example

ApplyMorita equivalence toM=k² andM=k²⊕k².

Their images arek andk², and the inverse tensor construction restores them. The functor preserves multiplicity but changes the presentation of the algebra. The two-coset induced example verifies dimensions and restriction compatibility exactly.

The mathematical content is now a pair of explicit equivalence/adjunction proofs rather than an unrestricted categorical slogan.

Scope

This does not prove a critical-level Kac-Moody/oper equivalence. Fundamental local equivalences have precise groups, levels and categories; the recovered formula suppresses data essential in primary statements.

Further development

For a proposed geometric correspondence, write its actual adjoint functors, units and counits and specify compactness/support conditions. Test their compositions on stated generators before claiming an equivalence of whole categories. The explicit inverse maps also preserve all morphisms, not only dimensions. A linear map eM→eN induces the A-linear map reconstructed by the matrix units, and restricting that reconstruction returns the original map. This full-faithfulness check is essential: two categories can have matching object counts while having different morphism spaces.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R039.

Y25-02-014 Monodromy-free means identity, not eigenvalues equal to one

2025-02-22 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which monodromy condition can be checked for a concrete second-order equation, and why is unipotent monodromy still nontrivial?

A constant-potential equation on a periodic parameter has a complete monodromy calculation. The zero-potential case has only eigenvalue one but a nonidentity Jordan block, exposing a condition that spectral eigenvalues alone miss.

Definitions and assumptions

Consider y″=λy over a parameter interval of length one and writev=(y,y′). Thenv′=A_λv withA_λ=[[0,1],[λ,0]].

The monodromy isM_λ=e^{A_λ}; monodromy-free in this model meansM_λ=I.

This scalar constant-coefficient test is newly specified; a general oper includes geometric and gauge data not recovered here.

Complete constant-potential monodromy test

For$\lambda=-4\pi^2n^2$ with nonzero integer$n, M_\lambda=I$. For$\lambda=0, M_0=[[1,1],[0,1]]\neq I$ although both eigenvalues are one.

Derivation or proof

$A_\lambda^2=\lambda I$. If$\lambda=-\omega^2$ with$\omega\neq0$, the exponential is$I\cos\omega+(A_\lambda/\omega)\sin\omega$. It is the identity when$\omega=2\pi n$, giving the stated family. At$\lambda=0$ the square of$A$ vanishes, so its exponential is$I+A$, proving the nontrivial unipotent case. For general complex$\lambda\neq0$, diagonalization gives eigenvalues$e^{\pm\sqrt{\lambda}}$. Both are one only when$\sqrt{\lambda}$ is a nonzero integer multiple of$2\pi i$, in which case the matrix is diagonalizable and is$I$. The exceptional$\lambda=0$ case is precisely where that diagonalization fails; replacing matrix equality by eigenvalue equality loses it.

Worked example

Compare initial data$v(0)=(0,1)$ at$\lambda=0$ and$\lambda=-4\pi^2$.

At zero potential,$y(t)=t$ and$v(1)=(1,1)$, so the initial vector is not restored.
At$\lambda=-4\pi^2, y(t)=\sin(2\pi t)/(2\pi)$ and$v(1)=(0,1)$, restoring every solution’s data after one period.

The example supplies an exact monodromy condition rather than a label attached to an unspecified spectral support.

Scope

It does not classify monodromy-free opers on arbitrary curves or justify the historical categorical equivalence. Gauge, singularity and global-bundle conditions matter.

Further development

For a piecewise constant potential, multiply the exact transfer matrices and solve the full matrix identity$M=I$. Compare it with$\operatorname{tr}M=2$, which also admits nontrivial unipotent matrices and is therefore weaker. The determinant of every monodromy matrix in this model is one because $\operatorname{tr} A_\lambda=0$ and $\det e^{A_\lambda}=e^{\operatorname{tr} A_\lambda}$. Nevertheless determinant one and trace two do not force identity: the zero-potential Jordan matrix satisfies both. A geometric condition stated through traces alone must therefore address this nonsemisimple exception.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R040.

Y25-02-015 A reflection deformation has an explicit polynomial representation

2025-02-24 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Can a Weyl algebra crossed with a reflection group have a real deformation parameter, contrary to a blanket rigidity claim?

A rank-one Dunkl operator gives a concrete family satisfying a deformed commutator. The known Hochschild theorem for Weyl crossed products identifies the corresponding degree-two deformation direction.

Definitions and assumptions

OnC[x], letX multiply byx andsf(x)=f(−x). DefineD_κf=f′+κ(f(x)−f(−x))/(2x).

The quotient is polynomial because the odd part of a polynomial is divisible byx.

The Weyl algebra convention is$[D_0,X]=I$. The reflection anticommutes with bothX andD_κ.

Exact deformed Weyl relation

$[D_\kappa,X]=I+\kappa s, s^2=I, sX=-Xs, sD_\kappa=-D_\kappa s$. For the Weyl algebra$A_1$ crossed with$\{\pm I\}$, the standard Hochschild theorem gives$HH^2\cong C$, not zero.

Derivation or proof

ApplyD_κ tomonomialx^n. The result is$[n+(\kappa/2)(1-(-1)^n)]x^{n-1}$: even degrees have coefficientn and odd degrees coefficientn+κ. SubtractXD_κx^n fromD_κXx^n. For evenn the difference is1+κ, and for oddn it is1−κ, exactly$(I+\kappa s)x^n$. Parity reversal of a derivative and of the odd-part quotient gives the anticommutation relations. These formulas prove a consistent represented deformation family. The cited general theorem computesHH^j(A(V)⋊G) by conjugacy classes withrank(g−I)=j for finite symplecticG; in dimension two the identity contributes degree zero and−I contributes degree two. That theorem is background, distinct from the polynomial certificate.

Worked example

Takeκ=2 and compareD_κ acting onx, x^2, x^3.

The outputs are$3, 2x, 5x^2$, whereas ordinary differentiation gives$1, 2x, 3x^2$. The commutator on1 is three and onx is−x, agreeing withI+2s. The parameter is not absent from the operator relations.

This provides constructive deformation data rather than inferring rigidity from an unsupported zero cohomology assertion.

Scope

The finite checks do not prove the full PBW or universal-deformation theorem. Hochschild rigidity also has no automatic implication for Berry curvature, which belongs to a specified parameterized state bundle.

Further development

Establish the PBW basisx^a s^εD^b for the abstract algebra using a confluence proof, then classify which parameter values admit finite-dimensional quotients of the polynomial representation. A special value can change the polynomial representation. If κ is a negative odd integer, D_κ annihilates x^{−κ} as well as the constant. This follows immediately from the odd-degree coefficient n+κ and signals an additional singular vector. Such exceptional parameters should be recorded before asserting uniform representation-theoretic behavior.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R041, R042.

Y25-02-017 Heat dimension and summability are definitions with calculable limits

2025-02-25 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What does a heat trace measure in a finite model, and which analytic conditions are needed before a proposed operator is a spectral triple?

A four-cycle gives an exact scale-dependent heat dimension. A separate diagonal model shows that compact resolvent and finite heat trace do not imply finite-power zeta summability.

Definitions and assumptions

For the four-cycle graph Laplacian eigenvalues are0,2,2,4. SetP(t)=Tr(e^{−tL})/4 andd_s(t)=−2dlogP/dlogt.

For a diagonal positive operatorD with eigenvaluen of multiplicityd_n=ceil(e^{√n}), each eigenspace is finite dimensional.

A spectral-triple claim needs a represented algebra, a self-adjointD, compact resolvent and bounded commutators; its summability properties require additional estimates.

Exact heat dimension and a summability distinction

> For the cycle,P(t)=(1+e^{−2t})²/4 andd_s(t)=8t/(e^{2t}+1), tending to zero at both ends. The diagonalD has compact resolvent andTr(e^{−tD})<∞ for everyt>0, butTr(D^{−p})=∞ for every finitep>0.

Derivation or proof

Sum the four heat eigenvalues and differentiate the logarithm to obtain the first formula. At smallt the numerator tends to zero; at larget exponential growth in the denominator dominates. ForD, eigenvalues tend to infinity with finite multiplicity, so the resolvent eigenvalues tend to zero and finite-rank truncations converge in norm. For fixedt>0,√n≤tn/2 for all sufficiently largen, givingd_ne^{−tn}≤2e^{−tn/2}, a convergent tail. Butd_n n^{−p} does not even tend to zero, so the power trace diverges. This example distinguishesθ-summability from finitep-summability without requiring a conjectural Monster spectrum.

Worked example

Evaluate the cycle dimension att=1/2.

It equals4/(e+1), approximately1.07577, while its large-time limit is zero because the constant zero mode dominates. A single reported dimension value therefore needs its scale, trace normalization and zero-mode convention.

The result provides analytic tests for a proposed spectral construction instead of treating a named algebra and a formalD as sufficient.

Scope

The model multiplicities are newly chosen and are not claimed to be the exact gradedMonster dimensions. A termlogL_0 is undefined on a zero eigenspace unless its domain or regularization is specified.

Further development

For a supplied graded multiplicity sequence, prove asymptotic bounds strong enough to determine heat and zeta convergence. Then check commutator domains and the actual spectral-action expansion before assigning physical constants. For the finite graph, removing the constant mode changes the large-time behavior completely: the remaining normalized heat trace decays as e^{−2t}, so its logarithmic derivative grows linearly in t rather than tending to zero. Zero-mode subtraction is therefore a definition-changing choice, not a harmless numerical cleanup.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R043.

Y25-02-018 A logarithmic L-derivative has a pole at a zero

2025-02-25 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What data make anL-value constant well defined, and what happens toL'/L when the proposed evaluation point is a zero?

A local factorization completely determines the singular part of a logarithmic derivative. It supplies a concrete analytic boundary to parameter formulas whose underlyingL-function, normalization or nonvanishing was not supplied.

Definitions and assumptions

Let$L$ be holomorphic near s_0, not identically zero, with a zero of order$m\geq 0$ there. Then$L(s)=(s-s_0)^m g(s), g(s_0)\neq 0$.

A value$L'(s_0)/L(s_0)$ is an ordinary finite quotient only when$m=0$.

A numerical expression$L(E,1), L(f,3/2)$ or$L'(g,1)/L(g,1)$ must identify the curve/form, analytic normalization, conductor and required continuation.

Exact local logarithmic-derivative decomposition

$L'(s)/L(s)=m/(s-s_0)+g'(s)/g(s)$. Thus a positive-order zero produces a simple pole with residue$m$, and multiplying$L$ by a nonzero constant does not change its logarithmic derivative.

Derivation or proof

Differentiate the product factorization and divide away from s_0. The quotient splits into the displayed terms. Since g is nonzero nearby, g'/g is holomorphic there. Multiplying L by $c \neq 0$ cancels in L'/L, but multiplying it by an exponential e^{as} adds a. Therefore completed versus uncompleted L-functions can have different logarithmic derivatives even when they encode related arithmetic. Their gamma and conductor factors cannot be omitted from a numerical identification. For a contour enclosing only s_0, integration gives $(2\pi i)^{-1} \oint L'/L = m$, directly counting the zero order.

Worked example

Take $L(s) = (s-1)^2 e^{3s}$.

Its logarithmic derivative is $2/(s-1) + 3$ and has no finite value at $s = 1$. Multiplying L by seven leaves that derivative unchanged; replacing it by $e^{-s}L$ changes the constant term from three to two. These exact examples separate scaling invariance from normalization sensitivity.

An arithmetic parameter formula now has a specific well-definedness and verification contract.

Scope

No source-defined curve or modular form was recovered for the historical α, β, κ constants, so their asserted decimal values are not newly certified. The analytic lemma does not prove their conjectural physical interpretation.

Further development

Supply the exact arithmetic object and completed-function convention, certify nonvanishing at the evaluation point and bound truncation/continuation error. If the point is a zero, use a stated regularized coefficient after subtracting the pole rather than an undefined quotient. The residue is stable under multiplication by any holomorphic nonvanishing factor because that factor contributes only a holomorphic logarithmic derivative. This makes zero order an invariant local quantity while the regular finite term remains normalization dependent. A proposed constant should specify which of those two pieces it intends to use.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R044.

Y25-02-019 A repetition code corrects specified flips but has quantum distance one

2025-02-25 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

Which error family is corrected by an explicit code, and why must code distance be an integer rather than a fitted real constant?

The three-qubit repetition subspace corrects any single bit flip, but it does not correct a single phase flip and has quantum distance one. The exact distinction prevents classical redundancy from being reported as universal quantum protection.

Definitions and assumptions

LetC=span{|000⟩,|111⟩}, with projectorP. Consider errorsE={I,X₁,X₂,X₃}.

Knill-Laflamme requiresPE_a†E_bP=c_abP for the declared correctable error family.

A stabilizer code distance is the minimum integer weight of a nontrivial logical Pauli operator; it is not an arbitrary real expression√C.

Exact correction and exact uncorrectable error

For the four bit-flip errors,PE_a†E_bP=δ_abP, so they are jointly correctable. ButPZ₁P acts as logicaldiag(1,−1), not a scalar. Hence the code has parameters[[3,1,1]], despite its classical bit-flip distance three.

Derivation or proof

Ifa=b, the error product isI and the condition givesP. Ifa≠b, the product flips one or two distinct bits. Applied to either codeword it lands in a basis state orthogonal to both codewords, making the projected product zero. The four resulting error subspaces are mutually orthogonal and fill the eight-dimensional ambient space; measuring their syndrome and undoing the indicated flip gives recovery. A phase flipZ₁ leaves the basis strings in place but reverses the phase of|111⟩, so its restriction is a nontrivial logicalZ. This weight-one logical operator proves distance one. The distinction follows from the actual error action, not a code-size analogy.

Worked example

ApplyX₂ andZ₁ separately to(|000⟩+|111⟩)/√2.

X₂ gives(|010⟩+|101⟩)/√2 in a recognizable orthogonal syndrome subspace and is reversed byX₂. Z₁ gives(|000⟩−|111⟩)/√2 inside the original codespace, changing its encoded state while leaving the bit-flip syndrome trivial.

A code claim is now tied to an encoding and a checked error family.

Scope

The source’s large parameter expression[[196884,24,√C]] does not provide an encoding, integer distance or recovery theorem. The quantum Singleton bound is a necessary parameter check, not an existence proof.

Further development

Construct a code correcting bothX andZ errors and verify all pairwise Knill-Laflamme products. Report the resulting integer distance and logical channel under a specified noisy recovery circuit.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R045.

Y25-02-020 Descent and factorization need the actual diagram

2025-02-27 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What finite gluing diagram and algebraic quotient illustrate the limit and colimit constructions being invoked?

Functions on a finite cover satisfy an explicit equalizer formula. A circle’s algebraic factorization example leads to$HH_0(A)=A/[A,A]$, which is calculable and different from merely naming a colimit.

Definitions and assumptions

For a finite set$U$, let$F(U)=k^U$ with restriction maps. A cover$U=U_1\cup U_2$ produces the equalizer of$F(U_1)\oplus F(U_2)\rightrightarrows F(U_1\cap U_2)$.

For an associative algebra$A$, the first Hochschild boundary is$b(a\otimes b)=ab-ba$, so$HH_0(A)=A/\mathrm{span}\{ab-ba\}$.

The general identification of factorization homology of a circle with Hochschild homology is background; the finite quotient below is proved directly.

Explicit gluing and trace quotient

> Compatible functions onU_1,U_2 glue uniquely. For$A=M_2(k)$ in characteristic zero,$HH_0(A)$ is one-dimensional and is detected by trace.

Derivation or proof

Define the glued function at$x$ by either local value. Compatibility on the overlap makes it well defined, and every point lies in one cover member, proving uniqueness. Thus the equalizer is genuinely isomorphic tok^U. For matrices, every commutator has trace zero by cyclicity. Conversely the traceless basis$E_{12},E_{21},E_{11}-E_{22}$ consists of commutators$[E_{11},E_{12}],[E_{22},E_{21}],[E_{12},E_{21}]$. Hence the commutator span is exactly the three-dimensional trace-zero subspace, and the quotient has dimension one. These are complete finite computations of two categorical constructions with all maps supplied.

Worked example

Use$U=\{1,2,3\},U_1=\{1,2\},U_2=\{2,3\}$.

Local pairs(a,b) and(c,d) glue precisely when$b=c$, giving the global triple(a,b,d). For matrices$\mathrm{diag}(2,-2)$ vanishes inHH_0, while$I$ represents a nonzero class because its trace is two.

A limit or colimit notation becomes mathematical content only after its indexing category and maps are fixed.

Scope

IncludingU itself as a terminal test object can make an unrestricted formulaF(U)=lim_{V→U}F(V) tautological rather than a descent theorem. The higher factorization-homology construction needs its algebraic and geometric hypotheses.

Further development

Specify a cover or disk-indexing diagram, construct its maps and compare the resulting derived limit/colimit with the intended object. Compute higher homology rather than identifying a single vector space with an entire persistence module. For a commutative algebra, every commutator is zero and $HH_0(A)=A$ itself; for k×k it has dimension two. Thus replacing $M_2(k)$ by an algebra with the same number of displayed matrix entries or a different decomposition can change the quotient. Algebra multiplication, not vector-space size alone, determines this invariant.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R046.

Y25-02-021 A CPTP update gives a quantitative mixing law

2025-02-27 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What does a specified completely positive update actually preserve, and which coherence decays under repetition?

The qubit depolarizing channel has an exact Kraus certificate and contraction factor. Trace preservation does not mean preservation of an input’s off-diagonal information or a claimed operational-coherence percentage.

Definitions and assumptions

For$0\le p\le 1$ define$E_p(\rho)=(1-p)\rho+pI/2$ on trace-one qubit matrices.

A Kraus representation$E(\rho)=\Sigma K_j\rho K_j\dagger$ is completely positive, and$\Sigma K_j\dagger K_j=I$ gives trace preservation.

Trace distance is$D(\rho,\sigma)=||\rho-\sigma||_1/2$. Write$\rho=(I+r\cdot\sigma)/2$ in Bloch coordinates.

Exact depolarizing contraction

Kraus operators are$\sqrt{(1-3p/4)}I$ and$\sqrt{(p/4)}X,Y,Z$. The channel sendsr to$(1-p)r$ and satisfies$D(E_p\^n\rho,E_p\^n\sigma)=(1-p)\^nD(\rho,\sigma)$.

Derivation or proof

The identity$X\rho X+Y\rho Y+Z\rho Z=2Tr(\rho)I-\rho$ follows by expandingρ in the Pauli basis. Substituting the four Kraus terms gives$(1-p)\rho+pI/2$, and their squared coefficients sum to one, proving the channel properties. On a difference of two states the identity term cancels, so$E_p(\rho-\sigma)=(1-p)(\rho-\sigma)$. Homogeneity of the trace norm proves the exact distance formula, with iteration giving its power. This is stronger and more specific than saying the update is coherent: it quantifies the information that is intentionally lost.

Worked example

Takep=1/5,ρ=|0⟩⟨0|,σ=|1⟩⟨1| andn=5.

Their initial trace distance is one; after five updates it is(4/5)^5=1024/3125≈0.32768. Each state approachesI/2. Off-diagonal input entries shrink by the same factor, so no nontrivial qubit coherence is exactly protected by this channel.

The update’s positivity, normalization and observability behavior have independent exact certificates.

Scope

A genericCPTP map need not have this contraction rate or a unique stationary state. The historical95% phrase has no mathematical meaning until its observable, norm, sample and confidence rule are defined.

Further development

For a specified multi-qubit channel, determine its fixed-point algebra and peripheral spectrum. State which information is recoverable by an encoding and recovery map rather than equating complete positivity with error correction. At p=0 every state is fixed and there is no mixing. At p=1 every input becomes I/2 in one step. For 0<p≤1 the fixed state is unique because a fixed Bloch vector must satisfy r=(1−p)r. These endpoint and uniqueness cases are part of the exact channel theorem.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R047.

Y25-02-024 Pentagon consistency is a path identity with a norm error bound

2025-02-27 · historical evidence A · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Recovered dated mathematical topic; newly revised and developed here.

Revised question

What finite identity is tested by an associativity pentagon, and what can a tolerance of0.07 actually imply?

For ordinary vector-space tensor products the five parenthesizations are canonically identified and the pentagon commutes exactly. Approximate unitary identifications accumulate error according to a stated norm, not an undefined coherence percentage.

Definitions and assumptions

The five parenthesizations of four ordered factors are(((AB)C)D),((A(BC))D),A((BC)D),A(B(CD)),(AB)(CD).

Each associator sends a pure tensor to the same ordered four-tuple of entries, changing only brackets.

For intended unitary mapsU_j and implemented unitaryV_j assume||V_j−U_j||≤ε_j in operator norm.

Exact pentagon and quantitative perturbation

The two directed associator paths from(((AB)C)D) toA(B(CD)) are equal. If their edge counts are three and two, their approximate products differ by at most the sum of the five edge errors.

Derivation or proof

Pure tensors span the tensor product. Along either path the elementary associators retain the same ordered entriesa,b,c,d and end at a⊗(b⊗(c⊗d)). The two linear maps are therefore equal. For approximate maps, compare each path product with its exact product using a telescoping sum. Unitary left and right factors have norm one, so one path error is at most the sum of its three edge errors and the other at most the sum of its two. Their exact middle products agree, and the triangle inequality gives the five-error bound. This proves a precise residual statement while avoiding any claim that arbitrary fusion-category matrices satisfy the pentagon.

Worked example

Take all five errors bounded by0.07.

The worst-case path discrepancy is at most0.35. This is an operator-norm upper bound, not a statement that65%,93% or95% of operations are coherent. A much smaller observed residual may be possible through cancellation, but would require actual matrix data and an independent numerical error bound.

The recovered tolerance now has a mathematical meaning tied to explicit maps and paths.

Scope

Nontrivial fusion categories require their actual F-symbols, admissible labels, basis conventions and pentagon equations. Checking only that one F-matrix is unitary or involutive is insufficient. No hardware success rate follows from these algebraic identities.

Further development

Given exact fusion rules, enumerate every admissible labeled pentagon and certify the maximum residual with interval arithmetic. Separate truncation error, arithmetic error and experimental error before selecting a tolerance.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R048.

Y25-02-023 Parity observations alone do not certify quantum coherence

Date unavailable · historical evidence T · Developed mathematical result with explicit hypotheses and checked example; historical claims repaired where necessary

Recovered starting point

Uploaded external February2025topological-qubit discussion; its participants and quoted questions are not attributed to the user.

Revised question

Which distinct states have identical parity observations, and what additional observable can distinguish their coherence?

An uploaded external discussion questioned the strength of topological-qubit evidence. A finite two-qubit calculation gives the relevant identifiability principle without attributing that external discussion to the user or adjudicating device claims.

Definitions and assumptions

Let$\rho_B=|\Phi^+\rangle\langle\Phi^+|$ with$|\Phi^+\rangle=(|00\rangle+|11\rangle)/\sqrt{2}$, and$\rho_M=(|00\rangle\langle00|+|11\rangle\langle11|)/2$.

Computational-basis measurements andZZ parity are diagonal in the same basis. A coherence witness can be off diagonal, such as$X\otimes X$.

The source is uploaded external text with its own February2025discussion dates; no date of a user conversation about it is recovered.

An exact observational collision

> ρ_B andρ_M have identical computational-basis probabilities, identical one-qubit marginals$I/2$ andZZ expectation one. But$\mathrm{Tr}(\rho_B XX)=1$ and$\mathrm{Tr}(\rho_M XX)=0$, and their trace distance is$1/2$.

Derivation or proof

Both states place probability one half on00 and11 and zero elsewhere. Every diagonal observable therefore has the same expectation. Partial trace removes the Bell cross terms, leaving the same marginal$I/2$. The operatorXX exchanges00 and11, so its expectation is the sum of those cross terms: one for the Bell state and zero for the mixture. Their difference is$(|00\rangle\langle11|+|11\rangle\langle00|)/2$, whose nonzero eigenvalues are$\pm1/2$. Its trace norm is one, giving distance one half. This proves that one class of observations can agree exactly while a different observable resolves the states.

Worked example

Use an equal mixture of the two states as a third candidate.

It retains every stated diagonal statistic but hasXX expectation$1/2$. In fact the entire family$\rho(q)$ with coherence$q/2$,$0\le q\le1$, shares those diagonal observations whileXX reads$q$. Repeating only the diagonal measurements at greater precision cannot identify$q$.

The mathematical issue is observation sufficiency, not merely the number of measurements.

Scope

This example makes no factual claim about Microsoft hardware or Majorana modes. Topological invariants, lifetimes, coherent control and error rates require different defined observations and models.

Further development

Specify a measurement family and compute the kernel of its linear map on Hermitian matrices. Add observables until the targeted parameter is identifiable, then derive finite-sample confidence bounds for those observables.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R047.

March 2025

Dated fragments recovered; complete transcript unavailable. 35 developed treatments.

Tensor approximation, spectral geometry, modular entropy, renormalization, higher gauge structures, phase noise and conservation bounds organize this month. Proposed cross-domain structures are developed through explicit calculations rather than accepted from their names.

ID · date	Concept and revised question	Result and mathematical core
Y25-03-001 2025-03-01 · A	A low-rank approximation has an exact error certificate What does a rank-k reduction preserve, and which error can be certified before assigning speed claims?	Truncated SVD optimality The minimum of ||A−B||_F² over rank(B)≤k isΣ_(j>k)s_j², attained by retaining the firstk singular terms.
Y25-03-003 2025-03-01 · A	Persistent topology starts with a chain complex whose boundary squares to zero Which boundary maps and filtration make the recovered persistent-homology notation correct?	An exact persistence interval OverQ, degree-one homology of this filtration is zero before1, one-dimensional for1≤t<2, and zero from2 onward. Its single interval is[1,2).
Y25-03-004 2025-03-01 · A	A QUBO and a one-edge QAOA instance can be solved exactly What exact optimization and quantum expectation are available before asserting a hybrid optimizer advantage?	Exact classical and quantum values On a triangle the maximum cut is2, attained by every nonconstant binary labeling. For a single quantum edge,⟨C⟩=1/2+(1/2)sin(4β)sinγ, reaching1 atβ=π/8,γ=π/2.
Y25-03-005 2025-03-01 · A	A Poisson response model gives an identifiable rate, not a latency theorem What assumptions justify the recovered event-response formula and how can its rate be checked?	Arrival probability and rate identification P(T≤t)=1−e^(−λt), E[T]=1/λ, andh(t)=λ. Given an exact nontrivial probabilityp at timet>0, the unique compatible rate isλ=−log(1−p)/t.
Y25-03-002 2025-03-01 · A	Tensor compression uses isometries, not an inverse of a rectangular map How should an effective operator be defined when a tensor-network coarse map changes dimension?	Compression and a residual identity H_eff is Hermitian and HZ−ZH_eff=R. IfH_effv=λv with||v||=1, then||(H−λ)Zv||=||Rv||, so a self-adjoint residual bound certifies spectral proximity by||Rv||.
Y25-03-029 2025-03-02 · U	A fractal probability measure has a quantitative weak-convergence proof What exact convergence theorem can be supplied for a specified fractal-measure sequence?	Quantitative weak convergence For everyL-Lipschitz real functionf, |∫fdμ−∫fdμ_n|≤L3^(−n). More generally a continuousf on[0,1] has error at most its modulus of continuity at3^(−n).
Y25-03-027 2025-03-03 · A	A stabilizer code needs commuting checks and an error model Which finite code supplies an actual correction theorem rather than an asserted fidelity?	Exact syndrome and failure probability The errorsI,X1,X2,X3 have syndromes(+,+),(−,+),(−,−),(+,−), respectively. Majority correction fails with probability3p²−2p³.
Y25-03-032 2025-03-03 · A	A spin-foam face cannot integrate a four-form as its surface action Which differential-form degrees make a proposed curvature action meaningful?	An exact degree and characteristic-number calculation The surface pullback ofTr(F∧F) iszero. OnT4, (1/2)∫(F/(2πi))²=1, whereas∫(F/(2πi))²=2.

ID · date	Concept and revised question	Result and mathematical core
Y25-03-034 2025-03-03 · A	Circle factorization has an explicit Hochschild chain model Which concrete circle calculation gives content to a factorization-homology symbol?	An exact Hochschild calculation HH0(A)=A has dimension2. EveryHH_n(A) forn≥1 has dimension1. After tensoring the resolution withA overA^e, its differentials alternate0 and multiplication by2ε.
Y25-03-028 2025-03-04 · A	Fibonacci fusion has a complete two-dimensional algebraic certificate Which exact fusion and diagonalization data can be proved before claiming complete categorical coherence?	Exact Fibonacci diagonalization Nτ²=I+Nτ, S²=I andNτ=S diag(φ,−1/φ)S. The ratiosS_(τ,j)/S_(1,j) are exactly these two eigenvalues.
Y25-03-030 2025-03-04 · A	A homotopy fiber retains paths that an ordinary preimage discards What explicit example distinguishes a homotopy fiber from an ordinary set-theoretic fiber?	An exact component classification π0(ΩS1)≅Z by winding number; each component is contractible. Thus the homotopy fiber has infinitely many components despite the one-point ordinary fiber.
Y25-03-031 2025-03-04 · A	A projection loss controls only the directions that its matrix sees What does a concrete gradient loss prove about convergence and invisible directions?	Exact dissipation and nullspace preservation Along the flow,L′=−||∇L||²≤0. Every kernel component ofx is constant, while positive eigenmodes ofWᵀW decay exponentially toward a least-squares solution.
Y25-03-033 2025-03-07 · A	The eta factors carry a calculable modular phase What transformation property can be checked directly for the recovered coherence coefficient?	Exact translation multiplier F_n(τ+1)=e^(πi(2n−1)/12)F_n(τ). In particular it is not translation-invariant for any integern≥1.
Y25-03-035 2025-03-07 · A	A square-root eigenvalue law determines a spectral dimension What mathematical consequences follow from a declared eigenvalue growth law?	Exact count and heat asymptotics D is positive self-adjoint with compact resolvent;N_D(Λ)=floorΛ². Its spectral zeta function isζ(s/2), initially convergent forRe s>2, and its heat trace is2/t²+O(1) ast→0+.
Y25-03-006 2025-03-09 · A	The recovered renormalization series has a conditional convergence theorem Which explicit coefficient and denominator bounds make this infinite expression well-defined?	A uniform tail estimate The oscillatory series converges uniformly. Its tail afterN is at mostM N^(a−3)/(3−a). The denominator-series tail is at most(M/φ)N^(a−5)/(5−a), and reciprocal errors are bounded by that tail divided by the product of denominator lower bounds.
Y25-03-007 2025-03-10 · A	An automorphic transformation law comes from reindexing a lattice sum What is a fully defined example of modular covariance, and why do names or convergence exponents not establish it?	Weight-four covariance The series converges absolutely and locally uniformly, is holomorphic, and satisfiesG4(γτ)=(cτ+d)^4G4(τ).
Y25-03-010 2025-03-13 · A	Connes distance can be computed exactly on two points What metric follows from an explicitly defined finite spectral triple?	Two-point spectral distance The distance equals1/|m|. Ifm=0 the distance is infinite.

ID · date	Concept and revised question	Result and mathematical core
Y25-03-011 2025-03-13 · A	Coherence is an identity of operations, and a spectral page still needs a differential Which algebraic data can be checked in a toy model before claiming an E∞ or spectral-sequence construction?	Exact coherence and a filtered cancellation The operationsm_n are symmetric and compatible with substitutions. In the filtered complex the associated graded differential is zero, but the next filtration-dropping differential takesv tow and kills both classes; the total complex is acyclic.
Y25-03-008 2025-03-14 · A	Spectral thresholds and curvature must survive changes of scale Which geometric statements are invariant, and what does the modified-flow notation actually require?	Exact scale laws and spherical flow Any fixed numerical cutoff forλ1 orR changes under metric scaling. For the round sphere,r(t)²=r(0)²−2(n−1)t up to its extinction time.
Y25-03-009 2025-03-14 · A	A categorical surface theory begins with a Frobenius algebra What finite algebraic model provides an actual gluing rule before higher-cobordism analogies are used?	A complete small Frobenius calculation The pairing matrix is[[0,1],[1,0]]. The coproduct isΔ(1)=1⊗ε+ε⊗1,Δ(ε)=ε⊗ε; it is coassociative and satisfies the Frobenius compatibility.
Y25-03-014 2025-03-17 · A	A modular cocycle obeys a twisted composition law How can a modular cocycle be verified exactly before using it as a transport or geometric flow?	Finite cocycle and intertwining The matricesu_t are unitary, satisfy the stated cocycle law, and obey αρ_t(A)=u_t ασ_t(A)u_t*.
Y25-03-016 2025-03-17 · A	A zero-dimensional Gaussian realizes an exact Wetterich flow What exact regulated model verifies the flow equation, and what does a zero beta function fail to determine?	Exact Gaussian effective-average action Γ_k(φ)=m²φ²/2+(1/2)log(m²+R_k)−(1/2)log(2π), so∂kΓ_k=(1/2)(Γ_k″+R_k)⁻¹∂kR_k. Its unique minimizing field isφ=0 for allk.
Y25-03-017 2025-03-17 · A	Spectral flow counts oriented crossings; odd-dimensional integrals need odd forms What is a correctly typed small spectral-flow calculation, and which form-degree mismatch must be repaired?	Two exact integer calculations ForD(t)=diag(t−1/4,t−3/4),0≤t≤1, spectral flow is2. ForU(t)=e^(2πit), the odd-character integral is1.
Y25-03-012 2025-03-18 · A	Finite modular flow distinguishes the state Hamiltonian from the standard-form generator What exactly are Δ and K in a finite faithful-state model?	Exact finite-dimensional modular structure For matrix unitsEij in aρ-eigenbasis with eigenvaluespi, ΔρEij=(pi/pj)Eij andKρEij=log(pj/pi)Eij. Modular flow on left observables isσ_tρ(A)=ρ^(it)Aρ^(−it).
Y25-03-013 2025-03-18 · A	The entropy first law is an exact first derivative at a faithful state Which variation makes the recovered first law valid, and what remains at second order?	First law and a positive second-order example Atε=0,dS(ρ(ε))/dε=Tr(Xhσ). Exactly, D(ρ‖σ)=Tr[(ρ−σ)hσ]−[S(ρ)−S(σ)]. For a diagonal binary perturbation its second-order coefficient is positive.
Y25-03-015 2025-03-18 · A	Wilson-loop words need a connection or a well-defined group assignment What group-valued construction replaces an undefined nested path-ordered integral?	Gauge covariance and an explicit noncommutative loop A based loop product transforms by conjugation at its basepoint, so its trace is invariant. ForU=iσx,V=iσz, the commutator is−I with trace−2, whereas commutingU,V givel with trace2.

ID · date	Concept and revised question	Result and mathematical core
Y25-03-019 2025-03-21 · U	A semidirect product requires an action, even when its factors are familiar What choices are missing from SU4⋉Q8, and which concrete actions produce well-defined groups?	An explicit untwisting theorem For a lifted inner action, F(g,q)=(g u_q,q) is a group isomorphism SU4⋊_αQ8→SU4×Q8. Another valid action uses a homomorphismQ8→C2 followed by complex conjugation onSU4.
Y25-03-018 2025-03-21 · A	A Maurer-Cartan equation needs graded brackets and a curvature calculation What exact flatness equation can be derived before invoking higher brackets?	An exact constant-connection flatness criterion ForA=Xdx+Ydy with constant matrices, F=[X,Y]dx∧dy. It is flat exactly whenX andY commute.
Y25-03-020 2025-03-21 · A	A projective multiplier gives a concrete H² cocycle What exact cocycle can replace an unspecified group-cohomology decoration?	An explicit nontrivial multiplier U_vU_w=ω(v,w)U_(v+w), whereω((a,b),(c,d))=(−1)^(bc). This is a normalized2-cocycle and is not a coboundary.
Y25-03-021 2025-03-21 · A	Cohomology-preserving blocking requires a chain homotopy certificate When does a coarse-graining map really preserve cycle information?	An exact chain-homotopy equivalence The mapsr,i commute with boundaries, ri=id_D, andir−id_C=∂h+h∂ for h(v0)=0,h(v1)=−e0,h(C1)=0. Therefore they induce inverse maps on integer homology.
Y25-03-026 2025-03-24 · A	An energy bound exposes incompatible power and thrust scalings Which elementary energy inequality must any claimed fixed-exhaust-speed scaling satisfy?	An exact necessary bound F≤2P/v. At fixedv, a force scalingF=cP^(1+δ) withc>0 andδ>0 cannot satisfy this bound for arbitrarily largeP.
Y25-03-025 2025-03-28 · U	Second-order correlations are factorial moments with distinguishable models What exact statistical quantity can be calculated for the recovered g² topic?	Exact model values A Poisson count hasg2=1; a deterministic countN=n≥1 hasg2=1−1/n; a geometric countP(N=k)=(1−q)q^k,0<q<1, hasg2=2.
Y25-03-022 2025-03-28 · A	A power-law phase-noise spectrum needs cutoffs to have finite variance When does the recovered spectral-density model define a finite-variance process?	Finite-band variance and low-frequency divergence For0<f_L<f_H, the variance isΣ_(i≠−1)h_i(f_H^(i+1)−f_L^(i+1))/(i+1)+h_(−1)log(f_ H/f_L). A nonzero nonnegativef^(−p) term withp≥1 diverges asf_L→0.
Y25-03-023 2025-03-28 · A	Eliminating a hidden variable derives a memory kernel and a source term Which terms must appear when a reduced equation is obtained from a larger system?	Exact reduced Volterra equation The observed coordinate satisfies x′(t)=−ax(t)+b e^(−ct)y0+bd∫₀ᵗe^(−c(t−s))x(s)ds. This is equivalent to the full system with the stated initial data.
Y25-03-024 2025-03-28 · A	The recovered self-bilinear Hirota equation loses its time derivative What equation does the displayed Hirota expression actually impose?	An exact reduction D_tτ·τ=0 andD_x²τ·τ=2(ττ_xx−τ_x²). Therefore(D_x²−D_t)τ·τ=0 is equivalent to(logτ)_xx=0, with solutionsτ=e^(a(t)x+b(t)) locally on a connectedx interval.

Y25-03-001 A low-rank approximation has an exact error certificate

2025-03-01T15:38:22Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does a rank-k reduction preserve, and which error can be certified before assigning speed claims?

The recovered tensor-processing proposal names SVD and rank reduction but gives no measured complexity certificate. The matrix problem has a sharp mathematical answer: singular values determine the best Frobenius error. This supplies a reproducible approximation statement independently of any hardware throughput.

Definitions and assumptions

Let A be an m×n real matrix with singular values s1≥s2≥…≥0. The Frobenius norm squares and sums all entries.

A rank-k approximation is a matrix B with rank at mostk; arithmetic cost and approximation error are different quantities.

Truncated SVD optimality

> The minimum of ||A−B||_F² over rank(B)≤k isΣ_(j>k)s_j², attained by retaining the firstk singular terms.

Derivation or proof

LetP be the orthogonal projection onto the column space ofB, with rank at mostk. Decompose A−B=(I−P)A+(PA−B); the two summands have orthogonal column spaces, so its squared norm is at least||(I−P)A||_F²=Tr(AAᵀ)−Tr(PAAᵀ). In an eigenbasis ofAAᵀ the latter trace isΣs_j²p_jj, where0≤p_jj≤1 andΣp_jj≤k. Its maximum is the sum of the k largest squared singular values. This proves the lower bound. TakingB=Σ_(j≤k)s_ju_jv_jᵀ attains it, completing the proof including ties.

Worked example

Use A=diag(5,3,1).

Its total squared norm is35. The optimal rank-one matrixdiag(5,0,0) has error9+1=10, and rank two has error1. A relative squared error below1/20 requires rank two, since10/35>1/20 but1/35<1/20.

The retained rank can now be chosen from a declared error target, rather than from a headline reduction factor.

Scope

The theorem concerns a matrix and the Frobenius norm. A tensor’s chosen unfolding can change the ranks, and a matrix error bound does not directly certify a downstream control or topology decision.

Further development

For the actual contraction, state dimensions and an operation-count model, bound the propagated contraction error with operator norms, and measure runtime separately.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R049.

Y25-03-003 Persistent topology starts with a chain complex whose boundary squares to zero

2025-03-01T15:38:22Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which boundary maps and filtration make the recovered persistent-homology notation correct?

The historical formula mixes upper-index cohomology with lower-index boundary notation. A small filtered triangle gives an exact replacement and distinguishes homology from cohomology before a differentiable surrogate is introduced.

Definitions and assumptions

For chains, ∂k:C_k→C_(k−1), H_k=ker∂k/im∂_(k+1). For cochains, δ^k:C^k→C^(k+1), H^k=kerδ^k/imδ^(k−1).

Orient edges12,23,31 and the filled face123 consistently. Put all vertices at time0, all edges at time1 and the face at time2.

An exact persistence interval

> OverQ, degree-one homology of this filtration is zero before1, one-dimensional for1≤t<2, and zero from2 onward. Its single interval is[1,2).

Derivation or proof

The vertex-edge boundary matrix has columns(−1,1,0),(0,−1,1),(1,0,−1). Its rank is2 and its kernel is spanned byc=(1,1,1). The face boundary is exactlyc, so∂1∂2=0. Before edges appear,C1=0. With all edges and no face,H1=ker∂1 is one-dimensional. When the face appears, its boundary fills that kernel, makingH1=0. Each inclusion sends the cycle to the same chain until it becomes a boundary, proving the stated persistence rather than merely counting dimensions. Cohomology gives the dual vector-space result overQ, but uses transposed coboundaries and reversed arrows.

Worked example

Perturb all assigned cell times by at mostε, while preserving the face-after-boundary rule.

The old sublevel complex at timet is contained in the new one at t+ε, and conversely. Thus there are explicitε-shift maps; their compositions are the original inclusions shifted by2ε. These are deterministic comparisons, not inferred from processing speed.

A topology computation now has valid indices, an incidence matrix and a finite exact barcode.

Scope

A smooth surrogate for a rank or argmax need not preserve this barcode. Changing adjacency or thresholding can alter the complex and requires its own comparison theorem.

Further development

Publish the actual filtration rule and certify cell-time errors. Then prove an interleaving before claiming that a compressed or learned approximation preserves homology.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R050.

Y25-03-004 A QUBO and a one-edge QAOA instance can be solved exactly

2025-03-01T15:44:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact optimization and quantum expectation are available before asserting a hybrid optimizer advantage?

The recovered hybrid proposal names QUBO, annealing and QAOA without a verified problem instance. A maximum-cut encoding provides an exact classical certificate and a completely solvable one-edge quantum circuit.

Definitions and assumptions

For binary$x_i \in \{0,1\}$, an edge cut indicator is$x_i + x_j - 2x_i x_j$. Minimize its negative to obtain a QUBO.

For one quantum edge set$C = (I - Z_1Z_2)/2$, $B = X_1 + X_2$ and$|\psi\rangle = e^{-i\beta B} e^{-i\gamma C} |{+}{+}\rangle$.

Exact classical and quantum values

On a triangle the maximum cut is2, attained by every nonconstant binary labeling. For a single quantum edge,$\langle C \rangle = 1/2 + (1/2)\sin(4\beta)\sin\gamma$, reaching1 at$\beta = \pi/8, \gamma = \pi/2$.

Derivation or proof

A triangle has three edges. A constant labeling cuts none; a nonconstant labeling places one vertex on one side and two on the other and cuts exactly the two incident cross edges. Enumeration of eight labels is complete. For the quantum formula, conjugateZ_1Z_2 by the two X rotations. The resulting terms are$\cos^2(2\beta) Z_1Z_2$, $\sin(2\beta)\cos(2\beta)(Z_1Y_2 + Y_1Z_2)$, and$\sin^2(2\beta) Y_1Y_2$. Conjugating the cross terms by the diagonal cost phase and evaluating in$|{+}{+}\rangle$ yields$\langle Z_1Z_2 \rangle = -\sin(4\beta)\sin\gamma$; the other contributions vanish. Substitution into$C$ gives the expression. Direct four-component complex arithmetic independently checks the sign convention.

Worked example

Take the stated angles on the single edge and compare with the uniform initial state.

The initial cut expectation is$1/2$. After the circuit the probability is supported on the two cut states01 and10, giving expectation1. On the triangle, the exact optimal negative QUBO energy is-2.

This is a full small-instance certificate, rather than an unsupported throughput or generic optimization claim.

Scope

An exact one-edge solution does not imply an advantage on large QUBOs. Encoding constraints may require penalty bounds, and hardware noise and parameter-search cost are separate issues.

Further development

For a specified finite family, publish the QUBO coefficients, exact lower bounds and circuit expectations, then compare complete computational costs at a fixed approximation guarantee.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R051.

Y25-03-005 A Poisson response model gives an identifiable rate, not a latency theorem

2025-03-01T15:44:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What assumptions justify the recovered event-response formula and how can its rate be checked?

The historical response probability is exactly the first-arrival law of a homogeneous Poisson process. It is a mathematical model with strong assumptions: independent stationary increments and a constant rate. A formula alone does not validate the claimed speed of an implementation.

Definitions and assumptions

Let$N(t)$ be a Poisson process of rate$\lambda>0$ and$T=\inf\{t:N(t)\geq 1\}$.

Its survival probability is$S(t)=P(T>t)$; the hazard rate is$h(t)=-d(\log S(t))/dt$.

Arrival probability and rate identification

> $P(T\leq t)=1-e^{(-\lambda t)}$, $E[T]=1/\lambda$, and$h(t)=\lambda$. Given an exact nontrivial probability$p$ at time$t>0$, the unique compatible rate is$\lambda=-\log(1-p)/t$.

Derivation or proof

The event$T>t$ is$N(t)=0$. The Poisson mass formula gives$P(N(t)=0)=e^{(-\lambda t)}$, establishing the distribution. The tail integral$E[T]=\int_0^\infty P(T>t)dt$ equals$1/\lambda$. Differentiating$\log S$ yields the constant hazard, and solving$S(t)=1-p$ gives the identification formula. The memoryless identity$P(T>s+t|T>s)=S(s+t)/S(s)=S(t)$ follows immediately. Conversely a claimed model whose conditional waiting probability depends on past waiting time is not this homogeneous arrival law.

Worked example

Use$\lambda=2$ in inverse-time units.

The median solvese^(−2t)=1/2, sot_med=(log2)/2, whereas the mean is1/2. At that median, the probability of at least two events is1−e^(−λt)(1+λt)=(1−log2)/2. First-arrival probability and event count are therefore different observables.

The model yields exact quantities that can be compared with data only after the event and time unit are defined.

Scope

Bursting, refractory periods, nonstationary rates and dependent arrivals violate the stated assumptions. None of the historical millisecond or energy numbers is certified by this distributional calculation.

Further development

Test the constant-hazard prediction on declared time bins; for a nonhomogeneous rateλ(t), replaceλt by the integrated intensity$\int_0^t \lambda(s)ds$ and derive the revised likelihood.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: .

Y25-03-002 Tensor compression uses isometries, not an inverse of a rectangular map

2025-03-01T17:37:28Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How should an effective operator be defined when a tensor-network coarse map changes dimension?

The recovered MERA expression uses an inverse that may not exist for a dimension-reducing map. An isometric embedding gives a precise compression. It preserves Hermiticity but generally loses an invariant subspace; the loss can be measured by an explicit residual.

Definitions and assumptions

LetZ:C^r→C^d satisfyZ*Z=I_r and letH=H* onC^d. DefineH_eff=Z*HZ andP=ZZ*.

The discarded coupling isR=(I−P)HZ; it vanishes exactly when the embedded subspace isH-invariant.

Compression and a residual identity

> H_eff is Hermitian and HZ−ZH_eff=R. IfH_effv=λv with||v||=1, then||(H−λ)Zv||=||Rv||, so a self-adjoint residual bound certifies spectral proximity by||Rv||.

Derivation or proof

Taking adjoints gives(Z*HZ)*=Z*HZ. NextZH_eff=ZZ*HZ=PHZ, and subtracting yieldsR. SinceZ preserves norms, Zv is a unit vector. The eigenvalue equation then gives(H−λ)Zv=Rv. For a HermitianH, diagonalize it and expandZv in its orthonormal eigenbasis: the residual norm squared is$\Sigma|c_j|^2(E_j-\lambda)^2$, at least$\mathrm{dist}(\lambda,\sigma(H))^2$. Thusdist≤||Rv||. No inverse ofZ is used. WhenR=0 the restriction is genuinely invariant and the compressed eigenvalues are exact eigenvalues; otherwise compression is a variational approximation.

Worked example

TakeH=diag(0,2) andZ=(1,1)ᵀ/√2, a2×1 isometry.

H_eff=1, whileHZ−ZH_eff=(−1,1)ᵀ/√2 has norm1. The candidate1 is exactly distance1 from the true spectrum{0,2}. Thus a valid Hermitian compression can produce an intermediate value rather than an exact level.

The example supplies both the correct effective operator and a quantitative certificate of what it loses.

Scope

An arbitrary tensor-network map need not be isometric, and repeated compressions accumulate errors. No MERA efficiency, emergent metric or arithmetic correspondence follows from this finite example.

Further development

For a sequence of blocking maps, track each discarded coupling and prove a cumulative bound on the target spectral cluster.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R049, R052.

Y25-03-029 A fractal probability measure has a quantitative weak-convergence proof

2025-03-02T01:53:39Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact convergence theorem can be supplied for a specified fractal-measure sequence?

The retrieved source names fractal convergence but supplies no usable measure construction. A BernoulliCantor measure gives a complete model with a deterministic error bound for test functions. It separates convergence from a verbal appeal to fractal moduli.

Definitions and assumptions

LetB_j be independent fair bits andX=Σ_(j≥1)2B_j/3^j. LetX_n be the firstn terms andμ_n its distribution.

μ_n is the uniform measure on2^n finiteCantor endpoints; μ is the distribution ofX.

Quantitative weak convergence

For everyL-Lipschitz real functionf, |∫fdμ−∫fdμ_n|≤L3^(−n). More generally a continuousf on[0,1] has error at most its modulus of continuity at3^(−n).

Derivation or proof

ConstructX andX_n on the same probability space. Their omitted tail is nonnegative and at mostΣ_(j>n)2/3^j=3^(−n). Therefore|f(X)−f(X_n)|≤L3^(−n), or≤ω_f(3^(−n)) for a continuousf. Taking expectations proves the estimates and weak convergence because the modulus tends0 by uniform continuity. The coupling also supplies aWasserstein-1 bound without requiring density functions; the limiting measure is supported on theCantor set. The theorem has actual sample variables, weights and a topology of convergence.

Worked example

Takef(x)=x andf(x)=x².

EX=1/2, whileEX_n=(1−3^(−n))/2, so the first-moment error is3^(−n)/2. AlsoVarX=Σ_(j≥1)1/9^j=1/8 andEX²=3/8; finite moments follow by truncating those geometric sums.

The exact measure construction supports convergence and moment calculations, not an unspecified higher-topos theorem.

Scope

The historical fractal-measure equations were not recovered fully. This is a newly specified mathematical model, not a reconstructed original proof or a validation of a claimed error rate.

Further development

For the intended measure, state its maps, weights and metric; construct an explicit coupling or prove tightness plus uniqueness of the limit.

Source receipt: months/03/root_coverage.json record 0. Mathematical references: R050.

Y25-03-027 A stabilizer code needs commuting checks and an error model

2025-03-03T01:13:00Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which finite code supplies an actual correction theorem rather than an asserted fidelity?

The recovered HQECC material states stabilizer constraints and numerical errors without a noise model. A three-qubit repetition code gives a complete limited theorem: it corrects one bit flip, while explicitly failing to protect against a general phase error.

Definitions and assumptions

Use commuting checksS1=Z1Z2 andS2=Z2Z3. Their common+1 eigenspace isspan{|000⟩,|111⟩}.

Assume independent bit-flip errors with probabilityp on each qubit, followed by ideal syndrome measurement and correction.

Exact syndrome and failure probability

The errorsI,X1,X2,X3 have syndromes(+,+),(−,+),(−,−),(+,−), respectively. Majority correction fails with probability$3p^2-2p^3$.

Derivation or proof

PauliZ anticommutes withX on the same qubit and commutes on distinct ones. Applying that rule to each check gives the four displayed syndrome pairs, so each single flip can be identified and inverted. The checks act identically on both logical basis vectors and do not measure their superposition coefficients. With two or three independent flips, the syndrome-based single-flip correction instead changes the logical bit. The total probability is$C(3,2)p^2(1-p)+p^3=3p^2-2p^3$. At the subspace level,$PE_a*E_bP=\delta_{ab}P$ for the four errors: diagonal products areI and distinct products take the code to an orthogonal syndrome subspace. This is the exact correction condition in this restricted error set.

Worked example

Take$p=1/100$.

The failure probability is$298/1000000=0.000298$. By contrast,Z1 has syndrome(+,+) and acts as a nontrivial logical phase, so it is undetectable by these checks. An unspecified general quantum noise rate cannot be assigned the same failure polynomial.

Checks, code space and a declared error distribution produce a real finite guarantee. The recovered0.012 is not validated.

Scope

Measurement faults, correlated errors and phase noise change the result. No large holographic-code or hardware fidelity theorem follows from this small code.

Further development

Give the actual stabilizer generators and noise channel, verify the fullKnill-Laflamme condition or a recovery bound, and report uncertainty separately from ideal-code probability.

Source receipt: months/03/root_coverage.json record 0. Mathematical references: R045.

Y25-03-032 A spin-foam face cannot integrate a four-form as its surface action

2025-03-03T21:07:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which differential-form degrees make a proposed curvature action meaningful?

The recovered expression integrates$\mathrm{Tr}(R\wedge R)$ over a face. IfR is a curvature2-form andf is a2-dimensional face, its wedge square has degree4 and is not a surface volume form. A correctly typed four-dimensional calculation gives a constructive replacement.

Definitions and assumptions

A curvatureF is a matrix-valued2-form. Tr(F∧F) is a4-form and pulls back to zero on a smooth2-manifold.

On the unit4-torus, letω1=dx1∧dx2,ω2=dx3∧dx4 anduse a line-bundle curvatureF=2πi(ω1+ω2).

An exact degree and characteristic-number calculation

The surface pullback ofTr(F∧F) iszero. OnT4, (1/2)∫(F/(2πi))²=1, whereas∫(F/(2πi))²=2.

Derivation or proof

An alternating4-form vanishes on a tangent space of dimension2, because every four tangent vectors are linearly dependent. It therefore cannot supply a nonzero ordinary face integral. OnT4, ω1²=ω2²=0 andω1∧ω2=ω2∧ω1 because both degrees areeven. Hence(ω1+ω2)²=2dx1∧dx2∧dx3∧dx4, whose integral is2. The factor1/2 gives the degree-four Chern-character normalization1. Existence of a line bundle with this integral first-Chern class is classical; the wedge arithmetic and dimension check are explicit. A two-dimensional action could instead use a specified2-form, but that defines a different model.

Worked example

Compare integration domainsT2 andT4 for the same formal curvature square.

OnT2 its pullback is identicallyzero. OnT4 it has the nonzero integral just computed. The change is not a small normalization correction: it changes which action is even well-defined as a top-degree integral.

Form degree and domain dimension give an immediate necessary consistency test before a partition integral is proposed.

Scope

Discrete spin foams can use holonomies and combinatorial weights rather than differential forms, but those weights must be defined separately. No continuum quantum-gravity theory is constructed here.

Further development

State the cell dimension and local fields, then derive either a typed continuum integral or a gauge-invariant combinatorial face weight.

Source receipt: months/03/root_coverage.json record 0. Mathematical references: R052.

Y25-03-034 Circle factorization has an explicit Hochschild chain model

2025-03-03T21:07:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which concrete circle calculation gives content to a factorization-homology symbol?

The source writes a disk-indexed colimit without defining the algebraic input. For an associative algebra, the classical circle model isHochschild homology. The dual-number algebra has a small exact resolution that can be worked out completely.

Definitions and assumptions

LetA=C[ε]/(ε²), andA^e=C[x,y]/(x²,y²), acting by left and rightε multiplication.

Use the periodic free resolution with alternating maps multiplication byu=x−y andv=x+y, followed by the multiplication augmentationA^e→A.

An exact Hochschild calculation

HH0(A)=A has dimension2. EveryHH_n(A) forn≥1 has dimension1. After tensoring the resolution withA overA^e, its differentials alternate0 and multiplication by2ε.

Derivation or proof

InA^e the productuv=x²−y² iszero. On the basis1,x,y,xy, multiplication byu has image span(u,xy) and kernel span(v,xy); multiplication byv has the reverse image and kernel. Both have rank2. The augmentation kernel isspan(u,xy), so the complex is exact before tensoring and is a free resolution. Tensoring setsx=y=ε; u becomes0 andv becomes2ε. HenceHH0=A. In odd positive degree the quotient isA/(2εA), one-dimensional. In even positive degree the kernel of2ε multiplication isspanε, again one-dimensional. The circle/factorization identification is a classical structural input; the algebraic chain computation itself is explicit.

Worked example

Compare degree0 and degree1.

HH0 has basis1,ε. HH1 has the class of1 moduloε, whileHH2 has basisε in a kernel. Equal positive-degree dimensions do not make their cycle and boundary representatives identical.

A genuine algebra and a chain model turn an abstract integration label into exact computable invariants.

Scope

The recoveredA_E8 was not defined as anassociative algebra or disk algebra, so this is an illustrative replacement, not a computation of that missing object.

Further development

Specify the intendedE_n algebra and manifold, then use a justified excision theorem or explicit bar construction to compute its factorization homology.

Source receipt: months/03/root_coverage.json record 0. Mathematical references: R053, R048.

Y25-03-028 Fibonacci fusion has a complete two-dimensional algebraic certificate

2025-03-04T15:53:49Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which exact fusion and diagonalization data can be proved before claiming complete categorical coherence?

The recovered Fibonacci rule has a standard constructive realization. Its fusion matrix and sine-freeS matrix can be verified by elementary algebra. These identities are real evidence for a fusion ring, while the full associator and braiding still require coherence.

Definitions and assumptions

Letφ=(1+√5)/2 and use labels1,τ withτ²=1+τ. The multiplication matrix isNτ=[[0,1],[1,1]].

SetS=(1/√(1+φ²))[[1,φ],[φ,−1]].

Exact Fibonacci diagonalization

Nτ²=I+Nτ, S²=I andNτ=S diag(φ,−1/φ)S. The ratiosS_(τ,j)/S_(1,j) are exactly these two eigenvalues.

Derivation or proof

The quadratic identity follows by multiplying the integer2×2 matrix. Sinceφ²=φ+1 andφ−1=1/φ, the vectors(1,φ) and(φ,−1) are orthogonal eigenvectors with eigenvaluesφ and−1/φ. Both have squared norm1+φ². Normalizing them givesS and proves its orthogonality and diagonalization. Hence theVerlinde diagonalization recovers the integer multiplication rather than merely fitting a number. The common real Fibonacci associator blockF=[[1/φ,1/√φ],[1/√φ,−1/φ]] is also orthogonal becauseφ⁻²+φ⁻¹=1, but orthogonality of one block is not the wholepentagon theorem.

Worked example

Compute repeated fusion multiplicities.

τ³=1+2τ andτ⁴=2+3τ, by repeatedly substitutingτ²=1+τ. Thus dimensions grow by Fibonacci recurrences. The positive dimension homomorphismdimτ=φ satisfiesφ²=1+φ exactly.

The fusion law andS diagonalization are fully checked, with a clear boundary between ring data and categorical realization.

Scope

The historical phasee^(2πic/7) and fidelity bounds have no demonstrated compatibility with this example. Associativity of a numerical ring does not verify allF/R coherence equations.

Further development

Fix the classicalF andR symbols and exhaust the nontrivialpentagon andhexagon equations in a declared gauge; then distinguish mathematical consistency from benchmark performance.

Source receipt: months/03/root_coverage.json record 0. Mathematical references: R048.

Y25-03-030 A homotopy fiber retains paths that an ordinary preimage discards

2025-03-04T15:53:49Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What explicit example distinguishes a homotopy fiber from an ordinary set-theoretic fiber?

The recovered HoTT discussion lists homotopy fibers without a defined map. The inclusion of a base point into the circle gives a simple exact example in which path data radically changes the answer. It is a classical topology model, not a formalization of an unspecified type theory.

Definitions and assumptions

Takef:{*}→S1 withf(*)=1. The ordinary fiber over1 is a single point.

The homotopy fiber consists of(*,γ), whereγ is a path fromf(*) to1; here it is the based loop spaceΩS1.

An exact component classification

π0(ΩS1)≅Z by winding number; each component is contractible. Thus the homotopy fiber has infinitely many components despite the one-point ordinary fiber.

Derivation or proof

Use the coveringp:R→S1,p(t)=e^(2πit). Every based loop has a unique lift starting at0, whose endpoint is an integern becausep sends allintegers to1. That integer is invariant under based homotopy by continuity and the discrete endpoint set. Conversely, two lifted paths with the same endpoints are linearly homotopic inR while fixing endpoints; projection gives a based homotopy. Within a fixedn, the lifted path space is contractible by straight-line contraction to the path t↦nt. This proves both the integer classification and the componentwise contractibility. The ordinary preimage never records these paths.

Worked example

Compare the constant loop withγ(t)=e^(2πit).

Their lift endpoints are0 and1. Both lie over the same source point, but they are in different homotopy-fiber components. The difference is path information rather than a numerical cardinality attached to the original set.

A homotopy construction has concrete extra data that must be stated, not inferred from a higher-category label.

Scope

General homotopy fibers require a topological or∞-categorical model and appropriate equivalences. The example does not prove newHoTT coherence rules or reconstruct a missing categorical map.

Further development

Specify the actual map and base point, then compute its long exact homotopy sequence or a model of the fiber before assigning an invariant.

Source receipt: months/03/root_coverage.json record 0. Mathematical references: R048.

Y25-03-031 A projection loss controls only the directions that its matrix sees

2025-03-04T15:56:30Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does a concrete gradient loss prove about convergence and invisible directions?

The recovered categorical-loss andE8-projection language needs a defined function. A least-squares loss gives a complete matrix theorem: energy decreases, but components in the kernel persist. It therefore does not automatically enforce every desired coherence condition.

Definitions and assumptions

LetL(x)=(1/2)||Wx−y||² for a real matrixW. Gradient flow isx′=−Wᵀ(Wx−y).

The observable subspace israngeWᵀ; the invisible subspace iskerW, its orthogonal complement.

Exact dissipation and nullspace preservation

Along the flow,L′=−||∇L||²≤0. Every kernel component ofx is constant, while positive eigenmodes ofWᵀW decay exponentially toward a least-squares solution.

Derivation or proof

The chain rule givesL′=∇L·x′=−||∇L||². Ifv∈kerW, thenv·x′=−(Wv)·(Wx−y)=0, so the projection onto the kernel is conserved. Diagonalize the symmetric positive semidefinite matrixWᵀW. Relative to any least-squares solutionx*, the error satisfiese′=−WᵀWe; its eigencomponent of eigenvalueλ evolves ase^(−λt). Thus positive modes decay and zero modes persist. Strict convexity and unique convergence occur only whenW has full column rank. A zero loss certifiesWx=y, not an unrelated associator or lattice constraint absent fromW.

Worked example

TakeW=[1,1] andy=2.

The sumx1+x2 obeyss′=−2(s−2), while the differencex1−x2 isconstant. Fromx(0)=(0,0), both coordinates tend1. From(2,−2), the difference remains4 and the limit is(3,−1); both limits have zero loss.

The model gives a precise convergence theorem and an exact identifiability limit.

Scope

The historical loss itself was not recovered. This does not establish anE8 embedding, a categorical training objective or any claimed benchmark.

Further development

Write the full loss and show that zero loss is equivalent to the intended relations. If it is not, add independent constraints and analyze the resulting Hessian rank.

Source receipt: months/03/root_coverage.json record 0. Mathematical references: R049.

Y25-03-033 The eta factors carry a calculable modular phase

2025-03-07T00:40:57Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What transformation property can be checked directly for the recovered coherence coefficient?

The recovered coefficient contains a product ofDedekind eta functions. Its modular behavior is not neutral: even translationτ↦τ+1 supplies a nontrivial phase unless additional factors compensate. A direct product calculation makes that test exact.

Definitions and assumptions

ForImτ>0, η(τ)=e^(πiτ/12)∏_(m≥1)(1−e^(2πimτ)). The product converges locally uniformly.

Forintegern≥1, simplify the eta factor toF_n(τ)=η(τ)^(n−1)η(nτ). Treat all other recovered factors asτ-independent for this conditional test.

Exact translation multiplier

F_n(τ+1)=e^(πi(2n−1)/12)F_n(τ). In particular it is not translation-invariant for any integern≥1.

Derivation or proof

Translationτ↦τ+1 leaves every factor1−e^(2πimτ) unchanged, while its prefactor multiplies bye^(πi/12). Thusη(τ+1)=e^(πi/12)η(τ), and iteration givesη(nτ+n)=e^(πin/12)η(nτ). Multiplyingn−1 copies of the first factor and one of the second yields the stated phase. For invariance,2n−1 would have to be divisible by24; it is odd, so this is impossible. The absolute values are translation-invariant, but the complex coefficient is not. A full automorphic law under a congruence subgroup additionally involves weight and multiplier-system data.

Worked example

Taken=2.

F2=η(τ)η(2τ) and its phase ise^(πi/4). Forn=1,F1=η(τ) withphasee^(πi/12). These are exact roots of unity, not numerical discrepancies.

The coefficient’s modular phase is a checkable condition that an intended invariant must address.

Scope

IfQ,Qn* orcn depend onτ, their transformations must also be given. The computation does not validate the entire recovered formula or anSL3/SL4 extension.

Further development

Specify everyτ-dependent factor and prove theS andT transformations with a consistent multiplier before callingCn a modular invariant.

Source receipt: months/03/root_coverage.json record 0. Mathematical references: R013.

Y25-03-035 A square-root eigenvalue law determines a spectral dimension

2025-03-07T00:42:53Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What mathematical consequences follow from a declared eigenvalue growth law?

The recovered passage asserts a square-root spectral law without an operator. A diagonal operator gives an exact model and reveals its counting, zeta-convergence and heat-trace consequences. These can be compared with any proposed geometric interpretation independently.

Definitions and assumptions

Onℓ²(N), defineDe_n=√n e_n, withdomain{a:Σn|a_n|²<∞}.

Its counting function isN_D(Λ)=#{n:√n≤Λ}; the heat trace isΣe^(−t√n),t>0.

Exact count and heat asymptotics

D is positive self-adjoint with compact resolvent;N_D(Λ)=floorΛ². Its spectral zeta function isζ(s/2), initially convergent forRe s>2, and its heat trace is2/t²+O(1) ast→0+.

Derivation or proof

The diagonal real operator with the stated square-summability domain is self-adjoint. The resolvent diagonal entries(√n+i)⁻¹ tend0, so finite-rank truncations converge in norm and its resolvent is compact. Counting givesfloorΛ² exactly. The zeta series isΣn^(−s/2), proving its convergence domain. For the heat trace, f(x)=e^(−t√x) is positive decreasing. Integral comparison boundsΣ_(n≥1)f(n) between∫₁∞f(x)dx and∫₀∞f(x)dx, with difference at most1. Substituteu=√x to obtain∫₀∞2u e^(−tu)du=2/t². Thus the stated asymptotic has a controlled bounded remainder.

Worked example

Apply a constant shiftD+aI witha≥0.

The count becomesfloor(max(Λ−a,0)²) and the heat trace multiplies bye^(−at). The leading2/t² remains unchanged. AnO(n^(−1/2)) perturbation of levels requires its own estimates but cannot be interpreted without specifying the sequence.

A spectral-growth assertion has definite analytic implications that a candidate operator must match.

Scope

This diagonal construction does not validate an economic, motivic or categorical interpretation of the recovered source. A fitted finite sequence does not prove an asymptotic law.

Further development

Define the actual operator, prove compact resolvent and compare its independent counting theorem with the proposed√n law before interpreting a spectral dimension.

Source receipt: months/03/root_coverage.json record 0. Mathematical references: R052, R013.

Y25-03-006 The recovered renormalization series has a conditional convergence theorem

2025-03-09T18:48:28Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which explicit coefficient and denominator bounds make this infinite expression well-defined?

The recovered expression uses the arithmetic choices4 and6 and is called self-dual without a demonstrated transformation law. Its first mathematical task is existence of the limit. Uniform absolute bounds give a clear sufficient theorem and an explicit truncation error.

Definitions and assumptions

Assume$|C_n(\tau)|\le M\, n^a$ uniformly on a domain, $|e^{(-E_n/T)}\cos(\Omega_n t)|\le 1$, and$a<3$.

Write$B(\tau)=1+\varphi^{-1}\Sigma_{(n\ge 1)}C_n(\tau)n^{(-6)}$, with$\varphi=(1+\sqrt{5})/2$, and assume$|B(\tau)|\ge\delta>0$; likewise bound truncated denominators away from0.

A uniform tail estimate

The oscillatory series converges uniformly. Its tail after$N$ is at most$M\, N^{(a-3)/(3-a)}$. The denominator-series tail is at most$(M/\varphi)N^{(a-5)/(5-a)}$, and reciprocal errors are bounded by that tail divided by the product of denominator lower bounds.

Derivation or proof

Take absolute values termwise. The first majorant is$M\Sigma n^{(a-4)}$, convergent because$a-4<-1$. The decreasing integral bound gives$\Sigma_{(n>N)}n^{(a-4)}\le N^{(a-3)/(3-a)}$. The second follows in the same way with exponent$a-6$. IfB_N is the partial denominator, $B_N^{-1}-B^{-1}=(B-B_N)/(B_N B)$, so lower bounds control inversion. These statements prove a limit; they do not establish modularity, duality, boundedness of unspecified operators or a fixed-point equation. Denominator nonvanishing is indispensable, because convergent denominator sums can still approach zero.

Worked example

Take$C_n=1, E_n=0, \Omega_n=0$.

The first series is$-\eta(4)=-(1-2^{(1-4)})\zeta(4)=-7\pi^4/720$. The denominator is$1+\zeta(6)/\varphi>1$, so its reciprocal is well-defined. The whole example is an explicit constant, with$\zeta(6)=\pi^6/945$ and the stated tails.

The limit and finite error are now justified in a concrete parameter regime. No new theorem follows merely from naming the constant self-dual.

Scope

The source does not recover usable bounds onC_n,E_n or a domain forτ. The theorem is therefore a newly specified sufficient model, not a validation of all historical instances.

Further development

Specify the intended duality$\tau\mapsto-1/\tau$ and prove the exact transformation of every factor. Then determine whether the expression is invariant, covariant of a weight or neither.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R013.

Y25-03-007 An automorphic transformation law comes from reindexing a lattice sum

2025-03-10T01:55:28Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What is a fully defined example of modular covariance, and why do names or convergence exponents not establish it?

The historical series was assigned several automorphic interpretations without an explicit action. A classical weight-four lattice sum gives an actual transformation proof. This constructive example isolates the data that a higher-rank replacement would have to provide.

Definitions and assumptions

Forτ in the upper half-plane define$G4(\tau)=\Sigma_{((m,n)\neq(0,0))}(m+n\tau)^{(-4)}$.

Let$\gamma=[[a,b],[c,d]]\in SL2(Z)$ act by$\gamma\tau=(a\tau+b)/(c\tau+d)$.

Weight-four covariance

The series converges absolutely and locally uniformly, is holomorphic, and satisfies$G4(\gamma\tau)=(c\tau+d)^4G4(\tau)$.

Derivation or proof

On any compact subset of the upper half-plane, |m+nτ| is bounded below by a positive constant times√(m²+n²). The fourth-power reciprocal is summable over the two-dimensional integer lattice, proving local uniform absolute convergence and holomorphy.
Nextm+nγτ=[(md+nb)+(mc+na)τ]/(cτ+d). The integer pair(md+nb,mc+na) is a bijective reindexing because its determinant isad−bc=1. Absolute convergence permits rearrangement, producing the claimed factor. The proof depends on the precise lattice and its group action, not on the appearance ofζ-values in a numerical constant.

Worked example

Check the two generatorsτ↦τ+1 andτ↦−1/τ.

The first uses(m,n)↦(m+n,n) and leavesG4 unchanged. The second uses(m,n)↦(n,−m) and givesG4(−1/τ)=τ⁴G4(τ). Atτ=i this factor is1, consistent with the fixed point; for a hypothetical weight2 sum the corresponding factor would be−1 and would force vanishing at that point if covariance held.

A transformation law is a proved identity with a defined group action. Merely addingSL3 orSL4 notation does not provide it.

Scope

This entry establishes anSL2 holomorphic example. Higher-rank Eisenstein series and Maass forms need different parameters, convergence and differential-eigenfunction conditions.

Further development

For the proposedC_n(τ), give a representation of the relevant group, verify the transformation law and prove analytic continuation separately from convergence.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R013.

Y25-03-010 Connes distance can be computed exactly on two points

2025-03-13T07:19:49Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What metric follows from an explicitly defined finite spectral triple?

The recovered spectral-triple proposal contains a formal commutator expansion but no concrete operator domain. A two-point triple has no domain ambiguity and yields a complete distance calculation. It also shows directly how rescalingD changes the metric.

Definitions and assumptions

RepresentA=C⊕C diagonally onH=C², and letD=[[0,m],[conj(m),0]] withm≠0.

For the two evaluation statesδ1,δ2, define d(δ1,δ2)=sup{|a−b|:a,b real,||[D,diag(a,b)]||≤1}.

Two-point spectral distance

The distance equals1/|m|. Ifm=0 the distance is infinite.

Derivation or proof

The commutator has off-diagonal entriesm(b−a) andconj(m)(a−b). Its adjoint times itself is|m|²(a−b)²I, so its operator norm is|m||a−b|. The constraint is therefore exactly|a−b|≤1/|m|, and equality is attained by a suitable real diagonal element. Ifm=0, every commutator vanishes while|a−b| is unbounded, giving infinite distance. All spectral-triple conditions in this finite setting are immediate: the algebra is represented by bounded operators,D is self-adjoint, and its resolvent is compact becauseH is finite-dimensional. A formal gradient approximation was unnecessary.

Worked example

Takem=2 and then replaceD by3D.

The distances are1/2 and1/6. The two-point set and its algebra are unchanged, yet the metric changes. Conversely, choosingm purely imaginary with the same modulus leaves the distance unchanged because the operator norm depends only on|m|.

The operator supplies concrete metric information that is not determined by the algebra’s name alone.

Scope

This does not create a continuum Dirac operator, a hyperbolic quotient or a motivic structure. Infinite-dimensional spectral triples require domains, bounded commutators and compactness hypotheses.

Further development

For a finite graph Dirac model, calculate the commutator norm and compare its induced metric with shortest-path distance; do not assume they agree without proof.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R052.

Y25-03-011 Coherence is an identity of operations, and a spectral page still needs a differential

2025-03-13T07:19:49Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which algebraic data can be checked in a toy model before claiming an E∞ or spectral-sequence construction?

The historical passage combines higher coherence and a motivic spectral sequence. Two small models make their roles concrete: strict commutative multiplication supplies compatible n-ary operations, while a filtered complex shows that an initial page alone does not determine final cohomology.

Definitions and assumptions

OnA=Q[ε]/(ε²), letm_n(a1,...,an)=a1⋯an. Associativity and commutativity imply substitution and permutation identities.

LetC⁰=Qv,C¹=Qw withdv=w. Givew filtration0 andv filtration1, using an increasing filtration preserved byd.

Exact coherence and a filtered cancellation

The operationsm_n are symmetric and compatible with substitutions. In the filtered complex the associated graded differential is zero, but the next filtration-dropping differential takesv tow and kills both classes; the total complex is acyclic.

Derivation or proof

For the operations, any iterated product obtained by substitutingm_r into one slot ofm_n has exactly the same ordered factors as the single productm_(n+r−1); associativity proves equality, and commutativity removes order dependence. A strict commutative algebra can be viewed through a chosen E∞-operad augmentation to the commutative operad, but a genuinely homotopy-coherent construction must state that operad and its maps. For the complex,d²=0 because there is no degree2 space. On each fixed filtration grade,d has zero same-grade component, so the first graded page has the two classes. The actual differential lowers filtration by1 and induces the next arrow v↦w, an isomorphism. Directly,ker(d:C⁰→C¹)=0 andcokerd=0, confirming acyclicity.

Worked example

Compare the same filtered vector spaces with the alternative differentiald=0.

The initial graded page remains the same, but final cohomology now has dimension1 in each degree. Thus the two possible totals differ despite equal initial tables.

Operations, coherence maps and differentials are decisive data. Naming a spectral-sequence target does not supply them.

Scope

The example does not prove the recovered motivic-to-K-theory spectral sequence for an unspecified derived object. Its existence and indexing require hypotheses that were not recovered.

Further development

Specify a concrete algebraic variety or ring, its filtration and coefficients. Calculate the first nonzero differential or prove collapse from support rather than treatingE2 as the final answer.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R048.

Y25-03-008 Spectral thresholds and curvature must survive changes of scale

2025-03-14T14:27:39Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which geometric statements are invariant, and what does the modified-flow notation actually require?

The recovered passage assigns absolute spectral and curvature thresholds to a model without specifying units or a metric. Scaling gives an exact diagnostic. A standard Ricci-flow solution also shows which part of the flow can be derived rather than inferred from named quantities.

Definitions and assumptions

For a Riemannian metric$g$, replace$g$by$c^2 g$with$c>0$. Laplacian eigenvalues and scalar curvature then scale byc^{-2}.

For a round$n$-sphere$g(t)=r(t)^2 g_{\text{unit}}$, ordinary Ricci flow is$\partial_t g=-2\mathrm{Ric}(g)$.

Exact scale laws and spherical flow

Any fixed numerical cutoff for$\lambda_1$or$R$changes under metric scaling. For the round sphere,$r(t)^2=r(0)^2-2(n-1)t$up to its extinction time.

Derivation or proof

In coordinates the inverse metric scales byc^{-2}, while the constant metric scaling does not change the Levi-Civita connection. The Laplace operator and the contraction defining scalar curvature thus each acquirec^{-2}. On the unit sphere$\mathrm{Ric}(g_{\text{unit}})=(n-1)g_{\text{unit}}$, and$\mathrm{Ric}(r^2 g_{\text{unit}})$as a covariant tensor is the same$(n-1)g_{\text{unit}}$. Substitution into the flow gives$(r^2)'=-2(n-1)$. A Lie derivative term$\mathcal{L}_X g$corresponds, under a suitably matched time-dependent pullback, to changing coordinates; an extraT_{ij}needs its own symmetry and regularity assumptions. None of these symbols determines an external numerical phase threshold.

Worked example

On a circle of radius$r$, the first positive scalar-Laplacian eigenvalue is$1/r^2$although intrinsic Ricci curvature is identically0.

Changing$r$from1to2moves$\lambda_1$from$1$to$1/4$without changing the topology or the zero Ricci tensor. On a two-sphere of initial radius2, the Ricci-flow formula gives$r^2=4-2t$and extinction at$t=2$.

Units and normalization must be fixed before thresholds are meaningful. Exact flow equations do not validate the recovered predictive percentages.

Scope

The derivation is geometric and does not assign any social or intelligence interpretation to curvature. General modified flows require existence theory and a defined tensor$T$.

Further development

Use a dimensionless quantity such asλ1·Vol^(2/n), state its hypotheses and test whether it separates explicit geometric examples.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R054.

Y25-03-009 A categorical surface theory begins with a Frobenius algebra

2025-03-14T16:49:48Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What finite algebraic model provides an actual gluing rule before higher-cobordism analogies are used?

The recovered cobordism and fair-division notation does not specify objects, morphisms or a functor. A two-dimensional oriented surface theory has a classical algebraic model. Its elementary calculations expose the multiplication, pairing and sewing identities that a gluing claim needs.

Definitions and assumptions

LetA=Q[ε]/(ε²), withunit1 and linear traceτ(1)=0,τ(ε)=1.

The Frobenius pairing is⟨a,b⟩=τ(ab). Its nondegeneracy permits a coproduct adjoint to multiplication.

A complete small Frobenius calculation

The pairing matrix is[[0,1],[1,0]]. The coproduct isΔ(1)=1⊗ε+ε⊗1,Δ(ε)=ε⊗ε; it is coassociative and satisfies the Frobenius compatibility.

Derivation or proof

The displayed pairing has determinant−1, so it is nondegenerate. The basis dual to(1,ε) is(ε,1), givingΔ(1)=1⊗ε+ε⊗1. Multiplying either tensor factor byε givesΔ(ε)=ε⊗ε. Checking the two basis vectors, both(Δ⊗id)Δ and(id⊗Δ)Δ map1 to1⊗ε⊗ε+ε⊗1⊗ε+ε⊗ε⊗1 andε toε⊗ε⊗ε. The identityΔ(ab)=(a⊗1)Δ(b)=Δ(a)(1⊗b) follows by the same basis check. These are explicit sewing constraints. The theorem identifying commutative Frobenius algebras with oriented2D TQFTs is a cited structural result, not a consequence of assigningBord to an undefined model.

Worked example

Compute the handle operatorh=μΔ and the first closed-surface valuesτ(h^g1).

h(1)=2ε andh(ε)=0. Hence the sphere value isτ(1)=0, torus valueτ(2ε)=2, and genus-two valueτ(0)=0. These values follow from the chosen trace and are not universal surface invariants independent of the theory.

An actual algebra and its identities give precise categorical content, while leaving the higher-dimensional proposal open.

Scope

This is a newly chosen2D model. It neither reconstructs a missing3- or4-category nor proves a theorem about division or conflict.

Further development

Specify the intended bordism category, generators and relations, then give functorial assignments satisfying every gluing relation before adding numerical observables.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R048.

Y25-03-014 A modular cocycle obeys a twisted composition law

2025-03-17T19:01:02Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How can a modular cocycle be verified exactly before using it as a transport or geometric flow?

The retrieved passage requests a modular-cocycle explanation but truncates the actual formula. A faithful finite-state example gives a precise replacement, including the twisted law that distinguishes a cocycle from an ordinary one-parameter group.

Definitions and assumptions

For faithful density matricesρ,σ defineu_t=ρ^(it)σ^(−it). Letασ_t(A)=σ^(it)Aσ^(−it).

A cocycle forασ satisfiesu_(s+t)=u_s ασ_s(u_t).

Finite cocycle and intertwining

The matricesu_t are unitary, satisfy the stated cocycle law, and obey αρ_t(A)=u_t ασ_t(A)u_t*.

Derivation or proof

Positive functional calculus makesρ^(it) andσ^(it) unitary. Multiplyingu_sασ_s(u_t) givesρ^(is)σ^(−is)σ^(is)ρ^(it)σ^(−it)σ^(−is)=ρ^(i(s+t))σ^(−i(s+t)), exactlyu_(s+t). Notice that commutingρ withσ was never assumed; only powers of the same matrix were combined. For intertwining, the adjacentσ factors cancel in u_t σ^(it)Aσ^(−it)u_t*, leavingρ^(it)Aρ^(−it). Ifρ andσ fail to commute, u_su_t need not equalu_(s+t); replacing the twisted composition by an ordinary group law is therefore an additional unjustified step.

Worked example

For a transparent commuting example takeρ=diag(3/4,1/4),σ=diag(1/2,1/2).

u_t=diag((3/2)^(it),(1/2)^(it)). Hereασ is trivial, so the cocycle happens to be an ordinary unitary group. Replacingσ by a nontrivially rotated diagonal state retains the twisted identity but generally loses that simplification.

The exact algebra supplies the transport law and identifies which simplification relies on commutativity.

Scope

This is the finite faithful-state case of a more general operator-algebraic construction. The historical categorical state space, morphisms and infinite-dimensional domains were not recovered.

Further development

Use two explicit noncommuting2×2 states, verify the cocycle numerically at several times with residual bounds, and calculate its derivative generator without assuming it is time independent.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R055.

Y25-03-016 A zero-dimensional Gaussian realizes an exact Wetterich flow

2025-03-17T19:01:02Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact regulated model verifies the flow equation, and what does a zero beta function fail to determine?

The historical source claims a completed RG toy derivation and a selected vacuum expectation. A Gaussian integral instead gives a fully transparent flow identity. It also shows why stationarity of a running coupling does not by itself fix a nonzero vacuum.

Definitions and assumptions

Let$m^2>0, R_k\geq 0$ and$Z_k(J)=\int_R \exp[-(m^2+R_k)x^2/2+Jx]dx$. Put$W_k=\log Z_k$.

Define$\varphi=\partial_J W_k$ and$\Gamma_k(\varphi)=\sup_J[J\varphi-W_k(J)]-R_k\varphi^2/2$.

Exact Gaussian effective-average action

$\Gamma_k(\varphi)=m^2\varphi^2/2+(1/2)\log(m^2+R_k)-(1/2)\log(2\pi)$, so$\partial_k\Gamma_k=(1/2)(\Gamma_k''+R_k)^{-1}\partial_k R_k$. Its unique minimizing field is$\varphi=0$ for all$k$.

Derivation or proof

Complete the square in the integral:$Z_k=\sqrt{(2\pi/(m^2+R_k))}\exp[J^2/(2(m^2+R_k))]$. Thus$\varphi=J/(m^2+R_k)$ and$J=(m^2+R_k)\varphi$. Substitute into the Legendre transform and subtract$R_k\varphi^2/2$ to obtain the formula. Twoφ derivatives give$\Gamma_k''=m^2$; differentiating its additive logarithmic term in k gives the right-hand side exactly. The curvature is positive, so the unique minimizer is0. The quadratic coupling does not run in this free model, yet no nonzero scale or mass ratio is selected from that fact.

Worked example

Take$m^2=2$ and$R_k=k^2$.

The action is$\varphi^2+(1/2)\log(2+k^2)-(1/2)\log(2\pi)$, and$\partial_k\Gamma=k/(2+k^2)$. At$k=1$ the flow is$1/3$. Removing the field-independent logarithm would hide this exact flow while leaving the minimizer unchanged; the two questions must be distinguished.

The regulated integral, transform and flow agree exactly. This is constructive RG mathematics without a claimed hierarchy or vacuum prediction.

Scope

A quantum field theory needs ultraviolet control, function spaces and a regulator prescription. The scalarGaussian example does not prove the full functional equation in every setting or validate the recoveredmass-ratio claim.

Further development

Add a positive quartic term and compute certified integral derivatives. Compare the exact Legendre curvature with a declared truncation to measure its approximation error.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R056.

Y25-03-017 Spectral flow counts oriented crossings; odd-dimensional integrals need odd forms

2025-03-17T19:01:02Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What is a correctly typed small spectral-flow calculation, and which form-degree mismatch must be repaired?

The recovered formula integrates a conventional even Chern-character component over an odd-dimensional sphere. Such an expression needs a specified odd character or transgression. A finite path and a unitary loop give exact examples of the two different constructions.

Definitions and assumptions

For a Hermitian path with invertible endpoints and transversal zero crossings, spectral flow counts negative-to-positive crossings minus positive-to-negative crossings.

The conventional curvature characterch_n(F) is a$2n$-form. For a unitary loop$U:S^1\to U1$, the odd degree-one form is$(2\pi i)^{-1}U^{-1}dU$.

Two exact integer calculations

ForD(t)=diag(t−1/4,t−3/4),0≤t≤1, spectral flow is2. ForU(t)=e^(2πit), the odd-character integral is1.

Derivation or proof

The first diagonal eigenvalue crosses zero att=1/4 and the second att=3/4. Each derivative is+1, so both crossings are positive and the total is2. Endpoints have no zero eigenvalue. For the loop,dU=2πiUdt, hence(2πi)$^{-1}$U^{-1}dU=dt and its integral over one parameter period is1. This integral is well-typed because its degree equals the dimension. A2n-form cannot be integrated as a top-degree form overS^(2n+1) to yield the claimed invariant without additional data. In infinite-dimensional Fredholm families, spectral-flow/index and odd-Chern formulas are genuine theorems with hypotheses, not consequences of matching notation.

Worked example

Reverse the finite path or traverse the loop twice.

D(1−t) has two positive-to-negative crossings, giving−2. The loopU(t)=e^(4πit) has winding2. A constant invertible matrix path has spectral flow0, and a constant unitary loop has odd integral0.

Crossing orientation and differential-form degree make the proposed invariants checkable.

Scope

The finite-dimensional spectral-flow example does not construct the unspecifiedD-family from the historical framework. No automatic equality between the two examples is asserted.

Further development

Define a closed Fredholm family and an explicit unitary representative, then state a precise index theorem and verify all compactness and regularity assumptions.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R052.

Y25-03-012 Finite modular flow distinguishes the state Hamiltonian from the standard-form generator

2025-03-18T01:23:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exactly are Δ and K in a finite faithful-state model?

The recovered modular expressions are classical, but several operators share the name modular Hamiltonian. A faithful matrix-state model separates the density-matrix logarithm from the standard-form generator on Hilbert-Schmidt operators.

Definitions and assumptions

Letρ be a positive definite density matrix. OnHilbert-Schmidt matricesX, defineΔρ(X)=ρXρ^{-1} andJ(X)=X*.

The density Hamiltonian ishρ=−logρ, whereas the standard-form generator isKρ=−logΔρ.

Exact finite-dimensional modular structure

For matrix unitsEij in aρ-eigenbasis with eigenvaluespi, ΔρEij=(pi/pj)Eij andKρEij=log(pj/pi)Eij. Modular flow on left observables isσ_tρ(A)=ρ^(it)Aρ^(−it).

Derivation or proof

Left multiplication byρ and right multiplication byρ^{-1} commute and are positive self-adjoint on Hilbert-Schmidt space. Their action onEij gives the eigenvaluepi/pj. Functional calculus yields the logarithm and imaginary powers, proving every formula. WithΩ=√ρ, the Tomita map initially sendsAΩ toA*Ω. WritingY=A√ρ givesS(Y)=ρ^(−1/2)Y*ρ^(1/2), which isJΔρ^(1/2)Y. AlsoΔρΩ=Ω, soKρΩ=0. By contrastTr(ρhρ) is the generally nonzero von Neumann entropy. Conflating these two expectations would invalidate a first-law calculation.

Worked example

Takeρ=diag(3/4,1/4).

Δρ has eigenvalues1,3,1/3,1 onE11,E12,E21,E22. Thusσ_tρ(E12)=3^(it)E12 andσ_tρ(E21)=3^(−it)E21. The density Hamiltonian has eigenvalueslog(4/3),log4, whileKρ has0,−log3,+log3,0.

The finite model realizes Tomita-Takesaki notation exactly and exposes its two distinct logarithms.

Scope

Infinite von Neumann algebras may have no density matrix or trace. Their modular theory is deeper; this finite derivation does not establish a categorical version or a geometric backreaction equation.

Further development

For a proposed categorical analogue, identify its algebra, faithful state, Hilbert-space representation and an actual modular operator before asserting covariance or a first law.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R055, R052.

Y25-03-013 The entropy first law is an exact first derivative at a faithful state

2025-03-18T01:23:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which variation makes the recovered first law valid, and what remains at second order?

The recovered first-law identity has a precise finite-state meaning. It is a linearized equality at a fixed faithful reference state, not equality of finite entropy changes. Relative entropy records the discrepancy beyond that first derivative.

Definitions and assumptions

Letσ>0 have trace1, letX=X* withTrX=0, andρ(ε)=σ+εX remain positive for smallε.

SetS(ρ)=−Trρlogρ andhσ=−logσ; relative entropy isD(ρ||σ)=Trρ(logρ−logσ).

First law and a positive second-order example

Atε=0,dS(ρ(ε))/dε=Tr(Xhσ). Exactly, D(ρ||σ)=Tr[(ρ−σ)hσ]−[S(ρ)−S(σ)]. For a diagonal binary perturbation its second-order coefficient is positive.

Derivation or proof

The trace derivative ofTrf(σ+εX) isTrf′(σ)X. In an eigenbasis, the diagonal derivative terms give this expression and off-diagonal first-order contributions have zero trace. Withf(u)=u logu, the entropy derivative is−TrX(logσ+I)=−TrXlogσ becauseTrX=0. Expanding the definition ofD gives the exact finite-change identity. Forσ=diag(p,1−p) andX=diag(1,−1), direct differentiation of the scalar relative entropy givesD′(0)=0 andD″(0)=1/p+1/(1−p)=1/[p(1−p)]>0. Thus finite changes agree only to first order in general.

Worked example

Takep=3/4 andε=1/20, soρ=diag(4/5,1/5).

The modular-energy change isεlog((1−p)/p)=−(log3)/20. The exact relative entropy is(4/5)log(16/15)+(1/5)log(4/5)>0. Its quadratic approximation is(8/3)ε²=1/150, while the exact finite identity gives the entropy change by subtractingD from the modular-energy change.

The first law is now a proved derivative with a declared reference operator. Its second-order remainder can be checked explicitly.

Scope

Faithfulness is essential for a bounded matrix logarithm; support changes can make relative entropy infinite. No Einstein equation or categorical entropy follows solely from this identity.

Further development

Derive the noncommutative Hessian using the integral representation of the logarithm, and test its positivity on an explicit off-diagonal perturbation.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R055.

Y25-03-015 Wilson-loop words need a connection or a well-defined group assignment

2025-03-18T01:23:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What group-valued construction replaces an undefined nested path-ordered integral?

The recovered nesting places a holonomy inside another exponential without stating a new connection or its gauge transformation. A graph holonomy is an exact model: edge elements multiply along a loop, and the trace of that word is gauge invariant.

Definitions and assumptions

AssignUij∈SU2 to each oriented edge withUji=Uij⁻¹. Vertex matricesgi act byUij↦giUijgj⁻¹.

For two based loops with holonomiesU,V, their commutator loop has holonomyUVU⁻¹V⁻¹.

Gauge covariance and an explicit noncommutative loop

A based loop product transforms by conjugation at its basepoint, so its trace is invariant. ForU=iσx,V=iσz, the commutator is−I with trace−2, whereas commutingU,V giveI with trace2.

Derivation or proof

Multiply the transformed edge elements in loop order: every adjacentgj⁻¹gj cancels and the endpoints leaveg_base Wg_base⁻¹. Cyclicity of trace proves invariance. In the Pauli exampleσxσz=−σzσx and bothU,V square to−I; thusUV=−VU. Multiplying this relation byU⁻¹V⁻¹ givesUVU⁻¹V⁻¹=−I. This is a genuine nested loop word: the topology specifies concatenation and reversal, and the group law specifies multiplication. It does not require treating a group-valued holonomy as a Lie-algebra-valued1-form under an undefined outer integral.

Worked example

Take the four-factor commutator and compare an abelianU1 assignment.

TheSU2 trace is−2. EveryU1 pair commutes, so its commutator is1. ForSU2 matricesU=e^(iθσx),V=e^(iφσz), direct multiplication givesTrcomm=2[1−2sin²θsin²φ], recovering both the identity case and the Pauli case.

A concrete holonomy word has a reproducible gauge-invariant observable. The formula records noncommutativity rather than an unproved higher-categorical effect.

Scope

Continuum path ordering requires a connection along a path and an ODE; higher holonomy needs higher connection data and coherence. The finite graph proof does not construct those objects.

Further development

Specify a small2-complex and a crossed module, then formulate surface holonomy with its precise boundary transformation law.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R053.

Y25-03-019 A semidirect product requires an action, even when its factors are familiar

2025-03-21T06:13:18Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What choices are missing from SU4⋉Q8, and which concrete actions produce well-defined groups?

The recovered user topic supplies the factorsSU4 andQ8 but not the action. There are several inequivalent possibilities. A precise construction shows that an inner action lifted by a homomorphism can even be untwisted to a direct product.

Definitions and assumptions

An actionα:Q8→Aut(SU4) defines(g,q)(h,r)=(gα_q(h),qr).

Ifu:Q8→SU4 is a homomorphism, its inner action isα_q(h)=u_q h u_q^{-1}.

An explicit untwisting theorem

For a lifted inner action, F(g,q)=(g u_q,q) is a group isomorphism SU4⋊_αQ8→SU4×Q8. Another valid action uses a homomorphismQ8→C2 followed by complex conjugation onSU4.

Derivation or proof

For the first claim,F((g,q)(h,r))=(g u_q h u_q^{-1}u_(qr),qr)=(g u_q h u_r,qr), usingu_(qr)=u_qu_r. This equalsF(g,q)F(h,r) in the direct product. The inverse is(a,q)↦(a u_q^{-1},q). To obtain a concreteu, use the standard faithful2-dimensional quaternion matrices inSU2 and embed them asdiag(u_q,I2) inSU4. For the second action, any nontrivial sign character ofQ8 factors through its abelianizationC2×C2. Complex conjugation has order2 and preservesSU4, so composing gives a homomorphism into automorphisms. It acts on the central elementiI4 by inversion, whereas every inner action fixes that center.

Worked example

Compare the action on the central order-four subgroup.

Under a lifted inner action, iI4 staysiI4 for everyq. Under the conjugation action, aq with negative character sendsiI4 to−iI4. Thus the two presentations carry genuinely different action data before any lattice model is defined.

The action is part of the group’s definition. A label containing two factors does not choose it.

Scope

The source identifies a project topic, not a recovered action or validated simulation. No categorical gauge theory, continuum limit or numerical lattice result is inferred.

Further development

State the intended action and representation, then derive the edge transformation laws and gauge-invariant loop observables for a finite lattice.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R048.

Y25-03-018 A Maurer-Cartan equation needs graded brackets and a curvature calculation

2025-03-21T06:14:55Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact flatness equation can be derived before invoking higher brackets?

The historical higher-categorical flow lists a Maurer-Cartan expression without bracket degrees or factorial conventions. A matrix-valued differential-form algebra gives a rigorous degree-one example, where flatness is a concrete commutator condition.

Definitions and assumptions

LetA be a matrix-valued1-form onR². In the differential graded algebra of forms defineF=dA+A∧A=dA+(1/2)[A,A].

The graded commutator uses the wedge product and matrix multiplication; for degree-oneA the two terms in[A,A] add.

An exact constant-connection flatness criterion

> ForA=Xdx+Ydy with constant matrices, F=[X,Y]dx∧dy. It is flat exactly whenX andY commute.

Derivation or proof

Constant coefficients give dA=0. ExpandingA∧A removes thedx∧dx anddy∧dy terms; the cross terms areXYdx∧dy+YXdy∧dx=(XY−YX)dx∧dy. Thus the displayed formula is exact. More generally, the derivative and graded commutator satisfy the graded Leibniz and Jacobi identities, which imply the Bianchi identitydF+[A,F]=0. In anL∞ convention one commonly writesΣ_(n≥1)(1/n!)ℓ_n(A,…,A)=0 withℓ_n of degree2−n; a source using other normalization must define it. The symbols themselves do not establish those identities or define a scale-dependent category.

Worked example

TakeX=[[0,1],[0,0]] andY=[[0,0],[1,0]].

[X,Y]=diag(1,−1), so the connection has nonzero curvature. ReplacingY by2X gives zero curvature. Both examples involve simple constant coefficients, showing that flatness is algebraic compatibility rather than a generic consequence of naming a gauge field.

The corrected equation has a definite degree, normalization and exact curvature test.

Scope

This is a differential graded Lie-algebra model, not a construction of the recovered2-crossed module or emergent metric. Those require additional objects and coherence maps.

Further development

Choose an explicit chain complex andhigher brackets, verify the firstL∞ identities, and test whether a proposed blocking map sendsMaurer-Cartan solutions to solutions.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R048, R053.

Y25-03-020 A projective multiplier gives a concrete H² cocycle

2025-03-21T06:14:55Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact cocycle can replace an unspecified group-cohomology decoration?

The recovered higher-gauge discussion namesH²(BG,U1). A projective representation of a finite group gives a fully calculable example of the relevant cocycle identity. It also exhibits a nontrivial class without claiming a full cohomology classification.

Definitions and assumptions

LetG=F2² with addition. Forv=(a,b), setU_v=X^aZ^b using PauliX,Z, which satisfyZX=−XZ.

A normalized2-cocycleω satisfiesω(v,w)ω(v+w,u)=ω(w,u)ω(v,w+u) andω(0,v)=ω(v,0)=1.

An explicit nontrivial multiplier

U_vU_w=ω(v,w)U_(v+w), whereω((a,b),(c,d))=(−1)^(bc). This is a normalized2-cocycle and is not a coboundary.

Derivation or proof

MoveZ^b pastX^c to obtain the sign(−1)^(bc); exponents add modulo2 becauseX²=Z²=I. The cocycle identity compares exponentsbc+(b+d)e withde+b(c+e), equal modulo2, so associativity is exact. A coboundary on an abelian group has formf(v)f(w)/f(v+w) and is symmetric inv,w. But forv=(1,0),w=(0,1), ω(v,w)=1 andω(w,v)=−1. Hence this cocycle cannot be a coboundary. This elementary obstruction proves a nonzero cohomology class in the specified convention.

Worked example

Enumerate all triples inG³ and compare the cocycle identity.

There are64 triples, and every one satisfies the identity. The associated central extension with multiplication(z,v)(z′,w)=(zz′ω(v,w),v+w) is associative. Its commutator on the two chosen generators is−1, recording the noncommutativity hidden by the abelian labels.

The coefficient group, cocycle and identity are all explicit. They give real group-cohomological content.

Scope

The extension built from these particular Pauli lifts is not automatically the quaternion groupQ8; the squares of lifts and coefficient subgroup matter. The result is not a computation for everyG.

Further development

For the intended lattice symmetry, write its actual multiplication table and construct cocycles or coboundaries explicitly. Then determine which projective representations realize them.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R048.

Y25-03-021 Cohomology-preserving blocking requires a chain homotopy certificate

2025-03-21T06:14:55Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

When does a coarse-graining map really preserve cycle information?

The recovered blocking proposal asserts cohomology preservation. That property can be proved by exhibiting chain maps and a chain homotopy. A subdivided circle gives a complete integer certificate, which also illustrates exactly what arbitrary edge deletion could lose.

Definitions and assumptions

LetC have verticesv0,v1 and oriented edgese0:v0→v1,e1:v1→v0. LetD have one vertexv and one loopa.

Define r(v0)=r(v1)=v,r(e0)=0,r(e1)=a; define i(v)=v0,i(a)=e0+e1.

An exact chain-homotopy equivalence

> The mapsr,i commute with boundaries, ri=id_D, andir−id_C=∂h+h∂ for h(v0)=0,h(v1)=−e0,h(C1)=0. Therefore they induce inverse maps on integer homology.

Derivation or proof

The boundary ofe0 isv1−v0 and ofe1 isv0−v1; bothr-boundaries vanish, so r is a chain map. The chaini(a)=e0+e1 has zero boundary, proving the same fori. The identityri is immediate. Onv1, ir−id givesv0−v1=∂(−e0), and onv0 it is zero. One0,e1 the difference is respectively−e0,+e0, equal toh∂ of those edges. This checks the homotopy formula on an integer basis. Ifc is a cycle, (ir−id)c=∂hc is a boundary, so ir acts as identity on homology; together withri=id this proves the claim. Dualizing gives the corresponding cohomology equivalence.

Worked example

The fine cyclec=e0+e1 is sent toa and returns toc.

Both complexes haveH0≅Z andH1≅Z. In contrast, deletinge1 without adding the coarse loop leaves an interval and destroysH1. Dimension reduction alone is therefore insufficient to preserve the invariant.

A short exact certificate establishes preservation in a declared model, rather than inferring it from a category label.

Scope

The maps do not preserve every metric, local observable or dynamical energy. Homology equivalence is a specific algebraic guarantee, not universal information preservation.

Further development

For the actual tensor blocking, give integer incidence matrices and chain maps, then solve the homotopy equation. If no such maps exist, quantify the cycles lost instead of asserting preservation.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R050.

Y25-03-026 An energy bound exposes incompatible power and thrust scalings

2025-03-24T06:01:49Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which elementary energy inequality must any claimed fixed-exhaust-speed scaling satisfy?

The recovered source combines a fixed effective speed with power and force projections. Restricting the discussion to its mathematics, kinetic-energy accounting gives a simple necessary inequality. It can reject an inconsistent set of numbers without constructing an engineering model.

Definitions and assumptions

For a mass flow$\dot{m} \geq 0$ leaving at speed$v > 0$, momentum flux is$F = \dot{m} v$ and kinetic power is$P_{kin} = (1/2)\dot{m} v^2$.

Assume the declared available power$P$ bounds that kinetic power, with no additional energy source hidden in the model.

An exact necessary bound

> $F \leq 2P/v$. At fixed$v$, a force scaling$F = cP^{(1+\delta)}$ with$c > 0$ and$\delta > 0$ cannot satisfy this bound for arbitrarily large$P$.

Derivation or proof

Eliminate$\dot{m}$ from the two definitions to obtain$P_{kin} = Fv/2$. The assumption$P_{kin} \leq P$ gives the inequality by division by$v/2$. For the power law, the ratio$Fv/(2P) = (cv/2)P^{\delta}$ grows without bound if$\delta > 0$, so some finite$P$ violates the energy condition. This is an asymptotic obstruction, not a universal ban on superlinear behavior over a bounded range or on changing exhaust speed. A quantitative proposal must declare how$v$ and efficiency depend on$P$. The dimensions also agree: watts divided by meters per second are newtons.

Worked example

Take$P = 100000$ watts and$v = 65000$ meters per second, as in the recovered numerical combination.

The bound is200000/65000=40/13≈3.076923 newtons. A force of2150newtons at that same speed would require at leastFv/2=69,875,000 watts of kinetic power, about698.75 times the declared100kW. At10kW, the upper bound is4/13newtons≈0.30769.

The necessary inequality is exact and makes the numerical inconsistency reviewable.

Scope

The result is conditional on the explicitly stated energy budget and momentum-flux model. It does not evaluate a device, give operational advice or prove a full fluid or electromagnetic model. Additional energy sources would change the hypothesis and must be quantified.

Further development

Rewrite all claimed scaling laws with dimensionless efficiencyη=P_kin/P≤1 and a declaredv(P), then identify a parameter range where every necessary inequality is satisfied.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: .

Y25-03-025 Second-order correlations are factorial moments with distinguishable models

2025-03-28T19:55:14Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact statistical quantity can be calculated for the recovered g² topic?

The recovered user narrative namesg^(2) without a probability model. The zero-delay count version is a normalized factorial moment. Poisson, fixed-count and geometric distributions yield three exact values, so a measured value must be interpreted through its declared sampling model.

Definitions and assumptions

For a nonnegative integer random variableN with0<EN<∞ and finite second moment, defineg2=E[N(N−1)]/(EN)².

SinceE[N(N−1)]=VarN+(EN)²−EN, g2=1+(VarN−EN)/(EN)².

Exact model values

> A Poisson count hasg2=1; a deterministic countN=n≥1 hasg2=1−1/n; a geometric countP(N=k)=(1−q)q^k,0<q<1, hasg2=2.

Derivation or proof

ForPoissonλ, differentiate its generating functionG(z)=exp[λ(z−1)] twice:EN=G′(1)=λ andE[N(N−1)]=G″(1)=λ². A fixed count has numeratorn(n−1) immediately. For the geometric lawG(z)=(1−q)/(1−qz), givingG′(1)=q/(1−q) andG″(1)=2q²/(1−q)². The ratio is2. These calculations also show why ordinary second moments would give different normalizations: factorial moments remove the self-count contributionEN. Independent binomial thinning with efficiencyη scales first and second factorial moments byη andη², leavingg2 unchanged when the assumptions hold.

Worked example

UsePoisson mean2, fixed count2 and geometric mean2(q=2/3).

All three haveEN=2, but their factorial second moments are4,2,8 respectively, so theirg2 values are1,1/2,2. Equal mean flux therefore cannot identify the correlation model.

The statistic distinguishes explicit count laws while remaining independent of an unsupported resonance story.

Scope

Time-dependentg2(τ), detector dead time, backgrounds and non-independent thinning require their own model. The formulas do not validate a hardware architecture or a nonclassical interpretation from a single number alone.

Further development

Give the count-window definition and estimate both factorial moments with uncertainty intervals. Test the thinning-invariance prediction before comparing datasets.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R057.

Y25-03-022 A power-law phase-noise spectrum needs cutoffs to have finite variance

2025-03-28T19:56:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

When does the recovered spectral-density model define a finite-variance process?

The historical phase-noise model is a polynomial in negative powers of frequency. Its integrability is a mathematical issue before any resonance or hardware interpretation. A finite bandwidth gives an exact variance formula and exposes the divergent idealization.

Definitions and assumptions

Use a one-sided spectral-density conventionVarφ=∫₀∞Sφ(f)df. On a declared band[f_L,f_H], assumeSφ≥0.

The coefficients have the units needed forSφ; negative values require a separate positivity check rather than termwise interpretation.

Finite-band variance and low-frequency divergence

For0<f_L<f_H, the variance isΣ_(i≠−1)h_i(f_H^(i+1)−f_L^(i+1))/(i+1)+h_(−1)log(f_H/f_L). A nonzero nonnegativef^(−p) term withp≥1 diverges asf_L→0.

Derivation or proof

The formula is direct integration of each power on a compact positive band. Forp=1 the integral is logarithmic; forp>1 it grows likef_L^(1−p)/(p−1). Therefore an ideal spectrum with such a positive leading low-frequency term cannot be the spectral density of a stationary finite-variance phase process over all frequencies. A flat positive term also diverges at infinite frequency unless a high-frequency cutoff or decay is supplied. These are consequences of finite covariance at zero, not a claim that finite-band models are unusable. Frequency noise or phase increments may admit different generalized-process interpretations.

Worked example

TakeS(f)=f^(−2) on[1,10] and zero elsewhere.

The variance is1−1/10=9/10. Extending the lower endpoint to1/100 gives100−1/10=999/10, a large change even though the high-frequency endpoint is fixed. TakingS(f)=1/f on[1,10] giveslog10 instead.

A finite declared band makes the model quantitative. Omitting cutoffs can conceal an infinite variance.

Scope

The calculation does not validate measured coefficients or predicted oscillator performance. PSD positivity and a chosen one-sided/two-sided convention must precede numerical comparison.

Further development

Fit coefficients on a declared band with positivity constraints, propagate their intervals through the exact integral, and test whether a stationary phase or an increment model matches the intended observable.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R057.

Y25-03-023 Eliminating a hidden variable derives a memory kernel and a source term

2025-03-28T19:56:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which terms must appear when a reduced equation is obtained from a larger system?

The recovered Langevin reference does not specify a kernel or noise law. A two-variable linear system gives an exact reduction. It proves that a memory term and a source depending on the hidden initial condition arise together; dropping the source changes the model.

Definitions and assumptions

Letx′=−ax+by andy′=−cy+dx, withreal coefficients andc>0.

Initial data arex(0)=x0,y(0)=y0. A memory kernel acts by convolution over pastx.

Exact reduced Volterra equation

The observed coordinate satisfies $x'(t)=-ax(t)+b\,e^{(-ct)}y0+bd\int_0^t e^{(-c(t-s))}x(s)ds$. This is equivalent to the full system with the stated initial data.

Derivation or proof

Multiply the second equation by$e^{(ct)}$ and integrate to obtain$y(t)=e^{(-ct)}y0+d\int_0^t e^{(-c(t-s))}x(s)ds$. Substitute into the first equation to get the reduced formula. Conversely, given a solutionx of that Volterra equation, definey by the integral expression; differentiating recovers$y'=-cy+dx$ and then$x'=-ax+by$. The initial values also agree. Thus the reduction is exact, with no assumption that the hidden coordinate is random. Ify0 is random, the decaying source inherits its law, and a fluctuation-dissipation identity requires additional equilibrium assumptions rather than following from the elimination alone.

Worked example

Take$a=0,b=1,c=1,d=-1,x0=1,y0=0$.

The full matrix has characteristic polynomial$\lambda^2+\lambda+1$.
Hence$x(t)=e^{(-t/2)}[\cos(\sqrt{3}t/2)+(1/\sqrt{3})\sin(\sqrt{3}t/2)]$. It obeys$x'(0)=0$ and$x''+x'+x=0$. The Volterra form is$x'=-\int_0^t e^{(-(t-s))}x(s)ds$, which differentiates to that same equation.

The memory kernel is derived and independently checkable, with its sign fixed by the hidden coupling.

Scope

This deterministic model is not a stochastic thermostat or a proof of a resonance phenomenon. Infinite reservoirs require convergence and a specified initial-state distribution.

Further development

For a block matrix or self-adjoint Hamiltonian, repeat the variation-of-constants argument and bound the kernel norm. State the hidden initial condition before interpreting reduced dissipation.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R052.

Y25-03-024 The recovered self-bilinear Hirota equation loses its time derivative

2025-03-28T19:56:25Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What equation does the displayed Hirota expression actually impose?

The historical assistant writes a self-bilinear equation resembling the heat equation. Hirota derivatives act on two slots, and odd derivatives vanish when both slots are the same. Expanding the definition yields an exact correction and a family of solutions.

Definitions and assumptions

D_x^mf·g=(∂x−∂x′)^mf(x)g(x′)|_(x′=x); defineD_t similarly.

Assumeτ(x,t)>0 isC² inx andC¹ int, so logτ is defined.

An exact reduction

D_tτ·τ=0 andD_x²τ·τ=2(ττ_xx−τ_x²). Therefore(D_x²−D_t)τ·τ=0 is equivalent to(logτ)_xx=0, with solutionsτ=e^(a(t)x+b(t)) locally on a connectedx interval.

Derivation or proof

The first derivative isτ_tτ−ττ_t=0. The second spatial derivative expands toτ_xxτ−2τ_xτ_x+ττ_xx, yielding the formula. Dividing by2τ² gives(logτ)_xx. Integrating twice inx giveslogτ=a(t)x+b(t), with the regularity required for the chosen functions. Thus there is no time-evolution constraint in the displayed self-bilinear equation. A genuine heat equation appears instead from(D_x²−D_t)f·1=f_xx−f_t=0, because the second slot is constant rather than equal tof.

Worked example

Takeτ=e^(x+t²).

The self-bilinear equation vanishes identically because(logτ)_xx=0. Butτ_xx−τ_t=(1−2t)τ, which is nonzero exceptatt=1/2. The samefunction therefore decisively separates the recovered equation from the ordinary heat equation.

The bilinear notation is now evaluated correctly, with a constructive solution family and an explicit counterexample to a likely interpretation.

Scope

Zeros or complex values ofτ need separate local treatment and a logarithm branch; the division argument uses strict positivity. No soliton or integrability theorem follows from this one equation.

Further development

State the intended nonlinear PDE, choose aτ substitution, and derive its correct even-order self-bilinear form or two-function bilinear system directly.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R058.

April 2025

Dated fragments recovered; complete transcript unavailable. 28 developed treatments.

The 384-dimensional cyclotomic proposal supplies a rich starting point for exact arithmetic, sine-transform fusion data, singularities, finite topology and braided categories. The distinction between field degree, categorical rank and global dimension is made explicit.

ID · date	Concept and revised question	Result and mathematical core
Y25-04-001 2025-04-01 · U	The number384 is a cyclotomic degree, with a computable Galois group What exact arithmetic follows from1680 and which categorical dimensions do not follow from it?	Exact degree and group structure [K:Q]=384 andGal(K/Q)≅C4²×C2³×C3. Its exponent is12, so it is not cyclic of order384. The maximal real subfield has degree192.
Y25-04-002 2025-04-01 · A	A finite characteristic polynomial cannot equal a zeta entire function What exact obstruction rules out the recovered finite determinant identity, and what valid finite determinant model remains?	A finite-degree obstruction and a valid product identity No ordinary finiteH and nonzeroc satisfy p_H(s)=cξ(s)^384 for allcomplexs. For a finite Hermitian matrix, the valid identity is p_H(s)=∏_j(s−λ_j), with realλ_j.
Y25-04-024 2025-04-01 · U	Dimensionful constants cannot be numerical predictions of unit-free integers alone Which proposed constants are invariant under unit changes, and what can an integer framework actually determine?	An exact dimension-rank test The only dimensionless monomial formed solely fromℏ,c,G is the trivial onea=b=d=0. Their three-dimensional unit-exponent matrix is invertible, so their numerical values can all be changed by a choice ofM,L,T units.
Y25-04-003 2025-04-10 · A	The sine matrix at π/840 is orthogonal only with the full839 indices Which index range makes the recovered sine matrix a unitary modular candidate?	Exact sine orthogonality Σ_(s=1)^(m−1)sin(πrs/m)sin(πts/m)=(m/2)δ_rt. HenceS is real symmetric withS²=I. The384×384 principal truncation at m=840 is not orthogonal.
Y25-04-004 2025-04-10 · A	Quantum-dimension probabilities must be normalized by their actual sum What is the correct normalization and which entropy values can be computed exactly?	An exact contradiction and a corrected law D_384>1960>1680. For the full839 indices,D_839=420/sin²θ andp_r=sin²(rθ)/420. Atm=4 the probabilities are(1/4,1/2,1/4), withH1=(3/2)log2 andH2=log(8/3).
Y25-04-009 2025-04-18 · U	Milnor numbers come from a finite Jacobian algebra What exact invariant is obtained if the recovered triple is interpreted as a Brieskorn polynomial?	Exact monomial basis Forf=x^a+y^b+z^c, the monomialsx^iy^jz^k with0≤i≤a−2,0≤j≤b−2,0≤k≤c−2 form a basis, givingμ=(a−1)(b−1)(c−1).
Y25-04-010 2025-04-18 · U	A Brieskorn link can have torsion invisible to its Betti numbers What link homology can be checked when a recovered Betti table omits torsion?	Exact torsion order for the recovered examples For odda and evenb withgcd(a,b)=1, |H1(Σ(2,a,b);Z)|=a. ThusΣ(2,5,12) hasH1≅Z/5 andΣ(2,41,840) hasH1≅Z/41, while their first and second rational Betti numbers vanish.

ID · date	Concept and revised question	Result and mathematical core
Y25-04-017 2025-04-18 · U	A rational Harper operator is a finite clock-shift problem What finite operator actually corresponds to a rational flux parameter?	An exact half-flux spectrum Atp=1,q=2,U=X,V=Z andH=2X+2Z, with eigenvalues±2√2. In general the declared model isq×q, soα=41/840 yields840 dimensions.
Y25-04-018 2025-04-18 · U	A gap-label Diophantine equation determines a residue class, not an open gap What exactly is solved by the integer gap-label equation?	Exact solutions for the recovered rationals Forp=41,q=840, t≡41r mod840 because41²≡1 mod840. Forp=5,q=12, t≡5r mod12. At r=1 the solutions include(t,s)=(41,−2) and(5,−2), respectively.
Y25-04-014 2025-04-18 · A	Hyperbolic volume growth prevents a global Euclidean doubling argument Which exact geometric fact controls the recovered large-radius comparisons?	Exact growth and nondoubling For n≥2,V_n(R) is asymptotic to[ω_(n−1)/((n−1)2^(n−1))]e^((n−1)R). HenceV_n(2R)/V_n(R)→∞. InH³,V_3(R)=π(sinh2R−2R).
Y25-04-015 2025-04-18 · A	Local hyperbolic Hardy atoms pair with the matching BMO norm What bounded pairing can be proved without importing an undefined oscillation threshold?	A directly proved pairing bound For a finite atomic sumf, |∫fb|≤||f||_(atomic)||b||_(BMO,R0). Every globally Lipschitz b with constantL has this local seminorm at most2LR0.
Y25-04-016 2025-04-18 · A	A hyperbolic spectral lower bound follows from a completed square What rigorous spectral estimate replaces an unexplained numerical transition inH^n?	An exact Poincaré inequality ∫|∇u|²dV≥((n−1)²/4)∫u²dV. Thus the nonnegative operator−Δ has no spectrum below this quadratic-form bound.
Y25-04-012 2025-04-19 · A	A logarithmic derivative is local analytic data, not an undefined regulator identity What does a claimed Ω-L-function identity need, and what exact model can actually be calculated?	An exact evaluation and singularity test L′(s)/L(s)=log2/(2^s−1). At s=1 this islog2, so imposinglogΩ=L′(1)/L(1) would giveΩ=2. At s=0 the logarithmic derivative has a simple pole.
Y25-04-013 2025-04-19 · A	Golden-ratio dilogarithms have exact functional-identity evaluations Which real dilogarithm value can be proved exactly instead of numerically fitting an unexplained constant?	Exact golden dilogarithms Li2(r)=π²/10−log²φ andLi2(r²)=π²/15−log²φ.
Y25-04-019 2025-04-20 · U	Golden-ratio contraction with two overlapping branches fills an interval When does the familiar two-map similarity-dimension formula fail?	An exact overlap obstruction Ifr≥1/2, the attractor is[0,1] and has dimension1. At r=1/φ, the similarity exponent islog2/logφ>1, so it cannot equal the dimension of this subset ofR.
Y25-04-020 2025-04-20 · U	A multiplicative attractor update is a linear recurrence with a sharp stability test Which mathematical stability statement follows from the recovered update rule?	Exact linear stability A_n=a^nA0. The equilibrium is asymptotically stable iff|1+λr|<1, equivalently−2/r<λ<0. Forλ>0 every nonzero magnitude grows exponentially.
Y25-04-021 2025-04-20 · U	Wavelet coherence needs smoothing to avoid an identically-one ratio What normalized coherence can be mathematically informative at a finite set of coefficients?	Exact coherence bound and degeneracy 0≤C≤1 byCauchy-Schwarz, with equality1 exactly when the weighted nonzero coefficient vectors are proportional. For a single coefficient the statistic is always1.

ID · date	Concept and revised question	Result and mathematical core
Y25-04-011 2025-04-20 · A	The recovered top-form quotient would vanish if all exact forms are removed Which quotient actually produces the finite Jacobian algebra, and why is the displayed extra denominator fatal?	Exact vanishing and the repaired finite quotient dΩ^(n−1)=Ω^n, soΩ^n/(df∧Ω^(n−1)+dΩ^(n−1))=0. Meanwhile$\Omega^n/(df\wedge\Omega^{n-1})\cong\mathbb{C}[x_1,\dots,x_n]/(\partial_1 f,\dots,\partial_n f)$ times the volume form.
Y25-04-022 2025-04-20 · A	A homotopy pushout needs a common category and actual arrows What concrete homotopy-colimit calculation clarifies the missing data in the recovered categorical synthesis?	An exact homotopy pushout The ordinary pushout is a single point, while the homotopy pushout is a circle with$H_1\cong\mathbb{Z}$.
Y25-04-028 2025-04-20 · U	The E8 dimension248 follows from an explicit root count Which exact part of the recoveredE8 data can be established independently of a categorical synthesis?	Exact root count and dimension The displayed set has240 roots of squared length2, spans$\mathbb{R}^8$, and gives$\dim E_8=8+240=248$.
Y25-04-025 2025-04-23 · A	A two-level gap produces an exact doubled-frequency observable What exact spectral calculation underlies a doubled energy-frequency scale?	Exact observable oscillation $X(t)=\cos(2Et/\hbar)X-\sin(2Et/\hbar)Y$. For the initial+X state, its expectation is$\cos(2Et/\hbar)$.
Y25-04-026 2025-04-23 · A	A Bogoliubov transformation must preserve the canonical commutator What exact algebraic condition is necessary before a claimed mode conversion is canonical?	Exact canonical condition and vacuum count The transformation is canonical iff$\lvert\alpha\rvert^2-\lvert\beta\rvert^2=1$. The mean output occupation in the input vacuum is$\langle 0\rvert b^\dagger b\lvert 0\rangle=\lvert\beta\rvert^2$. Choosing$\alpha=\cosh r,\beta=\sinh r$ satisfies the condition.
Y25-04-027 2025-04-23 · A	A switched oscillator gives exact parametric instability without a resonance slogan What fully solved oscillator model can demonstrate growth from periodic parameter variation?	Exact Floquet multipliers The one-cycle matrix$S_{(\omega_2)}(T_2)S_{(\omega_1)}(T_1)=\operatorname{diag}(-\omega_1/\omega_2,-\omega_2/\omega_1)$. Its determinant is1 and one multiplier has modulus>1 whenever$\omega_1\neq\omega_2$.
Y25-04-006 2025-04-23 · U	Müger nondegeneracy can be checked through an explicit bicharacter What does nondegenerate braiding mean in a concrete pointed model?	Exact triviality of the transparent subgroup Only$v=0$ is transparent. The normalized matrix$S_{vw}=B(v,w)/2$ is a real orthogonal4×4 matrix.
Y25-04-007 2025-04-23 · U	The quantum double of $\mathbb{Z}_2$ has distinct flux and charge labels What exact double construction can be worked out before claiming a large categorical unification?	Exact $\mathbb{Z}_2$ double data For$G=\mathbb{Z}_2$ the labels are$1,e,m,em$, all dimension1, with twists$1,1,1,-1$. The double braiding ofe andm is-1, and its$S$ matrix is the four-label bicharacter matrix of the preceding model.
Y25-04-005 2025-04-23 · A	Fusion coefficients are constrained by a recurrence and parity Which exact fusion rule is compatible with the recovered SU2 sine data?	An exact basic fusion identity $N_1^2=N_0+N_2$, so$N_{11}^{\wedge}1=0$ in this convention. For $k\geq4$, $N_2^2=N_0+N_2+N_4$. Allowed output labels have the same parity as$i+j$.

ID · date	Concept and revised question	Result and mathematical core
Y25-04-008 2025-04-23 · A	A Frobenius algebra and a Lagrangian dimension test require actual multiplication Which concrete algebra object demonstrates the meaning of a Lagrangian construction?	An explicit isotropic algebra and its dimension The multiplication is associative and braided-commutative onA. It has a nondegenerate Frobenius pairingτ(ab), withτ selecting the identity coefficient. Moreover(dimA)²=4, the Lagrangian dimension condition in this model.
Y25-04-023 2025-04-27 · U	Chromatic height is visible in a formal-group p-series What exact algebraic example gives meaning to a prime-dependent chromatic label?	Exact additive and multiplicative heights ForF_m(x,y)=x+y+xy, [p](x)=(1+x)^p−1, reducing tox^p in characteristicp, so its height is1. The additive lawF_a=x+y has[p]=px=0 in characteristicp and infinite height.

Y25-04-001 The number384 is a cyclotomic degree, with a computable Galois group

2025-04-01T15:59:45Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact arithmetic follows from1680 and which categorical dimensions do not follow from it?

The recovered user framework centers on the correct arithmetic identityφ(1680)=384. This can be developed into a precise field degree and a complete finite Galois-group calculation. Those facts do not select the rank of a fusion category or a spectral operator.

Definitions and assumptions

Write1680=$2^4 \cdot 3 \cdot 5 \cdot 7$ and$\zeta = e^{(2\pi i/1680)} = e^{(i\pi/840)}$.

For the cyclotomic field$K=Q(\zeta)$, classical cyclotomic theory gives$Gal(K/Q) \cong (Z/1680Z)^*$, acting by$\zeta \mapsto \zeta^a$.

Exact degree and group structure

[K:Q]=384 and$Gal(K/Q) \cong C4^2 \times C2^3 \times C3$. Its exponent is12, so it is not cyclic of order384. The maximal real subfield has degree192.

Derivation or proof

The totient product is$1680(1-1/2)(1-1/3)(1-1/5)(1-1/7)=384$. The Chinese remainder theorem decomposes the unit group into$U16 \times U3 \times U5 \times U7$. Here$U16 \cong C4 \times C2$, generated for example by3 and−1; the remaining factors areC2,C4,C6. SplittingC6 as$C2 \times C3$ gives the stated structure. The least common multiple of its cyclic orders is12, so no element can generate all384 automorphisms. Complex conjugation corresponds to−1, has order2, and is nontrivial becauseζ is nonreal. Its fixed real field therefore has half the degree by the Galois correspondence. The irreducibility of cyclotomic polynomials is a stated classical input; the arithmetic and group decomposition are checked explicitly.

Worked example

Compare the field degree with the angle denominator and element orders.

The primitive root has order1680, the field degree is384 and the largest automorphism order is12. These three integers measure different objects. Enumeration of all residues verifies384 units and maximum multiplicative order12.

The framework now has a genuine exact arithmetic foundation, with its numerical meanings distinguished.

Scope

No theorem identifies this degree with a category’s number of simple objects, total quantum dimension or a conjecture-resolving operator dimension.

Further development

Compute the real-subfield minimal polynomial or explicit subfields from the unit-group quotient, and state an actual functor if a categorical connection is intended.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R059.

Y25-04-002 A finite characteristic polynomial cannot equal a zeta entire function

2025-04-01T15:59:47Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact obstruction rules out the recovered finite determinant identity, and what valid finite determinant model remains?

The prior assistant asserted grand proofs from a finite384-dimensional determinant identity. IfH384 is an ordinary finite matrix andc≠0, the identity is impossible: a polynomial has finitely many zeros while the completed zeta function has infinitely many. A finite spectral model remains useful, but proves only its own spectral algebra.

Definitions and assumptions

For a384×384 matrixH, p_H(s)=det(sI−H) is a monic polynomial of degree384.

The completed xi function is entire and has the nontrivial zeta zeros; their infinitude is a classical theorem independent of RH.

A finite-degree obstruction and a valid product identity

No ordinary finiteH and nonzeroc satisfy p_H(s)=cξ(s)^384 for allcomplexs. For a finite Hermitian matrix, the valid identity is p_H(s)=∏_j(s−λ_j), with realλ_j.

Derivation or proof

The determinant expansion has leading terms^384 and therefore defines a nonzero polynomial with at most384 distinct roots. The right-hand side vanishes at every zero ofξ, an infinite set; equality would give infinitely many polynomial roots, a contradiction. Ifc=0, equality is also impossible becausep_H is monic. An infinite-dimensional regularized determinant is a different object and needs convergence, continuation and normalization; changing the word determinant does not supply those theorems. For the valid finite model, unitary diagonalizationH=Udiag(λ_j)U* and multiplicativity of determinant yield the product, whileHermiticity makesλ_j real.

Worked example

TakeH=diag(1,4).

p_H(s)=(s−1)(s−4)=s²−5s+4. The resolvent trace away from its poles is1/(s−1)+1/(s−4)=p_H′(s)/p_H(s). This exact identity encodes two eigenvalues and no assertion about all zeros of an arithmetic function.

The finite construction can be checked, while the impossible entire-function identity is discarded.

Scope

The historical definitions ofH384 and the categorical determinant were not recovered. No proof ofRH, BSD, Langlands, mass gap or fluid regularity is inherited from those claims.

Further development

If an infinite spectral determinant is intended, define the operator and its counting law, prove a regularization and derive an arithmetic identity without presupposing the desired zero location.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R060.

Y25-04-024 Dimensionful constants cannot be numerical predictions of unit-free integers alone

2025-04-01T20:19:02Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which proposed constants are invariant under unit changes, and what can an integer framework actually determine?

The recovered framework attempts to derive several dimensional constants from categorical integers and an angle. A precise unit-change argument distinguishes numerical conventions from dimensionless predictions. This is a mathematical consistency test, not a claim that constants cannot be related within a defined theory.

Definitions and assumptions

Use base dimensionsM,L,T. Then[ħ]=ML²T⁻¹,[c]=LT⁻¹,[G]=L³M⁻¹T⁻².

A dimensionless monomialħ^a c^b G^d has zero exponent in each base dimension.

An exact dimension-rank test

The only dimensionless monomial formed solely from$\hbar$,c,G is the trivial one$a=b=d=0$. Their three-dimensional unit-exponent matrix is invertible, so their numerical values can all be changed by a choice ofM,L,T units.

Derivation or proof

The dimension equations are$-d=0, 2a+b+3d=0, -a-b-2d=0$. The first gives$d=a$; substituting into the other two gives$b=-5a$ and$b=-3a$, so$2a=0$ and all exponents vanish. Equivalently the determinant of the exponent matrix is nonzero. Thus a bare integer or angle cannot specify a unit-independent numerical value of one of these dimensional constants without a scale convention. Planck combinations are dimensionful definitions:$\sqrt{(\hbar G/c^3)}$ has length,$\sqrt{(\hbar G/c^5)}$ has time and$\sqrt{(\hbar c/G)}$ has mass. Their numerical values become1 in corresponding units, which is a coordinate choice rather than a prediction.

Worked example

Check the length dimension of$\ell_P=\sqrt{(\hbar G/c^3)}$.

Its squared dimension is$(ML^2T^{-1})(L^3M^{-1}T^{-2})/(L^3T^{-3})=L^2$, so the square root has dimensionL. A proposed pure number forℓ_P would change under length-unit rescaling. A dimensionlessα could in principle be predicted, but that requires an independently defined equation and a proof, not this dimensional argument.

The unit-invariant content of a claimed derivation is now explicit.

Scope

This does not rule out relations among dimensionless constants or a theory with specified scales. The historical formulas for those constants were not fully recovered and are not validated.

Further development

List every parameter with its dimension, compute the nullspace of the dimension matrix and express candidate predictions only through the resulting invariant combinations.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: .

Y25-04-003 The sine matrix at π/840 is orthogonal only with the full839 indices

2025-04-10T00:59:30Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which index range makes the recovered sine matrix a unitary modular candidate?

The recovered source explicitly truncates the sine matrix to indices0 through383. The standard discrete sine transform with this angle has839 rows and columns. Its orthogonality can be proved directly, making the truncation failure an exact mathematical issue.

Definitions and assumptions

Letm≥2 andS_rs=√(2/m)sin(πrs/m), withr,s=1,…,m−1.

For the recovered anglem=840, the full index count ism−1=839, corresponding toSU2 levelk=m−2=838 in the standard convention.

Exact sine orthogonality

Σ_(s=1)^(m−1)sin(πrs/m)sin(πts/m)=(m/2)δ_rt. HenceS is real symmetric with$S^2=I$. The384×384 principal truncation at m=840 is not orthogonal.

Derivation or proof

Use2sin u sinv=cos(u−v)−cos(u+v), and sum the finite geometric series of exponentials on the2mth roots of unity. Orthogonality of the odd Fourier combinations gives the displayed identity; equivalently, the vectors are the complete eigenbasis of the path adjacency matrix with zero boundary values. For the truncation, its first-row squared norm is(2/840)Σ_(s=1)^384$\sin^2$(sπ/840). The full norm is1, and every omitted term for385≤s≤839 is strictly positive. Thus the truncated norm is strictly less than1, disproving orthogonality without any numerical tolerance. Keeping the prefactor and angle while changing the rank destroys the identity.

Worked example

Usem=4 as an exact small transform.

S=(1/2)[[1,√2,1],[√2,0,−√2],[1,−√2,1]]. Multiplication gives$S^2=I3$. Deleting the last row and column leaves first-row squared norm3/4, not1, exactly illustrating the same truncation mechanism.

The matrix is now a proven transform with an explicit index convention. Cyclotomic degree384 does not change its required size.

Scope

Orthogonality alone does not construct a modular tensor category; fusion integrality, twists and coherence require additional data. The source’s symbol$K^2=C$ was not defined and is not silently identified with this matrix.

Further development

If a384-object category is intended, supply its actualS,T and fusion coefficients and verify the consistency relations instead of taking a principal block of a larger transform.

Source receipt: months/04/root_followups.json record 0. Mathematical references: R048.

Y25-04-004 Quantum-dimension probabilities must be normalized by their actual sum

2025-04-10T00:59:30Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What is the correct normalization and which entropy values can be computed exactly?

The recovered dimension sum1680 is inconsistent with the displayed trigonometric formula and index range. A direct bound already disproves it. With the proper sum, the quantities become legitimate probabilities and their Rényi entropies can be calculated in a small exact example.

Definitions and assumptions

Letd_(r−1)=sin(rθ)/sinθ,θ=π/840, andD_N=Σ_(r=1)^Nd_(r−1)².

Setp_r=d_(r−1)²/D_N. Forq>0,q≠1, H_q=(1/(1−q))logΣp_r^q; theq→1 limit is Shannon entropy.

An exact contradiction and a corrected law

D_384>1960>1680. For the full839 indices,D_839=420/sin²θ andp_r=sin²(rθ)/420. Atm=4 the probabilities are(1/4,1/2,1/4), withH1=(3/2)log2 andH2=log(8/3).

Derivation or proof

For1≤r≤24, rθ≤π/2 andsin(rθ)≥2rθ/π, while sinθ≤θ. Hence d_(r−1)²≥4r²/π². Summing givesD_384≥(4/π²)Σ_(r=1)^24r²=19600/π²>1960, usingπ²<10. The omitted positive terms make the lower bound strict as needed. Full normalization follows from the sine orthogonality sumΣsin²(rθ)=840/2. For m=4, the dimensions are1,√2,1 and their squared sum is4. Direct substitution into the entropy definitions gives the displayed values, includingΣp²=3/8.

Worked example

Compare the recovered putativeμ weights with the corrected law.

The first24μ weights alone sum to more than1960/1680=7/6, so the full source list cannot be a probability distribution. Dividing by its actualD_384 repairs normalization but does not repair the truncatedS-matrix orthogonality.

The correction is not a small rounding adjustment; the claimed probabilities fail an exact total-mass test.

Scope

Quantum dimensions must come from a defined fusion structure. The calculation treats the given trigonometric numbers precisely but does not validate a new category or entropy-based prediction.

Further development

Compute exact normalization and entropy bounds for the actual category, then distinguish truncating an observable from truncating the underlying fusion theory.

Source receipt: months/04/root_followups.json record 0. Mathematical references: R048.

Y25-04-009 Milnor numbers come from a finite Jacobian algebra

2025-04-18T17:31:12Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact invariant is obtained if the recovered triple is interpreted as a Brieskorn polynomial?

The notationM(a,b,c) was not defined in the recovered source. A precise conditional interpretation isf=x^a+y^b+z^c witha,b,c≥2. Its Jacobian algebra gives a directly countable Milnor number and distinguishes the singularity fiber from its three-dimensional link.

Definitions and assumptions

For an isolated polynomial critical point, the Jacobian algebra isQ_f=C[x,y,z]/(f_x,f_y,f_z); its finite dimension is the Milnor number.

The Milnor fiberf=ε in a small ball has complex dimension2; the linkf=0 intersected with a small5-sphere has real dimension3.

Exact monomial basis

Forf=x^a+y^b+z^c, the monomialsx^iy^jz^k with0≤i≤a−2,0≤j≤b−2,0≤k≤c−2 form a basis, givingμ=(a−1)(b−1)(c−1).

Derivation or proof

The partial derivatives are nonzero scalar multiples ofx^(a−1),y^(b−1),z^(c−1), so their ideal is generated by those three pure powers. Every monomial with an exponent beyond a bound vanishes in the quotient; the remaining monomials are linearly independent because a monomial ideal contains no nonzero linear combination supported on standard monomials. Counting the independent exponent choices proves the formula. The critical point is isolated because all three derivatives vanish simultaneously only at0. Identifying this algebraic number with the bouquet-of-spheres count of the Milnor fiber is Milnor’s structural theorem, not a new consequence of the count.

Worked example

Use(a,b,c)=(5,12,2) and(41,840,2).

The first basis has4·11·1=44 elements. The second has40·839·1=33560. The Milnor fiber hasb2=μ and reduced homology in that middle degree under the classical theorem; it is not a closed3-manifold withb3=1.

The recoveredμ=44 is correct in this explicit interpretation, while the topology and dimension of the associated object must be kept separate.

Scope

The historicalM notation and the meaning ofΩ were not recovered. This conditional development does not identifyμ with a spectral gap, a regulator or a categorical rank.

Further development

Compute monodromy and intersection forms for the specified polynomial, then use those matrices to investigate its link rather than inferring link homology fromμ alone.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R061.

Y25-04-010 A Brieskorn link can have torsion invisible to its Betti numbers

2025-04-18T17:31:12Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What link homology can be checked when a recovered Betti table omits torsion?

The recovered Betti numbers resemble a rational homology sphere but do not determine integral homology. For the explicit linksΣ(2,a,b) witha odd,b even andgcd(a,b)=1, an exact determinant calculation detects nontrivial torsion. This is a useful refinement of the conditional Brieskorn interpretation.

Definitions and assumptions

LetΣ(2,a,b) be the link ofz²+x^a+y^b=0, oriented as a complex link. It is the double branched cover ofS³ over the(a,b) torus knot.

A classical branched-cover theorem states that the order ofH1 is|Δ_(a,b)(−1)| when nonzero, whereΔ is the knot Alexander polynomial normalized up to units.

Exact torsion order for the recovered examples

For odda and evenb withgcd(a,b)=1, |H1(Σ(2,a,b);Z)|=a. ThusΣ(2,5,12) hasH1≅Z/5 andΣ(2,41,840) hasH1≅Z/41, while their first and second rational Betti numbers vanish.

Derivation or proof

The torus-knot polynomial isΔ_(a,b)(t)=[(t^(ab)−1)(t−1)]/[(t^a−1)(t^b−1)]. At t=−1, both theab andb factors vanish, but their ratio tendsa(−1)^(ab−b)=a becauseb is even. The remaining ratio(t−1)/(t^a−1) tends1 becausea is odd. Hence the determinant is a, which is finite and nonzero. The branched-cover theorem gives a finite first homology of that order. In the two examples a is prime, so the only abelian group of that order isZ/a. RationalH1 therefore vanishes; Poincaré duality on the closed oriented link gives rationalH2=0 andb0=b3=1.

Worked example

Check the coprimality and determinant data.

gcd(5,12)=1 andgcd(41,840)=1. The corresponding orders are5 and41, not1. Thus the recovered Betti pattern can hold rationally while the statement “integral homology sphere” would be false for these links.

Torsion supplies mathematical information that a real-valued or rational Betti table discards.

Scope

The result uses the stated link interpretation and classical branched-cover/Alexander theorems. It does not classify the full fundamental group or validate an undefinedM model.

Further development

Construct an explicit Seifert presentation or surgery matrix and compute its Smith normal form as a second independent certificate of the integral group.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R061.

Y25-04-017 A rational Harper operator is a finite clock-shift problem

2025-04-18T17:31:12Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What finite operator actually corresponds to a rational flux parameter?

The recoveredΩ framework supplies rationalα and references Hofstadter gaps. A rational flux determines a finite projective translation relation after boundary phases are chosen. This gives a precise matrix model without identifying its size with a Milnor number.

Definitions and assumptions

Forcoprime p,q, letUe_j=e_(j−1) cyclically andVe_j=e^(2πipj/q)e_j, withj modq. ThenUV=e^(2πip/q)VU.

Define the Hermitian clock-shift matrixH=U+U*+V+V*. Other boundary phases give a family of Bloch matrices.

An exact half-flux spectrum

> Atp=1,q=2,U=X,V=Z andH=2X+2Z, with eigenvalues±2√2. In general the declared model isq×q, soα=41/840 yields840 dimensions.

Derivation or proof

The commutation relation follows by applyinge_j to both products and comparing the phase atj versusj−1. The adjoint pairs makeH Hermitian. Forq=2, $X^2=Z^2=I$ and$XZ=-ZX$, so$H^2=4(X+Z)^2=8I$. SinceTrH=0, its two eigenvalues are opposite and equal in magnitude to√8. This computes a genuine finite spectrum. The matrix dimension comes from the denominator of the rational projective phase and not fromφ(q),φ(2q) or(a−1)(b−1)(c−1) of a different polynomial.

Worked example

Compareα=1/2 andα=41/840.

The former has the exact2×2 matrix[[2,2],[2,−2]]. The latter requires an840×840 Bloch model in this convention. Different boundary phases can move finite eigenvalues, so the fullband problem is not one isolated matrix at an unspecified phase.

The recovered rational is now connected to a well-defined finite operator and a testable spectral problem.

Scope

This calculation does not prove gap opening at every label, a Hall-conductance statement or anΩ formula. Irrational flux is an infinite-operator problem with different limiting questions.

Further development

For a fixedp/q, track the eigenvalues over both boundary phases, enclose band edges and verify whether a claimed gap remains open.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R062.

Y25-04-018 A gap-label Diophantine equation determines a residue class, not an open gap

2025-04-18T17:31:12Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exactly is solved by the integer gap-label equation?

The recovered framework combines rational flux and spectral-gap numerics. The associated Diophantine equation has a precise elementary content: it determines an integer residue class if a gap with a given band count is under discussion. It cannot by itself prove that the gap exists.

Definitions and assumptions

Forcoprime p,q and integerband countr, consider r=pt+qs with unknownintegerst,s.

Reducing moduloq determines t moduloq. Any additional centered representative or topological interpretation is a separate convention or theorem.

Exact solutions for the recovered rationals

Forp=41,q=840, t≡41r mod840 because41²≡1 mod840. Forp=5,q=12, t≡5r mod12. At r=1 the solutions include(t,s)=(41,−2) and(5,−2), respectively.

Derivation or proof

Bezout’s identity gives an inverse ofp moduloq sincegcd(p,q)=1. Thus the congruencer≡pt has exactly one solution class moduloq. If(t0,s0) is one solution, all solutions are(t0+kq,s0−kp) forintegerk, as substitution verifies. Conversely, subtracting two solutions givesp(t−t0)=−q(s−s0); coprimality forcesq|(t−t0), proving completeness. Here41²=1681=1+2·840 and5²=25=1+2·12, supplying the inverses and the explicitr=1 pairs.

Worked example

Enumerate allr from0 throughq−1 for the first recovered rational.

Multiplication by41 permutes the840 residue classes, so everyband count has one algebraically allowed class. That fact alone cannot imply840 spectral gaps: a spectral model may have touching bands, and some labels may not occur as open intervals.

Integer solvability is now fully classified, while the spectral existence question remains correctly separate.

Scope

The physical identification oft with a Chern number requires a specific magnetic Bloch bundle and a proved gap theorem. The algebraic equation alone supplies neither.

Further development

Choose a candidate open gap of the actual Harper family, certify it over the Bloch torus, then compute the corresponding vector-bundle invariant and check the Diophantine relation.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R063.

Y25-04-014 Hyperbolic volume growth prevents a global Euclidean doubling argument

2025-04-18T17:35:51Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which exact geometric fact controls the recovered large-radius comparisons?

The recovered H1-BMO passage proposes exponential rates and thresholds without proofs. Constant-curvature hyperbolic space has an exact radial measure that explains exponential volume growth. This establishes a clear geometric fact while leaving any claimed mixing exponent unproved.

Definitions and assumptions

InH^n of curvature−1, polar volume isω_(n−1)sinh^(n−1)r dr. LetV_n(R) be the ballvolume.

A globally doubling measure would haveV_n(2R)≤CV_n(R) for one constantC and everyR.

Exact growth and nondoubling

For n≥2,V_n(R) is asymptotic to[ω_(n−1)/((n−1)2^(n−1))]e^((n−1)R). HenceV_n(2R)/V_n(R)→∞. InH³,V_3(R)=π(sinh2R−2R).

Derivation or proof

Write sinhr=(e^r/2)(1−e^(−2r)). Dividing the radial integrand bye^((n−1)r) gives a limit2^(−(n−1)). Integrating or applying l'Hôpital toV_n(R)/e^((n−1)R) yields the displayed constant. The volume ratio therefore grows likee^((n−1)R), disproving global doubling. Forn=3,ω2=4π andsinh²r=(cosh2r−1)/2; integration givesπsinh2R−2πR. At smallR theTaylor expansion is(4π/3)R³+O(R⁵), consistent with local Euclidean behavior. Thus local doubling can hold on bounded radii although a uniform all-radius constant fails.

Worked example

CompareH³ balls of radiiR and2R.

The exact ratio is[sinh4R−4R]/[sinh2R−2R]. At smallR it tends8, the Euclidean3D factor; at largeR it is asymptotic toe^(2R). One cannot use the small-radius constant at every scale.

Exponential geometric growth is established. It does not by itself determine a dynamical decay coefficient or a BMO transition.

Scope

The calculation assumes the exact curvature−1 model. Quotients, finite-volume cusps and variable curvature require additional hypotheses. Noλ≈20 transition follows.

Further development

Declare the radius range of an atomic or oscillation norm, then derive constants from the actual volume comparison valid on that range.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y25-04-015 Local hyperbolic Hardy atoms pair with the matching BMO norm

2025-04-18T17:35:51Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What bounded pairing can be proved without importing an undefined oscillation threshold?

The recovered notationH1 may denote Hardy space or a Sobolev space; these are different. On hyperbolic space, an explicit local atom convention gives an elementary pairing theorem with a matching local BMO seminorm. The proof needs cancellation and a specified ball radius.

Definitions and assumptions

FixR0>0. A local L∞ atom a is supported in a geodesic ballB of radius≤R0, has∫a=0 and||a||∞≤1/VolB.

Define||b||_(BMO,R0)=sup_(radB≤R0)(1/VolB)∫_B|b−b_B|. Define the atomic norm by the infimum ofΣ|λj| inΣλjaj.

A directly proved pairing bound

> For a finite atomic sumf, |∫fb|≤||f||_(atomic)||b||_(BMO,R0). Every globally Lipschitz b with constantL has this local seminorm at most2LR0.

Derivation or proof

For an atom, cancellation gives∫ab=∫_Ba(b−b_B). Its absolute value is at most(1/VolB)∫_B|b−b_B|, bounded by the stated seminorm. Sum the coefficient bounds and take the infimum over finite representations. For theLipschitz claim, |b(x)−b_B|≤(1/VolB)∫_B|b(x)−b(y)|dy≤2LR0 because any two points inB are at distance at most2R0. Integrating proves the bound. No global doubling constant was needed for this finite pairing estimate. A completion and a converse duality theorem need additional function-space and geometric work.

Worked example

Takeb(x)=distance(x,o), which is1-Lipschitz by the triangle inequality.

For every ball of radius≤1, the local BMO seminorm is at most2. Any canceled atom pairs withb by an absolute value at most2. A noncanceled bump is not covered: adding a constant tob changes its integral pairing, whereas the BMO seminorm ignores constants.

The exact hypotheses explain why cancellation, radius scale and the choice ofH1 are essential.

Scope

This proves one bounded pairing, not the entire Hardy-BMO duality theorem or a large-radius growth claim. The Sobolev norm$\int(|f|^2+|\nabla f|^2)$ is not the atomic norm used here.

Further development

Choose an actual local Hardy-space completion and prove its dual description under the locally doubling hypotheses, then compare a declared oscillation statistic with that norm.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y25-04-016 A hyperbolic spectral lower bound follows from a completed square

2025-04-18T17:35:51Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What rigorous spectral estimate replaces an unexplained numerical transition inH^n?

The recovered passage cites a hyperbolic Laplacian and numerical thresholds. A coordinate calculation gives a real theorem: the quadratic-form spectrum has a positive lower bound depending only on dimension and the curvature normalization.

Definitions and assumptions

Use the upper-half-space metric$g=y^{-2}(dx^2+dy^2)$,$y>0$,onH^n. The Laplacian sign is$\Delta=\operatorname{div}\operatorname{grad}$.

Letb=−logy. Direct calculation gives$|\nabla b|=1$ and$\Delta b=n-1$. Test functionsu are real smooth and compactly supported.

An exact Poincaré inequality

$\int|\nabla u|^2dV\geq((n-1)^2/4)\int u^2dV$. Thus the nonnegative operator$-\Delta$ has no spectrum below this quadratic-form bound.

Derivation or proof

The coordinate expression is$\Delta = y^2(\Delta_x + \partial_y^2) - (n-2)y\partial_y$; substitutingb gives$\Delta b = n-1$, and the inverse metric gives$|\nabla b|^2 = y^2(1/y^2) = 1$. For any reala, expand$\int |\nabla u + a\, u\nabla b|^2$. The cross term is$a\int \nabla(u^2)\cdot\nabla b = -a\int u^2 \Delta b$, with no boundary term becauseu is compactly supported. Therefore the nonnegative square equals$\int |\nabla u|^2 + [a^2 - a(n-1)]\int u^2$. Choosea=(n−1)/2 to minimize the bracket, yielding the inequality. The passage from the form inequality to the Friedrichs self-adjoint operator is standard; equality of the spectral bottom needs an additional approximating-sequence argument.

Worked example

InH^3 the lower bound is1; inH^2 it is1/4.

Scaling the metric to curvature$-\kappa^2$ multiplies the operator byκ^2 and the lower bound becomes$\kappa^2(n-1)^2/4$. Thus a bare threshold20 has no invariant significance without metric units and a specified operator.

The estimate is proved analytically and its scale dependence is explicit.

Scope

The positive generalized functions y^s are not square-integrable eigenvectors on allH^n, so substituting them alone would not prove a spectral theorem. Quotients may have additional low discrete eigenvalues and require separate analysis.

Further development

Construct compactly supported cutoff functions approaching the lower Rayleigh bound and prove spectral-bottom equality in this model, then study a declared quotient.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y25-04-012 A logarithmic derivative is local analytic data, not an undefined regulator identity

2025-04-19T20:21:13Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does a claimed Ω-L-function identity need, and what exact model can actually be calculated?

The recoveredΩ formula does not specify a cycle, regulator normalization, L-function or evaluation point. A logarithmic derivative is a function of a variable, usually meromorphic, whereaslogΩ is a number. A finite Euler factor gives a fully defined elementary model and exposes the missing data.

Definitions and assumptions

For a nonvanishing holomorphicL near s0, $(\log L)' = L'/L$ locally, independently of the additive logarithm branch.

ConsiderL(s)=1−2^(−s), a single entire Euler-factor polynomial in2^(−s), not a global arithmetic L-function.

An exact evaluation and singularity test

L′(s)/L(s)=log2/(2^s−1). At s=1 this islog2, so imposinglogΩ=L′(1)/L(1) would giveΩ=2. At s=0 the logarithmic derivative has a simple pole.

Derivation or proof

Differentiate2^(−s)=e^(−slog2):L′=(log2)2^(−s). Division gives the stated expression. At s=1 the denominator is1. Near zero,2^s−1=slog2+O(s²), soL′/L=1/s+O(1), displaying the pole arising from the zero ofL. More generally, ifL(s)=(s−s0)^m g(s) withg(s0)≠0, thenL′/L=m/(s−s0)+g′/g. Thus an evaluation at a zero is not an ordinary finite value unless a finite-part or leading-coefficient convention is specified. A regulator value additionally needs a defined cycle and map; no such data follows from this analytic identity.

Worked example

Compare twoevaluation points for the same explicitfactor.

At s=1 the imposedΩ would be2; at s=2 it would beexp[(log2)/3]=2^(1/3). The number changes with the point, showing why suppressing the argument is mathematically consequential.

A complete identity must name its objects, evaluation point and normalization. The historical decimalΩ is not established by the displayed equality.

Scope

The exactX37 L-value remains unverified becauseX37 and its L-function are not defined in the recovered excerpt. This model is clearly illustrative, not a replacement historical result.

Further development

Specify the motive or curve, the cycle class, the regulator map and the precise leading-value statement. Compute both sides independently with certified errors.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: .

Y25-04-013 Golden-ratio dilogarithms have exact functional-identity evaluations

2025-04-19T20:21:13Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which real dilogarithm value can be proved exactly instead of numerically fitting an unexplained constant?

The recovered passage mentions dilogarithmic approximations. A constructive replacement is an exact evaluation at the inverse golden ratio, linked to the same golden scaling used elsewhere that month. The proof uses functional identities derivable by differentiation.

Definitions and assumptions

For0≤x≤1 defineLi2(x)=Σ_(n≥1)x^n/n², withLi2′(x)=−log(1−x)/x on(0,1).

Letφ=(1+√5)/2 andr=1/φ, so r²=1−r and1+r=1/r.

Exact golden dilogarithms

Li2(r)=π²/10−log²φ andLi2(r²)=π²/15−log²φ.

Derivation or proof

The reflection identityLi2(x)+Li2(1−x)=π²/6−logxlog(1−x) follows by differentiating both sides and evaluating atx→0. AlsoLi2(x)+Li2(−x)=(1/2)Li2(x²) follows directly from the series, andLi2(x/(1+x))=−Li2(−x)−(1/2)log²(1+x) follows by differentiation and the value at0. PutA=Li2(r),B=Li2(r²),l=logr. Reflection givesA+B=π²/6−2l². The other two identities, usingr/(1+r)=r², giveA=(3/2)B+(1/2)l². Solving the two linear equations yieldsB=π²/15−l² andA=π²/10−l², equal to the stated forms because l²=log²φ.

Worked example

Evaluate both sides by a positive convergent series.

AfterN terms at0<x<1, the tail is at mostx^(N+1)/[(N+1)²(1−x)]. AtN=80 this bounds the error forx=r by less than10^(−19), enough to check many digits without an arbitrary fit.

The special value follows from exact identities and a reproducible error bound, giving a genuine alternative to an unexplained numerical coincidence.

Scope

These identities do not prove the recoveredΩ regulator formula or any physical constant relation. Real arguments avoid branch choices; complex dilogarithms need an explicit analytic continuation convention.

Further development

Identify the exact algebraic arguments in an actual period integral and reduce them using functional equations before proposing a numerical recognition relation.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R065.

Y25-04-019 Golden-ratio contraction with two overlapping branches fills an interval

2025-04-20T00:41:00Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

When does the familiar two-map similarity-dimension formula fail?

The recovered source applies a Cantor-style dimension formula to a contraction ratio near0.618. Two equal interval branches at that ratio overlap. The simplest declared iterated-function system has attractor the entire unit interval andHausdorff dimension1, so the displayed formula cannot be used as its dimension.

Definitions and assumptions

For0<r<1, definef0(x)=rx andf1(x)=rx+1−r onR. The attractor is the unique nonempty compactK withK=f0(K)∪f1(K).

The similarity exponent solves2r^s=1, giving s=log2/log(1/r); equality with Hausdorff dimension needs separation hypotheses.

An exact overlap obstruction

Ifr≥1/2, the attractor is[0,1] and has dimension1. At r=1/φ, the similarity exponent islog2/logφ>1, so it cannot equal the dimension of this subset ofR.

Derivation or proof

The images of[0,1] are[0,r] and[1−r,1]. They cover the entire interval exactly whenr≥1/2, so[0,1] is a fixed compact set. The Hutchinson set map contracts Hausdorff distance by at mostr; iteration or the contraction principle gives uniqueness of the compact attractor. Hence it is[0,1]. ItsHausdorff dimension is1. Sinceφ<2, logφ<log2 and the formal similarity exponent exceeds1. The failure is overlap, not a subtle new fractional dimension. Forr<1/2, the two branches are separated and the standard equal-ratio Cantor dimension theorem applies, but that is a different parameter regime.

Worked example

Compare r=1/3 withr=1/φ.

The first produces the middle-thirdCantor set withdimensionlog2/log3. The second produces[0,1] even though the contraction remains strict. A single scalar ratio does not determine a fractal dimension without the maps and overlaps.

The recovered golden scaling can define a legitimate contraction, but the claimed Cantor dimension needs a corrected model.

Scope

Different translations or higher-dimensional maps may yield other attractors. The conclusion is for the explicitly specified two-branch line system, not every construction involving the golden ratio.

Further development

Write the actual maps, analyze exact overlaps and verify an open-set or alternative dimension theorem before assigning an exponent.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R066.

Y25-04-020 A multiplicative attractor update is a linear recurrence with a sharp stability test

2025-04-20T00:41:00Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which mathematical stability statement follows from the recovered update rule?

The recovered recurrence is elementary enough to solve exactly, but calling its iterates an attractor does not make them bounded. A precise stability analysis separates a positive growth multiplier from a contracting map with an actual fixed point.

Definitions and assumptions

LetA_(n+1)=aA_n witha=1+λr, where0<r<1 andλ real.

An equilibrium0 is asymptotically stable exactly when every sufficiently small initial condition tends to0.

Exact linear stability

A_n=a^nA0. The equilibrium is asymptotically stable iff|1+λr|<1, equivalently−2/r<λ<0. Forλ>0 every nonzero magnitude grows exponentially.

Derivation or proof

Induction gives the closed form. If|a|<1, powers tend0 and remain bounded by|A0|, proving stability and attraction. If|a|>1, arbitrarily small nonzero initial values eventually leave any fixed neighborhood, so stability fails. Ata=1 the iterates are constant and ata=−1 they alternate; neither attracts allnearby points to0. The real inequality−1<1+λr<1 solves to the stated interval because r>0. A parameter namedβ≈0.673 would be a contraction factor only if it actually bounds the derivative or multiplier of the defined map.

Worked example

Taker=1/φ. Withλ=1, a=1+r=φ.

A_n=φ^nA0 diverges in magnitude forA0≠0. Withλ=−1, a=1−r=r², soA_n=r^(2n)A0 tends0. Adding a constantc givesA_(n+1)=aA_n+c; when|a|<1 its unique fixed point isc/(1−a), and subtracting that point reduces to the same contraction proof.

The proposed update has an exact parameter classification, not a generic stability guarantee.

Scope

A bounded nonlinear state space, saturation or projection would define another map and require a new proof. No empirical stability threshold is inferred from the recovered decimalβ.

Further development

Define the full nonlinear update and bound its derivative on an invariant interval. Use that bound to prove a fixed point and a quantitative convergence rate.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R066.

Y25-04-021 Wavelet coherence needs smoothing to avoid an identically-one ratio

2025-04-20T00:41:00Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What normalized coherence can be mathematically informative at a finite set of coefficients?

The recovered user source names wavelet coherence but does not define its smoothing operator. Pointwise normalization of a single product gives1 whenever both coefficients are nonzero. An averaged complex inner product supplies a nontrivial statistic with a complete inequality.

Definitions and assumptions

For complex coefficient sequencesx_j,y_j and nonnegative weightsw_j, defineC=|Σw_jx_jconj(y_j)|²/[(Σw_j|x_j|²)(Σw_j|y_j|²)], when both denominator terms are positive.

The indexj may represent a declared time-scale neighborhood of wavelet coefficients; its weights are part of the definition.

Exact coherence bound and degeneracy

0≤C≤1 byCauchy-Schwarz, with equality1 exactly when the weighted nonzero coefficient vectors are proportional. For a single coefficient the statistic is always1.

Derivation or proof

ApplyCauchy-Schwarz to the vectors(√w_jx_j) and(√w_jy_j). Their squared inner-product magnitude is at most the product of their squared norms, proving the bound. Equality inCauchy-Schwarz requires linear dependence, giving the stated criterion after ignoring zero weights. With one nonzero weighted slot, the numerator isw²|x|²|y|² and the denominator is identical. Therefore the unsmoothed pointwise ratio cannot distinguish two nonzero signals. The meaningful information comes fromphase and amplitude consistency across the averaging neighborhood, not merely from writing a normalized product.

Worked example

Takex=(1,1),y=(1,−1) with equal weights1.

The averaged cross term is1−1=0, soC=0, although each individual one-slot ratio is1. Replacingy by(2,2) makesC=1. Thus averaging reveals cancellation invisible to pointwise normalization.

The statistic is now defined, bounded and interpretable in an exact finite example.

Scope

The calculation does not validate any recovered empirical coherence value. Wavelet admissibility, edge handling, scale weights, missing data and significance tests need separate specifications.

Further development

Choose a precise wavelet transform and smoothing kernel, then test phase-shifted and independent synthetic controls with a declared uncertainty model.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: .

Y25-04-011 The recovered top-form quotient would vanish if all exact forms are removed

2025-04-20T20:11:26Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which quotient actually produces the finite Jacobian algebra, and why is the displayed extra denominator fatal?

The recovered Brieskorn-module formula includes all exact top forms in the denominator. For polynomial forms overC, every top form is exact, so that quotient is zero. A nearby correct quotient recovers the Milnor algebra; a genuine Brieskorn module uses a different denominator.

Definitions and assumptions

LetΩ^j denote polynomial differentialj-forms in n variables overC.

The Jacobian top-form quotient isΩ^n/(df∧Ω^(n−1)). A standard Brieskorn module instead usesdf∧dΩ^(n−2), with indexing conventions stated.

Exact vanishing and the repaired finite quotient

dΩ^(n−1)=Ω^n, soΩ^n/(df∧Ω^(n−1)+dΩ^(n−1))=0.
MeanwhileΩ^n/(df∧Ω^(n−1))≅C[x1,…,xn]/(∂1f,…,∂nf) times the volume form.

Derivation or proof

For a polynomialp, integrate it termwise inx1 to obtain a polynomialP with∂1P=p. Then d(Pdx2∧⋯∧dxn)=pdx1∧⋯∧dxn; terms differentiated in other variables wedge a repeated differential and vanish. Thus every top form is exact, proving the first assertion. For the second, wedge df with a general(n−1)-form. Its coefficient of the volume form is an arbitrary linear combination of the partial derivatives, so its image is precisely the Jacobian ideal times that form. Taking the quotient proves the identification. This does not identify that finite quotient with the full Gauss-Manin or Brieskorn module, which retains deformation-parameter information.

Worked example

Takef=x³+y² in two variables.

The Jacobian ideal is(x²,y), so the repaired top-form quotient has basisdx∧dy andx dx∧dy, dimension2. Both forms are nevertheless exact:dx∧dy=d(xdy) andx dx∧dy=d((x²/2)dy). Adding all exact forms kills the entire two-dimensional quotient.

The difference between the denominators changes the object from a meaningful invariant to zero; it is not a notation choice.

Scope

The proof is for polynomial or convergent local forms in characteristic0. A full Brieskorn-lattice theorem, freeness and period connection require further structure.

Further development

Choose one standard module convention, define its parameter action and calculate the Gauss-Manin connection for a small singularity before comparing periods.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R061.

Y25-04-022 A homotopy pushout needs a common category and actual arrows

2025-04-20T20:11:26Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What concrete homotopy-colimit calculation clarifies the missing data in the recovered categorical synthesis?

The recovered formula places a category and anE8 object in a span without defining the common ambient category or maps. A homotopy pushout is not an instruction to add labels. The simplest space-level span already differs from its ordinary pushout.

Definitions and assumptions

Take the span*←S0→*, whereS0 consists of two points and both arrows collapse to the corresponding point.

A homotopy pushout can be modeled by the double mapping cylinder: attachS0×[0,1] to the two target points at its ends.

An exact homotopy pushout

> The ordinary pushout is a single point, while the homotopy pushout is a circle withH1≅Z.

Derivation or proof

In the ordinary quotient, each point ofS0 identifies the two target points, leaving just one point. In the double mapping cylinder, the two components ofS0 create two separate intervals joining the same pair of vertices. Their union is a circle. Its chain complex hasC1=Z²,C0=Z²; orient both edges fromleft vertex toright vertex, so the boundary matrix has two identical columns(−1,1). Its rank is1 and its kernel is generated by the edge difference. There are no2-cells, givingH1≅Z andH0≅Z. This proves the homotopy distinction directly rather than relying on a picture. It also shows why the arrows and model category matter: replacing a homotopy construction by an ordinary quotient can erase a cycle.

Worked example

Compare numerical ranks in the two constructions.

The point hasβ1=0; the homotopy pushout hasβ1=1. Neither result follows by adding the cardinalities of the input spaces, and a number like384 cannot serve as the missing gluing map.

The operation has mathematical content only after objects, morphisms and equivalences are fixed.

Scope

The example does not define a homotopy pushout of a fusion category and a Lie algebra. AnE8 representation category could be a different eligible object, but the functors fromB would still need specification.

Further development

Choose a single∞-category and explicit span, state its universal property and compute a small invariant of its colimit before assigning physical interpretations.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R048.

Y25-04-028 The E8 dimension248 follows from an explicit root count

2025-04-20T20:11:26Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which exact part of the recoveredE8 data can be established independently of a categorical synthesis?

The recovered framework uses the correct number248 forE8. An explicit root-system count explains that number and separates it from the384 cyclotomic degree. The dimension formula is a classical Lie-theoretic input, while the root enumeration is elementary and reproducible.

Definitions and assumptions

InR8, theE8 roots are the112 vectors±ei±ej,i<j, and the128 vectors(±1/2,…,±1/2) with an even number of minus signs.

For a complex semisimple Lie algebra, its root-space decomposition has one Cartan direction per rank and one dimension per root.

Exact root count and dimension

The displayed set has240 roots of squared length2, spansR8, and givesdimE8=8+240=248.

Derivation or proof

The first family hasC(8,2) choices ofcoordinate pair and four sign choices, giving28·4=112. In the second family, exactly half of the2^8 sign patterns have even parity: flip one chosen sign to biject even and odd patterns. Thus it has128 elements. Each first-family vector has two unit coordinates, so squared length2; each half-vector has eight coordinates of square1/4, again total2. The first family spansR8 because sums and differences ofei±ej recover the coordinate directions. Thus rank8 is explicit. Using the classical root-space dimension formula gives248. Verifying that this root set realizes theE8 Lie algebra rather than merely an equal-sized set uses the standard root-system construction, cited separately.

Worked example

Compare the three central integers in the recovered framework.

248 is a Lie-algebra dimension,240 is its root count and384 is the degree ofQ(ζ1680). None is the sum of the other two or an automatically inherited category rank. The verifier enumerates all240 distinct root vectors and checks their squared lengths exactly.

TheE8 number has a genuine structural derivation, while an asserted relation to384 remains a separate question.

Scope

The count alone does not construct a functor betweenE8 representations and a proposed fusion category or determine a homotopy colimit.

Further development

Give an explicit representation or branching rule and compare its character dimensions with the proposed categorical data, stating the ambient category and tensor functor.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R067.

Y25-04-025 A two-level gap produces an exact doubled-frequency observable

2025-04-23T20:06:04Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact spectral calculation underlies a doubled energy-frequency scale?

The recovered passage invokesZitterbewegung andDCE but supplies no fullDirac state or operator domain. A two-level Hamiltonian gives a rigorous limited model: off-diagonal observables oscillate at the energy difference. It clarifies the factor2 without making a device or particle-trajectory claim.

Definitions and assumptions

LetH=E Z onC^2,E>0, with$\hbar$>0. In Heisenberg evolutionA(t)=e^(itH/$\hbar$)Ae^(−itH/$\hbar$).

The two energies are+E and−E; their difference is2E.

Exact observable oscillation

X(t)=cos(2Et/ℏ)X−sin(2Et/ℏ)Y. For the initial+X state, its expectation iscos(2Et/ℏ).

Derivation or proof

In theZ eigenbasis, e^(itH/ℏ)=diag(e^(iEt/ℏ),e^(−iEt/ℏ)). ConjugatingX makes its upper off-diagonal entrye^(2iEt/ℏ) and its lower entry the conjugate. Writing these ascos±isin gives the displayedX,Y combination with the stated sign. The+X vector hasexpectations⟨X⟩=1,⟨Y⟩=0, so its evolved expectation is the cosine. More generally an off-diagonal matrix unit between energiesEa,Eb acquires phasee^(it(Ea−Eb)/ℏ). This is a direct spectral difference, not an automatic spatial motion law.

Worked example

PutE=mc² as an illustrative energy parameter.

The angular frequency is2mc²/ℏ, matching the recovered scale. An energy eigenstate instead has zeroX expectation for alltime, so the existence of the gap does not force every state to show an oscillation.

The frequency factor is derived exactly and its state dependence is explicit.

Scope

A fullDirac position operator has momentum-dependent drift, positive/negative energy sectors and domain issues. The finite model does not reconstruct those formulas or prove aDCE resonance.

Further development

Specify the actual self-adjointDirac Hamiltonian and wave packet, then compute the relevant expectation with domains and localization assumptions stated.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R068.

Y25-04-026 A Bogoliubov transformation must preserve the canonical commutator

2025-04-23T20:06:04Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact algebraic condition is necessary before a claimed mode conversion is canonical?

The recoveredDCE discussion mentionsBogoliubov coefficients without specifying them. A single bosonic mode yields a complete algebraic test and an exactly computable vacuum excitation. This is a canonical-transformation calculation, not a claim about measured emission.

Definitions and assumptions

Let[a,a†]=1 anddefineb=αa+βa† withcomplexα,β.

The input vacuum obeysa|0⟩=0. Canonical means[b,b†]=1.

Exact canonical condition and vacuum count

The transformation is canonical iff$|\alpha|^2-|\beta|^2=1$. The mean output occupation in the input vacuum is$\langle 0|b^\dagger b|0\rangle=|\beta|^2$. Choosing$\alpha=\cosh r,\beta=\sinh r$ satisfies the condition.

Derivation or proof

Expand the commutator. Theaa and$a^\dagger a^\dagger$ contributions vanish, while$|\alpha|^2[a,a^\dagger]+|\beta|^2[a^\dagger,a]=|\alpha|^2-|\beta|^2$. This proves necessity and sufficiency. Next expand$b^\dagger b$; all terms vanish in the vacuum expectation except$|\beta|^2aa^\dagger$, whose expectation is1. The hyperbolic identity$\cosh^2 r-\sinh^2 r=1$ verifies the proposed family exactly. In quadrature coordinates, the corresponding real linear map preserves the standard symplectic form; determinant1 alone would suffice in dimension2, but higher-dimensional symplecticity is stronger than a determinant condition.

Worked example

Take$r=\log 2$.

$\cosh r=(2+1/2)/2=5/4$ and$\sinh r=(2-1/2)/2=3/4$. Thus$|\alpha|^2-|\beta|^2=25/16-9/16=1$ and the vacuum occupation is9/16. Coefficients$\alpha=\beta=1$ would fail the commutator test despite looking like a balanced mixture.

The coefficients now satisfy an exact structural invariant and yield a reproducible expectation.

Scope

The value of$r$ must come from a defined dynamical equation andboundary conditions; it is not selected by the commutator alone. Infinite-mode implementability requires additional summability.

Further development

For a mode family, compute the fullBogoliubov matrices, verify both canonical identities and test$\Sigma|\beta_{jk}|^2$ for unitary implementability.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R068.

Y25-04-027 A switched oscillator gives exact parametric instability without a resonance slogan

2025-04-23T20:06:04Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What fully solved oscillator model can demonstrate growth from periodic parameter variation?

The recovered resonance condition is not a proof of an instability band. A piecewise-constant oscillator has an exact monodromy matrix, letting symplecticity and exponential growth be checked without an uncontrolled averaging approximation.

Definitions and assumptions

For$q''+\omega^2 q=0$, the state(q,p) with$p=q'$ evolves by$S_\omega(t)=[[\cos\omega t,\sin\omega t/\omega],[-\omega\sin\omega t,\cos\omega t]]$.

Alternate frequencies$\omega_1,\omega_2>0$ for quarter periods$T_1=\pi/(2\omega_1),T_2=\pi/(2\omega_2)$, repeating the cycle.

Exact Floquet multipliers

The one-cycle matrix$S_{\omega_2}(T_2)S_{\omega_1}(T_1)=\operatorname{diag}(-\omega_1/\omega_2,-\omega_2/\omega_1)$. Its determinant is1 and one multiplier has modulus>1 whenever$\omega_1\neq\omega_2$.

Derivation or proof

Solve the constant-frequency equation with its two initial conditions to obtainS_ω. At a quarter period, it becomes$[[0,1/\omega],[-\omega,0]]$. Multiply the two matrices in chronological order to get the diagonal expression. Their reciprocal multipliers prove symplectic area preservation and, if the frequencies differ, exponential growth along one state axis under repeated cycles. The other axis contracts. This supplies an exact instability example without claiming that all periodically driven oscillators or all frequencies near a symbolic2ω condition behave identically.

Worked example

Take$\omega_1=1,\omega_2=2$.

The cycle duration is$3\pi/4$ and the monodromy is$\operatorname{diag}(-1/2,-2)$. Starting at$(q,p)=(0,1)$, after$n$ cycles the state is$(0,(-2)^n)$. The exponential growth rate per unit time is$\log 2/(3\pi/4)=4\log 2/(3\pi)$.

A defined periodic system has a proven Floquet instability and an exact rate.

Scope

The coefficient jumps require an external parameter change; this is not a closed-system energy-conservation claim. It does not prove a quantum particle-production rate or validate a recoveredDCE apparatus.

Further development

Replace the jumps with a specified smoothperiodic modulation and enclose the monodromy trace to certify stable and unstable parameter bands.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R068.

Y25-04-006 Müger nondegeneracy can be checked through an explicit bicharacter

2025-04-23T20:13:27Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does nondegenerate braiding mean in a concrete pointed model?

The recovered user source names Müger’s structural work. A pointed braided example gives a finite test of transparent objects and separates a symmetric category from a nondegenerate one. The general theorem is cited, while the finite calculation is explicit.

Definitions and assumptions

LetG=F2² label invertible simple objects. Usequadratic functionq(a,b)=(−1)^(ab).

Its associated double-braiding bicharacter isB(v,w)=q(v+w)/(q(v)q(w))=(−1)^(ad+bc) forv=(a,b),w=(c,d). A label is transparent ifB(v,w)=1 for allw.

Exact triviality of the transparent subgroup

Onlyv=0 is transparent. The normalized matrixS_vw=B(v,w)/2 is a real orthogonal4×4 matrix.

Derivation or proof

Ifv=(a,b) is transparent, choosew=(0,1) to obtain(−1)^a=1, soa=0. Choosingw=(1,0) similarly givesb=0. For orthogonality, the inner product of rowsv,v′ is(1/4)Σ_wB(v−v′,w). This character sum is1 ifv=v′ and0 otherwise: a nontrivial character has equally many positive and negative values, as pairingw withw+w0 flips its sign. ThusSSᵀ=I. In the pointed realization, trivial transparent subgroup is the finite algebraic form of nondegeneracy. In a symmetric category, by contrast, all double braidings are identity and every object is transparent.

Worked example

Order the labels00,10,01,11.

S=(1/2)[[1,1,1,1],[1,1,−1,−1],[1,−1,1,−1],[1,−1,−1,1]]. Its determinant is±1 and its square isI. The four categorical dimensions are1, so global dimension is4.

A precise double-braiding law yields a complete transparency certificate rather than a claim based on rank alone.

Scope

The realization as a braided fusion category uses standard pointed-category coherence data; an arbitrary table would still need that realization. The calculation does not establish a category of rank384.

Further development

For proposed fusion data, compute the transparent subcategory and determine whether modularization is required, stating all field and semisimplicity hypotheses.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R069, R048.

Y25-04-007 The quantum double of Z2 has distinct flux and charge labels

2025-04-23T20:13:27Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact double construction can be worked out before claiming a large categorical unification?

The user source names quantum doubles. The finite abelian exampleG=Z2 is completely calculable and produces four labels rather than merely repeating the original two. It is a constructive model of the distinction between an algebra and its center.

Definitions and assumptions

For a finite abelianG, simple sectors of its untwisted double are pairs(g,χ), withg∈G andχ a character.

Fusion is(g,χ)(h,ψ)=(g+h,χψ), and the twist isθ_(g,χ)=χ(g).

Exact Z2 double data

ForG=Z2 the labels are1,e,m,em, all dimension1, with twists1,1,1,−1. The double braiding ofe andm is−1, and itsS matrix is the four-label bicharacter matrix of the preceding model.

Derivation or proof

Z2 has two group elements and two charactersχ0,χ1, withχ1(1)=−1. Pairing them gives four labels. Componentwise fusion makes every nonidentity label square to1. The twist formula gives−1 only when both flux and charge are nontrivial. The monodromy between(g,χ) and(h,ψ) isχ(h)ψ(g), so choosing a pure charge and a pure flux gives−1. Character orthogonality makes the normalizedS matrix invertible, as can also be checked by squaring its explicit integer-sign matrix. These calculations are the finite instance of the general theorem that the Drinfeld center of a suitable fusion category is nondegenerate.

Worked example

CompareVec_Z2 and its center.

The original pointed fusion category has two simple labels and global dimension2. Its double has rank4 and global dimension4=2². Consequently passing to a center changes the object count and dimension; they are not inherited unchanged from the input.

The small double gives exact fusion, twist and braiding data with a clear origin.

Scope

The untwisted abelian calculation does not cover nonabelian conjugacy-class centralizers or cocycle-twisted doubles. It also does not identify an unspecifiedSU4/Q8 framework with this category.

Further development

Compute the double of a declared nonabelian group by listing conjugacy classes and irreducible centralizer representations, then verify the sum of squared dimensions.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R070.

Y25-04-005 Fusion coefficients are constrained by a recurrence and parity

2025-04-23T20:22:00Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which exact fusion rule is compatible with the recovered SU2 sine data?

The source supplies isolated fusion numbers without a category definition. If the intended structure is theSU2 family associated with the earlier sine matrix, its integer labels have a precise parity and truncation rule. A path matrix gives an elementary constructive realization of the fusion ring.

Definitions and assumptions

Letk≥2 and use labelsj=0,…,k. LetN1 be the adjacency matrix of the path withk+1 vertices.

DefineN0=I andN_(j+1)=N1N_j−N_(j−1), withN_(k+1)=0. EquivalentlyN_j=U_j(N1/2) for Chebyshev polynomialsU_j.

An exact basic fusion identity

> $N1^2=N0+N2$, so$N_11\hat{}1=0$ in this convention. For $k\geq4$, $N2^2=N0+N2+N4$. Allowed output labels have the same parity asi+j.

Derivation or proof

The first identity is the definition ofN2. Since$N2=N1^2-I$, square it to obtain$N2^2=N1^4-2N1^2+I$. The recurrence gives$N4=N1^4-3N1^2+I$, so$N0+N2+N4=N1^4-2N1^2+I$ as required. The sine eigenvectors diagonalize the path matrix with eigenvalues$2\cos(r\pi/(k+2))$; the recurrence therefore has the matching eigenvalue ratio$\sin((j+1)r\theta)/\sin(r\theta)$. Each multiplication byN1 changes the parity of a path vertex, so the polynomialN_j only connects vertices of parity shifted byj. This enforces$i+j+\ell$ even on a fusion coefficient. Thus the source's odd outputs in2×2 cannot be imported into this specific convention.

Worked example

Takek=4, giving a five-vertex path.

The fundamental product is$1\times1=0+2$ and$2\times2=0+2+4$. Direct integer matrix multiplication verifies both identities and all entries are nonnegative. The matrixN4 reverses the five vertices, so$N4^2=I$.

The ring structure supplies exact integer constraints and a reproducible replacement for unexplained coefficients.

Scope

A different category or a different label convention could have other fusion numbers. The historicalX37 L-value and unspecified categorical data remain unverified. A fusion ring alone does not determine associators or braiding.

Further development

Construct the actualF andR symbols or cite a precise realization, then verify pentagon and hexagon identities in addition to the integer ring.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R048.

Y25-04-008 A Frobenius algebra and a Lagrangian dimension test require actual multiplication

2025-04-23T20:22:00Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which concrete algebra object demonstrates the meaning of a Lagrangian construction?

The recovered passage names Cardy and Lagrangian constructions without multiplication maps. In theZ2 double, the two-label object1⊕e gives a finite algebra whose dimension test can be checked. This is a classical model, not a reconstruction of missing384-object data.

Definitions and assumptions

LetL={1,e} be the pure-charge subgroup in the four-label double. DefineA=1⊕e with multiplication induced bye²=1 and the ordinary group law.

The categorical dimension isdimA=2; global category dimension is4. The subgroup has trivial twist and trivial mutual braiding.

An explicit isotropic algebra and its dimension

> The multiplication is associative and braided-commutative onA. It has a nondegenerate Frobenius pairingτ(ab), withτ selecting the identity coefficient. Moreover(dimA)²=4, the Lagrangian dimension condition in this model.

Derivation or proof

Associativity is exactly associativity of the two-element group. All braidings among pure charges are trivial, so multiplication is commutative in the braided sense. In the basis(1,e),τ(1)=1,τ(e)=0 and the pairing isdiag(1,1), proving nondegeneracy. OverC the algebra splits into the idempotents(1±e)/2, establishing separability in this elementary realization. The trivial twist ensures isotropy. Its dimension squared equals that of the ambient nondegenerate category; together with the specified connected commutative separable structure, this is the classical Lagrangian condition. The general relation between such algebras and Drinfeld centers is an external theorem explicitly referenced, not proved by a dimension equality alone.

Worked example

Calculate the idempotents and the complementary flux interaction.

p_+=(1+e)/2 andp_-=(1−e)/2 satisfyp_±²=p_±,p_+p_-=0 andp_++p_-=1. The fluxm braids nontrivially withe, so addingm to this algebra would require a new compatibility analysis rather than merely increasing its dimension.

The example supplies the maps and compatibility that a dimension-only claim omits. Its ambient double is Witt-trivial by the classical center relation.

Scope

Not every object of dimension√D is a Lagrangian algebra. Multiplication, unit, separability, commutativity and twist conditions are essential. No historicalL-value is validated here.

Further development

For an actual category, list candidate summands and solve the associativity and braided-commutativity equations before applying a Lagrangian or Cardy theorem.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R071, R072.

Y25-04-023 Chromatic height is visible in a formal-group p-series

2025-04-27T04:14:14Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact algebraic example gives meaning to a prime-dependent chromatic label?

The recovered source namesTMF and transchromatic structures but supplies no spectrum or formal group. The multiplicative formal group gives a complete algebraic calculation ofheight1 at every prime. It distinguishes a real invariant from assigning a prime to a framework label.

Definitions and assumptions

A one-dimensional formal group lawF(x,y) is an associative commutative power series withidentity0. Its[p]_F(x) addsx to itselfp times.

Over a field of characteristicp, finite heighth means the first nonzero term of[p]_F has degreep^h; if[p]_F=0 the height is infinite.

Exact additive and multiplicative heights

ForF_m(x,y)=x+y+xy, [p](x)=(1+x)^p−1, reducing tox^p in characteristicp, so its height is1. The additive lawF_a=x+y has[p]=px=0 in characteristicp and infinite height.

Derivation or proof

The identity1+F_m(x,y)=(1+x)(1+y) proves associativity, commutativity and the p-fold formula by ordinary multiplication. The inverse is−x/(1+x), a formal power series with zero constant term. In characteristicp, the binomial theorem removes all intermediate coefficients becausep divides binomial(p,j) for0<j<p, leavingx^p. The additive calculation is immediate. Thus two simple formal groups over the same characteristic have different heights. An invertible coordinate changeg(x)=ux+O(x²),u≠0, conjugates the p-series and preserves the lowest nonzero exponent, so this distinction is intrinsic to the formal-group isomorphism class.

Worked example

Atp=3, compare the integral p-series before and after reduction.

ForF_m, [3](x)=3x+3x²+x³, which becomesx³ overF3. ForF_a it is3x, which becomes0. The same prime therefore does not determine the height without a formal group.

The chromatic terminology now has a defined algebraic invariant and an exact computation.

Scope

TMF involves a much deeper sheaf of spectra over elliptic moduli; this formal calculation does not build it or establish a new field theory. No scattering or black-hole prediction follows.

Further development

Specify an elliptic curve and prime, compute its formal p-series or Hasse invariant and distinguish ordinary fromsupersingular reduction before proposing a chromatic comparison.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R073.

May 2025

Dated fragments recovered; complete transcript unavailable. 24 developed treatments.

Infra-rational uncertainty, hybrid totient-weighted variation, periodic-grid spectra, multigrid and entanglement information connect arithmetic to concrete calculations. The new developments respect the historical request for a mathematically grounded framework.

ID · date	Concept and revised question	Result and mathematical core
Y25-05-001 2025-05-04 · T	The Mexican-hat Fourier pair needs a declared normalization Reconstruction: Which Gaussian derivative gives a consistent Mexican-hat Fourier pair, what moments vanish, and what admissibility integral follows?	A consistent pair, two vanishing moments and finite admissibility $F[\psi\sigma](\omega)=\omega^2 e^{\wedge}(-\sigma^2\omega^2/2)$, $\int\psi\sigma=\int x\psi\sigma=0$, and $\int_R\lvert F[\psi\sigma](\omega)\rvert^2/\lvert\omega\rvert d\omega=\sigma^{\wedge}(-4)$. The source coefficient $1/(2\pi\sigma^3)$ would instead give an extra factor $1/\sqrt{(2\pi)}$ with this transform convention.
Y25-05-002 2025-05-04 · T	Curl flux is a boundary quantity before it is a topological invariant Reconstruction: What does a curl-flux integral actually determine, and which extra hypotheses would make it independent of the chosen surface?	Stokes independence with fixed boundary, and a counterexample to topological invariance $\Phi_F(S)=\oint_C F\cdot dr$. Consequently two admissible surfaces with the same oriented boundary give the same integral for the same field. The value can nevertheless vary continuously under changes of F while the surface topology stays fixed.
Y25-05-003 2025-05-09 · A	A primitive fourth-power counterexample defeats the proposed exclusion Reconstruction: Which claimed Euler-type exclusions already fail by exact arithmetic, and what does the correct number $\varphi(1680)=384$ actually establish?	An exact primitive counterexample and a local parity restriction $95800^4+217519^4+414560^4=422481^4$ and the four integers have gcd one. Also $\varphi(1680)=384$. The latter number does not imply the absence of the former solution. Any primitive solution has exactly one odd number among a,b,c and has d odd.
Y25-05-004 2025-05-14 · A	The periodic prime-grid Laplacian has an exact Fourier spectrum Reconstruction: What are the sign, kernel and spectral gap of the stated periodic p^3-grid operator, and which asymptotics genuinely follow from its formula?	Complete eigenvalues and the gap asymptotic $L\chi_k=-4\Sigma_j \sin^2(\pi k_j/p)\chi_k$, so A is positive semidefinite with a one-dimensional constant kernel. Its smallest positive eigenvalue is $\gamma_p=4\sin^2(\pi/p)\sim 4\pi^2/p^2$. The contractive heat evolution is $e^{\wedge}(-tA)=e^{\wedge}(tL)$, not $e^{\wedge}(-tL)$.
Y25-05-005 2025-05-14 · A	Galerkin projection explains what a coarse correction can guarantee Reconstruction: Which transfer-map and Krylov hypotheses make a coarse correction an energy projection rather than a claimed automatic convergence accelerator?	An exact energy projection and its limitation A_c is positive definite, $\Pi^2=\Pi$ and Π is A-self-adjoint. The coarse correction $e\mapsto(I-\Pi)e$ satisfies $\lVert(I-\Pi)e\rVert_A=\min_z\lVert e-Pz\rVert_A\le\lVert e\rVert_A$. This alone need not contract errors in the A-orthogonal complement of range(P).

ID · date	Concept and revised question	Result and mathematical core
Y25-05-006 2025-05-14 · A	Functional-calculus and Newton error bounds need explicit domains Reconstruction: How does scalar approximation transfer to a symmetric matrix, and what local assumptions actually prove quadratic Newton convergence?	Spectral error transfer and a local quadratic recurrence $\|\|f(A)-r(A)\|\|_2=\max_{\lambda\in\text{spectrum}(A)}\|f(\lambda)-r(\lambda)\|\le\sup_{[a,b]}\|f-r\|$. Newton's update $x^+=x-F(x)/F'(x)$ satisfies $\|x^+-\alpha\|\le[M/(2m)]\|x-\alpha\|^2$ under the stated local bounds.
Y25-05-007 2025-05-18 · A	Tripartite information vanishes for every pure three-party state Reconstruction: Can tripartite information distinguish pure GHZ and W states, and what can be concluded without the missing family ψ(g)?	Pure-state cancellation and unequal entanglement spectra Every finite-dimensional pure three-party state has $I_3=0$. GHZ has all nontrivial one-party and two-party entropies equal to one; W has them equal to $h_2(1/3)\approx0.9182958341$. The two states therefore share I_3 despite different reduced spectra.
Y25-05-008 2025-05-18 · U	A filtration changes homology when simplices enter, not when labels sound singular Reconstruction: What finite Hodge-Laplacian certificate distinguishes a weighted perturbation from a genuine filtration event?	The harmonic edge disappears exactly when the face is added For the unfilled triangle, Δ_1 has eigenvalues 0,3,3 with harmonic vector (1,1,1). After adding the face, $\Delta_1=3I$ and its kernel is zero. This matches $\beta_1=1$ and $\beta_1=0$ respectively.
Y25-05-011 2025-05-21 · A	Derived tensor products record information an ordinary product erases Reconstruction: What concrete higher algebraic information justifies using a derived construction before speaking about derived motivic stacks?	An explicit resolution with a higher class in every degree For every $n\ge0$, Tor_n^A(k,k)≅k. Thus the derived tensor cannot be replaced by its degree-zero ordinary tensor without losing nonzero information in every positive degree.
Y25-05-009 2025-05-21 · A	Tropicalization retains valuation constraints and forgets unit data Reconstruction: What can the tropicalization of x+y+1=0 prove, and what information is lost before any proposed enriched decoder is added?	A tropical line and a strict information-loss example Every such point satisfies that min(u,w,0) is attained at least twice. Its possible finite valuation pattern lies on the three rays (u,0) with $u\ge0$, (0,w) with $w\ge0$, or (u,u) with $u\le0$. Valuations alone do not test an individual proposed lift.
Y25-05-010 2025-05-21 · A	A p-adic integral can be evaluated exactly by valuation shells Reconstruction: Which measure and convergence region make the proposed p-adic integral meaningful, and what exact rational function results for |x|_p^s?	An exact local zeta integral For $\text{Re}(s)>-1$, $Z_p(s)=\int_{Z_p}\|x\|_p^s\,d\mu=(1-p^{-1})/(1-p^{-s-1})$. The integral is absolutely convergent exactly in this half-plane; the displayed rational-in-p^{-s} expression also supplies a meromorphic continuation.
Y25-05-012 2025-05-22 · A	Energy ratios require a shared boundary and a complete denominator Reconstruction: Which arithmetic conclusions follow from the stated output and EROI numbers, and what information is missing before any performance improvement can be claimed?	Ratio audit, monotonicity and a duty-factor bound An output of11923kWh with ratio28.6 requires an invested denominator11923/28.6≈416.8881119kWh. The output alone does not determine EROI. If output is zero when off and at most P_max when on, then0≤CF≤duty fraction, with equality only at rated output almost everywhere during on-time.

ID · date	Concept and revised question	Result and mathematical core
Y25-05-013 2025-05-22 · A	The stated gravity-wave dispersion has perpendicular phase and group directions Reconstruction: For the stated linear dispersion relation, what frequency and group velocity are actually implied, and does that establish recoverable power?	An exact group-velocity formula and orthogonality $c_g=(Nm^2,-Nkm)/(k^2+m^2)^{(3/2)}$, and $(k,m)\cdot c_g=0$. Hence $c_{ph}\cdot c_g=0$. Also$0<\omega\le N$, with equality only when $m=0$.
Y25-05-014 2025-05-23 · U	A tolerance relation cannot serve as field equality Reconstruction: Can contextual numerical closeness be equality in a field, and what happens if transitivity is imposed after the fact?	Nontransitivity and collapse of the naive quotient The tolerance relation is not transitive. Its transitive closure on Q is the universal relation, so quotienting by that closure identifies every rational number, including zero and one.
Y25-05-015 2025-05-23 · U	Uncertainty propagation needs covariance and does not create a field Reconstruction: When is quadrature addition exact, what term is missing from the proposed multiplication uncertainty, and why does x−x expose lost provenance?	Exact variance identities and a missing second-order term $Var(X+Y)=\varepsilon^2+\eta^2+2Cov(X,Y)$. Under independence, $Var(XY)=v^2\eta^2+w^2\varepsilon^2+\varepsilon^2\eta^2$. The source's relative-error product formula retains only the first two terms and is undefined as written at zero means.
Y25-05-016 2025-05-23 · A	Contextual arithmetic needs restriction maps before sheaf claims Reconstruction: What simple context model really satisfies a gluing theorem, and which data are missing from the proposed fibred or sheaf semantics?	A concrete sheaf of contextual values Every compatible family $f_i\in F(U_i)$ glues to a unique $f\in F(U)$. Restrictions preserve pointwise addition and multiplication. Thus this model has exact contextual algebra without replacing equality by numerical tolerance.
Y25-05-017 2025-05-24 · U	Totient-weighted variation is a hybrid optimization problem Reconstruction: When does the weight φ(n)/σ(n) change Euler-Lagrange dynamics, and what junction law is required at a prescribed discrete mode switch?	Interior equation and weighted momentum continuity On each constant-mode interval, $d/dt(L_v)-L_q=0$. At a fixed switch t_0, a stationary path must satisfy $w_-L_v(q,\dot{q}_-,t_0)=w_+L_v(q,\dot{q}_+,t_0)$. Differentiating φ at an integer is neither needed nor defined by this model.
Y25-05-018 2025-05-24 · U	The recovered gcd coupling defines a valid graph, not an automatic rigidity law Reconstruction: What can be proved from the actual R_{ij} formula and coprime set, and how should an integer arithmetic state evolve in a hybrid model?	A graph energy identity and an integer-state obstruction L is positive semidefinite and $x^TLx=(1/2)\Sigma_{ij}R_{ij}(x_i-x_j)^2$. Its kernel is constant on each connected component of the positive-weight graph. Any continuous $n:I\to\mathbb{Z}$ on an interval is constant, so a nonzero formula for ndot requires a real relaxation or a jump model.
Y25-05-019 2025-05-24 · U	Finite spectral zeta and heat transforms do not imply a universal density law Reconstruction: Which exact identities connect a finite positive spectrum, its heat trace and spectral zeta, and what makes the claimed small-eigenvalue density nonnormalizable without a cutoff?	A finite Mellin identity and an integrability obstruction For $Re(s)>0$, $\Gamma(s)\zeta_A(s)=\int_0^\infty t^{(s-1)}H(t)dt$. The finite spectral zeta itself is entire in s. A nonzero density proportional to$\lambda^{(-1.528)}$ cannot be normalized on(0,b] without a cutoff or another modification.

ID · date	Concept and revised question	Result and mathematical core
Y25-05-021 2025-05-24 · U	A PAC-Bayes guarantee needs a sampling law, prior and bounded loss Reconstruction: What actual assumptions produce a generalization bound, and why does a reported prime-gap $R^2=.847$ not certify one?	A fixed-parameter PAC-Bayes-Hoeffding bound With probability at least$1-\delta$, simultaneously for every posterior Q, $L(Q) \le \hat{L}(Q)+[KL(Q\|\|P)+\log(1/\delta)]/\lambda+\lambda/(8m)$. This is a theorem under the declared iid bounded-loss model, not a consequence of R^2 or a spectral fit.
Y25-05-020 2025-05-24 · U	Multiplicative convolution has a precise divisor algebra Reconstruction: Which multiplicative convolution is intended, when do transforms turn it into a product, and what exact arithmetic identity can an implementation certify?	Convolution, inversion and a transform product Dirichlet convolution is associative and commutative with identity ε. The Möbius function satisfies $\mu*1=\varepsilon$. Whenever D_f, D_g converge absolutely, $D_{(f*g)}(s)=D_f(s)D_g(s)$. Multiplicative inputs give a multiplicative convolution.
Y25-05-022 2025-05-30 · A	A thermal pulse bound reverses the source area-to-volume factor Reconstruction: What is the dimensionally correct thermal-buffering flux bound, and what fraction of a first-order response occurs during a0.2ms pulse when the time constant is0.8s?	Energy-limited flux and a certified short-pulse response Ignoring heat losses, $q'' \le \rho c_p \Delta T (V/A)/t_p$ is necessary to keep the stored-energy rise within $C\Delta T$. The source's factor(A/V) has the wrong length dimension. With $\tau=.8$s and $t_p=.0002$s, the step fraction is$1-e^{(-1/4000)} \approx .000249968753$.
Y25-05-024 2025-05-31 · A	A bonded-layer stress calculation needs a compatibility model Reconstruction: Under what compatibility and force-balance assumptions can interface thermal stress be calculated, and what remains unverifiable in the stated numerical claims?	Compatible strain as a unique convex minimizer The common strain is$\varepsilon^*=\Delta T(\Sigma A_i E_i \alpha_i)/(\Sigma A_i E_i)$, with$\Sigma A_i \sigma_i=0$. For equal A and E, stresses are$\pm E(\alpha_2-\alpha_1)\Delta T/2$. Changing boundary constraints changes the result.
Y25-05-023 2025-05-31 · U	Effective conductivity depends on geometry and interface resistance Reconstruction: How do Fourier flux, series and parallel composites differ, and why do a filler fraction and two conductivities not determine one effective conductivity?	Two exact effective-conductivity formulas and a conservation identity Parallel geometry gives$\kappa_{parallel}=(1-f)\kappa_1+f\kappa_2$; series geometry gives$\kappa_{series}=[(1-f)/\kappa_1+f/\kappa_2]^{(-1)}$. For total thickness$\ell$, contact resistance changes the series value to$\kappa_{eff}=\ell/[\ell_1/\kappa_1+\ell_2/\kappa_2+R_c]$. Heat conservation is$\rho c_p T_t=-\operatorname{div} q+s$.

Y25-05-001 The Mexican-hat Fourier pair needs a declared normalization

2025-05-04 · historical evidence T · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which Gaussian derivative gives a consistent Mexican-hat Fourier pair, what moments vanish, and what admissibility integral follows?

The external May4 file contains a Mexican-hat formula and Fourier transform that do not match under the usual unnormalized forward transform. This repair fixes the convention before proving cancellation and admissibility. The file context is retained as T; no historical user prompt or completed wavelet theorem is invented.

Definitions and assumptions

Use $F[f](\omega)=\int_R f(x)e^{(-i\omega x)}dx$. Let $G\sigma(x)=(\sqrt{(2\pi)}\sigma)^{(-1)}e^{(-x^2/(2\sigma^2))}$, $\sigma>0$, so $F[G\sigma]=e^{(-\sigma^2\omega^2/2)}$.

Define $\psi\sigma=-G\sigma''=(\sqrt{(2\pi)}\sigma^3)^{(-1)}(1-x^2/\sigma^2)e^{(-x^2/(2\sigma^2))}$. These functions and their derivatives decay rapidly.

A consistent pair, two vanishing moments and finite admissibility

$F[\psi\sigma](\omega)=\omega^2e^{(-\sigma^2\omega^2/2)}$, $\int\psi\sigma=\int x\psi\sigma=0$, and $\int_R|F[\psi\sigma](\omega)|^2/|\omega|d\omega=\sigma^{(-4)}$. The source coefficient $1/(2\pi\sigma^3)$ would instead give an extra factor $1/\sqrt{(2\pi)}$ with this transform convention.

Derivation or proof

Integration by parts gives $F[f'']=-\omega^2F[f]$ when boundary terms vanish; apply it to $G\sigma$. Integrating $-G\sigma''$ directly gives zero. Integrating $-xG\sigma''$ by parts also gives zero because the remaining integral is a boundary difference of $G\sigma$. Thus the Fourier transform vanishes to second order at frequency zero. For admissibility, symmetry reduces the integral to $2\int_0\infty\omega^3e^{(-\sigma^2\omega^2)}d\omega$. With $u=\sigma^2\omega^2$ this is $\sigma^{(-4)}\int_0\infty ue^{(-u)}du=\sigma^{(-4)}$. All integrations converge absolutely. Finally the ratio of the source's spatial coefficient to the corrected one is $1/\sqrt{(2\pi)}$; linearity of the transform forces the same ratio on the frequency side. A different Fourier convention could change constants, but it must be stated consistently on both sides.

Worked example

Take $\sigma=1$ and compare the corrected wavelet with the retrieved spatial coefficient.

The corrected value at zero is $1/\sqrt{(2\pi)}$, and the retrieved value is $1/(2\pi)$. The corrected frequency response at $\omega=1$ is $e^{(-1/2)}$, whereas the retrieved spatial normalization transforms to $e^{(-1/2)}/\sqrt{(2\pi)}$. The admissibility integral is exactly one. The verifier evaluates Gaussian moments independently to check the two cancellations and checks the analytic normalization ratio.

A normalization error is repaired without promoting the file's completeness or novel-invariant claims into proved results.

Scope

This proves a continuous Fourier calculation, not completeness or stable inversion of an arbitrary discrete translation/dilation grid. The file's Bui-Laugesen attribution and numerical reconstruction constant need their exact theorem and sampling hypotheses.

Further development

For a specified discrete grid, estimate lower and upper frame bounds rather than assuming that continuous admissibility implies stable discrete reconstruction. The cited completeness results are separate theorems with their own spaces and scale conditions.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R074, R075.

Y25-05-002 Curl flux is a boundary quantity before it is a topological invariant

2025-05-04 · historical evidence T · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What does a curl-flux integral actually determine, and which extra hypotheses would make it independent of the chosen surface?

The external topic label combines a wavelet with curl-flux integrals and proposed topological invariants. A precise first step is Stokes' theorem. It explains what is surface independent, what changes with the vector field, and why an arbitrary real-valued integral is not automatically a topological invariant.

Definitions and assumptions

Let S be a compact oriented piecewise smooth surface in R^3 with oriented boundary C. Let F be C^1 on a neighborhood of S.

Define $\Phi_F(S)=\int_S(\operatorname{curl} F)\cdot n\, dA$. Boundary orientation is chosen by the usual right-hand convention. For a closed surface assume the additional smoothness needed for the divergence theorem.

Stokes independence with fixed boundary, and a counterexample to topological invariance

$\Phi_F(S)=\oint_C F\cdot dr$. Consequently two admissible surfaces with the same oriented boundary give the same integral for the same field. The value can nevertheless vary continuously under changes of F while the surface topology stays fixed.

Derivation or proof

Stokes' theorem is the explicit classical input identifying the surface integral with circulation. Applying it to S_1 and S_2 with the same C gives equal values, provided F is defined smoothly on both. For a closed surface bounding a region with a smooth field, divergence of curl is zero, so its flux is zero; this also follows from Stokes because the boundary is empty. To show that no nontrivial topological invariant follows merely from the integral, use $F_a(x,y,z)=a(-y,x,0)/2$. Its curl is (0,0,a). On the unit disk in the xy-plane the flux is aπ, which ranges over all real values as a changes. The disk's topology has not changed. The equality with circulation still holds, so the failure is in the proposed interpretation rather than in Stokes' theorem.

Worked example

Use the upward-oriented disk of radius two and $F=(-y,x,0)/2$.

The curl is (0,0,1), so the flux is the area 4π. Parameterize its boundary by $r(t)=(2\cos t,2\sin t,0)$. Then $F(r(t))=(-\sin t,\cos t,0)$ and $F\cdot r'=2$, giving $\int_0^{(2\pi)}2dt=4\pi$ independently. Reversing orientation changes the sign. Scaling F by three gives 12π without changing the disk or boundary shape.

The calculation gives a complete boundary certificate and identifies exactly why the adjective topological needs further structure.

Scope

Singular fields, punctured domains and nonbounding cycles can carry additional information; the smooth-neighborhood assumptions cannot be dropped silently. No new invariant of knots or wavelets is proved here.

Further development

Replace the arbitrary vector field by a closed differential form with a specified integral cohomology class. Then periods on homologous cycles agree, and integer normalization can give genuinely topological data. The closedness and integral-class assumptions are the required mathematical work.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R076.

Y25-05-003 A primitive fourth-power counterexample defeats the proposed exclusion

2025-05-09T19:22:59Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which claimed Euler-type exclusions already fail by exact arithmetic, and what does the correct number φ(1680)=384 actually establish?

The recovered assistant framework combined a correct totient computation with an exclusion of primitive fourth-power solutions. Exact integer arithmetic separates them immediately. A categorical dimension or a numerical spectral pattern cannot override an explicit counterexample to the underlying Diophantine statement.

Definitions and assumptions

A primitive positive solution of $a^4+b^4+c^4=d^4$ has $\gcd(a,b,c,d)=1$. Euler's classical exclusion of three fourth powers is false.

Euler's totient is $\varphi(n)=n\prod_{p\mid n}(1-1/p)$. The cyclotomic field $Q(\zeta_n)$ has degree $\varphi(n)$, a classical algebraic theorem.

An exact primitive counterexample and a local parity restriction

$95800^4+217519^4+414560^4=422481^4$ and the four integers have gcd one. Also $\varphi(1680)=384$. The latter number does not imply the absence of the former solution. Any primitive solution has exactly one odd number among a,b,c and has d odd.

Derivation or proof

The fourth-power equality and gcd are finite integer certificates, checked without rounding by the verifier. For the parity statement, every fourth power is zero or one modulo sixteen. Thus the number of odd summands on the left, an integer from zero through three, must equal zero or one modulo sixteen according to the parity of d. It is therefore zero or one exactly. If all three left entries were even, d would also be even, violating primitivity. Exactly one must be odd and d must be odd. For the totient, factor $1680=2^4\cdot3\cdot5\cdot7$ and multiply $1680(1/2)(2/3)(4/5)(6/7)=384$. The angle $\pi/840$ is indeed $2\pi/1680$, compatible with a primitive 1680th root. These correct cyclotomic facts do not create a logical implication about the quartic equation; an additional theorem would have to state and prove that bridge, and the explicit counterexample prevents the proposed exclusion.

Worked example

Check the displayed solution and its congruence pattern, then distinguish a fifth-power example.

The quartic solution has even, odd, even left entries and an odd right entry, exactly as required. Independently $27^5+84^5+110^5+133^5=144^5$ disproves the corresponding four-term exclusion for fifth powers. That example has four summands, so it neither proves nor disproves a separate assertion about exactly three fifth powers.

Term count, exponent and primitivity are tracked explicitly, preventing a correct nearby fact from supporting an incorrect conjecture.

Scope

The finite certificates do not classify all solutions or settle every higher-exponent equation. The historical proposed absence of three-term solutions for all $k\geq5$ is not established by its spectral or motivic labels.

Further development

Construct congruence filters and certify each surviving exact candidate, then study a specified parametrized quartic family. Any claimed complete exclusion must survive these known solutions before additional machinery is introduced.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R077, R078.

Y25-05-004 The periodic prime-grid Laplacian has an exact Fourier spectrum

2025-05-14T16:27:05Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What are the sign, kernel and spectral gap of the stated periodic p^3-grid operator, and which asymptotics genuinely follow from its formula?

The exact periodic operator is one of the grounded parts of the May14 framework. Its sign must be fixed before writing a heat semigroup or a spectral gap. Primality of p is unnecessary for the diagonalization: the same calculation works for every integer $p\geq3$.

Definitions and assumptions

On $G=(Z/pZ)^3$ let $(Lu)(x)=\Sigma_{j=1}^{3}[u(x+e_j)+u(x-e_j)-2u(x)]$. Define the positive operator $A=-L$.

For $k\in G$ use normalized Fourier characters $\chi_k(x)=p^{-3/2}\exp(2\pi ik\cdot x/p)$. Orthogonality is with the finite counting inner product.

Complete eigenvalues and the gap asymptotic

$L\chi_k=-4\Sigma_j \sin^2(\pi k_j/p)\chi_k$, so A is positive semidefinite with a one-dimensional constant kernel. Its smallest positive eigenvalue is $\gamma_p=4\sin^2(\pi/p)\sim 4\pi^2/p^2$. The contractive heat evolution is $e^{-tA}=e^{tL}$, not e^{-tL}.

Derivation or proof

A shift by $\pm e_j$ multiplies χ_k by $e^{\pm 2\pi ik_j/p}$. Adding the two shifts and subtracting two gives $2\cos(2\pi k_j/p)-2=-4\sin^2(\pi k_j/p)$. Summing proves the eigenvalue formula. The character inner products are products of geometric sums, zero unless the indices agree, so these p^3 characters form a full orthonormal basis. Every eigenvalue of A is nonnegative, and it vanishes only when all $k_j=0$ modulo p. The least positive sine square occurs at $k_j=\pm 1$ in one coordinate with the other coordinates zero. The limit $\sin z/z\to 1$ proves the asymptotic. In this eigenbasis the norm of e^{-tA} is one, while its restriction to mean-zero functions has norm $e^{-t\gamma_p}$. The opposite sign e^{-tL} amplifies nonconstant modes and is not a dissipative heat operator.

Worked example

Take $p=4$ as a transparent exact check, even though this grid size is not prime.

Each one-coordinate contribution to A is 0,2,4,2. Thus the three-dimensional eigenvalues are sums of three such entries, the kernel is constants and the gap is two. The eigenvalue at $k=(1,1,1)$ is six. The verifier constructs the 64-site matrix and Fourier-mode residuals independently, and checks the exact row sum and quadratic-form identity.

The actual operator yields a p^{-2} gap. A different logarithmic gap law needs a different, fully specified graph and cannot be inferred from arithmetic terminology.

Scope

This proves no GUE spacing law, Ext dimension, motivic correspondence or learned prime-gap performance. The source’s later correction explicitly required separating those claims from this elementary Fourier theorem.

Further development

For variable positive edge weights, retain the incidence-factorization $A=B^{T}WB$ and compare its Rayleigh quotient with the constant-weight operator. Uniform weight bounds give rigorous gap enclosures without guessing a new density exponent.

Source receipt: months/05/root_followups.json record 0. Mathematical references: R079, R080.

Y25-05-005 Galerkin projection explains what a coarse correction can guarantee

2025-05-14T16:28:34Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which transfer-map and Krylov hypotheses make a coarse correction an energy projection rather than a claimed automatic convergence accelerator?

A named interpolation or Krylov space does not by itself give a convergence theorem. For a positive definite operator the Galerkin construction does give a precise best-approximation property. This is the finite mathematical core shared by coarse-grid correction and projection onto a Krylov subspace.

Definitions and assumptions

Let $A \in R^{n\times n}$ be symmetric positive definite and $P \in R^{n\times r}$ have full column rank. Define $A_c = P^T A P$ and $\Pi = P\, A_c^{-1} P^T A$.

Use the energy inner product $\langle x,y\rangle_A = x^T A y$. A Krylov space $K_m(A,v)=\mathrm{span}(v,Av,\dots,A^{m-1}v)$ is a particular candidate subspace; choose independent columns before forming P.

An exact energy projection and its limitation

A_c is positive definite, $\Pi^2=\Pi$ and Π is A-self-adjoint. The coarse correction $e \mapsto (I-\Pi)e$ satisfies $\|(I-\Pi)e\|_A = \min_z \|e-Pz\|_A \le \|e\|_A$. This alone need not contract errors in the A-orthogonal complement of range(P).

Derivation or proof

For $z\neq 0$, $z^T A_c z = (Pz)^T A (Pz) > 0$ because P is injective. Multiplying the formula for Π by itself cancels $P^T A P = A_c$ and proves idempotence. Also $A\Pi = AP\, A_c^{-1} P^T A$ is symmetric, which is precisely energy self-adjointness. The normal equation $P^T A(e-Pz)=0$ has solution $z = A_c^{-1} P^T A e$. Splitting $e-Pz$ into its projected and orthogonal pieces proves the Pythagorean identity and the minimization claim. If $P^T A e = 0$ then $\Pi e = 0$, so the same error is unchanged. A multigrid rate therefore requires an additional smoothing/approximation argument; transfer dimensions and a subspace name cannot supply it.

Worked example

Choose $A=\mathrm{diag}(1,4)$, $P=(1,1)^T$ and $e=(1,0)^T$.

Then $A_c=5$, $\Pi=[[1/5,4/5],[1/5,4/5]]$, and $(I-\Pi)e=(4/5,-1/5)$. Its squared energy is $16/25+4/25=4/5$, less than the original one. The vector $(-4,1)$ is A-orthogonal to P and remains unchanged. The verifier checks idempotence, A-self-adjointness and both examples exactly with fractions.

The useful guarantee is an energy best approximation with a clearly identified untouched complement.

Scope

A periodic Laplacian has a constant kernel, so apply this positive-definite argument on its mean-zero subspace or fix a gauge first. Arbitrary transfer maps, singular coarse matrices or nonsymmetric operators require modified hypotheses.

Further development

Combine this projection with a specified smoother S and prove ||(I−Π)S||_A≤q<1. A bound uniform over grid size would then be a genuine multilevel convergence theorem, stronger than a finite residual decrease.

Source receipt: months/05/root_followups.json record 0. Mathematical references: R081.

Y25-05-006 Functional-calculus and Newton error bounds need explicit domains

2025-05-14T16:29:47Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: How does scalar approximation transfer to a symmetric matrix, and what local assumptions actually prove quadratic Newton convergence?

Rational approximation and nonlinear iteration were named in the adaptive framework, but the retrieved exponential approximation bound is truncated. Two modest exact statements replace that missing certificate: a spectral maximum bound and a local Newton remainder estimate. Neither relies on an unrecorded Zolotarev constant.

Definitions and assumptions

Let A be real symmetric with spectrum in [a,b]. Let r be a rational function with no pole on that interval and f continuous there.

For Newton’s method let F be C^2 near a root α, assume $|F'|\geq m>0$ and $|F''|\leq M$ on an interval containing α and the current iterate, and assume the next iterates remain in that interval.

Spectral error transfer and a local quadratic recurrence

$||f(A)-r(A)||_2=\max_{\lambda\in\text{spectrum}(A)}|f(\lambda)-r(\lambda)|\leq\sup_{[a,b]}|f-r|$. Newton’s update $x^+=x-F(x)/F'(x)$ satisfies $|x^+-\alpha|\leq[M/(2m)]|x-\alpha|^2$ under the stated local bounds.

Derivation or proof

An orthogonal diagonalization $A=U\operatorname{diag}(\lambda_i)U^T$ makes f(A)−r(A) orthogonally similar to the diagonal matrix of scalar errors; its Euclidean operator norm is the largest absolute diagonal entry. The pole exclusion makes r(A) well defined. For Newton, Taylor-expand $F(\alpha)=F(x)+F'(x)(\alpha-x)+F''(\xi)(\alpha-x)^2/2$ for some intermediate ξ. Rearranging and using F(α)=0 gives $x^+-\alpha=F''(\xi)(x-\alpha)^2/[2F'(x)]$, up to an irrelevant sign, proving the absolute bound. Remaining inside the interval is a real hypothesis: a local error estimate is not global convergence from every initial point.

Worked example

Use A=diag(1,2), f(λ)=1/λ and r(λ)=3/2−λ/2. Separately use F(x)=x²−2 and x₀=3/2.

The polynomial r is exact at both eigenvalues, so r(A)=A^(−1), although it is not exact throughout [1,2]. Newton gives x₁=17/12 and x₂=577/408. The exact residuals are x₁²−2=1/144 and x₂²−2=1/166464. These are also Pell-type rational certificates, computed without floating-point cancellation.

A finite spectrum can make a simple approximation exact, while iteration convergence requires a separate nonlinear argument.

Scope

No unrecovered uniform rational-approximation rate, prime-grid optimization complexity or arbitrary-start Newton guarantee is asserted. A residual-to-error bound additionally needs a derivative or inverse-conditioning estimate.

Further development

For a prescribed positive interval [a,b], optimize a pole-free rational approximation and certify its scalar uniform error. That certificate then transfers immediately to every symmetric matrix with spectrum in the interval, independently of dimension.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R082, R083.

Y25-05-007 Tripartite information vanishes for every pure three-party state

2025-05-18T02:28:11Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Can tripartite information distinguish pure GHZ and W states, and what can be concluded without the missing family ψ(g)?

The source’s GHZ endpoint is consistent, but a slightly negative tripartite-information caveat for a pure W endpoint is not. For a pure state of exactly three parties, the quantity vanishes identically. This is a structural obstruction to using it as the proposed transition marker, independent of numerical precision.

Definitions and assumptions

For a density matrix ρ define S(ρ)=−tr(ρlog₂ρ), with 0log0=0. For a pure state on A⊗B⊗C define I₃=S_A+S_B+S_C−S_AB−S_AC−S_BC+S_ABC.

GHZ=(|000⟩+|111⟩)/√2 and W=(|001⟩+|010⟩+|100⟩)/√3. The retrieved parameterized state ψ(g) was not supplied.

Pure-state cancellation and unequal entanglement spectra

Every finite-dimensional pure three-party state has $I_3=0$. GHZ has all nontrivial one-party and two-party entropies equal to one; W has them equal to $h_2(1/3)\approx0.9182958341$. The two states therefore share I_3 despite different reduced spectra.

Derivation or proof

The Schmidt decomposition across A versus BC gives identical nonzero eigenvalues for ρ_A and ρ_BC, so S_A=S_BC. The other two bipartitions give S_B=S_AC and S_C=S_AB. Purity gives S_ABC=0. Substituting cancels every term. For GHZ, tracing two qubits leaves diag(1/2,1/2). For W it leaves diag(2/3,1/3); permutation symmetry makes this true for each party. The two-party entropies follow from the same Schmidt argument, avoiding a larger eigenproblem. This proves exact zero rather than an approximately small or slightly negative value.

Worked example

As a declared new comparison family take $\psi(\theta)=\cos\theta|000\rangle+\sin\theta|111\rangle$ for $0\le\theta\le\pi/2$.

Every one-party entropy is $h_2(\sin^2\theta)$, every two-party entropy is the same, and $I_3=0$ throughout. At $\theta=0$ it is a product; at $\theta=\pi/4$ it is GHZ. The entropy changes continuously, while I_3 contains no information about this change. The verifier checks the reduced diagonal eigenvalues for GHZ/W and the exact entropy cancellations numerically.

A vanishing summary statistic does not imply absent entanglement and cannot locate the claimed g=1 or1.67 transitions without a specified state and observable.

Scope

Mixed three-party states can have nonzero I_3, and a subsystem of a larger pure system is generally mixed. The theorem is for a globally pure state of exactly A,B,C. No historical ψ(g) is reconstructed from endpoint labels.

Further development

Specify a full normalized state family and compare reduced-spectrum invariants, genuine multipartite entanglement measures and filtration changes separately. Any proposed transition must be defined before numerical thresholds are tested.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R084.

Y25-05-008 A filtration changes homology when simplices enter, not when labels sound singular

2025-05-18T02:28:11Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What finite Hodge-Laplacian certificate distinguishes a weighted perturbation from a genuine filtration event?

The historical state, weights and filtration were not recovered, so its claimed transition parameters cannot be recomputed. A completely declared triangle filtration nevertheless demonstrates the exact distinction needed: positive weights change eigenvalues, whereas adding a filling simplex can change homology. This is a new mathematical example, not an inferred $\psi(g)$.

Definitions and assumptions

Orient triangle edges 01,12,20 and use boundary $B_1=[[-1,0,1],[1,-1,0],[0,1,-1]]$. Without a face, B_2 is absent; with a face its boundary column is $(1,1,1)^T$.

Use the unweighted edge Hodge Laplacian $\Delta_1=B_1^TB_1+B_2B_2^T$. A filtration is an increasing sequence of simplicial complexes, including all faces of every entered simplex.

The harmonic edge disappears exactly when the face is added

> For the unfilled triangle, Δ_1 has eigenvalues 0,3,3 with harmonic vector (1,1,1). After adding the face, $\Delta_1=3I$ and its kernel is zero. This matches $\beta_1=1$ and $\beta_1=0$ respectively.

Derivation or proof

The boundary sum $B_1(1,1,1)^T$ is zero, so $B_1B_2=0$ and the complex is valid. Direct multiplication gives $B_1^TB_1=3I-J$, where J has every entry one. J has eigenvalue three on constants and zero on their orthogonal complement, producing the unfilled spectrum. Adding $B_2B_2^T=J$ yields 3I. More generally $x^T\Delta_1x=||B_1x||^2+||B_2^Tx||^2$; therefore its kernel consists of cycles orthogonal to boundaries and represents homology in this finite inner-product model. Replacing the inner products by positive definite weights preserves the dimension of that harmonic space; positive weights do not by themselves create a new simplex or a new homology class.

Worked example

Let all edges enter at threshold zero and the face enter at threshold two.

The degree-one persistence interval begins at zero and ends at two. Before two there is one independent edge cycle; afterwards it is the boundary of the new face. A weighted eigenvalue crossing zero caused only by a weight becoming singular would instead leave the positive-definite setting and requires a different explanation. The verifier checks both spectra through exact matrix identities and ranks.

The certificate separates a genuine combinatorial event from an unproved wall-crossing interpretation of an unspecified entropy curve.

Scope

A weighted hypergraph is not automatically a simplicial complex. Its boundary maps, orientations, filtration and inner products must be supplied before this Hodge argument applies. No historical transition value is validated.

Further development

Given the actual state family, define each simplex’s entry time from an explicit entropy statistic, prove face monotonicity, and compute persistent homology with exact or interval-certified thresholds. Then compare those events with the quantum observables rather than identifying them by terminology.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R085.

Y25-05-011 Derived tensor products record information an ordinary product erases

2025-05-21T01:13:59Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What concrete higher algebraic information justifies using a derived construction before speaking about derived motivic stacks?

The recovered manuscript description lists derived objects but supplies no defining equations. A small algebraic example shows exactly why deriving a functor can matter. It is deliberately an authored comparison, not a claimed construction of the absent stacks, sheaves or path integrals.

Definitions and assumptions

Let k be a field, $A=k[\varepsilon]/(\varepsilon^2)$, and regard $k=A/(\varepsilon)$ as an A-module. Ordinary tensor gives $k\otimes_A k\cong k$.

$\mathrm{Tor}_n^A(k,k)$ is the nth homology of a free resolution of one copy of k after tensoring with the other. The derived tensor $k\otimes_A^L k$ retains this whole homology sequence.

An explicit resolution with a higher class in every degree

For every $n\geq 0$, $\mathrm{Tor}_n^A(k,k)\cong k$. Thus the derived tensor cannot be replaced by its degree-zero ordinary tensor without losing nonzero information in every positive degree.

Derivation or proof

Use the augmented complex $\ldots\to A\to A\to A\to k\to 0$, with every positive-degree differential multiplication by ε and the final map the quotient. Consecutive differentials compose to multiplication by $\varepsilon^2=0$. In A, an element is $a+b\varepsilon$; multiplying by ε gives $a\varepsilon$, so the kernel and image of multiplication by ε are both the one-dimensional ideal εA. The kernel of $A\to k$ is also εA. Hence the complex is an exact free resolution. Tensoring with k sends ε to zero, so all resulting differentials vanish and each chain group is k. Its homology is therefore k in every nonnegative degree. This is a complete calculation rather than an invocation of derived language alone.

Worked example

Represent multiplication by ε on the basis $(1,\varepsilon)$ as $M=[[0,0],[1,0]]$.

Then $M^2=0$, $\mathrm{rank}(M)=1$ and $\ker(M)=\mathrm{span}(\varepsilon)=\mathrm{image}(M)$. Reducing coefficients modulo ε turns M into zero on the tensor complex. The verifier checks these finite linear identities and the first eight homology dimensions. The written periodic argument proves the all-degree statement.

Higher Tor is the information that ordinary tensor discards; a derived replacement must be justified by such an actual failure of exactness.

Scope

This is a module calculation over dual numbers, not a motivic cohomology computation, an E_∞ structure or a syntomic integration measure. Passing from complexes to those theories requires their own objects, maps and comparison theorems.

Further development

Build a small verified chain-complex library that certifies $d^2=0$, exactness and chain-homotopy equivalence. Only then add a geometric input functor and ask which derived invariants it computes, with explicit test objects and coefficient rings.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R086, R087.

Y25-05-009 Tropicalization retains valuation constraints and forgets unit data

2025-05-21T03:29:31Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What can the tropicalization of $x+y+1=0$ prove, and what information is lost before any proposed enriched decoder is added?

A valuation map has a precise and useful loss of information. Its tropical balancing condition follows from the nonarchimedean triangle inequality, but it does not determine the original coordinates or a reconstruction advantage. The source’s additional filtered object needs an independently defined filtration.

Definitions and assumptions

Let K be a nonarchimedean valued field with additive valuation v, so $v(xy)=v(x)+v(y)$ and $v(x+y)\geq\min(v(x),v(y))$. Set $v(0)=\infty$.

For nonzero x,y on $X=\{x+y+1=0\}$, define $\operatorname{trop}(x,y)=(u,w)=(v(x),v(y))$. Use the minimum convention.

A tropical line and a strict information-loss example

Every such point satisfies that $\min(u,w,0)$ is attained at least twice. Its possible finite valuation pattern lies on the three rays $(u,0)$ with $u\geq0$, $(0,w)$ with $w\geq0$, or (u,u) with $u\leq0$. Valuations alone do not test an individual proposed lift.

Derivation or proof

If one of three nonzero summands has strictly smaller valuation than the other two, the sum of the other two has larger valuation and cannot cancel it. Therefore $x+y+1=0$ forces a repeated minimum. Enumerating which two entries attain it gives the three rays. Conversely, interpreting the rays as the complete tropical variety requires a lifting theorem over a suitable valued algebraic closure; only the elementary necessary condition is used here. The loss of unit data is concrete: over Q_3, $(x,y)=(1,-2)$ lies on X and has valuations $(0,0)$, while $(1,1)$ has the same valuations but gives $x+y+1=3\neq0$. Thus one valuation pair supports both an actual solution and a nonsolution depending on the leading units.

Worked example

Compare Q_3 points $(1,-2)$, $(3,-4)$ and the proposed pair $(1,1)$.

The first has valuation $(0,0)$, the second $(1,0)$; each has a repeated minimum and exactly satisfies the equation. The third has the same tropical coordinates as the first but fails the original equation.

Multiplication also illustrates retained structure: $v_3(3\cdot(-4))=1+0=1$. These exact integer valuations are checked without a p-adic numerical library.

Tropicalization gives an exact coarse constraint, and the counterexample identifies the additional residue information that a reconstruction method would need.

Scope

The source's notation $\mathrm{Trop}_+(X)=(X^{\mathrm{trop}},F\bullet)$ does not specify $F\bullet$ or prove a decoder. No detection threshold, resource gain or motivic comparison follows from adding a superscripted filtration symbol.

Further development

Retain an initial-form equation over the residue field together with the tropical point. Prove which residue solutions lift under a stated nonsingularity condition; that gives a rigorous first refinement before attempting any higher filtered or motivic enhancement.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R088, R089.

Y25-05-010 A p-adic integral can be evaluated exactly by valuation shells

2025-05-21T03:29:31Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which measure and convergence region make the proposed p-adic integral meaningful, and what exact rational function results for $|x|_p^s$?

The source names an integral over p-adic integer points, but an integrand, measure and convergence claim still have to be specified. The monomial case supplies a complete local-zeta calculation: its rationality comes from a geometric series rather than from a speculative decoding interpretation.

Definitions and assumptions

Fix a prime p and normalize Haar measure $\mu(\mathbb{Z}_p)=1$. Then $\mu(p^n\mathbb{Z}_p)=p^{-n}$, and $|x|_p=p^{-v_p(x)}$ for $x\neq 0$.

For complex s define $|x|_p^s=\exp(s\log|x|_p)$. The singleton zero has measure zero and may be assigned any value when the integral is finite.

An exact local zeta integral

For $\mathrm{Re}(s)>-1$, $Z_p(s)=\int_{\mathbb{Z}_p}|x|_p^s\,d\mu=(1-p^{-1})/(1-p^{-s-1})$. The integral is absolutely convergent exactly in this half-plane; the displayed rational-in-p^{-s} expression also supplies a meromorphic continuation.

Derivation or proof

Partition nonzero Z_p into shells $A_n=p^nZ_p\setminus p^{n+1}Z_p$, $n\geq 0$. Their measures are $(1-p^{-1})p^{-n}$, and the integrand is constant p^{-ns} on each shell. Thus $Z_p(s)=(1-p^{-1})\Sigma_{n\geq 0}p^{-n(s+1)}$. Absolute convergence is equivalent to $|p^{-s-1}|<1$, or $\mathrm{Re}(s)>-1$, and summation gives the formula. If $\mathrm{Re}(s)\leq -1$, the positive integral of the absolute value diverges by the same series. Analytic continuation is a statement about the resulting function; it does not make the original divergent integral convergent outside its initial half-plane.

Worked example

Choose $p=3$ and $s=2$, then compare shell integration with a finite residue approximation.

The exact answer is $(2/3)/(1-1/27)=9/13$. The first N shells sum to $(9/13)(1-27^{-N})$, with omitted tail $(9/13)27^{-N}$. In a residue calculation modulo 3^N, each congruence class has measure 3^{-N}; valuations below N reproduce those shell counts exactly. The unresolved zero class contributes a bounded tail rather than an arbitrary replacement value.

The calculation includes its normalization, domain and an exact truncation error, making it suitable for a reproducible arithmetic implementation.

Scope

A general variety $X(Z_p)$, singular polynomial or oscillatory integrand needs additional geometry and convergence hypotheses. This one-variable proof is not a proof of the full rationality theory or of a physical detector.

Further development

For a polynomial f, count solutions $f(x)\equiv 0$ modulo p^N and relate those counts to the measure of valuation sublevel sets. A proved resolution or cell decomposition can then replace the elementary shell structure used here.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R090, R091.

Y25-05-012 Energy ratios require a shared boundary and a complete denominator

2025-05-22T03:48:58Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which arithmetic conclusions follow from the stated output and EROI numbers, and what information is missing before any performance improvement can be claimed?

The recovered energy report contains mutually inconsistent ratios and system boundaries. The mathematical repair is an accounting model, with energies measured over the same interval and with the same conversion convention. No harvesting performance or hardware design is inferred.

Definitions and assumptions

For a stipulated system boundary let $E_{out} \geq 0$ be useful returned energy and $E_{in} > 0$ invested energy in the same units and time horizon. Define $R = E_{out}/E_{in}$.

For a rated power $P_{max} > 0$ over duration T, define capacity factor $CF = E_{out}/(P_{max}T)$, assuming output never exceeds that rating. A duty fraction measures time on, which is a different quantity.

Ratio audit, monotonicity and a duty-factor bound

An output of11923kWh with ratio28.6 requires an invested denominator$11923/28.6 \approx 416.8881119$kWh. The output alone does not determine EROI. If output is zero when off and at most P_{max} when on, then$0 \leq CF \leq$duty fraction, with equality only at rated output almost everywhere during on-time.

Derivation or proof

The implied denominator follows by rearranging $R = E_{out}/E_{in}$; no engineering hypothesis is hidden in that arithmetic. For a time-varying output $P(t)$, energy is $\int_0^T P(t)dt$. If the on-set has measure dT and $0 \leq P \leq P_{max}$ there, then the integral is at most dTP_{max}, proving $CF \leq d$. The ratio R increases with E_{out} and decreases with E_{in}, so independent positive interval inputs yield the sharp rectangle enclosure $[E_{out,min}/E_{in,max}, E_{out,max}/E_{in,min}]$. Adding an omitted positive investment strictly decreases the ratio. Converting a thermal output to an electrical equivalent with factor $\eta \leq 1$ multiplies that output contribution by η; it cannot be left numerically unchanged while the denominator is changed.

Worked example

Use a constructed two-hour schedule with rated power ten units, on for one hour at half rating, and off otherwise.

Duty fraction is$1/2$, energy is five units and $CF = 5/(10 \cdot 2) = 1/4$. Calling a60% duty assumption a60% capacity factor would therefore require an additional rated-output assumption. For the source’s reported output, the denominator implied by35:1 would be$11923/35 \approx 340.6571$kWh, different from the denominator implied by28.6:1; neither is recovered as an actual measured input.

The audit identifies what can be solved algebraically and what remains an unsupported model or measurement claim.

Scope

Ratios0.28 thermal,5.67 electrical and above35 cannot be compared without explicit boundary and conversion definitions. This mathematical treatment supplies no lifecycle estimate, cost forecast or physical validation.

Further development

Represent a complete accounting boundary as a directed flow graph with tagged units and time horizons. Prove conservation at every node and generate ratio intervals only after excluding double-counted internal transfers.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R092.

Y25-05-013 The stated gravity-wave dispersion has perpendicular phase and group directions

2025-05-22T04:04:37Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: For the stated linear dispersion relation, what frequency and group velocity are actually implied, and does that establish recoverable power?

A dispersion relation can support an exact propagation calculation without supporting the reported energy gains. The nonrotating constant-stratification model has a particularly clear geometry: its frequency is homogeneous of degree zero in wavevector, forcing phase and group directions to be perpendicular.

Definitions and assumptions

Take a stipulated $N>0$, horizontal wavenumber $k>0$ and vertical wavenumber $m\in R$. Use $\omega(k,m)=Nk/\sqrt{(k^2+m^2)}$, the positive branch of $\omega^2=N^2k^2/(k^2+m^2)$.

Define group velocity $c_g=\nabla_{(k,m)}\omega$ and phase-velocity vector $c_{ph}=\omega(k,m)(k,m)/(k^2+m^2)$. This is the stated ideal linear nonrotating dispersion model.

An exact group-velocity formula and orthogonality

$c_g=(Nm^2,-Nkm)/(k^2+m^2)^{(3/2)}$, and $(k,m)\cdot c_g=0$. Hence $c_{ph}\cdot c_g=0$. Also$0<\omega\le N$, with equality only when $m=0$.

Derivation or proof

Differentiate $Nk(k^2+m^2)^{(-1/2)}$ in k to obtain $N[(k^2+m^2)-k^2]/(k^2+m^2)^{(3/2)}=Nm^2/(k^2+m^2)^{(3/2)}$. Differentiating in m gives $-Nkm/(k^2+m^2)^{(3/2)}$. Taking the wavevector dot product cancels the two numerator terms. Equivalently $\omega(sk,sm)=\omega(k,m)$ for $s>0$, so Euler's homogeneous-function identity forces that dot product to vanish. The frequency bound follows from $k\le\sqrt{(k^2+m^2)}$. At $m=0$ the group velocity vanishes in this idealized expression; this limiting degeneracy must not be mistaken for a general transport prediction.

Worked example

Use nondimensional values $N=5,k=3,m=4$.

Then $\omega=3$, $c_g=(16/25,-12/25)$, and $c_{ph}=(9/25,12/25)$. Their dot product is$144/625-144/625=0$. The wavevector dot product is$48/25-48/25=0$. If both wavenumbers are doubled to$(6,8)$, the frequency stays three while both group-velocity components are halved. This follows from differentiating a degree-zero homogeneous function and provides a second scaling check. The verifier checks the displayed fractions exactly; the all-parameter derivative proof supplies the general statement.

The geometry follows from a specified equation and can be certified independently of the harvesting proposal.

Scope

Wave amplitude, energy density, dissipation, forcing and coupling are absent from a dispersion equation. It cannot determine watts per patch, total output or EROI. Rotation, variable stratification and boundaries require a different model.

Further development

Supply a linear wave-energy density and boundary coupling operator, then prove an input-output energy inequality. Such an inequality would be the proper mathematical bridge from propagation to an upper bound on extractable energy.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R093, R094.

Y25-05-014 A tolerance relation cannot serve as field equality

2025-05-23T23:36:13Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Can contextual numerical closeness be equality in a field, and what happens if transitivity is imposed after the fact?

The user’s infra-rational proposal explicitly allows non-transitive equality. That can be a useful tolerance relation, but it is incompatible with algebraic equality. The obstruction is elementary and remains even before multiplication, contexts or uncertainty propagation are introduced.

Definitions and assumptions

Fix $\delta>0$ rational and define $x\sim y$ on Q by $|x-y|<\delta$. A field’s equality relation must be reflexive, symmetric and transitive and must respect both operations.

The transitive closure of a relation declares two points equivalent whenever a finite chain of related points connects them.

Nontransitivity and collapse of the naive quotient

> The tolerance relation is not transitive. Its transitive closure on Q is the universal relation, so quotienting by that closure identifies every rational number, including zero and one.

Derivation or proof

Take $x=0, y=3\delta/4, z=3\delta/2$. Both neighboring differences are$3\delta/4<\delta$, but $|x-z|=3\delta/2>\delta$. Thus the relation cannot be substituted for equality in a field. For any two rational a,b, choose an integer $N>|b-a|/\delta$. The rational chain $a+j(b-a)/N$, $j=0,\ldots,N$, has every step smaller thanδ. Transitive closure therefore equates a and b. A quotient with$0=1$ is not a field under the usual nontrivial-field axiom. Merely closing the relation under transitivity does not repair the intended numerical information; it erases it.

Worked example

Set δ=1 and use the explicit triple0,3/4,3/2.

The two true closeness tests and one false endpoint test are exact rational comparisons. A chain from zero to ten with twenty equal steps also shows why iterated tolerances can travel arbitrarily far. The verifier checks these examples and constructs chains between several rational endpoints.

Tolerance should be a separately named relation whose inferential rules are controlled; it cannot inherit equality substitution automatically.

Scope

This is a counterexample to a fixed positive tolerance. A context-dependent threshold may produce other relations, but a proposed quotient still needs an actual equivalence relation and congruence proof. At zero threshold the source’s strict inequality is not even reflexive.

Further development

Use exact equality for values and store error bounds as additional data, or choose a declared rounding map q and define x≈y iff q(x)=q(y). The latter is an equivalence, but its arithmetic compatibility must still be tested rather than assumed.

Source receipt: months/05/root_details.json record 0. Mathematical references: R095.

Y25-05-015 Uncertainty propagation needs covariance and does not create a field

2025-05-23T23:36:13Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: When is quadrature addition exact, what term is missing from the proposed multiplication uncertainty, and why does x−x expose lost provenance?

The proposed uncertainties resemble standard deviations of independent random errors. They are not universal interval laws, and ignoring dependence destroys algebraic cancellations. An exact independent-product variance calculation identifies a further term omitted by first-order propagation.

Definitions and assumptions

Let X,Y have means v,w and variances ε^2,η^2. For independent variables assume finite second moments. For dependent variables write Cov(X,Y).

A stored pair (mean,standard deviation) does not record the joint law or whether two appearances refer to the same random variable.

Exact variance identities and a missing second-order term

Var(X+Y)=$\varepsilon^2+\eta^2+2$Cov(X,Y). Under independence, Var(XY)=$v^2\eta^2+w^2\varepsilon^2+\varepsilon^2\eta^2$. The source’s relative-error product formula retains only the first two terms and is undefined as written at zero means.

Derivation or proof

Expand the centered square of X+Y and take expectations to obtain the covariance identity. For independent X,Y, $E[X^2Y^2]=E[X^2]E[Y^2]=(v^2+\varepsilon^2)(w^2+\eta^2)$, while $E[XY]=vw$. Subtracting v^2w^2 gives the exact product variance. The formula $|vw|\sqrt{[(\varepsilon/v)^2+(\eta/w)^2]}$ has squared value $w^2\varepsilon^2+v^2\eta^2$ when $v,w\neq0$, omitting the nonnegative product term. For cancellation, using $Y=-X$ gives covariance $-\varepsilon^2$ and exact zero variance. Treating those two appearances as independent instead reports$2\varepsilon^2$, demonstrating the lost dependence information.

Worked example

Take independent Rademacher errors Z,W with values±1 equally likely and $X=1+Z,Y=1+W$.

The product takes values0,0,0,4 with equal probability. Its mean is one, second moment four and variance three. The first-order formula gives variance two. For the same X, $X-X$ is exactly zero, whereas blind quadrature reports standard deviation$\sqrt{2}$. The verifier enumerates all four outcomes exactly.

The repaired formulas explain the approximation and preserve the difference between independent measurements and repeated use of one uncertain quantity.

Scope

Means and variances alone do not define a field. The random variable$1+Z$ is nonzero as an algebra element yet vanishes on an event of probability1/2, so it has no pointwise multiplicative inverse. Context labels cannot remove that obstruction without changing the algebra.

Further development

Use an explicit affine-error model $X=v+\Sigma a_jZ_j$ with shared noise symbols to preserve correlations under addition, and retain higher-order symbols or rigorously bounded remainders under multiplication. Then prove enclosure or moment bounds for each operation.

Source receipt: months/05/root_details.json record 0. Mathematical references: R096.

Y25-05-016 Contextual arithmetic needs restriction maps before sheaf claims

2025-05-23T23:52:46Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What simple context model really satisfies a gluing theorem, and which data are missing from the proposed fibred or sheaf semantics?

The source’s categorical expansion is truncated after its fibred-category claim. A context poset, compatible operations and a collapse symbol do not alone define a sheaf or fibration. A finite observation-domain model supplies a genuine gluing statement while making the missing morphisms visible.

Definitions and assumptions

Fix a finite set Ω. Contexts are subsets U⊆Ω, ordered by inclusion. Define F(U)=Q^U, the rational-valued functions on U.

For V⊆U, restriction res^U_V sends f to f|_V. A cover of U is a family U_i with union U. Compatibility means equal restrictions on every overlap U_i∩U_j.

A concrete sheaf of contextual values

Every compatible family f_i∈F(U_i) glues to a unique f∈F(U). Restrictions preserve pointwise addition and multiplication. Thus this model has exact contextual algebra without replacing equality by numerical tolerance.

Derivation or proof

Define f(ω)=f_i(ω) using any i withω∈U_i. Compatibility on overlaps makes the value independent of the choice. Each restriction equals the prescribed f_i. If another function has those same restrictions, its value at anyω is fixed by one covering set, proving uniqueness. Restriction identities res^U_U=id and res^V_W∘res^U_V=res^U_W follow directly from evaluating functions. Pointwise arithmetic commutes with restriction because both sides evaluate to the same rational sum or product. These are actual maps and equations; they are the data absent from a bare statement that contexts compose by a semilattice join.

Worked example

Take Ω={a,b,c}, U_1={a,b}, U_2={b,c}, with $f_1(a)=1, f_1(b)=2$ and $f_2(b)=2, f_2(c)=3$.

The unique glued function is(1,2,3). Changing $f_2(b)$ to5 destroys compatibility, and no global function can have both restrictions. Pointwise functions form a product ring Q^3, not a field: the nonzero functions(1,0,0) and(0,1,0) multiply to zero. The verifier checks gluing and this zero-divisor certificate exactly.

A genuine categorical context model can preserve local arithmetic while still lacking a global field structure.

Scope

This authored finite sheaf is not a reconstruction of the source’s unknown category IRat, its morphisms or its claimed topos. Units, provenance and error intervals would need additional well-typed structures; incompatible values cannot simply be glued by notation.

Further development

Build a sheaf of sets of admissible values or a stack of uncertainty models with explicit descent data. Prove how provenance and shared random variables restrict and glue before asserting equivalence with the original numerical proposal.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R097.

Y25-05-017 Totient-weighted variation is a hybrid optimization problem

2025-05-24T00:26:43Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: When does the weight φ(n)/σ(n) change Euler-Lagrange dynamics, and what junction law is required at a prescribed discrete mode switch?

The user explicitly restricted this proposal to grounded hybrid control and optimization. For a prescribed integer mode, the action weight is a positive arithmetic scalar. It cancels from the interior Euler-Lagrange equation when constant, but it matters at a switch and in the globally optimal allocation of velocity.

Definitions and assumptions

Let n(t) be a prescribed piecewise constant positive integer with finitely many fixed switch times. Put w(t)=φ(n(t))/σ(n(t))>0.

For a continuous piecewise C^2 path q, use S[q]=∫w(t)L(q,qdot,t)dt with fixed endpoints. Variations are continuous at switches; no jump penalty is included.

Interior equation and weighted momentum continuity

On each constant-mode interval, d/dt(L_v)−L_q=0. At a fixed switch t_0, a stationary path must satisfy w_-L_v(q,qdot_-,t_0)=w_+L_v(q,qdot_+,t_0). Differentiating φ at an integer is neither needed nor defined by this model.

Derivation or proof

Take the first variation ∫w(L_qδq+L_vδqdot)dt separately on every interval. Integration by parts gives the interior factor w[L_q−d(L_v)/dt], which vanishes because w is constant and positive. Boundary terms at a shared fixed switch combine as (w_-L_v^-−w_+L_v^+)δq(t_0). Arbitrary continuous variation there gives the momentum condition. For L=v^2/2, the entire optimization also follows from Cauchy-Schwarz: (q(T)−q(0))²=(∫sqrt(w)v·w^(−1/2)dt)²≤(∫wv^2dt)(∫w^(−1)dt). Equality requires wv constant, so S_min=(Δq)²/[2∫w^(−1)dt]. This is a global certificate for the stated convex problem.

Worked example

Set T=1, q(0)=0,q(1)=1; choose n=1 on the first half and n=2 on the second.

The weights are1 and1/3. Since ∫w^(−1)dt=2, the minimum action is1/4. The optimal velocities are1/2 and3/2, with weighted momentum1/2 on both halves. A constant velocity instead gives action1/3. The verifier checks the endpoint, junction and objective values with exact rational arithmetic.

The arithmetic weight influences the optimization through declared mode choices and switch conditions, without introducing speculative physical laws.

Scope

If switch times are free, additional transversality conditions arise. If n depends on the path or jumps by feedback, guards, resets and admissible controls must be specified. The source’s extra forcing term ΣR_ijλ_j requires its own constraint or forcing derivation.

Further development

Optimize over a finite mode set with switching penalties. Derive a Bellman recursion for the discrete choices and a convex continuous subproblem within each schedule, then prove a lower bound against which a proposed algorithm can be certified.

Source receipt: months/05/root_details.json record 0. Mathematical references: R098, R099.

Y25-05-018 The recovered gcd coupling defines a valid graph, not an automatic rigidity law

2025-05-24T00:26:43Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What can be proved from the actual R_ij formula and coprime set, and how should an integer arithmetic state evolve in a hybrid model?

The retrieved coupling is explicit enough for a graph theorem. Its nonnegative symmetric entries yield a dissipative graph Laplacian regardless of whether the coupling matrix itself is positive definite. A separate elementary obstruction shows why a continuously differentiable integer state cannot satisfy an arbitrary nonzero velocity law.

Definitions and assumptions

On a finite set of distinct positive labels define R_ij=φ(gcd(i,j))/max(i,j) when gcd(i,j)>1, and zero otherwise. Define L=diag(R1)−R.

For m≥1, the coprime residue set is E_m={k mod m:gcd(k,m)=1}, of sizeφ(m). The unrestricted positive-integer set of coprime k is infinite.

The Möbius function μ is zero on integers with squared prime factors and equals(−1)^r on a product of r distinct primes.

A graph energy identity and an integer-state obstruction

L is positive semidefinite and $x^{T}Lx=(1/2)\Sigma_{ij}R_{ij}(x_i-x_j)^2$. Its kernel is constant on each connected component of the positive-weight graph. Any continuous n:I→Z on an interval is constant, so a nonzero formula for ndot requires a real relaxation or a jump model.

Derivation or proof

The formula makes R symmetric and nonnegative. Expanding the square and using symmetry proves the energy identity; it vanishes precisely when coordinates agree along every positive edge, hence along each connected component. For the integer-state claim, continuity sends a connected interval to a connected subset of Z, and every connected subset of Z is a singleton. Thus an equation ndot=αΣ||qdot_i||²μ(d_i) cannot simultaneously describe an integer-valued differentiable n unless its right-hand side vanishes. The Möbius sign can be used as a specified discrete update, but its arithmetic definition alone does not produce a dynamical theorem. A valid arithmetic identity is Σ_{d|m}φ(d)=m, whose Dirichlet Möbius inversion is φ=μ*id; it should not be confused with time evolution.

Worked example

Use labels2,3,6 and ignore self-loops in the graph Laplacian.

The only off-diagonal weights are R_26=1/6 and R_36=1/3. The Laplacian has characteristic polynomial λ(λ²−λ+1/6), so its positive eigenvalues are(3±√3)/6. Coprime labels2 and3 have no direct edge but are linked through6. Modulo6 the coprime residue set is{1,5}, of sizeφ(6)=2.

The exact graph certificate supplies a meaningful network model while preserving the distinction between connectivity, controllability and the proposed integer update.

Scope

A graph gap is not a complete controllability criterion for an input pair(A,B). Mechanical rigidity and energy-barrier claims need their own definitions. The user’s explicit exclusion of speculative physics is retained.

Further development

Choose input vertices and form a declared linear control system on the mean-zero graph space. Compute its Kalman rank or controllability Gramian exactly, then study how the arithmetic label set changes that certificate.

Source receipt: months/05/root_details.json record 0. Mathematical references: R100, R101.

Y25-05-019 Finite spectral zeta and heat transforms do not imply a universal density law

2025-05-24T04:27:42Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which exact identities connect a finite positive spectrum, its heat trace and spectral zeta, and what makes the claimed small-eigenvalue density nonnormalizable without a cutoff?

The report combines a learned spectral fit with a zeta-like series and transform inversion. These are different levels of claim. A finite positive spectrum gives exact analytic formulas; an asymptotic law requires a specified operator family, limiting measure and normalization.

Definitions and assumptions

Let0<λ₁,...,λ_r be positive eigenvalues counted with multiplicity. Define heat trace H(t)=Σe^(−tλ_i), t>0, and ζ_A(s)=Σλ_i^(−s). Exclude zero eigenvalues explicitly.

A proposed density ρ(λ)=Cλ^(−α) on(0,b] is integrable near zero only ifα<1. A finite lower cutoff changes this condition.

A finite Mellin identity and an integrability obstruction

For Re(s)>0, Γ(s)ζ_A(s)=∫₀∞t^(s−1)H(t)dt. The finite spectral zeta itself is entire in s. A nonzero density proportional toλ^(−1.528) cannot be normalized on(0,b] without a cutoff or another modification.

Derivation or proof

For one positive λ, substitute u=tλ in the gamma integral to obtain ∫₀∞t^(s−1)e^(−tλ)dt=λ^(−s)Γ(s). A finite sum can be interchanged with the integral, proving the identity. Each λ^(−s)=exp(−slogλ) is entire, so the finite sum is entire; the integral representation has its own restricted convergence region. For the density claim, ∫_ερ(λ)dλ=C[b^(1−α)−ε^(1−α)]/(1−α) whenα≠1. Ifα>1 this diverges asε→0. Therefore the fitted exponent does not define a probability density all the way to zero. Neither a finite fit nor this transform identity establishes a limiting law for an undefined arithmetic graph.

Worked example

Use the positive path-three Laplacian spectrum{1,3}.

H(t)=e^(−t)+e^(−3t), ζ_A(1)=4/3 and ζ_A(2)=10/9. Direct integration give∫H=4/3 and∫tH=10/9 becauseΓ(1)=Γ(2)=1. For a cutoff density on[a,b], the normalization constant is(1−α)/(b^(1−α)−a^(1−α)), which is positive forα>1 as well; the dependence on a cannot be omitted.

The exact transform statements remain useful after the universal spectral and number-theoretic interpretations are removed.

Scope

An infinite spectrum requires convergence estimates before summing, integrating or analytically continuing. No λ₁ asymptotic, Riemann-type zero theorem or prime-gap prediction is obtained from a finite spectral definition.

Further development

Specify a sequence of graph operators and empirical spectral measures. Prove tightness and identify any limiting density on a stated scaling window, keeping the smallest eigenvalue and zero-mode mass separate.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R102, R103.

Y25-05-021 A PAC-Bayes guarantee needs a sampling law, prior and bounded loss

2025-05-24T04:27:42Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What actual assumptions produce a generalization bound, and why does a reported prime-gap $R^2=.847$ not certify one?

A fitted score and a theorem about unseen samples are different mathematical objects. The source's spectral PAC-Bayes claim does not provide its prior, posterior, sample law or loss bound. A standard finite-hypothesis theorem can be derived transparently and shows exactly where those assumptions enter.

Definitions and assumptions

Let H be finite, P a strictly positive prior fixed independently of an iid sample of size m, and $\ell(h,z)\in[0,1]$. Define population loss $L(h)$, empirical loss $\hat{L}(h)$, and $KL(Q||P)=\Sigma Q(h)\log[Q(h)/P(h)]$.

Fix $\lambda>0$ and$\delta\in(0,1)$ before seeing the sample. Write $L(Q),\hat{L}(Q)$ for posterior averages.

A fixed-parameter PAC-Bayes-Hoeffding bound

> With probability at least$1-\delta$, simultaneously for every posterior Q, $L(Q)\le\hat{L}(Q)+[KL(Q||P)+\log(1/\delta)]/\lambda+\lambda/(8m)$. This is a theorem under the declared iid bounded-loss model, not a consequence of R^2 or a spectral fit.

Derivation or proof

Hoeffding's bounded-variable lemma gives $E_{sample}\exp(\lambda[L(h)-\hat{L}(h)])\le\exp(\lambda^2/(8m))$ for each h. Average over the fixed prior and apply Markov: with probability at least$1-\delta$, $\log E_P\exp(\lambda[L-\hat{L}])\le\lambda^2/(8m)+\log(1/\delta)$. For any finite probability Q, Jensen's inequality applied after writing the prior sum using Q gives $\lambda E_Q(L-\hat{L})\le KL(Q||P)+\log E_P\exp(\lambda[L-\hat{L}])$. Combining proves the bound uniformly in Q on that one event. Zero posterior masses are handled by continuity. Optimizing λ after observing Q requires a separate uniformization or parameter grid; the fixed-λ theorem does not silently supply it.

Worked example

Choose two hypotheses with uniform prior, a point-mass posterior, $m=100,\delta=.05,\lambda=20$ and empirical loss.1.

Then $KL=\log2$ and the penalty is $\log40/20+20/800\approx.2094439727$, giving $L(Q)\le.3094439727$ on the stated event. Separately $R^2=1-SSE/SST$ only describes squared errors on its evaluated response set; the number.847 omits the response normalization, split and dependency assumptions needed to connect it with this theorem.

The proof makes its probability event, optimization domain and constants reviewable, while preserving the empirical report as a claim rather than a verified forecast.

Scope

Prime-indexed observations are not automatically iid. Unbounded squared loss is outside the bounded-loss theorem unless explicitly controlled. A data-dependent prior needs another theorem. No cryptographic or cross-domain benefit follows from this certificate.

Further development

Specify a dependency model, bounded or tail-controlled loss and a prior independent of the held-out data. A spectral hypothesis can enter only through a proved complexity or KL estimate, after which the actual numerical bound should be reported alongside the score.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R104, R105.

Y25-05-020 Multiplicative convolution has a precise divisor algebra

2025-05-24T04:44:21Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which multiplicative convolution is intended, when do transforms turn it into a product, and what exact arithmetic identity can an implementation certify?

The historical report’s multiplicative-function approximation and inversion claims lack their actual kernel and norms. The classical divisor convolution gives a well-defined replacement. It proves an exact transform product when absolute convergence permits regrouping, rather than asserting a universal approximation capability.

Definitions and assumptions

For arithmetic functions f,g:N→C, define(f*g)(n)=Σ_{d|n}f(d)g(n/d). Let1(n)=1 and ε(1)=1, ε(n)=0 for n>1.

A function is multiplicative if f(mn)=f(m)f(n) for coprime m,n and f(1)=1. Its Dirichlet series is D_f(s)=Σf(n)n^(−s) where absolutely convergent.

Convolution, inversion and a transform product

Dirichlet convolution is associative and commutative with identity ε. The Möbius function satisfies μ*1=ε. Whenever D_f,D_g converge absolutely, D_(f*g)(s)=D_f(s)D_g(s). Multiplicative inputs give a multiplicative convolution.

Derivation or proof

Associativity follows by writing either parenthesization as the finite sum overabc=n of f(a)g(b)h(c); commutativity swaps complementary divisors. For μ, summing μ(d) over divisors of n factors as∏_{p|n}(1−1), zero unless n=1. The Dirichlet-series product is the absolutely convergent double sumΣ_{a,b}f(a)g(b)(ab)^(−s); group terms by n=ab to obtain the formula. For coprime m,n, each divisor of mn decomposes uniquely into a divisor of m times a divisor of n, so the convolution sum factors. These are concrete algebraic and analytic hypotheses, not consequences of a neural architecture name.

Worked example

Take f=g=1, then invert the divisor-count function τ=1*1 by μ.

τ(12)=6 from divisors1,2,3,4,6,12. The identity μ*τ=1 gives at12 the sum τ(12)−τ(6)−τ(4)+τ(2)=6−4−3+2=1. Consequently the absolutely convergent series of τ is ζ(s)² for Re(s)>1. The verifier checks the convolution and inversion on every integer through120.

A finite implementation can certify these exact arithmetic identities while leaving approximation and extrapolation claims separate.

Scope

This does not prove exponential approximation of arbitrary multiplicative functions, inversion of a noisy learned transform, or a number-theoretic theorem from finite training. Other notions of multiplicative convolution, such as Haar convolution on R_+, need different definitions.

Further development

For a specified truncated divisor transform, compute its triangular matrix and exact inverse. Then derive a perturbation norm bound for noisy coefficients, making conditioning and truncation error explicit before claiming stable inversion.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R106, R099.

Y25-05-022 A thermal pulse bound reverses the source area-to-volume factor

2025-05-30T02:40:47Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What is the dimensionally correct thermal-buffering flux bound, and what fraction of a first-order response occurs during a0.2ms pulse when the time constant is0.8s?

The late-month source includes a heat-flux formula with A/V where conservation requires V/A. A short energy balance fixes that inversion. A separate first-order response calculation gives an exact scale comparison without interpreting the time-scale mismatch as measured device behavior.

Definitions and assumptions

For a uniform thermal capacity C=ρc_pV, a permitted temperature rise ΔT stores energy CΔT. Let A be the area receiving a uniform flux q″ for duration t_p.

For a stated lumped linear model, τTdot+T=T_input after normalizing the temperature deviation. A unit step starting from zero gives T(t)=1−e^(−t/τ).

Energy-limited flux and a certified short-pulse response

Ignoring heat losses, q″≤ρc_pΔT(V/A)/t_p is necessary to keep the stored-energy rise within CΔT. The source’s factor(A/V) has the wrong length dimension. With τ=.8s and t_p=.0002s, the step fraction is1−e^(−1/4000)≈.000249968753.

Derivation or proof

Incoming energy is q″At_p. Requiring it to be no more thanρc_pVΔT and dividing by positive At_p gives the bound. The units are[J/(m³K)]·K·m/s=W/m²; replacing V/A by A/V would instead give W/m⁴. For the response, solve τTdot+T=1 by an integrating factor to obtain1−e^(−t/τ). For0≤x≤1, integrating1−u≤e^(−u)≤1 from zero to x gives x−x²/2≤1−e^(−x)≤x. Taking x=1/4000 encloses the response between.00024996875 and.00025. This is a rigorous bound in the declared first-order model.

Worked example

Use a constructed normalized capacity densityρc_p=2, volume V=3, area A=6,ΔT=4 and pulse duration2.

Stored capacity is24 energy units and the allowed uniform flux is24/(6·2)=2. The source’s inverted geometry would produce8, four times larger in this example and with inconsistent dimensions. Separately τ/t_p=4000 for the retrieved time scales, so a full response within one pulse is not supplied by this single-state step model.

The conservation and exponential calculations are practical mathematical checks, independent of unvalidated performance percentages.

Scope

Spatial gradients, contact losses, phase changes and pulse trains require a PDE or additional states. The bound is necessary within its energy accounting, not a validated heat-load rating. The user’s rejection of impractical cohomological deployment claims is preserved.

Further development

For a nonnegative pulse train convolved with the exponential impulse response, derive worst-case temperature envelopes from pulse energy and spacing. Validate the identified time constant against measurements before using the envelope for any design conclusion.

Source receipt: months/05/root_late_month.json record 0. Mathematical references: R107, R092.

Y25-05-024 A bonded-layer stress calculation needs a compatibility model

2025-05-31T03:21:40Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Under what compatibility and force-balance assumptions can interface thermal stress be calculated, and what remains unverifiable in the stated numerical claims?

A stress value is not determined by thermal expansion coefficients alone. A simple bonded-bar model states its mechanical assumptions and produces an exact stress formula by energy minimization. It is an illustrative calculation rather than validation of the reported274.5MPa or an operational limit.

Definitions and assumptions

Two perfectly bonded axial layers share strainε and temperature changeΔT. Their areas are A_i>0, moduli E_i>0 and expansion coefficientsα_i. Assume no net external axial force, no bending and linear elasticity.

Their stresses areσ_i=E_i(ε−α_iΔT), and stored elastic energy per common length is $U(\varepsilon)=\Sigma A_i E_i(\varepsilon-\alpha_i\Delta T)^2/2$.

Compatible strain as a unique convex minimizer

> The common strain isε*=ΔT(ΣA_iE_iα_i)/(ΣA_iE_i), withΣA_iσ_i=0. For equal A and E, stresses are$\pm E(\alpha_2-\alpha_1)\Delta T/2$. Changing boundary constraints changes the result.

Derivation or proof

Differentiate U to obtain U′=ΣA_iE_i(ε−α_iΔT). Setting this to zero is exactly axial force balance. Since U″=ΣA_iE_i>0, the stationary point is the unique global minimizer. Substitution yields the strain formula and equal-layer stresses. If both layers were externally constrained toε=0, the stresses would instead be−E_iα_iΔT, showing why omitted compatibility data cannot be repaired by a more precise multiplication. The stress dimensions are modulus times dimensionless strain; expansion coefficients carry inverse-temperature units.

Worked example

Choose a declared model with equal layers, E=200GPa,$\alpha_2-\alpha_1=10^{(-5)}K^{(-1)}$ andΔT=100K.

The stress magnitudes are100MPa, opposite in sign. The result is exact for these stipulated decimals. Separately the source’s specific-mass figures7.48 and6.8kg/kW differ by exactly10% relative to6.8; without the mass and power denominator neither is established. A reported electrophoretic velocity likewise cannot be recomputed without its field and mobility model.

The example supplies a practical compatibility certificate and identifies which historical numbers lack sufficient inputs.

Scope

Actual multilayers can bend, relax, creep, delaminate or have nonuniform temperature. The bar model is not an interface-failure criterion or a substitute for measurements. No strength, flight or deposition recommendation follows.

Further development

For a specified laminate, minimize a quadratic energy over midplane strain and curvature and propagate interval material parameters to stress bounds. Compare those bounds with independently measured strength distributions rather than treating one nominal stress as a reliability probability.

Source receipt: months/05/root_late_month.json record 0. Mathematical references: R108.

Y25-05-023 Effective conductivity depends on geometry and interface resistance

2025-05-31T17:23:55Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: How do Fourier flux, series and parallel composites differ, and why do a filler fraction and two conductivities not determine one effective conductivity?

Practical vector calculus is enough to expose the missing geometry. Parallel layers and series layers made from identical fractions have different exact effective conductivities. An interface resistance adds another independent parameter, so neither a nominal filler percentage nor a single multiplication validates the reported heat flux.

Definitions and assumptions

Use Fourier’s law q=−κ∇T withκ>0. On a stationary one-dimensional layered sample, constant flux and temperature drops determine its effective conductivity.

Let fractions f and1−f have conductivitiesκ₂ andκ₁. Parallel paths share a gradient; series layers share a flux. An interface resistance R_c≥0 gives temperature jump R_cq″.

Two exact effective-conductivity formulas and a conservation identity

Parallel geometry givesκ_parallel=(1−f)κ₁+fκ₂; series geometry givesκ_series=[(1−f)/κ₁+f/κ₂]^(−1). For total thicknessℓ, contact resistance changes the series value toκ_eff=ℓ/[ℓ_1/κ₁+ℓ_2/κ₂+R_c]. Heat conservation isρc_pT_t=−div q+s.

Derivation or proof

In parallel, sum area-weighted fluxes−κ_i∇T; the common gradient factors out. In series, each layer drops q″ℓ_i/κ_i, so the total drop is q″Σℓ_i/κ_i. Defining q″=κ_effΔT/ℓ yields the harmonic formula and the contact version after adding the jump. The arithmetic value exceeds the harmonic one by the Cauchy-Schwarz inequality applied to positive weighted conductivities and reciprocals. Conservation follows by equating the rate of stored energy in any control volume with incoming boundary flux plus sources, then applying the divergence theorem. These arguments require no Hodge decomposition or unvalidated learned predictor.

Worked example

Stipulate$\kappa_1=320, f=.12$ as retrieved inputs, and choose$\kappa_2=1$ only as a new illustrative contrast.

The parallel value is281.72, while the series value is4000/491≈8.14664 in the same conductivity units. Ifκ_2 were idealized as zero, the parallel limit would be281.6, matching one source number only under that extra idealization; the series limit would instead be zero. The verifier checks the rational formulas and the contact-resistance monotonicity.

The same fractions support radically different exact values depending on transport geometry, so a reported κ_eff needs a declared model and data.

Scope

The illustrativeκ_2 is not a material measurement. Arbitrary microstructures need a homogenization theorem or experiment; anisotropy requires a positive-definite conductivity tensor. Numerical metrics such as96.8% prediction accuracy or±2.8% variation are not validated here.

Further development

Pose a periodic conductivity cell problem, solve it with certified energy bounds and compare the resulting tensor against series/parallel limits. Interface resistances and uncertainty intervals should enter the same variational formulation.

Source receipt: months/05/root_late_month.json record 0. Mathematical references: R107, R076.

June 2025

Dated fragments recovered; complete transcript unavailable. 24 developed treatments.

ABC examples, perfect-power residue sieves, prime-delta operators and motivic comparison proposals are developed with correct domains and coefficients. Finite searches, explicit certificates and general theorems have separate roles.

ID · date	Concept and revised question	Result and mathematical core
Y25-06-001 2025-06-01 · U	ABC quality starts with an exact triple audit Reconstruction: Which displayed ABC examples actually satisfy a+b=c, coprimality and their reported quality?	Three exact source audits For(2,6436341,6436343), rad(abc)=15042 and q≈1.6299116841. For(121,109375,109496), the radical is10538990 and q≈.7175768892, not1.5808. The displayed(24833,181307,22784) is not an ABC triple because the first two entries sum to206140.
Y25-06-002 2025-06-01 · A	A smoothed radical has computable derivatives but no automatic curvature Reconstruction: What follows from the declared rad_σ formula, and what additional geometry would be required to call another expression its curvature or spectral gap?	Monotonicity, concavity and the exact zero of the energy surrogate $F'(\sigma)=\Sigma p^{(-\sigma)}(\log p)^2>0$ and $F''(\sigma)=-\Sigma p^{(-\sigma)}(\log p)^3<0$. Also $r_0=1$ and $r_\sigma\to\mathrm{rad}(n)$ as$\sigma\to\infty$. The source's E_σ vanishes exactly at$\sigma=q$ and has constant negative derivative$-\log \mathrm{rad}(abc)$.
Y25-06-003 2025-06-03 · U	Power-residue filters prove exclusions only for their declared sequence Reconstruction: Which exact modular obstruction applies to n!−1=m^d, and why do finite density patterns not imply a universal5/6 law?	An all-n factorial exclusion and an exact residue count For every even d≥2 and n≥3, n!−1 is not a dth power of an integer. For a prime p, $\lvert R_d(p)\rvert=1+(p-1)/\gcd(d,p-1)$.
Y25-06-005 2025-06-03 · U	The tanh fixed-point model has a provable threshold atλ=1 Reconstruction: Which bifurcation and susceptibility exponents follow from m=tanh(λm+h), and does that determine a degree threshold d=7?	A complete zero-field fixed-point classification For$0\le\lambda\le1$ the only fixed point is zero. For$\lambda>1$ there are exactly three:0 and$\pm m_\lambda$ with$0<m_\lambda<1$. Near$\lambda=1^+$, $m_\lambda\sim\sqrt{[3(\lambda-1)]}$. On the zero branch for$\lambda<1$,$\chi=(1-\lambda)^{(-1)}$.
Y25-06-004 2025-06-04 · U	The factorial gcd ratio is an exact valuation calculation Reconstruction: What does the piecewise factorial-gcd quantity compute, and how can its branches be certified without an arithmetic-physics interpretation?	Legendre valuations decide the branch exactly $g=\prod_{p\mid k}p^{\min(a_p,\Sigma\mathrm{floor}(n/p^j))}$; k divides n! exactly when each a_p is at most that factorial valuation. In the divisible branchρ=n!/k; otherwiseρ is the reciprocal of the positive integer k/g.
Y25-06-006 2025-06-08 · A	A prime-delta operator needs a quadratic form and jump conditions Reconstruction: Which self-adjoint operator realizes the delta expression, what eigenvalue bracket is justified, and when is the first-order shift formula valid?	A closed positive form, spectral bracket and conditional perturbation derivative The form defines a positive self-adjoint operator with discrete spectrum. With r interaction points, $(n\pi/L)^2\le\lambda_n\le((n+r)\pi/L)^2$, hence$\lambda_n=(n\pi/L)^2+O(n)$. For a differentiable simple eigenbranch of $q_\varepsilon=q_0+\varepsilon\Sigma\alpha_j\lvert u(x_j)\rvert^2$, $\lambda_n'(0)=(2/L)\Sigma\alpha_j \sin^2(n\pi x_j/L)$.

ID · date	Concept and revised question	Result and mathematical core
Y25-06-007 2025-06-12 · A	A fibre functor reconstructs a finite group only with its whole category Reconstruction: What exact finite theorem models Grothendieck Galois reconstruction, and which naturality data replace a bare claim of modal verification?	Aut(F) is canonically G The map a∈G↦[η_X(x)=ax] is a group isomorphism G→Aut(F). Natural automorphisms are determined by their action on the identity element of the regular G-set.
Y25-06-013 2025-06-12 · A	Étale and crystalline comparisons require different base data Reconstruction: Which base-field, lifting and coefficient hypotheses make the two proposed comparison equations meaningful?	Typed comparison statements and a rank/Frobenius test Under the lift hypotheses, H_crys^i(X₀/W(k))⊗K₀≅H_dR^i(X_K₀/K₀). For P¹, both sides have rank one in degrees zero and two and vanish in degree one. Over F_p the crystalline Frobenius multiplies the hyperplane class by p.
Y25-06-008 2025-06-12 · A	A modal reflector can be exact while failing tensor preservation Reconstruction: What explicit adjunction makes an idempotent operation a reflector, and why does that not prove every claimed realization compatibility?	A concrete reflector and a tensor obstruction R is left adjoint to I through Hom(V_+,W)≅Hom_C₂(V,IW). The induced idempotent monad IR projects to the trivial summand and is exact. But R(V)⊗R(W)→R(V⊗W) need not be an isomorphism.
Y25-06-009 2025-06-12 · A	Projective motivic cohomology retains the base field in every summand Reconstruction: What is the correctly indexed projective-bundle formula, and why is a bidegree window returning onlyZ(q) insufficient?	Correct projective decomposition and the Chow diagonal H^{p,q}(P^n_k,Z)≅⊕_{i=0}^n H^{p−2i,q−i}(Spec k,Z). In particular CH^*(P^n_k)=Z[h]/(h^{n+1}); hence H^{2q,q}=Z for0≤q≤n and zero outside that Chow range. Other bidegrees are not specified by that diagonal rule.
Y25-06-010 2025-06-12 · A	Motivic realization is not the claimed universal Betti isomorphism Reconstruction: Which degree does the Betti cycle map use, and what elementary motivic group already disproves the stated all-bidegree comparison?	A counterexample at a point At X=Spec C and(p,q)=(1,1), the source comparison has nonzero left side C×⊗_ZC but zero right side H_sing¹(point,C). Consequently it cannot be a universal isomorphism, even under its own smooth-proper hypothesis.
Y25-06-011 2025-06-12 · A	Künneth signs and Tor corrections survive a tiny exact example Reconstruction: What does a tensor-complex implementation need to preserve, and what higher class is lost by multiplying only degree-zero ranks?	The tensor has an additional degree-one Z/2 For D=C⊗C, the chain groups are Z,Z²,Z in degrees two,one,zero, with d₂=(-2,2)ᵀ and d₁=(2,2). Hence H₀(D)=Z/2,H₁(D)=Z/2,H₂(D)=0.
Y25-06-012 2025-06-12 · A	Gysin localization and a torus example force codimension shifts Reconstruction: What shifts appear in localization, and which exact calculation checks a torus or blow-up implementation?	The torus units and an exceptional-divisor rank check H^{1,1}(G_m)=k×⊕Z, with the extra Z measured by the exponent of t. The codimension-one boundary map records that exponent. The classical blow-up formula adds CH^{r−1}(point) to CH^r(P²) when blowing up a point, so CH¹ gains one independent exceptional class.

ID · date	Concept and revised question	Result and mathematical core
Y25-06-015 2025-06-13 · U	Hamiltonian tuning dynamics need an energy and damping law Reconstruction: What does an explicit Hamiltonian tuning model conserve, and which additional term actually makes errors decay?	Conservation without damping and dissipation with it Hdot=−vᵀDv≤0 with v=M^(−1)p. If D=0, H is conserved and nontrivial trajectories need not converge. In the scalar model M=1,K=4,D=2, all solutions converge to the target with characteristic exponents−1±i√3.
Y25-06-014 2025-06-13 · U	Rational musical intervals have an exact logarithmic lattice map Reconstruction: Which algebraic properties do prime-factor intervals and cents satisfy, and what separation bound is justified on a finite search box?	An injective additive encoding and a finite height bound r(a+b)=r(a)r(b), c(a+b)=c(a)+c(b), and c is injective on Z^d. If A,B≤M and A≠B, then|c(a)−c(b)|≥1200/(Mlog2). Without a height bound there need not be a positive uniform octave-reduced separation.
Y25-06-018 2025-06-15 · U	Tripod overlap and sensor aliasing have exact finite certificates Reconstruction: What does duty factor.7 with half-cycle tripod offset imply, and which band-limit hypothesis is needed for a λ/2 sensor-spacing rule?	Support overlap and the spatial alias identity Forβ≥1/2 the two tripods cover the whole cycle and overlap for fraction2β−1. Atβ=.7 the overlap is.4 and each exclusive stance occupies.3. Spatial waves with wavenumbers k and k+2πj/h have identical samples for every integer j.
Y25-06-019 2025-06-15 · U	Resonance and regenerative accounting are separate mathematical claims Reconstruction: Where does a damped oscillator maximize displacement, and when is subtracting regenerated energy from electrical input a valid efficiency denominator?	A resonance location and a net-efficiency identity If c²<2km, the positive displacement peak occurs atω_r²=k/m−c²/(2m²). Otherwise the maximal amplitude is at zero frequency. The accounting efficiency is W/(E_g−E_r) under a positive, complete net denominator, not a theorem that energy has been created.
Y25-06-021 2025-06-16 · U	An invented coefficient cannot turn unlike quantities into a validated scalar Reconstruction: Which unit and identifiability requirements are missing from D(t)=Σε_i d_i(t), and how can free weights reverse a conclusion?	Unit invariance and arbitrary ranking under unconstrained weights D is invariant under that coordinated unit change. Without a constraint or calibration on positive weights, even the ranking of x=(1,0) and y=(0,1) is undetermined: it is the sign of w_1-w_2.
Y25-06-020 2025-06-17 · A	A strong correlation is not a causal policy proof Reconstruction: Why can a correlation of−.973 coexist with zero intervention effect, and what does a discounted30-year arithmetic model actually assume?	An exact confounding example and a geometric-sum formula Corr(X,Y)=−1/√(1+σ²), yet do(X=x) leaves the distribution of Y unchanged. Choosingσ²=(1000/973)²−1 gives correlation−.973 exactly. Also V_N=b[1−(1+r)^(−N)]/r−C.
Y25-06-016 2025-06-22 · U	A nonlinear dyad has a rank-one Jacobian, not a bilinear tensor law Reconstruction: Which rank, composition and conservative-field properties follow from F(x)=a f(b·x), and how does this differ from a Clifford product?	Derivative, composition and a potential criterion DF(x)=f′(bᵀx)abᵀ has rank at most one. F∘G(x)=a f((bᵀc)g(dᵀx)). If f′ is nonzero somewhere, F is a gradient field on R^d exactly when a is parallel to b.

ID · date	Concept and revised question	Result and mathematical core
Y25-06-017 2025-06-22 · U	Nonlocal dyad learning needs an adjoint in its variational update Reconstruction: What is the correct Euler equation for an L^2 reconstruction loss with H^1 regularization, and when is its minimizer unique?	A coercive normal equation and its gradient The finite model has unique minimizer$(K^*K+\alpha I)u=K^*y$. With an H^1 term its formal Euler equation is $K^*(Ku-y)+\alpha u-\beta\Delta u=0$, with the chosen natural or fixed boundary conditions.
Y25-06-022 2025-06-24 · A	Area, fourth-power temperature and efficiency ratios need the same balance Reconstruction: What follows from a fourth-power radiation model and the stated power ratios, and which claimed improvements remain assumptions?	An exact area-temperature scaling and denominator audit At fixed P, multiplying area by1.7 divides temperature by$1.7^{(1/4)}$. Multiplying both area by1.7 and temperature by$1.7^{(1/4)}$ instead multiplies P by$1.7^2=2.89$. The source's150/555 and150/277.5 ratios are≈.27027 and.54054 before auxiliary power.
Y25-06-024 2025-06-28 · A	Exhaustive finite tests prove a polynomial identity only with a degree bound Reconstruction: When can an81-point test on F_3^4 be a proof, and when is it merely evidence about a different polynomial claim?	A complete finite certificate and its exact counterexample If the reduced f vanishes on all81 points, then f is the zero polynomial. Without the bound, $x_1^3-x_1$ is a nonzero formal polynomial that vanishes at all81 points.
Y25-06-023 2025-06-28 · A	The mod-three B_4 action has five exact orbits Reconstruction: What group action is actually realized by signed coordinate reflections on F_3^4, and can it be identified with exceptional Weyl groups without further data?	Order, orbits and a faithful-representation obstruction $\lvert B_4\rvert=2^4\cdot 4!=384$. Its orbits are exactly the five Hamming-weight classes, of sizes1,8,24,32,16. A faithful linear action of $W(E_8)$ on F_3^4 is impossible because$\lvert W(E_8)\rvert=696729600$ exceeds$\lvert GL_4(F_3)\rvert=24261120$.

Y25-06-001 ABC quality starts with an exact triple audit

2025-06-01T03:08:28Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which displayed ABC examples actually satisfy a+b=c, coprimality and their reported quality?

A numerical quality is meaningful only after the integer identity and radical are checked. The recovered first example is valid, the second quality is wrong, and the third displayed triple fails the defining sum. Those finite corrections precede any curvature or asymptotic interpretation.

Definitions and assumptions

For positive coprime a,b with a+b=c, define rad(n)=∏_{p|n}p and q=log c/log rad(abc). Distinct prime factors are counted once.

The classical ABC statement quantified in the source allows a constant K_ε depending onε; a finite observed maximum is not a theorem bounding all qualities.

Three exact source audits

For(2,6436341,6436343), rad(abc)=15042 and q≈1.6299116841. For(121,109375,109496), the radical is10538990 and q≈.7175768892, not1.5808. The displayed(24833,181307,22784) is not an ABC triple because the first two entries sum to206140.

Derivation or proof

The first factors are2,3^10·109 and23^5. Their distinct prime product is2·3·109·23=15042; pairwise coprimality follows from these disjoint prime factors. In the second example121=11²,109375=7·5^6, and109496=2³·13687, with13687 prime, giving the stated radical. Trial division through its square root certifies that primality. Exact addition verifies both sums and Euclid verifies coprimality. In the third, addition alone invalidates the setup, so its reported q cannot serve as an ABC quality. Logs are evaluated only after these integer certificates. If q>1+ε, rearrangement gives c>rad(abc)^(1+ε); this is an equivalence for one valid triple, not a proof of a universal finiteness statement.

Worked example

Also evaluate the small triple3+125=128.

Its radical is2·3·5=30 and its quality is log128/log30≈1.426565. The repeated powers5³ and2⁷ affect c but contribute only5 and2 to the radical. The verifier factors every displayed input and checks the source claims before approximating any logarithm.

The audit preserves a genuine strong example while removing two unsupported entries from the evidence base.

Scope

No universal q_max≈1.63 or ABC proof follows from these finite examples. The arithmetic check does not adjudicate any claimed general proof of ABC.

Further development

Store each candidate with a+b−c, gcd, full prime factorization and a directed-rounding quality interval. A search can then report only certified records and an explicit finite search range.

Source receipt: months/06/root_followups.json record 0. Mathematical references: R078, R109.

Y25-06-002 A smoothed radical has computable derivatives but no automatic curvature

2025-06-01T05:43:50Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What follows from the declared rad_σ formula, and what additional geometry would be required to call another expression its curvature or spectral gap?

The proposed smoothed radical is an explicit positive function of a real parameter. Its calculus can be proved exactly. The source’s separate scalar-curvature and eigenvalue formulas are not consequences of that definition without a metric, domain and operator.

Definitions and assumptions

For fixed n>1 andσ≥0 define r_σ(n)=∏_{p|n}p^(1−p^(−σ)) and F(σ)=log r_σ=Σ_{p|n}(1−p^(−σ))log p.

For a valid ABC triple define E_σ=log c−σlog rad(abc). This uses the ordinary radical and is a separate affine function ofσ.

Monotonicity, concavity and the exact zero of the energy surrogate

F′(σ)=Σp^(−σ)(log p)²>0 and F″(σ)=−Σp^(−σ)(log p)³<0. Also r₀=1 and r_σ→rad(n) asσ→∞. The source’s E_σ vanishes exactly atσ=q and has constant negative derivative−log rad(abc).

Derivation or proof

Because the prime support is finite, differentiate each exponential term directly. All logarithms of primes are positive, giving the strict signs. Atσ=0 every exponent1−p^(−σ) vanishes; asσ increases each tends to one, proving the limits. Solving the affine equation E_σ=0 givesσ=log c/log rad(abc). Thus a crossing at the quality is built into the definition rather than independent evidence of a phase transition. A Riemannian scalar curvature depends on a declared metric and its first and second derivatives in coordinate directions; an eigenvalue depends on an operator domain and boundary conditions. Neither is determined merely by a scalar function F.

Worked example

Choose n=6 and the valid triple3+125=128.

Atσ=1, r₁(6)=2^(1/2)3^(2/3), whereas F′(1)=(log2)²/2+(log3)²/3 and F″(1)=−(log2)³/2−(log3)³/3. The triple’s E_σ crosses zero at its quality≈1.426565. The verifier checks derivative signs and the independent exponential identity numerically, with exact prime support.

The declared surrogate has a rigorous calculus, while its geometric and spectral interpretation remains a separate problem.

Scope

No blow-up exponent, q_max, universal relationλ₁=|R_σ|/4 or arithmetic phase transition is proved. Calling a scalar expression curvature does not specify its underlying geometry.

Further development

Choose an explicit parameter manifold and positive metric tensor, compute its actual curvature, and separately define a self-adjoint operator. Only a proved comparison inequality could then connect that spectrum with the radical surrogate.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R110.

Y25-06-003 Power-residue filters prove exclusions only for their declared sequence

2025-06-03T18:11:03Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which exact modular obstruction applies to n!−1=m^d, and why do finite density patterns not imply a universal5/6 law?

The analyzer names many rapidly growing sequences and exponents. A modular filter can be completely rigorous, but its scope is a particular equation, modulus and range. One simple obstruction already separates even and odd degrees rather than producing a universal threshold at seven.

Definitions and assumptions

For a modulus M≥2 define R_d(M)={x^d mod M:x∈Z/MZ}. If S(n)±k=m^d then its residue must belong to R_d(M).

Take S(n)=n!, k=1, n≥3. For a prime p, the nonzero dth powers form the image of the dth-power map on the cyclic group F_p×.

An all-n factorial exclusion and an exact residue count

> For every even d≥2 and n≥3, n!−1 is not a dth power of an integer. For a prime p, |R_d(p)|=1+(p−1)/gcd(d,p−1).

Derivation or proof

Since3 divides n!, n!−1≡−1≡2 mod3. Every even power modulo3 is zero or one, giving the contradiction for all n in the stated range. For the residue count, write F_p× as a cyclic group of order p−1. Its dth-power homomorphism has kernel size gcd(d,p−1), so its image has(p−1)/gcd(d,p−1) elements; add zero. This theorem counts residues for uniformly ranging bases modulo p, not solutions along a factorial or tower sequence. Different sequence residue distributions and different signs can therefore have different exclusion fractions.

Worked example

Compare d=8 and d=7 modulo3, and inspect n=4.

$R_8(3)=\{0,1\}$, so4!−1=23 is excluded. $R_7(3)=\{0,1,2\}$, so this modulus excludes nothing for odd degree seven, although it does not prove23 is a seventh power. The positive equation4!+1=25=5^2 shows that changing the sign matters. The verifier exhaustively constructs residue sets and uses an exact integer-root search for finite examples.

A modular rejection is a proof of impossibility; passing the filter is only a necessary condition and not a solution certificate.

Scope

This does not classify n!+1, Fibonacci, Catalan, primorial or tower equations. The historicalρ=5/6 and d*=7 claims lack a defined sampling measure and proof. They are not consequences of this residue theorem.

Further development

For a fixed named sequence and k,d, combine several coprime moduli by the Chinese remainder theorem and prove the sequence’s residue frequencies. Only then define and estimate a density with an explicit limiting procedure.

Source receipt: months/06/root_followups.json record 0. Mathematical references: R099.

Y25-06-005 The tanh fixed-point model has a provable threshold atλ=1

2025-06-03T20:46:33Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which bifurcation and susceptibility exponents follow from m=tanh(λm+h), and does that determine a degree threshold d=7?

The source imports a mean-field equation into an arithmetic narrative. The scalar equation itself has a classical exact analysis. Its threshold is a parameter valueλ=1; connecting that parameter to a Diophantine degree requires an additional justified mapλ(d).

Definitions and assumptions

Consider m∈(−1,1), λ≥0 and h∈R with m=tanh(λm+h). For h=0, symmetry permits m=0 and opposite pairs.

On a smooth solution branch define susceptibilityχ=∂m/∂h. Use the local expansion atanh m=m+m³/3+O(m⁵).

A complete zero-field fixed-point classification

For0≤λ≤1 the only fixed point is zero. Forλ>1 there are exactly three:0 and±m_λ with0<m_λ<1. Nearλ=1⁺, m_λ~√[3(λ−1)]. On the zero branch forλ<1,χ=(1−λ)^(−1).

Derivation or proof

For m>0, the equation isλ=atanh(m)/m. This ratio tends to one at zero and infinity at one. Its derivative has numerator m/(1−m²)−atanh m, positive because atanh m=∫₀^m(1−t²)^(−1)dt<m/(1−m²). Hence the ratio is strictly increasing and crosses eachλ>1 exactly once. Negative solutions follow by symmetry. The Taylor expansion givesλ−1=m²/3+O(m⁴), proving the square-root behavior. Differentiating m=tanh(λm+h) givesχ=(1−m²)[λχ+1], soχ=(1−m²)/[1−λ(1−m²)] when the denominator is nonzero. Setting m=0 proves the stated susceptibility.

Worked example

Chooseλ=2,h=0 and solve only the positive branch.

Bisection gives m≈.9575040241 and a direct residual below the declared numerical tolerance. The zero solution also remains present; a solver returning one root is not a classification. For the separately declared fixed-point iteration m_next=tanh(2m), the derivative at this nonzero root is$2(1-m^2)\approx.1664$, below one; at zero it is two. Thus the iterative stability differs across these branches. The verifier checks the root bracket andχ=2 on the zero branch atλ=1/2.

The exponents are consequences of this model, while an arithmetic critical degree has not been derived.

Scope

No universalρ_d, spectral centroid, correlation length or Diophantine transition follows. Ifλ(d)=1 at d=7 by definition, the location is stipulated rather than discovered.

Further development

Specify a rigorously derivedλ(d) and quantify its expansion near a crossing. Only then can the scalar asymptotics be transferred to d, with error terms and the domain where the reduction is valid.

Source receipt: months/06/root_followups.json record 0. Mathematical references: R111.

Y25-06-004 The factorial gcd ratio is an exact valuation calculation

2025-06-04T12:43:38Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What does the piecewise factorial-gcd quantity compute, and how can its branches be certified without an arithmetic-physics interpretation?

The source provides a piecewise ratio but not a justified physical interpretation. Its divisibility content is elementary and useful: prime-adic valuations determine exactly which branch applies and give a stable implementation without forming a gigantic factorial.

Definitions and assumptions

For positive integers n,k let g=gcd(k,n!). Defineρ(n,k)=n!/g if k divides n!, andρ(n,k)=g/k otherwise, following the retrieved branch convention.

For a prime p, v_p(n!)=Σ_{j≥1}floor(n/p^j), a finite sum. Factor k=∏p^{a_p}.

Legendre valuations decide the branch exactly

g=∏_{p|k}p^{min(a_p,Σfloor(n/p^j))}; k divides n! exactly when each a_p is at most that factorial valuation. In the divisible branchρ=n!/k; otherwiseρ is the reciprocal of the positive integer k/g.

Derivation or proof

Count each factor of p in1,…,n. Multiples of p contribute one, multiples of p^2 contribute an additional one, and so on, proving Legendre's sum. The valuation of a gcd is the minimum of the two valuations, yielding the product formula. Divisibility is equivalent to the componentwise inequalities. If k|n!, then g=k and the first branch simplifies as stated. In the other branch k/g is an integer greater than one, soρ=g/k∈(0,1). The two branches therefore occupy distinct numerical ranges, with equality one possible at k=n! in the first branch. This is a chosen arithmetic encoding, not a continuous response law.

Worked example

Take n=5, so n!=120, and compare k=12 with k=16.

For12=2^2·3, the factorial valuations3 and1 suffice, so g=12 andρ=10. For16=2^4, only three factors of2 occur, so g=8 andρ=1/2. At k=120 the value is one. The verifier compares the valuation formula with Euclid for every k through200 and n through12.

The formula can be certified completely while its proposed rigidity or thermodynamic meaning is withheld.

Scope

A large value in one branch is not a measured energy or spectral gap. The discontinuous branch choice supplies no universal phase transition or special role for k=1 beyond its elementary divisibility.

Further development

Use valuation vectors as the state of a finite divisibility automaton, and prove a termination or search-pruning bound. This would turn the ratio into a transparent algorithmic tool with a specified objective.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R112.

Y25-06-006 A prime-delta operator needs a quadratic form and jump conditions

2025-06-08T04:10:12Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which self-adjoint operator realizes the delta expression, what eigenvalue bracket is justified, and when is the first-order shift formula valid?

The interval and numerical setup of the reported eigenvalues were not recovered. A bounded-interval point-interaction model nevertheless supports precise domain, min-max and perturbation statements. Prime locations are just one admissible finite set of interior points; their arithmetic labels do not prove a zeta correspondence.

Definitions and assumptions

On(0,L), take finitely many distinct interior x_j and$\alpha_j \geq 0$. On H_0^1 define $q[u]=\int|u'|^2dx+\Sigma\alpha_j|u(x_j)|^2$.

The associated operator acts as$-u''$ between interaction points. Its domain has u continuous, piecewise H^2, Dirichlet endpoints and jumps $u'(x_j+)-u'(x_j-)=\alpha_j u(x_j)$.

A closed positive form, spectral bracket and conditional perturbation derivative

The form defines a positive self-adjoint operator with discrete spectrum. With r interaction points, $(n\pi/L)^2\le\lambda_n\le((n+r)\pi/L)^2$, hence$\lambda_n=(n\pi/L)^2+O(n)$. For a differentiable simple eigenbranch of $q_\varepsilon=q_0+\varepsilon\Sigma\alpha_j|u(x_j)|^2$, $\lambda_n'(0)=(2/L)\Sigma\alpha_j \sin^2(n\pi x_j/L)$.

Derivation or proof

Point evaluation is continuous on H_0^1, for example$|u(x)|\le\sqrt{L}||u'||_2$. Thus the form norm is equivalent to the H_0^1 norm and is closed; the closed-form representation theorem and compact embedding into L^2 are the stated classical functional-analytic inputs. Integration by parts on each subinterval gives the jump conditions. The lower bound is min-max monotonicity. For the upper bound, intersect the span of the first n+r free sine modes with the r evaluation kernels; its dimension is at least n, and on it the interaction term vanishes, giving the bound. For the derivative formula, differentiate the normalized eigenvalue identity; the terms from differentiating the eigenvector cancel by the eigen-equation and normalization, leaving the form derivative evaluated on$\sqrt{(2/L)}\sin(n\pi x/L)$.

Worked example

Take $L=\pi$ and one interaction of strength2 at$\pi/2$.

A symmetric ground mode has wavenumber $k\in(1,3/2)$ satisfying$2k\cot(\pi k/2)+2=0$. The endpoint signs are exactly positive at1 and negative at3/2, and monotonicity gives one root; $\lambda=k^2$ lies in$(1,9/4)$. Antisymmetric free modes vanishing at the midpoint remain unchanged. For strength2ε, the ground derivative at zero is$4/\pi$, also obtained by implicit differentiation of the scalar equation.

Domain and min-max arguments validate a real operator theorem without validating the source’s unreproducible eigenvalue table.

Scope

The finite positive model does not imply an explicit formula for prime zeros. The first-order shift is a derivative at zero coupling, not an exact finite-strength correction. An unspecified interval cannot support a unique numerical spectrum.

Further development

Record the full interval and interaction list, bracket each eigenvalue through transfer matrices with interval arithmetic, and compare discretization errors before drawing spacing conclusions.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R113, R114.

Y25-06-007 A fibre functor reconstructs a finite group only with its whole category

2025-06-12T05:44:12Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What exact finite theorem models Grothendieck Galois reconstruction, and which naturality data replace a bare claim of modal verification?

Grothendieck’s Galois theory is a substantial theorem about categories and fibre functors. A finite group version admits a short complete proof and identifies the essential naturality requirement. It does not assert that the historical modal implementation exists.

Definitions and assumptions

Let G be a finite group, FinGSet its category of finite left G-sets and equivariant maps, and F its forgetful functor to finite sets.

A natural automorphismη of F is a family of bijections$\eta_X:F(X)\to F(X)$ commuting with every equivariant map. Its components are not required to be equivariant themselves.

Aut(F) is canonically G

> The map $a\in G\mapsto[\eta_X(x)=ax]$ is a group isomorphism $G\to\mathrm{Aut}(F)$. Natural automorphisms are determined by their action on the identity element of the regular G-set.

Derivation or proof

For any a, equivariance of a map f gives $f(ax)=af(x)$, so the family is natural; the inverse family uses a^{-1}. Conversely letη be natural and let $a=\eta_G(1)$, where G carries its left regular action. For any G-set X and $x\in X$, the map $f_x:G\to X$, $g\mapsto gx$, is equivariant. Naturality gives$\eta_X(x)=\eta_X(f_x(1))=f_x(\eta_G(1))=ax$. Thus every natural family has the displayed form and its a is unique. Composition of families corresponds to multiplication ab, proving the group law. The proof uses all these maps f_x; isolated orbit sizes or a single matrix do not supply the reconstruction.

Worked example

Take $G=C_3$ acting regularly on$\{0,1,2\}$ by cyclic addition.

The three possible natural families restrict to the three cyclic shifts on the regular set. A transposition fails to commute with its equivariant right translations and therefore cannot be a natural automorphism. The verifier enumerates all six permutations and retains exactly three.

The finite model shows what is reconstructed and why categorical compatibility is mathematical data rather than a label.

Scope

The scheme-theoretic theorem needs a Galois category, a geometric fibre functor and a profinite group with continuity. No universal reconstruction of arbitrary schemes from an unspecified modality is claimed.

Further development

For a finite field, identify finite étale covers with finite sets carrying Frobenius permutation. Recover orbit lengths and extension degrees, then pass to the profinite limit only after verifying the relevant categorical hypotheses.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R115, R116.

Y25-06-013 Étale and crystalline comparisons require different base data

2025-06-12T05:50:33Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which base-field, lifting and coefficient hypotheses make the two proposed comparison equations meaningful?

The retrieved equations omit data that distinguish complex Betti comparison from crystalline-to-de-Rham comparison and from p-adic étale comparison. These are separate classical theorems, not interchangeable modal identities. A projective-line calculation provides a transparent consistency check.

Definitions and assumptions

For finite-type X over C, the classical comparison identifies étale Q_l cohomology with singular Q_l cohomology of X(C). For a smaller characteristic-zero field one must choose an embedding into C and base change.

For smooth proper X_0 over a perfect field k of characteristic p, put K_0=Frac W(k). A smooth proper liftX/W(k) gives a crystalline-de-Rham comparison after tensoring with K_0. Relating p-adic étale cohomology generally uses a period ring such as B_cris and additional good-reduction hypotheses.

Typed comparison statements and a rank/Frobenius test

> Under the lift hypotheses, H_crys^i(X_0/W(k))⊗K_0≅H_dR^i(X_K_0/K_0). For P^1, both sides have rank one in degrees zero and two and vanish in degree one. Over F_p the crystalline Frobenius multiplies the hyperplane class by p.

Derivation or proof

The two comparison theorems are cited classical inputs rather than re-proved here. Their fields and spaces are part of the statement: a positive-characteristic X has no X(C) without a different characteristic-zero object, and X_K is undefined until a lift or model is given. For P^1 in characteristic zero, the Čech de-Rham calculation on its two affine charts leaves the constant in degree zero and the class dx/x in total degree two; positive and negative Laurent powers other than x^(−1) are removed by the chart subcomplexes. Equivalently its hyperplane cycle gives the unique degree-two class. Pullback under the p-power map takes O(1) to O(p), so the first Chern class is multiplied by p. This independently fixes the expected Frobenius factor for the cellular test object.

Worked example

Use P^1 over F_5 with its standard smooth lift over Z_5.

The ranks are(1,0,1), and Frobenius on degrees zero and two is(1,5). The resulting trace count is1+5=6, matching#P^1(F_5). Over F_{5^2} it is1+25=26. The verifier checks these exact point counts and factors; it does not computationally certify the general comparison theorem.

Correct types and a simple functorial test prevent comparisons across incompatible spaces or coefficient fields.

Scope

A comparison isomorphism may carry filtration, Frobenius and Galois data that a vector-space rank table omits. Lean/Agda-style pseudocode or postulated axioms are not evidence that these theorems were implemented or machine checked.

Further development

Build an interface whose type requires the base embedding or lift and records every preserved structure. Add period-ring base changes explicitly before attempting a unified p-adic realization diagram.

Source receipt: months/06/root_details.json record 0. Mathematical references: R117, R118.

Y25-06-008 A modal reflector can be exact while failing tensor preservation

2025-06-12T13:25:59Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What explicit adjunction makes an idempotent operation a reflector, and why does that not prove every claimed realization compatibility?

The historical correction marked the étale modality and implementation incomplete. An elementary representation example supplies genuine unit, idempotence and adjunction data. It also demonstrates that exactness or finite-limit preservation does not automatically make a reflector strong monoidal for a tensor product.

Definitions and assumptions

Work over R with a finite-dimensional representation of C_2, specified by $S^2=I$. Let $V_+=\ker(S-I)$ and $V_-=\ker(S+I)$, with projections $P_\pm=(I\pm S)/2$.

Let I include trivial representations and $R(V)=V_+$. A map to a trivial representation must satisfy $fS=f$.

A concrete reflector and a tensor obstruction

R is left adjoint to I through $\mathrm{Hom}(V_+,W)\cong\mathrm{Hom}_{C_2}(V,IW)$. The induced idempotent monad IR projects to the trivial summand and is exact. But $R(V)\otimes R(W)\to R(V\otimes W)$ need not be an isomorphism.

Derivation or proof

$P_+^2=P_+$ follows from $S^2=I$, and $P_++P_-=I$ with $P_+P_-=0$ splits V into the two eigenspaces. If $fS=f$, then $fP_+=f$ and f kills V_-, so f factors uniquely through V_+. This gives the adjunction and its naturality directly. An equivariant exact sequence splits into independent plus and minus sequences by these natural projections, proving exactness of R. For the tensor claim, take both V and W to be the one-dimensional sign representation. Each invariant subspace is zero, so $R(V)\otimes R(W)=0$. Their tensor has action$(-1)(-1)=+1$, so $R(V\otimes W)=R$, disproving strong tensor preservation.

Worked example

Use $S=\mathrm{diag}(1,-1)$ and compare its plus projection with two sign factors.

$P_+=\mathrm{diag}(1,0)$, with rank one and $P_+^2=P_+$. A sign vector has zero plus projection, but the tensor of two sign vectors is invariant and nonzero. The verifier checks the matrices and tensor signs exactly. No postulate is counted as an implementation proof.

The example distinguishes categorical idempotence, exactness and monoidal compatibility, each of which has different obligations.

Scope

This finite ordinary-category reflector is not the historical higher étale modality. HoTT modalities require their own dependent elimination and stability conditions; a linear idempotent alone is not enough.

Further development

State the actual ambient category and realization functor, construct its unit and comparison morphisms, and prove each required limit or tensor law. Test conservativity and monoidal strength separately before asserting a unified comparison theorem.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R119.

Y25-06-009 Projective motivic cohomology retains the base field in every summand

2025-06-12T13:55:56Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What is the correctly indexed projective-bundle formula, and why is a bidegree window returning onlyZ(q) insufficient?

Several retrieved rules attempted to compute all motivic cohomology from a small inequality in p,q,n. The projective-bundle theorem instead shifts the motivic cohomology of the base field. On the Chow diagonal this simplifies beautifully, but it does not erase off-diagonal field information.

Definitions and assumptions

For a field k and standard integral motivic cohomology of smooth schemes, use $H^{p,q}(X,Z)=CH^q(X,2q-p)$.

Let h be the hyperplane class of bidegree(2,1). The classical projective-bundle theorem is the stated input below.

Correct projective decomposition and the Chow diagonal

H^{p,q}(P^n_k,Z)≅⊕_{i=0}^n H^{p−2i,q−i}(Spec k,Z). In particular CH^*(P^n_k)=Z[h]/(h^{n+1}); hence H^{2q,q}=Z for0≤q≤n and zero outside that Chow range. Other bidegrees are not specified by that diagonal rule.

Derivation or proof

The projective-bundle theorem states thatΣh^iπ^* gives the displayed isomorphism, withπ:P^n→Spec k. The shifts follow from deg(h^i)=(2i,i). On the diagonal, H^{2r,r}(k)=CH^r(Spec k), which is Z for r=0 and zero for r>0 because a point has no positive-codimension cycles. Thus only i=q survives, yielding the polynomial Chow ring and the relation h^{n+1}=0. Off the diagonal, H^{1,1}(k)=k× is generally nonzero, so a computer rule returning merely zero or one Tate symbol loses genuine base-field data. The general projective-bundle theorem is cited rather than claimed as proved by finite tests; the diagonal and grading consequences are derived explicitly.

Worked example

For P^2 compute its Chow ranks and the product h·h.

The groups in bidegrees(0,0),(2,1),(4,2) are each Z, h^2 is the point class, and $h^3=0$. For P^1, H^{1,1} contains k× even though its Chow ranks are only one in codimensions zero and one. The verifier implements only the declared truncated polynomial ring and checks degree shifts and products; it does not pretend to compute arbitrary field K-theory.

A symbolic Tate-basis engine can be correct on a specified cellular Chow subtheory without being a universal motivic-cohomology algorithm.

Scope

The source’s window formulas and parity tests are not accepted as general theorems. Coefficients, base field and cohomology model must be fixed before any computation is interpreted.

Further development

Represent field cohomology as an explicit coefficient object with certified operations, then apply the projective decomposition functorially. Extend to products only with all bidegree summands and coefficient Tor information preserved.

Source receipt: months/06/root_details.json record 0. Mathematical references: R120, R121.

Y25-06-010 Motivic realization is not the claimed universal Betti isomorphism

2025-06-12T14:38:37Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which degree does the Betti cycle map use, and what elementary motivic group already disproves the stated all-bidegree comparison?

The source asserts H_mot^{p,q}(X)⊗C≅H_sing^{2q−p}(Xan,C) for every smooth proper complex X. This is false. The higher-Chow index2q−p is not the singular cohomological degree, and realization can discard nontrivial arithmetic or cycle data.

Definitions and assumptions

For smooth complex X, integral motivic cohomology has a Betti cycle map to H_sing^p(Xan,Z(q)); the target degree is p. The map is not generally an isomorphism.

Use the classical low-degree identity H^{1,1}(Spec k,Z)=k×. The complex point Spec C is smooth and proper.

A counterexample at a point

> At X=Spec C and(p,q)=(1,1), the source comparison has nonzero left side C×⊗_ZC but zero right side H_sing¹(point,C). Consequently it cannot be a universal isomorphism, even under its own smooth-proper hypothesis.

Derivation or proof

A point is contractible, so its positive-degree singular cohomology vanishes. The motivic weight-one complex yields the stated units group. To see the tensor on the left is nonzero, log|·|:C×→R is a surjective homomorphism of abelian groups; tensoring gives a surjection onto R⊗_ZC, which is nonzero. For example, the multiplication map R⊗_ZC→C sends1⊗1 to1. Thus the source cannot repair its claim merely by tensoring with C. The valid higher-Chow formula H^{p,q}=CH^q(X,2q−p) uses2q−p as a chain index, which is a different role from Betti degree.

Worked example

Compare the weight-one group at a point with the Chow class on P^1.

H^{1,1}(point)=C× maps to zero singular degree one. In contrast $H^{2,1}(P^1)=CH^1(P^1)=Z$ maps to the degree-two generator of $H^2(P^1(C),Z(1))$. The source's formula would send that class to degree zero instead, disguising the wrong grading because both target groups happen to have rank one in this particular example.

Passing a small rank test on P^1 is therefore not sufficient to validate a universal comparison implementation.

Scope

The low-degree motivic identities are classical inputs, not proved by a JavaScript test. This counterexample refutes the stated universal isomorphism; it does not refute valid finite-coefficient comparison theorems in their stable ranges.

Further development

For each realization, encode its source category, coefficient change, grading shifts and theorem range as part of the type. Include tests with nontrivial units and Picard-zero classes, not only cellular ranks.

Source receipt: months/06/root_details.json record 0. Mathematical references: R120, R122.

Y25-06-011 Künneth signs and Tor corrections survive a tiny exact example

2025-06-12T15:00:47Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What does a tensor-complex implementation need to preserve, and what higher class is lost by multiplying only degree-zero ranks?

The source itself notes missing Künneth summands and signs. A two-term integer complex gives a compact counterexample to naive rank multiplication and a direct certificate for the required sign. This is a test of homological algebra rather than a claimed universal motivic solver.

Definitions and assumptions

Let C have Z in homological degrees one and zero, with differential multiplication by two. Thus $H_0(C)=Z/2$ and $H_1(C)=0$.

The tensor differential is d(x⊗y)=dx⊗y+(−1)^deg(x)x⊗dy. The classical Künneth framework uses derived tensor and Tor when coefficients are not flat.

The tensor has an additional degree-one Z/2

For D=C⊗C, the chain groups are Z,Z^2,Z in degrees two,one,zero, with $d_2=(-2,2)^T$ and $d_1=(2,2)$. Hence $H_0(D)=Z/2, H_1(D)=Z/2, H_2(D)=0$.

Derivation or proof

Choose the degree-one basis$(e_1⊗e_0, e_0⊗e_1)$. The graded sign gives $d(e_1⊗e_1)=2e_0⊗e_1−2e_1⊗e_0$, the stated d_2. Each degree-one generator maps to$2e_0⊗e_0$, giving d_1. Their composition is$−4+4=0$. The kernel of d_1 is generated by$(1,−1)$, while image(d_2) is twice that generator, so H_1 is Z/2. d_2 is injective, giving $H_2=0$, and image$(d_1)=2Z$ gives $H_0=Z/2$. Omitting the sign would make $d_1d_2=8$ instead of zero. Multiplying only H_0 groups misses the degree-one $Tor_1^Z(Z/2,Z/2)=Z/2$ contribution.

Worked example

Repeat the calculation with differential multiplication by three.

The same argument gives $H_0=Z/3, H_1=Z/3, H_2=0$. The verifier checks matrix compositions and kernel/image indices for both examples exactly. Over a field the corresponding torsion obstruction disappears because vector spaces are flat; that is an extra coefficient hypothesis, not a reason to omit Tor over Z.

A small torsion test catches both a broken differential and a missing Künneth summand.

Scope

This example does not assert that every motivic Künneth spectral sequence collapses or that an arbitrary product motive is determined by a rank table. Its domain is an explicit tensor of free complexes.

Further development

Use Smith normal forms and chain maps to certify integral homology and tensor products. A motivic layer should call that verified algebra only after providing its genuine complexes and comparison maps.

Source receipt: months/06/root_details.json record 0. Mathematical references: R123, R124.

Y25-06-012 Gysin localization and a torus example force codimension shifts

2025-06-12T15:00:47Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What shifts appear in localization, and which exact calculation checks a torus or blow-up implementation?

Mayer-Vietoris, Gysin and blow-up formulas are not interchangeable rank rules. Each has maps, shifts and geometric hypotheses. The one-dimensional torus supplies an exact low-degree test; a point blow-up of the plane supplies a second test for exceptional classes.

Definitions and assumptions

For a smooth X, smooth closed Z of codimension c and U=X\Z, motivic purity/localization gives …→H^{p−2c,q−c}(Z)→H^{p,q}(X)→H^{p,q}(U)→H^{p−2c+1,q−c}(Z)→… . This is a stated classical theorem.

For a field k, G_m=Spec k[t,t^(−1)] has units c t^n, with c∈k× and n∈Z.

The torus units and an exceptional-divisor rank check

H^{1,1}(G_m)=k×⊕Z, with the extra Z measured by the exponent of t. The codimension-one boundary map records that exponent. The classical blow-up formula adds CH^{r−1}(point) to CH^r(P^2) when blowing up a point, so CH^1 gains one independent exceptional class.

Derivation or proof

To prove the units assertion directly, write any nonzero Laurent polynomial with least and greatest exponents a≤b. Under multiplication over a field the extreme coefficients cannot cancel, so exponent width b−a adds. A product equal to one has width zero, forcing each unit to be one monomial c t^n. Multiplication adds exponents, giving k[t,t^(−1)]×≅k×⊕Z. The classical identity H^{1,1}(Y)=O(Y)× identifies this with the motivic group. For X=A^1,Z={0},c=1, localization therefore compares k× with k×⊕Z and its quotient Z, matching the shift to H^{0,0}(Z). For a smooth codimension-c blow-up, the classical Chow formula contributesΣ_{i=1}^{c−1}CH^{r−i}(Z); at a point of P^2 only one extra codimension-one generator appears. The theorem is an explicit input, while these indices and ranks are its checked consequences.

Worked example

Take the Laurent units $3t^2$ and5t^(−3), and a point blow-up of P^2.

Their product is15t^(−1), and the boundary exponents add2+(−3)=−1. The blow-up has Chow ranks1,2,1 rather than1,1,1. An algorithm returning the same rank table as P^2 would miss the exceptional divisor. The verifier checks exponent arithmetic and the shifted rank contribution.

The example gives concrete tests for maps and new classes, beyond a list of construction names.

Scope

Purity needs regular embeddings or the appropriate replacement theorem. Singular centers, arbitrary coefficients and nonflat products require additional hypotheses. No general computation of tori, quadrics or complete intersections is supplied by these two tests.

Further development

Encode localization and blow-up as exact triangles with verified shifts and maps. Derive Mayer-Vietoris by an actual cover and test exactness rather than using an unproved universal rank-addition function.

Source receipt: months/06/root_details.json record 0. Mathematical references: R121, R120.

Y25-06-015 Hamiltonian tuning dynamics need an energy and damping law

2025-06-13T15:09:39Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What does an explicit Hamiltonian tuning model conserve, and which additional term actually makes errors decay?

The source names Hamiltonian and symplectic tuning dynamics but gives no operator or energy. A declared quadratic model separates conservative motion from an optimization flow. This prevents a mathematically Hamiltonian label from being interpreted as automatic convergence to a preferred tuning.

Definitions and assumptions

Let q,p∈R^d, M,K positive definite symmetric matrices and target q*. Define $H=p^{T}M^{(-1)}p/2+(q-q^*)^{T}K(q-q^*)/2$.

Use qdot=M^(−1)p and pdot=−K(q−q*)−D M^(−1)p, with D symmetric positive semidefinite. The variables may be declared logarithmic tuning coordinates in a new model.

Conservation without damping and dissipation with it

Hdot=$-v^{T}Dv\le 0$ with v=M^(−1)p. If D=0, H is conserved and nontrivial trajectories need not converge. In the scalar model M=1,K=4,D=2, all solutions converge to the target with characteristic exponents$-1\pm i\sqrt{3}$.

Derivation or proof

Differentiate H along the trajectory. The kinetic derivative is $v^T pdot$, and the potential derivative is$(q-q^*)^T Kqdot$. Their K terms cancel, leaving$-v^T Dv$. In the undamped scalar case the positive energy level is an ellipse and motion remains on it, so the target cannot be reached asymptotically unless the initial energy is zero. With the specified damping, write $x=q-q^*$ to obtain $x''+2x'+4x=0$. Its characteristic polynomial$(s+1)^2+3$ has the stated roots; every real solution is e^(−t)(Acos√3t+Bsin√3t), proving convergence directly rather than invoking an unspecified optimizer.

Worked example

Take $q^*=0$,$q(0)=1$ and $qdot(0)=0$ in that scalar model.

Then q(t)=e^(−t)[cos(√3t)+sin(√3t)/√3] and qdot(t)=−4e^(−t)sin(√3t)/√3. Initial energy is two and $Hdot=-2qdot^2$. The verifier differentiates the analytic coefficients and checks the ODE and energy balance numerically at declared times.

Conservation, dissipation and optimization are now distinct proved behaviors rather than interchangeable descriptions.

Scope

The target, metric, stiffness and damping are newly stipulated, not recovered perceptual measurements. A symplectic integrator generally targets conservative structure; damping changes the system. No haptic or cultural interpretation follows from the ODE alone.

Further development

Fit a positive quadratic or geodesic cost from independent data, then prove stability of a discretization with explicit step-size bounds. An adjunction or coherence claim would require separately defined functors and natural transformations.

Source receipt: months/06/root_coverage.json record 0. Mathematical references: R125.

Y25-06-014 Rational musical intervals have an exact logarithmic lattice map

2025-06-13T15:42:40Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which algebraic properties do prime-factor intervals and cents satisfy, and what separation bound is justified on a finite search box?

The MUSICA source names logarithmic lattices and Baker bounds but supplies no certified constants. Prime-factor encoding already yields a rigorous group map, and a simple height bound gives a modest finite separation estimate. Neither requires a claimed perceptual or modular classifier.

Definitions and assumptions

For distinct primes $p_1,\ldots,p_d$ and $a\in\mathbb{Z}^d$, define $r(a)=\prod p_i^{a_i}>0$ and $c(a)=1200\log_2 r(a)$, measured in cents.

For two distinct ratios write $r(a)/r(b)=A/B$ in lowest positive terms. A finite exponent box bounds both A and B by a declared M.

An injective additive encoding and a finite height bound

$r(a+b)=r(a)r(b)$, $c(a+b)=c(a)+c(b)$, and c is injective on $\mathbb{Z}^d$. If $A,B\le M$ and $A\neq B$, then $|c(a)-c(b)|\ge 1200/(M\log 2)$. Without a height bound there need not be a positive uniform octave-reduced separation.

Derivation or proof

The product and logarithm identities prove the homomorphism laws. If $c(a)=0$, exponentiation gives $\prod p_i^{a_i}=1$, forcing all exponents zero by unique prime factorization; apply this to $a-b$ for injectivity. Assume $A>B$. Then $\log(A/B)=\int_B^A dt/t\ge(A-B)/A\ge 1/M$, and multiply by $1200/\log 2$; reverse A,B otherwise. A sufficient box bound for $|a_i|,|b_i|\le N$ is $M=\prod p_i^{2N}$. For octave reduction, $\log_2 3$ is irrational by prime factorization. Pigeonholing $N+1$ fractional parts $j\log_2 3$ into N equal intervals gives a nonzero multiple within $1/N$ of an integer. Thus distinct intervals can approach an octave arbitrarily closely when exponent size is unbounded.

Worked example

Compute the perfect fifth $3/2$ and the syntonic comma $81/80$.

Their cents are approximately 701.9550009 and 21.5062896. The fifth differs from seven equal-tempered semitones by about 1.955 cents. Multiplying intervals adds these values exactly before numerical approximation. The verifier checks rational prime encodings and logarithmic identities with declared tolerances.

The encoding is exact algebra; finite search separation and perceived distance are different questions.

Scope

No explicit historical Baker constant, psychoacoustic adjunction, cultural metric or classifier accuracy was recovered. A logarithmic embedding does not prove a perception model or canonical musical ∞-stack.

Further development

Fix a bounded interval vocabulary and an independently calibrated cost functional, then solve a finite nearest-lattice problem with interval-certified logarithms. A sharper linear-forms-in-logarithms bound can be added only with its actual constants and hypotheses.

Source receipt: months/06/root_coverage.json record 0. Mathematical references: R110, R126.

Y25-06-018 Tripod overlap and sensor aliasing have exact finite certificates

2025-06-15T11:40:21Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What does duty factor.7 with half-cycle tripod offset imply, and which band-limit hypothesis is needed for a λ/2 sensor-spacing rule?

The robotics source combines gait geometry, sampling, scaling and feasibility numbers. Two elementary calculations can be established without recovering an apparatus: overlap of two periodic stance intervals, and the exact aliasing identity behind spatial sampling. Neither validates the endurance or sensor-rate estimates.

Definitions and assumptions

Normalize a gait period to one. Let two tripod stance intervals each have durationβ∈[0,1], offset by1/2 on the circle.

Sample a spatial complex wave e^(ikx) on x_n=nh. A Nyquist spacing rule requires a known band limit|k|<π/h; endpoint equality needs separate conventions.

Support overlap and the spatial alias identity

Forβ≥1/2 the two tripods cover the whole cycle and overlap for fraction2β−1. Atβ=.7 the overlap is.4 and each exclusive stance occupies.3. Spatial waves with wavenumbers k and k+2πj/h have identical samples for every integer j.

Derivation or proof

Each interval has measureβ. Forβ≥1/2 their half-cycle offset leaves no gap, so union measure is one. Inclusion-exclusion gives overlap2β−1; subtracting it from each interval gives exclusive measure1−β. Forβ<1/2 the intervals are disjoint and the uncovered fraction is1−2β. For sampling, e^(i(k+2πj/h)nh)=e^(iknh)e^(2πijn)=e^(iknh), proving nonidentifiability without a band-limit prior. If the smallest modeled wavelength isλ_min and the model is genuinely band limited, choosing h<λ_min/2 prevents this aliasing within the admitted wavenumber interval. A finite set of samples does not itself establish that prior.

Worked example

Useβ=7/10 and a constructed four-sensor data stream with640×480 pixels, eight bits/pixel and100 frames/s.

The gait fractions are exactly4/10,3/10,3/10. The raw data rate is4·640·480·8·100=983040000bits/s before overhead or compression. This illustrative count shows why sensor dimensions, bit depth and frame rate are required before a reported200Gbps can be audited. The verifier checks the exact interval lengths, integer rate and alias samples.

Declared geometry and units produce real certificates, while physical load capacity and measured sensor performance remain separate.

Scope

Tripod contact does not by itself prove static stability; support polygons and the center of mass are missing. The source’s M^(3/4) power scaling, Reynolds number and endurance need their actual parameters and validation.

Further development

Specify foot trajectories and center-of-mass paths, then certify support-polygon margins over each phase interval. Pair the sampling prior with an anti-alias filter model and a complete bandwidth budget.

Source receipt: months/06/root_coverage.json record 0. Mathematical references: R127.

Y25-06-019 Resonance and regenerative accounting are separate mathematical claims

2025-06-15T12:22:14Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Where does a damped oscillator maximize displacement, and when is subtracting regenerated energy from electrical input a valid efficiency denominator?

A24Hz target, a lock interval and a power-reduction criterion are experimental specifications, not a resonance theorem. A linear oscillator supplies a precise frequency response. Energy accounting independently determines whether a regeneration term has already been subtracted by the meter.

Definitions and assumptions

Let $mx''+cx'+kx=F\cos\omega t$ with $m,k,c>0$. Its steady displacement amplitude is $A(\omega)=F/\sqrt{[(k-m\omega^2)^2+c^2\omega^2]}$.

Define gross electrical draw E_g, separately returned energy E_r and useful work W. Net input is E_g-E_r only when E_g is a gross measure; if the input is already net, subtracting E_r again double counts.

A resonance location and a net-efficiency identity

If $c^2<2km$, the positive displacement peak occurs at$\omega_r^2=k/m-c^2/(2m^2)$. Otherwise the maximal amplitude is at zero frequency. The accounting efficiency is $W/(E_g-E_r)$ under a positive, complete net denominator, not a theorem that energy has been created.

Derivation or proof

Minimize the squared amplitude denominator $D(\omega)=(k-m\omega^2)^2+c^2\omega^2$. Its derivative is$2\omega[2m^2\omega^2+c^2-2km]$. Thus an interior positive minimum exists precisely under the stated inequality, and its location is the displayedω_r. At high frequency D grows without bound, confirming the maximum. For accounting, write the control-volume balance $E_g-E_r=W+E_{loss}+\Delta E_{store}$. Under steady storage and nonnegative losses this gives $W\le E_g-E_r$. The inequality depends on including every source and return in the same boundary and horizon; an incomplete denominator can manufacture an apparent efficiency.

Worked example

Choose normalized $m=1,k=25,c=2,F=1$. Separately choose $E_g=100,E_r=20,W=60$.

The displacement peak is at$\sqrt{23}$, not the undamped frequency five. The squared denominator there is96, versus100 at$\omega=5$. Net efficiency is$60/80=.75$, while gross work ratio is.6; both are valid different ratios with declared denominators. If the meter had already reported80 net, subtracting20 again would incorrectly report one.

The frequency and energy calculations are explicit and do not certify the proposed wing testbed’s performance.

Scope

The linear oscillator omits nonlinear aerodynamic loading and controller dynamics. A frequency lock or40% reduction target requires measurements and a baseline definition. Positive regeneration numbers alone do not guarantee overall efficiency gain.

Further development

Identify the frequency response with uncertainty intervals, distinguish displacement and power peaks, and prove stability of the actual feedback loop. Report gross, returned and net energies separately so any claimed reduction is reproducible.

Source receipt: months/06/root_coverage.json record 0. Mathematical references: R128, R092.

Y25-06-021 An invented coefficient cannot turn unlike quantities into a validated scalar

2025-06-16T15:20:01Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which unit and identifiability requirements are missing from $D(t)=\Sigma\varepsilon_i\, d_i(t)$, and how can free weights reverse a conclusion?

The user explicitly identified invented coefficients and conversion factors as speculative. A weighted sum can be mathematically defined, but dimensional compatibility and independently justified weights are necessary before its numerical output has the proposed interpretation. This treatment preserves that criticism rather than repeating unsupported empirical totals.

Definitions and assumptions

Let $x=(x_1,\ldots,x_d)$ be measured components and $D_w(x)=\Sigma w_i x_i$. Either the components share a unit or each weight must convert its component into a declared common unit.

A change of component units $x_i'=c_i x_i$ with $c_i>0$ must be accompanied by $w_i'=w_i/c_i$ to preserve the scalar.

Unit invariance and arbitrary ranking under unconstrained weights

D is invariant under that coordinated unit change. Without a constraint or calibration on positive weights, even the ranking of $x=(1,0)$ and $y=(0,1)$ is undetermined: it is the sign of w_1-w_2.

Derivation or proof

Substituting $w_i'x_i'=(w_i/c_i)(c_i x_i)=w_i x_i$ proves invariance term by term. If weights are retained while units change, the same physical observation produces a different scalar, so the expression is not unit invariant. In the two-component example $D_w(x)=w_1$ and $D_w(y)=w_2$; both orders are attainable with positive weights, as is equality. A numerical matrix or entropy multiplier chosen after seeing a desired conclusion can therefore encode that conclusion rather than establish it. Moreover an entropy in dimensionless bits cannot be relabeled joules without a stated conversion model and physical conditions.

Worked example

Choose weights(2,1), then(1,2), for the same two observations.

The ranking reverses from$2>1$ to$1<2$. Rescale the first component by100 and its weight by1/100; the score remains unchanged, providing an exact unit-change test. The verifier checks these algebraic identities and does not repeat the source’s invented population or health coefficients.

The sum is a declared index only after its units and weights are supplied; its real-world meaning is not created by notation.

Scope

This result does not prohibit useful calibrated indices. It shows that free coefficients and unlike units leave conclusions unidentified. No empirical harm total or causal interpretation is inferred.

Further development

Define axioms for monotonicity, unit invariance and calibration, then determine the admissible weight set. Report only comparisons that hold for every admissible weight, producing a robust partial order rather than a fabricated precise score.

Source receipt: months/06/root_coverage.json record 0. Mathematical references: R129.

Y25-06-020 A strong correlation is not a causal policy proof

2025-06-17T17:13:49Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Why can a correlation of−.973 coexist with zero intervention effect, and what does a discounted30-year arithmetic model actually assume?

The source’s assistant called a transportation analysis a mathematical proof and treated correlations as causal certainty. A structural counterexample establishes the missing inference exactly. Discounted scenario sums can also be computed, but their inputs remain assumptions rather than proved consequences of correlation.

Definitions and assumptions

Let U,V be independent mean-zero, variance-one variables. Define X=U and Y=−U+σV. An intervention do(X=x) replaces only the equation for X.

For a stipulated constant annual quantity b, discount r>0, horizon N and initial quantity C in compatible units, define V_N=Σ_{t=1}^N b/(1+r)^t−C.

An exact confounding example and a geometric-sum formula

Corr(X,Y)=−1/√(1+σ²), yet do(X=x) leaves the distribution of Y unchanged. Choosingσ²=(1000/973)²−1 gives correlation−.973 exactly. Also V_N=b[1−(1+r)^(−N)]/r−C.

Derivation or proof

Cov(X,Y)=−Var(U)=−1, Var(X)=1 and Var(Y)=1+σ², proving the correlation. The equation for Y contains U,V but no X, so intervening on X does not change it. Observational association can therefore be arbitrarily close to−1 with zero causal effect. For the sum, set a=(1+r)^(−1); Σ_{t=1}^Na^t=a(1−a^N)/(1−a)=[1−(1+r)^(−N)]/r. Multiplication by b and subtraction of C gives the expression. Its precision cannot validate b or the assumed intervention effect.

Worked example

Use N=30,r=1/20,b=1,C=10 as a new normalized scenario.

The finite geometric sum can be checked exactly as a rational number, then compared with its closed form. If a forecasted benefit b were actually zero under the confounded intervention model, the result would instead be−10. The verifier checks the correlation variance ratio and both sum forms.

A causal model and an arithmetic projection are separate objects; a precise projection can still rest on an unjustified causal input.

Scope

No legal, fiscal, mortality or current-policy recommendation is supplied. The historical success probabilities and benefit estimates were not calibrated by this counterexample or sum identity.

Further development

Specify an identification strategy and uncertainty set for the intervention effect, then propagate that set through the finite scenario sum. Report sensitivity to unmeasured confounding and discount assumptions rather than replacing them with a correlation coefficient.

Source receipt: months/06/root_coverage.json record 0. Mathematical references: R130.

Y25-06-016 A nonlinear dyad has a rank-one Jacobian, not a bilinear tensor law

2025-06-22T16:55:09Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: Which rank, composition and conservative-field properties follow from F(x)=a f(b·x), and how does this differ from a Clifford product?

The g-dyad proposal gives an explicit nonlinear action. Its derivative and compositions can be developed without claiming that arbitrary nonlinear or hysteretic behavior is a tensor product. A conservative-field test also determines when the action can come from a scalar variational energy.

Definitions and assumptions

Let a,b∈R^d be fixed nonzero vectors and f:R→R be C^1. Define $F(x)=a\,f(b^Tx)$. For composition let $G(x)=c\,g(d^Tx)$.

A linear dyad ab^T is a matrix. The geometric product $uv=u\cdot v+u\wedge v$ instead belongs to a specified Clifford algebra; it is a different algebraic construction.

Derivative, composition and a potential criterion

$DF(x)=f'(b^Tx)ab^T$ has rank at most one. $F\circ G(x)=a\,f((b^Tc)g(d^Tx))$. If f' is nonzero somewhere, F is a gradient field on R^d exactly when a is parallel to b.

Derivation or proof

Apply the chain rule componentwise to obtain the Jacobian and its rank bound. Substitution gives the composition, which generally leaves the original activation family unless it is closed under this scalar nesting. If $F=\nabla V$ with V C^2, its Jacobian is symmetric, so $ab^T=ba^T$ wherever $f'\neq0$. For nonzero a,b this equality implies linear dependence: apply both sides to b to get $a||b||^2=b(a\cdot b)$. Conversely if $a=\kappa b$ and $F_0'=f$, then $V(x)=\kappa F_0(b^Tx)$ has gradient$\kappa bf(b^Tx)=F(x)$. Thus a variational energy exists under an explicit alignment condition, not for every chosen dyad. For f(s)=s the composition reduces to the familiar rank-one matrix identity$(ab^T)(cd^T)=(b^Tc)ad^T$.

Worked example

Choose a=(1,0),b=(0,1),$f(s)=s^2$.

$F(x,y)=(y^2,0)$, with Jacobian[[0,2y],[0,0]]. At $y\neq0$ it is nonsymmetric, so it cannot be the gradient of a C^2 potential. Choosing instead a=b=(1,0) gives $F=(x^2,0)=\nabla(x^3/3)$. The verifier checks the derivative matrices, rank and linear composition identity exactly.

The nonlinear action has useful precise structure, while bilinearity, energy and algebraic product claims need separate hypotheses.

Scope

Hysteresis is not a single-valued memoryless f unless an internal state is added. A nonlocal kernel requires its own function spaces. This example does not prove a nonlinear Koopman closure or equivalence with geometric algebra.

Further development

Define a finite sum of dyads, characterize symmetry of its full Jacobian, and derive a controlled low-rank approximation error. For state-dependent or hysteretic activations, enlarge the state space before analyzing spectra.

Source receipt: months/06/root_coverage.json record 0. Mathematical references: R076, R079.

Y25-06-017 Nonlocal dyad learning needs an adjoint in its variational update

2025-06-22T17:17:35Z · historical evidence U · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What is the correct Euler equation for an L^2 reconstruction loss with H^1 regularization, and when is its minimizer unique?

The source’s reconstruction and regularization terms are specific enough to motivate a rigorous linear inverse problem. Its update uses the adjoint of the forward operator, not the operator itself unless self-adjointness has been proved. Positivity then supplies a uniqueness certificate.

Definitions and assumptions

Let $K:H\to Y$ be a bounded linear map between real Hilbert spaces. In a finite-dimensional model define $J(u)=||Ku-y||^2/2+\alpha||u||^2/2$ with$\alpha>0$.

For a smooth field on a bounded domain, add$\beta\int|\nabla u|^2/2$ with$\beta\geq0$ and choose boundary conditions before integration by parts.

A coercive normal equation and its gradient

The finite model has unique minimizer$(K^* K+\alpha I)u=K^*y$. With an H^1 term its formal Euler equation is $K^*(Ku-y)+\alpha u-\beta\Delta u=0$, with the chosen natural or fixed boundary conditions.

Derivation or proof

Differentiate J(u+εv) at zero to obtain⟨Ku−y,Kv⟩+α⟨u,v⟩=⟨K*(Ku−y)+αu,v⟩. Thus the derivative vanishes for every v exactly at the normal equation. The quadratic form of K*K+αI is||Kv||²+α||v||²≥α||v||², so the finite matrix is positive definite and invertible, proving existence and uniqueness. For the gradient term, integration by parts givesβ∫∇u·∇v=−β∫Δu v plus boundary flux; it vanishes for fixed variations or yields a Neumann condition in the natural problem. In infinite dimensions a corresponding coercive bilinear-form theorem supplies existence under stated domain assumptions, rather than a matrix inverse being assumed automatically.

Worked example

Take K=[[1,1],[0,1]], y=(1,0),α=1 andβ=0.

The normal matrix is[[2,1],[1,3]], its determinant is five, and K*y=(1,1). Therefore u=(2/5,1/5). The forward residual is(−2/5,1/5), and K*residual+u=(0,0). For any perturbation h, J(u+h)−J(u)=(||Kh||²+||h||²)/2≥||h||²/2, an exact global optimum certificate.

The adjoint, coercivity constant and objective gap can all be verified independently of the proposed learning architecture.

Scope

Optimizing activation parameters or enforcing nonlinear spectral constraints generally destroys this convexity. A CP tensor factorization has scaling ambiguities and is not unique merely because a reconstruction error is small.

Further development

For a nonlinear forward map use its Fréchet derivative and adjoint in a Gauss-Newton step, then prove a local convergence region and a regularization-dependent error bound rather than extrapolating linear uniqueness globally.

Source receipt: months/06/root_coverage.json record 0. Mathematical references: R081.

Y25-06-022 Area, fourth-power temperature and efficiency ratios need the same balance

2025-06-24T05:21:32Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What follows from a fourth-power radiation model and the stated power ratios, and which claimed improvements remain assumptions?

The engine passage reports assumed heat-transfer and output changes. The mathematical repair fixes the dependent variables in a fourth-power model and keeps auxiliary input in the efficiency denominator. It does not validate the proposed integration or its performance.

Definitions and assumptions

For a declared ideal radiation relation use $P=cAT^4$ with $c>0$ fixed. Comparing two states gives $P_2/P_1=(A_2/A_1)(T_2/T_1)^4$.

For a complete steady energy boundary use$\eta=P_{out}/(Q_{source}+P_{aux})$. A heat moved per actuator power ratio is not automatically an energy-conversion efficiency.

An exact area-temperature scaling and denominator audit

> At fixed P, multiplying area by1.7 divides temperature by1.7^(1/4). Multiplying both area by1.7 and temperature by1.7^(1/4) instead multiplies P by$1.7^2=2.89$. The source's150/555 and150/277.5 ratios are≈.27027 and.54054 before auxiliary power.

Derivation or proof

Divide $P=cAT^4$ between states and solve for the unknown ratio. The fixed-power conclusion follows by setting $P_2/P_1=1$. If both source multipliers are inserted, their product is$1.7\cdot1.7$. For efficiency, conservation under a steady boundary requires all positive energy inputs in the denominator. Adding an auxiliary2.5 units changes the displayed ratios to150/557.5≈.26906 and150/280≈.53571. The arithmetic does not prove that the thermal input can be halved at unchanged useful output; that is a separate unverified premise. Similarly$\Delta Q/P_{acoustic}=100$ states a ratio of moved heat to actuator power, not100 times net generated energy.

Worked example

Apply the temperature multiplier to1800K only as a stipulated arithmetic example.

$1800\cdot1.7^{(1/4)}\approx2055.3450$K, while the fixed-power, increased-area value is about1576.38K. The source's approximate2052K is a rough numerical estimate, but the larger issue is which power is held fixed. The verifier checks the fourth-power identities, exact power ratios and numerical roots.

Correct algebra identifies the model assumption responsible for the apparent gain rather than presenting it as a demonstrated device result.

Scope

Real radiation, convection, combustion and storage require a complete transport model. No temperature, acoustic drive, safety limit or apparatus instruction is recommended. The source itself later called the integration unvalidated.

Further development

Write a closed input-output energy balance with uncertainty intervals and measured auxiliary power. Prove an upper bound under that boundary before fitting any claimed enhancement factor.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R092, R107.

Y25-06-024 Exhaustive finite tests prove a polynomial identity only with a degree bound

2025-06-28T00:08:43Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: When can an81-point test on F_3^4 be a proof, and when is it merely evidence about a different polynomial claim?

The source correctly distinguished computational verification from deductive proof. A finite-field identity theorem makes that distinction sharper: an exhaustive test can be a complete certificate within a specified reduced polynomial space. Without the degree restriction a nonzero polynomial can vanish at every test point.

Definitions and assumptions

Let $f \in F_3[x_1,x_2,x_3,x_4]$ have degree at most two in each separate variable. Evaluate it on every point of F_3^4.

The polynomial-function quotient is $F_3[x_1,\ldots,x_4]/(x_1^3-x_1,\ldots,x_4^3-x_4)$. Each class has a unique representative with those separate degree bounds.

A complete finite certificate and its exact counterexample

> If the reduced f vanishes on all81 points, then f is the zero polynomial. Without the bound, $x_1^3-x_1$ is a nonzero formal polynomial that vanishes at all81 points.

Derivation or proof

A univariate polynomial of degree at most two over a field with three distinct roots is zero. View f as a polynomial in x_4 with coefficients in the first three variables. For any fixed values of those three variables, its three x_4 evaluations force all three coefficients to vanish at that triple. Repeat the argument in the remaining variables; induction makes every coefficient zero. Existence of a reduced representative follows by repeatedly replacing x_i³ by x_i, lowering degree; uniqueness follows from the just-proved evaluation injectivity. There are3^81 reduced polynomials and3^81 functions on an81-point set, so evaluation is also surjective. The counterexample violates the degree restriction and is zero only in the quotient of polynomial functions.

Worked example

Test the identity$(x+y)^3=x^3+y^3$ over F_3 and distinguish it from the formal claim $x^3=x$.

The first is a true polynomial identity because intermediate binomial coefficients are divisible by three. The second is true as a function on F_3 but false in the polynomial ring. The verifier evaluates every point and independently compares coefficient dictionaries, showing both the complete bounded certificate and the danger of an unstated quotient.

A program can issue a deductive finite certificate when its theorem class and algebra are explicit.

Scope

Testing a finite domain does not prove arbitrary integer, real, analytic or number-theoretic conjectures. The recovered program description does not provide its actual proof certificates, source code or a formal soundness theorem.

Further development

Have the research engine emit a reduced polynomial, the reduction trace and all evaluation values, then check that certificate with a small independent verifier. General conjecture generation should remain separate from the trusted proof kernel.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R095.

Y25-06-023 The mod-three B_4 action has five exact orbits

2025-06-28T00:09:50Z · historical evidence A · Developed mathematical repair or classical result; speculative source claims delimited

Recovered starting point

Recovered dated mathematical source fragment; external file context only when evidence T

Revised question

Reconstruction: What group action is actually realized by signed coordinate reflections on F_3^4, and can it be identified with exceptional Weyl groups without further data?

The program description names B_4,E_6,E_7,E_8 and coordinate reflections. Signed coordinate permutations give a genuine B_4 action with an exact finite classification. Names of exceptional groups do not make them the same representation, and their orders already rule out some proposed faithful linear identifications.

Definitions and assumptions

Let B_4 be the group of signed4×4 permutation matrices over F_3. Each sign is±1, distinct in this field. It acts on the81 vectors in F_3^4.

Hamming weight w(x) is the number of nonzero coordinates. The reduced quadratic sumΣx_i^2 equals w(x) modulo three, a coarser invariant.

Order, orbits and a faithful-representation obstruction

$|B_4|=2^4\cdot 4!=384$. Its orbits are exactly the five Hamming-weight classes, of sizes1,8,24,32,16. A faithful linear action of $W(E_8)$ on F_3^4 is impossible because$|W(E_8)|=696729600$ exceeds$|GL_4(F_3)|=24261120$.

Derivation or proof

Choose a permutation of the four coordinates and an independent sign for each output coordinate; the resulting384 matrices are distinct. They preserve the number of nonzero coordinates. Conversely any vector of weight r can be moved to(1,...,1,0,...,0) by a permutation and sign flips, proving one orbit for each r. Choose its support in binomial(4,r) ways and its nonzero signs in2^r ways, giving the sizes. Their sum is81. For the obstruction, a faithful linear action embeds its group into $GL_4(F_3)$. Counting ordered independent columns gives(81−1)(81−3)(81−9)(81−27)=24261120, too small for $W(E_8)$. The quoted E_8 order is a classical root-system input; the cardinality contradiction is elementary.

Worked example

Compare vectors(1,0,0,0) and(1,1,1,1).

Both have quadratic sum one modulo three, but their Hamming weights one and four put them in different B_4 orbits. The verifier enumerates all81 vectors and all384 signed permutations, checking the orbit partition exactly. It also checks the GL order and the order obstruction.

A complete finite group-action result replaces a loose exceptional-group label without claiming a review of unrecovered source code.

Scope

A nonfaithful action or a higher-dimensional exceptional representation is not excluded. The finite orbit calculation establishes no L-function, modularity, mirror-symmetry or deductive proof capability.

Further development

Supply actual generator matrices for every claimed group and verify defining relations, group order and faithfulness. Compare orbit invariants only after that representation certificate is complete.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R131, R095.

July 2025

Dated fragments recovered; complete transcript unavailable. 35 developed treatments.

Arithmetic ascent, modular symbols, quadratic lattices, ribbon connections, geometric flows and finite proof models recur in the recovered passages. Exact examples expose which data determine a meaningful arithmetic or geometric construction.

ID · date	Concept and revised question	Result and mathematical core
Y25-07-030 2025-07-02 · U	A SAT avoidance complex does not make all 2-SAT instances contractible What does the literal avoidance complex encode, and is the claimed universal2-SAT topology correct?	Satisfying facets and a counterexample Full(n−1)-dimensional faces correspond exactly to satisfying assignments. With no clauses, the complex is the join of n two-point sets, the boundary of the n-dimensional crosspolytope, homeomorphic to S^(n−1). It is not contractible for n≥1.
Y25-07-031 2025-07-02 · A	A decision reduction does not automatically inject homology Which part of the3-SAT-to-CLIQUE construction is proved, and what extra condition would make its homology map injective?	The standard clique equivalence and the missing topological theorem The formula is satisfiable exactly when the occurrence graph has a k-clique. A simplicial map in general need not induce an injective homology map, even when it preserves an existence predicate.
Y25-07-033 2025-07-02 · U	A singleton gives an exact Frankl injection Which union-closed families admit a direct half-frequency proof, and what do small enumerations establish?	Singleton certificate If{a}∈F, then a belongs to at least|F|/2 members of F. The map A↦A∪{a} injects the members not containing a into those containing a.
Y25-07-032 2025-07-03 · A	Ollivier curvature is a transport calculation, not a hardness oracle Which graph measure defines the curvature, and can a computed value justify a complexity or cryptographic claim?	Exact curvatures of two small graphs For K_2 the unique edge has κ=0. For K_3 every edge has κ=1/2 under this measure convention.
Y25-07-035 2025-07-04 · U	A Mackey decomposition can be proved by orbit splitting What concrete finite-group identity supports the named induction/restriction framework?	An explicit double-coset decomposition Res_H^G Ind_H^G1≅1_H⊕Ind_{1}^{H}1. Over characteristic zero this is2·1_H⊕sign_H and has dimension three.
Y25-07-034 2025-07-06 · U	Bernoulli691 belongs to a precise coefficient identity What is the exact Bernoulli coefficient, and which obstruction claims follow only by definition?	The exact weight-twelve normalization $B_{12}=-691/2730$ and$-24/B_{12}=65520/691$, so $E_{12}=1+(65520/691)\Sigma\sigma_{11}(n)q^n$. For a finite-dimensional H^5, the stipulated energy is$(691/2730)\dim H^5$.
Y25-07-001 2025-07-08 · U	Modular symbols begin with a boundary relation What does a modular symbol encode, and which relations can be proved before invoking a modular-form computation?	Triangle presentation of cusp divisors The map {a,b}↦[b]−[a] identifies M with Δ_0. In particular {a,b}=−{b,a}, and every symbol is a difference of symbols based at one cusp.

ID · date	Concept and revised question	Result and mathematical core
Y25-07-002 2025-07-08 · A	Local Euler factors distinguish three coefficient sequences How does a good-prime point count determine Euler coefficients, and how do these differ from Frobenius power sums?	Good-prime recurrence For r≥2, a_{p^r}=a_pa_{p^{r−1}}−pa_{p^{r−2}}. If α,β solve X²−a_pX+p=0, then a_{p^r}=Σ_{j=0}^rα^jβ^{r−j}, whereas the Frobenius power sum is α^r+β^r.
Y25-07-003 2025-07-08 · A	Weierstrass invariants are a first step toward Tate data Which exact invariants certify smoothness and good reduction, and what still requires a local reduction algorithm?	Exact discriminant identity Δ=−b₂²b₈−8b₄³−27b₆²+9b₂b₄b₆ satisfies c₄³−c₆²=1728Δ. The Weierstrass cubic is smooth over a field exactly when its discriminant is nonzero there.
Y25-07-004 2025-07-08 · A	Selmer bounds require actual local conditions What exact sequence relates rational points, the n-Selmer group and the Tate-Shafarevich group?	Kummer-Selmer exact sequence There is an exact sequence0→E(K)/nE(K)→Sel_n(E/K)→Ш(E/K)[n]→0. For n=p prime, if the Selmer dimension is s and dim_Fp E(K)[p]=t, then rank E(K)≤s−t.
Y25-07-005 2025-07-08 · U	Height pairings turn independence into a Gram certificate How can canonical-height data certify independent points, and how does subgroup index affect a regulator?	Positive height Gram matrices and index Positive definiteness of G implies the P_i are independent modulo torsion. If their subgroup has finite index I in the full free Mordell-Weil lattice, detG is I² times the full regulator in this convention.
Y25-07-011 2025-07-10 · A	Normalized conformal growth preserves area, not every proposed entropy Which consequence follows from the stated surface-metric evolution, and which monotonicity claims need additional equations?	Area preservation under mean-zero conformal velocity The flow satisfies dA_g/dt=−2f dA_g and d Area(M,g)/dt=0. If K_ε is spatially constant, the metric is stationary.
Y25-07-012 2025-07-10 · A	A fixed-geometry heat model has a complete energy law Under which separate fixed-metric assumptions does the proposed diffusion equation actually dissipate its quadratic energy?	Dissipation of a Sobolev quadratic energy Θ′=−2∫|∇v|²−2λ∫(Δv)²−2μΘ≤0. A Laplacian eigenmode with−Δφ=κφ decays as e^(−(κ+μ)t).
Y25-07-013 2025-07-12 · A	The dimension of Ext is not the splitting class of one extension Can log(1+dim Ext¹) detect whether a particular exact sequence splits?	An exact counterexample to a dimension-based splitting test Ext¹_R(k,k)≅k. Both the split extension with M=k⊕k and the nonsplit extension with M=R have the same endpoints. Thus log(1+dim_k Ext¹)=log2 for both, although only the first splits.
Y25-07-006 2025-07-13 · U	Quaternion invariants obey a global reciprocity constraint Which local data define a global Brauer class, and what is checked by the invariant sum?	Two ramified places for Hamilton’s algebra For H=(−1,−1)_Q, the invariants are1/2 at2 and∞ and zero at every odd prime. Thus their sum is zero modulo one, while H itself is nontrivial.
Y25-07-007 2025-07-13 · U	Exact certificates can cross an untrusted solver boundary Which witness lets a small checker verify an optimization or rank answer returned by external code?	Matching primal and dual values certify optimality Feasible x,y with c^Tx=b^Ty are optimal. A matrix with a nonzero r-minor and a factorization through dimension r has exact rank r.

ID · date	Concept and revised question	Result and mathematical core
Y25-07-008 2025-07-15 · U	Vector clocks describe a partial order with a join Which merge law makes duplicate and reordered messages harmless, and what remains undecided for concurrent events?	Componentwise maximum is a least upper bound The merge ∨ is associative, commutative and idempotent, and is the least upper bound in the componentwise order. Every finite multiset of received clocks therefore has an order-independent merged value, unchanged by repetition.
Y25-07-009 2025-07-15 · A	Bayesian prediction differs from plug-in independence How do parameter uncertainty and simulation error differ in a finite reliability model?	Posterior moments govern joint predictions After three successes and one failure, E[p]=2/3 and E[p²]=10/21. Two future successes therefore have probability10/21, exceeding the plug-in value4/9 by2/63.
Y25-07-010 2025-07-15 · A	A scalar LQR controller has an exact value certificate Which feedback minimizes a stated quadratic cost, and what proves both its stability and optimum?	Stabilizing Riccati solution The feedback u=−Px gives x(t)=x₀e^(−√2t), achieves J=Px₀² and minimizes J in the declared admissible class.
Y25-07-014 2025-07-16 · U	The arithmetic-ribbon ODE is an exact scalar connection What follows from the recovered scalar ODE before adding motivic, quantum or topological interpretations?	Exact integration and modulus evolution T(t)=T(0)exp(iπΔn−(logp)Δq−2(logp)Δs+iΔθ). Hence |T(t)|/|T(0)|=p^(−Δq−2Δs). The corresponding scalar connection one-form is closed for fixed p.
Y25-07-015 2025-07-16 · A	Finite matrices cannot satisfy an exact canonical commutator Can the proposed ribbon operators satisfy[s,θ]=iℏI in a finite prototype, and what finite relation is actually possible?	Trace obstruction and the finite Weyl pair No finite matrices satisfy[A,B]=iℏI. The clock and shift instead satisfy ZX=ωXZ, X^N=Z^N=I, and are unitary.
Y25-07-016 2025-07-17 · U	The numbers 24 and 240 arise from different exact constructions Which count produces240, which q-product produces24, and what theorem is needed to connect either with moonshine?	Root count and the first discriminant coefficients The displayed E₈ root set has112+128=240 elements. The formal product gives Δ=q−24q²+252q³−1472q⁴+…. These are exact statements about different defined objects.
Y25-07-017 2025-07-17 · A	A multiple-zeta product has an exact combinatorial decomposition What concrete identity illustrates period multiplication before invoking a motivic Galois symmetry?	A stuffle identity ζ(2)ζ(3)=ζ(2,3)+ζ(3,2)+ζ(5). The identical decomposition holds for finite harmonic sums when all summation ranges are truncated consistently.
Y25-07-018 2025-07-17 · A	A two-point spectral triple computes a genuine distance Which algebra, Hilbert space and operator define the proposed spectral geometry, and what distance do they produce?	Exact two-point distance The commutator norm equals|m||a−b|, so the spectral distance is1/|m|. D is self-adjoint and has compact resolvent because H is finite-dimensional.
Y25-07-019 2025-07-17 · A	An sl₂ crystal records weights without inventing a quantum model Which combinatorial operators and weight multiplicities make a crystal claim precise?	A rank-one character decomposition For m≥1, χ_m(z)(z+z⁻¹)=χ_{m+1}(z)+χ_{m−1}(z). Raising and lowering change weights by±2 and are inverse wherever both are defined.

ID · date	Concept and revised question	Result and mathematical core
Y25-07-020 2025-07-23 · U	A genuine 7-adic obstruction explains the ternary exclusions Which claimed nonrepresentations follow from an exact congruence theorem for the recovered ternary form?	A local nonrepresentation family If N=7^(2k+1)u with7∤u and u a quadratic residue modulo seven, then F never represents N. In particular7,14,28,56,63,77 are excluded.
Y25-07-021 2025-07-23 · U	Cyclic decimals and the tuple identities require separate proofs Which exact identities hold for142857 and the tuple(384,840,1680,1840), and what inference do they permit?	Exact period, residues and corrected fractions 142857=(10⁶−1)/7, ord₇(10)=6, and(a,b,c,d)≡(−1,0,0,−1)mod7. But a/b+c/d=1103/805, and a(d−c)/b=512/7.
Y25-07-022 2025-07-25 · U	The finite ray class group of Q(√10) can be computed exactly What does the stated finite modulus actually mean, and which ray group follows without inventing infinite-place data?	Finite ray group and ordinary class group Cl(O)≅C₂ and Cl_m(O)≅Cl(O). Also narrow and ordinary class groups agree for modulus1 because the units realize all real sign patterns.
Y25-07-023 2025-07-25 · A	The claimed Hilbert class field has the wrong ramification Which quadratic extension of Q(√10) is unramified, and does the proposed polynomial generate it?	The unramified quadratic extension The Hilbert class field of K=Q(√10) is L=Q(√2,√5). The proposed K(i) is ramified and is not that field. The polynomial x⁴−18x²+121 is instead satisfied by√10+i.
Y25-07-024 2025-07-25 · A	A real quartic is not its D₄ splitting field What are the actual degree, normal closure and rational ray-group size in the recovered D₄ example?	Normal closure and ray-group audit Q(α) has degree four and is not normal. The splitting field of x⁴−2 is Q(α,i), of degree eight with dihedral group D₄. The ray class group of Q for136∞ is(Z/136Z)×, of order64, not four.
Y25-07-025 2025-07-27 · U	The quadratic sequence has a rational generating function What exact recurrence and q-series follow from the recovered g(n), and what modular claim remains unproved?	Exact rational series g(0)=3, g(n)−g(n−1)=16n−13, and F(q)=(312−552q+624q²)/(1−q)³. Also g(x)=8(x−5/16)²+71/32 for real x.
Y25-07-026 2025-07-27 · A	The proposed mock-modular shadow has incompatible weight and domain Does the asserted shadow identity have the correct weight and a convergent theta argument?	A type obstruction, with a local differential example At k=3/2 the shadow weight is1/2, not four. Moreover θ_E8(−4τ) is not defined by its positive-definite theta series for y>0, because its terms grow rather than decay.
Y25-07-027 2025-07-27 · A	A₂ and D₄ lattice data precede any Jacobi completion Which determinants and root counts follow from the actual Gram matrices, and which claimed modular structures need more?	Determinants and initial theta coefficients The A₂ form x²+xy+y² has determinant3 and six vectors of Q=1. The D₄ Gram determinant is4 and it has24 roots of Q=1. Their theta series begin1+6q+… and1+24q+… respectively.
Y25-07-028 2025-07-28 · U	A fitted density line and three beta values do not define a theory Which quantities are actually defined, and do the supplied numbers support the proposed exponential law?	A direct residual test and a nonuniqueness theorem The supplied line does not fit the first two densities closely: their residuals are about0.619 and−1.254 logarithmic units. Three beta values determine no unique function, even within smooth functions.

ID · date	Concept and revised question	Result and mathematical core
Y25-07-029 2025-07-28 · U	A quartic splitting field cannot have Mathieu Galois group What Galois groups are even possible for the two recovered quartic polynomials?	Irreducible quartics with small possible Galois groups Both f and h are irreducible over Q, but neither can have splitting-field group M_{12} or M_{24}. A separate construction would be needed to relate their roots to a Mathieu representation.

Y25-07-030 A SAT avoidance complex does not make all 2-SAT instances contractible

2025-07-02 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What does the literal avoidance complex encode, and is the claimed universal2-SAT topology correct?

The early July project proposed a homological complexity hierarchy. Its simplicial construction can be made coherent, but its asserted topological classification needs independent proof. Even the empty2-CNF formula provides a decisive counterexample to universal contractibility.

Definitions and assumptions

For n Boolean variables use vertices x_i,¬x_i. A face contains no opposite pair and no set consisting of all negated literals of a clause. A full face chooses exactly one literal per variable; it represents the corresponding truth assignment. Reduced homology uses an explicitly chosen coefficient field.

Satisfying facets and a counterexample

Full(n−1)-dimensional faces correspond exactly to satisfying assignments. With no clauses, the complex is the join of n two-point sets, the boundary of the n-dimensional crosspolytope, homeomorphic to S^(n−1). It is not contractible for n≥1.

Derivation or proof

A full face satisfies a clause precisely when it does not choose all the clause’s negations. Applying that test to every clause proves the assignment correspondence. With no clauses, only opposite-pair exclusions remain. Each variable contributes a choice of either endpoint of a0-sphere, and joining these choices gives the crosspolytope boundary. Its reduced homology in degree n−1 is one-dimensional. The formula with no clauses is a valid2-CNF formula and is trivially satisfiable, yet has nontrivial topology. Therefore neither satisfiability nor ease of decision implies contractibility for this construction. An entropy or Betti profile may still be studied as an observable, but cannot inherit a complexity theorem from its name.

Worked example

Use two variables and no clauses.

The vertices $x, \neg x, y, \neg y$ form a four-cycle with edges joining choices from different variables. It has four vertices,four edges and no triangles, so $\beta_1 = 1$. Adding suitable clauses can delete faces and change the topology without establishing a monotone decision-complexity scale.

The construction survives, while the universal2-SAT/HCH classification fails for an explicit simplest input.

Scope

This does not settle any relation between topology and worst-case complexity. A filtration and an allowed reduction class must be defined before persistence or hierarchy claims are meaningful.

Further development

Fix a functorial construction and test which reductions actually preserve or reflect the chosen homology; state resulting implications as theorems or explicit conjectures.

Source receipt: months/07/root_coverage.json record 0. Mathematical references: .

Y25-07-031 A decision reduction does not automatically inject homology

2025-07-02 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which part of the3-SAT-to-CLIQUE construction is proved, and what extra condition would make its homology map injective?

The July2 response conjectured homology injectivity from a standard decision reduction. Correctness of a reduction concerns the existence of solutions, whereas injectivity concerns every cycle and boundary. These are different statements, and a small simplicial map shows the gap.

Definitions and assumptions

For a k-clause formula, create one graph vertex per literal occurrence. Join vertices from different clauses when their literals are not contradictory. A simplicial map sends vertices to vertices and every face to a face; repeated images may lower dimension.

The standard clique equivalence and the missing topological theorem

> The formula is satisfiable exactly when the occurrence graph has a k-clique. A simplicial map in general need not induce an injective homology map, even when it preserves an existence predicate.

Derivation or proof

A satisfying assignment chooses a true literal in each clause. The chosen occurrences are pairwise noncontradictory and lie in different clauses, hence form a k-clique. Conversely a k-clique contains one occurrence from each clause, since no same-clause vertices are adjacent. Its literals are consistent and extend to a full satisfying assignment. Nothing in this argument constructs a chain-level retraction or proves a nonzero cycle survives. For a counterexample to automatic injectivity, map every vertex of a simplicial circle to one vertex. This is a valid simplicial map and induces the zero map $H_1(S^1)\to H_1(\text{point})=0$. An injective vertex map also would not suffice if the target fills a cycle with new two-simplices.

Worked example

Map the boundary of a triangle into the filled triangle by inclusion.

The source has $H_1\cong F$ and the target $H_1=0$, so even an inclusion of simplicial complexes can kill homology. A genuine homology embedding needs a separate argument such as a chain retraction or a suitable relative-homology criterion.

The decision reduction remains valid; the claimed obstruction hierarchy requires substantially more than that equivalence.

Scope

NoP-versus-NP separation or no-reduction theorem follows. Historical PHP Betti numbers are not adopted without their actual complex and boundary matrices.

Further development

Build the proposed chain maps explicitly and either construct a left inverse up to homotopy or exhibit a cycle that is killed.

Source receipt: months/07/root_coverage.json record 0. Mathematical references: .

Y25-07-033 A singleton gives an exact Frankl injection

2025-07-02 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which union-closed families admit a direct half-frequency proof, and what do small enumerations establish?

The July2 project studied Frankl’s conjecture with topological summaries and empirical counts. A simple injection gives a genuine theorem for a broad subclass. Exhaustive small-universe checks can then be certified, while correlations and reported Betti values remain distinct evidence.

Definitions and assumptions

A finite family F of subsets is union-closed if A∪B∈F whenever A,B∈F. Frankl’s conjecture asks whether every nontrivial finite such family has an element belonging to at least half its sets. Assume the union of the family is nonempty.

Singleton certificate

If{a}∈F, then a belongs to at least|F|/2 members of F. The map A↦A∪{a} injects the members not containing a into those containing a.

Derivation or proof

Union closure puts A∪{a} in F. If A and B both omit a and their unions with{a} agree, removing a gives A=B, proving injectivity. Thus the number omitting a is at most the number containing it, exactly the desired inequality. This proof makes no use of a topology or an entropy. It also explains its limitation: a family with no singleton does not admit this particular injection automatically. A complete search on a fixed finite universe can test all families by exact unions and frequencies, but it cannot prove the unbounded conjecture. Reported small Betti vectors additionally require the actual associated complex and coefficient convention.

Worked example

Use F={∅,{1},{1,2},{1,3},{1,2,3}}.

This family is union-closed and contains{1}. Four of its five sets contain1. The only set omitting1 is∅, whose injection image is{1}. A portable checker can independently enumerate every family on three labels and verify the frequency statement for that finite scope.

The exact injection is a theorem; finite enumeration is a bounded certificate; neither is replaced by a topology-complexity correlation.

Scope

No general solution of Frankl’s conjecture is claimed. The historical n=4,n=5 counts depend on an unrecovered critical-strip selection rule and are not reproduced as facts.

Further development

Specify the selected family class and seek an explicit injection or compression argument that applies beyond the singleton case.

Source receipt: months/07/root_coverage.json record 0. Mathematical references: .

Y25-07-032 Ollivier curvature is a transport calculation, not a hardness oracle

2025-07-03 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which graph measure defines the curvature, and can a computed value justify a complexity or cryptographic claim?

The July3-4 frameworks used curvature, spectral gaps and Betti numbers as proposed difficulty indicators. Those observables require precise constructions. A complete elementary transport calculation shows how curvature depends on the chosen neighborhood measure, while no security or hardness theorem is inferred.

Definitions and assumptions

For adjacent vertices x,y at graph distance one, choose probability measures m_x,m_y and define $\kappa(x,y)=1-W_1(m_x,m_y)$. W_1 minimizes$\Sigma d(u,v)\pi(u,v)$ over couplings with the declared marginals. Here use the uniform measure on neighbors, with no idleness.

Exact curvatures of two small graphs

For K_2 the unique edge has $\kappa=0$. For K_3 every edge has $\kappa=1/2$ under this measure convention.

Derivation or proof

In K_2, the neighbor measures are point masses at opposite endpoints, so transporting unit mass costs one. In K_3 with vertices x,y,z, the measures are$(\delta_y+\delta_z)/2$ and$(\delta_x+\delta_z)/2$. Match the common z-mass to itself and move the remaining half from y to x, costing1/2. This is optimal because the target has half a unit at x and the source has none there; all possible source points are distance one from x. Hence $W_1=1/2$. A different idleness convention changes the measures and can change κ, so comparison requires locked definitions. Similarly complex cochain coefficients do not by themselves create new quantum Betti numbers: for integer boundary matrices the ranks over Q and C agree.

Worked example

Compare the two degree patterns.

K_2 gives curvature zero and K_3 positive curvature despite both being trivial finite decision inputs. The value describes transport geometry of the chosen graph, not a proved attack cost or a worst-case complexity class.

A mathematically defined observable can be computed exactly while its intended predictive interpretation remains unsupported.

Scope

No cryptographic vulnerability estimate or attack procedure is provided. The historical hash-state complexes, datasets and generalization claims were not recovered sufficiently for validation.

Further development

Predefine graph construction and measures, then test a declared prediction theorem or held-out association while separating confounding degree effects from causal claims.

Source receipt: months/07/root_coverage.json record 0. Mathematical references: .

Y25-07-035 A Mackey decomposition can be proved by orbit splitting

2025-07-04 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What concrete finite-group identity supports the named induction/restriction framework?

The July4 material invoked spectral Mackey functors, chromatic theories and descent. The finite permutation case provides a complete classical Mackey calculation. It supplies an actual double-coset decomposition without pretending to construct the equivariant spectral machinery.

Definitions and assumptions

Let $G=S_3$ and $H=\{1,(12)\}$. The induced trivial representation $Ind_H^G 1$ is the permutation representation on the three left cosets G/H. Restrict it back to H and decompose by the H-orbits on those cosets.

An explicit double-coset decomposition

> $Res_H^G Ind_H^G 1 \cong 1_H \oplus Ind_{1}^{H} 1$. Over characteristic zero this is $2\cdot 1_H \oplus sign_H$ and has dimension three.

Derivation or proof

One coset is H itself and is fixed by H. The other two cosets are exchanged by (12), so their permutation module is the regular representation of C_2. A fixed orbit contributes a trivial line; a free two-element orbit contributes $Ind_1^H 1$. Equivalently $H\backslash G/H$ has representatives 1 and (13). The first intersection $H\cap H$ is H, and the second $H\cap (13)H(13)^{-1}$ is trivial, giving exactly the two terms. The two-dimensional regular module splits into symmetric and antisymmetric lines by the vectors $(1,1)$ and $(1,-1)$. This proves the representation identity directly and illustrates the orbit argument underlying the general Mackey formula.

Worked example

Evaluate characters at the identity and at (12).

The three-coset permutation character is $(3,1)$: the identity fixes all cosets, and the transposition fixes one. The right side $2\cdot\text{trivial}+\text{sign}$ gives $(3,2-1)=(3,1)$, matching the explicit orbit decomposition.

The named correspondence now has defined groups, spaces and maps with a complete finite proof.

Scope

This does not construct a functor between symmetric spectra and genuine G-spectra, prove Segal completion or identify a Morava-theoretic obstruction.

Further development

Specify the equivariant category and its transfer/restriction maps, then verify the relevant double-coset and coherence identities rather than importing them from analogy.

Source receipt: months/07/root_coverage.json record 0. Mathematical references: R132.

Y25-07-034 Bernoulli691 belongs to a precise coefficient identity

2025-07-06 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What is the exact Bernoulli coefficient, and which obstruction claims follow only by definition?

The user corrected a fabricated Bernoulli value in the July6 exchange. The correction is exact and can be reproduced rationally. The proposed obstruction energy is then a deliberately weighted dimension, not a new arithmetic explanation of cohomology or Kervaire phenomena.

Definitions and assumptions

Define Bernoulli numbers by t/(e^t−1)=ΣB_nt^n/n!. For k≥2 use the normalized Eisenstein convention E_{2k}=1−(4k/B_{2k})Σ_{n≥1}σ_{2k−1}(n)q^n. The proposed E_obs(H^n)=dimH^n·|B_{2n+2}| is a stipulated functional.

The exact weight-twelve normalization

$B_{12}=-691/2730$ and $-24/B_{12}=65520/691$, so $E_{12}=1+(65520/691)\Sigma\sigma_{11}(n)q^n$. For a finite-dimensional H^5, the stipulated energy is$(691/2730)\dim H^5$.

Derivation or proof

Multiplying the defining Bernoulli series by e^t−1 and comparing coefficients yieldsΣ_{j=0}^n C(n+1,j)B_j=0 for n≥1, with $B_0=1$. This recursively computes B_{12} exactly using rational arithmetic. Substitution into the declared Eisenstein normalization gives the coefficient. Substituting n=5 into the proposed energy gives its scalar multiple of dimension. It therefore vanishes precisely when H^5 has zero dimension, but says nothing about whether a particular class in a nonzero cohomology group vanishes. The integer691 appears because the definition chose this Bernoulli weight; a canonical arithmetic obstruction requires a separately constructed map or theorem.

Worked example

Take $\dim H^5=2$.

The chosen energy is691/1365. Replacing H^5 by another two-dimensional vector space leaves the energy unchanged regardless of its origin or any distinguished class. Likewise the list2,6,14,30,62,126 is generated by2^(j+1)−2 for j=1,…,6; that arithmetic observation is not an existence theorem for Kervaire-invariant classes.

The genuine modular coefficient is preserved while unsupported interpretations are kept separate.

Scope

No Gödel,P-versus-NP,Kervaire or quantum-error-correction implication is proved. Nonzero cohomology dimension alone does not detect a chosen obstruction class.

Further development

Construct the proposed arithmetic pairing on actual classes and prove its invariance before selecting a numerical weight.

Source receipt: months/07/root_coverage.json record 0. Mathematical references: R133.

Y25-07-001 Modular symbols begin with a boundary relation

2025-07-08 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What does a modular symbol encode, and which relations can be proved before invoking a modular-form computation?

The recovered request sought direct mathematics on modular symbols. The degree-zero cusp-divisor presentation supplies an exact starting point. It explains the triangle relation and the role of a base cusp, while keeping numerical integration and congruence-group equivariance as further requirements.

Definitions and assumptions

For $S=P^1(Q)$, let $\Delta_0=\ker[Z[S]\to Z]$, where the map sums coefficients. Let M be generated by {a,b}, with {a,a}=0 and {a,b}+{b,c}={a,c}. A weight-k modular symbol additionally uses a declared group action and coefficient module V_{k−2}.

Triangle presentation of cusp divisors

> The map $\{a,b\}\mapsto[b]-[a]$ identifies M with Δ_0. In particular {a,b}=−{b,a}, and every symbol is a difference of symbols based at one cusp.

Derivation or proof

The relations map to zero by telescoping. Fix $o\in S$ and send a degree-zero divisor $\Sigma n_a[a]$ to $\Sigma n_a\{o,a\}$. The triangle relation gives {a,b}={o,b}−{o,a}, so this is inverse on each generator. Conversely $\Sigma n_a([a]-[o])=\Sigma n_a[a]$ because $\Sigma n_a=0$. Thus both injectivity and surjectivity are proved. Equivariance can subsequently be imposed relative to a specified congruence subgroup; it does not follow from arbitrary cusp labels or a randomly populated matrix. Integration of an actual differential must likewise respect paths and convergence at cusps.

Worked example

Use o=∞ and cusps0,1,∞.

{∞,0}+{0,1}+{1,∞}=0 maps to([0]−[∞])+([1]−[0])+([∞]−[1])=0. Therefore {0,1}={∞,1}−{∞,0}. This exact relation is checkable without numerical periods.

The algebra identifies a genuine presentation rather than a theorem-shaped software record. A valid implementation must state which further quotient, coefficient representation and Hecke operators it computes.

Scope

This does not compute a cusp-form space or period lattice. A formal triangle relation alone does not certify the mocked modular-symbol routines mentioned in the historical code.

Further development

Fix $\Gamma_0(N)$, its coefficient representation and Manin relations, then compute the resulting finite module with independently checked Hecke action.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R134.

Y25-07-002 Local Euler factors distinguish three coefficient sequences

2025-07-08 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How does a good-prime point count determine Euler coefficients, and how do these differ from Frobenius power sums?

The July8 material connected modular forms, Galois representations and elliptic curves. The precise elementary bridge is a formal reciprocal quadratic. Once the local trace is declared, it gives an exact recurrence without proving the deeper modularity or representation-theoretic theorems.

Definitions and assumptions

At a good prime p put $a_p=p+1-\#E(F_p)$, and define $A(T)=\Sigma_{r\geq 0}a_{p^r}T^r=(1-a_pT+pT^2)^{-1}$. Its constant coefficient is1. All coefficient calculations take place formally, so analytic convergence is irrelevant.

Good-prime recurrence

For $r\geq 2$, $a_{p^r}=a_pa_{p^{r-1}}-pa_{p^{r-2}}$. If α,β solve $X^2-a_pX+p=0$, then $a_{p^r}=\Sigma_{j=0}^r\alpha^j\beta^{r-j}$, whereas the Frobenius power sum is $\alpha^r+\beta^r$.

Derivation or proof

Multiplying A by its denominator and comparing coefficients gives the recurrence and uniquely determines its formal inverse. Alternatively expand$(1-\alpha T)^{-1}(1-\beta T)^{-1}$ and collect degree r. In particular $a_{p^2}=a_p^2-p$, but $\alpha^2+\beta^2=a_p^2-2p$. The extension-field count uses the latter: $\#E(F_{p^2})=p^2+1-(\alpha^2+\beta^2)$. This distinction prevents an Euler-series coefficient from being silently substituted for a point-count trace. For a weight-k eigenform with character χ the determinant term instead is $\chi(p)p^{k-1}$.

Worked example

Take $E:y^2=x^3-x$ at p=5.

For x=0,1,2,3,4 the y-counts are1,1,2,2,1. Infinity makes eight points, so $a_5=-2$. The local reciprocal has coefficients1,−2,−1,12,−19 through degree4. The extension-field formula gives $\#E(F_{25})=26-(4-10)=32$, while the Euler coefficient a_{25} is−1.

An exact short calculation now separates local traces, prime-power Euler coefficients and extension-field counts.

Scope

Bad primes require different local factors. Matching a finite recurrence does not construct a Galois representation, verify an Euler system or prove BSD.

Further development

Enumerate several finite extensions for a named good prime and verify the trace relation separately from the formal Euler recurrence.

Source receipt: months/07/root_followups.json record 0. Mathematical references: R134.

Y25-07-003 Weierstrass invariants are a first step toward Tate data

2025-07-08 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which exact invariants certify smoothness and good reduction, and what still requires a local reduction algorithm?

The historical request included Tate’s algorithm and conductor data. A discriminant calculation is substantial but does not supply all of that algorithm. A complete example identifies good primes and a singular-prime gap without confusing a model invariant with a conductor exponent.

Definitions and assumptions

For $y^2+a_1xy+a_3y=x^3+a_2x^2+a_4x+a_6$, set $b_2=a_1^2+4a_2$, $b_4=2a_4+a_1a_3$, $b_6=a_3^2+4a_6$ and $b_8=a_1^2a_6+4a_2a_6-a_1a_3a_4+a_2a_3^2-a_4^2$. Set $c_4=b_2^2-24b_4$ and $c_6=-b_2^3+36b_2b_4-216b_6$.

Exact discriminant identity

$\Delta=-b_2^2b_8-8b_4^3-27b_6^2+9b_2b_4b_6$ satisfies $c_4^3-c_6^2=1728\Delta$. The Weierstrass cubic is smooth over a field exactly when its discriminant is nonzero there.

Derivation or proof

Substitution proves the polynomial invariant identity. The discriminant condition results by eliminating common zeros of the cubic and its partial derivatives. In characteristic different from two and three, for $y^2=x^3+Ax+B$ this reduces to $\Delta=-16(4A^3+27B^2)$: a repeated root r satisfies$3r^2+A=0$ and $r^3+Ar+B=0$, yielding$4A^3+27B^2=0$. An integral model with p-adic unit discriminant therefore has smooth reduction. A divisible discriminant still leaves minimality and reduction type to be checked, especially at small primes. Its valuation cannot simply be labeled the conductor exponent.

Worked example

Use E:$y^2=x^3-x$.

Here $b_2=0, b_4=-2, b_6=0, b_8=-1, c_4=48, c_6=0$ and $\Delta=64$. Thus $j=48^3/64=1728$ and every odd prime is a good-reduction prime for this model. The identity$48^3=1728\cdot64$ holds exactly. The prime two remains exceptional and is not resolved by this calculation.

The result gives a reproducible smoothness certificate with explicit limits on what a partial Tate routine can claim.

Scope

No full Tate-algorithm branch at two is executed here. Coordinate changes can multiply a model discriminant by twelfth powers, making minimality essential.

Further development

Return a certificate for each local coordinate change, residual polynomial and branch condition before reporting minimal discriminant and conductor data.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R134.

Y25-07-004 Selmer bounds require actual local conditions

2025-07-08 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What exact sequence relates rational points, the n-Selmer group and the Tate-Shafarevich group?

The historical skeleton used simplified cocycles rather than verified Euler-system arithmetic. The Kummer sequence states the genuine object a descent computation must certify and explains exactly why Selmer dimension is a rank bound rather than automatically the rank.

Definitions and assumptions

Let E/K be an elliptic curve over a number field and n≥2. Define Ш(E/K)=ker[$H^1(K,E)\to\prod_v H^1(K_v,E)$]. Sel_n consists of $H^1(K,E[n])$ classes whose image in every $H^1(K_v,E)$ is zero.

Kummer-Selmer exact sequence

There is an exact sequence0→E(K)/nE(K)→Sel_n(E/K)→Ш(E/K)[n]→0. For n=p prime, if the Selmer dimension is s and dim_Fp E(K)[p]=t, then rank E(K)≤s−t.

Derivation or proof

Multiplication by n on E over an algebraic closure is surjective with kernel E[n]. Its cohomology sequence gives0→E(K)/nE(K)→$H^1(K,E[n])\to H^1(K,E)[n]$→0, and the analogous local sequences hold. Taking preimages of locally trivial global classes proves the displayed sequence: every n-torsion class of Ш has a Kummer lift, and each local image of that lift is locally Kummer. Finite generation writes E(K)=Z^r⊕T, so E(K)/pE(K) has dimension r+t. Its injection into Sel_p yields the inequality. Equality requires ruling out the remaining Ш[p] contribution; a norm-square condition by itself need not enforce every local condition.

Worked example

Suppose independently checked certificate data give s=4 and t=1.

Then r≤3. Three rational points with a positive-definite height Gram matrix give r≥3, forcing rank three. With only two independent points known, rank two and rank three remain possible because Ш[p] may account for a Selmer dimension.

An actual rank proof needs both a valid local-condition upper bound and a certified independence lower bound.

Scope

These certificate dimensions are stipulated, not a recovered computation for a named curve. Mock cocycles and an FFI theorem name do not prove Selmer membership.

Further development

Specify an explicit descent map and every relevant local field; return exact local witnesses and the finite linear algebra computing the Selmer subgroup.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R134.

Y25-07-005 Height pairings turn independence into a Gram certificate

2025-07-08 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How can canonical-height data certify independent points, and how does subgroup index affect a regulator?

The request for height refinement and BSD quantities becomes precise once a quadratic height and free subgroup are fixed. A Gram determinant can certify independence, but saturation is a separate issue and changes the regulator by a square factor.

Definitions and assumptions

Let q be canonical height modulo torsion, with $q(nP)=n^2q(P)$ and the parallelogram law. Define $B(P,Q)=[q(P+Q)-q(P)-q(Q)]/2$, so $B(P,P)=q(P)$. For points P_i put $G_{ij}=B(P_i,P_j)$.

Positive height Gram matrices and index

> Positive definiteness of G implies the P_i are independent modulo torsion. If their subgroup has finite index I in the full free Mordell-Weil lattice, detG is I^2 times the full regulator in this convention.

Derivation or proof

The quadratic-height identities give bilinearity and $q(\Sigma c_iP_i)=c^TGc$. A torsion combination has height zero, whereas a positive-definite matrix gives a positive value whenever the integer vector c is nonzero. For the index formula express the P_i in a full lattice basis using an integer matrix C. Then $G_P=C^TG_{full}C$, so $\det G_P=(\det C)^2\det G_{full}$. The absolute determinant is the lattice index. The same equation shows why a numerically positive matrix does not alone certify a full regulator: it first needs rigorous error control and then an independent saturation argument.

Worked example

Use the stipulated Gram matrix[[2,1],[1,3]].

Its principal minors are2 and5. Its quadratic form is$2a^2+2ab+3b^2=2(a+b/2)^2+(5/2)b^2$, positive for any nonzero pair. Hence the two points would be independent and their subgroup regulator would be5. If the subgroup index were two, the full regulator would instead be5/4.

The proof explains which arithmetic conclusion a positive height certificate supports and exactly which information it leaves open.

Scope

No specific elliptic points or verified height intervals were recovered for this example. A regulator is only one factor in a BSD leading term and does not prove that conjecture.

Further development

Certify height intervals preserving positive definiteness and prove saturation at all candidate index primes before reporting the full regulator.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R134.

Y25-07-011 Normalized conformal growth preserves area, not every proposed entropy

2025-07-10 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which consequence follows from the stated surface-metric evolution, and which monotonicity claims need additional equations?

The July10 TAG response proposed a smoothed-curvature flow and an entropy-like expression. A precise area identity follows directly from the metric equation. It does not justify independently prescribing the curvature’s heat equation or declaring the proposed functional monotone.

Definitions and assumptions

Let M be a closed smooth surface with time-dependent metric g. Let $f=K_\varepsilon-\mathrm{mean}_g(K_\varepsilon)$, where the mean is the area-weighted integral divided by total area. Assume $g'=-2fg$. The smoothing operator and its kernel still require their own definitions.

Area preservation under mean-zero conformal velocity

The flow satisfies $dA_g/dt=-2f\,dA_g$ and $d\,\mathrm{Area}(M,g)/dt=0$. If K_ε is spatially constant, the metric is stationary.

Derivation or proof

For a metric variation h, differentiating the volume density gives(d/dt)dA=(1/2)tr_g(h)dA. In dimension two, tr_g(−2fg)=−4f, hence the density derivative is−2f. Integrating yields−2∫f dA=0 by the definition of the mean. This calculation needs no special smoothing theorem. By contrast, Gaussian curvature is determined by g, and under this conformal velocity its evolution is K′=Δ_g f+2Kf. An independently stipulated equation K_ε′=Δ_gK_ε−μK_ε is therefore not automatically compatible with the metric flow, especially when the smoothing operator itself depends on time. Differentiating a proposed entropy must include all these metric and operator variations.

Worked example

Compare a round sphere with a spatially varying formal curvature input.

On a round sphere K_ε is constant for any normalized symmetry-preserving smoothing, so f=0 and g′=0. For a varying input the local area densities may grow or shrink, but their total derivative remains zero. Area preservation alone does not imply pointwise curvature decay.

The actual proved geometric invariant is total area. Other desired dissipation claims remain named problems with missing hypotheses.

Scope

No existence theorem, long-time regularity or empirical growth model is established. A kernel must be normalized relative to the changing measure before its mean-preserving properties can be used.

Further development

Specify the smoothing operator on evolving metrics and derive the complete first variation of the proposed functional; then prove a sign or exhibit a counterexample.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-012 A fixed-geometry heat model has a complete energy law

2025-07-10 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Under which separate fixed-metric assumptions does the proposed diffusion equation actually dissipate its quadratic energy?

A fixed-domain heat equation gives a rigorous comparison for the historical entropy proposal. It separates a solvable linear evolution from a coupled geometric-flow claim. Both the zeroth-order and gradient energies can be differentiated exactly.

Definitions and assumptions

On a closed fixed Riemannian manifold use Δ=div grad, so ∫vΔv=−∫|∇v|². Let v_t=Δv−μv with μ≥0 and smooth real v. For λ≥0 define Θ=∫(v²+λ|∇v|²)dA.

Dissipation of a Sobolev quadratic energy

$\Theta' = -2\int|\nabla v|^2 - 2\lambda\int(\Delta v)^2 - 2\mu\Theta \le 0$. A Laplacian eigenmode with $-\Delta\varphi = \kappa\varphi$ decays as e^(−(κ+μ)t).

Derivation or proof

Differentiate $\int v^2$ to obtain $2\int v(\Delta v - \mu v)$, then integrate by parts. Differentiate the gradient term to get $2\lambda\int\nabla v\cdot\nabla(\Delta v - \mu v)$. A second integration by parts gives $-2\lambda\int(\Delta v)^2 - 2\lambda\mu\int|\nabla v|^2$. Adding the two identities proves the formula. Boundary terms vanish because the manifold is closed; periodic boundaries provide the same cancellation. Substituting $v = a(t)\varphi$ into the PDE gives $a' = -(\kappa+\mu)a$. If $\mu = 0$, the constant mode is conserved, showing why nonincreasing energy does not imply convergence to zero without a mean-zero condition or a spectral gap.

Worked example

Use a 2π-periodic circle, $v(x,0) = \cos x$, $\mu = 1$ and $\lambda = 2$.

Here $\kappa = 1$ and $v = e^{(-2t)}\cos x$. Both $\int\cos^2 x$ and $\int\sin^2 x$ equal π, so $\Theta = 3\pi e^{(-4t)}$ and $\Theta' = -12\pi e^{(-4t)}$, exactly matching the three terms in the energy identity.

A claimed entropy now has a declared norm, an explicit PDE and an equality proving its monotonicity.

Scope

The metric and measure are fixed here. Applying this proof to an evolving curvature smoothing would omit terms and is not justified by the historical formal analogy.

Further development

For a time-dependent metric, retain every volume and gradient variation and determine what extra conditions recover a dissipative functional.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-013 The dimension of Ext is not the splitting class of one extension

2025-07-12 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Can $\log(1+\dim \mathrm{Ext}^1)$ detect whether a particular exact sequence splits?

The July12 toy asserted entropy zero exactly when a triangle splits. That statement is generally false if entropy uses the dimension of the whole Ext group rather than the particular extension class. An elementary dual-number module gives both a split and a nonsplit sequence with identical endpoint Ext dimension.

Definitions and assumptions

Let $R = k[\varepsilon]/(\varepsilon^2)$ and let $k = R/(\varepsilon)$. Equivalence classes of short exact sequences $0\to k\to M\to k\to 0$ form $\mathrm{Ext}^1_R(k,k)$, whose zero class is the split extension. Group dimension counts all possible classes, not the chosen one.

An exact counterexample to a dimension-based splitting test

$\mathrm{Ext}^1_R(k,k)\cong k$. Both the split extension with $M=k\oplus k$ and the nonsplit extension with $M=R$ have the same endpoints. Thus $\log(1+\dim_k \mathrm{Ext}^1)=\log 2$ for both, although only the first splits.

Derivation or proof

A free resolution of k has every positive differential multiplication by ε: …→R→R→R→k→0. Its exactness follows because ker(ε)=εR=im(ε). Applying Hom_R(−,k) makes every differential zero, since ε acts trivially on k. Therefore $\mathrm{Ext}^1\cong k$. The sequence$0\to k\to R\to k\to 0$, with injection$1\mapsto\varepsilon$ and quotient reduction, cannot have an R-linear section: the image v of1 would satisfy εv=0, forcing $v\in\varepsilon R$, but then its quotient is zero. The direct-sum sequence splits by its coordinate inclusion. Their Ext dimension is identical because endpoints are identical.

Worked example

Compare the two middle modules R and $k\oplus k$.

In R the nilpotent ε acts nontrivially, linking the two factors; on $k\oplus k$ it acts as zero. Both have vector-space dimension two, so dimensions alone cannot distinguish the extension. The actual obstruction is the zero or nonzero class in Ext^1.

A useful obstruction functional must depend on the class and a justified zero test, not just the size of its ambient group.

Scope

Over ordinary vector spaces all short exact sequences split, so that category supplies an even weaker toy. This correction does not construct a derived-motivic or BSD entropy.

Further development

Represent the actual extension cocycle, compute whether it is a coboundary, and only then design a class-sensitive invariant with a proved splitting criterion.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-006 Quaternion invariants obey a global reciprocity constraint

2025-07-13 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which local data define a global Brauer class, and what is checked by the invariant sum?

The proposed Lean structures named cyclic algebras and local invariants but did not perform their arithmetic. Hamilton’s quaternion algebra provides genuine local data and an exact reciprocity check, separating an interface schema from a local splitting proof.

Definitions and assumptions

The quaternion algebra(a,b)_K has generators $i^2=a, j^2=b, ij=-ji$. A local split quaternion algebra has Brauer invariant0 and a division quaternion algebra has invariant1/2. The Albert-Brauer-Hasse-Noether theorem identifies global classes with finitely supported local invariant families summing to zero in Q/Z.

Two ramified places for Hamilton’s algebra

For $H=(-1,-1)_Q$, the invariants are1/2 at2 and∞ and zero at every odd prime. Thus their sum is zero modulo one, while H itself is nontrivial.

Derivation or proof

Over R its norm is a sum of four squares, anisotropic on nonzero vectors, so H is division. Over Q_2 a putative isotropic vector can be scaled to a primitive integral quadruple. Squares are0,1,4 modulo eight. If the number of odd coordinates is1,2,3 or4, their sum with the even squares cannot be0 modulo eight; hence the norm remains anisotropic. At an odd prime the local Hilbert-symbol formula for unit parameters gives$(-1,-1)_p=1$, hence splitting. The two nonsplit invariants sum to$1=0$ in Q/Z. The cited global injection detects the nontrivial class from either localization. This example checks the reciprocity constraint, rather than proving the full theorem from one case.

Worked example

Compare the actual family$\{2\mapsto 1/2, \infty\mapsto 1/2\}$ with a proposed family$\{\infty\mapsto 1/2\}$.

The first has total0 modulo one and occurs for H. The second totals1/2 and cannot be the complete localization list of any global central simple algebra. Individually valid local numbers therefore need not be globally compatible.

A correct checker needs both justified local calculations and the global compatibility condition.

Scope

This does not compute a Brauer-Manin set. That requires evaluating specified classes at adelic points, controlling all relevant places and proving those evaluations.

Further development

For a concrete cyclic algebra, return valuation and residue certificates at every possible ramified place, then verify reciprocity before forming an obstruction pairing.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R135.

Y25-07-007 Exact certificates can cross an untrusted solver boundary

2025-07-13 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which witness lets a small checker verify an optimization or rank answer returned by external code?

The historical pipeline mixed Lean, opaque foreign pointers and arithmetic engines. A binding alone is not a proof. Exact linear certificates illustrate how complex untrusted computation can propose data whose decisive properties are checked with elementary mathematics.

Definitions and assumptions

For min $c^{T}x$ subject to $Ax \geq b, x \geq 0$, the dual is max $b^{T}y$ subject to $A^{T}y \leq c, y \geq 0$. A rank-r witness can pair a nonzero $r \times r$ minor with an exact factorization $A=UV$ through an r-dimensional space.

Matching primal and dual values certify optimality

Feasible x,y with $c^{T}x=b^{T}y$ are optimal. A matrix with a nonzero r-minor and a factorization through dimension r has exact rank r.

Derivation or proof

For any feasible pair, $b^{T}y \leq (Ax)^{T}y = x^{T}A^{T}y \leq c^{T}x$ by nonnegativity. Equality at one pair excludes a smaller primal value and a larger dual value. For rank, a nonsingular r-minor gives r independent columns, while $A=UV$ bounds the image dimension by r. These proofs use only exact equalities, inequalities and dimensions. The search procedure may be complicated, but its correctness is unnecessary once the complete witness is independently checked. Conversely a solver’s status string, foreign pointer or unproved theorem declaration is not such a witness.

Worked example

Minimize $x+2y$ with $x+y \geq 3$ and $x,y \geq 0$.

The primal point(3,0) has value3. The dual scalar $\lambda=1$ satisfies $\lambda \leq 1, \lambda \leq 2$ and has the same value3. Therefore the optimum is exactly3. A floating answer2.999999 needs a feasibility/error certificate rather than being silently rounded into a theorem.

A meaningful mathematical API returns the witness and every hypothesis used by its checker.

Scope

A bounded model check proves only its finite domain. This linear certificate does not validate a nonlinear model or an elliptic-rank answer lacking genuine local and height hypotheses.

Further development

Specify exact rational reconstruction and certificate dimensions; measure checker soundness separately from search success.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R025.

Y25-07-008 Vector clocks describe a partial order with a join

2025-07-15 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which merge law makes duplicate and reordered messages harmless, and what remains undecided for concurrent events?

The platform discussion named operation DAGs and vector clocks. Their mathematical core is a join semilattice. A finite model proves the merge law while showing why it neither orders all events nor establishes arbitrary payload consistency.

Definitions and assumptions

Order N^d componentwise and set(x∨y)_i=max(x_i,y_i). Under a vector-clock discipline, a replica merges observed clocks and increments its own coordinate for a new local event. Incomparable clocks denote concurrency in that declared model.

Componentwise maximum is a least upper bound

> The merge ∨ is associative, commutative and idempotent, and is the least upper bound in the componentwise order. Every finite multiset of received clocks therefore has an order-independent merged value, unchanged by repetition.

Derivation or proof

Each identity reduces coordinatewise to the corresponding elementary maximum identity. If z bounds x and y, every z_i≥max(x_i,y_i), proving leastness. Repeated associativity and commutativity remove message ordering; idempotence removes duplicates. A total order selected later can linearize concurrent events, but adds a tie-breaking convention absent from the partial order. Further, these clock laws do not prove that every payload operation commutes: the payload must have a compatible monotone merge or separately justified operation semantics. A causal DAG and its topological order are consequently bookkeeping structures, not automatic certificates of an arithmetic computation.

Worked example

Take a=(2,0,1),b=(1,3,1),c=(0,2,4).

a and b are incomparable because each exceeds the other in one coordinate. Both(a∨b)∨c and a∨(b∨c) equal(2,3,4); replaying b changes nothing. By contrast, writing a register to7 then9 can differ from reversing those writes unless a conflict rule is specified.

The exact merge theorem identifies both the invariant state and the source of unresolved concurrency.

Scope

Vector clocks alone do not establish consensus, serializability, security or availability. Clock discipline and payload semantics are essential assumptions.

Further development

Define a payload lattice, prove every update inflationary and derive read semantics under an explicit message-delivery model.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-009 Bayesian prediction differs from plug-in independence

2025-07-15 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How do parameter uncertainty and simulation error differ in a finite reliability model?

The proposal named Bayesian risk and Monte Carlo disruption models without a recovered dataset. A fully stipulated Bernoulli model yields exact posterior predictions and reveals dependence caused by a shared unknown parameter.

Definitions and assumptions

Let p have Beta(1,1) prior and let observations be independent Bernoulli(p) conditional on p. After s successes and f failures the posterior is Beta(s+1,f+1). No actual reliability measurements are inferred here.

Posterior moments govern joint predictions

After three successes and one failure, $E[p]=2/3$ and $E[p^2]=10/21$. Two future successes therefore have probability10/21, exceeding the plug-in value4/9 by2/63.

Derivation or proof

The likelihood multiplies the uniform prior by $p^3(1-p)$, giving Beta(4,2). Beta-integral ratios give $E[p]=4/6$ and $E[p^2]=4\cdot5/(6\cdot7)$. Conditional independence yields $P(X_1=X_2=1|p)=p^2$; averaging uses $E[p^2]$, not$(E[p])^2$. The difference is $\mathrm{Var}(p)=2/63$ and represents parameter uncertainty. Separately, for N independent Bernoulli simulations at a fixed stipulated probability, Hoeffding's inequality gives $P(|\hat{p}-p|\geq\varepsilon)\leq2\exp(-2N\varepsilon^2)$. That bound controls simulation sampling error under the model, not the model's adequacy or posterior uncertainty. It cannot justify independence of actual failures.

Worked example

Compare exact prediction with a separate fixed-probability simulation target $\varepsilon=0.05,\delta=0.05$.

The posterior two-success probability is0.476190…, while plug-in independence gives0.444444…. The Hoeffding sample bound is $N\geq\mathrm{ceil}(\log40/(2\cdot0.05^2))=738$ for the separate simulation problem.

A simulation may be accurate about an assumed model while uncertainty in its parameter remains substantial.

Scope

No supply-chain or financial probability is estimated. The prior, likelihood and conditional independence are hypotheses; correlated observations require another model.

Further development

Compare a shared-parameter model with a specified dependence model and report posterior predictive and Monte Carlo uncertainty separately.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-010 A scalar LQR controller has an exact value certificate

2025-07-15 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which feedback minimizes a stated quadratic cost, and what proves both its stability and optimum?

LQR was named in the architecture list. A normalized scalar example can be solved completely rather than supplying unexplained gains. Completion of squares also exposes the terminal condition needed for an infinite-horizon argument.

Definitions and assumptions

Let $x'=x+u, x(0)=x_0$ and $J=\int_0^\infty (x^2+u^2)dt$. Compare controls of finite cost with $x(t)\to 0$. Put $P=1+\sqrt{2}$, the positive root of$2P-P^2+1=0$.

Stabilizing Riccati solution

The feedback $u=-Px$ gives $x(t)=x_0e^{(-\sqrt{2}t)}$, achieves $J=Px_0^2$ and minimizes J in the declared admissible class.

Derivation or proof

For $V=Px^2$, $V'=2Px^2+2Pxu$. Since$2P+1=P^2$, $x^2+u^2+V'=(u+Px)^2$. Integration to T gives $J_T=Px_0^2-Px(T)^2+\int_0^T (u+Px)^2dt$. The terminal term tends to zero in the stated class, proving the lower bound. The feedback makes the square vanish and has decay rate$1-P=-\sqrt{2}$, so it attains equality. The other Riccati root$1-\sqrt{2}$ gives unstable feedback and is not a minimizing infinite-horizon certificate. Thus choosing the stabilizing solution is part of the theorem, not a numerical preference.

Worked example

Take $x_0=1$ and compare with $u=-2x$.

The optimal cost is$1+\sqrt{2}\approx 2.414214$. Under $u=-2x$, $x=e^{(-t)}$ and $J=\int 5e^{(-2t)}dt=5/2$, strictly larger. The gap is the integral of the nonzero completed square.

The plant, cost, admissible class, stability rate and value are all explicit and independently checkable.

Scope

This scalar model assumes exact state observation and no constraints or disturbances. It gives no operational prescription or performance guarantee for an original apparatus.

Further development

Add a bounded-input constraint and derive the changed value equation and invariant region before comparing policies.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R136.

Y25-07-014 The arithmetic-ribbon ODE is an exact scalar connection

2025-07-16 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What follows from the recovered scalar ODE before adding motivic, quantum or topological interpretations?

The July16 ribbon equation can be integrated exactly when its variables and parameter are defined. That gives a legitimate classical connection and modulus law. It does not make integer labels differentiable or create a nonzero curvature from constant scalar coefficients.

Definitions and assumptions

Fix p>1. Let n,q,s,θ be differentiable real functions and let T′=T[iπn′−(logp)q′−2(logp)s′+iθ′]. If n is required to be integer-valued and continuous, it is locally constant.

Exact integration and modulus evolution

T(t)=T(0)exp(iπΔn−(logp)Δq−2(logp)Δs+iΔθ). Hence |T(t)|/|T(0)|=p^(−Δq−2Δs). The corresponding scalar connection one-form is closed for fixed p.

Derivation or proof

Define F=iπn−(logp)q−2(logp)s+iθ. Differentiation shows(T e^(−F))′=0, proving the formula even when T(0)=0. Taking real parts of F proves the modulus identity. Locally the connection is dF, so its curvature d(dF) is zero. If θ is an angular coordinate, dθ is locally exact but its loop integral can be2πm; exponentiation still gives unit holonomy e^(2πim)=1 in this normalization. Allowing p to vary adds−(q+2s)dp/p to dF, so the fixed-p formula cannot be reused unchanged.

Worked example

Take p=2,n=q=0,s=t and θ=3t.

The solution is T(t)=T(0)2^(−2t)e^(3it), so the modulus at t=1 is one quarter of its initial value. A separate target|z|=2^(−s) would require a different variable definition or coefficient; it is not the same modulus law automatically.

The equation produces an explicit amplitude and phase law, with every additional interpretation remaining separate.

Scope

No perfectoid tilting, Adams differential, motivic trace or physical measurement theorem follows from scalar integration. Derivatives of discrete labels require a specified interpolation or a difference equation.

Further development

Specify the base space, line bundle and transition functions, then compute genuine global holonomy and compare a chosen discrete version with the continuous solution.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-015 Finite matrices cannot satisfy an exact canonical commutator

2025-07-16 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Can the proposed ribbon operators satisfy[s,θ]=iℏI in a finite prototype, and what finite relation is actually possible?

The assistant promoted formal commutators into legitimate structures without an operator domain. A trace argument gives an immediate obstruction for finite matrices. Clock and shift matrices provide a precise alternative, with an explicit deformation phase rather than a false finite canonical pair.

Definitions and assumptions

For N×N matrices define[A,B]=AB−BA. Fix $\hbar \neq 0$. On C^N let ω=e^(2πi/N), Xe_j=e_{j+1 mod N} and Ze_j=ω^je_j.

Trace obstruction and the finite Weyl pair

> No finite matrices satisfy[A,B]=iℏI. The clock and shift instead satisfy ZX=ωXZ, X^N=Z^N=I, and are unitary.

Derivation or proof

Cyclicity gives tr(AB−BA)=0, while tr(iℏI)=iℏN≠0, a contradiction. For the finite pair, ZXe_j=ω^(j+1)e_{j+1} and XZe_j=ω^je_{j+1}, proving the Weyl relation on a basis. Each map permutes or phases an orthonormal basis, so it is unitary. Iterating N times returns every basis vector. This multiplicative relation does not contradict the trace obstruction and should not be silently differentiated into an exact additive commutator of finite logarithms. Infinite-dimensional canonical operators require dense invariant domains and careful treatment of unbounded products.

Worked example

Take N=3.

The clock is diag(1,ω,ω²), and the shift is the cyclic permutation matrix. Their order-dependent product differs by the cube root of unity ω. Any claimed three-state realization of[A,B]=iℏI fails before its eigenvalues are computed.

A prototype must choose a finite Weyl model, an approximation with quantified error, or a properly defined infinite-dimensional representation.

Scope

This does not rule out canonical commutation relations on suitable infinite-dimensional spaces. It rules out the exact finite identity, not all discretizations or physical models.

Further development

Choose an explicit discretization and bound its commutator error on a stated low-energy subspace instead of claiming an exact finite canonical relation.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-016 The numbers 24 and 240 arise from different exact constructions

2025-07-17 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which count produces240, which q-product produces24, and what theorem is needed to connect either with moonshine?

The July17 discussion connected E_8, Ramanujan coefficients, the Leech lattice and tmf. An exact root count and a formal product calculation retain real mathematics without treating repeated integers as evidence of an unspecified common mechanism.

Definitions and assumptions

The E_8 roots consist of vectors with two coordinates±1 and six zeros, together with half-integral vectors$(\pm1/2,\ldots,\pm1/2)$ having an even number of minus signs. Every root has squared length2. Define $\Delta(q)=q\prod_{m\geq1}(1-q^m)^24$.

Root count and the first discriminant coefficients

The displayed E_8 root set has$112+128=240$ elements. The formal product gives $\Delta=q-24q^2+252q^3-1472q^4+\cdots$. These are exact statements about different defined objects.

Derivation or proof

Choose the two nonzero integral positions in $C(8,2)=28$ ways and their signs in four ways, giving112 roots. Among the2^8 half-integral sign choices exactly half have even parity, giving128. For Δ, each coefficient depends on finitely many factors. Through q^3 in the product, use$(1-q)^24,(1-q^2)^24,(1-q^3)^24$; the binomial expansion gives$1-24q+252q^2-1472q^3$. The deeper classical identity $\theta_{E8}=E_4$ follows from theta modularity for the even unimodular rank-eight lattice and the one-dimensional weight-four modular-form space. It is a separate theorem; the root count checks its first nonconstant coefficient240 rather than proving every possible moonshine correspondence.

Worked example

Compare the initial E_4 and Δ expansions.

$E_4=1+240\sum_{n\geq1}\sigma_3(n)q^n$ starts$1+240q+2160q^2+6720q^3$. Δ starts $q-24q^2+252q^3$. A coefficient24 in one expression is not the E_8 root count240, and neither equals the Leech kissing number196560.

The constants acquire meaning through explicit counting and transformation laws, not arithmetic resemblance alone.

Scope

No VOA, tmf orientation or sporadic-group action is constructed. Those claims need actual representations, graded spaces and identities, not a list of familiar constants.

Further development

Construct the proposed graded representation and verify enough exact character data together with a uniqueness theorem, rather than inferring a module from a few dimensions.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R132.

Y25-07-017 A multiple-zeta product has an exact combinatorial decomposition

2025-07-17 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What concrete identity illustrates period multiplication before invoking a motivic Galois symmetry?

The historical synthesis mentioned multiple zeta values and motivic symmetry. A convergent double-sum identity is a rigorous local piece of that programme. It demonstrates an algebraic relation while keeping the motivic lift and its symmetry group as additional constructions.

Definitions and assumptions

For a,b>1 define ζ(a)=Σ_{m≥1}m^(−a) and ζ(a,b)=Σ_{m>n≥1}m^(−a)n^(−b). Finite truncations use the same upper bound N on both indices. These positive series converge absolutely.

A stuffle identity

ζ(2)ζ(3)=ζ(2,3)+ζ(3,2)+ζ(5). The identical decomposition holds for finite harmonic sums when all summation ranges are truncated consistently.

Derivation or proof

Expand the product as a sum over all ordered pairs(m,n). Partition the index set into m>n,n>m and m=n. On the first part the summand is m^(−2)n^(−3), giving ζ(2,3). Interchanging names on the second yields ζ(3,2). The diagonal gives Σm^(−5). Absolute convergence justifies partitioning and reindexing; alternatively perform the identity for finite N and use monotone convergence. This proof is an exact relation among analytic numbers. Calling them motivic periods requires a specified category and comparison map; the summation identity by itself does not construct the motivic Galois group or prove algebraic independence.

Worked example

Truncate at N=2.

$H_2^{(2)}=5/4$ and $H_2^{(3)}=9/8$, whose product is45/32. The m>n contributions are1/4 and1/8, while $H_2^{(5)}=33/32$. Their sum is(8+4+33)/32=45/32 exactly.

The example provides a replayable finite certificate and a convergent limiting proof, with no numerical-recognition step.

Scope

One product identity does not classify all multiple-zeta relations, determine transcendence degrees or establish a geometric Langlands correspondence.

Further development

Define the relevant iterated integrals, derive their shuffle relations as well, and compare the two algebra structures before proposing a motivic interpretation.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-018 A two-point spectral triple computes a genuine distance

2025-07-17 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which algebra, Hilbert space and operator define the proposed spectral geometry, and what distance do they produce?

The source named spectral triples and physics links without defining an operator. A complete two-point triple shows how an operator norm can encode a metric. It is a classical finite example, not a construction of the Standard Model or an arithmetic spectral realization.

Definitions and assumptions

Represent $A=C^2$ on $H=C^2$ by diag(a,b). Let $D=[[0,m],[\overline{m},0]]$ with $m\neq0$. For the two evaluation states δ_1,δ_2, define distance $\sup\{|a-b|:a,b \text{ real}, ||[D,\mathrm{diag}(a,b)]||\leq1\}$.

Exact two-point distance

> The commutator norm equals$|m||a-b|$, so the spectral distance is$1/|m|$. D is self-adjoint and has compact resolvent because H is finite-dimensional.

Derivation or proof

Matrix multiplication gives$[D,\mathrm{diag}(a,b)]=[[0,m(b-a)],[\overline{m}(a-b),0]]$. Its adjoint-times-itself is$|m|^2|a-b|^2I$, so both singular values equal$|m||a-b|$. The constraint is therefore$|a-b|\leq1/|m|$, and equality is attained by $a=1/|m|,b=0$. The supremum is thus exact, not merely bounded. Scaling D by a positive factor c divides the distance by c. If $m=0$ the constraint imposes no restriction on $a-b$ and the distance is infinite, illustrating why operator coupling is essential. More elaborate triples need domains, representation and regularity conditions that are trivial only in this finite case.

Worked example

Take $m=2$.

The allowed difference is at most1/2, and a=1/2,b=0 attains it. The eigenvalues of D are±2, while the distance is1/2; these are related by the chosen operator, not by a free numerical identification.

Every component of the metric construction is now supplied and its consequence proved.

Scope

No field theory, arithmetic L-function or higher-dimensional geometry follows from this two-point example. A named spectral triple without an algebra action and analytic domain is incomplete.

Further development

For a finite weighted graph triple, derive its distance constraints and determine exactly when the result agrees with a chosen path metric.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-019 An sl_2 crystal records weights without inventing a quantum model

2025-07-17 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which combinatorial operators and weight multiplicities make a crystal claim precise?

The July17 synthesis named crystal bases and quantum groups. The rank-one crystal supplies an explicit, checkable combinatorial structure and character identity. It does not turn a crystal diagram into a physical lattice or an arbitrary categorical symmetry.

Definitions and assumptions

For $m\geq 0$, $B(m)=\{0,\dots,m\}$ has weight $wt(k)=m-2k$. Define lowering $f(k)=k+1$ when $k<m$ and raising $e(k)=k-1$ when $k>0$, with an undefined symbol outside the chain. Its formal character is $\chi_m(z)=\Sigma_{k=0}^{m} z^{(m-2k)}$.

A rank-one character decomposition

For $m\geq 1$, $\chi_m(z)(z+z^{-1})=\chi_{m+1}(z)+\chi_{m-1}(z)$. Raising and lowering change weights by± 2 and are inverse wherever both are defined.

Derivation or proof

The operator assertions follow directly from the chain definitions. For the character identity multiply the finite geometric-weight sum by z and z^{-1}. The extreme weights $m+1$ and$-m-1$ occur once; every interior weight from $m-1$ to$-m+1$ occurs twice. χ_{m+1} supplies one copy of every weight in that range, and χ_{m-1} supplies the second interior copy. This proves the identity as a Laurent polynomial. It matches the classical sl_2 tensor decomposition $V_m\otimes V_1\cong V_{m+1}\oplus V_{m-1}$, but constructing a full quantum-group crystal basis entails a module, integral form and limiting procedure beyond this combinatorial verification.

Worked example

Use m=2.

B(2) has weights2,0,−2. Multiplication by$\chi_1=z+z^{-1}$ gives $z^3+2z+2z^{-1}+z^{-3}$, equal to$\chi_3+\chi_1$. Evaluating at z=1 gives3·2=4+2, the corresponding dimension check.

The diagram and coefficients are now defined objects with exact identities, rather than symbols standing in for unconstructed symmetry.

Scope

The rank-one example does not produce an E_8 crystal, a canonical basis for every representation or a link to the proposed ribbon operators.

Further development

Specify the Cartan data and tensor-product rule for a higher-rank module, then verify its crystal axioms and character against an independent representation computation.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R132.

Y25-07-020 A genuine 7-adic obstruction explains the ternary exclusions

2025-07-23 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which claimed nonrepresentations follow from an exact congruence theorem for the recovered ternary form?

The July23 form has a real arithmetic obstruction. It can be proved by completing squares modulo seven and descending through powers of49, without any speculative tmf or Langlands construction. This is stronger than checking a few missing values numerically, but it remains a necessary condition rather than a complete representation classification.

Definitions and assumptions

Let $F(x,y,z)=x^2+2y^2+3z^2+xz+yz$ on Z^3. For N≠0 write $v_7(N)$ for its7-adic valuation. The nonzero quadratic residues modulo seven are1,2,4.

A local nonrepresentation family

If N=7^(2k+1)u with7∤u and u a quadratic residue modulo seven, then F never represents N. In particular7,14,28,56,63,77 are excluded.

Derivation or proof

Modulo seven, $F=(x+4z)^2+2(y+2z)^2$. Because$-2\equiv5$ is not a square modulo seven, $F\equiv0$ forces $x\equiv3z$ and $y\equiv5z$. Write $x=3z+7a, y=5z+7b$. Expansion gives $F/7=10z^2+7az+21bz+7a^2+14b^2$, hence $F/7\equiv3z^2$. If $v_7(F)=1$, its unit part must lie in$\{3,5,6\}$, the nonresidues, not$\{1,2,4\}$. If49 divides F, this congruence forces $z\equiv0$ and then $x,y\equiv0$; dividing all coordinates by seven replaces F by F/49. Iterating reduces any odd valuation to the first case, proving the theorem. Positivity and a finite search bound follow independently from$8F=2(2x+z)^2+(4y+z)^2+21z^2$.

Worked example

Apply the criterion to the six historical numbers.

Their quotients by seven are$1,2,4,8,9,11$, reducing to$1,2,4,1,2,4$. Every valuation is one and every unit residue is excluded. Conversely the theorem does not claim every allowed local residue is globally represented.

The historical list has an exact local explanation, with a proved infinite family rather than an inferred conceptual analogy.

Scope

Local admissibility need not imply integral representation. Genus, spinor genus and possible exceptional square classes require additional work.

Further development

Compute all local conditions and investigate whether they are sufficient for this specific form, using a proved genus/spinor-genus analysis rather than finite extrapolation.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R059.

Y25-07-021 Cyclic decimals and the tuple identities require separate proofs

2025-07-23 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which exact identities hold for142857 and the tuple$(384,840,1680,1840)$, and what inference do they permit?

The assistant corrected two false rational equalities in the July23 exchange. The tuple’s modular pattern and the repeating decimal are real, but they do not restore those equalities or prove a bridge to an obstruction theory. Exact fractions give a complete audit.

Definitions and assumptions

The decimal period of$1/7$ is the multiplicative order of10 modulo7. Let $a=384, b=840, c=1680, d=1840$. Arithmetic below is over integers and rational numbers, with no inferred physical interpretation.

Exact period, residues and corrected fractions

142857=(10⁶−1)/7, ord₇(10)=6, and(a,b,c,d)≡(−1,0,0,−1)mod7. But a/b+c/d=1103/805, and a(d−c)/b=512/7.

Derivation or proof

The residues10^j mod7 for j=0,…,6 are1,3,2,6,4,5,1, so the first return occurs at six. Long division then gives1/7=0.overline142857. Multiplying a six-digit repeating block by10 rotates it modulo10⁶−1, explaining the cyclic pattern without a new geometric object. For the tuple reduce each integer modulo seven directly. Rational simplification gives384/840=16/35 and1680/1840=21/23; their sum is(368+735)/805=1103/805. Also384·160/840=512/7. These exact values differ from3/2 and2048 respectively; modular agreement of selected integers cannot force either rational identity.

Worked example

Check the first cyclic multiple and the historical target comparison.

2·142857=285714, a rotation. Meanwhile1103/805≈1.3701863354, and512/7≈73.14285714. The common appearance of a seventh does not identify these quantities with a BPS invariant or a categorical boundary.

Every retained arithmetic observation is justified individually, while the proposed structural analogy remains unproved.

Scope

A repeating decimal and congruence tuple do not define a genus-three curve, an entropy, a motivic lift or a tmf class.

Further development

If a geometric interpretation is intended, define the objects and the map producing these exact integers, then prove its compatibility with the displayed arithmetic rather than fitting an analogy afterward.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-022 The finite ray class group of Q(√10) can be computed exactly

2025-07-25 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What does the stated finite modulus actually mean, and which ray group follows without inventing infinite-place data?

The July25 arithmetic contains enough exact data for a real computation. Treat the modulus as the stated finite ideal, separately from later contradictory narrow-modulus assertions. This gives a class-group proof and an explicit ray-kernel calculation.

Definitions and assumptions

Let $K=Q(\sqrt{10})$, $O=Z[\sqrt{10}]$, discriminant40, $P_2=(2,\sqrt{10})$, and $m=(3,1+\sqrt{10})$. Its quotient is F_3 through $\sqrt{10}\mapsto -1$, so Norm(m)=3. The finite ray group uses principal generators congruent to1 modulo m, without a positivity condition.

Finite ray group and ordinary class group

$Cl(O)\cong C_2$ and $Cl_m(O)\cong Cl(O)$. Also narrow and ordinary class groups agree for modulus1 because the units realize all real sign patterns.

Derivation or proof

Minkowski gives a representative of every class with norm$\le\sqrt{40}/2<3.163$. Thus norm-two and norm-three primes generate. $P_2^2=(2)$, while P_2 is nonprincipal: an element of norm± 2 would require $a^2-10b^2=\pm 2$, impossible modulo five. The element$2+\sqrt{10}$ has norm-6 and factors as P_2 times the norm-three prime with $\sqrt{10}\equiv 1$. The conjugate norm-three prime has inverse class; hence both have the same order-two class as P_2. These exhaust the Minkowski generators, proving class number two. The ray exact sequence $O^\times\to(O/m)^\times\to Cl_m\to Cl\to 1$ has trivial kernel term because-1 maps to the nontrivial unit of F_3. Finally$3+\sqrt{10}$ has norm-1 and signs(+,-); together with-1 it realizes all four real sign patterns, proving narrow=ordinary.

Worked example

Audit the modulus norm and the residue-unit image.

The map $a+b\sqrt{10}\mapsto a-b$ modulo three has kernel m. The two residue units are1,2, both images of global units1,-1. No extra ray factor survives. A computation treating m as norm9 would use the wrong quotient ring.

The ray answer is now tied to a precise modulus and exact sequence rather than a guessed class field.

Scope

Adding real places to this finite modulus changes the ray condition and requires recomputing the joint residue/sign image; the separate narrow-modulus1 result does not settle every combined modulus.

Further development

Write the full finite and infinite modulus, compute its unit image, and construct its Artin map before claiming principalization or an inversion algorithm.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R059, R135.

Y25-07-023 The claimed Hilbert class field has the wrong ramification

2025-07-25 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which quadratic extension of $Q(\sqrt{10})$ is unramified, and does the proposed polynomial generate it?

The assistant identified K(i) as a Hilbert class field and later gave incompatible narrow data. Discriminants and real embeddings give decisive exact tests. The resulting correction is an actual class-field identification, using the independently proved class number two.

Definitions and assumptions

For a biquadratic extension of Q, its field discriminant is the product of the discriminants of its three quadratic subfields. For L/K, Disc(L/Q)=Disc(K/Q)^[L:K]·Norm_K/Q(disc(L/K)). Use the ordinary Hilbert class field, unramified also at real places.

The unramified quadratic extension

> The Hilbert class field of K=Q(√10) is L=Q(√2,√5). The proposed K(i) is ramified and is not that field. The polynomial x^4-18x^2+121 is instead satisfied by√10+i.

Derivation or proof

The quadratic discriminants for L are8,5,40, whose product is$1600=40^2$. Hence the relative discriminant has norm one and is the unit ideal, so no finite prime ramifies. L is totally real, so no real place ramifies either. It is a nontrivial degree-two extension of K. Since Cl(K) has order two, class field theory identifies it with the Hilbert class field. For K(i), the subfield discriminants are40,−4,−40, whose product is6400. Its relative discriminant norm is$6400/40^2=4$, already excluding finite unramifiedness; its complex embeddings also ramify the real places. Finally$(√10+i)^2=9+2i√10$, so squaring the shifted square gives the stated quartic.

Worked example

Compare two primitive-element polynomials.

√2+√5 satisfies $x^4-14x^2+9=0$, whereas√10+i satisfies $x^4-18x^2+121=0$. A polynomial defining a degree-four extension does not identify its ramification or make it the desired class field. Polynomial discriminants may also differ from field discriminants by an index square.

The correction distinguishes a genuine unramified extension from a plausible-looking but wrong defining polynomial.

Scope

The biquadratic discriminant theorem and global class-field correspondence are invoked as classical results. This does not compute arbitrary ray fields or Stark units.

Further development

Construct integral bases and exact relative discriminant ideals for any further proposed field, then verify its Artin behavior at several unramified primes.

Source receipt: months/07/root_details.json record 0. Mathematical references: R059, R135.

Y25-07-024 A real quartic is not its D₄ splitting field

2025-07-25 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What are the actual degree, normal closure and rational ray-group size in the recovered D₄ example?

A later July25 source explicitly describes an assistant-generated example, despite being nested in a user-fact retrieval line. It is therefore treated as assistant evidence. Exact field containment and elementary congruence groups refute its central identifications.

Definitions and assumptions

Let α=2^(1/4)>0. Since α²=√2, Q(α,√2)=Q(α). The proposed rational modulus is136∞, where the infinite place imposes positive principal generators.

Normal closure and ray-group audit

Q(α) has degree four and is not normal. The splitting field of x⁴−2 is Q(α,i), of degree eight with dihedral group D₄. The ray class group of Q for136∞ is(Z/136Z)×, of order64, not four.

Derivation or proof

Eisenstein at two proves x⁴−2 irreducible. Q(α) is real and therefore omits the root iα, so it is not a splitting field. Adjoining i doubles the degree. Automorphisms r:α↦iα,i↦i and s:α↦α,i↦−i obey r⁴=s²=1 and srs=r⁻¹, giving the eight-element dihedral group. For the ray group, Q has trivial ordinary class group, and positive principal generators leave the full residue-unit group modulo136=8·17. CRT gives orderφ(8)φ(17)=4·16=64. Without the infinite place the quotient by±1 would instead have order32. Neither convention gives C₂².

Worked example

Check the asserted split-prime17 behavior.

Modulo17, x⁴−2=(x²−6)(x²+6). Neither6 nor11 is a square modulo17, so both quadratics are irreducible and17 does not split into four degree-one primes in Q(α). The exact identityα⁴=2 also makes(α)^4=(2), but the prime ideal(α) is already principal.

Degree, normality, modulus and splitting are independent checks, all of which must precede an arithmetic interpretation.

Scope

No recovered numerical L-derivative or Stark-unit claim is validated. A nonabelian D₄ extension cannot itself be identified with an abelian ray class field over Q.

Further development

Compute explicit Frobenius cycle types and the abelian subextensions of the actual normal closure, with each modulus stated separately.

Source receipt: months/07/root_details.json record 0. Mathematical references: R059, R135.

Y25-07-025 The quadratic sequence has a rational generating function

2025-07-27 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What exact recurrence and q-series follow from the recovered g(n), and what modular claim remains unproved?

The July27 sequence is a well-defined elementary object. Its recurrence, positivity and generating function can be proved completely. Those statements do not establish mock modularity or a sporadic-group representation.

Definitions and assumptions

For n≥0 let g(n)=8n²−5n+3 and a_n=240+24g(n). Let F(q)=Σ_{n≥0}a_nq^n, formally or analytically for|q|<1.

Exact rational series

g(0)=3, g(n)−g(n−1)=16n−13, and F(q)=(312−552q+624q²)/(1−q)³. Also g(x)=8(x−5/16)²+71/32 for real x.

Derivation or proof

Subtract the two quadratic values to obtain the first difference. Completing the square proves positivity and explains the discriminant−71 without constructing a quadratic field or moonshine object. Since a_n=312−120n+192n², substitute the formal identitiesΣq^n=(1−q)⁻¹, Σnq^n=q/(1−q)² andΣn²q^n=q(1+q)/(1−q)³. Collecting numerators yields312−552q+624q². Multiplying the denominator back gives an independent third-difference recurrence. All of this is classical polynomial enumeration. A rational q-expression is not a modular transformation law, and a negative discriminant in a quadratic polynomial does not define the missing lift or representation.

Worked example

Compute the initial coefficients.

g(0),…,g(4) are3,6,25,60,111, so a_n begins312,384,840,1680,2904. In particular the historical tuple’s last value1840 is not a_4. The pole at q=1 has order three and matches quadratic coefficient growth.

The actual sequence and its exact analytic/formal structure are now certified, while the claimed mock completion remains a separate problem.

Scope

No CHL, Donaldson-Thomas, M_{24} or E_8 module is constructed. Finitely many coefficient coincidences cannot prove such an identification.

Further development

State a group, multiplier, weight and transformation law; determine whether a correctly typed completion can satisfy them with this holomorphic part.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-026 The proposed mock-modular shadow has incompatible weight and domain

2025-07-27 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Does the asserted shadow identity have the correct weight and a convergent theta argument?

The prior assistant claimed a weight3/2 completion with E_8 theta shadow and uniqueness. Before any coefficient fitting, the operator's transformation weight and the analytic domain must agree. Two elementary checks already expose the missing structure.

Definitions and assumptions

For $\tau=x+iy$ in the upper half-plane define $\xi_k F=2iy^k \overline{(\partial F/\partial\overline{\tau})}$. On appropriately transforming harmonic Maass forms it has weight$2-k$. A positive-definite rank-eight theta series has weight four and converges for positive imaginary argument.

A type obstruction, with a local differential example

> At $k=3/2$ the shadow weight is$1/2$, not four. Moreover $\theta_{E8}(-4\tau)$ is not defined by its positive-definite theta series for $y>0$, because its terms grow rather than decay.

Derivation or proof

The weight$2-k$ follows by applying the antiholomorphic chain rule to the weight-k transformation and conjugating; this is the standard ξ-operator covariance. Independently, a theta term $\exp(\pi i(-4\tau)||v||^2)$ has modulus $\exp(4\pi y||v||^2)$, so the terms do not even tend to zero along lattice vectors. Thus the literal historical right-hand side is not a holomorphic theta function on the chosen domain. To show the differential operator concretely, take $F(y)=y^{1-k}$. Then $\partial_{\overline{\tau}}F=(i/2)(1-k)y^{-k}$, so $\xi_k F=1-k$. The weight-k Laplacian also annihilates this local function, but that does not make it a global modular form.

Worked example

Set $k=3/2$ in the local example.

$F=y^{-1/2}$ gives $\xi_{3/2}F=-1/2$. This checks the derivative normalization and sign. It supplies no transformation law or cusp condition and therefore cannot repair the claimed global completion.

A shadow identity needs compatible weight, multiplier, representation, domain and growth conditions before uniqueness is meaningful.

Scope

These checks reject the literal assertion, not every possible vector-valued mock-Jacobi construction. A corrected theory may involve different theta kernels and operators.

Further development

Choose a valid weight1/2 shadow and specify principal parts and growth conditions, then formulate a genuine existence/ambiguity theorem for the completion.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R137.

Y25-07-027 A_2 and D_4 lattice data precede any Jacobi completion

2025-07-27 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which determinants and root counts follow from the actual Gram matrices, and which claimed modular structures need more?

The July27 lattice formulas retain useful exact mathematics. Gram matrices define quadratic forms, covolumes and theta coefficients, while a mock-Jacobi completion additionally requires elliptic and modular transformation laws not supplied by the matrix alone.

Definitions and assumptions

For an even positive lattice with Gram matrix S, set $Q(v)=v^{T}Sv/2$. The A_2 matrix is$[[2,1],[1,2]]$. The D_4 source matrix has diagonal2 and off-diagonal-1 along edges12,23,24.

Determinants and initial theta coefficients

The A_2 form x^2+xy+y^2 has determinant3 and six vectors of $Q=1$. The D_4 Gram determinant is4 and it has24 roots of $Q=1$. Their theta series begin$1+6q+\dots$ and$1+24q+\dots$ respectively.

Derivation or proof

For A_2,$4Q=(2x+y)^2+3y^2$, proving positivity and bounding all $Q=1$ solutions. They are$(\pm1,0),(0,\pm1),(1,-1),(-1,1)$. Its determinant is$4-1=3$. For D_4, reorder the central node last: the Schur complement of the three diagonal2 leaves is$2-3/2=1/2$, so the determinant is$2^3\cdot1/2=4$ and positivity follows. Its standard isometric realization is$\{v\in Z^4:\Sigma v_i \text{ even}\}$, with roots$\pm e_i\pm e_j$. Choosing the pair of positions and four signs gives$6\cdot4=24$. Theta coefficients count vectors of a fixed norm by definition.

Worked example

Compare lattice rank with ordinary theta weight.

The classical theta transformation has weight rank/2, giving weight one for A_2 and weight two for D_4, with the appropriate level/representation. A claimed weight-two Jacobi object with an A_2-related shadow needs an explicitly different construction; the Gram matrix itself does not prove it.

The exact lattice data survive independently of unproved VOA and moonshine assignments.

Scope

No completion, Eichler integral, elliptic variable or multiplier was recovered in a verifiable form. Determinant and root count do not classify every lattice or its automorphism action.

Further development

Construct the discriminant group and vector-valued theta representation, then formulate the precise Jacobi or mock transformation problem.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R132.

Y25-07-028 A fitted density line and three beta values do not define a theory

2025-07-28 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which quantities are actually defined, and do the supplied numbers support the proposed exponential law?

The July28 framework assigned densities to Lie/sporadic pairs, cohomology degrees and a beta function. The numerical and logical claims can be audited without endorsing a new geometry. Definitions and independent predictions must precede a fitted law.

Definitions and assumptions

Use the stated ranks4,6,7,8 and densities9240,1320,80960,1020096. Let y_i=log₁₀density_i and proposed prediction0.514r_i+1.291. The symbol Λ_g and any density denominator still require definitions.

A direct residual test and a nonuniqueness theorem

> The supplied line does not fit the first two densities closely: their residuals are about0.619 and−1.254 logarithmic units. Three beta values determine no unique function, even within smooth functions.

Derivation or proof

Evaluate the line at each rank and subtract from the logarithm of the stated density. A residual of−1.254 means the proposed value is too large by a factor about10^1.254, so it cannot be hidden by decimal rounding. For the beta claim, let P be the quadratic interpolant through its three stipulated inputs1/2,1,2. Every P(x)+c(x−1/2)(x−1)(x−2), c∈R, agrees at all three points and generally differs elsewhere. Thus finitely many values do not identify a generating mechanism. Likewise2+3+4+5=14 adds degree labels; it is not a computation of dimH²+dimH³+dimH⁴+dimH⁵, whose summands are Betti numbers that must be obtained from actual spaces.

Worked example

Use the supplied beta samples1.32132546,0.13591270,0.00178552.

They can be interpolated exactly as rational decimals, but adding the cubic vanishing factor preserves every sample. The count14 remains a correct integer sum without proving a fourteen-dimensional variety or cohomology decomposition.

The audit separates exact arithmetic, parameter fitting and mathematical construction.

Scope

No statistical uncertainty, independently defined Λ_g, held-out predictions or geometric beta construction was recovered. These numbers cannot establish emergent integers or sporadic symmetry.

Further development

Define each observable before collecting values, state a finite model class and test independent predictions with declared error and null models.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: .

Y25-07-029 A quartic splitting field cannot have Mathieu Galois group

2025-07-28 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What Galois groups are even possible for the two recovered quartic polynomials?

The July28 framework associated quartics with M_{12} and M_{24}. A degree bound already rules out a literal identification of their splitting-field Galois groups. The quartics themselves are legitimate algebraic objects whose irreducibility can be checked exactly.

Definitions and assumptions

Let $f=x^4-4x^3+1$ and $h=x^4-6x^3+4x^2+2x+1$. A characteristic-zero splitting-field group acts faithfully by permutations of the four roots, hence embeds in S_4 and has order at most24.

Irreducible quartics with small possible Galois groups

> Both f and h are irreducible over Q, but neither can have splitting-field group M_{12} or M_{24}. A separate construction would be needed to relate their roots to a Mathieu representation.

Derivation or proof

A rational root of either monic polynomial with constant1 must be±1, and direct substitution excludes both. A remaining factorization would be two monic integral quadratics by Gauss’s lemma. Their constant terms must both be $e=1$ or both $e=-1$. Write$(x^2+ax+e)(x^2+bx+e)$. Its cubic coefficient is $a+b$ and linear coefficient $e(a+b)$. For f these would have to be-4 and0, impossible for $e=\pm1$. For h they would have to be-6 and2, likewise impossible. Hence both are irreducible. Independently, any automorphism fixing all four roots fixes the splitting field they generate, proving the faithful S_4 embedding. The Mathieu groups have orders greater than24, excluding them without numerical root fitting.

Worked example

Check the constant/cubic/linear factor constraints.

For f, e(a+b)=±4 can never be0. For h it is±6 and never2. This proof does not determine whether the actual transitive subgroup is S_4,A_4,D_4,C_4 or V_4; discriminant and resolvent tests are still needed.

A degree-four formula may participate in a larger representation, but its literal root-permutation group obeys an unavoidable degree bound.

Scope

No Mathieu action or field discriminant was recovered. Polynomial discriminants, splitting fields and unrelated Pell fields must remain distinct.

Further development

Compute exact discriminants and cubic resolvents, then identify each actual transitive subgroup and state any additional representation map explicitly.

Source receipt: months/07/root_receipts.json record 0. Mathematical references: R059.

August 2025

Dated fragments recovered; complete transcript unavailable. 40 developed treatments.

The month includes obstruction complexes, solver estimates, graph spectra, Gaussian extension, lattice refinement, invariant theory and calibrated dynamical models. Finite linear algebra and operator estimates provide constructive foundations for further work.

ID · date	Concept and revised question	Result and mathematical core
Y25-08-909 2025-08-01 · U	A calibrated search has a computable multiplicity cost How does searching many modular-coordinate fits change the interpretation of the best apparent anomaly?	Finite-search familywise control Pr(p_adj<=alpha)<=alpha. If the p-values are independent uniform variables, Pr(min pj<=u)=1-(1-u)^M exactly.
Y25-08-905 2025-08-02 · U	An Eisenstein correction has an exact modular transformation law Which combination involving E2(9*tau) is genuinely modular, and why does its anomalous term cancel?	Cancellation of the quasimodular anomaly F_N transforms with weight2 on Gamma0(N). Its holomorphic modular-form extension at the cusps is the standard weight-two Eisenstein construction.
Y25-08-901 2025-08-02 · U	An exact KP line soliton from a traveling-wave reduction Which tanh ansatz solves a precisely normalized KP equation, and how do amplitude, width and transverse slope determine speed?	Exact traveling-wave and tau-function identity u(x,y,t)=2k² sech²(kX) solves KP when c=4k²+3s*ell². It also equals2 partial_x² log(1+exp(2kX)).
Y25-08-001 2025-08-03 · U	Spirograph closure is a commensurability theorem Which frequency ratios make the stated trochoid close, and which extra data define a three-dimensional version?	Rational frequency ratio gives a closed curve If R/r=a/b in lowest positive terms,2πb is a period. If R/r is irrational, the function has no positive period. Degenerate zero-coefficient cases must be treated separately.
Y25-08-002 2025-08-03 · U	SO(3) rotations have a precise infinitesimal commutator What does the exponential of a skew matrix prove, and why do finite rotations about different axes fail to commute?	Rodrigues formula and Lie bracket exp(θK)=I+(sinθ)K+(1−cosθ)K² lies in SO(3). The generators satisfy[J_x,J_y]=J_z, with cyclic variants. Small rotations therefore commute only up to a second-order bracket term.
Y25-08-908 2025-08-03 · U	The classical Heegner hypothesis is a local test Does the proposed pair E:y²=x³-x and K=Q(i) satisfy the classical split-prime Heegner hypothesis?	Failure at2 and a split-prime alternative The source pair fails the classical split-prime hypothesis because2 ramifies in Q(i). In Q(sqrt(-7)),2 splits, so the prime condition for level32 is satisfied.
Y25-08-907 2025-08-04 · U	Filtered Frobenius membership needs a precision certificate What can a finite p-adic residual prove about a proposed filtered Frobenius splitting condition?	Exact obstruction and finite-precision distinction For v=(x,y), R(v)=4x. Exact splitting is equivalent to x=0; a zero residue modulo5^N proves only x=0 modulo5^N.
Y25-08-903 2025-08-04 · U	A nonprincipal ideal gives an actual Picard obstruction What exact ideal-class obstruction survives after finite-field multiplicative cocycles have vanished?	An explicit nonzero class of order2 I has norm2, satisfies I²=(2), and is not principal. Its ideal class is a nonzero element of Pic(O)[2].

ID · date	Concept and revised question	Result and mathematical core
Y25-08-902 2025-08-04 · U	Finite-field Hilbert90 as an exact descent calculation Can a norm-one element be written explicitly as v/Frobenius(v), and which part of this computation constitutes descent?	Explicit finite-field Hilbert90 Every norm-one u in F49-star is v/sigma(v). The element u=5-a is obtained from v=1+a.
Y25-08-904 2025-08-05 · U	Clutching produces a computable cohomological obstruction Which explicit obstruction distinguishes bundles with identical base dimension and structure group?	Clutching degree and curvature certificate The resulting line bundle has first Chern number m and is trivial exactly when m=0. Fixed dimensions therefore do not determine its obstruction.
Y25-08-910 2025-08-05 · A	Compounding distinguishes price growth from purchasing-power decay What mathematical relation connects an annual inflation factor, an annual purchasing-power loss and a thirty-year cumulative ratio?	Exact reciprocal compounding (1-d)=1/(1+pi), so d=pi/(1+pi). After T years the price factor is(1+pi)^T and the purchasing-power factor is(1-d)^T.
Y25-08-906 2025-08-08 · A	A Mellin split gives a reproducible central L-value formula What exact functional-equation identity can replace an unstable central-value quadrature?	Exact central-value series with an explicit tail L(1,f)=(1+epsilon)sum(n>=1)(a_n/n)exp(-c*n). Truncation after M terms has absolute tail at most2|1+epsilon|exp[-c(M+1)]/(1-exp(-c)).
Y25-08-003 2025-08-09 · A	A Temperley-Lieb trace needs a positivity check Do the finite TL relations and a declared trace actually define a positive inner product?	Two-strand positivity criterion For x=a+be, tr(x*x)=|a|²+2Re(overline a b)/δ+|b|². It is positive semidefinite exactly when|δ|≥1, and positive definite when|δ|>1.
Y25-08-004 2025-08-09 · A	Perron data define a Markov measure and a pressure value Which normalization turns an adjacency matrix into a stochastic transition matrix, and why does its entropy equal logλ?	Perron Markov chain and entropy maximization P is stochastic,π is stationary, and its entropy rate is logλ. Every stationary Markov chain Q supported on the same allowed edges has entropy rate at most logλ.
Y25-08-007 2025-08-09 · U	Cubic KKT equations do not certify a global maximum What hypotheses justify the cubic multiplier equations, and what can a stationary solution actually prove?	First-order stationarity and its scope A regular constrained extremum satisfies c_j+3x_j²S_j=0. If S_j≠0, real solutions require−c_j/(3S_j)≥0 and generally have both signs. Stationarity alone implies neither existence nor global optimality.
Y25-08-008 2025-08-09 · A	Consistent Gaussian coordinates need not live in the Hilbert space Which covariance condition turns cylinder data into an actual Hilbert-valued random variable?	Trace-summable construction and identity-covariance failure IfΣq_n<∞, X converges almost surely in H and in mean square, with characteristic function exp(−Σq_nh_n²/2). For q_n=1 the coordinate law givesℓ^2 probability zero.
Y25-08-005 2025-08-10 · A	The Perron projection must carry the Perron eigenvalue Why is M=Π+R inconsistent with a Perron eigenvalueφ when Πr=r and Rr=0?	Correct spectral projection decomposition With R=M−λΠ, one has Π²=Π,ΠR=RΠ=0 and M^k=λ^kΠ+R^k for k≥1. Therefore(M/λ)^k→Π whenρ(R)<λ. Defining R=M−Π instead gives Rr=(λ−1)r.
Y25-08-006 2025-08-10 · A	Chebyshev acceleration needs an interval and a norm Which spectral assumptions justify a Chebyshev residual bound, and why is a generic preconditioner label insufficient?	Uniform residual polynomial bound For e_m=p_m(A)e₀, ||e_m||₂≤|T_m(c/d)|⁻¹| |e₀||₂≤2((√κ−1)/(√κ+1))^m||e₀||₂,κ=b/a.

ID · date	Concept and revised question	Result and mathematical core
Y25-08-020 2025-08-10 · U	BFGS positivity requires curvature, not Armijo alone Which algebraic condition preserves a positive inverse-Hessian estimate, and what does the recovered quadratic test actually verify?	Curvature condition and positive definiteness If $y^Ts>0$, then H^+ is positive definite. An Armijo decrease test alone does not guarantee this curvature condition.
Y25-08-021 2025-08-10 · A	ISTA and conjugate gradients need different mathematical contracts Which assumptions justify a proximal step or a residual certificate, rather than a generic solver-success label?	Descent and a residual error bound For$0<t\le 1/L$, the proximal step satisfies $F(x^+)\le F(x)-(1/t-L/2)\|\|x^+-x\|\|^2$, $F=f+g$. For an approximate SPD solve with residual $r=b-Ax$, $\|\|x-x^*\|\|_A\le \|\|r\|\|_2/\sqrt{a}$.
Y25-08-009 2025-08-12 · A	Čech obstruction transfer requires a chain equivalence What exact map preserves a first-cohomology obstruction, and why are cover-preserving labels alone insufficient?	Homotopy equivalence preserves obstruction classes A cochain homotopy equivalence induces an isomorphism on H^1. An arbitrary cochain map, even between three-patch constructions, need not be injective or surjective.
Y25-08-016 2025-08-13 · A	Gauge transformations and Picard-Fuchs coordinates must be normalized Which operator annihilates the factorial period series, and how does a square-root gauge change its local monodromy?	Factorial recurrence and gauge shift F solves$[\theta^4-5z(5\theta+1)(5\theta+2)(5\theta+3)(5\theta+4)]F=0$ formally. Gauge conjugation gives $z^{(-\alpha)}\theta z^{\alpha}=\theta+\alpha$.
Y25-08-010 2025-08-15 · A	Translated cellular-automaton recurrence is a relative periodic orbit What exactly does $S^{(-T)}f^k(x)=x$ prove, and how does it differ from ordinary periodicity?	Iteration of a relative period If $f^k(x)=S^Tx$ and f commutes with translations, then $f^{(mk)}(x)=S^{(mT)}x$ for every $m\ge 0$. On a finite periodic lattice, some multiple becomes an ordinary period.
Y25-08-028 2025-08-15 · U	Complete-graph labels must obey the edge-count bound Which of the recovered edge counts can describe a simple graph on the stated number of vertices?	Edge feasibility and corrected examples A complete graph K_n has exactly $n(n-1)/2$ edges. Thus K_{23} has253 edges, not231; K_{10} has45 edges, so78 cannot be its simple-graph edge count. The counts528,946,4371 are correct for K_{33},K_{44},K_{94}.
Y25-08-018 2025-08-16 · U	The denominator5760 follows from a characteristic power series What mathematical normalization produces the Â polynomial, and what does it not determine about physical constants?	Degree-eight Â polynomial $Â=1-p_1/24+(7p_1^2-4p_2)/5760$+terms of higher degree.
Y25-08-019 2025-08-17 · U	A B_5 symmetry label does not force empirical rank five What does the signed-permutation group actually imply, and what would establish a five-dimensional feature model?	Generic orbits versus degenerate orbits and feature rank A vector with nonzero pairwise-distinct absolute coordinates has orbit3840. The vector(1,1,0,0,0) has orbit40. Neither group action determines the rank of a separately constructed data matrix.
Y25-08-011 2025-08-20 · A	Measurability and L^p convergence need different hypotheses What does the Carathéodory criterion prove, and why does pointwise convergence not imply norm convergence?	Carathéodory measurability and an L^p spike The measurable sets form a sigma-algebra on which μ^* is countably additive. On[0,1], $f_n=n^{(1/p)}1_{(0,1/n)}$ converges to zero almost everywhere while$\|\|f_n\|\|_p=1$.

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Y25-08-012 2025-08-20 · A	A torus delta integral requires a coarea factor Does δ(g)dx equal the induced surface measure, or is a gradient factor missing?	Exact torus normalization On g=0,|∇g|=2r. Hence∫fδ(g)dx=(1/(2r))∫_TfdA, whileδ(g)|∇g|dx represents dA. In particular∫δ(g)dx=2π²R.
Y25-08-013 2025-08-20 · U	Hardy atoms bound BMO pairings with an explicit constant Which cancellation and oscillation estimates support the named Hardy-BMO connection?	Atomic pairing and bounded perturbation |∫ab|≤osc_I(b). If b∈BMO and||e||∞≤ε, then b+e∈BMO and| ||b+e||_BMO−||b||_BMO |≤2ε.
Y25-08-014 2025-08-20 · A	A Landau quartic has a precise minimizer set Which conclusions follow from the stated quartic potential before invoking a field theory or phase transition?	Exact quartic minima If r≥0,V has unique minimum atφ=0. If r<0, its minimum is−r²/(4u), attained exactly on|φ|=√(−r/(2u)).
Y25-08-015 2025-08-20 · A	Hall coefficients count subobjects even when Ext¹ vanishes Which actual counting problem defines the Hall product, and what does a small example say about extension obstructions?	Gaussian binomial coefficients and flag associativity [a]*[b]=GaussianBinomial(a+b,b)_q[a+b]. In particular[1]*[1]=(q+1)[2], and[1]^3=(q+1)(q²+q+1)[3].
Y25-08-017 2025-08-20 · A	Phase consistency is a cycle condition, not a missing matrix certificate What can be verified from the recovered edge labels when the original eleven-edge coefficient matrix is unavailable?	Cycle criterion for graph potentials The graph equations are solvable exactly when every signed cycle sum vanishes. A connected incidence matrix has rank|V|−1. The recovered representatives are inconsistent in this model, but deleting the c/s edge leaves ten consistent edges.
Y25-08-023 2025-08-23 · A	The Heawood spectrum follows from Fano-plane incidence Which exact incidence identity proves the graph spectrum and the displayed Laplacian zeta function?	Exact Heawood spectrum and zeta A has eigenvalues3,−3, and±√2 each with multiplicity six. Thus ζ_L(s)=6(3−√2)^(−s)+6(3+√2)^(−s)+6^(−s).
Y25-08-024 2025-08-23 · A	Paley Laplacian moments have closed rational values Under which finite-field hypotheses do the Paley spectral formulas apply, and what do their inverse moments count?	First two Laplacian inverse moments The positive Laplacian eigenvalues are(q±√q)/2, each with multiplicity d. Therefore ζ_L(1)=2 and ζ_L(2)=4(q+1)/(q(q−1)).
Y25-08-025 2025-08-24 · U	CRT and finite entropy do not prove decimal normality Which six-kernel statistics are exact, and what can sixteen residues actually establish?	CRT reconstruction and finite entropy ceiling r≡33a+16b mod48. Among sixteen ternary samples, entropy is maximized by counts(6,5,5) up to permutation, giving H/log3≈0.9965. This does not prove equidistribution or normality of x.
Y25-08-027 2025-08-25 · U	The Riccati flow and its wall times are exact within the model Which identities genuinely follow from μ′=bμ², and which entropy or winding interpretations are extra assumptions?	Flow invariant and half-integer walls μ=μ₀/(1−bμ₀ℓ),1/μ+bℓ−1/μ₀=0,F′(ℓ)=−ncμ/2 and N′=μ/6. The time satisfying N(ℓ_k)=k/2 isℓ_k=(1−e^(−3bk))/(bμ₀).
Y25-08-026 2025-08-26 · U	The lattice symbol has a proved dispersion and anisotropy bound Which refinement discrepancy is expected from the discrete Laplacian, and when is a residue check circular?	Exact symbol and continuum-error bound For real k,0≤k²−σ_h(k)≤h²k⁴/12. In two dimensions,0≤|k|²−σ≤(h_x²k_x⁴+h_y²k_y⁴)/12. Sampled modes alias under k↦k+2πm/h.

ID · date	Concept and revised question	Result and mathematical core
Y25-08-029 2025-08-31 · U	A rank-one CRO matrix has an exact spectrum and inverse What can the recovered matrix S(t) actually prove before invoking a crystalline interpretation?	Spectrum, inverse and positivity S has eigenvalue1+nt on the constant line and1 on its orthogonal complement. detS=1+nt; for t≠−1/n, $S^{-1}=I-[t/(1+nt)]11^{T}$. It is positive definite exactly when t>−1/n.
Y25-08-030 2025-08-31 · A	A residual for one vector cannot certify vanishing of H^1 What does zero numerical residue actually imply, and what exact algebra would establish a cohomology-vanishing theorem?	Residual scope and logarithmic sensitivity A zero Laplacian residual can occur for a nonzero H^1 class. If E_k=e^(−αt_k+η_k) with|η_k|≤ε, then|αhat−α|≤2ε/Δt. A hash ledger certifies neither cohomology nor the truth of its logged claims.
Y25-08-022 date unava · T	Optimizer hypergradients must follow the entire state recursion What derivative should a one-step AdamW outer loss have, and why is a zero reported gradient not a success certificate?	One-step outer derivatives $\partial_\eta\theta_1=-u_0-\lambda\theta_0$ and$\partial_\lambda\theta_1=-\eta\theta_0$. Thus$\partial_\eta L=(\theta_1-y)\cdot(-u_0-\lambda\theta_0)$ and$\partial_\lambda L=(\theta_1-y)\cdot(-\eta\theta_0)$.

Y25-08-909 A calibrated search has a computable multiplicity cost

2025-08-01T16:13:11Z · historical evidence U · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

How does searching many modular-coordinate fits change the interpretation of the best apparent anomaly?

The source moonshine interpretation selected fitted coordinates and reported a significance value. A finite family of tests gives an exact mathematical benchmark for distinguishing one prespecified statistic from a selected maximum.

Definitions and assumptions

Let p1,...,pM be valid null p-values, so Pr(pj<=u)<=u for each j.

Define p_adj=min(1,M*min_j pj). Independence is not assumed for the familywise bound.

Finite-search familywise control

Pr(p_adj<=alpha)<=alpha. If the p-values are independent uniform variables, Pr(min pj<=u)=1-(1-u)^M exactly.

Derivation or proof

The event p_adj<=alpha is contained in the union of events pj<=alpha/M. The union bound gives probability at most sum_j alpha/M=alpha, proving the adjusted test. Under independence, the complementary event pj>u for all j has probability(1-u)^M, giving the exact formula. The theorem concerns the whole specified search family: fitting extra parameters, choosing observables after inspection or retaining only successful seeds enlarges that family or requires a different null calibration. A held-out evaluation can avoid reusing the same observations for both selection and confirmation.

Worked example

As a newly stipulated illustration, take M=1000 independent null tests and u=0.001.

The chance of at least one such small p-value is1-0.999^1000, approximately0.6323. Thus a selected value0.001 does not carry the interpretation of one prespecified test.

The cost of selection can be computed without asserting any physical mechanism or interpreting fitted couplings as predictions.

Scope

The exact independence formula is illustrative. The historical search size, data covariance and original analysis are unavailable, so no corrected historical significance is claimed.

Further development

Freeze the observables and coordinate map, simulate the complete selection procedure under a stated correlated null, and compare a separately held-out statistic with that calibrated distribution.

Source receipt: months/08/root_coverage.json record0. Mathematical references: .

Y25-08-905 An Eisenstein correction has an exact modular transformation law

2025-08-02T00:45:27Z · historical evidence U · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

Which combination involving E2(9*tau) is genuinely modular, and why does its anomalous term cancel?

The CP-moonshine manuscript used E2(9*tau) as an analytic lift. A scalar weight-two example gives an exact covariance benchmark before introducing unknown triplet representations or fitted physics parameters.

Definitions and assumptions

Use E2(tau)=1-24 sum(n>=1)sigma1(n)q^n with q=exp(2pi*i*tau).

For N>1 define F_N(tau)=E2(tau)-N*E2(N*tau). For gamma in Gamma0(N), c is divisible by N.

Cancellation of the quasimodular anomaly

F_N transforms with weight2 on Gamma0(N). Its holomorphic modular-form extension at the cusps is the standard weight-two Eisenstein construction.

Derivation or proof

The classical formula is E2(gamma*tau)=(c*tau+d)²E2(tau)+[6/(pi*i)]c(c*tau+d). Since c is divisible by N, the integral matrix gamma_N=(a,Nb;c/N,d) lies in SL2(Z) and satisfies N*gamma*tau=gamma_N*(N*tau). Applying the same transformation to E2(N*gamma*tau) gives anomaly[6/(pi*i)](c/N)(c*tau+d). Multiplication by N makes this anomaly identical to the first one, so their difference vanishes. Holomorphy at cusps follows from the standard Eisenstein-series construction; the displayed calculation proves the covariance directly. It also shows why E2(N*tau) alone does not have the claimed homogeneous weight-two law.

Worked example

Take N=9 and compute the constant and q^9 coefficients.

The constant is1-9=-8. Since sigma1(9)=13, the q^9 coefficient is-24*13+216*1=-96; the q coefficient is-24. The nonzero constant term means this example is not a cusp form.

The normalization and exact covariance are fixed independently of numerical success percentages.

Scope

No unprovided Delta(27) representation, mock completion or shadow integral has been certified. Finite sampled transformation tests cannot establish a global modular identity.

Further development

Specify exact S and T matrices and prove their group relations, then derive the completed vector-valued transformation law with a stated shadow and convergence domain.

Source receipt: months/08/root_coverage.json record0. Mathematical references: R138.

Y25-08-901 An exact KP line soliton from a traveling-wave reduction

2025-08-02T05:10:45Z · historical evidence U · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

Which tanh ansatz solves a precisely normalized KP equation, and how do amplitude, width and transverse slope determine speed?

The source proposed a symbolic tanh solver. A complete solution begins by fixing the PDE normalization and deriving the algebraic constraints. The single-soliton family provides an exact benchmark for a symbolic solver and a numerical residual.

Definitions and assumptions

Use (u_t+6uu_x+u_xxx)_x+3s u_yy=0 with s=+1 or -1.

Let X=x+ell*y-c*t, k>0, and impose decay of U(X), U-prime and U-double-prime at both infinities.

Exact traveling-wave and tau-function identity

u(x,y,t)=2k² sech²(kX) solves KP when c=4k²+3s*ell². It also equals2 partial_x² log(1+exp(2kX)).

Derivation or proof

Substitution gives [-cU-prime+6UU-prime+U-triple-prime]-prime+3s*ell² U-double-prime=0. Integrate twice using the stated decay to obtain U-double-prime+(3s*ell²-c)U+3U²=0. For f=sech²(kX), direct differentiation yields f-double-prime=4k²f-6k²f². Setting U=2k²f therefore gives U-double-prime=4k²U-3U²; the reduced ODE vanishes exactly at the stated speed. Differentiating log(1+exp(2kX)) twice gives k²sech²(kX), proving the tau representation. The tanh form is U=2k²[1-tanh²(kX)], so degree balancing M=2 becomes a verified consequence in this example.

Worked example

Choose k=1/2, ell=1 and s=+1.

The amplitude is1/2, speed c=4 and profile U(X)=(1/2)sech²(X/2). At X=0, U=1/2 and U-double-prime=-1/4, so U-double-prime- U+3U²=-1/4-1/2+3/4=0. Its line-integrated mass is integral U dX=4k=2.

All solution parameters and the PDE residual are determined analytically.

Scope

A KP solution alone establishes neither a tsunami source model nor accuracy near shorelines, breaking waves or varying bathymetry. Those require additional equations and data.

Further development

Extend the symbolic procedure to a two-exponential tau function and derive the interaction coefficient from the bilinear KP identity; verify the one-soliton limit before numerical comparisons.

Source receipt: months/08/root_coverage.json record0. Mathematical references: R139.

Y25-08-001 Spirograph closure is a commensurability theorem

2025-08-03 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which frequency ratios make the stated trochoid close, and which extra data define a three-dimensional version?

The August3 spirograph discussion linked geometry to Lie groups and speculative applications. The exact mathematical kernel is a sum of rotating complex vectors. Closure can be proved from its frequencies without endorsing an application or assuming every spatial curve is a group orbit.

Definitions and assumptions

For R>r>0 and d>0 define z(t)=(R−r)e^(it)+d e^(−i(R−r)t/r). Assume its two distinct frequencies have nonzero coefficients. A period means z(t+T)=z(t) for every t, not merely one accidental self-intersection.

Rational frequency ratio gives a closed curve

If R/r=a/b in lowest positive terms,2πb is a period. If R/r is irrational, the function has no positive period. Degenerate zero-coefficient cases must be treated separately.

Derivation or proof

The two exponentials return together at T=2πb because their phases change by2πb and−2π(a−b). Conversely equality of the two trigonometric sums for every t implies that each distinct-frequency coefficient has the same phase before and after translation; linear independence follows, for example, by evaluating the function and its derivative. Thus e^(iT)=1 and e^(−i(R−r)T/r)=1. The first gives T=2πm, and the second makes m(R−r)/r an integer, impossible for irrational R/r. This is a theorem about the declared parametrization. A third coordinate such as hsin(ωt) adds a further frequency whose commensurability must also be checked.

Worked example

Use R=5,r=2,d=1.

The frequencies are1 and−3/2, so T=4π works. A3D lift with vertical frequency1/3 has common period12π, whereas an irrational vertical frequency destroys periodicity.

Closure is governed by all active frequencies, rather than by a visual resemblance or an arbitrary Lie-group label.

Scope

The proof gives no fabrication, biological or drug-design result. Self-intersection, knot type and embeddedness need further analysis.

Further development

For a specified spatial lift, classify self-intersections and prove whether the closed image is an embedding before assigning knot invariants.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-002 SO(3) rotations have a precise infinitesimal commutator

2025-08-03 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What does the exponential of a skew matrix prove, and why do finite rotations about different axes fail to commute?

The August3 source named skew matrices, quaternions and BCH. Rodrigues' formula supplies an exact operator, while a basis commutator identifies the first noncommutative correction. Neither implies a speculative model of an external system.

Definitions and assumptions

For a unit vector $a\in R^3$ let $Kv=a\times v$. Then $K^T=-K, K^2=aa^T-I$ and $K^3=-K$. Let J_x, J_y, J_z be the corresponding unit-axis matrices.

Rodrigues formula and Lie bracket

$\exp(\theta K)=I+(\sin\theta)K+(1-\cos\theta)K^2$ lies in SO(3). The generators satisfy $[J_x,J_y]=J_z$, with cyclic variants. Small rotations therefore commute only up to a second-order bracket term.

Derivation or proof

The vector triple-product identity gives $K^2v=a(a\cdot v)-v$ and hence $K^3=-K$. Separate the exponential series into its odd and even powers to obtain Rodrigues' expression. Since K is skew, $\exp(\theta K)^T\exp(\theta K)=I$; its determinant is $\exp(\operatorname{tr}\theta K)=1$. For the bracket, apply the two compositions to v and use $a\times(b\times v)-b\times(a\times v)=(a\times b)\times v$. Expanding exponentials gives $e^{(\varepsilon Jx)}e^{(\varepsilon Jy)}-e^{(\varepsilon Jy)}e^{(\varepsilon Jx)}=\varepsilon^2 Jz+O(\varepsilon^3)$, making noncommutativity quantitative. A quaternion representation must use the half-angle if it is to reproduce a spatial rotation of angleθ.

Worked example

Rotate e_z successively by right-angle rotations around x and y.

$R_x(\pi/2)R_y(\pi/2)e_z=e_x$, whereas $R_y(\pi/2)R_x(\pi/2)e_z=-e_y$. The order matters even though both factors individually preserve lengths and orientation.

The group action, matrix convention and order of composition are now explicit.

Scope

This calculation establishes rotation geometry, not a PDE, energy functional or physical coupling.

Further development

Specify the representation and dynamics, then derive any symmetry reduction from actual commuting operators rather than axis labels.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: R132.

Y25-08-908 The classical Heegner hypothesis is a local test

2025-08-03T03:56:33Z · historical evidence U · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

Does the proposed pair $E: y^2=x^3-x$ and $K=Q(i)$ satisfy the classical split-prime Heegner hypothesis?

A formal q-deformation must begin with a correctly specified classical construction. The conductor and quadratic field already decide whether the standard X0(N) Heegner-point setup applies.

Definitions and assumptions

Use the classical Heegner hypothesis: every rational prime dividing the conductor N splits in the imaginary quadratic field K.

For the source curve N=32. Q(i) has discriminant-4; a prime dividing the discriminant is ramified rather than split.

Failure at2 and a split-prime alternative

> The source pair fails the classical split-prime hypothesis because2 ramifies in Q(i). In Q(sqrt(-7)),2 splits, so the prime condition for level32 is satisfied.

Derivation or proof

In Z[i],2=-i(1+i)², proving ramification directly. In the ring of integers Z[(1+sqrt(-7))/2], the generator theta satisfies theta²-theta+2=0. Modulo2 its polynomial is X(X-1), with two distinct roots. Dedekind factorization, or the corresponding two distinct norm-two ideals, therefore gives splitting of2. These calculations check the exact local premise of the classical modular-curve construction. They do not supply the height formula, prove a point has infinite order or define a q-deformation. A generalized Heegner construction allowing ramification would require its own different hypotheses.

Worked example

Compare discriminants-4 and-7 at the unique prime dividing32.

The first has discriminant divisible by2 and a repeated factor; the second has discriminant congruent1 modulo8 and distinct factors over F2. No numerical period calculation is required for this decision.

The starting field can be chosen to meet a precise classical condition before any formal deformation is attempted.

Scope

A satisfied Heegner hypothesis alone does not imply a nonzero L-derivative or a non-torsion point. Formal power-series coordinates are not automatically points on the original elliptic curve.

Further development

Specify the modular parametrization and CM ideal, compute a genuine classical Heegner point, verify its curve equation, and only then formulate an algebraic deformation with a proved specialization map.

Source receipt: months/08/root_coverage.json record0. Mathematical references: R140.

Y25-08-907 Filtered Frobenius membership needs a precision certificate

2025-08-04T02:41:44Z · historical evidence U · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

What can a finite p-adic residual prove about a proposed filtered Frobenius splitting condition?

The proposed Generator5 returns a vanishing code from phi(v)-p^(-1)v in Fil1. Even before a genuine syntomic realization is supplied, a filtered-vector-space example explains which finite-precision conclusions are valid.

Definitions and assumptions

Let V=Q5², Fil1=span(e2), phi=diag(1,5), and pi:V→V/Fil1 be first-coordinate projection.

Define the integral-scaled obstruction R(v)=pi[(5phi-I)v]. Membership phi(v)-v/5 in Fil1 is equivalent to R(v)=0.

Exact obstruction and finite-precision distinction

For v=(x,y), R(v)=4x. Exact splitting is equivalent to x=0; a zero residue modulo5^N proves only x=0 modulo5^N.

Derivation or proof

The first coordinate of(5phi-I)v is4x, while the second is24y and disappears in the quotient. Since4 is a5-adic unit, multiplication by4 preserves the5-adic valuation. Therefore an observed residual of valuation k<N certifies a nonzero obstruction, while a residual represented by0 modulo5^N only bounds its valuation below by N. It cannot distinguish exact0 from5^N times a nonzero unit. This conclusion uses the exact filtered matrix and the declared input precision; it does not depend on an empirical threshold. The denominator p^(-1) must also be accounted for when converting a residual precision back to the original equation.

Worked example

Choose v=(25,1).

R(v)=100 has5-adic valuation2. It is0 modulo25 but100 modulo125, so the precision-two verdict cannot establish vanishing and the precision-three verdict disproves it.

The example produces an explicit precision-stable nonvanishing decision once sufficient digits are available.

Scope

This is a stipulated filtered linear-algebra model, not a computed syntomic cohomology group or a Selmer certificate for an elliptic curve.

Further development

For an actual filtered(phi,N) module, compute quotient matrices over a valuation ring, certify their elementary divisors and propagate input precision through each denominator and inverse.

Source receipt: months/08/root_coverage.json record0. Mathematical references: R141.

Y25-08-903 A nonprincipal ideal gives an actual Picard obstruction

2025-08-04T15:22:05Z · historical evidence U · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

What exact ideal-class obstruction survives after finite-field multiplicative cocycles have vanished?

The proposed torsor-descent framework becomes concrete over the integers of Q(sqrt(-5)). An ideal can have a principal square while itself remaining nonprincipal; that is a genuine Picard-group phenomenon.

Definitions and assumptions

Let O=Z[a], $a^2=-5$, the ring of integers of Q(sqrt(-5)). Let I=(2,1+a).

For the fppf topology the Kummer sequence gives$0\to O^\times/(O^\times)^2\to H^1(\mathrm{Spec}\ O,\mu_2)\to \mathrm{Pic}(O)[2]\to 0$.

An explicit nonzero class of order2

I has norm2, satisfies $I^2=(2)$, and is not principal. Its ideal class is a nonzero element of Pic(O)[2].

Derivation or proof

Reduction modulo I sends a to1 in F2 and identifies O/I with F2, so the ideal norm is2. The square I^2 is generated by4,2+2a and-4+2a. All are divisible by2, and(2+2a)-(-4+2a)-4=2; hence $I^2=(2)$. If I were generated by r+s*a, its norm would be $|r^2+5s^2|=2$, impossible for integers r,s. Thus[I] is nonzero and2[I]=0. The fppf Kummer exact sequence places this obstruction in the image of a mu2 torsor. More explicitly, the isomorphism I-tensor-I→O given by multiplication followed by division by2 supplies a square trivialization of the invertible module I.

Worked example

Compare the norm equation for a generator with the explicit generators of the squared ideal.

For s=0, $r^2=2$ has no integral solution; for |s|>=1 the norm is at least5. Nevertheless the displayed integral combination of generators of I^2 equals2 exactly.

The obstruction and its order are proved without an unprovided Selmer computation.

Scope

At the prime2 the mu2 torsor need not be étale; the fppf topology is essential here. No claim about the complexity of general torsor descent follows.

Further development

Compute the local trivializations of O direct-sum I with multiplication(v,w)→vw/2, then verify the transition cocycle and its Kummer boundary explicitly.

Source receipt: months/08/root_coverage.json record0. Mathematical references: R142, R143.

Y25-08-902 Finite-field Hilbert90 as an exact descent calculation

2025-08-04T15:37:10Z · historical evidence U · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

Can a norm-one element be written explicitly as v/Frobenius(v), and which part of this computation constitutes descent?

The source requested a concrete Hilbert90 example rather than an unspecified obstruction. A quadratic finite field makes the cocycle, norm and coboundary entirely visible.

Definitions and assumptions

Let F49=F7[a]/(a²-3);3 is not a square in F7.

The nontrivial Galois automorphism is sigma(a)=a^7=-a. A multiplicative1-cocycle for this cyclic group is specified by u with u*sigma(u)=1.

Explicit finite-field Hilbert90

> Every norm-one u in F49-star is v/sigma(v). The element u=5-a is obtained from v=1+a.

Derivation or proof

The multiplicative group F49-star is cyclic of order48. Its norm is x maps to x^8, whose image has order6 and whose kernel has order8. The homomorphism v maps to v/sigma(v)=v^(-6) has kernel F7-star of order6, hence image of order8. Every image element has norm1 by cancellation, so image and norm kernel coincide. For the explicit example, (1+a)/(1-a)=(1+a)²/(1-a²)=(4+2a)/5=5+6a=5-a in F7. Its norm is(5-a)(5+a)=25-3=22=1 modulo7. Thus the cocycle is a coboundary with an explicit witness.

Worked example

Represent field elements by pairs(r,s), meaning r+s*a, with multiplication(r,s)(t,v)=(rt+3sv,rv+st) modulo7.

Enumeration gives48 nonzero elements,8 norm-one elements and8 distinct ratios v/sigma(v). The displayed ratio and norm both hold exactly in this pair arithmetic.

The descent claim is certified by finite algebra, and its witness can be independently replayed.

Scope

Hilbert90 eliminates this particular multiplicative cocycle obstruction. It does not solve arbitrary discrete logarithms or identify every arithmetic obstruction with a finite-field cocycle.

Further development

Generalize the enumerator to F(q^n), compare the norm kernel with the image of v/sigma(v), and then separate linear-algebra cost from a discrete-logarithm representation of the same elements.

Source receipt: months/08/root_coverage.json record0. Mathematical references: R141.

Y25-08-904 Clutching produces a computable cohomological obstruction

2025-08-05T06:22:34Z · historical evidence U · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

Which explicit obstruction distinguishes bundles with identical base dimension and structure group?

A proposed universal degree delta=d+s leaves the actual obstruction group and class unspecified. Circle bundles on the same two-sphere give a complete small model in which every integer obstruction can occur.

Definitions and assumptions

Cover S^2 by northern and southern discs; on their equatorial overlap glue complex lines by g_m(exp(i*phi))=exp(i*m*phi), m an integer.

Use the convention c1=[F/(2pi)]; reversing the clutching convention reverses both signs. In polar coordinates take A_N=(m/2)(1-cos theta)dphi and A_S=-(m/2)(1+cos theta)dphi.

Clutching degree and curvature certificate

> The resulting line bundle has first Chern number m and is trivial exactly when m=0. Fixed dimensions therefore do not determine its obstruction.

Derivation or proof

On the overlap A_N-A_S=m*dphi, matching the logarithmic derivative of the transition function under the stated convention. Both local curvatures equal F=(m/2)sin(theta)dtheta-wedge-dphi. Its normalized integral is(2pi)^(-1) integral F=m. Changing a local trivialization on a disc multiplies the equatorial transition by a loop that extends across that disc and has winding0, so it cannot change m. Conversely, a degree-zero map $S^1 \to U(1)$ is null-homotopic; its extension across a hemisphere removes the transition and trivializes the bundle. Thus the integer is both an obstruction and a complete invariant in this clutching problem.

Worked example

Keep the base S^2 and group U(1) fixed and choose m=0,1,2.

The curvature integrals are0,2pi,4pi. All three models have identical base and group dimensions, while only the first bundle is trivial.

A meaningful obstruction requires coefficients, degree and a class, all visible in this example.

Scope

This calculation does not classify loop groups, gerbes or higher stacks. Their obstruction theories need their own coefficient objects and descent data.

Further development

Construct a gerbe on a three-sphere from explicit triple-overlap data and determine the corresponding integral degree-three class, keeping the cocycle identities separate from dimensional labels.

Source receipt: months/08/root_coverage.json record0. Mathematical references: R144.

Y25-08-910 Compounding distinguishes price growth from purchasing-power decay

2025-08-05T15:20:07Z · historical evidence A · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

What mathematical relation connects an annual inflation factor, an annual purchasing-power loss and a thirty-year cumulative ratio?

The source challenged an invented annual rate. The mathematical repair is to state a hypothetical rate transparently and distinguish the two reciprocal quantities before inserting any empirical index.

Definitions and assumptions

Let the annual price multiplier be1+pi>0 and the annual purchasing-power multiplier be1-d>0.

Purchasing power is reciprocal to the price index for a fixed nominal amount and a fixed basket.

Exact reciprocal compounding

(1-d)=1/(1+pi), so d=pi/(1+pi). After T years the price factor is(1+pi)^T and the purchasing-power factor is(1-d)^T.

Derivation or proof

If one basket costs P0 initially and PT=P0(1+pi)^T later, a fixed nominal amount buys PT/P0 times fewer baskets; its relative purchasing power is P0/PT. Equating the annual reciprocal factor with1-d gives the stated conversion. For variable annual rates pi_t, replace the power by the product of(1+pi_t); averaging rates arithmetically does not in general reproduce that product. The constant equivalent annual rate is exp[T^(-1)sum log(1+pi_t)]-1. These are index identities, not estimates of any country or historical period.

Worked example

Stipulate d=0.033 for T=30; do not treat it as a measured historical rate.

The purchasing-power factor is0.967^30, approximately0.3654, and the reciprocal price factor is about2.7366. The corresponding price inflation rate is0.033/0.967, approximately3.4126 percent, rather than3.3 percent.

The example separates a mathematical scenario from the missing empirical rate and keeps numerator and denominator conventions consistent.

Scope

Actual purchasing power depends on a declared basket, index construction, time window and source series. No current inflation or investment conclusion is inferred from this hypothetical example.

Further development

Given a specified historical index series, compute log growth by interval, chain the factors exactly to the available decimal precision, and quantify how alternative basket definitions change the result.

Source receipt: months/08/root_coverage.json record0. Mathematical references: .

Y25-08-906 A Mellin split gives a reproducible central L-value formula

2025-08-08T23:35:01Z · historical evidence A · Developed classical result with a scoped worked example

Recovered starting point

Dated mathematical fragment recovered from the August2025 coverage sweep

Revised question

What exact functional-equation identity can replace an unstable central-value quadrature?

The historical approximate-functional-equation outputs disagree even in sign. For a weight-two form, splitting the Mellin integral at the Fricke fixed point gives a rapidly convergent center formula whose normalization can be checked directly.

Definitions and assumptions

Let f(tau)=sum(n>=1)a_n exp(2pi*i*n*tau) be a weight-two cusp form of level N.

Assume f(i/(N*t))=epsilon*N*t²*f(i*t), epsilon=+1 or -1, and |a_n|<=2n. Put c=2pi/sqrt(N).

Exact central-value series with an explicit tail

L(1,f)=(1+epsilon)sum(n>=1)(a_n/n)exp(-c*n). Truncation after M terms has absolute tail at most2|1+epsilon|exp[-c(M+1)]/(1-exp(-c)).

Derivation or proof

The Mellin identity gives L(1,f)=2pi integral(0,infinity)f(i*t)dt. Set A=1/sqrt(N) and substitute t=1/(N*u) in the lower integral. The Fricke identity turns it into epsilon integral(A,infinity)f(i*u)du. Thus the full integral equals(1+epsilon) times the upper integral. Termwise integration is justified by absolute exponential convergence on t>=A and gives the stated series. The coefficient bound yields |a_n/n|<=2, so a geometric series bounds the omitted tail. In particular epsilon=-1 forces the center value to vanish. For an elliptic newform, the bound follows from |a_n|<=d(n)sqrt(n) and the elementary inequality d(n)<=2sqrt(n).

Worked example

For N=11 use f=eta(tau)²eta(11*tau)², whose Fricke sign is epsilon=+1.

The q product q*product(1-q^n)²(1-q^(11n))² yields exact integer coefficients. Twenty terms give approximately0.2538418608559107, with analytic tail below3e-17 under the coefficient bound. The

numerical approximation has its own rounding error; the tail formula is independent of quadrature-node choices.

The local normalization, functional equation and truncation error are exposed rather than hidden in a generic AFE routine.

Scope

The theorem assumes genuine modular coefficients and the stated Fricke law. Applying it to arbitrary surrogate traces would not construct a modular L-function.

Further development

Implement interval exponential evaluation and retain the exact Fourier coefficients plus the analytic tail to produce a completely rounded central-value enclosure.

Source receipt: months/08/root_coverage.json record0. Mathematical references: R145, R013.

Y25-08-003 A Temperley-Lieb trace needs a positivity check

2025-08-09 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Do the finite TL relations and a declared trace actually define a positive inner product?

The August9 synthesis used Temperley-Lieb, Markov traces and thermodynamics together. A two-strand calculation isolates a necessary positivity condition. It does not claim that a condition sufficient for this small algebra is sufficient for every level of a TL tower.

Definitions and assumptions

Let $TL_2(\delta)$ have basis1,e with $e^*=e$ and $e^2=\delta e$,δ real and nonzero. Declare tr(1)=1,tr(e)=1/δ. The trace inner product is$\langle x,y\rangle=\mathrm{tr}(x^*y)$.

Two-strand positivity criterion

> For x=a+be, $\mathrm{tr}(x^*x)=|a|^2+2\mathrm{Re}(\text{overline a b})/\delta+|b|^2$. It is positive semidefinite exactly when$|\delta|\geq 1$, and positive definite when$|\delta|>1$.

Derivation or proof

Expand x^*x using $e^2=\delta e$ and apply the declared trace. In the basis1,e the Gram matrix is$[[1,1/\delta],[1/\delta,1]]$, with eigenvalues$1\pm 1/\delta$. Both are nonnegative exactly when$|\delta|\geq 1$. If$\delta=\pm 1$ there is a nonzero null vector and a quotient is needed for a Hilbert-space realization. At higher TL levels additional positivity restrictions arise; passing one finite Gram test cannot justify a full C*-tower or KMS identification. The algebraic relation and a trace functional are distinct data, and positivity must be proved after choosing them.

Worked example

Compare δ=2 and δ=1/2.

Atδ=2 the Gram eigenvalues are3/2 and1/2. Atδ=1/2, x=1−e has tr(x*x)=2−2/δ=−2, so this cannot be a squared Hilbert norm.

The example converts a named trace into a verifiable mathematical condition and identifies a concrete failure.

Scope

The full Markov-trace tower, Jones parameter classification and graph-C*-algebra dynamics are not constructed by this two-dimensional computation.

Further development

Compute Gram matrices at successive diagram levels and identify the null ideal or parameter restrictions before asserting a positive tower representation.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-004 Perron data define a Markov measure and a pressure value

2025-08-09 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which normalization turns an adjacency matrix into a stochastic transition matrix, and why does its entropy equal logλ?

The August9 PF/KMS discussion can be grounded in a finite primitive adjacency matrix. Its Perron eigenvectors define an exact stationary Markov chain. A relative-entropy calculation proves the relevant variational statement for this specified class of chains.

Definitions and assumptions

Let A be a primitive0-1 matrix with positive right/left PF vectors Ar=λr,ℓ^{T}A=λℓ^{T},ℓ^{T}r=1. Set P_ij=A_ij r_j/(λr_i) and π_i=ℓ_i r_i.

Perron Markov chain and entropy maximization

> P is stochastic,π is stationary, and its entropy rate is logλ. Every stationary Markov chain Q supported on the same allowed edges has entropy rate at most logλ.

Derivation or proof

The PF equations giveΣ_jP_ij=1 andΣ_iπ_iP_ij=π_j. On allowed edges logP_ij=logr_j−logr_i−logλ. Averaging with stationary edge weightsπ_iP_ij cancels the r terms, giving−Σπ_iP_ijlogP_ij=logλ. For another stationary chain(Q,ν), nonnegativity of the rowwise KL divergencesΣ_iν_iΣ_jQ_ijlog(Q_ij/P_ij) yields h(Q)≤−Σν_iQ_ijlogP_ij=logλ, again by stationarity. This proves the finite Markov variational problem without silently extending it to every dynamical or C*-algebraic setting.

Worked example

Use A=[[1,1],[1,0]] and φ=(1+√5)/2.

Take r=(φ,1),ℓ=r/(φ²+1). Then P=[[1/φ,1/φ²],[1,0]], andπ=(φ²,1)/(φ²+1). The entropy rate is logφ. A KMS inverse-temperature interpretation requires a separately specified algebra and time evolution.

The normalization, stationary law and pressure value are explicit rather than conflated.

Scope

The full pressure variational principle and a particular KMS theorem need additional categories, measures and dynamics. A PF eigenvalue alone does not provide those identifications.

Further development

Introduce a declared edge potential W_ij>0 and repeat the eigenvector construction and KL proof for weighted pressure.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-007 Cubic KKT equations do not certify a global maximum

2025-08-09 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What hypotheses justify the cubic multiplier equations, and what can a stationary solution actually prove?

The August9 maximality solver repeatedly produced KKT roots and square-root formulas. Their algebra is a necessary first-order condition under regularity, not an automatic global certificate. A one-dimensional cubic constraint makes both sign branches and missing optima explicit.

Definitions and assumptions

Consider f(x)=cᵀx and equality constraints g_i(x)=Σ_jA_ijx_j³−b_i=0. With L=f+Σλ_ig_i, set S_j=Σ_iλ_iA_ij. Assume the constraint gradients are independent at the candidate.

First-order stationarity and its scope

A regular constrained extremum satisfies c_j+3x_j²S_j=0. If S_j≠0, real solutions require−c_j/(3S_j)≥0 and generally have both signs. Stationarity alone implies neither existence nor global optimality.

Derivation or proof

The multiplier theorem put∇f in the span of the constraint gradients, yielding the displayed equations. Solving for x_j² gives the square-root condition; taking only the positive root excludes legitimate candidates unless x_j≥0 was separately imposed. If S_j=0, the equation requires c_j=0 and cannot be divided by S_j. Even all valid stationary points need not cover singular optima. Finally a continuous objective reaches a maximum on a nonempty compact feasible set, but arbitrary affine algebraic sets need not be compact; a cubic parametrization does not supply boundedness.

Worked example

Minimize or maximize x subject to $x^3=-1$.

The only feasible point is $x=-1$. Its gradient constraint is3 and stationarity uses $\lambda=-1/3$. The positive-root-only expression would produce+1 and miss the feasible point. Separately $f(x)=x$ on the unconstrained real line has no maximum at all.

A solver needs feasibility, regularity/singular-case handling and a global comparison certificate after computing multipliers.

Scope

Random surrogate restriction matrices and reported H^1 dimensions do not prove that a KKT optimum solves a geometric problem. The source matrices were not recovered.

Further development

Restrict to a proved compact semialgebraic domain and certify candidate completeness and objective bounds with exact elimination or interval methods.

Source receipt: months/08/root_followups.json record 0. Mathematical references: .

Y25-08-008 Consistent Gaussian coordinates need not live in the Hilbert space

2025-08-09 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which covariance condition turns cylinder data into an actual Hilbert-valued random variable?

The August9 synthesis invoked Sazonov and Minlos. A diagonal Hilbert-space construction proves a concrete sufficient condition and exhibits the failure of identity covariance. This keeps a genuine measure theorem separate from formal consistency or category names.

Definitions and assumptions

Let $H=\ell^2$ with basis e_n and independent standard normals Z_n. For $q_n\geq 0$ consider $X=\Sigma\sqrt{q_n}Z_ne_n$. The covariance has diagonal entries q_n.

Trace-summable construction and identity-covariance failure

If$\Sigma q_n<\infty$, X converges almost surely in H and in mean square, with characteristic function $\exp(-\Sigma q_nh_n^2/2)$. For $q_n=1$ the coordinate law givesℓ^2 probability zero.

Derivation or proof

The nonnegative random sum S=Σq_nZ_n² has expectationΣq_n by monotone convergence. Finite expectation implies S<∞ almost surely, which is exactly square summability of the coordinates. The tail mean-square error equalsΣ_{n>N}q_n, proving L² convergence. Finite Gaussian characteristic functions multiply, and dominated convergence yields the displayed limit. The resulting Borel probability on the Polish spaceℓ² is Radon. By contrast the strong law gives N⁻¹Σ_{n≤N}Z_n²→1 almost surely when q_n=1, forcing divergence of the squared norm. Consistency of all finite marginals remains true, but their product measure is supported outsideℓ².

Worked example

Choose q_n=2^(−n),n≥1.

The covariance trace and E||X||² are one. Truncation after N has squared mean error2^(−N). Identity-covariance truncations have expected squared norm N and their probability of lying in any fixed-radius ball tends to zero.

Support and tightness, not just finite consistency, are the decisive missing hypotheses.

Scope

This proves the diagonal case, not the general Sazonov or nuclear-space Minlos theorem. Cameron-Martin vectors and typical samples also have different meanings.

Further development

Specify the actual topological vector space and covariance, then prove the continuity/tightness assumptions of the chosen extension theorem.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-005 The Perron projection must carry the Perron eigenvalue

2025-08-10 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Why is M=Π+R inconsistent with a Perron eigenvalueφ when Πr=r and Rr=0?

The August10 source simultaneously claimed λ≈1.618,Π=rℓᵀ,R=M−Π,Rr=0 and M^k→Π. These assertions cannot all hold. The correct decomposition is elementary and makes the normalization error decisive.

Definitions and assumptions

Let Mr=λr,ℓᵀM=λℓᵀ,ℓᵀr=1 and Π=rℓᵀ. Assume λ is simple and all other eigenvalues have modulus belowλ>0.

Correct spectral projection decomposition

With R=M−λΠ, one has Π²=Π,ΠR=RΠ=0 and M^k=λ^kΠ+R^k for k≥1. Therefore(M/λ)^k→Π whenρ(R)<λ. Defining R=M−Π instead gives Rr=(λ−1)r.

Derivation or proof

The normalization gives Π²=r(ℓᵀr)ℓᵀ=Π and MΠ=ΠM=λΠ. SubtractingλΠ yields the zero cross-products, so binomial expansion has only the two pure terms. Powers of R/λ tend to zero in finite dimension when its spectral radius is below one, including possible polynomial Jordan factors. The erroneous definition subtracts only r from Mr, leaving(λ−1)r, which directly contradicts the claimed annihilation unlessλ=1. Also a number1−ρ(R) is not a universal transient norm budget for a nonnormal matrix.

Worked example

Use the symmetric Fibonacci matrix M=[[1,1],[1,0]].

Its eigenvalues areφ and−1/φ. The orthogonal PF projection isΠ=(1/(φ²+1))[[φ²,φ],[φ,1]]. Here R=−φ⁻¹(I−Π), and normalized errors decay exactly asφ^(−2k). Unnormalized M^k grows along r rather than converging.

A single exact eigenvector calculation exposes the historical inconsistency and supplies the corrected convergence theorem.

Scope

General nonsymmetric matrices may have large transient constants. Spectral radius alone does not certify a numerical solver or perturbation margin.

Further development

Add a proved norm or resolvent bound and distinguish asymptotic rate from finite-step amplification before claiming a stability budget.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-006 Chebyshev acceleration needs an interval and a norm

2025-08-10 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which spectral assumptions justify a Chebyshev residual bound, and why is a generic preconditioner label insufficient?

The solver narrative named Chebyshev spectrum shaping without recovered operators. A positive-definite matrix with a known interval admits an exact polynomial bound. The assumptions explain why the same formula cannot be transferred blindly to a nonnormal operator.

Definitions and assumptions

Let A be real symmetric positive definite with spectrum in[a,b],0<a<b. Set c=(a+b)/2,d=(b−a)/2 and p_m(λ)=T_m((c−λ)/d)/T_m(c/d), where T_m(cosθ)=cos(mθ). Then p_m(0)=1.

Uniform residual polynomial bound

For $e_m=p_m(A)e_0$, $||e_m||_2 \le |T_m(c/d)|^{-1}||e_0||_2 \le 2((\sqrt{\kappa}-1)/(\sqrt{\kappa}+1))^m||e_0||_2, \kappa=b/a$.

Derivation or proof

The affine map sends[a,b] to[−1,1], where|T_m|≤1. The spectral theorem turns the matrix norm into the maximum scalar modulus on its spectrum. For z>1, T_m(z)=(w^m+w^(−m))/2 with w=z+√(z²−1)>1, so its reciprocal is at most2w^(−m). At z=c/d, w=(√κ+1)/(√κ−1), giving the formula. A nonsymmetric right preconditioner can destroy symmetry in the Euclidean inner product; eigenvalue placement alone then does not justify the same norm equality. One must identify an appropriate energy inner product or use a different resolvent-based argument.

Worked example

Take A=diag(1,4),m=1.

Here c/d=5/3 and $p_1(\lambda)=1-2\lambda/5$. Its eigenvalue multipliers are3/5 and−3/5, so the exact worst-case contraction is3/5. The general asymptotic bound is looser but valid.

The interval, polynomial normalization and norm are part of the theorem, not implementation details.

Scope

No source preconditioner matrix or spectrum certificate was recovered. This does not validate the historical solver’s claimed performance.

Further development

Construct a symmetric positive preconditioner with certified spectral bounds, then compare finite-step residual norms against the exact polynomial certificate.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-020 BFGS positivity requires curvature, not Armijo alone

2025-08-10 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which algebraic condition preserves a positive inverse-Hessian estimate, and what does the recovered quadratic test actually verify?

The August10 narrative compiler included L-BFGS scaling and Armijo backtracking. A complete quadratic example checks its line search and scale. The inverse-update formula identifies the separate curvature safeguard required for positive-definite search directions.

Definitions and assumptions

Let H be symmetric positive definite,$s=x^{+}-x, y=\nabla f(x^{+})-\nabla f(x), \rho=(y^{T}s)^{-1}$. The inverse-BFGS update is $H^{+}=(I-\rho sy^{T})H(I-\rho ys^{T})+\rho ss^{T}$.

Curvature condition and positive definiteness

If $y^{T}s>0$, then H^{+} is positive definite. An Armijo decrease test alone does not guarantee this curvature condition.

Derivation or proof

For any $z\neq 0$, $z^{T}H^{+}z=w^{T}Hw+\rho(s^{T}z)^{2}$ with $w=(I-\rho ys^{T})z$. Both terms are nonnegative. If $s^{T}z\neq 0$ the second is positive; if it is zero then $w=z$ and the first is positive. This proves the claim. Armijo bounds $f(x+\alpha p)$ against $f(x)+c_{1}\alpha\nabla f^{T}p$, but does not bound the new directional derivative; a curvature/Wolfe condition or an explicit $y^{T}s$ safeguard is additional. The scale $H_{0}=(s^{T}y)/(y^{T}y)I$ is positive only under the same sign requirement and a nonzero denominator.

Worked example

Use $f=(x_{0}-1)^{2}+(x_{1}+2)^{2}$ at $x=(4,-6)$.

The gradient is$(6,-8)$, initial objective25. A unit steepest-descent step leaves objective25 and fails Armijo at $c_{1}=10^{-4}$. Step1/2 reaches$(1,-2)$ with objective0. Then $s=(-3,4), y=(-6,8), s^{T}y=50, y^{T}y=100$, so the initial inverse scale is1/2.

The test checks real derivative and update identities rather than merely a solver status.

Scope

No global L-BFGS convergence theorem follows from one quadratic. Nonsmooth objectives and skipped or damped updates require their own hypotheses.

Further development

Certify descent, curvature and sufficient-decrease conditions at every update, then compare finite-memory outputs with a direct inverse update on a small exact problem.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: R146.

Y25-08-021 ISTA and conjugate gradients need different mathematical contracts

2025-08-10 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which assumptions justify a proximal step or a residual certificate, rather than a generic solver-success label?

The August10 compiler listed several algorithms with validators. The mathematical contract must depend on the chosen solver. A proximal descent proof and an SPD residual bound show two distinct valid certificates, neither of which follows merely from matching array shapes.

Definitions and assumptions

Let f have L-Lipschitz gradient and let g be convex. A proximal step minimizes $g(z)+(1/(2t))||z-x+t\nabla f(x)||^2$. Separately, for Ax=b assume A symmetric positive definite and $\lambda_min(A)\geq a>0$.

Descent and a residual error bound

For$0<t\leq 1/L$, the proximal step satisfies $F(x^+)\leq F(x)-(1/t-L/2)||x^+-x||^2$, F=f+g. For an approximate SPD solve with residual r=b−Ax, $||x-x^*||_A\leq||r||_2/\sqrt{a}$.

Derivation or proof

Optimality of the proximal minimizer gives $s=-\nabla f(x)-(x^+-x)/t\in\partial g(x^+)$. Convexity of g bounds its change by$-\nabla f(x)\cdot d-||d||^2/t$, where $d=x^+-x$. The gradient descent lemma bounds f's change by$\nabla f(x)\cdot d+(L/2)||d||^2$. Adding proves descent. For the linear solve, $e=x^*-x=A^{-1}r$, so$||e||_A^2=r^TA^{-1}r\leq||r||^2/a$ by the spectral theorem. Nonsymmetric or indefinite matrices invalidate this SPD energy argument, even when an iterative routine returns a small-looking number.

Worked example

Take $f(x)=||x-(3,-1)||^2/2, g=||x||_1$ and t=1.

L=1 and the proximal step is coordinate soft-thresholding of(3,−1) at one, giving(2,0), the separable global optimum. A CG symmetry warning is not a certificate that an unrelated operator meets this proximal contract.

Algorithm-specific hypotheses yield checkable decrease or error statements.

Scope

The source implementation and all test matrices were not recovered. No universal convergence claim is made for the narrative compiler.

Further development

Attach an exact problem class and stopping certificate to each solver, and reject inputs whose assumptions cannot be verified.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-009 Čech obstruction transfer requires a chain equivalence

2025-08-12 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What exact map preserves a first-cohomology obstruction, and why are cover-preserving labels alone insufficient?

The August12-13 assistant asserted literal equivalence across four domains. A finite cochain complex gives a valid obstruction test, while an explicit counterexample shows why arbitrary adapters do not induce isomorphisms. The historical random surrogate matrices are not adopted as geometric data.

Definitions and assumptions

A cochain complex has $C^0 \to \delta^0 C^1 \to \delta^1 C^2$ with $\delta^1\delta^0=0$. Its H^1 is $\ker\delta^1/\mathrm{im}\delta^0$. A cochain map F satisfies $F\delta=\delta F$. A cochain homotopy equivalence has an inverse up to maps of the form $\delta h+h\delta$.

Homotopy equivalence preserves obstruction classes

> A cochain homotopy equivalence induces an isomorphism on H^1. An arbitrary cochain map, even between three-patch constructions, need not be injective or surjective.

Derivation or proof

A cochain map sends cocycles to cocycles and coboundaries to coboundaries, hence induces cohomology maps. If $GF-I=\delta h+h\delta$, then on a cocycle z the difference $GFz-z=\delta hz$ is a coboundary. Thus the induced composite is the identity; the other homotopy proves the inverse composite. Without these hypotheses the zero map between identical nontrivial complexes is already a counterexample to injectivity. Cover preservation and finite-limit preservation by themselves are not a substitute for the specific descent and comparison hypotheses of a sheaf-cohomology theorem.

Worked example

Use the triangle graph complex $C^0=Q^3, C^1=Q^3, C^2=0$ with oriented edge differences.

The vertex-to-edge matrix has rank two, so H^1 has dimension one. The constant1 edge circulation is a nonzero class. The identity preserves it and the zero chain map kills it. Filling the triangle adds a rank-one δ^1 condition and removes this class entirely.

The domain adapters need actual complexes, restriction maps and a proved comparison, not shared names for a nonzero number.

Scope

No CA/relativity/urban/category equivalence or persistence correspondence is established. The source’s claimed $H^1=15/0$ values lacked recovered genuine restriction matrices.

Further development

Return all boundary matrices, prove $\delta^2=0$ and construct explicit comparison maps with inverse homotopies before transporting an obstruction.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-016 Gauge transformations and Picard-Fuchs coordinates must be normalized

2025-08-13 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which operator annihilates the factorial period series, and how does a square-root gauge change its local monodromy?

The August13 synthesis mixed ribbon forms with mirror-quintic, BPZ and Seiberg-Witten equations. A formal factorial series and a gauge-conjugation identity provide exact mathematics. They do not construct the claimed inter-domain adapters or their monodromy comparisons.

Definitions and assumptions

Let θ=z d/dz and a_n=(5n)!/(n!)⁵. Put F(z)=Σa_nz^n. A scalar gauge y=z^αv is interpreted on a chosen branch over a punctured neighborhood.

Factorial recurrence and gauge shift

F solves[θ⁴−5z(5θ+1)(5θ+2)(5θ+3)(5θ+4)]F=0 formally. Gauge conjugation gives z^(−α)θz^α=θ+α.

Derivation or proof

The factorial quotient gives n⁴a_n=5(5n−4)(5n−3)(5n−2)(5n−1)a_{n−1}; these are exactly the coefficient equations of the displayed operator. Its first coefficients are1,120,113400. An operator with−ζ rather than−5z uses a rescaled coordinateζ=5z and correspondingly rescaled coefficients; the two conventions cannot be mixed. The product rule givesθ(z^αv)=z^α(θ+α)v, proving conjugation for any polynomial inθ, with coefficient multiplication retained in its proper order. A loop around zero multiplies z^α by e^(2πiα), so a half-power gauge contributes−1 to monodromy.

Worked example

Use α=1/2 and the coefficient at z^1.

The recurrence gives $a_1=5\cdot1\cdot2\cdot3\cdot4=120$. The gauge shifts everyθ by1/2 and changes a local monodromy matrix by its scalar branch factor. It is not a declaration that two unmodified differential operators are identical.

Variables, operator order and branch factors are essential data in any claimed transport between differential systems.

Scope

No integral symplectic basis, geometric period map or actual AGT/SW adapter was recovered. Formal annihilation does not validate physical or arithmetic interpretations.

Further development

Supply connection matrices, path ordering and comparison covectors, then verify the proposed gauge relation coefficientwise and compare monodromy in the same basis.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: R147.

Y25-08-010 Translated cellular-automaton recurrence is a relative periodic orbit

2025-08-15 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What exactly does $S^{(-T)}f^k(x)=x$ prove, and how does it differ from ordinary periodicity?

The August15 response supplied blinker and glider examples. The mathematical invariant is relative periodicity under a translation symmetry. Its iteration law can be proved without assuming that every repeated pattern is a new cohomological obstruction.

Definitions and assumptions

Let a deterministic map f commute with a translation action $S^{T,T\in Z^d}$. A state x is relatively periodic when $f^k(x)=S^Tx$ for some $k>0$. Ordinary periodicity is the special case $T=0$.

Iteration of a relative period

> If $f^k(x)=S^Tx$ and f commutes with translations, then $f^{(mk)}(x)=S^{(mT)}x$ for every $m\geq 0$. On a finite periodic lattice, some multiple becomes an ordinary period.

Derivation or proof

Induct on m. If the relation holds for m, apply f^k and commute it past $S^{(mT)}$ to get $S^{(mT)}f^k(x)=S^{((m+1)T)}x$. On a finite translation group, T has finite order r, so $f^{(rk)}(x)=x$. On an infinite lattice, nonzero translation drift generally prevents an ordinary period unless the state has a matching spatial stabilizer. This makes time covers and moving frames precise. An observed visual repetition is not sufficient: the equality must hold for the full state under an explicitly defined update and boundary condition.

Worked example

Use the historical formal glider data $k=4,T=(1,1)$ on an $L\times L$ periodic torus.

The iteration law gives $f^{(4m)}(x)=S^{(m,m)}x$. Since $S^{(L,L)}=I$,4L is an ordinary period bound, possibly not minimal if the pattern has extra symmetry. A blinker with $k=2,T=0$ is already periodic after two steps.

The exact recurrence distinguishes drift, temporal period and boundary-induced return.

Scope

No historical code bytes or full configurations were recovered here. The proof validates the implication from a checked equality, not the unverified claim that a particular implementation satisfies it.

Further development

Specify the local rule and finite support/boundaries, verify the entire relative-period equality, and compute the translation stabilizer to certify minimal return time.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-028 Complete-graph labels must obey the edge-count bound

2025-08-15 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which of the recovered edge counts can describe a simple graph on the stated number of vertices?

The August15 project reported strong triangular-number fits and a high R^2. Exact combinatorics already finds inconsistent labels. A fit statistic cannot repair a value outside the feasible simple-graph range or an off-by-one triangular convention.

Definitions and assumptions

For a simple undirected graph on $n \geq 2$ vertices,$0 \leq e \leq C(n,2)$. Define completeness $C = e/C(n,2)$ and residual $r = e - C(n,2)$. Use the standard triangular number $T_k = k(k+1)/2$.

Edge feasibility and corrected examples

A complete graph K_n has exactly $n(n-1)/2$ edges. Thus K_{23} has253 edges, not231; K_{10} has45 edges, so78 cannot be its simple-graph edge count. The counts528,946,4371 are correct for K_{33},K_{44},K_{94}.

Derivation or proof

Every edge is an unordered pair of distinct vertices, and there are C(n,2) such pairs. A simple graph chooses a subset, proving both the maximum and equality criterion. For the stated examples direct multiplication gives$23 \cdot 22/2 = 253$ and$10 \cdot 9/2 = 45$. With $e = 231$ at $n = 23$, the residual is-22 and completeness is$21/23$; it is a possible incomplete graph, not K_{23}. With $e = 78$ at $n = 10$, completeness is$26/15 > 1$, impossible in the stated model. The value78 is the complete edge count for $n = 13$. Calling both values triangular without fixing the indexing convention can hide this mismatch.

Worked example

Compare the two problematic records.

The first could describe a graph missing22 edges. The second needs a corrected vertex count, multigraph semantics or a data correction. They cannot be treated as the same kind of small measurement noise.

Feasibility is an exact prerequisite to regression, cross-domain scaling and interpretation.

Scope

The full42-pair dataset, extraction rule and claimed81%/71% statistics were not recovered, so they are not recomputed or endorsed.

Further development

Publish the raw vertex/edge extraction and graph model, audit every pair, and fit only after separating invalid data and indexing errors.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-018 The denominator5760 follows from a characteristic power series

2025-08-16 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What mathematical normalization produces the Â polynomial, and what does it not determine about physical constants?

The August16 framework used5760 alongside proposed gravitational and coupling formulas. The characteristic-class coefficient has an exact formal derivation. Its arithmetic does not imply those external formulas without an independent geometric and physical construction.

Definitions and assumptions

Use formal Pontryagin roots x_i with $p_1=\Sigma x_i^2$ and $p_2=\Sigma_{i<j}x_i^2x_j^2$. The Â multiplicative sequence is$\prod_i[(x_i/2)/\sinh(x_i/2)]$. The variables are formal characteristic roots, not observed coupling constants.

Degree-eight Â polynomial

$$\hat{A}=1-p_1/24+(7p_1^2-4p_2)/5760+\text{terms of higher degree.}$$

Derivation or proof

Expanding sinh gives$(x/2)/\sinh(x/2)=1-x^2/24+7x^4/5760+O(x^6)$. In the product, the quadratic-root term is$-\Sigma x_i^2/24$. The quartic-root term is$7\Sigma x_i^4/5760+\Sigma_{i<j}x_i^2x_j^2/576$. Substitute$\Sigma x_i^4=p_1^2-2p_2$ to obtain$(7p_1^2-14p_2+10p_2)/5760$. This proves the stated polynomial using only formal series multiplication. Integrating the appropriate degree over a compact oriented manifold requires actual characteristic classes; integrality of the Dirac index additionally uses a spin structure and the index theorem. An integer denominator in a universal polynomial is not by itself a derivation of a running coupling or a Planck scale.

Worked example

Use two formal roots x,y.

The x^2y^2 coefficient in the product is$1/576$. In the Pontryagin expression it is$(14-4)/5760=1/576$, an exact independent cross-term check.

The normalization and coefficient are classical and reproducible, while the proposed physical identifications remain separate hypotheses.

Scope

No source manifold, bundle or compactification data justify the historical numerical physical formulas. No current measurement or prediction is endorsed.

Further development

Specify an actual spin manifold and bundle, compute its characteristic numbers, and distinguish the proved index from any subsequent model-dependent identification.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-019 A B_5 symmetry label does not force empirical rank five

2025-08-17 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What does the signed-permutation group actually imply, and what would establish a five-dimensional feature model?

The August17-18 source alternated between rank-three/rank-four results and unsupported rank-five claims. The group order is exact, but orbit size and design-matrix rank require separate calculations. A finite counterexample prevents symmetry terminology from manufacturing empirical evidence.

Definitions and assumptions

The Weyl groups B_5 and C_5 act by arbitrary coordinate permutations and independent sign changes on R^5. Their order is$2^5 \cdot 5! = 3840$. An orbit has size group order divided by stabilizer order. A log-monomial model is linear regression after taking declared dimensionless logarithms.

Generic orbits versus degenerate orbits and feature rank

A vector with nonzero pairwise-distinct absolute coordinates has orbit3840. The vector(1,1,0,0,0) has orbit40. Neither group action determines the rank of a separately constructed data matrix.

Derivation or proof

There are$5!$ permutations and2^5 sign choices. Under the stated generic condition no nonidentity choice fixes the vector, so the stabilizer is trivial. For(1,1,0,0,0), choose the two nonzero positions in ten ways and their signs in four ways; permuting equal entries or flipping zeros changes nothing, giving40. For feature rank, a constant column becomes zero after centering. More generally exact linear dependence between declared features bounds the rank independently of their labels. An invariant polynomial ring can have five algebraically independent generators while a particular finite dataset explores fewer independent directions.

Worked example

Take five centered columns with the fifth equal to twice the first.

The matrix rank is at most four even if its columns are named after five physical axes. Conversely a group orbit count3840 does not equal the number of independent observations or a held-out error statistic.

Symmetry order, effective rank and predictive accuracy are different quantities with different certificates.

Scope

The historical physical datasets, train/test split and claimed MSE were not recovered. They remain assistant claims, not validated empirical results or mass predictions.

Further development

Freeze dimensionless features and data provenance, publish singular values and exact dependencies, then calibrate a predeclared held-out test against a justified null model.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: R132.

Y25-08-011 Measurability and Lᵖ convergence need different hypotheses

2025-08-20 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What does the Carathéodory criterion prove, and why does pointwise convergence not imply norm convergence?

The August20 continuum combined outer measure and several function spaces. A precise construction and a spike counterexample separate those issues. Neither a measurability theorem nor a small pointwise error controls an integral norm without additional hypotheses.

Definitions and assumptions

An outer measure μ* is monotone,countably subadditive and zero on∅. A set E is Carathéodory measurable if μ*(A)=μ*(A∩E)+μ*(A\E) for every A. For p≥1, ||f||_p=(∫|f|^p)^(1/p).

Carathéodory measurability and an Lᵖ spike

> The measurable sets form a sigma-algebra on which μ* is countably additive. On[0,1], f_n=n^(1/p)1_(0,1/n) converges to zero almost everywhere while||f_n||_p=1.

Derivation or proof

For the measure theorem, the criterion is invariant under complements. Applying it successively proves finite disjoint splitting and closure under finite unions. For a disjoint measurable sequence E_j, split an arbitrary A by the first N sets; monotonicity and the finite splitting give the required lower bound byΣ_{j≤N}μ*(A∩E_j). Passing to the supremum and combining countable subadditivity proves measurability of the union and countable additivity on measurable sets. For the spike, every fixed x>0 eventually lies outside(0,1/n), but∫|f_n|^p=n·(1/n)=1. Thus norm convergence fails despite pointwise convergence. A common integrable dominating pth power would repair this through dominated convergence.

Worked example

Compare fixed amplitude with growing amplitude.

For g_n=1_(0,1/n),||g_n||_p=n^(−1/p)→0. Multiplying by n^(1/p) preserves its p-norm. The support size alone therefore does not determine convergence.

Outer-measure construction, mode of convergence and uniform integrability are separate mathematical requirements.

Scope

No genomic or hyperspectral interpretation is inferred from the function-space labels. SobolevH¹ and HardyH¹ are different spaces.

Further development

State the measure, norm and a domination or uniform-integrability condition before transferring pointwise approximation claims into an integral error estimate.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-012 A torus delta integral requires a coarea factor

2025-08-20 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Does δ(g)dx equal the induced surface measure, or is a gradient factor missing?

The August20 torus equation supports an exact correction. A delta distribution on a regular level set is weighted by the reciprocal gradient norm. The resulting factor is constant for this particular defining function, but changes when the equation is rescaled.

Definitions and assumptions

Let R>r>0,ρ=√(x²+y²),g=(R−ρ)²+z²−r². Its zero set is a smooth torus. Coarea gives∫fδ(g)dx=∫_{g=0}f/|∇g|dA for suitable test functions.

Exact torus normalization

On g=0,|∇g|=2r. Hence∫fδ(g)dx=(1/(2r))∫_TfdA, whileδ(g)|∇g|dx represents dA. In particular∫δ(g)dx=2π²R.

Derivation or proof

Differentiate in cylindrical coordinates:∇g has radial component2(ρ−R) and vertical component2z. Its squared norm on the torus is4[(ρ−R)²+z²]=4r², so the coarea factor is constant2r. Parametrize by((R+r cosv)cosu,(R+r cosv)sinu,r sinv). The area density is r(R+r cosv); integrating u,v over[0,2π] gives4π²Rr. Dividing by2r proves the delta integral. If g is replaced by cg,δ(cg)=δ(g)/|c|, whereas the zero set and its geometric area are unchanged. The compensating gradient factor is therefore indispensable.

Worked example

Take R=3,r=1.

The area is12π² but the unweighted delta integral is6π². Using2g as a defining equation halves that delta integral again while preserving the torus itself.

A geometric observable must be invariant under harmless rescaling of its defining equation.

Scope

The delta is a distribution, not an ordinary pointwise function. Regularity fails for other level sets if∇g vanishes, and the present formula then needs replacement.

Further development

Derive the appropriate coarea formula for the actual level-set model and test invariance under reparametrization before interpreting surface integrals.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-013 Hardy atoms bound BMO pairings with an explicit constant

2025-08-20 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which cancellation and oscillation estimates support the named Hardy-BMO connection?

The August20 synthesis named Fefferman duality and controlled irregularity. A local atomic inequality can be proved completely and is useful without pretending to prove the full global duality theorem or identify Sobolev and Hardy spaces.

Definitions and assumptions

For an interval I define b_I=|I|$^{-1}$∫_Ib and osc_I(b)=|I|$^{-1}$∫_I|b−b_I|. A real Hardy atom a is supported on I, has mean zero and||a||∞≤1/|I|. The BMO seminorm is sup_I osc_I(b).

Atomic pairing and bounded perturbation

|∫ab|≤osc_I(b). If b∈BMO and||e||∞≤ε, then b+e∈BMO and| ||b+e||_BMO−||b||_BMO |≤2ε.

Derivation or proof

Mean zero gives∫ab=∫a(b−b_I); bounding a by1/|I| proves the pairing. For perturbations, osc_I(b+e)≤osc_I(b)+osc_I(e) by the triangle inequality. Since|e_I|≤ε, the second oscillation is at most2ε. Taking suprema and exchanging b with b+e proves the finite two-sided bound. These arguments identify exactly which norms control cancellation and perturbation. Full(H^{1})*=BMO involves atomic decompositions and functional representation beyond this single estimate; merely choosing a function called entropy does not establish that theorem’s hypotheses.

Worked example

Take I=[−1,1],a=sign(x)/2 and b=x.

The atom has zero mean and sup norm1/2. Here b_I=0,osc_I(b)=1/2 and∫ab=(1/2)∫|x|dx=1/2, so the local constant is attained. The function x on all of R is not claimed globally BMO.

The cancellation, coefficient convention and perturbation class make the irregularity claim mathematically testable.

Scope

No physiological or urban-model inference follows from this local estimate. HardyH¹ is not the derivative-based SobolevH¹.

Further development

Specify the intended function space and atoms, then prove the required global decomposition or embedding with constants uniform in the proposed parameters.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: R148.

Y25-08-014 A Landau quartic has a precise minimizer set

2025-08-20 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which conclusions follow from the stated quartic potential before invoking a field theory or phase transition?

The categorical August20 discussion also named a Landau-Ginzburg energy. Its pointwise potential is solvable exactly. Separating that minimization from the gradient energy and statistical model prevents a formal minimum circle from being presented as a proved physical phase transition.

Definitions and assumptions

Let $\varphi \in C, u>0$ and $V(\varphi)=r|\varphi|^2+u|\varphi|^4$ with r real. For a spatial field one may add $\int|\nabla\varphi|^2$, but boundary conditions and function space must then be stated separately.

Exact quartic minima

If $r \geq 0$,V has unique minimum at$\varphi=0$. If $r<0$, its minimum is$-r^2/(4u)$, attained exactly on$|\varphi|=\sqrt{(-r/(2u))}$.

Derivation or proof

Set $s=|\varphi|^2 \geq 0$. Complete the square:$rs+us^2=u(s+r/(2u))^2-r^2/(4u)$. If $r<0$ the square vanishes at an allowed positive s, giving the circle. If $r \geq 0$ the function is increasing for $s \geq 0$ and minimized at zero. Along a nonzero minimum the phase direction costs no pointwise potential, while radial perturbations do. For a field with unconstrained periodic boundaries, constant minimizers also minimize the nonnegative gradient term. Nonconstant boundary conditions, topological sectors or fluctuations change the variational problem and require additional analysis.

Worked example

Use $r=-2, u=1$.

$V=|\varphi|^4-2|\varphi|^2=(|\varphi|^2-1)^2-1$. Every unit complex number minimizes it with value$-1;\varphi=0$ has value0 and is not a minimum. Fixing the field to zero on a boundary prevents a nonzero constant from meeting that boundary condition.

The minimizer manifold is a proved feature of a declared polynomial, with the remaining physical interpretation separated.

Scope

No quantum field theory, finite-temperature transition or categorical Ext obstruction is established by this elementary potential calculation.

Further development

Specify a domain and boundary conditions and prove existence and stability of minimizers for the full gradient-plus-potential energy.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-015 Hall coefficients count subobjects even when Ext¹ vanishes

2025-08-20 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which actual counting problem defines the Hall product, and what does a small example say about extension obstructions?

The August20 synthesis connected Hall algebras, Ext² and spectral sequences. Finite vector spaces provide a complete Hall counting example. It also distinguishes counting subobjects from detecting a nonsplit extension.

Definitions and assumptions

Work in finite-dimensional vector spaces over F_q. Let[n] denote an isomorphism class of dimension n. Use the Hall convention[a]*[b]=Σ_c g^a,b_c[c], where g counts b-dimensional subspaces with a-dimensional quotient.

Gaussian binomial coefficients and flag associativity

[a]*[b]=GaussianBinomial(a+b,b)_q[a+b]. In particular[1]*[1]=(q+1)[2], and[1]^3=(q+1)(q²+q+1)[3].

Derivation or proof

Only a space of dimension a+b can contribute. Count ordered linearly independent b-tuples in F_q^(a+b):there are∏_{j=0}^{b−1}(q^(a+b)−q^j). Divide by the number∏_{j=0}^{b−1}(q^b−q^j) of ordered bases of a fixed b-plane. This yields the Gaussian binomial. Associativity counts the same flags with two successive quotient dimensions in either parenthesization. For three one-dimensional factors, choose a line and then a plane containing it, giving(q²+q+1)(q+1). Every short exact sequence of vector spaces splits, so Ext¹=0 here despite nontrivial Hall coefficients. In a spectral sequence, classes can similarly disappear under later differentials; a page dimension is not automatically an abutment dimension.

Worked example

Take q=2.

There are three lines in F_2^2, so[1]*[1]=3[2]. In F_2^3 there are seven lines and three planes through each line, giving21 complete line-plane flags and[1]^3=21[3].

A named algebra becomes an exact finite counting structure, distinct from an obstruction group.

Scope

No derived Hall algebra or specific Ext^2 obstruction is constructed. Spectral-sequence claims require actual differentials and convergence data.

Further development

Move to a specified finite quiver category, enumerate extensions and automorphism factors, and verify associativity with the chosen normalization.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-017 Phase consistency is a cycle condition, not a missing matrix certificate

2025-08-20 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What can be verified from the recovered edge labels when the original eleven-edge coefficient matrix is unavailable?

The August20 source reports ranks and maximal exact subsets but does not supply its full coefficient matrix. A clearly defined graph-potential model can still be analyzed exactly. It must not be silently identified with that missing model, because it gives different subset counts.

Definitions and assumptions

For a directed ratio edge u/v with label $b_{uv}\in F_5$ define the equation $x_u-x_v=b_{uv}$. On a connected graph, potentials are defined up to adding a constant. The published mod60 representatives can be reduced modulo five, but this does not recover every original axis constraint.

Cycle criterion for graph potentials

The graph equations are solvable exactly when every signed cycle sum vanishes. A connected incidence matrix has rank$|V|-1$. The recovered representatives are inconsistent in this model, but deleting the c/s edge leaves ten consistent edges.

Derivation or proof

Necessity follows by telescoping potential differences around a cycle. For sufficiency choose a root potential zero and integrate labels along a spanning tree. Two paths give the same value precisely when their difference cycle has zero sum. Tree edges are independent and all remaining incidence rows lie in their span, proving the rank. In the source list every mod5 label is zero except c/s=17≡2. Yet c/τ=55,τ/μ=0 and s/μ=55 force x_c=x_τ=x_μ=x_s, contradicting x_c−x_s=2. If that one edge is removed, the zero potential satisfies all remaining ten equations. Thus the source’s claimed maximum-eight family cannot be reconstructed from edge phases alone; its original matrix contains missing structure or different semantics.

Worked example

Follow the cycle c→τ→μ→s→c.

The first three differences are zero and the last contributes−2, giving a nonzero cycle. CRT modulo60 combines residues modulo3,4,5 only after each component is consistent; it does not repair a contradiction modulo five.

The exact obstruction is real for the declared graph model, while unrecovered ranks and subset statistics remain qualified historical claims.

Scope

The original eleven-edge exponent/axis matrix and seed table were not recovered. This entry does not certify its reported maximal subsets or a physical mass relation.

Further development

Publish the exact matrix and right-hand side, then enumerate feasible subsets and provide explicit left-nullspace inconsistency witnesses.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-023 The Heawood spectrum follows from Fano-plane incidence

2025-08-23 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which exact incidence identity proves the graph spectrum and the displayed Laplacian zeta function?

The August23 testbeds named Ramanujan graphs, graph zeta functions and primitive geodesics. A finite incidence construction gives a complete spectral certificate. It distinguishes the ordinary Laplacian zeta from Ihara’s nonbacktracking zeta.

Definitions and assumptions

Let B be the7×7 point-line incidence matrix of the Fano plane: each point lies on three lines and each distinct pair shares one line. Let A=[[0,B],[B^{T},0]] and L=3I−A. Remove the zero Laplacian eigenvalue from ζ_L(s).

Exact Heawood spectrum and zeta

A has eigenvalues3,−3, and±√2 each with multiplicity six. Thus ζ_L(s)=6(3−√2)^(−s)+6(3+√2)^(−s)+6^(−s).

Derivation or proof

The incidence property gives $BB^T=2I+J$. On the constant line this has eigenvalue9, while the six-dimensional orthogonal complement has eigenvalue2. The symmetric block adjacency matrix therefore has the positive and negative singular values3 and√2 with the stated multiplicities. Subtracting from3 gives Laplacian eigenvalues0,6,3±√2. Their power sum proves the zeta formula. The nontrivial adjacency values satisfy the cubic Ramanujan bound2√(3−1), but that does not identify graph zeta zeros with arithmetic L-function zeros. Ihara's determinant formula uses $\det(I-uA+2u^2I)$ and a separate factor$(1-u^2)^{21-14}$, not the Laplacian inverse-power sum.

Worked example

Evaluate ζ_L(1).

The conjugate reciprocal sum is1/(3−√2)+1/(3+√2)=6/7, so ζ_L(1)=36/7+1/6=223/42. The graph has14 vertices and21 edges; its girth-six property is separately checked from its incidence structure.

A complete finite operator identity supports the exact graph claims and keeps the two zeta constructions distinct.

Scope

This does not prove a number-field Riemann hypothesis, spectral universality or a statistical law for arbitrary graphs.

Further development

Construct the nonbacktracking matrix and verify Bass's determinant identity directly, including multiplicities and the first cycle lengths.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-024 Paley Laplacian moments have closed rational values

2025-08-23 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Under which finite-field hypotheses do the Paley spectral formulas apply, and what do their inverse moments count?

The August23 source supplied striking exact zeta values. They follow algebraically from a specific strongly regular graph spectrum, not from a general anomaly law. The smallest example is a five-cycle and can be checked without approximate eigenvalue fitting.

Definitions and assumptions

Let q be an odd prime power with q≡1 mod4. The Paley graph joins elements of F_q whose difference is a nonzero square. Its degree is d=(q−1)/2; the classical nontrivial adjacency eigenvalues are(−1±√q)/2, each with multiplicity d.

First two Laplacian inverse moments

The positive Laplacian eigenvalues are(q±√q)/2, each with multiplicity d. Therefore ζ_L(1)=2 and ζ_L(2)=4(q+1)/(q(q−1)).

Derivation or proof

Subtract the adjacency eigenvalues from d. Their positive Laplacian pair has sum q and product q(q−1)/4. The sum of reciprocals is4/(q−1); multiply by d to get two. The sum of reciprocal squares is(sum²−2product)/product²; simplifying and multiplying by d gives the second formula. The underlying adjacency spectrum is a classical quadratic-character/Gauss-sum theorem and is stated as such, rather than proved from these moments. The moments do not uniquely identify an arbitrary graph, since different spectra can share a few power sums.

Worked example

Use q=5.

The Paley graph is C_5. Its positive Laplacian values are(5±√5)/2, each twice. Hence ζ_L(1)=2 and ζ_L(2)=6/5. Kirchhoff’s matrix-tree formula gives five spanning trees, agreeing with the elementary fact that removing any one edge of a five-cycle produces a tree.

The exact rational moments are explained by conjugate algebraic eigenvalues.

Scope

The q≡1 mod4 condition ensures an undirected graph. These formulas are not a universal spectral signature or an arithmetic L-function theorem.

Further development

Derive the adjacency spectrum from a specified additive-character convention and verify both moment and spanning-tree formulas independently.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-025 CRT and finite entropy do not prove decimal normality

2025-08-24 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which six-kernel statistics are exact, and what can sixteen residues actually establish?

The August24 source reported a16-row seed, a near-maximal ternary entropy and a PASS verdict. The full table was not recovered. CRT reconstruction and the maximal possible entropy of sixteen ternary

observations can still be proved, exposing why a finite diagnostic is not an asymptotic randomness theorem.

Definitions and assumptions

For r∈Z/48 let a=r mod16 and b=r mod3. For counts $n_0+n_1+n_2=16$ define $H=-\Sigma(n_i/16)\log(n_i/16)$. The source sequence r_n=floor(48{10^nx}) uses x=(φπ)^(2/9); exact evaluation requires sufficient precision.

CRT reconstruction and finite entropy ceiling

r≡33a+16b mod48. Among sixteen ternary samples, entropy is maximized by counts(6,5,5) up to permutation, giving H/log3≈0.9965. This does not prove equidistribution or normality of x.

Derivation or proof

The CRT coefficients satisfy33≡1 mod16,33≡0 mod3,16≡0 mod16 and16≡1 mod3, proving reconstruction and uniqueness. For entropy, the function−t logt is concave. If two integer counts differ by at least two, moving one from the larger to the smaller increases the sum. Repeating yields the balanced integer partition6,5,5. Thus the reported entropy scale can arise simply from the best-balanced count pattern at this sample size. Decimal normality requires limiting frequencies of all blocks; neither sixteen residues nor a cryptographic manifest proves it. Binary floating multiplication by10^n can also lose fractional information and must not be treated as exact real arithmetic.

Worked example

Take a=7,b=2.

The reconstructed residue is33·7+16·2=263≡23 mod48; indeed23 mod16=7 and23 mod3=2. The capped valuation convention $v_2(0)=4$ means $\min(v_2(r),4)$, not the ordinary valuation of zero.

The exact finite checks survive, while the unrecovered seed verdict and asymptotic interpretation remain qualified.

Scope

No seed table, null calibration or proof of normality was recovered. A near-maximal marginal entropy can coexist with completely periodic data.

Further development

Publish exact residues with interval precision certificates, predeclare block statistics and compare against a stated null without treating finite acceptance as a theorem about the infinite sequence.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-027 The Riccati flow and its wall times are exact within the model

2025-08-25 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which identities genuinely follow from $\mu' = b\mu^2$, and which entropy or winding interpretations are extra assumptions?

The August25 source gives a calibrated scalar flow with several derived quantities. The ODE and hitting-time formulas can be verified exactly. Their correctness does not establish the proposed information-geometric or A_2 interpretation.

Definitions and assumptions

Let $b, \mu_0 > 0$ and $\mu(0) = \mu_0, \mu' = b\mu^2$. Define $F(\mu) = -nc \log \mu/(2b), N(\ell) = \log[1/(1 - b\mu_0\ell)]/(6b)$, on $\ell < 1/(b\mu_0)$.

Flow invariant and half-integer walls

$\mu = \mu_0/(1 - b\mu_0\ell), 1/\mu + b\ell - 1/\mu_0 = 0, F'(\ell) = -nc\mu/2$ and $N' = \mu/6$. The time satisfying $N(\ell_k) = k/2$ is$\ell_k = (1 - e\text{^}(-3bk))/(b\mu_0)$.

Derivation or proof

Differentiate$1/\mu$ to obtain$-b$ and integrate. This proves both the rational solution and invariant; the denominator gives the finite forward pole. Chain differentiation gives $F' = -nc\mu'/(2b\mu) = -nc\mu/2$. Differentiating N gives$\mu/6$. Solving its logarithmic equation at $k/2$ yields the wall formula. For $k \geq 0$ the walls increase toward the pole and never cross it. If one defines an entropy as$\int \mu' d\ell = \mu - \mu_0$, that identity is simply the fundamental theorem of calculus; it does not identify Fisher information or physical entropy without a probability model.

Worked example

Use the stipulated $b = 0.0455 = 91/2000$ and $\mu_0 = 0.1$.

The pole is$20000/91 \approx 219.78022$. At$\ell = 100, \mu = 20/109 \approx 0.183486$, and the invariant is exactly$109/20 + 91/20 - 10 = 0$. Wall values use the same fixed parameters and can be checked by substitution.

The calibrated ODE is mathematically complete within its declared domain, while its external interpretation is not promoted.

Scope

The truncated tuple later in the source was not recovered. No Fisher, winding or particle-model claim is established by scalar integration.

Further development

Specify an actual likelihood or geometric connection and prove the proposed invariant relation before interpreting F,N or the pole physically.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-026 The lattice symbol has a proved dispersion and anisotropy bound

2025-08-26 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which refinement discrepancy is expected from the discrete Laplacian, and when is a residue check circular?

The August26 HLR work corrected its discrete symbol and reported very small numerical residues. Fourier analysis supplies the exact operator and an explicit approximation bound. A self-consistency residual using the same symbol is distinguished from independent validation.

Definitions and assumptions

For mesh h, let$-\Delta_h u_j=(2u_j-u_{j+1}-u_{j-1})/h^2$. On $u_j=e^{ikjh}$, its symbol is$\sigma_h(k)=4h^{-2}\sin^2(kh/2)$. The two-dimensional rectangular symbol is the sum over coordinates.

Exact symbol and continuum-error bound

For real k,$0\le k^2-\sigma_h(k)\le h^2k^4/12$. In two dimensions,$0\le|k|^2-\sigma\le(h_x^2k_x^4+h_y^2k_y^4)/12$. Sampled modes alias under $k\mapsto k+2\pi m/h$.

Derivation or proof

Substitute the exponential into the three-point stencil and use$2-2\cos(kh)=4\sin^2(kh/2)$. The inequalities$1-u^2/2\le\cos u\le1-u^2/2+u^4/24$ give the error bounds after multiplication by$2/h^2$. Adding coordinate bounds proves the rectangular case. Taylor expansion further gives$\sigma_h=k^2-h^2k^4/12+O(h^4k^6)$. On a square grid the fourth-order term depends on $k_x^4+k_y^4$, so rotational anisotropy is expected even though the continuum Laplacian is isotropic. The periodic cosine also proves aliasing. A code evaluating a mode with exactly this symbol may produce a ratio numerically one by construction; that does not test an independent continuum or renormalization hypothesis.

Worked example

Take $h=1,k=\pi$.

The symbol is4 while the continuum value isπ^2. Their difference is positive and below$\pi^4/12$. Modesπ and$-\pi$ have the same grid samples and symbol, illustrating Nyquist ambiguity.

Refinement and anisotropy now have explicit expected scales instead of relying on a roundoff-sized pass statistic.

Scope

The historical implementation, mode tables and error gates were not recovered. No physical renormalization theorem follows from stencil consistency.

Further development

Use an independent continuum reference and a stable approximation norm; separate truncation, aliasing and floating-point errors in a refinement certificate.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-029 A rank-one CRO matrix has an exact spectrum and inverse

2025-08-31 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What can the recovered matrix S(t) actually prove before invoking a crystalline interpretation?

The August31 source combines a concrete rank-one matrix with sheaf and Frobenius terminology. The matrix itself is completely analyzable. Its spectral facts neither identify crystalline cohomology nor turn finite decimal constants into special transcendental invariants.

Definitions and assumptions

For n≥2 let1∈R^n be the all-ones vector and $S(t)=I+t11^{T}$. Put $P=11^{T}/n$, the orthogonal projection onto the constant line. Then S=(I−P)+(1+nt)P.

Spectrum, inverse and positivity

S has eigenvalue1+nt on the constant line and1 on its orthogonal complement. detS=1+nt; for $t\neq-1/n$, $S^{-1}=I-[t/(1+nt)]11^{T}$. It is positive definite exactly when t>−1/n.

Derivation or proof

Since$1^{T}1=n$,$P^{2}=P$ and P*=P. The direct-sum expression gives the eigenvalues and determinant immediately. Invert separately on the two orthogonal subspaces to obtain$(I-P)+(1+nt)^{-1}P$; expanding P gives the displayed rank-one inverse. Positive definiteness is equivalent to positivity of both scalar eigenvalues. The spectral norm is max(1,|1+nt|), and the condition number follows from the ratio of largest to smallest singular values. No sheaf or cohomology assumption is used anywhere. The source decimals α=0.0333996828 and β=0.6732050807568877 are rational numbers when taken literally as finite strings.

Worked example

Use n=3,t=1/6.

The eigenvalues are3/2,1,1, determinant3/2 and inverse$I-(1/9)11^{T}$. Multiplying S by that inverse gives I exactly. At t=−1/3 the constant vector lies in the kernel.

The concrete matrix has precise classical meaning; any additional geometry needs an independently defined construction.

Scope

No recovered Frobenius action, integral lattice or comparison theorem identifies this matrix with crystalline cohomology. A fitted decimal does not prove a universal constant.

Further development

Specify the base field, module and Frobenius-semilinear map, then prove which matrix invariant survives a change of basis.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-030 A residual for one vector cannot certify vanishing of H^1

2025-08-31 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What does zero numerical residue actually imply, and what exact algebra would establish a cohomology-vanishing theorem?

The August31 CRO response asserted equivalence between a runtime residue and vanishing H^1 while retaining stubs. A minimal cochain example rejects that equivalence. A separate decay-rate bound shows what the surviving logarithmic diagnostic can legitimately certify.

Definitions and assumptions

For a finite cochain complex, $H^1=\ker\delta^1/\mathrm{im}\delta^0$. In an inner-product complex the Hodge Laplacian is $L_1=\delta^0\delta^{0*}+\delta^{1*}\delta^1$. For positive observed energy set $\hat{\alpha}=-(\log E_k-\log E_{k-1})/\Delta t$.

Residual scope and logarithmic sensitivity

> A zero Laplacian residual can occur for a nonzero H^1 class. If $E_k=e^{-\alpha t_k+\eta_k}$ with$|\eta_k|\le\varepsilon$, then$|\hat{\alpha}-\alpha|\le 2\varepsilon/\Delta t$. A hash ledger certifies neither cohomology nor the truth of its logged claims.

Derivation or proof

Take $C^0=C^1=R, C^2=0$ and$\delta^0=\delta^1=0$. Then $H^1=R$ and $L_1=0$, so every vector has zero Laplacian residual although cohomology is nonzero. More generally Hodge theory identifies $\ker L_1$ with cohomology, so a nonzero harmonic vector is precisely the wrong witness for global vanishing. For the rate bound, substitute the noisy log energy:$\hat{\alpha}=\alpha-(\eta_k-\eta_{k-1})/\Delta t$ and apply the triangle inequality. Smaller time steps amplify bounded log noise. Finally a finite digest records a function of bytes; correctness of the bytes' mathematical assertion is a separate predicate and is not implied by digest equality.

Worked example

Use $\Delta t=1/10$ and log-noise bound$\varepsilon=1/1000$.

The rate error can be as large as$1/50=0.02$. Meanwhile the zero map in the one-dimensional complex has zero residual and H^1 dimension one. Neither a small rate discrepancy nor a SHA manifest changes that algebra.

The diagnostic becomes useful only when its actual implication and sensitivity are stated.

Scope

The original residue complex, norms and completed Lean proof were not recovered. No vanishing equivalence or crystalline-descent theorem is accepted from a stub or a logged PASS.

Further development

Publish exact boundary maps, verify $\delta^2=0$ and prove a quantitative equivalence between the chosen residual and the specific class property being tested; retain certificate provenance separately.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

Y25-08-022 Optimizer hypergradients must follow the entire state recursion

Date unavailable · historical evidence T · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What derivative should a one-step AdamW outer loss have, and why is a zero reported gradient not a success certificate?

This topic comes from a later recovery file attributing an OptComp project to August12, not a recovered original transcript. The distinction is retained as T evidence with no exact chat date. A newly derived one-step calculation tests the chain rule without claiming to recover the old program bytes.

Definitions and assumptions

For fixed θ_0,u_0 define $\theta_1=\theta_0-\eta u_0-\eta\lambda\theta_0$. The direction u_0 is held fixed only for this partial-derivative example. Let $L=||\theta_1-y||^2/2$. In a multi-step optimizer, momentum and adaptive moments are part of the differentiated state.

One-step outer derivatives

$\partial_\eta\theta_1=-u_0-\lambda\theta_0$ and$\partial_\lambda\theta_1=-\eta\theta_0$. Thus$\partial_\eta L=(\theta_1-y)\cdot(-u_0-\lambda\theta_0)$ and$\partial_\lambda L=(\theta_1-y)\cdot(-\eta\theta_0)$.

Derivation or proof

Differentiate the explicit affine update while holding the stated inputs fixed, then apply the ordinary chain rule to the quadratic loss. For a full state recursion $s_{t+1}=U_\varphi(s_t)$, its Jacobian obeys $J_{t+1}=(\partial_s U)J_t+\partial_\varphi U$, not just the last-step partial. Removing a dependency with a detached value can make an automatic-differentiation gradient zero even when the mathematical derivative is nonzero. Conversely a true stationary point can also give zero; the value must be checked against the actual graph and an independent derivative calculation.

Worked example

Use $\theta_0=(1,-2),u_0=(1/2,-1/4),\eta=1/10,\lambda=1/5,y=0$.

Then $\theta_1=(93/100,-387/200)$. The exact gradients are$\partial_\eta L=-1527/800=-1.90875$ and$\partial_\lambda L=-12/25=-0.48$. Centered finite differences of this quadratic reproduce them exactly in rational arithmetic.

A nonzero analytic test is available, while the historical zero-gradient report remains unverified evidence.

Scope

The later file’s audit and finite-difference work are not original successful tuning results. Real AdamW directions generally depend on parameters and previous state; those paths must be included.

Further development

Specify the entire optimizer-state graph and compare its Jacobian-vector products with exact or interval finite differences on a small deterministic problem.

Source receipt: months/08/root_receipts.json record 0. Mathematical references: .

September 2025

Dated fragments recovered; complete transcript unavailable. 21 developed treatments.

The tau-Crystal threads concern invariants, cocycles, fixed points, arithmetic experiments and proof provenance. Genuine elliptic-curve and Hecke calculations are developed alongside precise algebraic conservation and error bounds.

ID · date	Concept and revised question	Result and mathematical core
Y25-09-001 2025-09-01 · A	A compositional contraction certificate with an exact fixed point Which complete invariant domain and Lipschitz bounds certify the entire composition, and what error does its residual rigorously control?	A compositional contraction certificate with an exact fixed point A self-map with contraction constant q has a unique fixed point x*. For every x, d(x,x*)≤d(x,Tx)/(1−q). For d=2,t=1/4 and T(x)=(1/4)S_t x+(1,0), the exact constant is3/8 and x*=(22/15,2/15).
Y25-09-019 2025-09-02 · A	A modular minor certifies a rational rank lower bound Which exact minor proves the rank claim, and what independent data certify the upper bound?	A modular minor certifies a rational rank lower bound rank_pA≤rank_QA. If a minor is nonzero modulo p and a rational rank-r factorization is verified exactly, the two certificates prove rank_QA=r. Conversely rank_pA=0 does not imply rational rank zero.
Y25-09-020 2025-09-06 · A	Adjacent exact inequalities prove full monotonicity of a finite clock What does monotonicity mean over the exact stored type, and does the proposed proof establish all index pairs?	Adjacent exact inequalities prove full monotonicity of a finite clock A sequence in a transitive ordered set is nondecreasing if and only if every adjacent inequality t_k≤t_(k+1) holds. Over Q, positive adjacent increments imply strict increase, and t_j−t_i is their telescoping sum.
Y25-09-021 2025-09-08 · A	A genuine KdV soliton requires a PDE identity, not a checkpoint metaphor Which field and equation make the tau function a soliton construction, and can the traveling wave be verified directly?	A genuine KdV soliton requires a PDE identity, not a checkpoint metaphor The field u=F(z) equals2∂²_xlogτ and solves the declared KdV equation exactly. It travels at speed4k², has amplitude2k² and integral∫_R u dx=4k.
Y25-09-005 2025-09-09 · A	A trace invariant follows from local rewrite equations by induction Which precise transition relation preserves which algebraic quantity, and what statement must the formal proof actually establish?	A trace invariant follows from local rewrite equations by induction If wᵀr_j=0 for every allowed rewrite, τ has the same value at both ends of every finite trace. Conversely, conservation for all one-step traces forces all these equations. Thus exact linear conservation laws are the integer solutions of Rᵀw=0, where R contains the rewrite columns.

ID · date	Concept and revised question	Result and mathematical core
Y25-09-018 2025-09-10 · A	A geometric checkpoint clock estimates a decay rate without being a soliton What mathematical quantity does the checkpoint schedule estimate, and which formula remains invariant when the time spacing changes?	A geometric checkpoint clock estimates a decay rate without being a soliton For exact exponential data $\alpha_hat_k=\alpha$ for every k, regardless of q. The number of positive checkpoints through horizon $T\geq t_0$ is$1+\mathrm{floor}(\log(T/t_0)/\log q)$. If each log measurement has absolute error at mostδ, then $\lvert\alpha_hat_k-\alpha\rvert\leq 2\delta/(t_k-t_(k-1))$.
Y25-09-015 2025-09-14 · A	The square-root smoothing map has dynamics but is not a linear Hecke operator What does the actual surrogate map do, and which exact endpoint tests can reject a claimed commutation law?	The square-root smoothing map has dynamics but is not a linear Hecke operator h is increasing and concave, has exactly the fixed points0 and1, and every orbit starting in$(0,1]$ converges to1. It is not linear and is not globally Lipschitz on$[0,1]$. Any commuting θ must send each of0,1 to one of0,1.
Y25-09-007 2025-09-15 · A	The symmetric-square factor needs all three eigenvalue products Which normalization of the local parameters defines Sym^2, and what information is lost by squaring only the trace?	The symmetric-square factor needs all three eigenvalue products The unramified symmetric-square denominator is$(1-\alpha^2t)(1-pt)(1-\beta^2t)=1-(a^2-p)t+p(a^2-p)t^2-p^3t^3$. For the exact $p=2,a=-2$ input, it equals$1-2t+4t^2-8t^3$.
Y25-09-008 2025-09-15 · A	FFT derivatives are exact for an interpolant, not for an unknown function Which frequency band and aliasing bound justify the derivative estimate used by the zero detector?	FFT derivatives are exact for an interpolant, not for an unknown function DFT differentiation exactly recovers derivatives of the declared band-limited polynomial. Without a frequency or analytic-tail bound, no finite sample grid gives a uniform derivative-error bound: $h(x)=\cos(Nx)-1$ vanishes on the grid but $h''(0)=-N^2$.
Y25-09-009 2025-09-15 · A	A contour count can certify two zeros without certifying a double zero Does the certificate count nearby zeros with multiplicity, or prove exact multiplicity at a specified point?	A contour count can certify two zeros without certifying a double zero For $f_\varepsilon(z)=(z-1)^2+\varepsilon$ and$0<\lvert\varepsilon\rvert<r^2$, the circle $\lvert z-1\rvert=r$ encloses exactly two zeros, but both are simple. Its midpoint has $f'_\varepsilon(1)=0$ and $f''_\varepsilon(1)=2$, although it is not a zero.
Y25-09-010 2025-09-15 · U	A μ-invariant is a valuation, not a real-valued twist statistic Which p-adic power series defines μ and λ, and which algebraic operations preserve those invariants?	A μ-invariant is a valuation, not a real-valued twist statistic For nonzero F,G, $\mu(FG)=\mu(F)+\mu(G)$ and $\lambda(FG)=\lambda(F)+\lambda(G)$. In particular multiplication by a unit preserves both invariants. These statements follow directly from the integral-domain property of $F_p[[T]]$.
Y25-09-016 2025-09-15 · A	Integer micro-arithmetic needs a proved rounding envelope Which real map does the integer routine approximate, and how do its local rounding errors propagate through a commutator?	Integer micro-arithmetic needs a proved rounding envelope For every integer x in the domain,$0\leq h(x/M)-h_M(x)/M<1/M$. Globally, $\lvert h(u)-h(v)\rvert\leq(\lvert u-v\rvert+\sqrt{\lvert u-v\rvert})/2$. Consequently an input uncertainty of one micro-unit can produce an output uncertainty of order M^(−1/2), rather than order M^{-1}.

ID · date	Concept and revised question	Result and mathematical core
Y25-09-006 2025-09-17 · A	Genuine local arithmetic replaces a synthetic Hecke trace Which elliptic curve and good-reduction counts produce the coefficient sequence, and which exact recurrences extend it?	Genuine local arithmetic replaces a synthetic Hecke trace Exact enumeration gives $\#E(F_2)=5,\#E(F_3)=5,\#E(F_5)=5,\#E(F_7)=10$, hence $a_2=-2,a_3=-1,a_5=1,a_7=-2$. The Euler factor implies a_(p^r)=a_pa_(p^(r−1))−pa_(p^(r−2)) for r≥2.
Y25-09-013 2025-09-19 · U	Fractional ownership is a partial algebra, not duplicable permission Which partial composition law enforces the claimed ownership budget, and which logical rules are still needed for a race-freedom theorem?	Fractional ownership is a partial algebra, not duplicable permission This operation is a partial commutative monoid with cancellation. A full share has no compatible positive remainder and cannot be composed with a second full share. Any positive rational share can be split into smaller compatible rational shares.
Y25-09-014 2025-09-19 · U	Bisimulation preserves observable traces under a stated transition relation Which transitions and observations are matched, and are latency steps observable or intentionally hidden?	Bisimulation preserves observable traces under a stated transition relation Related states in a strong bisimulation have identical sets of finite labeled traces. The relation pairing a two-state alternating machine with a one-state a-loop is a bisimulation, although their concrete state counts differ.
Y25-09-017 2025-09-19 · U	Chebyshev polynomials give exact normalized Hecke recurrences Which recurrence connects genuine local coefficients to a bounded spectral parameter, and what stability bound follows from it?	Chebyshev polynomials give exact normalized Hecke recurrences b_r=p^(r/2)U_r(x). If $\lvert a\rvert\le 2\sqrt{p}$, then $\lvert b_r\rvert\le(r+1)p^{r/2}$. For $\lvert x\rvert<1$, $U_r(\cos\theta)=\sin((r+1)\theta)/\sin\theta$, with continuous endpoint values U_r(±1)=(±1)^r(r+1).
Y25-09-003 2025-09-20 · U	Transport defects satisfy a cocycle law and a sharp norm bound What are the source and target groups of each defect, and which compatibility makes the claimed scalar invariant survive transport?	Transport defects satisfy a cocycle law and a sharp norm bound $\Delta_{g\circ f}=G\Delta_f+\Delta_g$. If source and target weights satisfy $w_m^{T}F=w_n^{T}$, then $\lvert\tau_{w_m}(b)-\tau_{w_n}(a)\rvert\le\lVert w_m\rVert_\infty\lVert\Delta_f\rVert_1$. The constant $\lVert w_m\rVert_\infty$ is the best uniform coefficient for this inequality.
Y25-09-004 2025-09-20 · U	Cokernel certificates detect torsion that real residuals can miss Which exact integer quotient encodes the obstruction, and how does its Smith form differ from a small floating residual?	Cokernel certificates detect torsion that real residuals can miss coker diag(2,3) is $\mathbb{Z}/2\oplus\mathbb{Z}/3\cong\mathbb{Z}/6$. Its order is six although F is invertible over Q and R. Every homomorphism from this finite cokernel to the additive group Z is zero.
Y25-09-011 2025-09-20 · U	A cross-ratio produces an auditable map into elliptic-curve moduli Which projective invariants determine the curve family, and which distinct parameter values map to the same moduli point?	A cross-ratio produces an auditable map into elliptic-curve moduli The j-value is unchanged by $\lambda\mapsto 1-\lambda$ and $\lambda\mapsto 1/\lambda$, hence by the six usual cross-ratio transforms. For λ=3 these six distinct values are3,−2,1/3,−1/2,3/2,2/3, all with j=21952/9. A probability law on λ therefore has a well-defined pushforward to j, with collisions that can be counted exactly.

ID · date	Concept and revised question	Result and mathematical core
Y25-09-012 2025-09-21 · A	A Legendre-symbol certificate has a precise and limited arithmetic meaning Which finite-field symbol is evaluated, and what additional data would be required to interpret it as a Hilbert or local geometric invariant?	A Legendre-symbol certificate has a precise and limited arithmetic meaning For p=163 and a=6 the Legendre symbol is1. An explicit square root modulo163 certifies this directly. More generally, the set of nonzero squares is a subgroup of index two, and the Legendre symbol is a multiplicative homomorphism F_p×→{±1}.
Y25-09-002 2025-09-29 · A	The cosine iteration has a rigorous tail certificate Which uniform interval bound validates the cosine fixed point and the proposed summable-tail condition?	The cosine iteration has a rigorous tail certificate There is one real solution r=cos r, every real initial condition converges to r, and R_N≤|x_(N+1)−x_N|/(1−sin1) for N≥1. In particular the increment tail is absolutely summable. The sequence x_n² need not decrease.

Y25-09-001 A compositional contraction certificate with an exact fixed point

2025-09-01 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which complete invariant domain and Lipschitz bounds certify the entire composition, and what error does its residual rigorously control?

The contraction-verifier project has a substantial mathematical core: a product of verified operator bounds can certify a unique fixed point and provide an a posteriori error bar. The certificate must apply to the actual composed map, rather than to labels assigned to its stages.

Definitions and assumptions

A map T on a complete metric space is a contraction if d(Tx,Ty)≤q d(x,y) for one uniform q<1 and T maps that space into itself.

For T=A∘R∘Γ∘S, Lipschitz constants multiply. On R^d, $S_t=I+t11^T$ has Euclidean norm max(1,|1+td|).

A compositional contraction certificate with an exact fixed point

A self-map with contraction constant q has a unique fixed point x*. For every x, d(x,x*)≤d(x,Tx)/(1−q). For d=2,t=1/4 and T(x)=(1/4)S_t x+(1,0), the exact constant is3/8 and x*=(22/15,2/15).

Derivation or proof

Starting at x_0, the successive increments satisfy $d(x_{n+1},x_n)\le q^n d(x_1,x_0)$. Their geometric sum makes the iterates Cauchy; completeness gives a limit, and Lipschitz continuity makes it fixed. Two fixed points have distance at most q times their distance, forcing equality. The triangle inequality gives $d(x,x^*)\le d(x,Tx)+q\,d(x,x^*)$, proving the residual bound. For S_t, the all-ones direction has eigenvalue $1+td$ and its orthogonal complement eigenvalue1; symmetry turns their maximum absolute value into the operator norm. The worked affine map has matrix$[[5/16,1/16],[1/16,5/16]]$, with eigenvalues3/8,1/4. Solving$(I-M)x=(1,0)$ gives the stated rational fixed point.

Worked example

Certify the zero initial state without first using the closed-form answer.

Its residual norm is one, so the general bound gives $||x^*||_2\le 8/5$. Direct substitution of(22/15,2/15) verifies the fixed-point equation exactly. After n iterations the initial increment bound gives error at most$(8/5)(3/8)^n$. An SVD-based certificate can be replaced here by exact eigenvalues, avoiding uncertainty about numerical singular-value tolerances.

The verifier now has an executable theorem: a named operator, a proved contraction margin, an exact solution and a certified iteration error.

Scope

A computed ε or a sampled Jacobian does not certify a global bound. Completeness, domain invariance and all stage bounds remain necessary; a nonnormal matrix can have spectral radius below one while its chosen norm exceeds one.

Further development

For a block-coupled system, construct a nonnegative matrix of cross-Lipschitz bounds and choose positive weights making its weighted maximum norm less than one.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R149.

Y25-09-019 A modular minor certifies a rational rank lower bound

2025-09-02 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which exact minor proves the rank claim, and what independent data certify the upper bound?

The obstruction pipeline proposed modular rank certificates. Their useful theorem is asymmetric: nonzero reduction gives a reliable lower bound over Q, while a rank drop modulo a prime can be accidental. A full certificate needs both sides.

Definitions and assumptions

For an integer matrix A let rank_QA denote its rational rank and rank_pA its rank after reduction over the prime field F_p.

A nonzero r×r minor proves rank at least r; a factorization A=BC with B having r columns proves rank at most r over the chosen field.

A modular minor certifies a rational rank lower bound

rank_pA≤rank_QA. If a minor is nonzero modulo p and a rational rank-r factorization is verified exactly, the two certificates prove rank_QA=r. Conversely rank_pA=0 does not imply rational rank zero.

Derivation or proof

If an r×r determinant is nonzero modulo p, its integer determinant is not zero, hence the corresponding rational rows and columns are independent. Equivalently every minor larger than the rational rank is the zero integer and stays zero after reduction. The factorization bound follows because every column of BC lies in the span of the r columns of B. Both checks are finite exact equations. For failure of the converse, the one-entry matrix[p] is nonzero over Q but reduces to zero modulo p.

Worked example

Use A=[[1,2,3],[2,4,6],[0,1,1]].

The minor from rows1,3 and columns1,2 has determinant one, so every prime certifies rank at least two. The second row is twice the first, proving rank at most two, hence exact rank two. A null vector is(−1,−1,1), checked by matrix multiplication. The certificate therefore includes a field-independent nonzero minor and a rational dependence, rather than an unsupported prime-rank number.

This is an implementable rank proof suitable for an obstruction complex or formal verifier.

Scope

The modulus must actually be prime before finite-field Gaussian elimination applies. A historical integer merely labeled p is not accepted as prime without a certificate. Rank also does not determine torsion in integral homology.

Further development

Emit pivot minors, exact row relations and Smith data together, separating rational rank from integer torsion and checking every claimed transformation matrix.

Source receipt: months/09/root_coverage.json record 0. Mathematical references: R150.

Y25-09-020 Adjacent exact inequalities prove full monotonicity of a finite clock

2025-09-06 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

What does monotonicity mean over the exact stored type, and does the proposed proof establish all index pairs?

The clock scaffold left monotonicity admitted while displaying decimal sample times. A finite order theorem is enough to close this mathematical obligation. Exact rational timestamps avoid confusing an intended step with a floating representation.

Definitions and assumptions

A finite sequence $t_0,\dots,t_n$ is nondecreasing if $i\le j$ implies $t_i\le t_j$, and strictly increasing if $i<j$ implies $t_i<t_j$.

For the recovered six-value pattern use exact rationals $t_k=3k/50$ for $k=0,\dots,5$.

Adjacent exact inequalities prove full monotonicity of a finite clock

> A sequence in a transitive ordered set is nondecreasing if and only if every adjacent inequality $t_k\le t_{(k+1)}$ holds. Over Q, positive adjacent increments imply strict increase, and t_j-t_i is their telescoping sum.

Derivation or proof

The forward direction follows by choosing $j=i+1$. For the reverse, induct on $j-i$. Difference zero is reflexivity; for positive difference use the induction hypothesis from i to $j-1$ and transitivity with the final adjacent inequality. In Q, telescoping gives $t_j-t_i=\Sigma_{(k=i)^{(j-1)}}(t_{(k+1)}-t_k)$, which is positive when $i<j$ and every summand is positive. This supplies the exact statement a list-inductive proof should establish, including the empty and singleton cases.

Worked example

Verify the six timestamps and the minimum gap.

The sequence is$0,3/50,3/25,9/50,6/25,3/10$. Every adjacent gap is$3/50$, so $t_j-t_i=3(j-i)/50$ and the end-to-end difference is$3/10$. The minimum positive separation is$3/50$, not the average slope of an unspecified continuous interpolant. If the values are rounded with absolute error$\delta<3/100$, strict order is retained because each perturbed gap exceeds$3/50-2\delta>0$. Exact positivity therefore proves strict monotonicity.

The clock obtains a complete finite-order proof and a quantitative robustness margin.

Scope

NaN values in a floating type do not obey the ordinary total-order assumptions used here. A computed sequence does not prove a soliton conservation law, and no successful completion of the historical admitted theorem is claimed.

Further development

Encode rational timestamps and certified error intervals, prove the adjacent-gap lower bound by recursion, and derive global monotonicity from that lemma rather than asserting it as an axiom.

Source receipt: months/09/root_coverage.json record 0. Mathematical references: R151.

Y25-09-021 A genuine KdV soliton requires a PDE identity, not a checkpoint metaphor

2025-09-08 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which field and equation make the tau function a soliton construction, and can the traveling wave be verified directly?

The September discussion named KdV and Hirota tau functions alongside the software clock. An actual classical soliton supplies a constructive comparison. It has a nonlinear PDE, a traveling profile and a precise logarithmic tau formula; those are additional mathematical data.

Definitions and assumptions

Use KdV $u_t+6uu_x+u_{xxx}=0$ on $R\times R$. For $k>0$ set $z=x-4k^2t-x_0$ and $F(z)=2k^2\operatorname{sech}^2(kz)$.

Define the positive tau function $\tau(x,t)=1+\exp(2kz)$, and $u=2\partial^2_x\log\tau$.

A genuine KdV soliton requires a PDE identity, not a checkpoint metaphor

> The field $u=F(z)$ equals$2\partial^2_x\log\tau$ and solves the declared KdV equation exactly. It travels at speed$4k^2$, has amplitude$2k^2$ and integral$\int_R u\,dx=4k$.

Derivation or proof

Differentiating $\log(1+e^{(2kz)})$ twice gives$4k^2e^{(2kz)}/(1+e^{(2kz)})^2=k^2\operatorname{sech}^2(kz)$, proving the tau formula after multiplying by two. The profile identity is $F''=4k^2F-3F^2$, obtained by differentiating sech^2 or its equivalent tanh expression. Differentiating that identity gives $F'''=4k^2F'-6FF'$. Since $u_t=-4k^2F'$ and $u_x=F'$, the three KdV terms cancel exactly. Finally$\int\operatorname{sech}^2(kz)dz=2/k$, yielding mass$4k$.

Worked example

Take $k=1$ and $x_0=0$.

The profile $u=2\operatorname{sech}^2(x-4t)$ moves right with speed four while retaining amplitude two. Its maximum occurs at $x=4t$, and its integral remains four by translation. The tau function itself grows exponentially on one spatial side, but its second logarithmic derivative is localized; therefore tau and the physical profile are different objects even in this exact construction. The speed and amplitude are not independently adjustable within this family: speed is twice amplitude. The translation parameter changes neither the integral nor this amplitude-speed relation.

The soliton terminology is now attached to a verified nonlinear-wave solution, giving a positive mathematical model rather than a software analogy.

Scope

This proves one traveling-wave solution, not general inverse-scattering theory, multi-soliton interactions or any relation to hash ledgers. A geometrically spaced list of timestamps lacks the field and PDE needed for this theorem.

Further development

Construct the two-soliton Hirota tau expression with its interaction coefficient and verify its PDE identity symbolically, then analyze the asymptotic phase shift as a separately stated result.

Source receipt: months/09/root_coverage.json record 0. Mathematical references: R139.

Y25-09-005 A trace invariant follows from local rewrite equations by induction

2025-09-09 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which precise transition relation preserves which algebraic quantity, and what statement must the formal proof actually establish?

The Lean scaffold named transport and soliton conservation but retained an admitted proof. There is a straightforward positive theorem available: homomorphisms annihilating each allowed rewrite are conserved along every finite trace. It is independent of whether the states are called solitons.

Definitions and assumptions

Let states be $x \in \mathbb{Z}^n$ and allowed rewrite increments $r_1,\dots,r_k$. A transition has $x' = x + r_j$ for some j.

For integer weights w, let $\tau(x) = w^{\mathsf{T}} x$. A finite trace is a list of states linked by allowed transitions.

A trace invariant follows from local rewrite equations by induction

> If $w^{\mathsf{T}} r_j = 0$ for every allowed rewrite, τ has the same value at both ends of every finite trace. Conversely, conservation for all one-step traces forces all these equations. Thus exact linear conservation laws are the integer solutions of $R^{\mathsf{T}} w = 0$, where R contains the rewrite columns.

Derivation or proof

For a one-step trace $\tau(x + r_j) - \tau(x) = w^{\mathsf{T}} r_j = 0$. Induct on trace length: the empty trace has identical endpoints; appending one allowed transition preserves the already conserved value. Conversely, apply the claimed conservation law to the one-step trace $0 \to r_j$. This yields $w^{\mathsf{T}} r_j = 0$ for every j. The argument identifies the exact proposition to encode: every transition and every finite closure of that relation preserves τ. A theorem of type True or a runtime message saying VERIFIED does not establish this quantified statement.

Worked example

Use three leaf types with rewrite $r = (1, -2, 1)$.

Both $w = (1,1,1)$ and $v = (1,2,3)$ annihilate r. Starting at $x = (0,4,0)$, one rewrite gives$(1,2,1)$ and two give$(2,0,2)$. Total count remains4, while the weighted quantity remains8. Solving $w_1 - 2w_2 + w_3 = 0$ gives $w_1 = 2w_2 - w_3$, so every integer conservation law is an integer combination of$(2,1,0)$ and$(-1,0,1)$. If nonnegative states are required, transitions leaving that domain must be forbidden; the same proof still applies to the remaining traces.

The formalization target is a complete induction theorem with a computable family of all linear invariants, not merely an artifact hash.

Scope

This theorem establishes no nonlinear PDE soliton, physical energy or collision resistance. It also does not certify that the historical admitted Lean scaffold was successfully completed.

Further development

Compute the integer nullspace for a larger declared rewrite system and prove that its chosen basis generates all linear conservation laws; then formalize the trace induction against that exact basis.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R152.

Y25-09-018 A geometric checkpoint clock estimates a decay rate without being a soliton

2025-09-10 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

What mathematical quantity does the checkpoint schedule estimate, and which formula remains invariant when the time spacing changes?

The geometric τ-clock was eventually described as a sampling cadence rather than a PDE mechanism. That modest interpretation still has a complete mathematical development: exponential decay can be recovered from unequally spaced samples, and a geometric schedule has a transparent resolution-versus-horizon tradeoff.

Definitions and assumptions

Choose times $t_k=t_0q^k$ with $t_0>0, q>1$, and a positive signal $E(t)=C\exp(-\alpha t)$, $C>0$.

For consecutive samples define $\alpha_hat_k=-[\log E(t_k)-\log E(t_{(k-1)})]/[t_k-t_{(k-1)}]$.

A geometric checkpoint clock estimates a decay rate without being a soliton

For exact exponential data $\alpha_hat_k=\alpha$ for every k, regardless of q. The number of positive checkpoints through horizon $T\geq t_0$ is $1+\operatorname{floor}(\log(T/t_0)/\log q)$. If each log measurement has absolute error at most δ, then $|\alpha_hat_k-\alpha|\leq 2\delta/(t_k-t_{(k-1)})$.

Derivation or proof

Take logarithms: $\log E(t)=\log C-\alpha t$. Subtracting two samples cancels $\log C$ and gives $-\alpha$ times their time difference, proving the rate identity. The checkpoint count follows by solving $t_0q^k\leq T$ for integers $k\geq 0$. If the measured logarithms are $\log E(t_k)+e_k$ with $|e_k|\leq\delta$, the estimated slope differs by $-(e_k-e_{(k-1)})/(t_k-t_{(k-1)})$; the triangle inequality gives the error bound. Thus larger gaps reduce this particular absolute log-noise amplification, while they also discard information about changes within those gaps.

Worked example

Let $t_0=1, q=2, C=3, \alpha=1/2$ and $T=16$.

The checkpoints are$1,2,4,8,16$, exactly five. Their log values are $\log 3-1/2, \log 3-1, \log 3-2, \log 3-4, \log 3-8$. Every consecutive secant gives$1/2$. With log-error bound$\delta=1/100$, the first rate uncertainty is at most$1/50$, and the last is at most$1/400$. A nonexponential signal could share all five values yet differ between them, so this schedule alone does not identify a differential equation. A dimensionless geometric ratio controls the schedule; the decay rate still has reciprocal-time units. Rescaling time by a positive constant rescales the estimated rate inversely, while leaving the geometric checkpoint count through a similarly rescaled horizon unchanged.

The clock has an exact estimation law and an explicit sampling cost, without importing unsupported soliton terminology.

Scope

The historical caching speedup was workflow-dependent and is not a mathematical performance guarantee. The theorem concerns exact or bounded-error log samples, not hash validity or physical energy.

Further development

Assume a bounded second derivative of $\log E$ and derive an interpolation-error certificate between checkpoints, then choose q to balance sample count, noise amplification and missed curvature.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R110.

Y25-09-015 The square-root smoothing map has dynamics but is not a linear Hecke operator

2025-09-14 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

What does the actual surrogate map do, and which exact endpoint tests can reject a claimed commutation law?

The recovered micro-space map is concrete enough to analyze after normalizing its unit interval. Its positive mathematical content is a concave dynamical map; assigning it a Hecke label does not make it a linear arithmetic correspondence.

Definitions and assumptions

Normalize $x\in[0,M]$ by $u=x/M$. The real-valued smoothing becomes $h(u)=(u+\sqrt{u})/2$ on$[0,1]$.

Let $\theta(u)=\mathrm{clamp}(su+t,0,1)$, where $\mathrm{clamp}(v,0,1)=\min(1,\max(0,v))$. Exact reciprocity in this finite model means $h\circ\theta=\theta\circ h$ on the stated domain.

The square-root smoothing map has dynamics but is not a linear Hecke operator

h is increasing and concave, has exactly the fixed points0 and1, and every orbit starting in(0,1] converges to1. It is not linear and is not globally Lipschitz on[0,1]. Any commuting θ must send each of0,1 to one of0,1.

Derivation or proof

For0<u<1, √u>u, so u<h(u)<1. Iterates are increasing and bounded, hence have a limit; continuity makes it a fixed point. Solving h(u)=u gives√u=u and therefore u∈{0,1}; a positive starting value excludes zero. Concavity follows from h″(u)=−1/(8u^(3/2))<0 in the interior. The ratio[h(u)−h(0)]/u=(1+u^(−1/2))/2 is unbounded as u→0, ruling out a global Lipschitz constant. If h and θ commute and e is an endpoint fixed by h, then h(θ(e))=θ(h(e))=θ(e), so θ(e) must itself be an endpoint.

Worked example

Apply the endpoint test to a newly specified affine-clipped example s=3/10,t=−1/10.

Here θ(0)=0 but θ(1)=1/5. Thus h(θ(1))=(1/5+1/√5)/2 differs from θ(h(1))=1/5. The commutator already fails at the endpoint, independently of a dense grid or an optimizer. Nonlinearity can be checked from h(1/4)=3/8, whereas linear homogeneity would require h(1/4)=(1/4)h(1)=1/4.

The surrogate has a fully understood interval dynamics and simple necessary reciprocity tests. Those are meaningful mathematical results without an arithmetic reinterpretation.

Scope

The recorded micro-witness numbers are not independently certified: the exact integer rounding, scale, scan grid and definition of Δ were not all recovered. The new example does not claim to reproduce their run.

Further development

Classify affine maps preserving[0,1] that commute with h. The endpoint condition reduces the nonconstant unclipped case to identity or reflection; one interior test excludes reflection, leaving only identity and the two endpoint constants.

Source receipt: months/09/root_details.json record 0. Mathematical references: R110.

Y25-09-007 The symmetric-square factor needs all three eigenvalue products

2025-09-15 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which normalization of the local parameters defines Sym², and what information is lost by squaring only the trace?

The detector used factors of the form$(1-a_p p^{(-s)})^{-1}$. That is not the symmetric-square factor of a two-dimensional unramified arithmetic parameter. Writing its three eigenvalues explicitly yields a complete finite correction and an exact worked factor.

Definitions and assumptions

Use arithmetic parameters α,β satisfying $\alpha+\beta=a$ and $\alpha\beta=p$. On the symmetric square, the eigenvalues are$\alpha^2,\alpha\beta,\beta^2$.

Let $t=p^{(-s)}$. This unnormalized convention keeps the middle eigenvalue p. Dividing α,β by$\sqrt{p}$ instead gives the unitary normalization and changes the variable scaling.

The symmetric-square factor needs all three eigenvalue products

The unramified symmetric-square denominator is$(1-\alpha^2 t)(1-pt)(1-\beta^2 t)=1-(a^2-p)t+p(a^2-p)t^2-p^3t^3$. For the exact $p=2,a=-2$ input, it equals$1-2t+4t^2-8t^3$.

Derivation or proof

The trace of the three-dimensional representation is $\alpha^2+\alpha\beta+\beta^2=(\alpha+\beta)^2-\alpha\beta=a^2-p$. The pairwise eigenvalue products sum to $p(\alpha^2+\beta^2)+p^2=p(a^2-2p)+p^2=p(a^2-p)$. Their product is$(\alpha\beta)^3=p^3$. Expanding the three linear factors proves the cubic formula without choosing roots numerically. For $p=2,a=-2$, the roots of X^2+2X+2 are$-1\pm i$, whose squares are$\mp 2i$. Their quadratic contribution is$1+4t^2$, and the middle factor is$1-2t$, giving the displayed product.

Worked example

Compare the local coefficient sequences of the corrected and historical factors.

Inverting$1-2t+4t^2-8t^3$ gives coefficients$1,2,0,0,16,\ldots$; direct recurrence or multiplication checks these exactly. The rank-one surrogate$(1-4t)^{-1}$ instead gives$1,4,16,64,256,\ldots$, already disagreeing at the first local coefficient. The distinction is representation-theoretic: the trace of Sym^2 is a^2-p, not the square a^2 of the original trace. A finite Euler product over selected primes additionally has only the analytic properties of that finite rational expression. At $t=0$ every local factor has constant coefficient one. This normalization is part of the representation: multiplying or averaging arbitrary observed values changes the local factor rather than preserving its arithmetic definition.

The functorial-lift theme acquires an actual local representation calculation rather than a mean-square heuristic.

Scope

This is an unramified local factor calculation. It does not construct ramified factors, analytic continuation, a global functional equation or the automorphy of a symmetric-square lift. Those are separate deep inputs.

Further development

Starting from genuine curve traces, include correct ramified local data and a known global functional equation, then prove a truncation bound before interpreting any critical-line scan.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R153.

Y25-09-008 FFT derivatives are exact for an interpolant, not for an unknown function

2025-09-15 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which frequency band and aliasing bound justify the derivative estimate used by the zero detector?

The historical pipeline proposed DFT derivatives for a critical-line zero hunt. Fourier differentiation can be exact and useful, provided its object is identified: it differentiates the trigonometric interpolant selected by the sample grid. Finite samples alone cannot bound derivatives of an unrestricted analytic function.

Definitions and assumptions

Sample a2π-periodic function at x_j=2πj/N, j=0,…,N−1. The DFT assigns coefficients to one representative of each frequency class modulo N.

A band-limited trigonometric polynomial with frequencies |k|<N/2 has derivative coefficients ikc_k. Modes k and k+N have identical samples.

FFT derivatives are exact for an interpolant, not for an unknown function

DFT differentiation exactly recovers derivatives of the declared band-limited polynomial. Without a frequency or analytic-tail bound, no finite sample grid gives a uniform derivative-error bound: h(x)=cos(Nx)−1 vanishes on the grid but h″(0)=−N².

Derivation or proof

Discrete orthogonality gives N⁻¹Σ_jexp(i(k−ℓ)x_j)=1 when N divides k−ℓ and0 otherwise, by the finite geometric-sum formula. If all frequencies lie in a single nonaliasing band, this recovers each coefficient exactly. Differentiating the finite Fourier expansion multiplies its coefficient by ik, so the derivative is exact for that expansion. For the counterexample, Nx_j=2πj makes every value of h zero, while differentiating twice gives h″(x)=−N²cos(Nx). Multiplying h by any positive constant enlarges the derivative discrepancy without changing any sample. Analyticity alone therefore does not remove finite-grid nonidentifiability.

Worked example

Use N=8 and the known polynomial f(x)=sin x+2cos2x.

Its nonzero frequencies are±1,±2, all inside the nonaliasing band. The DFT differentiates it to cos x−4sin2x and gives f′(0)=1 exactly in symbolic arithmetic. Adding3(cos8x−1) changes none of the eight samples or the first derivative at zero, but changes the second derivative at zero by−192. A double-zero diagnostic relying on both numerical derivatives needs an error estimate that excludes precisely this kind of invisible high-frequency component.

The Fourier method becomes a rigorous tool when accompanied by its bandwidth or tail certificate.

Scope

A grid derivative and a bootstrap interval do not by themselves certify the derivative of a global L-function. Floating-point error is additional to aliasing and must be handled separately.

Further development

Assume an explicit coefficient bound |c_k|≤Ce^(−a|k|), derive a tail bound forΣk²|c_k| outside the sampled band, and combine it with rounding bounds to certify derivative signs.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R154.

Y25-09-009 A contour count can certify two zeros without certifying a double zero

2025-09-15 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Does the certificate count nearby zeros with multiplicity, or prove exact multiplicity at a specified point?

The double-zero search correctly described its status as heuristic. The distinction can be made mathematically sharp: a contour comparison robustly certifies a total count, whereas a double zero requires exact vanishing of the value and first derivative at the same point.

Definitions and assumptions

A holomorphic function has a zero of multiplicity two at z_0 exactly when $f(z_0)=f'(z_0)=0$ and $f''(z_0)\neq 0$.

Rouché's theorem says that $|h|<|g|$ on a simple closed contour implies g and g+h have the same total zero count inside, including multiplicities, provided they are holomorphic on the enclosed region.

A contour count can certify two zeros without certifying a double zero

For $f_\varepsilon(z)=(z-1)^2+\varepsilon$ and$0<|\varepsilon|<r^2$, the circle $|z-1|=r$ encloses exactly two zeros, but both are simple. Its midpoint has $f'_\varepsilon(1)=0$ and $f''_\varepsilon(1)=2$, although it is not a zero.

Derivation or proof

On the contour, $|(z-1)^2|=r^2>|\varepsilon|$, so Rouché compares f_ε with$(z-1)^2$ and gives total count two. Alternatively its roots are$1\pm\sqrt{(-\varepsilon)}$, both strictly inside the circle. When $\varepsilon\neq 0$ they are distinct, and the derivative$2(z-1)$ is nonzero at either root, proving simplicity. At $\varepsilon=0$ the roots merge and the first two vanishing equations hold exactly. The strict contour gap $r^2-|\varepsilon|$ is robust under additional perturbations smaller than that gap, while exact equality $f(1)=0$ is not stable under arbitrary small constant perturbations.

Worked example

Take $\varepsilon=1/100$ and $r=1/2$.

The roots are$1\pm i/10$. On the contour the unperturbed modulus is$1/4$ and the perturbation is$1/100$, leaving margin$6/25$. A certified evaluation error below that margin preserves the total count, but not a

claim that either root is repeated. The midpoint values are $f(1)=1/100, f'(1)=0, f''(1)=2$. A tolerance treating the first value as zero can therefore misclassify two nearby simple zeros as one double zero.

A useful certificate states exactly whether it establishes existence, total multiplicity, isolation or an exact repeated root.

Scope

A full Turing count for an arithmetic L-function has additional hypotheses and global input; this elementary Rouché example does not supply that machinery. Empty candidate lists establish no nonexistence theorem.

Further development

Combine a rigorous analytic approximation with an explicit contour error bound, then isolate each zero in disjoint disks and apply a derivative lower bound to certify simplicity.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R060.

Y25-09-010 A μ-invariant is a valuation, not a real-valued twist statistic

2025-09-15 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which p-adic power series defines μ and λ, and which algebraic operations preserve those invariants?

The September sampler used cosine aggregates and bootstrap intervals under an Iwasawa label. A genuine elementary analogue starts with a power series over Z_p. Its invariants come from coefficient valuations and reduction modulo p, and can be proved invariant under multiplication by a unit.

Definitions and assumptions

For nonzero $F(T)=\Sigma a_n T^n \in Z_p[[T]]$, define $\mu(F)=\min_n v_p(a_n)$ and $\lambda(F)=\min\{n: v_p(a_n)=\mu(F)\}$.

Write $F=p^{\mu}F_0$, where at least one coefficient of F_0 is a unit. A power-series unit U has constant coefficient in $Z_p\times$.

A μ-invariant is a valuation, not a real-valued twist statistic

For nonzero F,G, $\mu(FG)=\mu(F)+\mu(G)$ and $\lambda(FG)=\lambda(F)+\lambda(G)$. In particular multiplication by a unit preserves both invariants. These statements follow directly from the integral-domain property of $F_p[[T]]$.

Derivation or proof

Divide F and G by their lowest p-powers, obtaining primitive series F_0,G_0 whose reductions modulo p are nonzero. Their product remains nonzero because a formal series ring over a field is a domain: the first nonzero coefficients multiply to the first nonzero coefficient of the product. Thus no extra p divides every coefficient of F_0G_0, proving the μ identity. The orders in T of the two nonzero reductions are λ(F),λ(G); their product has the sum of those orders. A unit has μ=λ=0, so multiplication by it preserves both values. The argument uses the complete power series, not a real distributional interpretation.

Worked example

Use p=3 and $F(T)=9(3+T^2+3T^3)$.

The coefficient valuations are at least two, with the first valuation-two coefficient at degree two; hence μ=2 and λ=2. Multiplying by$1+T$ gives$9(3+3T+T^2+4T^3+3T^4)$. After division by9 and reduction modulo3 this is $T^2+T^3=T^2(1+T)$, retaining the same invariants. Conversely, a finite truncation before degree two would incorrectly suggest all known coefficients have valuation at least three; unseen coefficients must be controlled before certifying the infinite-series minimum.

The arithmetic meaning is now explicit and supplies exact certificates for a declared algebraic object.

Scope

This elementary definition does not prove that a sampled series is the characteristic series of an Iwasawa module or a p-adic L-function. Real cosine twists, confidence intervals and numerical regularity alone do not establish that connection.

Further development

Construct a genuine measure on Z_p, prove integrality of its Amice transform, and bound all unseen coefficients sufficiently to certify its μ invariant from a finite computation.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R155.

Y25-09-016 Integer micro-arithmetic needs a proved rounding envelope

2025-09-15 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which real map does the integer routine approximate, and how do its local rounding errors propagate through a commutator?

The laboratory reported micro-unit reciprocity witnesses. Exact integer arithmetic removes floating roundoff but does not remove approximation error: flooring a square root changes the mathematical map. A transparent quantization model gives an explicit certificate and explains why a one-unit local error can be amplified near zero.

Definitions and assumptions

Fix an integer M≥1 and integer0≤x≤M. Define h_M(x)=floor((x+floor√(xM))/2), and h(u)=(u+√u)/2.

For normalized quantities use the error |h_M(x)/M−h(x/M)|. On[δ,1], h has Lipschitz bound L_δ=1/2+1/(4√δ), while on the full interval it is only Hölder controlled near zero.

Integer micro-arithmetic needs a proved rounding envelope

For every integer x in the domain,0≤h(x/M)−h_M(x)/M<1/M. Globally, |h(u)−h(v)|≤(|u−v|+√|u−v|)/2. Consequently an input uncertainty of one micro-unit can produce an output uncertainty of order M^(−1/2), rather than order M^{-1}.

Derivation or proof

Write r=√(xM) and k=floor r, so0≤r−k<1. Replacing r by k changes the numerator half by less than1/2. Since x+k is an integer, its division by two loses either0 or1/2 under flooring. The total unnormalized loss is strictly below one; division by M proves the bound. For0≤v≤u, √u−√v=(u−v)/(√u+√v)≤√(u−v), proving the global inequality. On a domain bounded below by δ, the derivative formula gives the stated uniform Lipschitz coefficient. This separates a local exact rounding lemma from its potentially nonuniform propagation.

Worked example

Take $M=10^6$ and compare normalized inputs0 and1/M.

The integer square root of M is1000, so h_M(1)=floor1001/2=500. Thus the normalized output difference is500/M=1/2000 even though the input difference is$1/10^6$, an amplification factor500. The real output is$1001/(2\cdot10^6)$, and its difference from the integer output is exactly$1/(2\cdot10^6)$, within the proved envelope.

A legitimate micro-unit certificate must state both the integer equality and the real-map error bound; the example makes the missing scale dependence explicit.

Scope

This independently specified quantizer is not asserted to be the exact historical Bash implementation. A claimed Δ of two units cannot be interpreted without the grid, flooring conventions and domain where error propagates.

Further development

For a specified clipped θ quantizer, prove a commutator-error envelope using the Hölder bound globally or L_δ on a verified positive subdomain, then compare the observed integer defect with that envelope.

Source receipt: months/09/root_details.json record 0. Mathematical references: R110.

Y25-09-006 Genuine local arithmetic replaces a synthetic Hecke trace

2025-09-17 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which elliptic curve and good-reduction counts produce the coefficient sequence, and which exact recurrences extend it?

The September repair explicitly moved from smoothing arbitrary floats to the curve conventionally labeled11a1. This is a strong constructive turn: its finite-field equation independently determines local traces, and the unramified Euler factor determines prime-power coefficients without fitting them.

Definitions and assumptions

Take E:$y^2+y=x^3-x^2-10x-20$. For a good prime p, define a_p=p+1−#E(F_p), including the point at infinity.

The local generating function is$\Sigma_{r\geq 0} a_{p^r} t^r=(1-a_p t+pt^2)^{-1}$, with $a_1=1$. At coprime indices use a_(mn)=a_ma_n.

Genuine local arithmetic replaces a synthetic Hecke trace

> Exact enumeration gives $\#E(F_2)=5, \#E(F_3)=5, \#E(F_5)=5, \#E(F_7)=10$, hence $a_2=-2, a_3=-1, a_5=1, a_7=-2$. The Euler factor implies a_(p^r)=a_pa_(p^(r−1))−pa_(p^(r−2)) for r≥2.

Derivation or proof

For each listed prime enumerate x and y and test the polynomial equation modulo p, then add one point at infinity; this is a finite arithmetic certificate rather than a fitted trace. Multiplying the generating series by$1-a_p t+pt^2$ and comparing its constant, linear and higher coefficients proves the recurrence. The input primes are good: direct Weierstrass discriminant calculation gives-11^5, so only11 is excluded from these particular factors. Hasse’s inequality can check the resulting counts, but it does not select the counts by itself. A construction assigning arbitrary data within that bound still needs a curve or another arithmetic source.

Worked example

Generate the next coefficients from the exact p=2 trace.

The recurrence gives $a_4=(-2)^2-2=2$, $a_8=(-2)\cdot 2-2(-2)=0$, and $a_{16}=(-2)\cdot 0-2\cdot 2=-4$. Coprime multiplicativity gives $a_6=a_2a_3=2$ and $a_{10}=a_2a_5=-2$. The generating factor begins$1-2t+2t^2+0t^3-4t^4+\cdots$. This coefficient sequence is a consequence of the declared factor; the modular interpretation of the curve is a separate classical theorem.

The local input is now reproducible from an actual equation, and the extension to prime powers has an exact algebraic justification.

Scope

These finite calculations establish no BSD formula, analytic rank or zero certification. Bad-prime factors require their own reduction analysis. The historical repair passage was truncated after the proposed special-value computations.

Further development

Construct a rigorously bounded modular-form special-value computation using a stated functional equation and tail estimate, keeping the local arithmetic and analytic continuation certificates separate.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R156, R157.

Y25-09-013 Fractional ownership is a partial algebra, not duplicable permission

2025-09-19 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which partial composition law enforces the claimed ownership budget, and which logical rules are still needed for a race-freedom theorem?

The second-descent passage included rational fractional ownership and race-freedom goals. The underlying algebra can be proved completely in a small model. It explains why full ownership cannot be duplicated, while keeping the separate operational and logical assumptions visible.

Definitions and assumptions

Let resource shares be rationals $q \in [0,1]$. Define $q \oplus r = q + r$ only when $q + r \leq 1$; zero is the identity.

A compatible family of shares is one whose sum is at most one. A full share is1. The associated separation-style composition is partial, so an undefined sum is not another resource.

Fractional ownership is a partial algebra, not duplicable permission

> This operation is a partial commutative monoid with cancellation. A full share has no compatible positive remainder and cannot be composed with a second full share. Any positive rational share can be split into smaller compatible rational shares.

Derivation or proof

Commutativity and the zero law follow from rational addition. A three-share composite exists exactly when $q + r + s \leq 1$; in that case either parenthesization exists and has the same sum, proving the appropriate partial associativity law. If $q \oplus r = q \oplus s$, cancellation in Q gives $r = s$. If$1 \oplus r$ is defined with $r \geq 0$, the defining inequality$1 + r \leq 1$ forces $r = 0$. Thus$1 \oplus 1$ is undefined. For any rational$0 < q \leq 1$ and rational$0 < t < 1$, the shares tq and$(1-t)q$ are positive, compatible and sum to q. These facts precisely encode a conserved ownership budget.

Worked example

Split one permission into three equal readers and recombine it.

The three shares1/3,1/3,1/3 have total one, and any two compose to2/3. Adding a fourth positive third is undefined because4/3 exceeds one. Similarly1/3 and2/3 compose to a full share, whereas retaining either part after consuming the recombined full resource would require an additional rule that duplicates the budget. The algebra is exact over rational numbers and its validity requires no numerical tolerance.

The ownership idea has a concrete conservation law that can support a formal program logic.

Scope

A data-race-freedom theorem additionally requires a language semantics and rules connecting read/write actions to these resources. The partial algebra alone proves no compiler, garbage collector, constant-time bound or bootloader property. It also does not identify the exact historical Iris modality denoted by◇.

Further development

Specify a small concurrent language, give reads fractional permission and writes full permission, and prove that two enabled writes or a write/read conflict cannot hold compatible resource assertions.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R158.

Y25-09-014 Bisimulation preserves observable traces under a stated transition relation

2025-09-19 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which transitions and observations are matched, and are latency steps observable or intentionally hidden?

The latency-parametric garbage-collection passage named bisimulation without returning complete operational semantics. A finite labeled model develops the exact mathematical claim available from that idea and distinguishes strong step matching from an unproved constant-time assertion.

Definitions and assumptions

A labeled transition system has states S and relations $x \to^{a} x'$ for labels a. A relation R is a strong bisimulation if every transition on either side can be matched with the same label and related target.

A finite observable trace is a label word obtained along a path. Declaring an internal label τ invisible leads to a different, weak-bisimulation definition.

Bisimulation preserves observable traces under a stated transition relation

Related states in a strong bisimulation have identical sets of finite labeled traces. The relation pairing a two-state alternating machine with a one-state a-loop is a bisimulation, although their concrete state counts differ.

Derivation or proof

Prove trace preservation by induction on word length. The empty word is possible from every state. If xRy and x has a path beginning with label a, the forth condition supplies a matching a-transition from y to a related state; apply induction to the remaining word. The back condition gives the reverse inclusion. For the example let $x_0 \to^a x_1$ and $x_1 \to^a x_0$, and $y \to^a y$. The relation$\{(x_0,y),(x_1,y)\}$ matches the unique a-transition from each state in both directions, so it is a bisimulation. Every finite trace in either model is a power a^n.

Worked example

Add a silent preparatory step to only one implementation.

If $x_0 \to^{tau} x_1 \to^a x_0$ while y has only an a-loop, the previous relation is not a strong bisimulation because y cannot match the τ label. A weak relation may intentionally erase that internal step, but then it has proved equivalence only after that abstraction. If τ consumes measurable time, erasing it cannot by itself prove equal latency, constant running time or an18-cycle bound.

The semantic matching relation is a constructive abstraction theorem once its labels and observability policy are explicit.

Scope

No historical garbage-collector transition relation or timing model was recovered. The example therefore supplies a newly defined finite theorem rather than claiming that the original implementation had been verified. A modal symbol must likewise be interpreted before any formula such as$\Diamond$false$\to\bot$ can be assessed.

Further development

Use a cost-labeled transition relation and prove a cost-preserving simulation, or an explicit bounded-cost refinement, so latency conclusions follow from the actual matched data.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R159.

Y25-09-017 Chebyshev polynomials give exact normalized Hecke recurrences

2025-09-19 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which recurrence connects genuine local coefficients to a bounded spectral parameter, and what stability bound follows from it?

The platform suggestions named both Chebyshev recurrence and Hecke data. Their precise connection is classical and productive: the unramified degree-two local recurrence becomes the second-kind Chebyshev recurrence after normalization. This provides a testable algebraic bridge rather than a new correspondence claim.

Definitions and assumptions

Set $U_0(x)=1, U_1(x)=2x$ and $U_r(x)=2xU_{(r-1)}(x)-U_{(r-2)}(x)$.

For arithmetic parameters a,p, define $b_0=1, b_1=a$ and $b_r=a\,b_{(r-1)}-p\,b_{(r-2)}$. Put $x=a/(2\sqrt{p})$.

Chebyshev polynomials give exact normalized Hecke recurrences

$b_r=p^{(r/2)}U_r(x)$. If $|a|\le 2\sqrt{p}$, then $|b_r|\le (r+1)p^{(r/2)}$. For $|x|<1$, $U_r(\cos\theta)=\sin((r+1)\theta)/\sin\theta$, with continuous endpoint values $U_r(\pm 1)=(\pm 1)^{r(r+1)}$.

Derivation or proof

Substitute $b_r=p^{(r/2)}c_r$ into its recurrence to obtain $c_r=(a/\sqrt{p})c_{(r-1)}-c_{(r-2)}=2xc_{(r-1)}-c_{(r-2)}$, with the same initial data as U_r. Induction proves the identity. The sine addition formula proves the trigonometric recurrence and initial values, giving the expression in the interior. For a bound that remains stable at the endpoints, use the equivalent sum $U_r(\cos\theta)=e^{(i r\theta)}+e^{(i(r-2)\theta)}+\cdots+e^{(-ir\theta)}$, a sum of $r+1$ unit complex numbers. The triangle inequality gives $r+1$ uniformly, and evaluation at $\theta=0,\pi$ gives the endpoint values.

Worked example

Use the genuine $p=2, a=-2$ input from11a1.

The first U-polynomials are$1, 2x, 4x^2-1, 8x^3-4x$. At $x=-1/\sqrt{2}$, multiplying by the prescribed powers of$\sqrt{2}$ gives $b_0=1, b_1=-2, b_2=2, b_3=0, b_4=-4$, matching exact arithmetic recurrence. The bound for $r=4$ is$|-4|\le 5\cdot 4=20$. It is a coarse certified bound, not a prediction that every coefficient reaches the maximum. The same sum representation proves the endpoint limit without dividing by a vanishing sine. It also shows why a small denominator in the quotient formula is a numerical issue rather than a singularity of the underlying polynomial. For exact rational a and p, the unnormalized recurrence remains integral throughout.

The relationship provides exact normalization checks and endpoint-safe coefficient bounds across the arithmetic and spectral descriptions.

Scope

The Hasse-type input $|a|\le 2\sqrt{p}$ must come from a proved arithmetic source. An arbitrary float converted into x is not thereby a Hecke eigenvalue. This recurrence does not prove global automorphy or endoscopy.

Further development

Use recurrence-based interval arithmetic to evaluate long prime-power coefficient blocks and compare its certified error growth with unstable evaluation of the sine quotient near $x=\pm 1$.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R160.

Y25-09-003 Transport defects satisfy a cocycle law and a sharp norm bound

2025-09-20 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

What are the source and target groups of each defect, and which compatibility makes the claimed scalar invariant survive transport?

The user supplied unusually concrete formalization goals: free abelian leaf groups, cokernel defects, a composition law and a linear τ functional. A finite-rank model proves the cocycle and stability statements directly and exposes the compatibility required of τ.

Definitions and assumptions

Represent a finite free abelian leaf group by Z^n. For transport $F:Z^n\to Z^m$, source state a and target state b, define $\Delta_f=b-Fa\in Z^m$.

For a later transport $G:Z^m\to Z^k$ with target c, define $\Delta_g=c-Gb$. A scalar functional is $\tau_w(x)=w^Tx$ for integer weights w.

Transport defects satisfy a cocycle law and a sharp norm bound

$\Delta_{(g\circ f)}=G\Delta_f+\Delta_g$. If source and target weights satisfy $w_m^TF=w_n^T$, then $|\tau_{(w_m)}(b)-\tau_{(w_n)}(a)|\le||w_m||_\infty||\Delta_f||_1$. The constant $||w_m||_\infty$ is the best uniform coefficient for this inequality.

Derivation or proof

Expand $G(b-Fa)+(c-Gb)=c-GFa$; the intermediate state cancels exactly. The compatibility rewrites the scalar difference as $w_m^T(b-Fa)=w_m^T\Delta_f$. The triangle inequality gives $|\Sigma w_i\Delta_i|\le\Sigma|w_i||\Delta_i|\le\max|w_i|\Sigma|\Delta_i|$. To prove sharpness, choose Δ to be a unit coordinate vector at an index maximizing $|w_i|$, with its sign chosen accordingly; equality holds. Zero defect therefore certifies conservation under the explicit weight compatibility, while a nonzero defect can still be invisible to a chosen functional.

Worked example

Take $F=[[1,1],[0,1]]$, $a=(1,2)$, $b=(4,1)$, and target weight $w=(2,-1)$.

The transported state is $Fa=(3,2)$, so $\Delta=(1,-1)$, with $\text{norm}_1=2$. The source weight is $F^Tw=(2,1)$, giving $\tau_{\text{target}}(b)=7$ and $\tau_{\text{source}}(a)=4$, a difference3 bounded by$2\cdot2=4$. A subsequent map $G=[1,1]$ sends this defect to zero: the two coordinate discrepancies cancel under that observation. Thus a conserved coarse scalar need not imply that the full transport defect vanished.

This supplies a meaningful formal invariant and a quantitative error certificate, including an exact explanation of observational blindness.

Scope

A cokernel class is a further quotient, not the same object as the raw vector. A functional descends to cokerF only when it annihilates imF, a different condition from the transport compatibility above. Hash equality supplies neither condition.

Further development

For a transport network, impose compatible weights at all vertices and derive a pathwise bound by propagating each local defect through downstream operators; optimize weights to minimize the resulting certificate.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R161.

Y25-09-004 Cokernel certificates detect torsion that real residuals can miss

2025-09-20 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which exact integer quotient encodes the obstruction, and how does its Smith form differ from a small floating residual?

The free-abelian formalization naturally leads to integer cokernels. Their torsion is a concrete and computable obstruction. Passing to real vectors can erase it entirely, so an approximate linear solve is not a replacement for the integral certificate.

Definitions and assumptions

For an integer matrix $F:Z^n\to Z^m$, its cokernel is $Z^m/\mathrm{im}F$. Smith normal form uses unimodular integer row and column changes to diagonal entries $d_1|d_2|\cdots$.

For $F=\mathrm{diag}(2,3)$, its image is$2Z\times 3Z$; a class is zero exactly when the first coordinate is even and the second is divisible by three.

Cokernel certificates detect torsion that real residuals can miss

> coker diag(2,3) is $Z/2\oplus Z/3\cong Z/6$. Its order is six although F is invertible over Q and R. Every homomorphism from this finite cokernel to the additive group Z is zero.

Derivation or proof

The quotient map sends(a,b) to$(a \bmod 2, b \bmod 3)$, and its kernel is exactly the image of F. The Chinese remainder map from Z/6 to this quotient, $k\mapsto(k \bmod 2, k \bmod 3)$, is bijective: injectivity follows because divisibility by both2 and3 implies divisibility by6, and both sets have six elements. Consequently the Smith invariant factors are1,6, consistent with determinant6 and gcd of all entries1. A homomorphism h from a finite group to Z sends every element x of finite order r to an integer with $rh(x)=0$; torsion-freeness of Z forces $h(x)=0$. Thus an integer-valued τ on this cokernel cannot detect its nonzero torsion classes.

Worked example

Test the obstruction vector d=(1,1).

It has order six because kd lies in the image exactly when2|k and3|k. The equation Fx=d has the rational solution x=(1/2,1/3), and therefore a real solver reports residual zero. Nevertheless there is no integer solution. Six times the class vanishes, and smaller positive multiples do not. Reduction modulo2 and modulo3 provides a finite exact witness of the obstruction; no numerical tolerance enters the conclusion.

The cokernel is an actual arithmetic invariant, and the example shows exactly which information a real-valued norm or integer scalar functional loses.

Scope

A raw defect vector, its norm, its rational solvability and its integral quotient class answer different questions. A vanishing scalar summary does not imply a vanishing class without a separating family of tests.

Further development

Construct explicit unimodular transformation matrices U,V with UFV in Smith form, so the certificate includes the change of coordinates and not merely a diagonal list claimed by a program.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R162.

Y25-09-011 A cross-ratio produces an auditable map into elliptic-curve moduli

2025-09-20 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which projective invariants determine the curve family, and which distinct parameter values map to the same moduli point?

The historical account eventually identified an auditable pushforward distribution on elliptic-curve moduli as the defensible mathematical object. This is more precise than treating a change of input source as a new Langlands correspondence. The Legendre family provides an exact finite model of that pushforward.

Definitions and assumptions

For four ordered distinct points on P^1, a Möbius transformation places three at$0,1,\infty$; the fourth is a cross-ratio $\lambda\notin\{0,1\}$.

The Legendre curve is $E_\lambda: y^2=x(x-1)(x-\lambda)$, with $j(\lambda)=256(1-\lambda+\lambda^2)^3/[\lambda^2(1-\lambda)^2]$.

A cross-ratio produces an auditable map into elliptic-curve moduli

The j-value is unchanged by $\lambda\mapsto 1-\lambda$ and $\lambda\mapsto 1/\lambda$, hence by the six usual cross-ratio transforms. For $\lambda=3$ these six distinct values are$3,-2,1/3,-1/2,3/2,2/3$, all with $j=21952/9$. A probability law on λ therefore has a well-defined pushforward to j, with collisions that can be counted exactly.

Derivation or proof

Möbius invariance follows by cancellation in the cross-ratio formula after substituting(az+b)/(cz+d). For j, replacing λ by1−λ preserves the quadratic1−λ+λ² and the denominator factors. Replacing λ by1/λ multiplies the numerator byλ⁻⁶ and the denominator byλ⁻⁶, leaving the ratio unchanged. These two transformations generate the six-point orbit, so all its values share j. Substitution at λ=3 gives256·7³/(9·4)=21952/9. For any measure ν, define the pushforward by j_*ν(B)=ν(j⁻¹(B)); this directly makes the collision structure part of the induced distribution.

Worked example

Assign equal probability to the six listed rational parameters.

Their parameter distribution has six atoms, but its j-pushforward is a single atom at21952/9. Thus apparent parameter diversity can collapse entirely after passage to moduli. The j-invariant classifies elliptic curves over an algebraic closure, but equality of j over Q allows distinct quadratic twists, so an arithmetic distribution must retain twist data when appropriate. Conductor, rational isogeny and reduction traces are additional arithmetic statistics rather than consequences of the j histogram alone.

The meaningful output is a transparent, testable map with explicit fibers and sampling choices.

Scope

No new Galois representation or modularity theorem follows from the provenance of λ. The example does not infer sensitive geometric or biological properties from the arithmetic; it treats only abstract input parameters.

Further development

Define a finite rational parameter sample, retain twist classes and compute exact good-prime trace vectors; quantify separately collisions in j, isomorphism class and rational isogeny class.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R163.

Y25-09-012 A Legendre-symbol certificate has a precise and limited arithmetic meaning

2025-09-21 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which finite-field symbol is evaluated, and what additional data would be required to interpret it as a Hilbert or local geometric invariant?

The historical face-plumbing module attached Maslov and Tamagawa names to a Legendre-symbol proxy. The useful mathematical component is exact quadratic residuosity. Its input, proof and finite certificate can be stated independently of those unsupported interpretations.

Definitions and assumptions

For an odd prime p and p∤a, the Legendre symbol(a/p) is1 if a is a nonzero square modulo p and−1 otherwise.

Euler’s criterion is a^((p−1)/2)≡(a/p) mod p. It concerns the multiplicative group of F_p, not a general integer-valued Maslov index or elliptic Tamagawa number.

A Legendre-symbol certificate has a precise and limited arithmetic meaning

For p=163 and a=6 the Legendre symbol is1. An explicit square root modulo163 certifies this directly. More generally, the set of nonzero squares is a subgroup of index two, and the Legendre symbol is a multiplicative homomorphism F_p×→{±1}.

Derivation or proof

The squaring map on F_p× has kernel{1,−1}, because $x^2=1$ factors as(x−1)(x+1) and the field has odd characteristic. Its image therefore has(p−1)/2 elements and is a subgroup. The quotient has order two, proving multiplicativity of its sign character. To prove Euler’s criterion, use cyclicity of F_p×: for a generator g, a=g^k and a^((p−1)/2)=(-1)^k, which is1 exactly for even k, namely the squares. For the stated numerical case, a direct modular square root and modular exponentiation supply mutually consistent finite certificates.

Worked example

Distinguish a quadratic residue result from an unrestricted local-symbol claim.

The roots are13 and150:$13^2=169\equiv 6$ mod163 and150≡−13. The verifier independently enumerates them and checks$6^{81}\equiv 1$ mod163. In an odd-p Hilbert symbol formula, a unit u paired with the uniformizer p gives(u,p)_p=(u/p); this explains one genuine relationship. But a general Hilbert symbol(a,b)_p depends on two square classes and their valuations. A single Legendre value cannot supply those missing arguments, and neither label produces the component group of an elliptic curve’s Néron model.

The returned sign can be certified exactly, while its semantic interpretation remains a separate theorem requiring separately defined objects.

Scope

The historical build failed before completing the claimed formal pipeline; emitted JSON alone does not validate a formal proof. No anatomical or physical conclusion is drawn from this arithmetic example.

Further development

Specify a genuine quadratic extension of Q_p and verify norm-solvability or the full odd-prime Hilbert-symbol formula against exact valuation and residue data.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R135.

Y25-09-002 The cosine iteration has a rigorous tail certificate

2025-09-29 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which uniform interval bound validates the cosine fixed point and the proposed summable-tail condition?

The τ-Reflection fragment proposed τ_(k+1)=cosτ_k, a tail norm and a local derivative indicator. This example supports a complete theorem, but a derivative at one iterate must be replaced by a bound on a forward-invariant interval.

Definitions and assumptions

Let x_(n+1)=cos x_n for a real initial value. After the first iteration $x_1 \in [-1,1]$.

On I=[−1,1], cos maps I into[cos1,1]⊂I and has Lipschitz constant q=sin1<1. Define the increment tail R_N=Σ_(k=N)^∞|x_(k+1)−x_k|.

The cosine iteration has a rigorous tail certificate

There is one real solution r=cos r, every real initial condition converges to r, and R_N≤|x_(N+1)−x_N|/(1−sin1) for N≥1. In particular the increment tail is absolutely summable. The sequence x_n^2 need not decrease.

Derivation or proof

The mean-value theorem bounds |cos x−cos y| by(sin1)|x−y| on I. The interval is complete and invariant, so the contraction theorem gives its unique fixed point and convergence after the first step. Any real fixed point already lies in[−1,1], so this is also global uniqueness on R. Consecutive increments decrease at least geometrically with factor q; summing proves the tail estimate. The pointwise number |sin x_N| can be smaller than the supremum derivative on a containing interval, so it alone is not a uniform certificate. Starting at zero gives $x_0^2=0$ and $x_1^2=1$, disproving a blanket decreasing-energy interpretation of $H_n=x_n^2$.

Worked example

Construct a narrow certified interval and estimate a concrete tail.

Alternating Taylor bounds for cosine show cos(739/1000)>739/1000 and cos(740/1000)<740/1000. Since cos x−x is strictly decreasing on this positive interval, the unique fixed point lies between these rationals. The same Taylor bounds can be implemented with rational arithmetic. Once two consecutive iterates are known within error δ, the tail formula gives a bound using their difference enlarged by2δ, divided by1−sin1; this makes rounding uncertainty explicit.

The proposed convergence statistic now follows from a uniform contraction theorem and an explicit geometric tail, rather than from a threshold written into a metadata record.

Scope

A sheaf carrying fields named tailL1,rho and eps does not thereby satisfy a descent or fixed-point theorem. The result concerns this scalar iteration and does not construct a categorical embedding.

Further development

Use interval arithmetic to propagate an invariant rational interval and replace the conservative sin1 bound by the maximum derivative on that interval, producing sharper certified stopping rules.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R149.

October 2025

Dated fragments recovered; complete transcript unavailable. 36 developed treatments.

The recovered material connects arithmetic regulators, index and residue formulas, logarithmic differential fields, chain-complex cones and spectral models. The developments specify the hypotheses needed for each comparison and retain missing numerical certificates as gaps.

ID · date	Concept and revised question	Result and mathematical core
Y25-10-001 2025-10-01 · U	The RG leaf has an exact first integral, and its two-loop extension introduces a logarithm Which differential equation defines the leaf, and what field of functions contains its two-loop solution?	The RG leaf has an exact first integral, and its two-loop extension introduces a logarithm The one-loop solution is $\mu(\ell)=\mu_0/(1-b\mu_0\ell)$, with invariant $\mu^{-1}+b\ell=\mu_0^{-1}$. Its annihilating one-form is $d\mu/\mu^2-b\,d\ell$. For $b\neq0$, an antiderivative of$1/[\mu^2(b+c\mu)]$ is $F(\mu)=-1/(b\mu)+(c/b^2)\log\lvert(b+c\mu)/\mu\rvert$.
Y25-10-002 2025-10-01 · U	Whitening and signed-permutation invariance impose different equations Which Gramian is positive definite, and which additional condition makes a quadratic form invariant under the whole B_5 group?	Whitening and signed-permutation invariance impose different equations $W^{T}\Sigma W=I$. A real symmetric matrix κ satisfies $P^{T}\kappa P=\kappa$ for every $P\in B_5$ if and only if $\kappa=cI$. Thus whitening alone does not prove that the original Gramian was B_5 invariant.
Y25-10-003 2025-10-01 · A	Newton lifting supplies an exact p-adic certificate rather than a digest comparison Which congruence, derivative unit and precision doubling justify the proposed p-adic realization?	Newton lifting supplies an exact p-adic certificate rather than a digest comparison If v_p(f(a))≥m≥1 and f′(a) is a unit, the Newton step has v_p(f(a′))≥2m and a′≡a modp^m. Repetition converges to the unique root in a+pZ_p. For f=X²−2,p=7, the residue3 lifts to a genuine7-adic square root.
Y25-10-004 2025-10-03 · A	The mapping cone turns an echo equivalence into an acyclicity test Which chain map is being compared, and does its cone have zero homology in every degree?	The mapping cone turns an echo equivalence into an acyclicity test Cone(id_C) is contractible by h(y,x)=(0,y). More generally Cone(f) is acyclic if and only if f is a quasi-isomorphism.
Y25-10-005 2025-10-03 · A	Transition functions satisfy a cocycle equation, and gauge changes preserve it Are the overlap functions genuine descent data, and which gauge-invariant obstruction is actually computed?	Transition functions satisfy a cocycle equation, and gauge changes preserve it The triple product $c=g_{12}g_{23}g_{31}$ is invariant under these abelian gauge changes. It equals one exactly when the logarithmic edge values sum to zero around the triangle. A nonunit product cannot be repaired by changing local trivializations.
Y25-10-006 2025-10-03 · A	A pushout glues compatible maps by a universal property Which maps are being glued, and what uniquely characterizes the proposed sealing object?	A pushout glues compatible maps by a universal property For every pair u:B→V,v:C→V with ui=vj, there is one linear map w:P→V with wβ=u,wγ=v. Thus P is a pushout. If i and j are injective, dimP=dimB+dimC−dimA.

ID · date	Concept and revised question	Result and mathematical core
Y25-10-007 2025-10-05 · A	A rank-two Mordell-Weil group forces a4-Selmer group of at least sixteen elements Which claims are actual group cardinalities, and do they satisfy the Kummer exact sequence before any BSD conclusion is attempted?	A rank-two Mordell-Weil group forces a4-Selmer group of at least sixteen elements If rankE(Q)=2, then $\lvert Sel_4\rvert \geq 16$. If additionally $\lvert Sel_2\rvert=4$, then Sha[2]=0 and the entire2-primary torsion subgroup Sha[2^∞] is zero. These are conditional deductions from genuine certified rank and Selmer groups, not validation of the historical computations.
Y25-10-008 2025-10-05 · A	A regulator is a Gram determinant tied to a generator basis Which height pairing and independent generators define the regulator, and what changes under a finite-index basis replacement?	A regulator is a Gram determinant tied to a generator basis $detG' = (detM)^2 detG$. A unimodular change preserves the regulator, while a finite-index sublattice of index m multiplies it by m^2. Linear dependence makes the Gram determinant zero.
Y25-10-009 2025-10-09 · A	A real quadratic L-derivative is exactly the logarithm of a unit Which character, conductor and regulator normalization make the proposed Stark comparison an actual exact identity?	A real quadratic L-derivative is exactly the logarithm of a unit $L(0,\chi_5)=0$ and $L'(0,\chi_5)=\log\varphi$. The element φ is an algebraic unit of norm−1, satisfying $\varphi^2-\varphi-1=0$. This is a precisely normalized regulator-sized logarithm, not a proof of every proposed Stark realization.
Y25-10-011 2025-10-09 · A	A heat supertrace equals an Euler index because nonzero modes pair Which grading and differential make the claimed protected index independent of heat time?	A heat supertrace equals an Euler index because nonzero modes pair For every t≥0 in this finite-dimensional complex, $Str(e^{-tD^2}) = dimker_d - dimcok er_d = dimC^0 - dimC^1$. More generally a finite complex has heat supertrace equal to its alternating cohomology dimension.
Y25-10-013 2025-10-09 · A	A theta heat equation and an elliptic transformation can be proved term by term Which q and theta conventions make the proposed heat equation and Jacobi covariance agree?	A theta heat equation and an elliptic transformation can be proved term by term $\partial_\tau\theta = (1/(4\pi i))\partial^2_z\theta$. Also $\theta(\tau,z+1)=\theta(\tau,z)$ and $\theta(\tau,z+\tau)=\exp(-\pi i\tau-2\pi iz)\theta(\tau,z)$. The last factor is the half-index convention for this theta function.
Y25-10-014 2025-10-10 · U	An index does not determine Yukawa singular values Which invariant is protected, and which additional metric data determine the unprotected singular values?	An index does not determine Yukawa singular values For $Y_t: C \to C^2$ defined by $z \mapsto (tz,0)$, the index is−1 for every complex t, while its sole singular value is$\lvert t\rvert$. At t=0 the kernel and cokernel dimensions both jump, but their difference remains−1.
Y25-10-015 2025-10-10 · A	A Tor group requires modules and a resolution before it can model an obstruction Which ring and modules define the claimed derived interaction, and what does its first Tor group actually calculate?	A Tor group requires modules and a resolution before it can model an obstruction $Tor_0^R(M,M)\cong k$ and $Tor_1^R(M,M)\cong k$, while higher Tor groups vanish for this resolution. By contrast $Tor_1^R(R,M)=0$ because R is free.

ID · date	Concept and revised question	Result and mathematical core
Y25-10-012 2025-10-12 · U	A Picard-Fuchs basis comparison is determined by values and derivatives at one regular point Do both proposed period constructions solve the same normalized ODE, and what initial data fix their constant change of basis?	A Picard-Fuchs basis comparison is determined by values and derivatives at one regular point The Wronskian is constant. Any two fundamental matrices for the same equation differ by a constant invertible matrix. At u=0 the normalized solutions have expansions $y_1=1+u^2/8+3u^4/128+\cdots$ and $y_2=u+u^3/24+5u^5/384+\cdots$.
Y25-10-016 2025-10-14 · U	The E_8 roots, its Lie dimension and the totient384 are different exact counts Which finite set or vector space is being counted by each dimension in the framework?	The E_8 roots, its Lie dimension and the totient384 are different exact counts The displayed E_8 root set has112+128=240 norm-squared-two roots spanning R^8. The associated complex simple Lie algebra has dimension8+240=248. Also φ(1680)=384, since1680=$2^4\cdot3\cdot5\cdot7$.
Y25-10-017 2025-10-14 · A	Semisimplicity kills ordinary extension classes Which cohomology theory measures the proposed categorical deformation, and does ordinary Ext vanish under the stated semisimplicity assumption?	Semisimplicity kills ordinary extension classes In a semisimple abelian category with the usual Ext construction, $Ext^n(C,A)=0$ for n>0. For finite-dimensional vector spaces, a split extension is explicitly B≅A⊕C.
Y25-10-018 2025-10-14 · U	A four-object pointed model has fully checkable F, R and S data What actual coherence and nondegeneracy equations does the proposed modular data satisfy?	A four-object pointed model has fully checkable F, R and S data The pentagon and hexagon equations hold; S is real orthogonal with $S^2=I$, and its Verlinde coefficients are N_(a,b)^c=1 when c=a+b and zero otherwise. The global squared dimension is four.
Y25-10-019 2025-10-14 · A	A Pareto frontier contains points that no linear selector can choose Which objectives define dominance, and what information is lost when a two-objective frontier is replaced by H−λd?	A Pareto frontier contains points that no linear selector can choose Every strict optimizer of a positive weighted selector is nondominated, but not every nondominated point optimizes such a selector. The predictor simplifies to $A=248D^2d^{-2}e^{-H}$, independent of r.
Y25-10-010 2025-10-16 · A	Heat residues follow from a Mellin transform and need not be integer indices Which operator, heat expansion and Mellin normalization define the proposed residue?	Heat residues follow from a Mellin transform and need not be integer indices ζ_A(s)=2(L/(2π))^(2s)ζ_R(2s), so its residue at s=1/2 is L/(2π). Equivalently K(t) has leading term L/(√(4πt)), whose Mellin pole divided by Γ(1/2) gives the same residue. It is not generally an integer.
Y25-10-029 2025-10-19 · U	Fiber integration commutes with d on closed product fibers in a fixed convention What are the degrees and signs of the operators in the proposed pushforward-curvature identity?	Fiber integration commutes with d on closed product fibers in a fixed convention In this convention d_Bπ!ω=π!dω for smooth product forms on closed fibers. The commutator has degree1−m, whereasπ!(Ω_L∧ω) has degree2−m; the proposed equality therefore cannot hold as an identity of ordinary graded forms without changing an operator or adding extra structure.

ID · date	Concept and revised question	Result and mathematical core
Y25-10-030 2025-10-19 · U	The Bloch-Wigner correction has coefficient one, and a scalar residue leaves a level curve Which logarithm branches and normalization preserve the dilogarithm symmetry, and can one real residue identify a complex point?	The Bloch-Wigner correction has coefficient one, and a scalar residue leaves a level curve The normalized function satisfies $D(1-z)=-D(z)$. A coefficient1/2 in its correction fails this relation in general. Near any real $x\in(0,1)$, the equation $R(z)=\log x$ defines a local real one-dimensional level curve, rather than uniquely determining z.
Y25-10-020 2025-10-21 · A	Character orthogonality gives an exact frame bound before a large-sieve claim Which character family and sampling operator determine the best quadratic bound?	Character orthogonality gives an exact frame bound before a large-sieve claim $\Sigma_k\|S_k\|^2=m\Sigma_r\|a_r\|^2$. The optimal general constant is the largest eigenvalue of A*A. Repeating a character changes the frame bound, so the number and multiplicity of characters cannot be omitted.
Y25-10-021 2025-10-21 · A	Projecting the updated state has a descent proof; clipping the gradient is a different operation Which projection is used, and which smoothness and feasibility hypotheses imply the claimed stability?	Projecting the updated state has a descent proof; clipping the gradient is a different operation $f(x^+)\le f(x)-(1/\eta-L/2)\|\|x^+-x\|\|^2$. Hence$0<\eta<2/L$ gives descent unless the step is zero. The alternative $x-\eta\Pi_C(\nabla f(x))$ need not stay in C.
Y25-10-027 2025-10-21 · A	Trace class is contained in Hilbert-Schmidt, and strong convergence alone preserves neither Which operator topology is certified, and are the claimed trace-class and Hilbert-Schmidt properties mutually consistent?	Trace class is contained in Hilbert-Schmidt, and strong convergence alone preserves neither Every trace-class operator is Hilbert-Schmidt, with$\|\|A\|\|_{HS}\le\|\|A\|\|_1$. Compactness does not imply either summability condition. A strong limit of finite-rank operators need not even be compact.
Y25-10-036 2025-10-22 · T	Product-variable reduction gives exact zeta integrals and a controlled fractional-part transform Which exact integrand is known, and what general reduction can be proved without inventing the indexed paper’s missing input?	Product-variable reduction gives exact zeta integrals and a controlled fractional-part transform $\int_{[0,1]^d}F(x_1\cdots x_d)dx=(1/(d-1)!)\int_0^1F(t)(-\log t)^{d-1}dt$. In particular$\int_{[0,1]^4}[1-x_1x_2x_3x_4]^{-1}dx=\zeta(4)=\pi^4/90$. For $s>1$, $A(s)=1/(s-1)-\zeta(s)/s$.
Y25-10-022 2025-10-23 · A	A finite prime oscillatory sum has exact calculus but no automatic zero correspondence Which analytic identities follow from the actual finite sum, and what additional theorem would connect it to zeta zeros?	A finite prime oscillatory sum has exact calculus but no automatic zero correspondence $\lambda_X(0)=0$, $\lambda_X'(0)=1/2$ and $\lambda_X''(0)=-2\Sigma_{p\le X}(\log p)^3/(1+p)<0$. Its derivative is$1/2-2\Sigma_{p\le X}(\log p)^2\sin(t\log p)/(1+p)$. These identities do not force zeros at any prescribed ordinates.
Y25-10-033 2025-10-23 · U	Prime-log tones have exact beat differences but no common nonzero period Which frequencies and decay law define the waveform, and what can its beats prove mathematically?	Prime-log tones have exact beat differences but no common nonzero period log2/log3 is irrational, so the two-tone function has no positive common period. Its product formula has difference frequency $f_0\log(3/2)$, while the damped envelope has half-life(log2)/c.

ID · date	Concept and revised question	Result and mathematical core
Y25-10-034 2025-10-27 · U	Second and fourth mean values expose multiplicative resonances exactly Which finite mean-value identity distinguishes diagonal energy from genuine fourth-order resonance?	Second and fourth mean values expose multiplicative resonances exactly lim_(T→∞)M_T(|D|²)=Σ|a_n|². The fourth mean isΣ_(mn=rs)a_ma_n conjugate(a_ra_s), summed over indices at most N. Multiplicative coincidences, not merely pairwise orthogonality, govern this fourth moment.
Y25-10-035 2025-10-27 · A	The asset recurrence has an exact stability threshold independent of its interpretation Which parameter combinations determine a stable fixed point, and what additional closure is required for the proposed feedback law?	The asset recurrence has an exact stability threshold independent of its interpretation For q≠1, A_t=q^tA₀+c(1−q^t)/(1−q). Its fixed point c/(1−q) is globally attracting exactly when|q|<1. For M>0, γ>0 exactly when λ<M/(M+1).
Y25-10-028 2025-10-29 · U	Cahn-Hilliard flow conserves mass and dissipates a declared energy Which free energy and boundary conditions make the proposed fourth-order equation a gradient flow?	Cahn-Hilliard flow conserves mass and dissipates a declared energy The total mass∫ψ is constant and dE/dt=−∫M|∇μ|²≤0. For constant M and f(ψ)=aψ²/2, a Fourier mode k evolves at rate−M|k|²(a+K|k|²).
Y25-10-031 2025-10-30 · U	Finite-group symmetry permits frequency ratios but does not usually force one Which representation and equivariant operator make30/13 a necessity rather than a chosen pair of coefficients?	Finite-group symmetry permits frequency ratios but does not usually force one K commutes with everyρ(g)⊕ρ(g) for every positive pair a,b. Hence A₅ equivariance alone allows every positive frequency ratio b/a, including30/13. It does not select that ratio.
Y25-10-032 2025-10-30 · U	Damping ratio, decay and Q must use the same oscillator convention Which ODE defines the reported damping quantities, and do their numerical values satisfy its exact relations?	Damping ratio, decay and Q must use the same oscillator convention The amplitude multiplier after one damped period is exp[−2πζ/√(1−ζ²)]. For ζ=697/1000, the model Q is500/697, not1.43×10⁹. These conclusions are exact consequences of the declared ODE.
Y25-10-023 2025-10-31 · U	The sine quotient has an exact parity integral and a finite quadrature certificate Why does the integral distinguish odd and even positive integers, and when is a discrete midpoint rule exact?	The sine quotient has an exact parity integral and a finite quadrature certificate I(a)=aπ for odd a and zero for even a. For odd a=2m+1, the N-point midpoint rule in θ is exact whenever m<N. For even a its symmetric midpoint sum is zero for every N.
Y25-10-024 2025-10-31 · U	The logarithmic sine integral separates into an elementary term and Catalan’s constant What is the exact value of the recovered logarithmic integral, including its integrable endpoint singularity?	The logarithmic sine integral separates into an elementary term and Catalan’s constant Ψ=a log(1+a²)−2a+2arctana−10a log2−5G, with a=π/4.
Y25-10-025 2025-10-31 · U	The recovered fourth-power logarithmic integral has an exact elementary evaluation Can the claimed closed form be derived from the exact input, with both endpoint limits controlled?	The recovered fourth-power logarithmic integral has an exact elementary evaluation I=[18π+90log2−33log20−12arctan(1/2)]/[1152(logπ)⁴].

ID · date	Concept and revised question	Result and mathematical core
Y25-10-026 2025-10-31 · A	Cyclotomic trace orthogonality fixes the coefficient p, not p+1 Which embeddings define the trace, and are the coefficients integral character multiplicities or arbitrary rational weights?	Cyclotomic trace orthogonality fixes the coefficient p, not p+1 $Tr_{Q(\zeta)/Q}(X^2)=pC_0-A^2$. If all a_k are integers, this trace is an integer. The same integrality holds for every algebraic-integer X, including ordinary finite-group character values contained in this field.

Y25-10-001 The RG leaf has an exact first integral, and its two-loop extension introduces a logarithm

2025-10-01 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which differential equation defines the leaf, and what field of functions contains its two-loop solution?

The RG capsule has a clear mathematical core: a first-order ODE with a rational invariant. The historical differential form and the invariant require a sign check. The two-loop equation can also be integrated explicitly, although its inverse is generally not rational in the parameters.

Definitions and assumptions

Let $\mu'=b\mu^2$ with $\mu(0)=\mu_0\neq 0$. A prime denotes differentiation with respect to ℓ.

For a two-loop model use $\mu'=\mu^2(b+c\mu)$, on a connected interval avoiding $\mu=0$ and $b+c\mu=0$.

The RG leaf has an exact first integral, and its two-loop extension introduces a logarithm

The one-loop solution is $\mu(\ell)=\mu_0/(1-b\mu_0\ell)$, with invariant $\mu^{-1}+b\ell=\mu_0^{-1}$. Its annihilating one-form is $d\mu/\mu^2-b\,d\ell$. For $b\neq 0$, an antiderivative of$1/[\mu^2(b+c\mu)]$ is $F(\mu)=-1/(b\mu)+(c/b^2)\log|(b+c\mu)/\mu|$.

Derivation or proof

Differentiate $\mu^{-1}+b\ell$: its derivative is$-\mu'/\mu^2+b=0$. This also shows that the historical plus-sign form $d\mu/\mu^2+b\,d\ell$ would instead vanish on $\mu'=-b\mu^2$, so the form and invariant cannot share the same sign without correction. For the two-loop integrand, partial fractions give$1/[\mu^2(b+c\mu)]=1/(b\mu^2)-c/(b^2\mu)+c^2/[b^2(b+c\mu)]$. Integrating yields F; thus $F(\mu(\ell))-F(\mu_0)=\ell$ until a singularity or branch boundary is reached. The logarithm is essential unless $c=0$ or another degeneracy removes it.

Worked example

Use the recovered rational capsule parameters $\mu_0=3/7$ and $b=5/2$ in the one-loop equation.

The solution is $\mu(\ell)=6/(14-15\ell)$, with pole at$14/15$. At $\ell=1/5$ it equals$6/11$, and$1/\mu+b\ell=11/6+1/2=7/3=1/\mu_0$. This exact rational check validates the leaf on its nonsingular interval. For the two-loop inversion, Newton iteration is meaningful only after choosing a real branch and checking that F' stays nonzero there.

The capsule now carries a verifiable invariant, a corrected differential form and an explicit scope for the implicit logarithmic extension.

Scope

No arithmetic motive, physical RG law or global continuation is obtained merely by hashing these parameters. The pole excludes part of the nominal time axis, and a solution branch must not be continued through it by an algebraic cancellation.

Further development

Bracket a branch of the two-loop implicit equation and prove a Newton or interval-Newton convergence certificate using explicit derivative bounds.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R110.

Y25-10-002 Whitening and signed-permutation invariance impose different equations

2025-10-01 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which Gramian is positive definite, and which additional condition makes a quadratic form invariant under the whole B_5 group?

The October framework combined normalized Gramians, whitening and a B_5 chamber index. These operations have precise independent meanings. Whitening produces an orthonormal metric, while invariance under all coordinate sign flips and permutations forces a quadratic form to be scalar.

Definitions and assumptions

Let Σ be a real symmetric positive-definite5×5 matrix and W=Σ^(−1/2), the positive spectral inverse square root.

The signed-permutation group B_5 consists of matrices P with one±1 in each row and column. Its order is$2^5 5!=3840$.

Whitening and signed-permutation invariance impose different equations

$W^T \Sigma W = I$. A real symmetric matrix κ satisfies $P^T \kappa P = \kappa$ for every $P \in B_5$ if and only if $\kappa = cI$. Thus whitening alone does not prove that the original Gramian was B_5 invariant.

Derivation or proof

By the spectral theorem write $\Sigma = QDQ^T$ with Q orthogonal and positive diagonal D. Then W=QD^(−1/2)Q^T, and multiplication gives the identity. For the group claim, flip the sign of coordinate i. An off-diagonal entry κ_ij with j≠i changes sign under conjugation, so invariance forces it to be zero. Permutations then exchange diagonal entries, forcing them all equal. Conversely a scalar multiple of the identity is fixed by every orthogonal signed permutation. Counting the group requires an arbitrary permutation and five independent signs, giving the stated order.

Worked example

Take $\Sigma=\mathrm{diag}(1,4,9,16,25)$.

Its whitening matrix is $\mathrm{diag}(1,1/2,1/3,1/4,1/5)$, and direct rational multiplication yields I. But swapping the first two coordinates changes Σ, so the original metric lacks B_5 symmetry. A generic vector with distinct nonzero absolute coordinates has3840 distinct signed-permutation images; ties or zeros introduce stabilizers and shorten its orbit. A claimed chamber index therefore needs a tie convention and an admissibility test.

The geometry separates normalization, symmetry and orbit counting, making each claim independently checkable.

Scope

A singular Gramian has no inverse square root on the full space; a pseudoinverse instead whitens only its range. The number3840 is not the totient value384 or the dimension248 of E_8.

Further development

For a measured Gramian, quantify the symmetry defect $\max_P\|P^{T}\Sigma P-\Sigma\|$ and compare it with the exact group average, which is$(\mathrm{tr}\Sigma/5)I$.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R164.

Y25-10-003 Newton lifting supplies an exact p-adic certificate rather than a digest comparison

2025-10-01 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which congruence, derivative unit and precision doubling justify the proposed p-adic realization?

The Rust scaffold named Newton/Hensel and p-adic logarithms. A small exact lift gives those words mathematical substance. It also clarifies why reducing a cryptographic digest modulo p is unrelated to comparing p-adic realizations of an arithmetic object.

Definitions and assumptions

For $f\in Z_p[X]$, a simple root modulo p is a residue a_0 with $f(a_0)\equiv 0 \bmod p$ and $f'(a_0)$ not divisible by p.

A Newton step $a'=a-f(a)/f'(a)$ is defined in Z_p when $f'(a)$ is a unit. Congruence precision is measured by v_p.

Newton lifting supplies an exact p-adic certificate rather than a digest comparison

If $v_p(f(a))\geq m\geq 1$ and $f'(a)$ is a unit, the Newton step has $v_p(f(a'))\geq 2m$ and $a'\equiv a \bmod p^m$. Repetition converges to the unique root in $a+pZ_p$. For $f=X^2-2, p=7$, the residue3 lifts to a genuine7-adic square root.

Derivation or proof

Taylor expansion gives f(a+h)=f(a)+f′(a)h+h²g for integral g. Choose h=−f(a)/f′(a), whose valuation is at least m; the first two terms cancel and the remainder has valuation at least2m. The derivative remains a unit because a′≡a modp. The successive corrections have valuations tending to infinity, so completeness of Z_p gives a root. If two roots x,y share a modulo p, then f(x)−f(y)=(x−y)u with u a unit, by the same derivative argument; hence x=y.

Worked example

Lift the square root by ordinary congruence arithmetic.

Modulo7 the root is3. Write a=3+7k. The congruence a²−2≡7+42k mod49 requires1+6k≡0 mod7, giving k=1 and a=10 mod49. Write a=10+49k; modulo343 its defect is98+980k, yielding2+20k≡0 mod7 and k=2, so a=108 mod343. Indeed108²−2=11662=34·343. Each check is an exact integer divisibility statement.

The p-adic value is specified by compatible congruences and a uniqueness theorem, not a floating decimal or a hash residue.

Scope

The ordinary p-adic logarithm needs a stated convergence domain such as1+pZ_p for odd p. Comparing real and p-adic periods additionally requires actual comparison maps; no such map is supplied by the lift alone.

Further development

Implement precision-doubling lifts together with exact polynomial residual valuations and prove compatibility of all emitted congruence certificates.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R059.

Y25-10-004 The mapping cone turns an echo equivalence into an acyclicity test

2025-10-03 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which chain map is being compared, and does its cone have zero homology in every degree?

The corrected Morphic Echo proposal replaced informal equality with a chain map and its cone. This is a precise mathematical improvement: quasi-isomorphism has an exact obstruction complex, and the identity map admits an explicit contracting homotopy.

Definitions and assumptions

Use cochain complexes with d increasing degree. For a cochain map f:C→D, Cone(f)ⁿ=Dⁿ⊕C^(n+1), with differential d(y,x)=(d_Dy+fx,−d_Cx).

A quasi-isomorphism induces an isomorphism on every cohomology group. A complex is contractible if dh+hd is its identity for some degree−1 map h.

The mapping cone turns an echo equivalence into an acyclicity test

Cone(id_C) is contractible by h(y,x)=(0,y). More generally Cone(f) is acyclic if and only if f is a quasi-isomorphism.

Derivation or proof

The cone differential squares to zero because d_Df=fd_C: its first component has the two cross terms cancel, while the second is d_C²x. For f=id, dh(y,x)=(y,−dy) and hd(y,x)=(0,dy+x); their sum is(y,x). Contractibility forces cohomology to vanish because every closed v equals d(hv). For general f, the short exact sequence0→D→Cone(f)→C[1]→0 gives a long exact cohomology sequence whose connecting map is H(f). Exactness shows that vanishing cone cohomology is equivalent to H(f) being injective and surjective in every degree. This classical exact-sequence input is stated explicitly.

Worked example

Let C and D both be Q concentrated in degree zero, with f multiplication by2.

The cone is Q in degree−1 mapped by2 to Q in degree zero, so its kernel and cokernel are both zero. Thus multiplication by2 is a quasi-isomorphism over Q. Over Z the same cone has cokernel Z/2, so it is not acyclic. The coefficient ring changes the certificate even though the displayed matrix is unchanged.

An echo claim becomes a concrete complex, a coefficient choice and an exact homology calculation.

Scope

A hash comparison does not establish a chain-map equation. Acyclicity and contractibility are also distinct for general complexes over arbitrary rings; only the identity-cone case here is given an explicit contraction.

Further development

For bounded free integer complexes, certify d²=0 and the chain-map identities first, then use Smith forms to compute the cone homology including torsion.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R165.

Y25-10-005 Transition functions satisfy a cocycle equation, and gauge changes preserve it

2025-10-03 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Are the overlap functions genuine descent data, and which gauge-invariant obstruction is actually computed?

The corrected curvature proposal used c_ijk=g_ijg_jkg_ki. For abelian transition functions this supplies a finite, exact descent condition. It should be distinguished from a differential curvature two-form and from the hash attached to a record.

Definitions and assumptions

On a three-patch cover use positive real transition functions g_{ij}, with $g_{ji}=g_{ij}^{-1}$. Compatible line-bundle descent requires $g_{12}g_{23}g_{31}=1$ on the triple overlap.

A change of local trivialization by positive h_i sends g_{ij} to $g'_{ij}=h_i^{-1}g_{ij}h_j$. For constant positive functions, logarithms turn multiplication into addition.

Transition functions satisfy a cocycle equation, and gauge changes preserve it

The triple product $c=g_{12}g_{23}g_{31}$ is invariant under these abelian gauge changes. It equals one exactly when the logarithmic edge values sum to zero around the triangle. A nonunit product cannot be repaired by changing local trivializations.

Derivation or proof

Multiply the three transformed functions: $h_1^{-1}g_{12}h_2\cdot h_2^{-1}g_{23}h_3\cdot h_3^{-1}g_{31}h_1=h_1^{-1}ch_1=c$. Taking logarithms gives $\log c=\log g_{12}+\log g_{23}+\log g_{31}$. On a filled triangle, a closed additive edge cochain is exact: choose local potentials $v_1=0, v_2=\log g_{12}, v_3=\log g_{12}+\log g_{23}$; the third edge equation holds precisely when the sum vanishes. This is a finite descent calculation with every overlap specified.

Worked example

Set $g_{12}=2, g_{23}=3$ and first choose $g_{31}=1/6$.

The product is one, and local weights$(1,2,6)$ exhibit $g_{ij}=h_j/h_i$ around the cycle. Replacing the final value by$1/5$ gives $c=6/5$. No gauge choices h_i can change that value to one. In logarithmic coordinates the defect is $\log(6/5)$, a real cocycle-failure measurement, while an integer coefficient model would require a different group and differential.

The descent condition and its invariant failure have explicit algebraic meaning, so a proposed curvature detector has a precise target.

Scope

For nonabelian transition groups the product transforms by conjugation rather than remaining literally fixed. A connection curvature and an anomaly class require further differential or cohomological data; this three-number example does not establish those identifications.

Further development

Build a larger cover with specified intersection groups and verify every coboundary identity, then determine which gauge-invariant obstruction classes survive refinement.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R166.

Y25-10-006 A pushout glues compatible maps by a universal property

2025-10-03 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which maps are being glued, and what uniquely characterizes the proposed sealing object?

The sealing proposal invoked a pushout and verifier replay. A finite vector-space pushout provides an actual mathematical model: it glues two interfaces along their shared part. Its universal property expresses compatibility independently of file names, hashes or motives.

Definitions and assumptions

For maps $i:A\to B$ and $j:A\to C$ over Q, define $P=(B\oplus C)/\{(i(a),-j(a)):a\in A\}$.

The canonical maps $\beta:B\to P$ and $\gamma:C\to P$ send b to $[(b,0)]$ and c to $[(0,c)]$. They satisfy $\beta i=\gamma j$.

A pushout glues compatible maps by a universal property

> For every pair $u:B\to V, v:C\to V$ with $ui=vj$, there is one linear map $w:P\to V$ with $w\beta=u, w\gamma=v$. Thus P is a pushout. If i and j are injective, $\dim P=\dim B+\dim C-\dim A$.

Derivation or proof

Define a map on the direct sum by $(b,c)\mapsto u(b)+v(c)$. On a relation vector $(i(a),-j(a))$ its value is $ui(a)-vj(a)=0$, so it descends to the quotient. Its composites have the prescribed values. Every quotient class is a sum of an image from B and one from C, so these composites determine w uniquely. For injective i and j, the relation subspace is an injected copy of A; subtracting its dimension gives the formula.

Worked example

Take $A=Q, B=C=Q^2$ and let both maps send a to $(a,0)$.

The pushout has dimension three, with coordinates that can be written $[(b_1,b_2;c_1,c_2)]\mapsto(b_1+c_1,b_2,c_2)$. Its kernel consists exactly of $(a,0;-a,0)$, the relation subspace. Compatible maps $u(b)=b_1+2b_2$ and $v(c)=c_1+3c_2$ glue to $w(x,y,z)=x+2y+3z$. Changing one map's shared-coordinate coefficient would break compatibility and prevent that factorization. The dimension count can therefore be checked without guessing an ambient arithmetic interpretation: four coordinates lose exactly one independent shared-interface relation.

The construction gives sealing an exact interface-gluing interpretation and a finite certificate of compatibility.

Scope

A software manifest must specify a category and actual maps before its merge is a pushout. Universal constructions do not automatically supply arithmetic motives, semantic correctness or replay equality. Those require independent definitions and verification.

Further development

Specify a category of typed finite execution records and prove that the proposed merge satisfies this universal property, including how conflicting shared fields are rejected.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R167.

Y25-10-007 A rank-two Mordell-Weil group forces a4-Selmer group of at least sixteen elements

2025-10-05 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which claims are actual group cardinalities, and do they satisfy the Kummer exact sequence before any BSD conclusion is attempted?

The389a1 passage reported rank two and both2- and4-Selmer cardinality four. If interpreted literally, these quantities conflict with elementary consequences of the Kummer exact sequence. The positive result is a compact exact-sequence audit that arithmetic receipts must pass.

Definitions and assumptions

For an elliptic curve E/Q and n≥2, the Kummer sequence gives0→E(Q)/nE(Q)→Sel_n(E/Q)→Sha(E/Q)[n]→0.

By the Mordell-Weil structure theorem write $E(Q)\cong Z^r\oplus T$ with finite torsion T. Consequently $|E(Q)/nE(Q)|=n^r|T/nT|$.

A rank-two Mordell-Weil group forces a4-Selmer group of at least sixteen elements

If rankE(Q)=2, then $|Sel_4|\geq 16$. If additionally $|Sel_2|=4$, then Sha[2]=0 and the entire2-primary torsion subgroup Sha[2^∞] is zero. These are conditional deductions from genuine certified rank and Selmer groups, not validation of the historical computations.

Derivation or proof

The injection in the exact sequence gives $|Sel_n|\geq|E/nE|\geq n^r$, proving the lower bound. At n=2,r=2, the lower bound is four. Equality forces both |T/2T|=1 and |Sha[2]|=1 because their product with four equals four. If an abelian group has a nonzero element of order2^k, multiplying it by2^(k−1) produces a nonzero element of order two. Therefore absence of2-torsion eliminates every finite2-power-order element. No finiteness assumption on all of Sha is needed for this last elementary implication.

Worked example

Check the free-rank model Z^2 at the two levels.

Its quotient modulo2 has four elements, represented by coordinate pairs in$\{0,1\}^2$. Its quotient modulo4 has sixteen, represented by$\{0,1,2,3\}^2$. The reduction map modulo4→modulo2 has four-element fibers. An identity-looking2×2 Smith matrix between unspecified lattices cannot replace these quotient cardinalities or prove that a purported4-Selmer group has only four elements.

The exact sequence supplies both a contradiction test and a valid route from a minimal2-Selmer group to a2-primary Sha conclusion when its premises are genuinely proved.

Scope

The historical rank, regulator and analytic intervals have not been independently certified here. A number labeled Sel_4 might denote another invariant; that ambiguity must be resolved before interpreting its value.

Further development

Produce explicit descent maps, local solubility certificates and quotient presentations, then verify the Kummer injection and all group orders with exact integer algebra.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R134.

Y25-10-008 A regulator is a Gram determinant tied to a generator basis

2025-10-05 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which height pairing and independent generators define the regulator, and what changes under a finite-index basis replacement?

The reported narrow regulator interval is meaningful only after fixing the points, height convention and error bounds. A positive-definite lattice model gives the exact change-of-basis law and shows why determinant data alone cannot certify that the chosen points generate the full lattice.

Definitions and assumptions

For independent generators $P_1,\dots,P_r$ with symmetric height-pairing matrix G, the regulator is $\det G$ in the declared normalization.

Replacing the generator column by an integer matrix M changes the Gram matrix to $G'=M^{T}GM$. A unimodular matrix has determinant ±1.

A regulator is a Gram determinant tied to a generator basis

$\det G'=(\det M)^2\det G$. A unimodular change preserves the regulator, while a finite-index sublattice of index m multiplies it by m^2. Linear dependence makes the Gram determinant zero.

Derivation or proof

The determinant identity follows directly from multiplicativity of determinants and $\det M^{T}=\det M$. For a full-rank integer coordinate matrix, the index of the generated sublattice is $|\det M|$, as can be proved by Smith form. Thus its squared covolume scales by the square of the index. Positive definiteness makes independent-vector Gram determinants positive. If the vectors are dependent, their coordinate matrix has a nontrivial kernel and so does their Gram matrix, forcing determinant zero.

Worked example

Take $G=[[2,1],[1,3]]$, whose determinant is five.

The unimodular matrix M=[[1,1],[0,1]] gives G′=[[2,3],[3,7]], still with determinant five. Replacing the second generator by twice itself gives diag(1,2)ᵀGdiag(1,2)=[[2,2],[2,12]], whose determinant is twenty, four times five. The latter pair spans an index-two sublattice. A numerical determinant near twenty would therefore not by itself identify the full-lattice regulator.

The regulator becomes an invariant with explicit basis and saturation conditions rather than a decimal detached from its generators.

Scope

An elliptic canonical-height computation additionally needs normalization, rigorous truncation bounds and a proof of independence. A Smith matrix from another complex does not certify those inputs. None of the historical decimal intervals is assumed verified here.

Further development

Given candidate points, certify the height matrix by intervals, bound its smallest eigenvalue away from zero, and prove saturation at each relevant prime before calling its determinant the full Mordell-Weil regulator.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R134.

Y25-10-009 A real quadratic L-derivative is exactly the logarithm of a unit

2025-10-09 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which character, conductor and regulator normalization make the proposed Stark comparison an actual exact identity?

The Stark reframing supplied Q(√5), its golden unit and L′(0,χ). There is an elegant complete classical model: use the primitive even quadratic Dirichlet character modulo five. Hurwitz-zeta and gamma identities then evaluate its derivative exactly, without invoking an unproved general regulator theorem.

Definitions and assumptions

Let $\chi_5(1)=\chi_5(4)=1,\chi_5(2)=\chi_5(3)=-1$ and $\chi_5(5k)=0$. Put $\varphi=(1+\sqrt{5})/2$.

For Re(s)>1, L(s,χ₅)=5^(−s)Σ_(a=1)^4χ₅(a)ζ(s,a/5), then extend analytically using Hurwitz zeta.

A real quadratic L-derivative is exactly the logarithm of a unit

$L(0,\chi_5)=0$ and $L'(0,\chi_5)=\log\varphi$. The element φ is an algebraic unit of norm−1, satisfying $\varphi^2-\varphi-1=0$. This is a precisely normalized regulator-sized logarithm, not a proof of every proposed Stark realization.

Derivation or proof

Use the classical identities $\zeta(0,x)=1/2-x$ and $\zeta'(0,x)=\log\Gamma(x)-(1/2)\log(2\pi)$. The weighted constant and linear sums vanish, giving $L(0)=0$; thus differentiating5^(−s) contributes no extra term. The character sum is zero, so the log(2π) terms also cancel. Therefore $L'(0)=\log[\Gamma(1/5)\Gamma(4/5)/(\Gamma(2/5)\Gamma(3/5))]$. Euler's reflection formula turns the ratio into $\sin(2\pi/5)/\sin(\pi/5)=2\cos(\pi/5)=\varphi$. The special-function identities are the explicit classical inputs; the cancellation and unit comparison are derived here. The conjugate unit is$(1-\sqrt{5})/2=-\varphi^{-1}$, giving norm−1.

Worked example

Separate this character from unspecified ray-class data over $\mathbb{Q}(\sqrt{5})$.

The displayed χ_5 is a character over Q of conductor five. It cuts out the real quadratic field, but is not by itself a complete choice of ray-class character over that field. Its derivative exponentiates to φ in the stated real embedding; the other embedding has negative value, and absolute logarithms or totally positive-unit conventions can introduce a factor two. Squaring φ gives a norm-one unit and doubles the logarithm.

The recovered idea has an actual special-value identity and an exact algebraic unit certificate.

Scope

No general toric motivic class, mixed-signature Eisenstein cocycle or p-adic comparison is constructed. The universal claim of unconditional validity for all totally real or CM situations is not adopted.

Further development

Specify a genuine ray-class character, its infinity type and vanishing order, then compare a rigorously evaluated L-leading term with a candidate unit using a stated regulator map and norm relations.

Source receipt: months/10/root_details.json record 0. Mathematical references: R168, R169.

Y25-10-011 A heat supertrace equals an Euler index because nonzero modes pair

2025-10-09 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which grading and differential make the claimed protected index independent of heat time?

The index tables have a rigorous finite analogue that explains the mechanism behind protection: a differential pairs nonzero even and odd modes, so their heat contributions cancel. An ordinary heat trace does not have this cancellation.

Definitions and assumptions

Take a two-term finite-dimensional real or complex Hilbert complex $C^0\to^d C^1$. Let $D=d+d^*$ on $C^0\oplus C^1$, graded even/odd.

The heat supertrace is Str(e^(−tD²))=Tr_(C⁰)e^(−td*d)−Tr_(C¹)e^(−tdd*).

A heat supertrace equals an Euler index because nonzero modes pair

For every t≥0 in this finite-dimensional complex, Str(e^(−tD²))=dimkerd−dimcokerd=dimC⁰−dimC¹. More generally a finite complex has heat supertrace equal to its alternating cohomology dimension.

Derivation or proof

The positive spectra of d*d and dd* agree including multiplicities: if d*d v=λv with λ>0, then dv≠0 and dd*(dv)=λdv, with inverse correspondence λ⁻¹d* on the eigenspaces. Hence e^(−tλ) cancels between the two graded traces. The remaining zero modes are kerd in degree zero and ker d*=cokerd in degree one. Rank-nullity gives their dimension difference dimC⁰−dimC¹. The same pairing on each degree proves the finite-complex statement, equivalently by decomposing into harmonic spaces and two-term exact pairs.

Worked example

Use d=[[1,−1],[0,0]] between two copies of R².

Both d*d and dd* have eigenvalues0,2. Each ordinary heat trace is1+e^(−2t), so the total ungraded trace is2+2e^(−2t), which changes with time. Their supertrace is zero for every t. The cohomology dimensions are one in each degree, also giving1−1=0. This explicitly demonstrates why the grading and the differential identity are part of the index theorem’s mechanism.

A protected quantity can be constructed and checked, but its protection comes from a proved pairing of modes.

Scope

The finite theorem is not the Atiyah-Singer theorem for a differential operator on a manifold. Twists, boundary conditions and APS corrections require their own hypotheses. A generic residue trace has no reason to share this cancellation.

Further development

Construct a finite discretization that preserves d²=0 and grading exactly, then prove that its supertrace index matches a chosen topological cochain complex before taking any continuum limit.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R170.

Y25-10-013 A theta heat equation and an elliptic transformation can be proved term by term

2025-10-09 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which q and theta conventions make the proposed heat equation and Jacobi covariance agree?

The October tables included Jacobi forms, heat equations and modular period comparisons. The elementary theta series supplies a completely normalized case. It also shows why an index or coefficient bound cannot be copied between conventions without checking its transformation law.

Definitions and assumptions

For Imτ>0, set θ(τ,z)=Σ_(n∈Z)exp(πin²τ+2πinz). This uses q=exp(πiτ).

On compact subsets with Imτ bounded below, Gaussian decay dominates every fixed derivative in τ or z, allowing termwise differentiation.

A theta heat equation and an elliptic transformation can be proved term by term

∂_τθ=(1/(4πi))∂²_zθ. Also θ(τ,z+1)=θ(τ,z) and θ(τ,z+τ)=exp(−πiτ−2πiz)θ(τ,z). The last factor is the half-index convention for this theta function.

Derivation or proof

Differentiating one term with respect to τ multiplies it byπin². Differentiating twice in z multiplies it by−4π²n²; division by4πi givesπin², proving the heat equation. Shifting z by one contributes exp(2πin)=1. For the τ shift, n²+2n=(n+1)²−1; reindexing n+1 extracts exp(−πiτ−2πiz) and leaves the original series. The convergent Gaussian bounds justify both the differentiations and the index shift, so these are identities of holomorphic functions on their stated domain.

Worked example

Inspect the first nonconstant modes and a prospective finite-coefficient certificate.

At fixed τ the n=±1 terms are2qcos(2πz). Their z-second derivative divided by4πi is2πi qcos(2πz), exactly their τ derivative. But agreement of finitely many Fourier modes does not identify an unrestricted function: adding q^N changes none of the lower coefficients. A genuine Sturm-type certificate needs a specified finite-dimensional modular or Jacobi space, its weight, level, index and a correct theorem bounding vanishing.

The exact transformation and heat identities are a solid base for a normalized modular-gauge comparison.

Scope

No Jacobi bound B=2 or cutoff K=2 is inferred from this theta function. Scalar modular, half-index theta and index-four Jacobi forms have different spaces and coefficient conventions. Nekrasov/PF matching requires its own normalization data.

Further development

Specify a finite-dimensional target space and construct a coefficient matrix of a proved basis; exact rank then determines how many coefficients suffice to identify an element within that space.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R171.

Y25-10-014 An index does not determine Yukawa singular values

2025-10-10 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which invariant is protected, and which additional metric data determine the unprotected singular values?

The October program tried to link masses or Yukawa data to a protected index pushforward. Finite linear algebra gives a decisive constraint on any such bridge: an index can remain fixed while every nonzero singular value varies. A theorem relating them would need extra structure beyond index equality.

Definitions and assumptions

For a linear map $Y:C^n\to C^m$, its finite Fredholm index is $\dim\ker Y-\dim\operatorname{coker} Y=n-m$.

The singular values are square roots of the eigenvalues of Y^*Y and depend on the chosen Hermitian metrics. They are not determined by the index.

An index does not determine Yukawa singular values

For $Y_t:C\to C^2$ defined by $z\mapsto(tz,0)$, the index is-1 for every complex t, while its sole singular value is$|t|$. At $t=0$ the kernel and cokernel dimensions both jump, but their difference remains-1.

Derivation or proof

If $t\neq0$, the map is injective and its range has dimension one, so ker dimension is zero and coker dimension one. At zero, the kernel has dimension one and the cokernel dimension two. Both give index-1. Meanwhile $Y_t^*Y_t$ is the1×1 matrix$|t|^2$, proving the singular-value formula. The index equality also follows from rank-nullity for any finite map, regardless of its entries. Thus an index theorem at this level cannot recover a continuous scale such as$|t|$ without another equation that restricts t.

Worked example

Compare $t=1$ with $t=100$ and with zero.

All three maps have the same index. Their singular values are$1,100,0$, and their operator norms have those same values. A proposed formula multiplying the singular value by a separately specified scale can consequently vary by a factor100 while every index datum remains unchanged. This is an exact family, not a numerical near-counterexample. Gauge or basis changes that are unitary preserve singular values, whereas arbitrary invertible basis changes need not.

The distinction identifies the genuine mathematical burden of connecting a protected invariant to metric observables.

Scope

No physical mass convention is adopted from the recovered factor v/2, and no biological or particle prediction follows. A topological index may constrain zero modes or differences of dimensions without setting nonzero spectral scales.

Further development

State a metric or variational principle that restricts Y inside a fixed index class, then prove quantitative bounds on its singular values; only those added hypotheses can support a scale prediction.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R172.

Y25-10-015 A Tor group requires modules and a resolution before it can model an obstruction

2025-10-10 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which ring and modules define the claimed derived interaction, and what does its first Tor group actually calculate?

The Higgs/torsor synthesis invoked Tor_1 without a specified resolution. A small exact model demonstrates what a derived tensor product detects: failure of a relation to remain injective after tensoring. This is meaningful homological mathematics but does not automatically identify a physical field.

Definitions and assumptions

Let $R=k[x]$ over a field k and $M=R/(x)$. A free resolution is$0\to R\to^{x} R\to M\to 0$.

Tor_i^R(M,N) is the i-th homology of a free resolution of M tensored with N. Ordinary tensor product is its degree-zero part.

A Tor group requires modules and a resolution before it can model an obstruction

$\mathrm{Tor}_0^R(M,M)\cong k$ and $\mathrm{Tor}_1^R(M,M)\cong k$, while higher Tor groups vanish for this resolution. By contrast $\mathrm{Tor}_1^R(R,M)=0$ because R is free.

Derivation or proof

Multiplication by x on R is injective, since the polynomial ring over a field is a domain; its cokernel is M, proving exactness of the resolution. Tensor with M. On M the element x acts as zero, so the resulting two-term complex is $M\to^{0} M$. Its kernel in degree one is M and its cokernel in degree zero is M, with no other terms. The one-term free resolution of R instead has no degree-one term, giving zero first Tor. The difference is structural: the relation x becomes zero after imposing x=0, leaving a degree-one class.

Worked example

Replace the second module by $N=R/(x-1)$.

Multiplication by x acts as multiplication by one on N, so the tensored resolution is $N \to {}^{1}N$, an isomorphism. Both its kernel and cokernel vanish. Thus $\mathrm{Tor}_1(M,N)=0$ and $M \otimes N=0$. The supports $x=0$ and $x=1$ do not meet, in contrast to the self-intersection example. This calculation supplies an explicit input-dependent obstruction, not a universal nonzero derived field.

The categorical language now has an exact ring, modules, maps and homology certificate.

Scope

Nothing here identifies Tor_1 with a Higgs bundle, a particle mass or a determinant pushforward. Those interpretations would require a specified functor and a theorem preserving the relevant structures, not just matching notation.

Further development

For a declared intersection of algebraic subschemes, compute a Koszul resolution and determine when higher Tor measures nontransversality, then state precisely which geometric obstruction it represents.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R173.

Y25-10-012 A Picard-Fuchs basis comparison is determined by values and derivatives at one regular point

2025-10-12 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Do both proposed period constructions solve the same normalized ODE, and what initial data fix their constant change of basis?

The period comparison returned genuinely mismatched certified leaves. A constant scale or symplectic mixing is a hypothesis to prove, not a consequence of the word period. The recovered second-order equation gives an exact local uniqueness test and a useful series certificate.

Definitions and assumptions

Consider $(1-u^2)a''-a/4=0$ on a simply connected domain avoiding ± 1. A fundamental solution matrix records two solutions and their first derivatives.

For scalar solutions y_1, y_2, their Wronskian is $W=y_1y_2'-y_1'y_2$.

A Picard-Fuchs basis comparison is determined by values and derivatives at one regular point

The Wronskian is constant. Any two fundamental matrices for the same equation differ by a constant invertible matrix. At $u=0$ the normalized solutions have expansions $y_1=1+u^2/8+3u^4/128+\cdots$ and $y_2=u+u^3/24+5u^5/384+\cdots$.

Derivation or proof

Rewrite the equation as $a''=a/[4(1-u^2)]$. Differentiating W gives $y_1y_2''-y_1''y_2=0$. Both solution matrices satisfy the same first-order system $F'=AF$; hence$(F^{-1}G)'=-F^{-1}AF$ $F^{-1}G+F^{-1}AG=0$. The constant is determined by the value-and-derivative matrices at any regular point. For $a=\Sigma c_nu^n$, coefficient comparison yields $c_(n+2)=[n(n-1)+1/4]c_n/[(n+2)(n+1)]$. Substituting initial data(1,0) and(0,1) gives the displayed rational coefficients and $W(0)=1$.

Worked example

Compare a claimed second basis with $z_1=2y_1+y_2$ and $z_2=y_1+y_2$.

The change matrix is[[2,1],[1,1]], determinant one, so it preserves the two-dimensional standard alternating form. Its value-and-derivative data at zero are exactly that matrix. A mere scale/sign fit is a smaller class than all such changes; conversely six matching sampled values do not prove equality unless the ODE and initial data are already certified. A hash mismatch may reflect normalization or a genuinely different function; the differential test distinguishes them.

The comparison becomes a finite initial-data and differential-equation certificate with a declared normalization.

Scope

The equation alone does not identify a specific Seiberg-Witten differential, period cycle or global monodromy basis. In rank one, WDVV commutator conditions are scalar and cannot replace these missing differential constraints.

Further development

Certify each numerical solution’s residual and initial-data enclosure, then use a Grönwall estimate for the first-order system to bound their discrepancy across a regular comparison domain.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R174.

Y25-10-016 The E_8 roots, its Lie dimension and the totient384 are different exact counts

2025-10-14 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which finite set or vector space is being counted by each dimension in the framework?

The pre-geometric framework combined E_8, rank384 and φ(1680). These numbers can all be checked, but they count different mathematical objects. An explicit root construction gives a positive algebraic certificate without claiming that the resulting counts establish a spacetime model.

Definitions and assumptions

In R^8 take roots of two kinds: vectors with two entries±1 and six zeros, and vectors with all entries±1/2 with an even number of minus signs.

Euler’s totient φ(n) counts invertible residues modulo n. The signed-permutation group B_5 has order3840, separately.

The E_8 roots, its Lie dimension and the totient384 are different exact counts

The displayed E_8 root set has112+128=240 norm-squared-two roots spanning R^8. The associated complex simple Lie algebra has dimension8+240=248. Also φ(1680)=384, since$1680=2^4\cdot3\cdot5\cdot7$.

Derivation or proof

There are choose(8,2) ways to choose the nonzero coordinates of an integer root and four sign choices, yielding112. There are2^8 half-sign patterns, half with even parity, yielding128. Every vector has norm squared two. Integer roots e_i−e_j span the coordinate-sum-zero subspace, and e_1+e_2 is outside it, proving full rank eight. For a complex simple Lie algebra the root decomposition consists of an eight-dimensional Cartan subalgebra and one-dimensional spaces for each root, giving248; that structural root-decomposition fact is the classical input. The totient formula gives$1680(1-1/2)(1-1/3)(1-1/5)(1-1/7)=384$.

Worked example

Compare root count, Lie rank, Lie dimension and an independently chosen categorical rank.

They are240,8,248 and, if separately constructed, a possible number384 of simple objects. No equality forces a category with384 simples or global dimension1680. Reflection closure of the displayed root set is additionally checked exactly by the verifier: for norm-two α, s_α(β)=β−(α·β)α must remain in the set for every ordered pair. This verifies root-system structure beyond merely counting short vectors.

The numbers become concrete algebraic invariants with distinct definitions, and the root structure receives an exact finite certificate.

Scope

These calculations do not construct F-symbols, scattering amplitudes, an emergent metric or a physical coupling. Matching numerical counts across domains supplies no functor or equivalence by itself.

Further development

Specify a genuine category or representation construction from the root data and prove its rank, dimensions and fusion rules rather than using the totient as a substitute for that construction.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R175.

Y25-10-017 Semisimplicity kills ordinary extension classes

2025-10-14 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which cohomology theory measures the proposed categorical deformation, and does ordinary Ext vanish under the stated semisimplicity assumption?

The October audit correctly questioned using positive-degree Ext in a semisimple setting. The exact reason is elementary: every short exact sequence splits, and every object is projective. A deformation theory must therefore identify another complex or weaken semisimplicity.

Definitions and assumptions

An abelian category is semisimple here when every short exact sequence splits. In finite-dimensional vector spaces, every subspace has a complement.

$\mathrm{Ext}^1(C,A)$ classifies short exact sequences$0\to A\to B\to C\to 0$ up to equivalence; higher Ext can be computed from projective resolutions when available.

Semisimplicity kills ordinary extension classes

> In a semisimple abelian category with the usual Ext construction, $\mathrm{Ext}^n(C,A)=0$ for $n>0$. For finite-dimensional vector spaces, a split extension is explicitly $B\cong A\oplus C$.

Derivation or proof

Choose a section $s:C\to B$ of the surjection $p:B\to C$. Then$(a,c)\mapsto i(a)+s(c)$ is an isomorphism $A\oplus C\to B$: injectivity follows after applying p, and surjectivity follows by subtracting $s(p(b))$ from b. Hence every extension represents the zero class in Ext^1. More generally, given an epimorphism and a map from any object, a splitting of the epimorphism lifts that map; therefore every object is projective. A resolution concentrated in degree zero computes all higher Ext as zero. This is a theorem about this specific derived functor, not every cohomology assigned to a tensor category.

Worked example

Compare vector spaces with modules over the dual numbers $k[\varepsilon]/(\varepsilon^2)$.

In the latter category the sequence$0\to k\to k[\varepsilon]/(\varepsilon^2)\to k\to 0$, where the first map sends1 to ε, does not split as a module sequence. A hypothetical lift of1 from the quotient would be$1+c\varepsilon$, but multiplication by ε sends it to $\varepsilon\neq 0$, whereas ε acts as zero on k. Thus nontrivial extensions arise once the semisimple hypothesis is removed.

The framework must select an obstruction theory that survives its own assumptions, rather than assigning nonzero Ext classes where they are forced to vanish.

Scope

Davydov-Yetter, Hochschild and other deformation cohomologies have their own definitions; this argument does not identify them or claim they all vanish. A software field called cohomology does not fix which functor is intended.

Further development

Specify the tensor-deformation cochain complex and prove that its differential squares to zero, then compute a low-degree example before interpreting it as geometric curvature.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R176.

Y25-10-018 A four-object pointed model has fully checkable F, R and S data

2025-10-14 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

What actual coherence and nondegeneracy equations does the proposed modular data satisfy?

The speculative categorical framework lacked a genuine source for its F,R,S arrays. A small pointed braided model shows what a complete finite certificate looks like. Its fusion, associator, braiding and modular matrix can all be verified from bilinearity.

Definitions and assumptions

Let $A=F_2^2$ label four simple objects, with fusion $a\otimes b=a+b$, unit zero and all dimensions one. Use associator $F=1$.

Define $R(a,b)=(-1)^{(a_1b_2)}$, quadratic twist $q(a)=R(a,a)$, and $b(a,b)=R(a,b)R(b,a)$. Set $S_{(a,b)}=b(a,b)/2$.

A four-object pointed model has fully checkable F, R and S data

> The pentagon and hexagon equations hold; S is real orthogonal with $S^2=I$, and its Verlinde coefficients are $N_{(a,b)}^{c}=1$ when $c=a+b$ and zero otherwise. The global squared dimension is four.

Derivation or proof

A constant associator one satisfies every pentagon product identically. The two hexagon equations reduce to multiplicativity of R in each argument, which follows from bilinearity of a_1b_2 over F_2. The symmetric bicharacter is$(-1)^{(a_1b_2+a_2b_1)}$; its pairing is nondegenerate because any nonzero a has a b giving exponent one. Finite-character orthogonality gives$\Sigma_x b(a,x)b(c,x)=4\delta_{(a,c)}$, proving $S^2=I$. Since $S_{(0,x)}=1/2$, the Verlinde sum becomes$(1/4)\Sigma_x b(a+b+c,x)$, which is one precisely when $a+b+c=0$. Finally $q(a+b)=q(a)q(b)b(a,b)$, checking the twist compatibility.

Worked example

List the twist values and compare their data requirements with a proposed384-object model.

In order00,10,01,11 the twists are1,1,1,−1, and every S entry is±1/2. Four one-dimensional simples give$\Sigma d_a^2=4$. The verifier checks all64 fusion triples, all pairings and all finite hexagon instances exactly. A larger claimed rank does not determine any of these arrays; it increases the number of coherence equations that need a source or proof.

This finite model demonstrates constructive categorical mathematics instead of using numerical dimensions as proxies for coherence.

Scope

The example is a pointed modular category related to the toric-code data; it is not the proposed $E_8/384$ category, a semiconductor scattering model or a physical realization. Those require separately specified functors and dynamics.

Further development

Starting from a genuine larger fusion rule, solve or derive its coherence data and verify a modular nondegeneracy certificate before computing RT-type invariants.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R176.

Y25-10-019 A Pareto frontier contains points that no linear selector can choose

2025-10-14 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which objectives define dominance, and what information is lost when a two-objective frontier is replaced by $H-\lambda d$?

The dashboard selected high entropy and low maximal displacement, then inserted its selection into a coupling predictor. The optimization itself has a precise finite theory. Weighted scalarization can miss nondominated points, and an inserted normalization factor can cancel algebraically.

Definitions and assumptions

For pairs(d,H), smaller d and larger H are preferred. A point is dominated if another has no larger d and no smaller H, with at least one strict improvement.

For $\lambda \geq 0$ a linear selector maximizes $H-\lambda d$. The recorded predictor has form $A=(D^2/d^2)(248/r)(r/e\hat{}H)$, with $d>0, r\neq 0$.

A Pareto frontier contains points that no linear selector can choose

> Every strict optimizer of a positive weighted selector is nondominated, but not every nondominated point optimizes such a selector. The predictor simplifies to $A=248D^2d\hat{}(-2)e\hat{}(-H)$, independent of r.

Derivation or proof

If a selected point were dominated, the dominating point would improve or preserve each weighted objective and strictly improve at least one with positive weight, contradicting strict optimality. For failure of the converse, use(0,0),(1,2/5),(2,2). All three are nondominated. For the middle point to beat the first requires $\lambda<2/5$; to beat the last requires $\lambda>8/5$. No λ satisfies both. The predictor cancellation follows by multiplication, and logarithmic differentiation gives $d\log A=-2d\log d-dH$ when D is fixed. This separates actual sensitivity from a redundant displayed rank parameter.

Worked example

Use the three-point frontier to audit the dashboard rule.

At $\lambda=1$, the endpoint scores are zero and zero, while the middle score is$-3/5$. The middle remains a valid tradeoff despite never winning any nonnegative linear weight. If d is multiplied by two at fixed H, the simplified predictor falls by a factor four; increasing H bylog2 halves it. These exact sensitivities do not constitute a fitted physical law.

The selection rule and predictor now have explicit limitations and algebraic invariants that can be checked independently of presentation.

Scope

The stored H may not be Shannon entropy unless its definition says so. No value of a physical fine-structure constant or causal relation follows from choosing a point on this frontier.

Further development

Enumerate the whole nondominated finite set and use a declared constraint $d \le d_0$ to select among unsupported points; propagate uncertainty through the simplified predictor rather than its redundant factorization.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R025.

Y25-10-010 Heat residues follow from a Mellin transform and need not be integer indices

2025-10-16 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which operator, heat expansion and Mellin normalization define the proposed residue?

The Global Residue Index proposal combined spectral residues, complex residues and integer indices. A circle Laplacian gives a complete countercheck and a positive calculation: its heat leading term determines a zeta residue that varies continuously with the circle length.

Definitions and assumptions

On a circle of circumference L, remove the zero mode from $-d^2/dx^2$. The nonzero eigenvalues are$(2\pi n/L)^2, n \in Z$ minus$\{0\}$.

Let $K(t)=\Sigma_{(n\neq 0)} e^{(-t\lambda_n)}$ and $\zeta_A(s)=\Sigma_{(n\neq 0)}\lambda_n^{(-s)}$. For $Re(s)>1/2$, $\Gamma(s)\zeta_A(s)=\int_0^\infty t^{(s-1)}K(t)dt$.

Heat residues follow from a Mellin transform and need not be integer indices

$\zeta_A(s)=2(L/(2\pi))^{(2s)}\zeta_R(2s)$, so its residue at $s=1/2$ is $L/(2\pi)$. Equivalently $K(t)$ has leading term $L/(\sqrt{(4\pi t)})$, whose Mellin pole divided by $\Gamma(1/2)$ gives the same residue. It is not generally an integer.

Derivation or proof

The spectral series factors immediately into a scale and$2\Sigma n^{(-2s)}$. The Riemann zeta function has residue one at argument one, so $\zeta_R(2s)$ has residue$1/2$ at $s=1/2$; multiplication gives $L/(2\pi)$. For the Mellin identity, integrate each positive heat term using the substitution $u=t\lambda_n$, giving $\Gamma(s)\lambda_n^{(-s)}$; absolute convergence permits summation. Splitting the heat integral at one shows that a term $a\, t^{(-1/2)}$ contributes $a/(s-1/2)$ to $\Gamma(s)\zeta_A(s)$. With $a=L/(2\sqrt{\pi})$, division by$\sqrt{\pi}$ gives $L/(2\pi)$. This explains the normalization rather than equating different residues by terminology.

Worked example

Take circles of lengths2π andπ.

Their zeta residues are respectively one and one-half, while the topology is the same. Scaling the length by c multiplies the residue by c. An integer Fredholm index cannot track this continuous noninteger variation without a different definition or an additional normalization theorem. Complex contour residues likewise depend on the particular meromorphic function whose coefficient is being taken.

The spectral residue is fully computed and its relationship to a heat coefficient is proved.

Scope

The heat expansion and meromorphic continuation for general elliptic operators require analytic hypotheses. This example does not identify the proposed logarithmic, categorical and topological indices, nor prove GRIT.

Further development

Choose one elliptic family and state its supertrace, zero-mode subtraction and twisting class; compare the resulting integer index with separately normalized zeta residues before seeking an equality.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R177.

Y25-10-029 Fiber integration commutes with d on closed product fibers in a fixed convention

2025-10-19 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

What are the degrees and signs of the operators in the proposed pushforward-curvature identity?

The recovered identity$[d,\pi_+]\omega=\int\Omega_L\wedge\omega$ requires a type check before a geometric interpretation. With ordinary exterior derivative and a curvature two-form, its sides have different degrees. A closed-product-fiber calculation supplies the correct elementary comparison.

Definitions and assumptions

Let$\pi:B\times F\to B$ with F a closed oriented m-manifold. Write horizontal forms before vertical forms and define$\pi!$ by integrating the vertical top-degree coefficient.

Then$\pi!$ lowers form degree by m, d raises it by one, and wedge multiplication by a curvature two-form raises it by two.

Fiber integration commutes with d on closed product fibers in a fixed convention

In this convention$d_B\pi!\omega=\pi!d\omega$ for smooth product forms on closed fibers. The commutator has degree$1-m$, whereas$\pi!(\Omega_L\wedge\omega)$ has degree$2-m$; the proposed equality therefore cannot hold as an identity of ordinary graded forms without changing an operator or adding extra structure.

Derivation or proof

For a product $\alpha \wedge \beta$ with α horizontal and β vertical of top degree, d contributes $d_B\alpha \wedge \beta$ plus a vertical derivative that has no higher vertical degree; integrating gives$(d_B\alpha)\int\beta$, the derivative of$\pi!\omega$. Terms of vertical degree $m-1$ can contribute$\pi!d_F\beta$, but their integral is zero by Stokes because F has no boundary. All other vertical degrees integrate to zero. Linearity and a local coefficient expansion prove the statement. The degree comparison follows simply by adding the operator degree shifts, independently of any dynamics.

Worked example

Use $B=R, F=S^1$ and $\omega=(x^2+\sin\theta)d\theta$.

Integration gives$\pi!\omega=2\pi x^2$. Its derivative is$4\pi x\,dx$. Meanwhile $d\omega=2x\,dx\wedge d\theta$, whose fiber integral is also$4\pi x\,dx$. There is no extra curvature term. If the fiber had a boundary, Stokes would supply an explicitly signed boundary contribution instead, not an arbitrary line-bundle curvature insertion.

The pushforward idea becomes a well-typed theorem with a direct example and a clear location for genuine boundary corrections.

Scope

Other orientation conventions move signs, and covariant or superconnection pushforwards are different constructions. The calculation does not rule out correctly defined index-theoretic curvature formulas; it rules out copying this displayed degree-mismatched formula literally.

Further development

Specify the connection and graded complex for a twisted pushforward, then derive its commutator formula with every degree and boundary term shown.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R178.

Y25-10-030 The Bloch-Wigner correction has coefficient one, and a scalar residue leaves a level curve

2025-10-19 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which logarithm branches and normalization preserve the dilogarithm symmetry, and can one real residue identify a complex point?

The recovered formula used half the standard logarithmic correction in the Bloch-Wigner function. The normalized function has a useful exact functional relation. Adding it to $\log|z|$ gives a legitimate real observable, but generally only one constraint on a two-dimensional variable.

Definitions and assumptions

On compatible branches away from0 and1 use $D(z)=\mathrm{Im}\,\mathrm{Li}_2(z)+\log|z|\arg(1-z)$, extended as the standard single-valued Bloch-Wigner function.

Define $R(z)=\log|z|+D(z)$. At a real $0<x<1$, $\mathrm{Li}_2(x)$ is real and $D(x)=0$.

The Bloch-Wigner correction has coefficient one, and a scalar residue leaves a level curve

The normalized function satisfies $D(1-z)=-D(z)$. A coefficient $1/2$ in its correction fails this relation in general. Near any real $x\in(0,1)$, the equation $R(z)=\log x$ defines a local real one-dimensional level curve, rather than uniquely determining z.

Derivation or proof

Spence's identity $\mathrm{Li}_2(z)+\mathrm{Li}_2(1-z)=\pi^2/6-\log z\log(1-z)$ has imaginary part equal to $-[\log|z|\arg(1-z)+\log|1-z|\arg z]$. Adding the two full correction terms cancels it, proving antisymmetry on compatible branches and then by continuation. Half corrections leave half the uncanceled expression. Along the real interval $R(x)=\log x$, so its derivative in the real direction is $1/x\neq 0$. The real implicit-function theorem therefore gives a smooth one-dimensional local level set of R in the complex plane.

Worked example

Take $z=i$ and compare it with $1-i$.

The series $\mathrm{Im}\,\mathrm{Li}_2(i)=\Sigma_{n\geq 1}\sin(n\pi/2)/n^2$ equals Catalan's constant G, and $\log|i|=0$, so $D(i)=G$. Antisymmetry gives $D(1-i)=-G$. For the half-normalized formula, the sum of the two values instead equals $-\pi\log 2/8$, since $\log|1-i|=(\log 2)/2$ and $\arg(i)=\pi/2$. This exact nonzero defect exposes the normalization error. Thus a small residual value is not equivalent to a unique recovered point in the parameter plane.

The regulator-sized function has a correct identity, while its observation map has a precise local nonidentifiability statement.

Scope

The result does not construct a motivic regulator, a hyperbolic 3-manifold or the proposed physical curvature. Global branch behavior and integral lattice normalizations require their own definitions.

Further development

Add an independently defined second real observable and compute the 2×2 Jacobian; a nonzero determinant would give local identifiability of z, whereas one scalar residue cannot.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R179.

Y25-10-020 Character orthogonality gives an exact frame bound before a large-sieve claim

2025-10-21 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which character family and sampling operator determine the best quadratic bound?

The proposed q-large-sieve optimizer asserted an arithmetic inequality and automatic stability. A finite character transform supplies a rigorous foundation: its quadratic bound is an operator-norm

statement. Orthogonality proves an exact special case and reveals the missing dependence on the chosen family.

Definitions and assumptions

For the cyclic group C_m use characters χ_k(r)=exp(2πikr/m), k,r=0,...,m−1.

For complex coefficients a_r, define S_k=Σ_r a_rχ_k(r). More generally a family has analysis matrix A with entries character values and bound ||Aa||²≤C||a||².

Character orthogonality gives an exact frame bound before a large-sieve claim

Σ_k|S_k|²=mΣ_r|a_r|². The optimal general constant is the largest eigenvalue of A*A. Repeating a character changes the frame bound, so the number and multiplicity of characters cannot be omitted.

Derivation or proof

Expand the squared modulus: Σ_kΣ_(r,s)a_r conjugate(a_s)exp(2πik(r−s)/m). The inner geometric sum is m when r=s and zero otherwise, proving Parseval. For a general matrix, the Rayleigh quotient a*A*Aa/(a*a) is maximized at its largest eigenvalue by the Hermitian spectral theorem. If a character row is repeated, its rank-one contribution is added again to A*A and can increase that eigenvalue. The same finite algebra governs multiplicative characters after identifying a finite cyclic unit group; primitive-character restrictions or sums over multiple moduli change the family and require another bound.

Worked example

Take m=4 and coefficients(1,2,0,−1).

Their squared norm is six, so the transformed squared norm is exactly twenty-four. At m=4 the transform uses only1,−1,i,−i and can be checked with exact integer pairs. A gradient vector in the same space satisfies this identity, but that identity alone specifies no learning rate, objective smoothness or decrease theorem.

The arithmetic transform has an exact finite certificate, and the claimed inequality becomes a concrete spectral question.

Scope

This does not prove the recovered multi-modulus weighted large-sieve inequality, its q^(ω(n)) factors or any datacenter speedup. A frame inequality does not automatically establish optimization stability.

Further development

Specify the precise character matrix and weights, compute or bound its Gram spectrum, then combine that bound with a separate smooth-objective convergence theorem.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R180.

Y25-10-021 Projecting the updated state has a descent proof; clipping the gradient is a different operation

2025-10-21 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which projection is used, and which smoothness and feasibility hypotheses imply the claimed stability?

The proposed update projected a gradient into a budget set. A standard projected-gradient theorem applies instead to projection of the proposed new state. Distinguishing the two operations yields both a rigorous descent certificate and a one-dimensional counterexample to automatic constraint preservation.

Definitions and assumptions

Let $C \subset \mathbb{R}^n$ be nonempty closed and convex, $x \in C$, and f have L-Lipschitz gradient. Set $x^+ = \Pi_C(x - \eta \nabla f(x))$.

Euclidean projection is characterized by $\langle z - \Pi_C z, y - \Pi_C z \rangle \le 0$ for all $y \in C$.

Projecting the updated state has a descent proof; clipping the gradient is a different operation

$f(x^+) \le f(x) - (1/\eta - L/2)\|x^+ - x\|^2$. Hence $0 < \eta < 2/L$ gives descent unless the step is zero. The alternative $x - \eta \Pi_C(\nabla f(x))$ need not stay in C.

Derivation or proof

Use the projection characterization with $z = x - \eta g$, $g = \nabla f(x)$, and $y = x$. Writing $s = x^+ - x$ gives $\langle g, s \rangle \le -\|s\|^2/\eta$. The smoothness inequality $f(x+s) \le f(x) + \langle g, s \rangle + (L/2)\|s\|^2$ then proves the descent bound. The projection characterization itself follows by differentiating the squared distance along each segment from the minimizer to y. Convexity of C makes those segments feasible and strict convexity of squared distance gives uniqueness of the projection.

Worked example

Compare the two updates on $C = [-1, 1]$, $f(x) = (x-2)^2/2$ and $x = 1$.

The gradient is -1. Projecting the new state gives $\Pi_C(1 + \eta) = 1$, the constrained minimizer. Projecting the gradient into C leaves -1, so the alternative gives $1 + \eta$, outside the feasible set for every $\eta > 0$. For an interior example $x = 0$ and $\eta = 1/2$, the correct update gives 1 and decreases f from two to one-half, consistent with the theorem $L = 1$. This proof uses feasibility of the old state when testing the projection inequality. Omitting that premise removes the stated descent conclusion.

A complete stability statement now rests on a specific update and explicit constants, rather than on the arithmetic name of the constraint.

Scope

Nonconvex feasible sets can have multivalued projections, and a changing or stochastic objective needs additional analysis. No reported training acceleration or memory reduction is verified by this finite theorem.

Further development

For a specified ellipsoidal budget, derive the exact projection through its scalar Lagrange-multiplier equation and certify that the numerical solver’s error preserves the descent margin.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R025.

Y25-10-027 Trace class is contained in Hilbert-Schmidt, and strong convergence alone preserves neither

2025-10-21 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which operator topology is certified, and are the claimed trace-class and Hilbert-Schmidt properties mutually consistent?

The operator report asserted a compact trace-class but non-Hilbert-Schmidt limit. Under the standard definitions this is impossible. The useful development is an exact hierarchy of operator ideals and a set of examples distinguishing singular-value summability from strong convergence.

Definitions and assumptions

For a compact operator A on a Hilbert space, let $s_n(A) \geq 0$ be its singular values. Trace class means $\Sigma s_n < \infty$; Hilbert-Schmidt means $\Sigma s_n^2 < \infty$.

Strong convergence $A_N \to A$ means $A_N x \to Ax$ for every fixed vector x, without requiring convergence in operator norm.

Trace class is contained in Hilbert-Schmidt, and strong convergence alone preserves neither

> Every trace-class operator is Hilbert-Schmidt, with $\|A\|_{HS} \leq \|A\|_1$. Compactness does not imply either summability condition. A strong limit of finite-rank operators need not even be compact.

Derivation or proof

For any finite nonnegative sequence, $\Sigma s_n^2 \leq (\Sigma s_n)^2$; monotone limits give the norm inequality when the trace norm is finite. For diagonal operators on ℓ^2, compactness is equivalent to diagonal entries tending to zero, and the two ideal conditions reduce to the displayed scalar series. Thus $\mathrm{diag}(1/n)$ is Hilbert-Schmidt but not trace class, while $\mathrm{diag}(1/\sqrt{n})$ is compact but not Hilbert-Schmidt. Coordinate projections P_N have finite rank and converge strongly to identity because the ℓ^2 tail of every fixed vector tends to zero. Identity is not compact on an infinite-dimensional space, since its image of the orthonormal unit vectors has no norm-convergent subsequence.

Worked example

Certify the genuinely trace-class diagonal A=diag(1/n²).

Its finite-rank truncation has trace-norm errorΣ_(n>N)n⁻²≤1/N and Hilbert-Schmidt squared errorΣ_(n>N)n⁻⁴≤1/(3N³), by integral comparison. These are actual convergence bounds in specified norms. A persistent empirical ratio near0.8 is neither one of these bounds nor a proof of strong convergence.

The hierarchy rejects an inconsistent claim while supplying concrete norm certificates a spectral approximation could genuinely satisfy.

Scope

Singular values, not signed eigenvalue sums, define these ideals for general operators. A fitted exponent on finitely many residuals proves no asymptotic summability theorem.

Further development

Derive a target-independent singular-value envelope for the proposed prime kernel, then integrate its tail to certify the strongest operator ideal and topology the construction actually supports.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R181.

Y25-10-036 Product-variable reduction gives exact zeta integrals and a controlled fractional-part transform

2025-10-22 · historical evidence T · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which exact integrand is known, and what general reduction can be proved without inventing the indexed paper’s missing input?

The sourcebook identifies an October22 four-dimensional fractional-part integral but does not return its full formula or a conversation transcript. The following newly specified examples develop its product-variable theme and retain that attribution limit explicitly.

Definitions and assumptions

For an integrable function F of a product of d independent unit-interval variables, set t=x₁⋯x_d.

For s>0 define A(s)=∫₀¹{1/t}t^(s−1)dt, with fractional part{u}=u−flooru. The bounded integrand makes this transform well defined.

Product-variable reduction gives exact zeta integrals and a controlled fractional-part transform

∫_[0,1]^dF(x₁⋯x_d)dx=(1/(d−1)!)∫₀¹F(t)(−logt)^(d−1)dt. In particular∫_[0,1]^4[1−x₁x₂x₃x₄]⁻¹dx=ζ(4)=π⁴/90. For s>1, A(s)=1/(s−1)−ζ(s)/s.

Derivation or proof

Substitute x_j=e^(−y_j), so the product is e^(−u) with u=Σy_j and the measure contributes e^(−u). The volume density of the nonnegative simplex slice is u^(d−1)/(d−1)!, proved by iterated integration or induction. Substituting t=e^(−u) gives the reduction. For the zeta example, expand1/(1−t)=Σ_(n≥0)tⁿ and use nonnegative termwise integration; each product monomial integrates to1/(n+1)^d. For A, use floor(1/t)=Σ_(n≥1)1_(t≤1/n); its weighted integral isζ(s)/s, while1/t contributes1/(s−1).

Worked example

Apply the reduction to the newly defined four-dimensional fractional-part example.

Its value is(−1/6)A‴(1), where the apparent pole in the displayed zeta expression cancels at s=1. Differentiation under the bounded fractional-part integral is justified by integrability of|logt|³ near zero. This gives an exact derivative representation without claiming it was the original indexed paper’s equation. The simpler ζ(4) example supplies an independently checkable closed form. For d=4 the zeta integrand is integrable even at the all-ones corner, as the convergent nonnegative series proves directly.

The product-variable method is a real theorem, and the missing source expression stays missing rather than being silently reconstructed.

Scope

The sourcebook date is an indexed publication/upload attribution, not a recovered original chat date. The log-cotangent and ArcCoth paper formulas likewise were not returned and are not claimed evaluated here.

Further development

Recover a precise fractional-part integrand, apply this reduction, and justify any zeta-derivative continuation with dominated convergence or an explicit cancellation at the relevant point.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R182.

Y25-10-022 A finite prime oscillatory sum has exact calculus but no automatic zero correspondence

2025-10-23 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which analytic identities follow from the actual finite sum, and what additional theorem would connect it to zeta zeros?

The Ribbon-Lakeland draft supplied an explicit prime sum and derivative but also interpreted a nonvanishing defect as flatness. The finite formula can be analyzed exactly. That analysis should precede any operator or RH interpretation.

Definitions and assumptions

For $X \geq 2$ set $\lambda_X(t)=\Sigma_{(p\leq X)}(\log p)/(1+p)[2\cos(t\log p)-2]+t/2$.

The sum is finite, so it is a real entire function of real t extended holomorphically to complex t. It is not by definition an operator eigenvalue or a regularized determinant.

A finite prime oscillatory sum has exact calculus but no automatic zero correspondence

$\lambda_X(0)=0$, $\lambda_X'(0)=1/2$ and $\lambda_X''(0)=-2\Sigma_{(p\leq X)}(\log p)^3/(1+p)<0$. Its derivative is $1/2-2\Sigma_{(p\leq X)}(\log p)^2\sin(t\log p)/(1+p)$. These identities do not force zeros at any prescribed ordinates.

Derivation or proof

Differentiate each finite summand using the chain rule. The first derivative introduces a second logarithmic factor, and the second introduces a third with cosine. At zero the sine terms vanish and every cosine equals one, giving the stated values. Taylor expansion then gives $\lambda_X(t)=t/2-Ct^2+O(t^4)$, where $C=\Sigma(\log p)^3/(1+p)>0$. The nonzero derivative at zero makes that particular zero simple. A point selected independently need not make the prime sum or its proposed flatness defect vanish.

Worked example

Keep only $p=2$, giving a fully explicit one-prime model.

$\lambda_2(t)=(2\log 2/3)(\cos(t\log 2)-1)+t/2$. At $t=2\pi/\log 2$ the cosine returns to one, so $\lambda_2(t)=\pi/\log 2$, not zero. Its periodic prime part and linear drift therefore do not jointly vanish merely because the phase returns. Increasing X changes the function; it is not a proved convergence approximation until a tail theorem is supplied.

The exact calculus is a valid mathematical result, while the arithmetic correspondence remains a separate burden.

Scope

A self-adjoint operator already has real point spectrum, so that property alone cannot be equivalent to RH without a proved correspondence encoding all nontrivial zeros. Finite PASS bounds or target-seeded matrices do not supply that theorem. A flatness defect near5.7 is not zero.

Further development

Define one operator with domain and a target-independent construction, prove its spectral counting law, and only then seek an analytic identity linking its determinant or trace to an arithmetic function.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R060.

Y25-10-033 Prime-log tones have exact beat differences but no common nonzero period

2025-10-23 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which frequencies and decay law define the waveform, and what can its beats prove mathematically?

The prime audio construction is a well-defined finite signal. Its useful mathematics concerns irrational frequency ratios, difference frequencies and an exponential envelope. None of these properties alone gives a zeta-zero or spectral-operator theorem.

Definitions and assumptions

Take the undamped two-tone signal $y(t)=\cos(2\pi f_0 \log 2 \cdot t)+\cos(2\pi f_0 \log 3 \cdot t)$, with $f_0>0$.

The proposed damped version is $x(t)=e^{(-ct)}y(t)$, $c>0$. A common period must reproduce both phases for all t.

Prime-log tones have exact beat differences but no common nonzero period

log2/log3 is irrational, so the two-tone function has no positive common period. Its product formula has difference frequency $f_0 \log(3/2)$, while the damped envelope has half-life(log2)/c.

Derivation or proof

A rational equality $\log 2/\log 3=m/n$ would imply$2^n=3^m$ after exponentiation, impossible by unique prime factorization for nonzero positive integers m,n. A period of a finite sum of distinct exponential frequencies must make each phase factor one: linear independence follows by differentiating enough times and solving the Vandermonde system. Thus the two frequencies would have rational ratio if they shared a period. The cosine addition formula rewrites y as$2\cos[\pi f_0 \log 6 \cdot t]\cos[\pi f_0 \log(3/2)\cdot t]$, displaying the difference modulation. Solving $e^{(-ct)}=1/2$ gives the half-life.

Worked example

Use c=1/100 as in the recovered waveform.

Its envelope half-life is$100\log 2$. Increasing prime depth adds more logarithmic frequencies, but does not change this imposed envelope. A claimed beat period5.7 needs an actual f_0 and a definition of whether signed modulation or absolute amplitude is measured; the exact difference-frequency formula supplies that conversion. Rescaling f_0 rescales every temporal beat, so an isolated beat number is not an arithmetic invariant.

The signal is mathematically interpretable and its parameters have separate, verifiable roles.

Scope

Finite audio spectra are not eigenvalue certificates for an infinite operator. A fitted log-log exponent or curvature amplitude on one plot does not prove an asymptotic theorem.

Further development

Specify an observation window and sampling rate, prove an aliasing bound for its highest retained prime frequency, and estimate which difference frequencies can be resolved within the window.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R080.

Y25-10-034 Second and fourth mean values expose multiplicative resonances exactly

2025-10-27 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which finite mean-value identity distinguishes diagonal energy from genuine fourth-order resonance?

The curvature-cancellation passage named Dirichlet-polynomial large values and fourth-order chaos without returning its full formulas. A newly specified finite polynomial supplies exact second and fourth moment identities, making cancellation and multiplicative coincidences precise.

Definitions and assumptions

Let D(t)=Σ_(n=1)^N a_n n^(−it), with fixed complex coefficients. Define the long average M_T(F)=(1/(2T))∫_(−T)^T F(t)dt.

For finite sums, the mean of exp(itω) tends to zero for nonzeroω and equals one forω=0.

Second and fourth mean values expose multiplicative resonances exactly

lim_(T→∞)M_T(|D|²)=Σ|a_n|². The fourth mean isΣ_(mn=rs)a_ma_n conjugate(a_ra_s), summed over indices at most N. Multiplicative coincidences, not merely pairwise orthogonality, govern this fourth moment.

Derivation or proof

Integrating one exponential gives sin(Tω)/(Tω) whenω≠0, proving the mean rule. Expanding the squared modulus produces frequencies log(m/n), whose zero condition is m=n. Expanding the fourth power produces frequencies log(rs/(mn)), whose zero condition is mn=rs. Every sum is finite, so termwise integration and passage to the limit need no uniform infinite-series argument. Nonzero-frequency errors are explicitly bounded by1/(T|ω|), though nearly equal products can make finite-window convergence slow.

Worked example

Take D(t)=1+2^(−it)+4^(−it).

Writing z=2^(−it), its square is1+2z+3z²+2z³+z⁴. Distinct powers have nonzero frequency differences, so the fourth mean is1²+2²+3²+2²+1²=19. The second mean is three. The central coefficient three records the resonance1·4=2·2=4·1. This is a concrete fourth-order interaction that a diagonal-only approximation would miss. More generally, collect the coefficients of D squared by the product mn. The fourth mean is the sum of squared moduli of those grouped coefficients, so collisions add coherently before the final square.

The source’s cancellation theme has a rigorous finite model with an exact resonance certificate.

Scope

This does not establish a new large-values bound, zero-density estimate or generic quintic solution. Such claims require the unrecovered inequalities, parameter ranges and analytic estimates.

Further development

For a specified coefficient family, bound its multiplicative energy and the nonzero product-gap errors uniformly as N grows, then compare the resulting theorem with any proposed q-sieve improvement.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R183.

Y25-10-035 The asset recurrence has an exact stability threshold independent of its interpretation

2025-10-27 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which parameter combinations determine a stable fixed point, and what additional closure is required for the proposed feedback law?

The recovered EMARDA equations can be treated purely as affine dynamics. Their recurrence and capture-gap inequality have exact solutions. No economic or applied advice is needed to state the stability conditions, and an unspecified feedback observable cannot close the system.

Definitions and assumptions

Write $A_{t+1}=qA_t+c$, where $q=1+\rho-\delta$ and $c=\alpha W[1+\pi(M-1)]$ are fixed real constants.

The recovered gap is $\gamma=(1-\lambda)M-\lambda$. For a feedback variable α, a law $\alpha'=-\kappa(G-G^*)$ is closed only after G is specified as a function of the state.

The asset recurrence has an exact stability threshold independent of its interpretation

For $q\neq1$, $A_t=q^tA_0+c(1-q^t)/(1-q)$. Its fixed point $c/(1-q)$ is globally attracting exactly when $|q|<1$. For $M>0$, $\gamma>0$ exactly when $\lambda<M/(M+1)$.

Derivation or proof

Subtract the fixed point to obtain $A_{t+1}-A^*=q(A_t-A^*)$, and iterate. This gives the formula and convergence when $|q|<1$. For $|q|>1$ a nonzero initial displacement grows; $q=1$ gives $A_t=A_0+tc$, while $q=-1$ gives an alternating displacement and generally no convergence. For the gap, $\gamma=M-\lambda(M+1)$; division by positive $M+1$ proves its threshold. If one independently sets $G=g\alpha+h$ with $g>0$ and $\kappa>0$, then $\alpha-\alpha^*$ decays as $\exp(-\kappa gt)$, but that conclusion is unavailable without the specified dependence.

Worked example

Choose $q=3/4, c=2, A_0=0$.

The fixed point is eight and $A_t=8[1-(3/4)^t]$. The first values are $0, 2, 7/2, 37/8$, each verified by the recurrence. With $M=3$, the gap is $3-4\lambda$ and changes sign at $\lambda=3/4$. These are exact parameter statements rather than empirical estimates of the named quantities. At the boundary $q=-1$, deviations alternate forever unless the initial state already equals the fixed point.

The recurrence, threshold and feedback closure have distinct mathematical roles and can each be checked independently.

Scope

Parameters are treated as constants in the recurrence theorem; time-varying coefficients require products and new bounds. No prediction about wealth, policy or real systems follows from the algebra alone.

Further development

For bounded time-dependent q_t,c_t, derive a uniform contraction and input-to-state bound, then close a specified feedback law and analyze the resulting coupled Jacobian.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R184.

Y25-10-028 Cahn-Hilliard flow conserves mass and dissipates a declared energy

2025-10-29 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which free energy and boundary conditions make the proposed fourth-order equation a gradient flow?

The recovered dissipative equation has a genuine variational structure when its mobility and boundary terms are specified. The resulting mass and energy identities are strong mathematical content, but do not imply a universal controller or transfer the model to unrelated applications.

Definitions and assumptions

On a periodic domain set $E[\psi]=\int[f(\psi)+(K/2)|\nabla\psi|^2]dx$, with $K\geq 0$ and smooth f. Its variational derivative is $\mu=f'(\psi)-K\Delta\psi$.

Let $\psi_t=\nabla\cdot(M\nabla\mu)$ with smooth nonnegative mobility M; assume enough regularity to differentiate E and integrate by parts.

Cahn-Hilliard flow conserves mass and dissipates a declared energy

The total mass $\int\psi$ is constant and $dE/dt=-\int M|\nabla\mu|^2\leq 0$. For constant M and $f(\psi)=a\psi^2/2$, a Fourier mode k evolves at rate $-M|k|^2(a+K|k|^2)$.

Derivation or proof

Integrating the divergence over a periodic domain gives zero, proving mass conservation. Differentiating the energy and integrating its gradient term by parts gives $dE/dt=\int\mu\psi_t$. Substituting the PDE and integrating once more gives the negative mobility-weighted gradient norm. Derivatives of M create no extra term in this divergence-form integration identity. For the quadratic potential and Fourier mode $e^{(ik\cdot x)}$, Δ contributes $-|k|^2$, so $\mu=(a+K|k|^2)\psi_{mode}$ and its evolution has the stated rate. When $a<0$, low-frequency modes can grow despite global energy dissipation.

Worked example

On a 2π-periodic line take $a=K=M=1$ and $\psi(x,t)=m+\varepsilon e^{(-2t)}\cos x$.

Its mean is m and μ=m+2εe^(−2t)cosx. The PDE gives ψ_t=−2εe^(−2t)cosx exactly. Its energy isπm²+πε²e^(−4t), whose derivative is−4πε²e^(−4t). The dissipation integral∫|μ_x|² has exactly that positive value. This checks both conservation and decay without discretization. The constant Fourier mode has zero decay rate, exactly matching the mass conservation law.

The gradient-flow claim now has a complete energy certificate and an explicit solution.

Scope

Boundary fluxes, negative mobility or insufficient regularity change the proof. The identity does not make holonomy equal to thermodynamic loss, and a feedback law requires a separately defined state and stability analysis.

Further development

Discretize with a symmetric graph Laplacian and a discrete variational derivative, then prove exact mass conservation and a time-step energy inequality before comparing implementations.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R185.

Y25-10-031 Finite-group symmetry permits frequency ratios but does not usually force one

2025-10-30 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which representation and equivariant operator make30/13 a necessity rather than a chosen pair of coefficients?

The October acoustic-Casimir program attached30/13 to A_5 symmetry. A genuine symmetry constraint can be proved, but it normally leaves independent scalar parameters on distinct blocks. An explicit equivariant oscillator family shows why the ratio needs an additional operator construction.

Definitions and assumptions

Letρ be the orthogonal permutation representation of A_5 on R^5. On V⊕V use the same group action in both blocks.

For positive a,b let K=diag(a^2I_5,b^2I_5). The conservative oscillator is x″+Kx=0, with frequencies a and b.

Finite-group symmetry permits frequency ratios but does not usually force one

K commutes with everyρ(g)⊕ρ(g) for every positive pair a,b. Hence A_5 equivariance alone allows every positive frequency ratio b/a, including30/13. It does not select that ratio.

Derivation or proof

Each block of K is a scalar matrix, which commutes with every linear map on that block. Direct block multiplication therefore proves equivariance without needing a character table. The oscillator separates into five copies of $x''+a^2x=0$ and five of $y''+b^2y=0$, so their frequencies are exactly a,b. Since the same representation and symmetry hold as a,b vary continuously, no single ratio is forced by this symmetry condition. Schur’s lemma can make an operator scalar on an irreducible block, but it does not fix its scalar without specifying the actual central element or dynamical law.

Worked example

Choose a=13,b=30 and then change only b to31.

Both stiffness matrices have integer entries and exactly the same A_5 action. Their frequency ratios are respectively30/13 and31/13. An experimental ratio matching the former cannot distinguish a symmetry necessity from a fitted parameter unless the stiffness operator is independently derived. If a prescribed group-algebra element supplies the operator, its character values and normalization can then be calculated as extra data.

The next mathematical task is an operator derivation, rather than a numerical coincidence between a ratio and a group label.

Scope

This example does not model a particular resonator or accept the reported Q, loss or decay measurements. Perfectoid descent and Stark regulators provide no automatic map to this real stiffness matrix.

Further development

Specify an A_5-equivariant geometric differential operator and boundary conditions, decompose its representation and prove an exact relation between the relevant eigenvalues; that would make a ratio theorem substantive.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R014.

Y25-10-032 Damping ratio, decay and Q must use the same oscillator convention

2025-10-30 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which ODE defines the reported damping quantities, and do their numerical values satisfy its exact relations?

The acoustic discussion reported damping$\zeta\approx0.697$ alongside an enormous Q. A conditional oscillator calculation checks consistency without treating any historical physical measurement as established. Different experimental definitions require an explicit conversion rather than a shared label.

Definitions and assumptions

Use $x''+2\zeta\omega_0 x'+\omega_0^2 x=0$, with$\omega_0>0$ and$0<\zeta<1$. Define the model quality parameter $Q=1/(2\zeta)$.

The damped angular frequency is$\omega_d=\omega_0\sqrt{(1-\zeta^2)}$, and a solution's amplitude envelope decays as $\exp(-\zeta\omega_0 t)$.

Damping ratio, decay and Q must use the same oscillator convention

The amplitude multiplier after one damped period is $\exp[-2\pi\zeta/\sqrt{(1-\zeta^2)}]$. For $\zeta=697/1000$, the model Q is$500/697$, not1.43×10^9. These conclusions are exact consequences of the declared ODE.

Derivation or proof

The characteristic equation has roots$-\zeta\omega_0\pm i\omega_0\sqrt{(1-\zeta^2)}$, giving the exponential envelope and oscillatory factor. One damped period is$2\pi/\omega_d$; substituting it in the envelope gives the multiplier. The Q value follows from the stated conventional parameter definition. For the mechanical energy $E=(x'^2+\omega_0^2x^2)/2$, differentiating and using the ODE gives $E'=-2\zeta\omega_0x'^2\le0$, which is the actual dissipative identity. The envelope and energy are related but not identical pointwise functions.

Worked example

Evaluate the dimensionless relations at the reported damping scale.

Here$1-\zeta^2=514191/10^6$, so the period multiplier is $\exp(-1394\pi/\sqrt{514191})$, approximately$0.0022$. A value$0.738$ could describe a different time interval or observable, but it is not this full-period amplitude multiplier. Conversely a model Q near10^9 requiresζ near5×10^{-10} under the same definition. The discrepancy cannot be resolved by changingω_0, which cancels from both relations. The dimensionless period multiplier is independent of the chosen time unit and natural-frequency scale.

A small exact ODE audit identifies which reported quantities could belong to one common mathematical model.

Scope

No claim is made about an actual device's measured Q, bandwidth or physical damping mechanism. At stronger damping, alternative experimental Q definitions need care; the definition used here is explicit.

Further development

Define the measured transfer function and derive its bandwidth and ringdown parameters from one model, propagating uncertainty before comparing them with arithmetic constants.

Source receipt: months/10/root_coverage.json record 0. Mathematical references: R186.

Y25-10-023 The sine quotient has an exact parity integral and a finite quadrature certificate

2025-10-31 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Why does the integral distinguish odd and even positive integers, and when is a discrete midpoint rule exact?

The parity integral is one of the clearest exact mathematics problems recovered in October. Its apparent endpoint singularities are removable for integer a, and a finite trigonometric expansion proves both the integral and a useful sampling theorem.

Definitions and assumptions

For a positive integer a define I(a)=∫₀^(aπ)sinx/sin(x/a)dx, interpreted by continuous endpoint limits.

Substitute x=aθ to get I(a)=a∫₀πU_(a−1)(cosθ)dθ, where U_(a−1)(cosθ)=sin(aθ)/sinθ.

The sine quotient has an exact parity integral and a finite quadrature certificate

> I(a)=aπ for odd a and zero for even a. For odd a=2m+1, the N-point midpoint rule in θ is exact whenever m<N. For even a its symmetric midpoint sum is zero for every N.

Derivation or proof

For odd a=2m+1, the quotient equals1+2Σ_(k=1)^mcos(2kθ); for even a=2m it equals2Σ_(k=1)^mcos((2k−1)θ). Both follow from a finite geometric sum of complex exponentials. Integrating each nonconstant integer cosine over[0,π] gives zero, leavingπ only in the odd case. The expansion also removes the endpoint singularities. At midpoint nodes θ_j=(j+1/2)π/N, the sum of cos(2kθ_j) vanishes when1≤k<N, by a geometric sum. Every odd-frequency cosine sum vanishes by pairing θ_j withπ−θ_j, proving the even case.

Worked example

Take a=5 and N=3.

The quotient is1+2cos2θ+2cos4θ. Both cosine sums vanish at the three midpoint nodes, so the midpoint integral equalsπ and I(5)=5π exactly. If each sampled quotient value has absolute error at mostδ, the propagated integration error is at most aπδ, because the positive quadrature weights sum to aπ. This is a mathematical error bound; hardware specifications alone do not imply a particular δ.

The exact parity identity and its discrete certificate follow from the same finite Fourier structure.

Scope

The theorem concerns positive integer a. Noninteger values may introduce interior poles and require a different integration prescription. No energy conversion, superconductivity or self-calibration claim follows from the integral identity.

Further development

Derive an end-to-end interval certificate combining removable-endpoint evaluation, sample errors and quantized weights, retaining the exact Fourier cutoff that makes the ideal midpoint rule exact.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R160.

Y25-10-024 The logarithmic sine integral separates into an elementary term and Catalan’s constant

2025-10-31 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

What is the exact value of the recovered logarithmic integral, including its integrable endpoint singularity?

The exact input is $\Psi=\int_0^{\pi/4}\log[(1+x^2)\sin^{10}x]dx$. Separating its logarithms produces a complete evaluation with a familiar special constant, and prevents later large integer factors from obscuring the original expression.

Definitions and assumptions

Let $a=\pi/4$ and $G=\Sigma_{n\geq0}(-1)^n/(2n+1)^2$, Catalan’s constant.

On$0<x\leq a$, $\log[(1+x^2)\sin^{10}x]=\log(1+x^2)+10\log\sin x$. Near zero the singularity is$10\log x$ plus a bounded function, hence integrable.

The logarithmic sine integral separates into an elementary term and Catalan’s constant

$$\Psi=a\log(1+a^2)-2a+2\arctan a-10a\log2-5G, \text{ with } a=\pi/4.$$

Derivation or proof

An elementary primitive is$\int\log(1+x^2)dx=x\log(1+x^2)-2x+2\arctan x$, verified by differentiation. For the sine term use the Abel-regularized Fourier expansion $\log(2\sin x)=-\Sigma_{n\geq1}\cos(2nx)/n$ on$(0,\pi)$. Integrating its convergent radial regularizations from0 to a and taking the limit gives$\int_0^a\log\sin x\,dx=-a\log2-(1/2)\Sigma_{n\geq1}\sin(2na)/n^2$. At $a=\pi/4$, the sine values cycle1,0,−1,0, so this sum is G. Multiplying by ten and adding the elementary part gives the formula. Absolute convergence of the integrated$1/n^2$ series makes its final evaluation unambiguous.

Worked example

Give a rigorous series error bar without numerical integration through the logarithmic endpoint.

Truncating G after N odd-denominator terms has alternating-series error at most$1/(2N+1)^2$. Its contribution to Ψ is at most$5/(2N+1)^2$. Bounds forπ, logarithms and arctangent can then be propagated separately. The exact expression has no intrinsic16384 factor; any such factor in a rearranged later formula must come from an explicitly shown algebraic normalization. To justify the Abel limit near zero, one can split the interval at a small positiveε. Uniform convergence applies on the remaining compact interval, while the integral of$|\log x|$ over$(0,\varepsilon)$ tends to zero. The integrated Fourier series is absolutely summable, so its final Catalan specialization is independent of the approach.

The problem receives a closed form, an endpoint-convergence argument and a direct numerical certification route.

Scope

The derivation concerns the recovered Ψ alone. It does not reconstruct the separate unrecovered golden-ratio integral or an unspecified rational antiderivative from the broad assistant list.

Further development

Extend the same calculation to$\int_0^a \log[(1+x^2)\sin^{(2m)}x]dx$ for rational multiples a/π, expressing the trigonometric part through explicitly normalized Clausen or Dirichlet L-values.

Source receipt: months/10/root_specifics.json record 0. Mathematical references: R065.

Y25-10-025 The recovered fourth-power logarithmic integral has an exact elementary evaluation

2025-10-31 · historical evidence U · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Can the claimed closed form be derived from the exact input, with both endpoint limits controlled?

The recovered input is $I=\int_0^1[(\sin(\log\sqrt{t}-\pi/4)+\cos(\pi/4-\log\sqrt{t}))/(\log\pi\cdot\log(t^2))]^4dt$. Its numerator simplifies before any integration. A Laplace transform and three integrations by parts then verify the reported closed form exactly.

Definitions and assumptions

Set $v=-(1/2)\log t>0$. The numerator becomes$-\sqrt{2}\sin v$, and $dt=-2e^{(-2v)}dv$.

Define $K(a)=\int_0^\infty e^{(-av)}\sin^4 v/v^4\ dv$ for $a>0$. The sine quotient tends to one at zero and exponential decay controls infinity.

The recovered fourth-power logarithmic integral has an exact elementary evaluation

$$I=[18\pi+90\log 2-33\log 20-12\arctan(1/2)]/[1152(\log\pi)^4].$$

Derivation or proof

The substitution gives $I=K(2)/[32(\log\pi)^4]$. Use$8\sin^4 v=3-4\cos 2v+\cos 4v$ and set $H(v)=e^{(-av)}(3-4\cos 2v+\cos 4v)$. H and its first three derivatives vanish at zero, while they decay at infinity. Three integrations by parts give$\int H/v^4=(1/6)\int H'''/v$. Applying the elementary Frullani identity$\int_0^\infty (e^{(-\lambda v)}-e^{(-\mu v)})/v\ dv=\log(\mu/\lambda)$, valid for positive real parts with continuous logarithm, yields $K(a)=(1/48)\mathrm{Re}[3a^3\log a-4(a-2i)^3\log(a-2i)+(a-4i)^3\log(a-4i)]$. All divergent individual terms cancel because the combined derivatives vanish at zero.

Worked example

Evaluate this formula at a=2.

$(2-2i)^3=-16-16i$ and$(2-4i)^3=-88+16i$. Their logarithms have real parts$(\log 8)/2,(\log 20)/2$ and arguments$-\pi/4,-\arctan 2$. Collecting real parts gives $K(2)=\pi/2+(5/2)\log 2-(11/12)\log 20-(1/3)\arctan(1/2)$, using $\arctan 2=\pi/2-\arctan(1/2)$. Division by$32(\log\pi)^4$ gives precisely the displayed numerator and denominator. Frullani’s identity follows by

differentiating with respect toλ in the half-plane Reλ>0: the derivative of the integral is−1/λ, equal to the derivative of−logλ. Matching the value atλ=μ fixes the constant. In the combined expression all coefficients sum to zero, allowing a common reference exponential to be subtracted before integration. This supplies a convergent identity at every stage rather than assigning values to divergent individual integrals. The original t-integrand has finite limit1/[64(logπ)⁴] as t approaches one, because sinv/v tends to one; the exponential substitution makes both endpoint bounds explicit.

This is a genuine verified historical closed form, supported by a complete analytic derivation rather than by accepting an assistant decimal.

Scope

The real logarithms and fourth power matter. The original expression has removable behavior at t=1; numerical routines evaluating it literally can create0/0 unless that limit is handled.

Further development

Derive the analogous transform for other even powers sin^(2m)v/v^(2m), proving endpoint cancellation before using logarithmic or complex-power expressions.

Source receipt: months/10/root_specifics.json record 0. Mathematical references: R110.

Y25-10-026 Cyclotomic trace orthogonality fixes the coefficient p, not p+1

2025-10-31 · historical evidence A · Classical mathematics with newly authored proof or worked finite certificate

Recovered starting point

Recovered dated mathematical subject; attribution limits are preserved.

Revised question

Which embeddings define the trace, and are the coefficients integral character multiplicities or arbitrary rational weights?

The retrieved cyclotomic expression used a coefficient p+1 and reported30/13 in a character setting. Summing the actual embeddings gives an exact trace identity and a useful integrality test, while leaving the unrecovered original coefficient data unspecified.

Definitions and assumptions

Let p be prime, ζ=exp(2πi/p), and X=Σ_(k=0)^(p−1)a_kζ^k. For the trace formula first allow a_k∈Q.

The embeddings of Q(ζ) send ζ toζ^r for r=1,…,p−1. Put A=Σa_k and C_0=Σ_(k+ℓ≡0 modp)a_ka_ℓ.

Cyclotomic trace orthogonality fixes the coefficient p, not p+1

Tr_(Q(ζ)/Q)(X²)=pC_0−A². If all a_k are integers, this trace is an integer. The same integrality holds for every algebraic-integer X, including ordinary finite-group character values contained in this field.

Derivation or proof

The embedding sumΣ_(r=1)^(p−1)ζ^(rm) is p−1 if p divides m and−1 otherwise: include the r=0 term and use a geometric series. Expand X^2 and apply this sum to every pair(k,ℓ). Congruence-zero pairs contribute$(p-1)C_0$, and the other pairs contribute$-(A^2-C_0)$, giving pC_0-A^2. For integral coefficients the formula is visibly integral. More generally the field trace of an algebraic integer is a rational algebraic integer, hence an ordinary integer. Thus a noninteger30/13 cannot be that trace without changing the object or its normalization.

Worked example

Take p=5 and $X=\zeta+\zeta^{-1}$.

Here A=2 and $C_0=2$, so the trace is$5\cdot 2-4=6$. Directly $X^2=\zeta^2+2+\zeta^{-2}$, whose trace is$-1+8-1=6$. Replacing p by p+1 would give eight and therefore fails this exact test. Repeated coefficients or a different rational normalization must be stated before applying an integrality conclusion. This test uses the complete embedding trace, not a selected subset of conjugates.

The trace is now an explicit finite embedding sum with a sharp algebraic consistency check.

Scope

The actual A_5 coefficient list and embedding into $Q(\zeta_{13})$ were not recovered. This entry does not invent them or identify an arbitrary rationally normalized sum with an ordinary character trace.

Further development

Recover a genuine character expansion, verify its field of values and integral lattice, then compute all Galois traces and compare them with representation-theoretic multiplicity formulas.

Source receipt: months/10/root_specifics.json record 0. Mathematical references: R187.

November 2025

Dated fragments recovered; complete transcript unavailable. 31 developed treatments.

Analytic recursion, special-function integrals, elliptic spectral theory, modular operators and subdivision dynamics are central. Exact identities and concrete finite models separate established mathematics from interpretive spectral or physical claims.

ID · date	Concept and revised question	Result and mathematical core
Y25-11-001 2025-11-01 · U	The cosine-quartic integral has an exact closed form Can parity remove the apparently asymmetric exponential denominator before contour integration?	Parity reduction and quartic transform I(a)=π/(4√2)·[1+e^(−√2|a|)(cos(√2|a|)+sin(√2|a|))].
Y25-11-006 2025-11-01 · U	Exact forms have zero periods; winding and entropy need different data Can an exact differential carry the claimed nonzero winding periods and entropy?	Period obstruction and a correct circle model Every closed-loop integral of χ is zero. In contrast η=(N/(2π e))d θ has period∫_γη=n N/e for a loop of windingn. The expressionlog(N/e)=1 holds only when N=e², not for arbitrary N.
Y25-11-003 2025-11-02 · U	A positive kernel is not necessarily positive definite What extra condition is required before a scalar response is a reproducing kernel or a covariance?	An exact quartic Gram counterexample The pointwise positive even function f(t)=1/(1+t⁴) is not positive definite on R. Its Gram matrix at points−1/2,0,1/2 has negative determinant−157/1156.
Y25-11-004 2025-11-02 · A	A bridge functional with an exact variance How does conditioning a Gaussian process change the calibration integral?	Gaussian bridge-functional variance J_f is centered Gaussian and Var(J_f)=∫f²dt−T^(−1)(∫fdt)². Consequently a constantf produces zero, rather than Wiener variance T f².
Y25-11-005 2025-11-02 · U	Boundary conditions determine a beam operator and its stability What does a hinged beam’s boundary domain contribute to its spectrum and buckling threshold?	Hinged spectrum and positive-form threshold The sine modes have eigenvalues D(n π/L)⁴−P(n π/L)². The form⟨w, Aw⟩=D||w″||²−P||w′||² is positive if P<D π²/L². The Fourier response denominator is Dk⁴−Pk²−m ω²−i ηω under the stated convention.
Y25-11-002 2025-11-05 · U	Three integral checks from the integer-breath atlas Which atlas constants survive exact substitution and normalization checks?	Exact corrected evaluations ∫R cos²(π ax)/(1+π²x²)dx=(1+e^(−2|a|))/2. Also∫₀¹∫₀¹xy/(1+x²+y²)dxdy=(3log3−4log2)/4, and∫₀^(π/6)tan³x/(1+sin²x)dx=[2+3log(3/5)]/24.
Y25-11-007 2025-11-05 · U	The proposed resonator spectrum grows factorially, not geometrically What operator can rigorously realize the displayed eigenvalues, and what is their actual ratio?	Diagonal realization and exact ratio Ω is positive self-adjoint with compact inverse, and λ_(n+1)/λ_n=π e n. In particular the ratio diverges;8π e occurs only atn=8.

ID · date	Concept and revised question	Result and mathematical core
Y25-11-011 2025-11-05 · U	A radial heat kernel and a convergent Volterra expansion Which parts of the master-kernel and nonlinear recursion have a direct PDE proof?	Hankel formula and first nonlinear response $\int_0^\infty ke^{-Dtk^2}J_0(kr)dk=(2Dt)^{-1}e^{-r^2/(4Dt)}$. The ODE has exact solution$x(t)=ae^{-t}/[1-\varepsilon\, a(1-e^{-t})]$, so$x_0=ae^{-t}$, $x_1=a^2e^{-t}(1-e^{-t})$.
Y25-11-012 2025-11-06 · A	A Hecke relation is not an arbitrary prime-index power ratio What exact coefficient identity should be tested before interpreting prime-index correlations as Hecke symmetry?	Exact prime-power coefficient recurrence For every prime p and integern≥1, $\sigma_3(pn)+p^3\sigma_3(n/p)=(1+p^3)\sigma_3(n)$. Thus $\sigma_3(pn)/\sigma_3(n)=1+p^3$ only when the correction term vanishes.
Y25-11-010 2025-11-07 · U	Wallis recurrences and what a hypergeometric operator really proves Can the recovered Wallis-Apéry bridge be replaced by an exact recurrence and a correctly normalized differential equation?	Two exact recurrence certificates $W_n=((n-1)/n)W_{(n-2)}$forn≥2. Also$[\Theta(\Theta-1/4)(\Theta-3/4)-z(\Theta+1)^3]F=0$as a formal series, with radius of convergenceone.
Y25-11-008 2025-11-10 · U	Lamé’s first finite-gap example from Jacobi identities Which Lamé eigenfunctions and spectral edges can be checked directly in the smallest nontrivial case?	Three exact eigenfunctions and positivity Hdn=m dn, Hcn=cn, Hsn=(1+m)sn. Since dn is positive on the real axis, $H-m=A^*A\geq 0$ with $A=d/dx-dn'/dn$, so the lowest periodic eigenvalue ism.
Y25-11-009 2025-11-10 · U	The product locus is Humbert discriminant one, not four What does a diagonal genus-two period matrix prove, and which Humbert discriminant describes the product locus?	Diagonal periods give the H_1 product model If $\Omega=\mathrm{diag}(\tau_1, \tau_2)$, with Im $\tau_i>0$, then A_Ω is the product $E_{(\tau_1)}\times E_{(\tau_2)}$with product principal polarization. The relation $\Omega_{12}=0$has coefficients$(a, b, c)=(0,1,0)$and $D=b^2-4ac=1$.
Y25-11-017 2025-11-10 · A	An inflection ridge is not automatically a phase transition What does the recovered threshold$\lvert E_{SS}\rvert<\varepsilon$ detect, and what extra structure would make it a phase boundary?	Smooth ridge versus a true variational singularity For $E(S, T)=S^3+TS^2$, the zero-second-derivative locus is the smooth line $T=-3S$ even though E is polynomial everywhere. For $F(m, a)=m^4/4-am^2/2$, $\min_m F$ is0 fora≤0 and$-a^2/4$ fora≥0, which is C^1 but not C^2 ata=0.
Y25-11-016 2025-11-11 · A	Kronecker flow is densely recurrent and has zero entropy What rigorous dynamics underlies an irrational phase texture, and is dense motion maximal entropy?	Unique ergodicity and zero topological entropy Time averages along this flow converge to Haar-space averages for every continuous function, so every orbit is dense. Every fixed-time translation map has topological entropyzero.
Y25-11-013 2025-11-12 · U	The scalar Laplacian variation under Ricci flow What is the correct commutator of time differentiation and the scalar Laplace-Beltrami operator under Ricci flow?	Correct scalar commutator For a smooth scalarf,$[\partial_t, \Delta_g]f=2Ric^{ij}\nabla_i\nabla_jf$. More generally $\delta\Delta f=-h^{ij}\nabla_i\nabla_jf-(divh-(1/2)d\,trh)\cdot\nabla f$.
Y25-11-014 2025-11-12 · A	Commutator traces, matrix words, and the missing regularization Which algebraic invariants of the recovered commutator words can be proved before interpreting numerical heights?	Finite trace and determinant constraints X is Hermitian and Tr X=0. Also W=[X, Y][Y, Z] anddet W=1. Thus a nonzero complex number cannot be the ordinary finite trace of$[\rho, \rho^*]$.

ID · date	Concept and revised question	Result and mathematical core
Y25-11-015 2025-11-13 · A	Circular concentration measures evidence, not a motivic sector How can the recovered phase clustering be tested without confusing a fitted cone with a mathematical classification?	Rotation invariance and a mixture ambiguity R_n is invariant under a common phase rotation and equalsone exactly when all phases agree modulo2π. Equal populations at phases0 and π have $R_n=0$ despite being maximally concentrated in two separate directions.
Y25-11-018 2025-11-15 · A	Tomita-Takesaki theory in a faithful finite matrix model Can the modular operator and conjugation be computed completely before extrapolating to QFT or holography?	Finite modular operator and flow $JX=X^*$, $\Delta X=\rho X \rho^{(-1)}$, and $S=J\Delta^{(1/2)}$. The modular automorphism of a left observable A is $\sigma_t(A)=\rho^{(it)}A\rho^{(-it)}$; the modular generator on H has eigenvalues$-\log(p_i/p_j)$.
Y25-11-019 2025-11-15 · A	Spectral multiplicity defeats the sealed-commutant claim When is an operator commuting with Xactually a scalar function of X, and what extra data defines a spectral triple?	Multiplicity and a two-point distance For $X=\mathrm{diag}(0,0,1)$, its commutant is $M_2(\mathbb{C})\oplus\mathbb{C}$, whereas$f(X)=\mathrm{diag}(f(0)I_2, f(1))$. In the two-point spectral triple, Connes'distance between the evaluation states is$1/m$.
Y25-11-027 2025-11-16 · A	IMP projection equals an iteration limit only under spectral hypotheses When can repeated refinement be replaced by a single orthogonal spectral projection?	Exact contraction limit and a counterexample to an unrestricted SVD rule If every other eigenvalue is at most$1-\gamma$ with $\gamma>0$, then$\|M^n-P\|\le(1-\gamma)^n$. For a general contraction, theunit-singular subspace can contain rotating directions and need not be the fixed space.
Y25-11-029 2025-11-18 · U	Low-rank computation has a measurable error, not a universal speedup What computational saving and approximation error follow from a genuine rank-dfactorization?	Optimal spectral residual and a product error bound $\|W-W_d\|_2=\sigma_{(d+1)}$. Applying $UV^T$to a vector costs about2ndmultiply-adds instead ofn^2. For products with$\|A_j\|,\|B_j\|\le M$and$\|A_j-B_j\|\le\varepsilon$, the difference of L-layerlinear products is at most $L\,\varepsilon\,M^{(L-1)}$.
Y25-11-031 2025-11-18 · U	Poisson-consistent smoothing follows from commuting convolution When does smoothing a potential preserve its Poisson equation?	Poisson equation commutes with mollification $\Delta(\eta_\varepsilon*\Phi)=\eta_\varepsilon*\rho$ in distributions. On a torus, a solution of $\Delta\Phi=\rho$ requires mean $\rho=0$, and smoothing preserves that compatibility condition.
Y25-11-030 2025-11-19 · A	A Riccati equation with a complete optimal-control proof Which concrete Hamilton-Jacobi and Riccati statements can replace the unsupported viability theorem?	Exact value and optimal feedback The minimum is $V(0, x_0)=P(0)x_0^2/2$, and the unique optimal control satisfies$u=-P(t)x/r$. The function Pobeys $P'=P^2/r-q$with $P(T)=0$.
Y25-11-028 2025-11-20 · A	The recovered curvature block is unstable and not the standard convex stencil What does the explicitly recovered two-by-two block prove on its own?	Exact characteristic polynomial and instability $\mathrm{tr}\,B=-61/48$, $\det B=3/64$, and $\chi_B(t)=t^2+(61/48)t+3/64$. Its eigenvalues are$(-61\pm\sqrt{3289})/96$, with one strictly below-1.

ID · date	Concept and revised question	Result and mathematical core
Y25-11-021 2025-11-21 · U	Levi adjacency is controlled by singular values What exact spectrum can be extracted from a weighted bipartite incidence matrix?	Bipartite singular-value spectrum For every positive singular value σ of B, Ahas eigenvalues±σ with the same multiplicity; its remaining eigenvalues arezero. In particular Tr A=0and A²=diag(B*B, BB*).
Y25-11-020 2025-11-24 · A	Catmull-Clark Fourier blocks derived from the actual stencil What spectrum follows from the standard face and edge rules, without assigning motivic constants by analogy?	Exact nonconstant Fourier block For a nonzero modez=e^(2π ik/v), P=0, E_j=Ez^j, F_j=Fz^j, the block on(E, F)is M(z)=[[(3+c)/8,(1+z^(−1))/16],[(1+z)/4,1/4]], c=cos(2π k/v). Its eigenvalues are(5+c±√((c+9)(c+1)))/16.
Y25-11-024 2025-11-24 · T	The recovered S_6 has square-zero Jordan mixing Does the explicit matrix's raw upper triangle equal its commuting nilpotent part?	Exact commuting semisimple-nilpotent decomposition S=S_s+Nwith S_s=diag(B, B), N=[[0, I_2],[0,0]],[S_s, N]=0, N²=0, N≠0. Since Bhas distinct eigenvalues, S_sis diagonalizable, and S is similar to $J_2(a)\oplus J_2(b)$.
Y25-11-025 2025-11-24 · A	The plastic constant is a genuine Pisot recurrence eigenvalue Which dominant/decaying-mode structure follows rigorously from the Padovan recurrence?	Pisot property and asymptotic mode splitting ψ>1is the unique real root ofp, and its two other roots are complex conjugates of modulus ψ^(−1/2)<1. The companion matrix has characteristic polynomialp; generic solutions are A ψ^nplus an exponentially decaying oscillatory term.
Y25-11-022 2025-11-25 · U	Discrete Hodge splitting and the correct cohomology transport group What can harmonic blocks and cohomology transport mean rigorously for a finite mesh?	Finite Hodge decomposition and symplectic necessity C^k=im d_(k−1)⊕ker Δ_k⊕im d_k*orthogonally, andker Δ_krepresents H^k. Any orientation-preserving surface-induced transport Msatisfies M^TJM=J, hence lies in Sp(2g, Z), not merely SL(2g, Z).
Y25-11-023 2025-11-25 · U	Forced block dynamics and a genuine Schur-complement correction How can a decaying curvature block be destabilized by harmonic forcing, and what does a Schur complement actually certify?	Exact forcing law and positive Schur criterion Ifa≠d, κ_n=a^n κ_0+c(d^n−a^n)α_0/(d−a). Ifa=d, the forcing term iscn a^(n−1)α_0. Also S≥0if and only if A−BD^(−1)B^T≥0.
Y25-11-026 2025-11-29 · U	Cancelling quadratic drift requires an expansion, not a benchmark label What explicit assumption would justify the claimed improvement from quadratic to quartic refinement drift?	Explicit order-four extrapolation G(h)=F_0−d h⁴/4+R_G(h), where|R_G(h)|≤(17/48)M h⁶. Thus theh²term cancels exactly and the remaining leading drift is quartic when d≠0.

Y25-11-001 The cosine-quartic integral has an exact closed form

2025-11-01 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can parity remove the apparently asymmetric exponential denominator before contour integration?

The recovered integral was described as lacking an elementary closed form. Its asymmetry disappears under x↦−x, leaving the classical Fourier transform of a quartic rational function. The resulting formula is exact for every real modulation parameter.

Definitions and assumptions

Let a∈R and I(a)=∫R cos²(ax)/[(1+ix²)(1+4^x)(1−ix²)]dx.

Write w(x)=1/(1+4^x); then w(x)+w(−x)=1.

Parity reduction and quartic transform

$$I(a)=\pi/(4\sqrt{2})\cdot[1+e^{(-\sqrt{2}|a|)}(\cos(\sqrt{2}|a|)+\sin(\sqrt{2}|a|))].$$

Derivation or proof

The conjugate factors multiply to1+x⁴. The remaining factor f(x)=cos²(ax)/(1+x⁴) is even and integrable. Substitution x↦−x gives I=∫f(x)w(−x)dx, so adding yields2I=∫f. Now cos²(ax)=(1+cos2ax)/2. For b>0 close the contour for e^(ibz)/(1+z⁴) in the upper half-plane; the poles are e^(i π/4) and e^(3i π/4), with residues e^(ibz)/(4z³). Their sum, multiplied by2π i, is F(b)=π e^(−b/√2)[cos(b/√2)+sin(b/√2)]/√2. The large arc vanishes by the exponential factor and quartic denominator; b=0 follows directly or by dominated convergence. Evenness supplies F(−b)=F(b). Therefore I=(F(0)+F(2a))/4, proving the formula. Every exchange is justified by domination with1/(1+x⁴).

Worked example

Evaluate a=0, compare large|a|, and check local curvature at the origin.

I(0)=π/(2√2), while I(a)→π/(4√2) as|a|→∞. Differentiating under the integral twice is allowed since x²/(1+x⁴) is integrable, giving I″(0)=−π/√2. The third derivative cannot be justified by the same absolute-moment argument; the closed form indeed has an|a|³ term near zero.

The recovered prototype is solved exactly, and its smoothness is linked to the decay of the quartic weight.

Scope

This identity proves no eigenvalue law for an unspecified integral operator and no physical realization. A finite precision plot should be checked against this formula rather than treated as evidence for a new constant.

Further development

Replace4^x by any positive-base exponential c^x with c>0, c≠1: the same symmetry gives exactly the same integral for every even integrable numerator.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R188.

Y25-11-006 Exact forms have zero periods; winding and entropy need different data

2025-11-01 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can an exact differential carry the claimed nonzero winding periods and entropy?

The recovered model simultaneously declared Jconstant, defined χ=(e/N)d J, and assigned nonzero quantized integrals. These statements are incompatible. A genuine circle one-form gives a constructive replacement and makes clear what hash entropy would mean mathematically.

Definitions and assumptions

Let N>0 and J be a single-valued smooth real function; χ=(e/N)d J is exact.

On S^1 the angular differentiald θ is a global closed one-form even though θ is only locally a single-valued real function.

Period obstruction and a correct circle model

> Every closed-loop integral of χ is zero. In contrast η=(N/(2π e))d θ has period$\int_\gamma \eta = n\ N/e$ for a loop of windingn. The expressionlog(N/e)=1 holds only when $N=e^2$, not for arbitrary N.

Derivation or proof

For a parametrized closed loop γ, the chain rule gives$\int_\gamma d\ J = J(\gamma(1)) - J(\gamma(0)) = 0$. If Jis constant, χ vanishes pointwise, making the contradiction stronger. For the circle model lift γ to an angle Θ(t) on an interval. Its endpoint difference is$2\pi\ n$, so the integral of η is Nn/e. There is no globally single-valued real Jwithd J=d θ, because such a function would force the nonzero circle period to vanish. Finallylog(N/e)=log N−1, so setting it equal toone fixes $N=e^2$. If N denotes the number of independent uniform hash bits, Shannon entropy is Nlog2 nats, whereaslog N is the entropy of N equiprobable outcomes; these are different sample spaces.

Worked example

Take N=320 and a once-around circle loop.

The exact-form model gives zero regardless of N. The corrected angular form gives320/e. The claimed entropy becomeslog320−1≈4.76832nats, while320uniform bits have320log2≈221.807nats. Merely rescaling384to320 cannot preserve an equation asserting the same fixed entropy.

Winding can be modeled consistently, but it requires a non-exact cohomology class and a clearly specified probability space.

Scope

No cryptographic deck transformation or physical energy quantization follows from these definitions. A hash digest is finite discrete data and does not automatically define a smooth covering-space action.

Further development

Specify a principal U(1)bundle and connection, then compare gauge-invariant holonomy with integral cohomology and a separately defined entropy model.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R007.

Y25-11-003 A positive kernel is not necessarily positive definite

2025-11-02 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What extra condition is required before a scalar response is a reproducing kernel or a covariance?

The recovered kernel discussion moved from positive integrands to kernel methods and a conjectured spectral decay. A covariance or reproducing kernel must satisfy all finite Gram-matrix inequalities, a much stronger condition than having positive values.

Definitions and assumptions

A Hermitian kernel k(x, y) is positive definite if Σ overline c_i c_j k(x_i, x_j)≥0 for every finite point set and coefficients.

For a real translation kernel k(x, y)=f(x−y), Hermitian symmetry requires f(−t)=f(t).

An exact quartic Gram counterexample

The pointwise positive even function f(t)=1/(1+t⁴) is not positive definite on R. Its Gram matrix at points−1/2,0,1/2 has negative determinant−157/1156.

Derivation or proof

At adjacent separation1/2, f=16/17; at separation1, f=1/2. The Gram matrix is[[1, a, b],[a,1, a],[b, a,1]] witha=16/17, b=1/2. Expanding its determinant gives(1−b)(1+b−2a²). Here1+b−2a²=3/2−512/289=−157/578, so the determinant is−157/1156. A positive semidefinite Hermitian matrix has nonnegative determinant, proving failure. An even more immediate obstruction affects the original real asymmetric logistic factor: a translation kernel using it is not even, hence not Hermitian. On the other hand the quartic Fourier transform F(b)=∫e^(ibx)/(1+x⁴)dx is positive definite as a function of b: the Gram quadratic form equals∫|Σ c_j e^(ib_jx)|²/(1+x⁴)dx≥0. The response and its Fourier transform therefore play different roles.

Worked example

Compare f itself with F(b)=π e^(−|b|/√2)[cos(|b|/√2)+sin(|b|/√2)]/√2.

F can take negative values, yet its Gram forms remain nonnegative by the integral-square representation. Thus pointwise nonnegativity is neither necessary nor sufficient for positive definiteness. A valid covariance can have negative correlations, whereas a positive-valued response can fail the covariance test.

The exact three-point certificate distinguishes two often-confused meanings of “positive kernel.”

Scope

No spectrum λ_n can be assigned until an operator, domain, measure and boundary conditions are supplied. In particular the recovered(n logn)^(−4/3) claim is not certified by a scalar formula.

Further development

Define an integral operator on a compact measured domain, prove self-adjointness and compactness, and derive its eigenvalue asymptotics from regularity and geometry rather than from a numerical fit.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R189.

Y25-11-004 A bridge functional with an exact variance

2025-11-02 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How does conditioning a Gaussian process change the calibration integral?

The recovered stochastic calibration used a kernel inside an OU/Wiener functional without a complete process specification. A Brownian bridge example gives an exact definition and reveals the covariance correction that is lost if a bridge is treated as an unconstrained Wiener process.

Definitions and assumptions

Let W_t be standard Brownian motion on[0, T] and B_t=W_t−(t/T)W_T.

For deterministicf∈L²(0, T), define its bridge functional J_f=∫₀ᵀf(t)d W_t−(W_T/T)∫₀ᵀf(t)dt.

Gaussian bridge-functional variance

J_f is centered Gaussian and Var(J_f)=∫f²dt−T^(−1)(∫fdt)². Consequently a constantf produces zero, rather than Wiener variance T f².

Derivation or proof

Approximatef by step functions to define the Wiener integral in mean square. Its mean is zero, its variance is∫f² by the Itô isometry, and its covariance with W_T=∫1d W is∫f. Expanding the variance of the displayed linear combination gives∫f²−2T^(−1)(∫f)²+T^(−1)(∫f)², proving the formula. It is nonnegative by Cauchy-Schwarz and vanishes exactly for almost-everywhere constantf. Gaussianity follows because the approximating functionals are finite linear combinations of a jointly Gaussian process and converge in L². This definition also avoids incorrectly using the original Brownian filtration to treat the anticipative representation of B as an ordinary Itô integrator.

Worked example

Take T=1 andf(t)=t, then comparef(t)=1.

Forf=t the variance is1/3−(1/2)²=1/12. Forf=1, J_f=W_1−W_1=0. An unconstrained Wiener functional∫₀¹td W has variance1/3, so conditioning removes a substantial component. For an OU bridge the same Gaussian conditional-covariance rule is C_br(s, t)=C(s, t)−C(s, T)C(T, t)/C(T, T), with the OU covariance explicitly supplied first.

The stochastic object has a complete probability model, a proved variance, and an exact nontrivial example.

Scope

Random or adapted kernel parameters require separate integrability and filtration conditions. A complex-valued kernel needs its Hermitian covariance convention. This finite-time construction does not validate a device or an unspecified stochastic calibration.

Further development

For a deterministic observation kernel and a specified OU covariance, evaluate the double covariance integral and compare bridge and unconditioned uncertainties with rigorous quadrature errors.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R190.

Y25-11-005 Boundary conditions determine a beam operator and its stability

2025-11-02 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What does a hinged beam’s boundary domain contribute to its spectrum and buckling threshold?

The recovered discussion correctly emphasized boundary conditions but assigned response formulas without fixing a time convention. A constant-coefficient hinged beam supplies an exact spectrum, a domain, and a sign check for its frequency kernel.

Definitions and assumptions

On[0, L], imposew(0)=w(L)=w″(0)=w″(L)=0 and let A=D∂ₓ⁴+P∂ₓ², with D>0.

For a damped equationmw_tt+η w_t+Aw=0 use the time convention e^(−i ω t), withm, η≥0.

Hinged spectrum and positive-form threshold

The sine modes have eigenvalues D(n π/L)⁴−P(n π/L)². The form⟨w, Aw⟩=D||w″||²−P||w′||² is positive if P<D π²/L². The Fourier response denominator is Dk⁴−Pk²−m ω²−i ηω under the stated convention.

Derivation or proof

Integrating the fourth derivative twice and using both hinged endpoint conditions removes boundary terms. Integrating the second derivative once gives the negative gradient norm, proving the form. The sine basis satisfies both endpoint conditions and differentiates to the displayed eigenvalues. Since$||w''||^2 \geq (\pi/L)^2 ||w'||^2$ by sine expansion, the threshold gives a strictly positive form. Conversely the first sine mode has zero form at $P = D\,\pi^2/L^2$ and negative form above it, so the threshold is sharp. Finally substitutinge^(ikx−i ω t) gives$\partial_x^4 = k^4, \partial_x^2 = -k^2, \partial_t^2 = -\omega^2, \partial_t = -i\,\omega$. A plusi ηω term would correspond to the opposite time convention, not a different damping law.

Worked example

Take $D=1$, $L=\pi$, $m=1$ and $P=1/2$.

The first two stiffness eigenvalues are$1-1/2=1/2$ and$16-2=14$. At $P=1$ the first mode becomes neutral. With $\eta>0$ and positive stiffness, each modal energy $E_n=(|a_n'|^2+\lambda_n|a_n|^2)/2$ satisfies $E_n' = -\eta|a_n'|^2 \leq 0$.

The boundary domain, stiffness spectrum, threshold and damping sign are all fixed by the equation rather than an analogy.

Scope

Variable coefficients, sixth-order terms, and different endpoints define different operators. The recovered numerical residue and engineering specifications lack the full response function needed for independent verification.

Further development

For variable $D(x)>0$, start from the form$\int D(x)|w''|^2 - P|w'|^2$ and prove coercivity before deriving a Green function or assigning resonance poles.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R186.

Y25-11-002 Three integral checks from the integer-breath atlas

2025-11-05 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which atlas constants survive exact substitution and normalization checks?

The recovered atlas contains a correct tangent integral together with unsupported Lorentzian and gamma normalizations. Direct substitutions turn these into elementary benchmarks, useful precisely because they do not rely on a proposed physical interpretation.

Definitions and assumptions

Use $a \in \mathbb{R}$ and the standard real branches of logarithm and arctangent.

The gamma reflection identity is $\Gamma(x)\Gamma(1-x) = \pi/\sin \pi\, x$ for$0<x<1$.

Exact corrected evaluations

∫R cos²(π ax)/(1+π²x²)dx=(1+e^(−2|a|))/2. Also∫₀¹∫₀¹xy/(1+x²+y²)dxdy=(3log3−4log2)/4, and∫₀^(π/6)tan³x/(1+sin²x)dx=[2+3log(3/5)]/24.

Derivation or proof

For the first formula substitute u=π x and use cos²(au)=(1+cos2au)/2. The elementary contour transform∫R e^(ibu)/(1+u²)du=π e^(−|b|) gives the result after the Jacobian1/π. In the double integral put u=x², v=y²; the Jacobian converts xy dxdy todu dv/4. Integrating1/(1+u+v) on the unit square gives(3log3−4log2)/4. For the tangent integral set t=tanx. Since sin²x=t²/(1+t²) and dx=dt/(1+t²), the integrand becomes t³/(1+2t²)dt. An antiderivative is t²/4−log(1+2t²)/8; evaluate between0 and1/√3. Finally the recovered gamma-normalized integral reduces pointwise by reflection to π∫₀¹e^{x(1−x)}dx, so it is strictly greater than π and cannot equal1.

Worked example

Take a=1/4, and audit the claimed gamma constant without numerical quadrature.

The Lorentzian integral equals(1+e^(−1/2))/2≈0.803265, not π/4. For the gamma integral,0<x(1−x)≤1/4 gives π<T≤π e^(1/4). Completing the square gives the optional exact special-function form T=π^(3/2)e^(1/4)erf(1/2). The tangent expression is positive because it is the integral of a positive integrand.

The atlas yields reproducible exact test cases, with one historical tangent identity retained and two normalization errors corrected.

Scope

No claimed integer value of a separate Bessel sum or complex gamma integral is inferred; its summation domain, convergence and normalization must be independently established.

Further development

Use these formulas as unit-independent analytic benchmarks, then supply absolute-convergence bounds before rearranging any infinite sum of related kernels.

Source receipt: months/11/root_followups.json record 0. Mathematical references: R169, R191.

Y25-11-007 The proposed resonator spectrum grows factorially, not geometrically

2025-11-05 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What operator can rigorously realize the displayed eigenvalues, and what is their actual ratio?

The recovered spectrum λ_n=π^n e^(n−1/2)Γ(n) was said to have ratio limit8π e. The gamma recurrence gives an exact contradiction. A diagonal construction nevertheless realizes the sequence as a legitimate self-adjoint operator without endorsing unspecified noncommuting products.

Definitions and assumptions

Let{e_n}_{n≥1} be an orthonormal basis, for example a normalized Laguerre basis of the stated weighted L²space.

Define ΩΣ c_ne_n=Σλ_nc_ne_n on the domain Σλ_n²|c_n|²<∞.

Diagonal realization and exact ratio

Ω is positive self-adjoint with compact inverse, and λ_(n+1)/λ_n=π e n. In particular the ratio diverges;8π e occurs only atn=8.

Derivation or proof

The gamma recurrence Γ(n+1)=n Γ(n) cancels every factor in the ratio except π e n. This calculation needs no asymptotic approximation. For the operator, finite basis sums lie in the domain, making it dense. In coefficient space Ω is multiplication by a real positive sequence. Its adjoint has the same coordinatewise action and the same square-summability condition, proving self-adjointness. Since λ_n→∞, the inverse is bounded and is a norm limit of finite-rank diagonal truncations, with tail normsup_{n>N}1/λ_n→0. Thus the inverse is compact. Stirling additionally gives λ_n~√(2π/e)π^n n^(n−1/2), confirming growth much faster than geometric. The productnotation π^W e^P Γ(D) is not justified by this construction unless W, P, Dare separately defined on compatible domains and have the required commutation properties.

Worked example

Compare successive ratios atn=1,8,16.

They are π e,8π e,16π e. The spectrum starts atn=1 because Γ(0) is a pole. A manufactured diagonal realization shows consistency of a sequence as spectral data, but it has no independent prediction: the desired eigenvalues were inserted by definition.

The sequence defines a valid operator and an exact growth law, while the historical finite ratio limit is false.

Scope

No ring experiment or acoustic realization is inferred. A physical or differential operator must be independently specified before spectral agreement carries explanatory value.

Further development

Define a concrete differential or integral operator with its boundary domain and prove its spectral asymptotics independently; only then compare with the proposed sequence.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R169, R192.

Y25-11-011 A radial heat kernel and a convergent Volterra expansion

2025-11-05 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which parts of the master-kernel and nonlinear recursion have a direct PDE proof?

The recovered atlas used Laplace-Hankel kernels and recursive nonlinear responses. A Gaussian heat kernel and a scalar semilinear example give exact derivations. They also show why the radial measure and small-parameter assumptions cannot be omitted.

Definitions and assumptions

For D, t>0, the two-dimensionalheatkernel is G_t(x)=(4π Dt)^(−1)e^(−|x|²/(4Dt)).

Forx′=−x+ε x²withx(0)=a, writean expansionx=x₀+ε x₁+O(ε²).

Hankel formula and first nonlinear response

∫₀∞ke^(−Dtk²)J₀(kr)dk=(2Dt)^(−1)e^(−r²/(4Dt)). The ODE has exact solutionx(t)=ae^(−t)/[1−ε a(1−e^(−t))], sox₀=ae^(−t), x₁=a²e^(−t)(1−e^(−t)).

Derivation or proof

The Cartesian Fourier transform of a two-dimensional Gaussian factors into two one-dimensional Gaussian integrals. Converting the inverse transform to polar coordinates gives the angularintegral2π J₀(kr)and the radial measurekdk, proving the Hankel identity with itsnormalization. For the ODE, sety=e^t x; theny′=ε e^(−t)y². Differentiating1/yand integrating gives1/y=1/a−ε(1−e^(−t)), which yields the exact formula wherever itsdenominatoris nonzero. Expanding itsgeometricseries isvalid uniformlyont≥0if|ε a|<1. Alternativelymatchingpowers of ε givesx₀′=−x₀andx₁′=−x₁+x₀²withx₁(0)=0. Duhamel'sformula givesx₁(t)=∫₀ᵗe^(−(t−s))a²e^(−2s)ds, the same expression.

Worked example

Takea=1and|ε|≤1/4; comparethe exact solutionwith the first two terms.

The ratioz=ε(1−e^(−t))has|z|≤1/4. Thegeometricremainderafter1+zisbounded by|z|²/(1−|z|), so|x−x₀−ε x₁|≤e^(−t)ε²/(1−|ε|)≤(4/3)ε²e^(−t). This is an actual convergence/errorcertificate rather than an unqualified recursion slogan.

The heat transform and nonlinear response both have exact normalizations and explicit hypotheses.

Scope

The historical master integral usesa differentpowerofkandadditionaldispersion/cutoff factors; it is not automatically the two-dimensionalheatkernel. A general Volterra expansion needs bounds foritsnonlinearityandsemigroup.

Further development

For a Banach-space equationu′=Au+ε N(u), usea contraction estimate on the Duhamelmapandderivean actualradiusofconvergence orcontrolledfinite-orderremainder.

Source receipt: months/11/root_followups.json record 0. Mathematical references: R193.

Y25-11-012 A Hecke relation is not an arbitrary prime-index power ratio

2025-11-06 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What exact coefficient identity should be tested before interpreting prime-index correlations as Hecke symmetry?

The recovered CMB discussion reported prime-coherence significance and a Hecke analogy. A classical divisor-sum example supplies a complete arithmetic identity and shows the missing correction when the prime divides the index. Statistical claims additionally require the original data and covariance model.

Definitions and assumptions

Let $\sigma_3(n)=\Sigma_{d|n} d^3$ and put $\sigma_3(n/p)=0$ whenp does not dividen.

The normalized weight-four Eisenstein eigenvalue at primep is$1+p^3$.

Exact prime-power coefficient recurrence

> For every prime p and integern≥1, $\sigma_3(pn)+p^3\sigma_3(n/p)=(1+p^3)\sigma_3(n)$. Thus $\sigma_3(pn)/\sigma_3(n)=1+p^3$ only when the correction term vanishes.

Derivation or proof

Write n=p^a m withgcd(p, m)=1. Divisor multiplicativity gives $\sigma_3(n)=\sigma_3(m)\Sigma_{j=0}^a p^{3j}$. For a=0 the formula follows immediately. For a≥1, add $\Sigma_{j=0}^{a+1}p^{3j}$ and$p^3\Sigma_{j=0}^{a-1}p^{3j}$; each interior power occurs twice, matching$(1+p^3)\Sigma_{j=0}^ap^{3j}$. This proves the identity without assuming a physical interpretation. It is the Fourier-coefficient form of the weight-four Hecke relation for E_4. Replacing these coefficients by an arbitrary angular power spectrum does not preserve the relation; the relevant modular form, weight, coefficient normalization, and prime-divisibility correction must be identified independently.

Worked example

Usep=2 atn=1 andn=2, then consider multiple prime tests.

$\sigma_3(1)=1$, $\sigma_3(2)=9$, $\sigma_3(4)=73$. For n=2 the identity is$73+8=9\cdot9$, but the raw ratio is$73/9$ rather than9. If m separate null tests are searched, a Bonferroni bound gives P(any false rejection)≤Σ p_i; a single-test significance is not automatically a familywise significance. Shared data can also correlate the tests.

The arithmetic claim has an exact benchmark, while the reported observational significance remains an unverified historical result.

Scope

No raw CMB data, pipeline, null ensemble, covariance matrix, or selection history was recovered. Neither the coefficient identity nor a fitted effective dimension proves a physical Hecke action, RH, or an E_8 correspondence.

Further development

Predeclare the coefficient extraction and null model, test the full prime-power recurrence includingp-divisible indices, and reserve independent data for validation.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R106, R194.

Y25-11-010 Wallis recurrences and what a hypergeometric operator really proves

2025-11-07 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the recovered Wallis-Apéry bridge be replaced by an exact recurrence and a correctly normalized differential equation?

The recovered analytic-recursion program named Wallis integrals, a third-order hypergeometric operator, and an unverified odd-zeta ladder. Exact recurrences are available, but identifying a solution with ζ(3)requires an independent evaluation and connection data.

Definitions and assumptions

Let $W_n=\int_0^{(\pi/2)}\sin^n x\,dx$ for integers$n\geq 0$. Let $\Theta=z\,d/dz$.

Define $F(z)=\Sigma\ c_nz^n$ with$c_0=1$and$c_n=(n!)^2/[(3/4)_n(1/4)_n]$, where$(a)_n$ is the rising factorial.

Two exact recurrence certificates

$W_n=((n-1)/n)W_{(n-2)}$forn≥2. Also$[\Theta(\Theta-1/4)(\Theta-3/4)-z(\Theta+1)^3]F=0$as a formal series, with radius of convergenceone.

Derivation or proof

Integrate W_nby parts using one factor sinx and the remaining sin^(n−1)x. The boundary term vanishes and gives W_n=(n−1)(W_(n−2)−W_n), hence the Wallis recurrence. Its initial values are $W_0=\pi/2$, $W_1=1$. For the differential equation, the coefficient ofz^nforn≥1isn(n−1/4)(n−3/4)c_n−n³c_(n−1), which vanishes exactly by the displayed rising-factorial ratio. The constant coefficient also vanishes. The ratio c_n/c_(n−1)=n²/[(n−1/4)(n−3/4)]tends toone, so the radius is one. This verifies the operator but not a ζ value. In fact the coefficients arepositive andgrow asymptotically likea constant timesn, so the unmodifiedseriesdiverges atz=1; a claimed finite value there needs an explicitly different solution, regularization or connection formula.

Worked example

Compute W_2, $W_4$and the firstthreehypergeometriccoefficients.

$W_2=\pi/4$, $W_4=3\pi/16$. Meanwhile$c_1=16/3$and$c_2=(16/3)\cdot 64/35=1024/105$. The coefficient equation atn=2is$2(7/4)(5/4)c_2-8c_1=0$, which can be checked exactly. None of these rational coefficients alone singles out ζ(3).

The reconstructed recurrences are genuine mathematics and expose the missing normalization in the asserted bridge.

Scope

Apéry’s irrationality theorem is established mathematics, but the recovered unspecified binomial sum and ζ(5+)ladder are not proved by this differential equation. Analytic continuation does not by itself prove a special-value identity.

Further development

Specify the exact Apéry-type series and establish its evaluation by an integral or telescoping certificate; then derive a rigorous remainder bound before using it as an odd-zeta calculator.

Source receipt: months/11/root_followups.json record 0. Mathematical references: R195, R196.

Y25-11-008 Lamé’s first finite-gap example from Jacobi identities

2025-11-10 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which Lamé eigenfunctions and spectral edges can be checked directly in the smallest nontrivial case?

The recovered Lamé equation has a precise classical meaning independent of the speculative RH visualization. Takingn=1 yields three exact elliptic eigenfunctions and a factorization that proves a ground-state bound on a stated periodic domain.

Definitions and assumptions

Let$0<k<1$, $m=k^2$, and $K=K(k)$. Define $H=-d^2/dx^2+2m\,\mathrm{sn}^2(x,k)$ on4K-periodic H^2functions.

Jacobi identities aresn′=cn dn, cn′=−sn dn, dn′=−m sn cn, $\mathrm{cn}^2+\mathrm{sn}^2=1$, $\mathrm{dn}^2+m\,\mathrm{sn}^2=1$.

Three exact eigenfunctions and positivity

Hdn=m dn, Hcn=cn, Hsn=(1+m)sn. Since dn is positive on the real axis, H−m=A*A≥0 with A=d/dx−dn′/dn, so the lowest periodic eigenvalue ism.

Derivation or proof

Differentiate the Jacobi derivative relations. They givesn″=−(1+m)sn+2m sn³, cn″=(2m sn²−1)cn, anddn″=(2m sn²−m)dn. Substitution into Hproves the three identities. Letw=dn′/dn, a smooth real periodic function because dn≥√(1−m)>0. The adjoint of A=∂−w on the periodic domain is−∂−w, and A*A=−∂²+w′+w²=−∂²+dn″/dn=H−m. Hence⟨f,(H−m)f⟩=||Af||²≥0. Since Adn=0, the lower bound is attained. The functionsdn, cn, sn are all4K-periodic; dn has the shorter2Kperiod whilecn andsn change sign after2K. This fixes an otherwise consequential periodic-versus-antiperiodic convention.

Worked example

Takem=1/2 andcompare the three displayed levels.

The energies are1/2,1,3/2. The factorization excludes any lower4K-periodic eigenvalue. In the limitm→0, the potential tends tozero, sn→sin, cn→cos, dn→1, and the two excited levels merge at1, matching the free periodic operator after K→π/2.

This is a complete exact elliptic spectral calculation with a valid domain and a proved lowest level.

Scope

The three eigenfunctions alone are not a proof of every finite-gap theorem. No Lamé-to-zeta-zero map or counterfactual “off-critical catastrophe” establishes RH.

Further development

Compute the monodromy matrix over2K and its discriminant, distinguishing periodic and antiperiodic band edges before discussing the full real-line spectrum.

Source receipt: months/11/root_details.json record 0. Mathematical references: R197, R198.

Y25-11-009 The product locus is Humbert discriminant one, not four

2025-11-10 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What does a diagonal genus-two period matrix prove, and which Humbert discriminant describes the product locus?

The recovered source identified the product locus with H_4using a restricted Humbert relation. The standard principal-polarization convention assigns decomposable products to H_1. A lattice factorization proves the basic distinction and prevents a discriminant label from being silently changed.

Definitions and assumptions

Siegel upper half-space H_2 consists of symmetric2×2complex matrices Ω with positive definite imaginary part.

The principally polarized complex torus is $A_\Omega=C^2/(Z^2+\Omega Z^2)$; a primitive Humbert relation has integer coefficients and discriminant D.

Diagonal periods give the H_1 product model

If $\Omega=\mathrm{diag}(\tau_1, \tau_2)$, with $\mathrm{Im}\ \tau_i>0$, then A_Ω is the product $E_{\tau_1}\times E_{\tau_2}$with product principal polarization. The relation $\Omega_{12}=0$has coefficients(a, b, c)=(0,1,0)and $D=b^2-4ac=1$.

Derivation or proof

The period lattice is$(Z+\tau_1 Z)\oplus(Z+\tau_2 Z)$, so the coordinate map $C^2\to C/(Z+\tau_1 Z)\times C/(Z+\tau_2 Z)$has exactly that kernel and induces the product isomorphism. The standard symplectic alternating form on the lattice splits into the sum of the two elliptic principal forms, identifying the product polarization. The indicated Humbert relation is primitive because its coefficients have gcdone. Writing$2\Omega_{12}=0$would yield a formal discriminant4but multiplies the entire relation bytwo; it is not primitive and does not relabel the same locus as H_4. More general Humbert relations include quadratic termsd$(\Omega_{12}^2-\Omega_{11}\Omega_{22})$+e, so a three-coefficient relation is only a special chart of the full definition.

Worked example

Take $\tau_1=i$and $\tau_2=2i$, and compare an elliptic subtorus with its polarization degree.

The lattice is$(Z+i\,Z)\oplus(Z+2i\,Z)$. The first factor is an elliptic subtorus with principal restricted polarization of degreeone. In the standard convention H_{n^2}instead describes a degree-nelliptic subcover/restricted polarization condition; n=2leads to $H_4$and does not mean the generic product polarization.

The product statement follows from explicit lattices and the discriminant correction follows from primitiveness.

Scope

The full moduli-theoretic characterization of Humbert surfaces is cited background. A period matrix or Humbert condition supplies no dynamics or zeta-zero theorem by itself.

Further development

For a genus-two Jacobian with an actual degree-two elliptic quotient, construct its induced polarization and verify the primitive discriminant-four relation rather than multiplying a discriminant-one equation.

Source receipt: months/11/root_details.json record 0. Mathematical references: R199.

Y25-11-017 An inflection ridge is not automatically a phase transition

2025-11-10 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What does the recovered threshold|E_SS|<ε detect, and what extra structure would make it a phase boundary?

The recovered phase-diagram script drew ridges from a small second derivative. Such a set can be a useful visualization, but it need not be singular, critical in the optimization sense, or thermodynamic. A pair of elementary models separates these possibilities.

Definitions and assumptions

For a smooth surface E(S, T), an inflection locus may be defined by E_SS=0.

A nonanalytic minimized free energy can arise after optimizing over an order parameterm.

Smooth ridge versus a true variational singularity

For E(S, T)=S³+TS², the zero-second-derivative locus is the smooth line T=−3S even though E is polynomial everywhere. For F(m, a)=m⁴/4−am²/2, min_m F is0 fora≤0 and−a²/4 fora≥0, which is C¹ but not C² ata=0.

Derivation or proof

Differentiate the polynomial E twice in S to get6S+2T; its zero set is a line with nonvanishing gradient(6,2). The implicit-function theorem confirms it is a regular smooth submanifold, not a singularity of E. Replacing equality by an ε-threshold merely thickens the line. For F, stationarity givesm(m²−a)=0. Ifa≤0, the sum of nonnegative quartic and quadratic terms is minimized atzero. Ifa>0, the critical points±√a give−a²/4, below the valuezero at m=0. The resulting minimized function has first derivativezero on the left and−a/2 on the right, continuous ata=0; second derivatives0 and−1/2 disagree. The singularity arises from minimization and a change of minimizer, not from an arbitrary curvature threshold.

Worked example

Compare the two definitions near their putative critical sets.

At(S, T)=(0,0), E_SS=0 but E and every derivative are finite. At a=0, the minimizer of F changes from one point to two symmetric nonzero points as a becomespositive, producing the proved nonanalyticity in the minimized objective.

The visualization can honestly represent an inflection band, while a phase-transition claim needs an actual order parameter and limiting or minimization argument.

Scope

No historical E, Gor Ricci-Perelman model was specified sufficiently to derive its phase boundary. A memory-efficient animation or a fitted ridge is not a theorem about its dynamics.

Further development

Define the governing evolution or free energy, state the control parameters, and prove which observable or derivative becomes singular before labeling a critical manifold.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R200.

Y25-11-016 Kronecker flow is densely recurrent and has zero entropy

2025-11-11 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What rigorous dynamics underlies an irrational phase texture, and is dense motion maximal entropy?

The recovered animation was interpreted as a dense torus flow and a maximal-entropy texture. The first claim can be true under rational independence; the second fails for a torus translation. The distinction is accessible through characters and the isometry property.

Definitions and assumptions

On $T^2=R^2/Z^2$ consider $\varphi_t(x)=x+t(1, \varphi) \bmod Z^2$, where $\varphi=(1+\sqrt{5})/2$.

Characters aree_k(x)=e^(2π ik·x), k∈Z²; use the flat translation-invariant metric.

Unique ergodicity and zero topological entropy

> Time averages along this flow converge to Haar-space averages for every continuous function, so every orbit is dense. Every fixed-time translation map has topological entropyzero.

Derivation or proof

For a nonconstant charactere_k, the time average is e_k(x)[e^(2π i T(k₁+k₂φ))−1]/[2π i T(k₁+k₂φ)]. Irrationality ensures the denominator coefficient is nonzero for k≠0, so this tends tozero uniformly in x. Constants retain their average. Trigonometric polynomials are uniformly dense in continuous functions, proving the averaging statement for every continuous function. A nonempty open set contains a nonnegative continuous function of positive Haar integral; its positive time average forces every orbit to enter that set, proving density. For entropy, a translation is an isometry, so the distance between two points under any number of iterates is their original distance. The number of ε-separated orbit segments is bounded by the finite ε-packing number of the torus, independent of segment length; logarithmic growth per iterate is thereforezero.

Worked example

Let an RGB texture use three continuous periodic scalar functions of the two phase angles.

Following the irrational orbit samples their joint torus distribution, with no exact period. This does not make the RGB vector lie on S^2: it belongs to R^3 or a color cube unless explicitly normalized, and normalization requires its norm never vanish. Clipping can also create flat regions and change the pushed-forward distribution.

Dense quasiperiodic sampling is rigorously explained without confusing visual complexity with positive entropy.

Scope

A torus embedding in R^3 does not alter the translation dynamics. An animation’s finite frames cannot by themselves certify irrationality or ergodicity.

Further development

For a specified color map, compute Fourier coefficients and quantify finite-time averaging error using lower bounds on$|k_1+k_2\varphi|$ for its active modes.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R201.

Y25-11-013 The scalar Laplacian variation under Ricci flow

2025-11-12 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What is the correct commutator of time differentiation and the scalar Laplace-Beltrami operator under Ricci flow?

The recovered manuscript included a gradient term in the Ricci-flow Laplacian commutator without the companion term that cancels it. The general metric-variation formula and contracted Bianchi identity produce a clean corrected statement.

Definitions and assumptions

Use $\Delta f=g^{ij}\nabla_i\nabla_j f$ and Ricci flow$\partial_t g_{ij}=-2\mathrm{Ric}_{ij}$.

For a metric variation$h=\partial_t g$, let$(\mathrm{div}h)_j=\nabla^i h_{ij}$ and$\mathrm{tr}h=g^{ij}h_{ij}$.

Correct scalar commutator

For a smooth scalar$f,[\partial_t, \Delta_g]f=2\mathrm{Ric}^{ij}\nabla_i\nabla_j f$. More generally $\delta\Delta f=-h^{ij}\nabla_i\nabla_j f-(\mathrm{div}h-(1/2)d\,\mathrm{tr}h)\cdot\nabla f$.

Derivation or proof

Differentiate$g^{ij}g_{jk}=\delta^i_k$ to obtain $\delta g^{ij}=-h^{ij}$. In local coordinates, differentiate $\Delta f=g^{ij}(\partial_i\partial_j f-\Gamma^k_{ij}\partial_k f)$. The connection variation is $\delta\Gamma^k_{ij}=(1/2)g^{k\ell}(\nabla_i h_{j\ell}+\nabla_j h_{i\ell}-\nabla_\ell h_{ij})$. Contracting i, j gives$(\mathrm{div}h)^k-(1/2)\nabla^k\mathrm{tr}h$, proving the general formula. Substituting$h=-2\mathrm{Ric}$ gives$2\mathrm{Ric}:\mathrm{Hess}f+(2\mathrm{div}\,\mathrm{Ric}-d\,R)\cdot\nabla f$. Contracted Bianchi says$\mathrm{div}\,\mathrm{Ric}=(1/2)d\,R$, so the entire gradient coefficient cancels. If f depends on time, the $\Delta\partial_t f$ term cancels in the commutator definition; the same formula remains.

Worked example

Take a unit round two-sphere with$g(t)=(1-2t)g_0, 0\le t<1/2$.

Its connection is unchanged by the spatially constant scale and $\Delta_t=(1-2t)^{-1}\Delta_0$. Hence$\partial_t \Delta_t=2(1-2t)^{-2}\Delta_0$. Raising the indices of $\mathrm{Ric}=g_0$ withg(t) yields $\mathrm{Ric}^{ij}=(1-2t)^{-2}g_0^{ij}$, giving exactly the same right-hand side.

The corrected commutator is an actual tensor identity with a complete variation derivation and an explicit evolving-metric check.

Scope

No zeta-regularized trace, anomaly coefficient, cosmological constant, or motivic statement follows from this differential identity alone. Operator traces need separate domains and summability hypotheses.

Further development

Combine the identity with$\partial_t \mathrm{dvol}=-R\,\mathrm{dvol}$ when differentiating heat traces, retaining both metric and measure variations before exchanging derivatives and traces.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R054.

Y25-11-014 Commutator traces, matrix words, and the missing regularization

2025-11-12 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which algebraic invariants of the recovered commutator words can be proved before interpreting numerical heights?

The recovered ladder mixed $X=[\rho, \rho^*]$, complex-valued outputs, inverse words, and motivic interpretations. Ordinary finite-dimensional traces impose exact constraints, while regularized traces require an explicitly different definition. A word identity gives a constructive invariant that can actually be checked.

Definitions and assumptions

For a finite complex matrix ρ let $X=\rho\rho^*-\rho^*\rho$.

For invertible X, Y, Z define $W=XYX^{-1}ZY^{-1}Z^{-1}$, using ordinary matrix multiplication.

Finite trace and determinant constraints

X is Hermitian and Tr X=0. Also W=[X, Y][Y, Z] anddet W=1. Thus a nonzero complex number cannot be the ordinary finite trace of$[\rho, \rho^*]$.

Derivation or proof

Taking adjoints leaves ρρ* and ρ*ρ unchanged, proving Hermiticity. Cyclicity of finite trace gives Tr ρρ*=Tr ρ*ρ, so the trace is zero. Multiplying the two group commutators gives XYX^(−1)Y^(−1)·YZY^(−1)Z^(−1)=W after cancelling Y^(−1)Y. Determinants multiply and commute as scalars, so each group commutator has determinantone and so does W. Neither statement fixes Tr W or turns the word into an intrinsic “height-three” object. If a weighted or zeta-regularized trace is intended, it must be defined separately and cannot inherit cyclicity without proof.

Worked example

Take ρ=[[0,1],[0,0]] and a diagonal weight Q=diag(1,2).

X=diag(1,−1), with tracezero and real eigenvalues. For real s, Tr(Q^(−s)X)=1−2^(−s), generally nonzero, while at s=0 it is zero. This finite example shows exactly how a weight changes a trace functional. It does not produce a zeta anomaly because the finite expression is already entire in s.

The output type and invariant are explicit: ordinary commutator trace, weighted trace, determinant and group word are different quantities.

Scope

The historical complex ladder values and stubbed higher heights have no recovered definitions sufficient for independent numerical reproduction. No multiple-zeta or motivic classification is inferred from their magnitudes or phases.

Further development

Provide the exact matrices and trace regularizer, prove meromorphic continuation, and calculate any finite part with a stated subtraction scheme before discussing a trace defect.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R202.

Y25-11-015 Circular concentration measures evidence, not a motivic sector

2025-11-13 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How can the recovered phase clustering be tested without confusing a fitted cone with a mathematical classification?

The recovered plots were described as rigid motivic splitting across seeds. A circular resultant is a genuine invariant of phase concentration, but it cannot identify a motivic class without an independent map from the data to that class. A finite exact example exposes the statistic’s scope.

Definitions and assumptions

For phases θ_j define Z_n=n^(−1)Σ e^(i θ_j) and R_n=|Z_n|∈[0,1].

The von Mises model has densityexp(κ cos(θ−μ))/(2π I_0(κ)); it is an assumed family, not a consequence of plotting angles.

Rotation invariance and a mixture ambiguity

R_n is invariant under a common phase rotation and equalsone exactly when all phases agree modulo2π. Equal populations at phases0 and π have R_n=0 despite being maximally concentrated in two separate directions.

Derivation or proof

A common rotation multiplies Z_n by a unit complex number and leaves its modulus unchanged. The triangle inequality gives R_n≤1. Equality requires all nonzero unit summands to have the same argument, proving the equality characterization. For a two-point equal mixture, e^(i0)+e^(i π)=1−1=0, proving the ambiguity. The second harmonic Z_n^(2)=n^(−1)Σ e^(2i θ_j) is one for the same sample, detecting axial concentration. Thus a first-harmonic statistic can entirely miss a pair of opposing cones. Fisher-information or Cramér-Rao formulas require an actual parametric sampling model, independence assumptions, and differentiation under its likelihood; an empirical angular standard deviation is not automatically such a bound.

Worked example

Take ten observations, five at0 andfive at π; compare ten identical phases at π/4.

The first sample has R_1=0 and second-harmonic resultant R_2=1. The second has R_1=R_2=1. A tiny perturbation of each phase changes these values continuously but does not establish a discrete underlying category. Growth regression$\log_{10}$|C_h|≈α h+β similarly estimates a numerical trend and supplies no algebraic relation among the C_h.

The same data can support a concentration statement while leaving the proposed categorical interpretation completely unproved.

Scope

No raw seeds or exact coefficient-generation map was recovered. Reported κ, Rand uncertainty estimates cannot be independently recomputed here, and labels such as mixed-Tate or elliptic remain unsupported.

Further development

Predefine a circular mixture model, compare first and higher harmonics on held-out seeds, and test any independently defined algebraic label separately from phase clustering.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R203.

Y25-11-018 Tomita-Takesaki theory in a faithful finite matrix model

2025-11-15 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the modular operator and conjugation be computed completely before extrapolating to QFT or holography?

The recovered request explicitly demanded real mathematics after fictional spectral-universe claims. A faithful matrix state gives an exact standard representation and modular flow, with the important density-matrix ratios visible rather than hidden.

Definitions and assumptions

Let H=M_d(C)with Hilbert-Schmidt inner product and M_d(C)act by left multiplication.

Choose ρ>0 with Tr ρ=1 and cyclic separating vector Ω=ρ^(1/2). Define S(A Ω)=A*Ω.

Finite modular operator and flow

JX=X*, Δ X=ρ X ρ^(−1), and S=J Δ^(1/2). The modular automorphism of a left observable A is σ_t(A)=ρ^(it)A ρ^(−it); the modular generator on H has eigenvalues−log(p_i/p_j).

Derivation or proof

Every Xcan be written A ρ^(1/2)because ρ is invertible, so Ω is cyclic. If A Ω=0then A=0, so it is separating. In terms of X, the Tomita map is SX=ρ^(−1/2)X*ρ^(1/2). The positive operator Δ^(1/2)acts as X↦ρ^(1/2)X ρ^(−1/2); applying Jgives exactly SX. In an eigenbasis ρ e_i=p_ie_i, the matrix unit E_ijis an eigenvector of Δ with positive eigenvaluep_i/p_j, proving positivity and the logarithmic generator formula. Conjugating a left-multiplication operator by Δ^(it)gives left multiplication by ρ^(it)A ρ^(−it), while the right factors cancel. This explicitly checks the modular automorphism theorem in the model.

Worked example

Take ρ=diag(2/3,1/3).

Δ has eigenvalues1,2,1/2,1on E_{11}, E_{12}, E_{21}, E_{22}. The off-diagonal observable E_{12}acquires phase2^(it), while diagonal observables are fixed. Thus the modular operator is not simply ρ on this Hilbert space: it compares left and right weights.

The requested modular identities have exact definitions, domains and a complete two-level example.

Scope

An infinite algebra can be type III and have no density matrix or trace of this kind. Modular inclusions, DHR categories, JLMS, RT and entanglement-derived field equations require additional hypotheses and are not proved by the finite calculation.

Further development

For a genuine algebraic-QFT model, specify the observable net, representation, state and standard subspaces before claiming geometric modular flow or a holographic identity.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R204.

Y25-11-019 Spectral multiplicity defeats the sealed-commutant claim

2025-11-15 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

When is an operator commuting with Xactually a scalar function of X, and what extra data defines a spectral triple?

The recovered assertion{X}′={f(X)}is false in the presence of spectral multiplicity. A finite counterexample and a two-point spectral metric show both the missing condition and the information needed beyond a list of eigenvalues.

Definitions and assumptions

For self-adjoint X, its commutant consists of all bounded operators Awith AX=XA.

On the two-point algebra C⊕C use π(a, b)=diag(a, b)and D=[[0, m],[m,0]], withm>0.

Multiplicity and a two-point distance

For X=diag(0,0,1), its commutant is $M_2(C)\oplus C$, whereasf(X)=diag(f(0)I_2, f(1)). In the two-point spectral triple, Connes’distance between the evaluation states is1/m.

Derivation or proof

Commuting with a diagonal matrix forces A_ij=0whenever the two eigenvalues differ, but imposes no constraint within an equal-eigenvalue block. This gives the stated commutant. Scalar functional calculus assigns the same scalar to every vector in one eigenspace and therefore gives only the center of that commutant. For the two-point metric,[D, π(a, b)]=[[0, m(b−a)],[m(a−b),0]], whose norm ism|a−b|. The spectral distance is the supremum of|a−b|over self-adjoint(a, b)with commutator norm≤1, so it is exactly1/m. Finite dimensions make the resolvent compact automatically, and all commutators are bounded. This is a valid spectral-triple model, but its distance comes from Dand the represented algebra, not from an isolated spectral list.

Worked example

Choose Ato swap the first two coordinates in the Xexample, and takem=2in the two-point model.

The swap commutes with Xbut is notf(X), since its upper2×2block is not scalar. The two-point distance is1/2. Changingmchanges the distance while leaving the algebra C⊕Cunchanged, exhibiting the independent geometric role of D.

The historical universal reduction fails already in three dimensions; a legitimate noncommutative-geometric model requires more structure.

Scope

For a simple finite spectrum, commutant equality does hold by interpolation, but this special case cannot be assumed for arbitrary observables. No universe or QFT reconstruction follows.

Further development

For an infinite self-adjoint operator, describe its direct-integral spectral multiplicity and identify exactly when the generated abelian von Neumann algebra is maximal abelian.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R043.

Y25-11-027 IMP projection equals an iteration limit only under spectral hypotheses

2025-11-16 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

When can repeated refinement be replaced by a single orthogonal spectral projection?

The recovered IMP narrative identified a slow singular subspace with a subdivision limit. A symmetric contraction has exactly such a limit, but general singular vectors do not determine eigenvalue dynamics. A rotation and a nonnormal stochastic matrix show why the assumptions matter.

Definitions and assumptions

Let Mbe a real symmetric matrix with all eigenvalues in[0,1]. Let Pproject onto its1-eigenspace.

A singular-value decomposition describes one-step norm amplification; it is not generally an eigendecomposition.

Exact contraction limit and a counterexample to an unrestricted SVD rule

> If every other eigenvalue is at most1−γ with γ>0, then||M^n−P||≤(1−γ)^n. For a general contraction, theunit-singular subspace can contain rotating directions and need not be the fixed space.

Derivation or proof

The spectral theorem writes M=Σλ_j P_jwith orthogonal projections. Powers give M^n−P=Σ_(λ_j<1)λ_j^n P_j, whose operator norm is the largest surviving|λ_j|^n, proving the estimate. For a planar rotation R_θ with θ not a multiple of2π, both singular values areone and the retained unit-singular subspace is the whole plane, yet R_θ^ngenerally fails to converge and its fixed subspace may bezero. Nonnormal convergence can also yield an oblique projector:the row-stochastic matrix T=[[1,0],[a,1−a]],0<a<1, has T^n=[[1,0],[1−(1−a)^n,(1−a)^n]], so its limit is[[1,0],[1,0]], which is idempotent but not symmetric.

Worked example

Take M=diag(1,1/2,1/4)and compare Tata=1/2.

M^nconverges todiag(1,0,0)with exact error2^(−n). T^nconverges to an oblique projection, whose action sends(x, y)to(x, x). Neither the singular vectors nor an orthogonal projector chosen from a one-step tolerance alone identifies that latter limit.

A one-step replacement can be rigorously justified in the stated symmetric model, and its approximation error is explicit.

Scope

This proves no universal equality between an IMP/SVD construction and a subdivision limit surface, transformer, quantum channel or diffusion sampler. Each operator requires its own invariant subspaces and peripheral-spectrum analysis.

Further development

For a specific refinement matrix, prove eigenvalueoneis semisimple and all other eigenvalues lie strictly inside the unit disk, then compute its possibly oblique spectral projector exactly.

Source receipt: months/11/root_coverage.json record 0. Mathematical references: R205.

Y25-11-029 Low-rank computation has a measurable error, not a universal speedup

2025-11-18 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What computational saving and approximation error follow from a genuine rank-dfactorization?

The recovered transformer discussion estimated enormous projection speedups from a small retained dimension. A rank factorization gives an arithmetic-count reduction for a specified linear map; whole-network speed, accuracy and energy claims require independent evidence.

Definitions and assumptions

Let W∈R^(n×n)have singular values $\sigma_1 \geq \ldots \geq \sigma_n$, and let $W_d$be its truncated SVD.

A rank-dfactorization $W_d = UV^T$usesn×dmatrices U, V.

Optimal spectral residual and a product error bound

$||W - W_d||_2 = \sigma_{d+1}$. Applying $UV^T$to a vector costs about2ndmultiply-adds instead ofn². For products with$||A_j||, ||B_j|| \leq M$and$||A_j - B_j|| \leq \varepsilon$, the difference of L-layerlinear products is at most $L \varepsilon M^{L-1}$.

Derivation or proof

Singular-value diagonalization shows the residual norm is σ_{d+1}. Optimality follows because anyrank-dapproximation has a nonzero kernel vector within the span of the first$d+1$right singular vectors; on that vector Wacts with norm at least σ_{d+1}. The two matrix-vector multiplies V^Txand Uyuse2ndoperations to leading order. For products, insert one factor difference at a time:$\Pi A - \Pi B = \Sigma A_L \ldots A_{j+1}(A_j - B_j)B_{j-1}\ldots B_1$. Submultiplicativity bounds each term by εM^{L-1}, and summing proves the estimate. Thus small per-layer error can accumulate with depth and large operator norms.

Worked example

Taken=8192, d=16, and compare one linear application.

n^2=67108864while2nd=262144, giving an idealized factor256for that isolated map under the same multiply-add convention. Including additional projections, attention, activations, memory movement or a different baseline changes the ratio. A formula12n/d=6144uses a different baseline workload and is not the same operation count.

The saving and the residual are explicit mathematical quantities; they do not imply a universal neural-network acceleration.

Scope

No actual retained subspace, layer residuals, hardware timing or energy benchmark was recovered. The low-rank hypothesis must be tested on the intended model and inputs.

Further development

Estimate σ_(d+1)and downstream Lipschitz factors for each actual layer, then compare the certified approximation error with measured latency and task accuracy.

Source receipt: months/11/root_coverage.json record 0. Mathematical references: R049.

Y25-11-031 Poisson-consistent smoothing follows from commuting convolution

2025-11-18 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

When does smoothing a potential preserve its Poisson equation?

The recovered universal smoothing proposal connected Barneskernels, subdivision, neural layers and gravitational potentials. A precise convolution theorem is useful and provable, but it requires regularity, translation invariance and a correctly normalized kernel.

Definitions and assumptions

On R^dlet η_ε be a smooth nonnegative mollifier of total massone.

Assume Φ and ρ are distributions satisfying ΔΦ=ρ, with convolution η_ε*Φ well-defined.

Poisson equation commutes with mollification

Δ(η_ε*Φ)=η_ε*ρ in distributions. On a torus, a solution of ΔΦ=ρ requires mean ρ=0, and smoothing preserves that compatibility condition.

Derivation or proof

Distributional differentiation commutes with convolution by a test function:testing against ψ and integrating by parts transfers Δ to ψ and then to Φ, with no boundary term on R^dor the torus. Thus Δ(η_ε*Φ)=η_ε*(ΔΦ)=η_ε*ρ. Integrating a torus Laplacian gives zero, proving the necessarymeancondition. Since η has massone, the mean of η*ρ equals the mean of ρ. In Fourier coordinates on a torus, the equation is$-|k|^2\hat{\Phi}(k)=\hat{\rho}(k)$for $k\neq0$; both smoothed coefficients are multiplied by $\hat{\eta}_\varepsilon(k)$, so the identity is visible mode by mode. The zero potential mode remains an arbitrary additive constant.

Worked example

Take ρ(x)=cosx on a2π periodic circleand Φ(x)=−cosx. Use a heat mollifier with multiplier e^(−ε²k²).

The smoothed density ise^(−ε²)cosxand the smoothed potential is−e^(−ε²)cosx; the second derivative of the latter is exactly the former. If ρ=1instead, there is no periodic Poissonsolution because its mean is nonzero.

The legitimate operator-smoothing relation is a precise PDE commutation theorem with an exact example.

Scope

An arbitrary Barnesorlearnedkernelneed not have these properties, and smoothing a nonlinear operator generally does not commute with its nonlinearity. No universal computing or physical equivalence follows.

Further development

For a proposed kernel, prove normalization, regularity and Fourier bounds, and calculate the commutator with any variable-coefficient or nonlinear operator rather than assuming it vanishes.

Source receipt: months/11/root_coverage.json record 0. Mathematical references: R206.

Y25-11-030 A Riccati equation with a complete optimal-control proof

2025-11-19 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which concrete Hamilton-Jacobi and Riccati statements can replace the unsupported viability theorem?

The recovered passage invoked Pontryagin, HJB, Riccatiand broad system viability. A scalar linear-quadratic control problem provides a complete finite-horizon theorem with no unstated economic or geometric interpretation.

Definitions and assumptions

Let$x'=u$, $x(0)=x_0$, and minimize $J=(1/2)\int_0^T(qx^2+ru^2)dt$withq, r>0and no terminal cost.

Set $P(t)=\sqrt{(qr)}\tanh(\sqrt{(q/r)}(T-t))$.

Exact value and optimal feedback

The minimum is $V(0, x_0)=P(0)x_0^2/2$, and the unique optimal control satisfies$u=-P(t)x/r$. The function Pobeys $P'=P^2/r-q$with $P(T)=0$.

Derivation or proof

Differentiate the displayedhyperbolictangentto verify the Riccatiequation and terminal condition. For any admissible control, $d(Px^2/2)/dt=P'x^2/2+Pxu$. Add this derivative to the running cost and substitute $P'=P^2/r-q$; the result is$(r/2)(u+Px/r)^2$. Integrating and using $P(T)=0$gives $J=P(0)x_0^2/2+(r/2)\int(u+Px/r)^2dt$. The nonnegative remainder proves the lower bound, and the feedback equation makes itzero. The feedback ODE has a unique solution because P is continuous, so the optimum is attained and unique up to almost-everywhere equality. This also derives the quadratic HJB ansatz rather than merely asserting it.

Worked example

Take$q=r=1$and $T=1$.

$P(0)=\tanh1$, so the optimal cost is$(\tanh1)x_0^2/2$. The feedback weakens tozeroas $t\to T$because there is no terminal penalty. In the infinite-horizon limit $P\to1$and the feedback tends to$u=-x$, the familiar stable closed loop.

The control theorem has exact hypotheses, an explicit solution, and a proof by completing squares.

Scope

This does not certify the recovered campus or AGI viability inequalities, nor a Tannakiangroup. Their states, costs, constraints and uncertainty model were not specified sufficiently for a theorem.

Further development

For a constrained nonlinear model, derive the actual HJBor Pontryaginconditions from its defined state and cost, then distinguish necessary conditions from a global sufficiency proof.

Source receipt: months/11/root_coverage.json record 0. Mathematical references: R207.

Y25-11-028 The recovered curvature block is unstable and not the standard convex stencil

2025-11-20 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What does the explicitly recovered two-by-two block prove on its own?

The late recovered source gives a concrete complex block and two negative eigenvalues. Its characteristic polynomial can be checked exactly; one eigenvalue has modulus greater thanone. This contradicts the claim that every atlas mode contracts, and it prevents identifying the block with a restriction of a standard nonnegative row-stochastic refinement matrix without further explanation.

Definitions and assumptions

Let B=[[-7/48,(3/32)(1+i√3)],[(5/16)(1−i√3),−9/8]].

The proposed geometric scalar is R=3√3/32; it is independent of the characteristic-polynomial calculation.

Exact characteristic polynomial and instability

tr B=−61/48, det B=3/64, and χ_B(t)=t²+(61/48)t+3/64. Its eigenvalues are(−61±√3289)/96, with one strictly below−1.

Derivation or proof

The diagonal product is21/128. The off-diagonal product is(15/512)(1+3)=15/128, so their difference is3/64. The trace is immediate. The discriminant is(61/48)²−4(3/64)=3289/2304, giving the stated roots. Since3289>35², the negative root(−61−√3289)/96is less than−1. Equivalently χ_B(−1)=1−61/48+3/64=−43/192<0while χ_B(t)→+∞as t→−∞, forcing a root below−1. A nonnegative row-stochastic matrix has induced∞-normone and spectral radius at mostone; every genuine invariant restriction inherits its eigenvalues. Therefore this Bcannot be such a restriction under a similarity change unless the recovered object differs from that standard operator.

Worked example

Compute the separately proposed logarithmic regulator R(μ)=∫₀¹log|1−μ t|dt.

For μ≠0, substitutionu=1−μ tgives R(μ)=−1−[(1−μ)/μ]log|1−μ|, with R(0)=0by continuity and R(1)=−1. For small positive μ it is negative, approximately−μ/2. Thus a positive geometric scalar cannot automatically equal this integral for arbitrary positive μ.

The concrete block has a complete exact spectral certificate, and a separate regulator formula has a derived normalization.

Scope

The claimed13×13source matrix and extraction map were not recovered, so the block’s Catmull-Clark provenance is not certified. Algebraic entries alone do not establish a mixed-Tate object or a universal latent dimension.

Further development

Recover the full matrix and Fourier injection, then verify SM=MB entrywise before transferring eigenvalue claims between the two operators.

Source receipt: months/11/root_coverage.json record 0. Mathematical references: R208.

Y25-11-021 Levi adjacency is controlled by singular values

2025-11-21 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What exact spectrum can be extracted from a weighted bipartite incidence matrix?

The recovered incidence construction uses a genuine block operator. Its spectrum is determined by singular values of the incidence block; diagonal conjugation preserves it, but arbitrary reweighting does not. This gives a useful finite theorem without inventing arithmetic period data.

Definitions and assumptions

Let Bbe anm×ncomplex matrix and A=[[0, B*],[B,0]]on C^n⊕C^m.

A similarity A↦LAL^(−1)requires invertible L; a congruence LAL*is a different operation.

Bipartite singular-value spectrum

> For every positive singular value σ of B, Ahas eigenvalues±σ with the same multiplicity; its remaining eigenvalues arezero. In particular Tr A=0and A²=diag(B*B, BB*).

Derivation or proof

Block multiplication gives the square formula and Hermiticity. If Bv=σ uand B*u=σ vwith unit singular vectorsu, v, then A(v, u)=σ(v, u)and A(v,−u)=−σ(v,−u). Null vectors of Bin the first block and B*in the second give the remaining kernel. The singular-value decomposition supplies a complete orthogonal decomposition, proving the full spectrum. Similarity preserves the characteristic polynomial bydet(t I−LAL^(−1))=det(L(t I−A)L^(−1)), whereas congruence need not; even A=Iand L=2Ichanges Iinto4Iunder congruence.

Worked example

Take B=[[1,1],[0,1]].

B*B=[[1,1],[1,2]]has characteristic polynomialt²−3t+1and eigenvalues(3±√5)/2. Its singular values are φ and1/φ, so Ahas eigenvalues±φ,±1/φ and characteristic polynomialt⁴−3t²+1. This algebraic appearance of φ comes from the chosen finite matrix and is fully explained by its polynomial.

The incidence spectrum can be computed exactly and used in a real refinement or graph analysis.

Scope

An algebraic eigenvalue does not by itself define a Frobenius action, motive, regulator or L-function. The recovered characteristic recurrences and evaluator need their actual matrices before further claims can be checked.

Further development

Construct Bfrom the documented mesh incidences and weights, then compare its singular vectors with the actual refinement operator rather than identifying the two by shape alone.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R205.

Y25-11-020 Catmull-Clark Fourier blocks derived from the actual stencil

2025-11-24 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What spectrum follows from the standard face and edge rules, without assigning motivic constants by analogy?

The recovered classical rewrite invoked Fourier blocks but asserted complex curvature eigenvalues. Deriving the ordinary Catmull-Clark block from its stencil gives a real discriminant. This is a constructive replacement that can be checked independently of any proposed atlas.

Definitions and assumptions

Around a valence-vvertex, label edge neighbors E_jand opposite face vertices F_jin cyclic order. A face point is F′_j=(P+E_j+F_j+E_(j+1))/4.

The edge rule is E′_j=(P+E_j+F′_(j−1)+F′_j)/4.

Exact nonconstant Fourier block

For a nonzero modez=e^(2π ik/v), P=0, E_j=Ez^j, F_j=Fz^j, the block on(E, F)is M(z)=[[(3+c)/8,(1+z^(−1))/16],[(1+z)/4,1/4]], c=cos(2π k/v). Its eigenvalues are(5+c±√((c+9)(c+1)))/16.

Derivation or proof

Substitute the modal values into the face rule to get F′=[(1+z)E+F]/4. In the edge rule, F′_(j−1)+F′_jcontributes(1+z^(−1))F′, so its E-coefficient is1/4+(2+z+z^(−1))/16=(3+c)/8and its F-coefficient is(1+z^(−1))/16. This proves the block. Its trace is(5+c)/8and determinant1/16. The quadratic formula gives the stated eigenvalues. Sincec∈[−1,1], the radicand(c+9)(c+1)is nonnegative; for this standard block the eigenvalues are real. Atz=−1the off-diagonal entries vanish and M=I/4, so the repeated eigenvalue there is semisimple.

Worked example

Takev=4, k=1, andv=6, k=1,2.

Forv=4, c=0, the eigenvalues are1/2and1/8. Forv=6, thek=1values are(11±√57)/32; thek=2values are(9±√17)/32. These exact formulas differ from the recovered ad hoc constants. Sixth roots of unity may put√3in matrix entries, but that does not force complex eigenvalues or a motivic interpretation.

The spectrum now follows from explicit subdivision rules and a two-by-two characteristic polynomial.

Scope

This is the nonconstant one-ring block, not a complete proof of global surface regularity or every extended stencil. Different modified schemes can have different blocks and must supply their own rules.

Further development

Derive the constant-mode vertex block and characteristic map, then verify the eigenvalue ordering and regularity conditions required by a subdivision-smoothness theorem.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R208, R209.

Y25-11-024 The recovered S_6 has square-zero Jordan mixing

2025-11-24 · historical evidence T · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Retrieved file passage carrying an explicit historical date

Revised question

Does the explicit matrix’s raw upper triangle equal its commuting nilpotent part?

The retrieved dated file supplies an explicit $S_6$and a cube-zero nilpotent claim. The strictly upper-triangular part is indeed cube-zero, but it fails to commute with the diagonal. The actual Jordan-Chevalley nilpotent part is square-zero. Distinguishing these decompositions makes the finite algebra correct.

Definitions and assumptions

Seta=53/60, b=(−61+5√3)/96, r=3√3/32and B=[[a, r],[0, b]].

The recovered4×4matrix is S=[[B, I_2],[0, B]], witha≠b.

Exact commuting semisimple-nilpotent decomposition

S=S_s+Nwith S_s=diag(B, B), N=[[0, I_2],[0,0]],[S_s, N]=0, N^2=0, N≠0. Since Bhas distinct eigenvalues, S_sis diagonalizable, and S is similar to $J_2(a)\oplus J_2(b)$.

Derivation or proof

Direct block multiplication gives[S_s, N]=0and N^2=0. A two-by-two triangular matrix with distinct diagonal values is diagonalizable; explicitly T=[[1, r/(b−a)],[0,1]]satisfies BT=Tdiag(a, b). Conjugating Sbydiag(T, T)diagonalizes both Bblocks while leaving I_2unchanged. Reorder the basis to pair the twoa-coordinates and the twob-coordinates; the result is $J_2(a)\oplus J_2(b)$. This proves the commuting Jordan decomposition and its two block sizes. By contrast write S=D_0+Uwith D_0=diag(a, b, a, b)and Uits raw strict upper triangle. U^2has entry2rin position(1,4), so $U^2\neq 0$, U^3=0. But$[D_0, U]_{12}=(a-b)r\neq 0$, so D_0+Uis not a commuting semisimple-nilpotent decomposition.

Worked example

Compare exponentials using the two decompositions.

The correct commuting form givesexp(t S)=diag(e^(t B), e^(t B))·(I+t N). The raw Uhasexp(t U)=I+t U+$t^2U^2/2$, butexp(t S)cannot be replaced bye^(t D_0)e^(t U)because $D_0$and Udo not commute. A legitimate step-two unipotent example is E_{12}, $E_{23}$with central commutator E_{13}, where the finite Baker-Campbell-Hausdorff correction is explicit.

The dated matrix supports a complete exact Jordan certificate, and the conflicting nilpotency statements are reconciled by specifying which decomposition is meant.

Scope

The matrix was retrieved from a file passage carrying the historical date, not derived from a recovered Catmull-Clark stencil. Its algebra does not establish motivic, BSDor wall-crossing identifications.

Further development

For any proposed block atlas, compute its commuting Jordan decomposition exactly and test whether the claimed nilpotent generators are extracted from it or merely chosen separately.

Source receipt: months/11/root_followups.json record 0. Mathematical references: R205.

Y25-11-025 The plastic constant is a genuine Pisot recurrence eigenvalue

2025-11-24 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which dominant/decaying-mode structure follows rigorously from the Padovan recurrence?

The recovered refinement narrative invoked the plastic constant and a Pisot-Frobenius archetype. The Padovan companion matrix provides a complete classical instance:one expanding real algebraic unit and two contracting conjugates. That structure does not identify an arbitrary subdivision matrix with it.

Definitions and assumptions

Letp(x)=x^3-x-1andlet ψ be its positive real root.

For P_(n+3)=P_(n+1)+P_n, the state(P_n, P_(n+1), P_(n+2))evolves by C=[[0,1,0],[0,0,1],[1,1,0]].

Pisot property and asymptotic mode splitting

ψ>1is the unique real root ofp, and its two other roots are complex conjugates of modulus ψ^(−1/2)<1. The companion matrix has characteristic polynomialp; generic solutions are A ψ^nplus an exponentially decaying oscillatory term.

Derivation or proof

The polynomial discriminant is$-4(-1)^3-27(-1)^2=-23<0$, so it has one real root and a nonreal conjugate pair. Sincep(1)=−1andp(3/2)>0, the real root lies in(1,3/2). The product of all three roots is one, so ifz, overline zare the other roots, ψ$|z|^2$=1, proving the modulus. The rational-root test excludes±1and shows irreducibility over Q, so ψ is an algebraic integer of degree three with all other conjugates inside the unit circle:a Pisot number. Expandingdet(t I−C)givesp(t); the roots are distinct, so diagonalization gives the three-mode recurrence solution. The real dominant coefficient Acan vanish for specially chosen initial data, but not for a positive Perron eigencomponent.

Worked example

Useinitial values $P_0=P_1=P_2=1$.

The sequence begins1,1,1,2,2,3,4,5,7,9,12,16. Its ratios need not be monotone, but after dividing by ψ^n, the oscillatory contribution decays at relative rate ψ^(−3n/2). The eigenvalue ψ≈1.324717957is unrelated to53/60, which is rational and less thanone.

The claimed archetype has a rigorous example with a proved algebraic polynomial and quantitative decay.

Scope

A dominant real eigenvalue alone does not imply the Pisot property; its algebraic conjugates must be controlled. No arithmetic regulator or subdivision identification is inferred from resemblance.

Further development

For a documented integer refinement matrix, compute its irreducible characteristic factors and test their conjugate moduli before assigning a Pisot or unit interpretation.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R210.

Y25-11-022 Discrete Hodge splitting and the correct cohomology transport group

2025-11-25 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What can harmonic blocks and cohomology transport mean rigorously for a finite mesh?

The recovered subdivision program used harmonic/curvature blocks and an SL(2g, Z)transport. A finite cochain complex has an exact Hodge decomposition, but an oriented surface’s integral intersection form restricts its cohomology transport more strongly than determinantone.

Definitions and assumptions

For finite inner-product cochain spaces with differentiald and$d^2=0$, let $\Delta=d^*d+dd^*$.

On an oriented genus-gsurface, the first integral cohomology carries a nondegenerate skew intersection matrix J.

Finite Hodge decomposition and symplectic necessity

C^k=im d_(k−1)⊕ker Δ_k⊕im d_k*orthogonally, andker Δ_krepresents H^k. Any orientation-preserving surface-induced transport Msatisfies M^TJM=J, hence lies in Sp(2g, Z), not merely SL(2g, Z).

Derivation or proof

The two image spaces are orthogonal because⟨du, d*v⟩=⟨d²u, v⟩=0. Also⟨x, Δ x⟩=||dx||²+||d*x||², so harmonic vectors are exactlykerd∩kerd*. The orthogonal complement of the two image spaces is precisely that intersection, proving the decomposition. Every cocycle is an exact part plus a harmonic part, and a harmonic exact vector has zero norm, so harmonic representatives are unique. For transport, the cup-product/intersection pairing is preserved by an orientation-preserving homeomorphism; writing this invariance in a basis gives M^TJM=J. In dimensiontwo this is equivalent todet M=1, but in higher dimensions it is stronger.

Worked example

For a three-edge cycle oriented consistently, take the incidence differentiald:C³→C³. Forg=2compare a determinant-one matrix with J=[[0, I₂],[−I₂,0]].

The harmonic edge space isspan(1,1,1), and its period is the sum of edge values. Let A=[[1,1],[0,1]]and M=diag(A, I₂). Then det M=1but M^TJM has upper-right block A^T≠I₂, so it is not symplectic and cannot represent the claimed oriented-surface transport.

Harmonic degrees of freedom and allowed transport have exact finite definitions and an explicit obstruction.

Scope

A numerical block labeled harmonic must be tied to an actual cochain complex and inner product. No recovered modified subdivision map was proved to commute withd or preserve this pairing.

Further development

For a refinement map R, verify Rd=d Rat the appropriate degrees and prove its induced map on harmonic representatives before interpreting a continuum transport limit.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R007.

Y25-11-023 Forced block dynamics and a genuine Schur-complement correction

2025-11-25 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How can a decaying curvature block be destabilized by harmonic forcing, and what does a Schur complement actually certify?

The recovered program described unstable transport feeding a contracting curvature sector and proposed flattening the off-diagonal blocks. Both mechanisms have exact elementary models. Their consequences depend on the actual blocks, not on labeling them cohomological.

Definitions and assumptions

Consider κ_(n+1)=a κ_n+c α_n and α_(n+1)=d α_n.

For a real symmetric block matrix S=[[A, B],[B^T, D]], assume Dpositive definite.

Exact forcing law and positive Schur criterion

Ifa≠d, κ_n=a^n κ₀+c(d^n−a^n)α₀/(d−a). Ifa=d, the forcing term iscn a^(n−1)α₀. Also S≥0if and only if A−BD^(−1)B^T≥0.

Derivation or proof

Iterating the scalar recurrence gives κ_n=a^n κ₀+c Σ_{j=0}^{n−1}a^(n−1−j)d^j α₀. The finite geometric sum is(d^n−a^n)/(d−a); when the two values coincide allnterms equal a^(n−1). Thus|a|<1alone does not guarantee curvature decay if|d|>1andc α₀≠0. For the Schur statement, complete the quadratic form:x^TAx+2x^TBy+y^TDy=x^T(A−BD^(−1)B^T)x+(y+D^(−1)B^Tx)^TD(y+D^(−1)B^Tx). Positivity of the two right terms proves sufficiency. Choosingy=−D^(−1)B^Txproves necessity. This is an energy statement for a symmetric matrix, distinct from the evolution eigenvalues of an arbitrary block update.

Worked example

Takea=1/2, d=2, c=1, κ₀=0, α₀=1. Separately take A=2, B=1, D=1.

κ_n=(2^n−2^(−n))/(3/2), which grows despite the contractinga. Settingc=0removes forcing and restores κ_n=(1/2)^n κ₀. The symmetric matrix[[2,1],[1,1]]has Schur complement1and is positive definite; increasing the coupling to B=2makes the Schur complement−2and destroys positivity.

The recovered stability mechanism has an exact solution and a sharp matrix criterion that can guide a real block analysis.

Scope

Flattening a block can change the intended geometry or dynamics; no universal optimizer speedup follows. The historical curvature eigenvalues and claimed ML gains were not independently supplied by stencils or experiments.

Further development

For varying blocks, prove uniform contraction and bound the forcing convolution; for a feedback correction, quantify the difference between the original and corrected operators.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R205.

Y25-11-026 Cancelling quadratic drift requires an expansion, not a benchmark label

2025-11-29 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What explicit assumption would justify the claimed improvement from quadratic to quartic refinement drift?

The recovered cohomological-correction discussion reported ε²to ε⁴improvement and an exact-rationalk=35validation. No underlying certificate was recovered. A standard extrapolation theorem shows a constructive mechanism with the exact assumptions and residual coefficient visible.

Definitions and assumptions

Suppose $F(h)=F_0+c\,h^2+d\,h^4+R(h)$, with$|R(h)|\le M\,h^6for0<h\le h_0$.

Define the corrected estimate $G(h)=[4F(h/2)-F(h)]/3$.

Explicit order-four extrapolation

$G(h)=F_0-d\,h^4/4+R_G(h)$, where$|R_G(h)|\le(17/48)M\,h^6$. Thus theh^2term cancels exactly and the remaining leading drift is quartic when $d\neq0$.

Derivation or proof

Substitute the expansion into$4F(h/2)-F(h)$. The constant coefficient is$3F_0$; the quadratic coefficient is$4c(h^2/4)-ch^2=0$; the quartic coefficient is$4d(h^4/16)-dh^4=-3dh^4/4$. Dividing bythree gives the formula. The remainder obeys$|R_G|\le[4M(h/2)^6+Mh^6]/3=(17/48)Mh^6$. This is a complete cancellation proof. It also identifies a necessary structural assumption:the samec, dand F_0must govern the two scales. If the correction changes the problem or if an oddh^3term is present, the claimed quartic order does not follow.

Worked example

Take $F(h)=7+3h^2+5h^4$exactly, and compareh=1andh=1/2.

$F(1)=15$, $F(1/2)=7+3/4+5/16=129/16$. Hence $G(1)=[4\cdot129/16-15]/3=23/4=7-5/4$. The uncorrected quadratic coefficient3has disappeared, and $G(h)-7=-5h^4/4$for everyh. Every value is rational, so no floating-point ambiguity is involved.

The improvement is a proved conditional approximation result with an exact example, rather than a claim inferred from a benchmark table.

Scope

The historicalk=35certificate, Loopperiod collapse, and specific corrected stencil were not supplied, so their advertised accuracy remains unverified. Benchmarks measure a chosen finite experiment, not an automatic theorem for every mesh.

Further development

Derive a uniform local error expansion for the actual corrected subdivision rule, then prove the cancellation operator preserves its geometric constraints and bound the remainder across valences.

Source receipt: months/11/root_receipts.json record 0. Mathematical references: R211.

December 2025

Dated fragments recovered; complete transcript unavailable. 27 developed treatments.

The final month contains stochastic actions, weighted triangular operators, umbral transforms, oscillator realizations, cyclic decimal dynamics, optimization geometry and vorticity estimates. The strongest treatments include derivations, exact finite examples and clearly scoped extensions.

ID · date	Concept and revised question	Result and mathematical core
Y25-12-024 2025-12-02 · U	Weighted threshold rank needs a tensorial transformation law When does the recovered weighted-translation matrix have a basis-invariant rank, and what capacity bound follows?	Exact rank invariance and a Hadamard counterexample rank(LTR)=rank Tfor invertible L, R. But transforming both μ and ΔΦ by the same invertible Ldoes not generally transform μ⊙ΔΦ by L, and can change its rank.
Y25-12-025 2025-12-03 · U	A spherical portal can cancel only specified spectral components What precise cancellation theorem applies to the recovered spherical-harmonic portal functional?	Exact spectral cancellation and a stability bound J annihilates every harmonic of odd degree and every degreeabovefour. For anyf∈L², J(f)=J(P_Vf), and|J(f)−J(g)|≤||c||₂||P_V(f−g)||₂.
Y25-12-026 2025-12-03 · U	Greedy spectral deflation converges to zero in trace norm Which clearly specified deflation process has a complete finite-rank convergence theorem?	Exact trace-norm tail formula After nsteps, Σ_n=Σ_(j>np)λ_je_j⊗e_jand||Σ_n||₁=Σ_(j>np)λ_j→0. In finite dimension the process terminates after at mostceiling(d/p)steps.
Y25-12-008 2025-12-04 · A	Freidlin-Wentzell action as an exact linear control problem Which part of the recovered rare-event formalism admits a complete exact minimization?	Exact minimum action For targetx_Tandd=x_T−e^(AT)x₀, the minimum cost is(1/2)d^TW_T^(−1)d. An optimal control isu*(s)=B^Te^(A^T(T−s))W_T^(−1)d.
Y25-12-007 2025-12-06 · A	Euler-Maruyama and an exact Ornstein-Uhlenbeck comparison Which drift, noise and step-size conventions make the recovered SDE chain reproducible?	Exact OU variance and discrete stability The continuous stationary variance is εσ²/(2λ). The discrete chain has a finite stationary variance exactly when0<λ h<2, and then it is εσ²/(2λ−λ²h).
Y25-12-010 2025-12-07 · U	A constant rectangular metric cannot make a detour shorter Can the same translation-invariant norm assign a detour action0.4and the straight path action1.6between identical endpoints?	Endpoint lower bound and the LC unit check Length_F(γ)≥F(γ(T)−γ(0)), with equality for the constant-speed straight segment. Forℓ=0.2andc=7.5×10^(−10), ω₀≈81649.658rad/sandf₀≈12994.947Hz.
Y25-12-012 2025-12-09 · U	Abel-Liouville controls volume, not every mode What does Abel's identity prove for the recovered two-component stress model?	Liouville determinant and sharp two-mode stability det X(t)=exp(∫₀ᵗtr A(s)ds). For α, β>0, the constant symmetric model is asymptotically stable exactly when κ²<αβ.
Y25-12-001 2025-12-10 · U	The Stirling-Bernoulli matrix is exact and singular What does A_nk=s(n, k)B_k actually do, and can it serve as an invertible change of coordinates?	Exact rows and a noninvertibility obstruction A=S₁diag(B₀, B₁,…), so its first rows are(1),(0,−1/2),(0,1/2,1/6),(0,−1,−1/2,0),(0,3,11/6,0,−1/30). Every finite truncation including index3is singular.

ID · date	Concept and revised question	Result and mathematical core
Y25-12-002 2025-12-10 · A	Bernoulli inversion and the corrected Blissard product Which umbral identities follow from the stated generating functions, with independence and multiplication specified?	Finite Bernoulli inverse and product identity M=(e^D−1)/D and M^(−1)=D/(e^D−1)=Σ_(k=0)^N B_k D^k/k!. Meanwhile B(t)E(t)=2t/(e^(2t)−1)=B(2t), not G(t).
Y25-12-003 2025-12-10 · U	Weighted ℓ^1 bounds use weighted columns What exact inequality should the recovered Lean structure prove, and how does it produce a causal envelope?	Weighted column theorem Tf is absolutely defined forf∈ℓ^1_Rand‖Tf‖_(1, R)≤C‖f‖_(1, R). For causal T_ij=Aq^(i−j) when i≥j and zero otherwise, assume A≥0 and 0≤q<1/R; then C=A/(1−q R).
Y25-12-004 2025-12-10 · A	Finite differences are genuine FIR annihilators How do Pascal and Stirling transforms lead to an exact finite filter rather than an undefined chaos correspondence?	Polynomial annihilation and inverse basis changes Δ(x)_n=n(x)_(n−1). Consequently Σ_jh_jp(x+j)=0for every polynomial of degree<m, and the value form-degreepwith leading coefficientais m!a. Signed and unsigned Stirling matrices are inverse changes between powers and falling powers.
Y25-12-005 2025-12-10 · A	Composition by $x+x^d$ is triangular but not uniformly banded What do degree growth and truncation say about the proposed finite-band composition operator?	Explicit degree growth and the missing conjugacy C_φ x^n=Σ_(j=0)^n C(n, j)x^(n+(d−1)j). Thus columnncontainsn+1nonzero terms spread over degreesnthroughdn, and there is no degree-independent finite bandwidth.
Y25-12-009 2025-12-10 · U	Polyhedral Fenchel duality and the drift-action distinction How do rectangular norms, Fenchel duality, and geometric rare-event actions fit together with correct hypotheses?	Rectangular dual and a scalar hinge Lagrangian The dual norm is $F^*(p)=a\lvert p_1\rvert+b\lvert p_2\rvert$. For $H(p)=p^2/2+\lvert p\rvert$, the conjugate is $L(v)=(1/2)(\max(\lvert v\rvert-1,0))^2$.
Y25-12-011 2025-12-10 · U	Escape probability inversion needs a rate reference scale What assumptions make the recovered probability-to-barrier inversion valid?	Exact rate and barrier inversion If0<P_t<1, then λ=−log(1−P_t)/tand W=−ε log[−log(1−P_t)/(t λ_0)]. The factor $\lambda_0$cannot be omitted unless a dimensionless time convention fixes it toone.
Y25-12-027 2025-12-10 · U	Cumulant evolution is triangular for an actual OU process What exact cumulant theorem can replace the proposed unproved first-cumulant/eigenvalue controller?	Exact OU cumulant equations $\kappa_1'=-\lambda\kappa_1+b$, $\kappa_2'=-2\lambda\kappa_2+\varepsilon\sigma^2$, and κ_n′=−n λκ_nforn≥3. In particular changingbonly shifts the mean and leaves λ and the variance-relaxation rate2λ unchanged.
Y25-12-006 2025-12-12 · A	The first logistic period doublings can be derived exactly What part of the Feigenbaum discussion can be proved directly without an unsupported operator identification?	Fixed point and two-cycle stability The nonzero fixed point1−1/ris stable for1<r<3. Forr>3, the two-cycle points arex_±=[r+1±√((r−3)(r+1))]/(2r); their multiplier is$-r^2+2r+4$, so the cycle is stable for$3<r<1+\sqrt{6}$.

ID · date	Concept and revised question	Result and mathematical core
Y25-12-013 2025-12-12 · A	Decimal remainders and the exact Koopman period When does a decimal remainder orbit havep−1modes, and how does its spectrum differ from an FFT fit?	Cycle length and Koopman roots of unity Every nonzero remainder orbit has lengthd, and F_p×splits into(p−1)/dcycles. On one cycle, the Koopman operator Uf=f∘Thas eigenvalues e^(2π ik/d), k=0,..., d−1.
Y25-12-016 2025-12-14 · U	Quantization error and Layer Norm require a positive denominator Which deterministic bounds can guard the recovered quantization and normalization pipeline?	Quantization and global normalization bounds Coordinatewise|Q_s(x_i)−x_i|≤s/2 without clipping. The Jacobian norm of LN_ε is at most1/√ε, so||LN_ε(x)−LN_ε(y)||≤||x−y||/√ε.
Y25-12-015 2025-12-15 · A	Mahalanobis proximal steps and what a tube projector preserves What exact optimization problem underlies a tube-aligned proximal step?	Exact diagonal proximal solution and projection obstruction If M=diag(m_i), the minimizer isd_i=−(η/m_i)sign(g_i)max(|g_i|−λ,0). If a gradient lies inker P, a projected update Pg is zero even when the objective is not stationary.
Y25-12-017 2025-12-20 · U	A noisy pitchfork has an exact barrier and a conditional Kramers law Which barrier and prefactor follow from the recovered quartic potential, and what remains a model of grokking?	Exact symmetric barrier The minima arem_±=±√α, the saddle ism_s=0, and Δ U=α²/4. The overdamped one-dimensional Kramers prefactor gives E τ~(√2π/α)e^(α²/(4D))under its standard small-noise assumptions for a well-to-well transition, or an absorbing boundary strictly beyond the saddle.
Y25-12-014 2025-12-22 · U	Softmax Fisher geometry and a uniform fixed-point diagnostic Which diagnostics of uniform and concentrated predictions follow exactly from softmax geometry?	Fisher null direction and local KL F is positive semidefinite with F1=0. At the uniform distribution, p_i=1/m, F=(1/m)(I−11^T/m), and H=logm. For a small logit perturbationh, KL(p(u)||p(u+h))=(1/2)h^TFh+O(||h||³).
Y25-12-018 2025-12-22 · U	The recovered oscillator has minimal continuous state dimension eight Can the exact eight-mode claim be proved from the recovered curve rather than declared by a certifier?	Exact order-eight annihilator and minimality Each coordinate is annihilated by P(D)=(D²+1)(D²+4)(D²+16)(D²+64)=D⁸+85D⁶+1428D⁴+5440D²+4096. The derivative-Hankel rank ofxiseight, so no lower-dimensional constant linear realization exists.
Y25-12-019 2025-12-22 · U	Exact windings of the recovered oscillator Can the reported indices−4and−6be certified without trusting a sampled phase unwrap?	Polynomial root counts and indices γ′=(i/2)u^(−8)P(u), with P=−9+2u⁴−3u⁶+u⁷+3u⁹−u¹⁰+6u¹²−3u¹⁶. Also w=(1/16)u^(−8)Q(u), with Q=81−20u⁴+36u⁶−16u⁷+4u¹⁰−36u¹²+21u¹⁶. P has six zeros in the unit disk, Q has two, and neither has a unit-circle zero.
Y25-12-021 2025-12-24 · U	A rigorous two-point-to-gradient inequality What precisely follows from the recovered finite-difference disorder statistic?	Exact weighted FTC-Jensen bound A(r)≤4r²∫||∇τ(y)||²J_r(y)dy, where J_r(y)=ess sup_(s∈[0,1])∫w(y−sh)w(y+(1−s)h)ρ_r(h)dh. If J_r≤C_int w² almost everywhere for a finite C_int>0, then ∫w²||∇τ||²≥A(r)/(4r²C_int).

ID · date	Concept and revised question	Result and mathematical core
Y25-12-023 2025-12-24 · U	A conditional proof chain is not a verified closure What does a theorem proved from named regularity axioms actually establish?	Countermodel test for an alleged unconditional conclusion The implication(P∧Q∧R)⇒Gcan be true even when Gis false, because the antecedent may be false. Likewise verifying eigenvalues of a hard-coded matrix certifies that matrix, not its provenance from an unavailable source operator.
Y25-12-022 2025-12-25 · A	Misalignment is nonnegative but cannot universally control gradient energy Can a two-point directional-disorder integral give an unconditional coercive bound for vorticity gradients?	Exact directional-disorder counterexample For everypairx, y, M(ω(x), ω(y))=0, but∫||∇ω||²is strictly positive. Hence no positive constantccan make a weighted two-point integral of Monly dominatec∫||∇ω||²for all smooth divergence-free velocity fields.
Y25-12-020 2025-12-31 · U	A capped-addition monoid with an exact trace ideal What general monoid theorem survives when the historical29×29operation table is unavailable?	Associative capped product and absorbing trace layer M is a monoid with identity(1_S,0), andtr(xy)=min(trx+try, k). The top layer S×{k}is a two-sided ideal. If S is a group andk>0, the idempotents are exactly(1_S,0)and(1_S, k).

Y25-12-024 Weighted threshold rank needs a tensorial transformation law

2025-12-02 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

When does the recovered weighted-translation matrix have a basis-invariant rank, and what capacity bound follows?

The recovered RG framework usedt_i=μ_i⊙ΔΦ_iandrank Tas an intrinsic memory count. Ordinary rank is invariant under invertible matrix changes, but a componentwise product is coordinate-dependent unless its factors have a compatible transformation law. Sparsity identifiability requires still more information.

Definitions and assumptions

Let Tbe a matrix whose columns aret_i. Rank is the dimension of their linear span.

A linear observation map J=BT factors through that span and therefore hasrank J≤rank T.

Exact rank invariance and a Hadamard counterexample

rank(LTR)=rank Tfor invertible L, R. But transforming both μ and ΔΦ by the same invertible Ldoes not generally transform μ⊙ΔΦ by L, and can change its rank.

Derivation or proof

Invertible multiplication preserves the dimensions of the image and kernel, provingrankinvariance for LTR. Image(BT)⊆B(image T)gives the rank bound. For the counterexample, take one column with μ=(1,0), ΔΦ=(0,1), so the Hadamard product is zeroandrank T=0. Use L=[[1,1],[1,−1]], which is invertible. The transformed factors are(1,1)and(1,−1), whose Hadamard product is(1,−1), so the newrankis one. Thus a claimed scheme-invariant UV count needs a tensor or coordinate rule beyond applying the same matrix to both factors. Also rankralone does not ensure unique recovery ofr-sparse coefficients:the1×2matrix[1,1]hasrank onebut sends both distinctone-sparse vectorse_1, $e_2$to the same datum.

Worked example

Compare exact rank with a numerical singular-value cutoff.

T=diag(1,10^(−13))has exactrank two, but a relative threshold10^(−12)reportsone. Under a perturbation of norm ε, the singular values can shift by at most ε; stable numericalrankrequires an explicit gap around the threshold, not merely a chosen tolerance.

The valid linear factorization theorem is preserved while the coordinate and identifiability assumptions are made explicit.

Scope

No recoveredproof establishes the proposed SM/seesawrankor B_5capacity interpretation. A physical threshold map needs independently specified fields, weights and reparametrization rules.

Further development

Write the weighted map as an actual tensor contraction, then prove its covariance and analyze sparse recovery through column dependencies or spark, not rankalone.

Source receipt: months/12/root_coverage.json record 0. Mathematical references: R205.

Y25-12-025 A spherical portal can cancel only specified spectral components

2025-12-03 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What precise cancellation theorem applies to the recovered spherical-harmonic portal functional?

The recovered portal construction sought weights killing odd spherical modes and all degrees abovefour, then extended the idea to graph Laplacians. This is a valid projection problem once the kernel is defined. It does not require an unexplained golden-ratio determinant.

Definitions and assumptions

Use orthonormal spherical harmonics Y_l^m on $S^2$and the L^2inner product.

Letcbelong to the finite-dimensional space V=span{Y_l^m:l=0,2,4}, and define J(f)=⟨c, f⟩.

Exact spectral cancellation and a stability bound

J annihilates every harmonic of odd degree and every degreeabovefour. For anyf∈L², J(f)=J(P_Vf), and|J(f)−J(g)|≤||c||₂||P_V(f−g)||₂.

Derivation or proof

Orthogonality gives⟨Y_l^m, Y_j^n⟩=δ_lj δ_mn. Expandingcin degrees0,2,4therefore makes all excluded harmonic coefficients vanish in the pairing. Decomposef=P_Vf+(I−P_V)f; the second component is orthogonal toc, proving the identity. Cauchy-Schwarz yields the error bound. Antipodal symmetryc(−x)=c(x)alone kills odd degrees because Y_l^m(−x)=(−1)^l Y_l^m(x), but does not kill even degreesabovefour; the finite spectral cutoff is a separate condition. The same proof works on a finite graphwith a real symmetric Laplacianand an orthonormal eigenbasis.

Worked example

Takec=Y_0^0+2Y_2^0andf=3Y_0^0+5Y_2^0+7Y_3^1+11Y_6^0.

J(f)=3+2·5=13. The odddegree-threeand evendegree-sixterms cancel for different reasons:parity and the explicit finite support. On a graph, choosingcfrom the firstthreeeigenvectors similarly annihilates all other eigenspaces; the numerical ordering must account for repeated eigenvalues.

The portal becomes a reproducible linear functional with exact cancellation and a sharp perturbation estimate.

Scope

The full historical4×4matrix field and annulus integral were not recovered, so no claimed φ cancellation or numerical portal table is inferred. A changing graphrequires tracking its eigenspaces and spectral gaps.

Further development

Provide the actual matrix-valuedfield, define its scalar observable independently, and compute its harmonic coefficients before designing the projection kernel.

Source receipt: months/12/root_coverage.json record 0. Mathematical references: R212.

Y25-12-026 Greedy spectral deflation converges to zero in trace norm

2025-12-03 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which clearly specified deflation process has a complete finite-rank convergence theorem?

The recovered iteration chose a top eigenspace of an accumulated covariance, which is different from simply removing the current largest eigenvalues. A precise current-covariance model has an exact theorem and clarifies why a nonzero residual or a fixed limiting subspace cannot be assumed.

Definitions and assumptions

Let $\Sigma_0 \geq 0$be trace class with orthonormal eigenvectorse_jand decreasing eigenvalues $\lambda_j \geq 0$.

At each step let P_nproject onto the nextpuncut eigenvectors and define $\Sigma_{(n+1)}=(I-P_n)\Sigma_n(I-P_n)$.

Exact trace-norm tail formula

After nsteps, $\Sigma_n=\Sigma_{(j>np)}\lambda_je_j\otimes e_jand||\Sigma_n||_1=\Sigma_{(j>np)}\lambda_j\to 0$. In finite dimension the process terminates after at mostceiling(d/p)steps.

Derivation or proof

The projection is diagonal in the same eigenbasis as Σ_nand removes exactly the selectedpterms. Induction gives the displayed expansion. For a positive trace-classoperator, the trace norm equals the sum of its nonnegative eigenvalues, so thetailformula follows. Summability of $\Sigma\lambda_j$makes that tailtend tozero. This proves a stronger convergence than strong-operator convergence. If the operator is merely compact, the largest remaining eigenvalue tends tozeroand operator-norm convergence still follows, but trace-norm statements need not make sense. Selecting the top eigenspace of an accumulatedsumcan repeatedly select already removed modes or requiretie rules; the samebatch formula cannot be copied to that different algorithm.

Worked example

Take $\Sigma_0=\mathrm{diag}(1,1/2,1/4,1/8,\ldots)$, p=1.

After nsteps, the trace norm is $\Sigma_{(j>n)}2^{(1-j)}=2^{(1-n)}$. Every residual direction is eventually removed, and the limit is zero. The selectedrank-oneprojectorsmove through the basis rather than converging in operator normto a fixedrank-oneprojector; their pairwise norm distance is one.

The revised model has exact convergence and shows which extra claims about a limiting subspace need separate proofs.

Scope

This is explicitly a newly specifiedcurrent-spectrum deflation model, not a proof that the recovered cumulative-history iteration obeys it. The latter needs its full discrete/continuous history convention and tie-breaking rule.

Further development

For the actual history-based iteration, derive scalar score dynamics under simultaneous diagonalizability and prove eventual removal of every positive mode before claiming an operator or Grassmannianlimit.

Source receipt: months/12/root_coverage.json record 0. Mathematical references: R202.

Y25-12-008 Freidlin-Wentzell action as an exact linear control problem

2025-12-04 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which part of the recovered rare-event formalism admits a complete exact minimization?

The recovered action is standard for small noise$\sqrt{\varepsilon}$ with a specified covariance. In a linear system it is the energy of a control that reaches a target, and the controllability Gramian solves the minimization exactly. This supplies a constructive result rather than merely a formal analogy.

Definitions and assumptions

Let$x' = Ax + Bu$and cost $S(u) = (1/2)\int_0^T ||u||^2 dt$, with initial$x_0$.

Let $W_T = \int_0^T e^{(A(T-s))} BB^T e^{(A^T(T-s))} ds$and assume $W_T$is invertible.

Exact minimum action

For target$x_T$and$d = x_T - e^{(AT)} x_0$, the minimum cost is$(1/2) d^T W_T^{(-1)} d$. An optimal control is$u^*(s) = B^T e^{(A^T(T-s))} W_T^{(-1)} d$.

Derivation or proof

Variation of constants gives$d = Ku$, where $Ku = \int e^{(A(T-s))} Bu(s) ds$. The adjoint is $K^* v(s) = B^T e^{(A^T(T-s))} v$, and $KK^* = W_T$. Thus$u^* = K^* W_T^{(-1)} d$satisfies $Ku^* = d$. Every other feasible control is$u^* + v$with $Kv = 0$. Orthogonality follows from$\langle u^*, v \rangle = \langle W_T^{(-1)} d, Kv \rangle = 0$, so$||u^* + v||^2 = ||u^*||^2 + ||v||^2$and the minimizer is unique.
Finally$||u^*||^2 = d^T W_T^{(-1)} K K^* W_T^{(-1)} d = d^T W_T^{(-1)} d$. For an SDE with$\sqrt{\varepsilon}$ B, this deterministic cost appears divided by ε in the large-deviation exponential; that asymptotic theorem needs its own hypotheses.

Worked example

Take the scalar stable system$x' = -x + u$, $x_0 = 0$, target$a$.

$W_T = (1 - e^{(-2T)})/2$, so $S_{min} = a^2/(1 - e^{(-2T)})$. As $T \to \infty$the cost tends$to a^2$, the quasipotential for the OU convention$dX = -X dt + \sqrt{\varepsilon}\, dW$. Its stationary density is proportional to$e^{(-x^2/\varepsilon)}$, consistent with the same exponent.

The action, the noise normalization and the minimum have a complete proof and an exact stochastic consistency check.

Scope

If $W_T$is singular, only reachable targets have finite energy and an inverse cannot be inserted without restricting to its range. General nonlinear exit-time prefactors are not proved by this linear result.

Further development

For a documented nonlinear system, compare local Gramian costs with a separately solved Hamilton-Jacobi quasipotential and quantify the region where linearization is accurate.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R213.

Y25-12-007 Euler-Maruyama and an exact Ornstein-Uhlenbeck comparison

2025-12-06 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which drift, noise and step-size conventions make the recovered SDE chain reproducible?

The recovered treatise connected ItôSDEs, Euler-Maruyamaand rare-event actions. A linear diffusion permits an exact comparison of the continuous and discrete stationary variances, so the numerical model can be audited without accepting unrelated spectral or particle ratios.

Definitions and assumptions

Letd X_t=−λ X_tdt+√εσ d W_twith λ>0.

Euler-Maruyama withsteph>0is X_(n+1)=(1−λ h)X_n+√(ε h)σξ_n, where ξ_nare independent standard normals.

Exact OU variance and discrete stability

> The continuous stationary variance is εσ²/(2λ). The discrete chain has a finite stationary variance exactly when0<λ h<2, and then it is εσ²/(2λ−λ²h).

Derivation or proof

Multiply the SDE bye^(λ t)and integrate to obtain X_t=e^(−λ t)X₀+√εσ∫₀ᵗe^(−λ(t−s))d W_s. The Itôisometry givesvariance εσ²(1−e^(−2λ t))/(2λ), whose limit is the stated value. For the discrete chain, seta=1−λ h. Independence givesv_(n+1)=a²v_n+ε h σ². A finite stable solution exists when|a|<1and equals ε h σ²/(1−a²)=εσ²/(2λ−λ²h). If|a|≥1, the accumulated noise variance Σ a^(2j)does not converge. The ratio to the continuous variance is1/[1−λ h/2], showing a systematic positive discretization bias.

Worked example

Take λ=σ=1andh=1/2.

The continuous variance is ε/2. The discrete variance is2ε/3, one third larger. As h→0the discrete formula tends to ε/2. A nonlinear rare-event calculation performed withh=1/2would therefore already inherit a known stationary error in this simplest test.

The SDE and its discretization have exact conventions, a proved stability region, and a quantitative bias.

Scope

The calculation does not prove a general rare-event asymptotic or validate a fitted rank-three throat, octonionic ratio, or neutrino claim. State-dependent diffusion needs the chosen Itô/Stratonovich convention and regularity assumptions.

Further development

For a nonlinear confining diffusion, derive a Lyapunov bound uniform in h and compare invariant measures before estimating exponentially rare transitions.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R214.

Y25-12-010 A constant rectangular metric cannot make a detour shorter

2025-12-07 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the same translation-invariant norm assign a detour action0.4and the straight path action1.6between identical endpoints?

The recovered calibration specified rectangular weights and desired actions for several paths. A norm obeys a sharp endpoint lower bound:the straight segment is always a shortest path in a constant norm. Thus such a calibration needs different endpoints, position dependence, a drift term, or a different functional.

Definitions and assumptions

For an absolutely continuous path $\gamma:[0, T]\to R^2$, set $\text{Length}_F(\gamma)=\int_0^T F(\gamma'(t))dt$, where Fis a fixed norm.

For uncoupled LC matrices $L=\ell I$, $C=c\ I$, the angular frequency is $\omega_0=1/\sqrt{(\ell c)}$.

Endpoint lower bound and the LC unit check

$\text{Length}_F(\gamma)\geq F(\gamma(T)-\gamma(0))$, with equality for the constant-speed straight segment. For$\ell=0.2$and$c=7.5\times10^{-10}$, $\omega_0\approx81649.658$rad/sand$f_0\approx12994.947$Hz.

Derivation or proof

The endpoint difference is$\int\gamma'dt$. The integral triangle inequality, or convex Jensen inequality applied to F, gives $F(\int\gamma'dt)\leq\int F(\gamma')dt$. For $\gamma(t)=\gamma(0)+t\ \Delta/T$, positive homogeneity gives $\text{Length}=T\cdot F(\Delta/T)=F(\Delta)$, proving attainability. Therefore no path between the same endpoints can have one quarter the straight-segment length under the same constantnorm. For LC units, the scalar equation$\ell q''+q/c=0$gives $\omega_0^2=1/(\ell c)$, and ordinary frequency is $\omega_0/(2\pi)$. The characteristic impedance is$\sqrt{(\ell/c)}$, a different dimension and operation.

Worked example

Take $F(v)=\max(|v_1|/a,|v_2|/b)$, with displacement(a, b).

The straight length is one. A path first moving by(a,0)and then(0, b)has lengthtwo, not less thanone. Multiplying the entire functional by κ scales both values by κ and cannot reverse their ordering. The recovered5800Hzdrive divided by the natural frequency is about0.446327, but this ratio alone does not determine a nonlinear path action.

The geometric incompatibility is exact, and the elementary oscillator units are independently reproducible.

Scope

This does not rule out drift-dependent or position-dependent actions, which can prefer nonstraight paths. Their additional terms must be specified before fitting numerical calibration targets.

Further development

Given the actual three paths and action density, compute their integrals symbolically or with certified bounds, and test whether the claimed endpoint and norm assumptions are truly shared.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R215.

Y25-12-012 Abel-Liouville controls volume, not every mode

2025-12-09 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What does Abel’s identity prove for the recovered two-component stress model?

The recovered application correctly described Abel-Liouvilleas structural rather than predictive. A two-dimensional linear system gives an exact determinant law and demonstrates why volume contraction does not imply stability of every direction.

Definitions and assumptions

Let $X'=A(t)X$with $X(0)=I$and continuous A.

For a constant symmetric model use $A=[[-\alpha, \kappa],[\kappa,-\beta]]$.

Liouville determinant and sharp two-mode stability

> $\det X(t)=\exp(\int_0^t \operatorname{tr} A(s)ds)$. For $\alpha, \beta>0$, the constant symmetric model is asymptotically stable exactly when $\kappa^2<\alpha\beta$.

Derivation or proof

Jacobi’s determinant formula gives$d(\det X)/dt=\det X\cdot\operatorname{tr}(X^{-1}X')=\det X\cdot\operatorname{tr} A$as long as $X$is invertible. Solving the scalar equation gives the displayed positive exponential, which also guarantees invertibility for the full interval and completes the argument. For the symmetric matrix, the eigenvalues are$-(\alpha+\beta)/2\pm\sqrt{((\alpha-\beta)^2/4+\kappa^2)}$. Both are negative exactly when the square root is smaller than$(\alpha+\beta)/2$, equivalently $\kappa^2<\alpha\beta$. The trace is negative regardless of κ, so a negative volume-growth exponent alone does not exclude an unstable positive eigenvalue.

Worked example

Take α=0.015, β=0.020and κ=0.010in one consistent inverse-time unit.

κ²=0.0001<αβ=0.0003, so both eigenvalues are negative anddet X=e^(−0.035t). If κ is increased to0.030, the trace and determinant law remain the same but κ²=0.0009>0.0003and one eigenvalue is positive. This distinction is exact and can be checked without simulation.

The identity supplies volume evolution and a separate spectral test supplies directional stability.

Scope

These coefficients are historical illustrative assumptions, not verified fault parameters or a rupture forecast. Nonlinear switching, forcing and constitutive laws can invalidate conclusions drawn from one linearization.

Further development

For time-varying coefficients, seek a common positive Lyapunov matrix or an exponential-dichotomy bound; the determinant alone cannot supply a uniform stability theorem.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R186.

Y25-12-001 The Stirling-Bernoulli matrix is exact and singular

2025-12-10 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What does A_nk=s(n, k)B_k actually do, and can it serve as an invertible change of coordinates?

The recovered matrix has a precise polynomial meaning and reproducible rational rows. Its zero odd Bernoulli numbers, however, make it singular from degree three onward. It cannot be used as an unrestricted coordinate conjugacy or automatically transport probability moments.

Definitions and assumptions

Signed Stirling numbers satisfy(x)_n=Σ_ks(n, k)x^k, where(x)_n=x(x−1)…(x−n+1).

Use Bernoulli numbers defined byt/(e^t−1)=Σ B_nt^n/n!, so $B_1=-1/2$.

Exact rows and a noninvertibility obstruction

A=S_1diag(B_0, B_1,…), so its first rows are(1),(0,−1/2),(0,1/2,1/6),(0,−1,−1/2,0),(0,3,11/6,0,−1/30). Every finite truncation including index3is singular.

Derivation or proof

The Stirling recurrence s(n+1, k)=s(n, k−1)−ns(n, k) follows by multiplying(x)_nby(x−n); starting withs(0,0)=1generates the stated rows. Multiplying each columnkby B_kgives A. The Stirling matrix is unit lower triangular, but its Bernoulli diagonal factor has $B_3=0$, so column3is zero and the truncation is singular. This is a structural obstruction to an expression A^(−1)BAon that full space. As a linear functional construction, if$\ell(x^k)=B_k\ \mu_k$, then$\ell((x)_n)=\Sigma\ s(n,\ k)B_k\ \mu_k=(A\ \mu)_n$; that is exact formal algebra. It does not say the output is a valid moment sequence.

Worked example

Take μ=(1,1/10,3/100,1/100), and compare a deterministic input X=10.

A μ=(1,−1/20,11/200,−23/200), matching the corrected historical example. If μ_k=10^k, the fourth transformed value is3·10+(11/6)·100−10000/30=−120. A probability measure cannot have a negative fourth raw moment, so this matrix is not a general moment-preserving stochastic map.

The algebraic core is exact, the table is corrected, and two overextensions fail by explicit finite certificates.

Scope

No Feigenbaum eigenvalue, cumulant-response law, or escape probability follows merely from this matrix. Finite triangularity is not boundedness on an infinite sequence space.

Further development

Restrict to a specified invariant subspace with nonzero Bernoulli diagonal, or use an actually invertible Sheffer basis transform; then prove the relevant weighted operator bounds before conjugating an infinite operator.

Source receipt: months/12/root_details.json record 0. Mathematical references: R133, R216.

Y25-12-002 Bernoulli inversion and the corrected Blissard product

2025-12-10 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which umbral identities follow from the stated generating functions, with independence and multiplication specified?

The recovered Blissard claims confused a product of exponential generating functions with the Genocchi generating function. Formal power series make the correction exact. A separate integration operator has a genuine Bernoulli inverse on every bounded-degree polynomial space.

Definitions and assumptions

Let D=d/dx on polynomials of degree≤N, so D^(N+1)=0.

Define M f(x)=$\int_0^1 f(x+t)dt$ and B(t)=t/(e^t−1), E(t)=2/(e^t+1), G(t)=2t/(e^t+1).

Finite Bernoulli inverse and product identity

M=(e^D−1)/D and M^(−1)=D/(e^D−1)=Σ_(k=0)^N B_k D^k/k!. Meanwhile B(t)E(t)=2t/(e^(2t)−1)=B(2t), not G(t).

Derivation or proof

Taylor’s formula is finite on this polynomial space:f(x+t)=Σ t^k D^kf/k!. Integrating t^k gives M=Σ D^k/(k+1)!, which is the formal quotient(e^D−1)/D. Its constant term is Iand the rest is nilpotent, so a finite geometric inverse exists. The Bernoulli generating identity identifies that inverse coefficient by coefficient. For the product, simply multiply denominators(e^t−1)(e^t+1)=e^(2t)−1; the claimed Genocchi function has a different denominator. If umbrae B, Eare independent in the usual product-functional sense, the coefficients of(B+E)^nare therefore2^n B_n, not G_n.

Worked example

Apply M^(−1)tox³and compare the first-degree product coefficient.

M^(−1)x³=x³−(3/2)x²+(1/2)x. Its average underx↦x+tis exactlyx³:integrating the expanded cubic cancels all lower terms. Atdegreeone, B(t)E(t)=1−t+…, whereas G(t)=t+…has zero constant term. The two formal series cannot be equal even before any numerical test.

The revised umbral development has an actual invertible finite operator and a decisive generating-function correction.

Scope

The Bernoulli differential operator here is not the singular matrix A_nk=s(n, k)B_kof the previous entry. Sharing Bernoulli coefficients does not identify the two maps.

Further development

Choose a Banach space of analytic functions and estimate the Bernoulli series there; the finite nilpotent proof alone does not justify an infinite differential operator.

Source receipt: months/12/root_details.json record 0. Mathematical references: R133.

Y25-12-003 Weighted ℓ^1 bounds use weighted columns

2025-12-10 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What exact inequality should the recovered Lean structure prove, and how does it produce a causal envelope?

The recovered formalization correctly centered a weightedℓ^1space but alternated between row and column bounds. For theℓ^1norm, the relevant quantity is a weighted column sum. This gives a complete theorem and a sharp geometric-tail example.

Definitions and assumptions

For R>0, set||f||_(1, R)=Σ_j R^j|f_j|.

Let T_ijbe a finite or countable matrix with C=sup_j Σ_i R^(i−j)|T_ij|<∞.

Weighted column theorem

Tf is absolutely defined forf∈ℓ¹_Rand||Tf||_(1, R)≤C||f||_(1, R). For causal T_ij=Aq^(i−j) when i≥j and zero otherwise, assume A≥0 and 0≤q<1/R; then C=A/(1−q R).

Derivation or proof

Apply the triangle inequality and Tonelli to nonnegative terms:Σ_i R^i|Σ_j T_ijf_j|≤Σ_j|f_j|Σ_i R^i|T_ij|≤C Σ_j R^j|f_j|. Finite right side proves absolute convergence as well as the bound. For the causal kernel, putm=i−j≥0; every weighted column sum is A Σ_(m≥0)(q R)^m=A/(1−q R). Positivity shows this norm is attained on a coordinate vector in the infinite model, so the bound is sharp. Composing Loperators with the same bound gives norm at most C^Lby repeated application; this is worst-case amplification, not a prediction for typical vectors. A finite bandwidth Mchanges the sum to A[1−(q R)^(M+1)]/(1−q R).

Worked example

Take A=1, q=1/2, R=1, and a three-row truncation with inpute₀.

Te₀=(1,1/2,1/4), whoseℓ¹norm is7/4. The finite column bound is7/4and the infinite bound is2. Atq R=1the infinite column sum diverges, although every finite truncation is a valid finite matrix. A row bound alone would instead naturally control anℓ∞norm.

The desired theorem is proved with exactly the summation hypothesis needed by a portable finite checker or a formal proof.

Scope

No end-to-end Lean compilation was recovered. An axiom namedcolumn_bounddoes not establish that a particular kernel satisfies it, and numerical#evaldoes not prove the infinite estimate.

Further development

Formalize the Tonelli inequality and the concrete kernel separately, then prove truncation convergence using a tail bound uniform in the retained columns.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R217.

Y25-12-004 Finite differences are genuine FIR annihilators

2025-12-10 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How do Pascal and Stirling transforms lead to an exact finite filter rather than an undefined chaos correspondence?

The recovered program named Pascal, Stirling and FIR filters. Their reliable common core is discrete polynomial calculus:shifts give Pascal coefficients, falling powers make differences triangular, and a finite binomial filter annihilates low-degree polynomial trends.

Definitions and assumptions

Define Ef(x)=f(x+1), Δ=E−I, and(x)_n=x(x−1)…(x−n+1).

An ordermfinite impulse-response filter has coefficientsh_j=(−1)^(m−j)C(m, j), j=0,…, m.

Polynomial annihilation and inverse basis changes

Δ(x)_n=n(x)_(n−1). Consequently Σ_jh_jp(x+j)=0for every polynomial of degree<m, and the value form-degreepwith leading coefficientais m!a. Signed and unsigned Stirling matrices are inverse changes between powers and falling powers.

Derivation or proof

Factor(x+1)_n−(x)_nby the common productx(x−1)…(x−n+2); the remaining difference is(x+1)−(x−n+1)=n. Iterating gives Δ^m(x)_n=n(n−1)…(n−m+1)(x)_(n−m), which vanishes forn<m. Since falling powers form a basis with leading coefficientone, the result extends to every polynomial and gives the top-degree constantm!a. The binomial theorem for commuting Eand Iexpands Δ^minto the displayed FIR coefficients. Expressing the same polynomial successively in the two triangular bases proves the inverse Stirling relation; it is a finite identity and needs no analytic convergence claim.

Worked example

Takem=3andp(x)=x²orx³.

The filter is(−1,3,−3,1). Applied tox²atx=0it gives0+3−12+9=0. Applied tox³it gives0+3−24+27=6=3!. Its frequency response is(e^(i ω)−1)³, whose magnitude is8|sin(ω/2)|³; low frequencies are suppressed cubically while high-frequency noise can be amplified.

The filter and basis calculus have exact identities and a transparent frequency interpretation.

Scope

Trend annihilation is not stochastic stability or Feigenbaum universality. A differencing filter can increase noise, and its performance depends on the signal and noise model.

Further development

Combine exact polynomial moment constraints with a norm-minimizing filter design, then prove its noise variance under a specified covariance matrix.

Source receipt: months/12/root_details.json record 0. Mathematical references: R216.

Y25-12-005 Composition by $x+x^d$ is triangular but not uniformly banded

2025-12-10 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What do degree growth and truncation say about the proposed finite-band composition operator?

The recovered warm-up composed functions withx+x^dand suggested conjugation by the Stirling-Bernoulli matrix. Polynomial algebra gives exact triangular coefficients, but its bandwidth grows with degree, and the proposed Bernoulli matrix is singular.

Definitions and assumptions

Fixintegerd≥2and φ(x)=x+x^d. Let C_φ f=f∘φ.

Use the monomial basis and distinguish a finite-degree truncation from an operator on all formal or analytic series.

Explicit degree growth and the missing conjugacy

C_φ x^n=Σ_(j=0)^n C(n, j)x^(n+(d−1)j). Thus columnncontainsn+1nonzero terms spread over degreesnthroughdn, and there is no degree-independent finite bandwidth.

Derivation or proof

Apply the ordinary binomial theorem to(x+x^d)^n=x^n(1+x^(d−1))^n. The lowest degree isnand the highest isdnwith nonzero coefficientone, proving both triangularity in degree and growing support. A projection to degrees≤Nproduces a finite triangular matrix, but its entries and omitted tail depend on N. A composition with an analytic nonpolynomial φ can have infinitely many nonzero coefficients even in a single column. An exact finite-band statement therefore needs an invariant basis and a proof, rather than an observation about one truncation. Further, the matrix A_nk=s(n, k)B_khas a zero column at k=3and no inverse on the full polynomial truncation, so A^(−1)B_φ Ais undefined there.

Worked example

Taked=2andn=3.

C_φ $x^3=(x+x^2)^3=x^3+3x^4+3x^5+x^6$. A cutoff atdegreefourretains$x^3+3x^4$and discards$3x^5+x^6$; on|x|≤rthe discarded magnitude is at most$3r^5+r^6$. This explicit tail is small only under a stated small-domain norm; the formal equality is not recovered by merely increasing matrix precision.

The composition map is completely explicit, with a concrete truncation error and a decisive obstruction to the claimed unrestricted conjugacy.

Scope

No exact Feigenbaum operator orcompact-perturbation theorem follows. The true renormalization derivative depends on a specified fixed point, normalization and analytic function space.

Further development

Choose a weighted analytic coefficient space, estimate C_φ on a disk mapped into its domain, and prove operator-norm convergence of the actual truncations before comparing spectra.

Source receipt: months/12/root_details.json record 0. Mathematical references: R218.

Y25-12-009 Polyhedral Fenchel duality and the drift-action distinction

2025-12-10 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How do rectangular norms, Fenchel duality, and geometric rare-event actions fit together with correct hypotheses?

The recovered engine combined polyhedral Finslergeometry and a Hamiltonian H(p)=p²/2+|p|. Convex duality makes that model exact, but it does not automatically identify a finite-temperature or drift-dependent stochastic action with a norm.

Definitions and assumptions

Forpositivea, b, set F(v)=max(|v₁|/a,|v₂|/b).

For a convex L, its Fenchelconjugate is H(p)=sup_v(p·v−L(v)).

Rectangular dual and a scalar hinge Lagrangian

The dual norm is F*(p)=a|p₁|+b|p₂|. For H(p)=p²/2+|p|, the conjugate is L(v)=(1/2)(max(|v|−1,0))².

Derivation or proof

The Funitballis the rectangle|v₁|≤a,|v₂|≤b. Maximizingp·vchooses each coordinate endpoint according to the sign ofp, giving the dual formula. For the scalar conjugate, symmetry reduces tov≥0, p≥0or comparison withp=0. There the objective ispv−p²/2−p=p(v−1)−p²/2. Ifv≤1its maximum onp≥0is zero atp=0; otherwise the maximizer isp=v−1and the value is(v−1)²/2. Reflectingvproves the formula. This Lhas a zero-cost interval and is not a positive norm. For the Freidlin-Wentzell geometric action, eliminating time gives||b||||v||−b·vfor identity covariance, which vanishes along positive drift directions; it too is not a strongly positive Finslernorm without extra conditions.

Worked example

Comparev=1/2,2, and a rectangular path displacement(a, b).

L(1/2)=0, L(2)=1/2, and F(a, b)=1. A positive energy penalty E>0in the free-time action(1/2)||x′−b||²+Ereplaces the geometric factor||b||by√(||b||²+2E), producing√(||b||²+2E)||v||−b·v, which is strictly positive forv≠0by Cauchy-Schwarz.

The duality is exact and the missing positivity assumption in the Finslerlabel is explicit.

Scope

Polyhedralnorms are nonsmooth, whereas classical smooth Finslergeometry assumes additional regularity. No exact rare-event calibration follows from choosing side lengths after seeing desired actions.

Further development

For a specified covariance metric, derive the free-time action by minimizing over local speed and verify positivity and convexity before calling it a Randersor Finslerstructure.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R215.

Y25-12-011 Escape probability inversion needs a rate reference scale

2025-12-10 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What assumptions make the recovered probability-to-barrier inversion valid?

The recovered scalarengine inverted an exponential exit law and identified its result with a geometric barrier. This is exact only for an assumed constant-hazard model, and a logarithm of a dimensional rate requires a reference prefactor.

Definitions and assumptions

For a constant rate $\lambda>0$, an exponential waiting time satisfies $P_t=P(\tau\le t)=1-e^{(-\lambda t)}$.

An Arrheniusmodel is $\lambda=\lambda_0 e^{(-W/\varepsilon)}$, with $\lambda_0$and λ having the same time-inverse units.

Exact rate and barrier inversion

If$0<P_t<1$, then $\lambda=-\log(1-P_t)/t$and $W=-\varepsilon\log[-\log(1-P_t)/(t\lambda_0)]$. The factor $\lambda_0$cannot be omitted unless a dimensionless time convention fixes it toone.

Derivation or proof

Solving$1-P_t=e^{(-\lambda t)}$by logarithms gives the rate. Dividing λ by $\lambda_0$in the Arrheniusrelation gives a dimensionless quantity whose logarithm is$-W/\varepsilon$. This proves the formula, with no approximation. The exponential distribution has constant hazard because its survival ratio $S(t+s)/S(t)=e^{(-\lambda s)}$depends only ons. A general first-passage distribution need not have this memoryless property, and a small-noise exit asymptotic does not assert an exact exponential law at every time. The derivative of Wwith respect to $P_t$is$-\varepsilon/[t\lambda_0\cdot(-\log(1-P_t)/(t\lambda_0))\cdot(1-P_t)]$, showing ill-conditioning near the endpoints.

Worked example

Take$t=2$, $\lambda_0=1$in chosen reciprocal-time units, and $P_t=1-e^{(-1)}$.

Then $\lambda=1/2$and $W=\varepsilon\log 2$. If the time unit changes, the numericalrate and $\lambda_0$both rescale, leaving $\lambda/\lambda_0$and Wunchanged. Omitting $\lambda_0$would make the inferred barrier depend on a unit choice.

The scalar inversion is a valid model calculation once the hazard and prefactor are stated.

Scope

No general identity Δ V=W/√detgwas recovered from a coordinate-invariant action; detgchanges under reparametrization. A first-cumulant correction requires a proved response map, not a dimensional guess.

Further development

Fit and test the full survival curve against a constant-hazard model, then propagate uncertainty through the logarithmic inversion with explicit prefactor uncertainty.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R213.

Y25-12-027 Cumulant evolution is triangular for an actual OU process

2025-12-10 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What exact cumulant theorem can replace the proposed unproved first-cumulant/eigenvalue controller?

The recovered engine used formal Stirling-Bernoullimaps as if they automatically described stochastic cumulants. For a specifiedlinear diffusion, the cumulants do obey an exact triangular system, but changing its mean does not alter its relaxation eigenvalue. This provides a clear positive model and a sharp limit.

Definitions and assumptions

Letd X=−λ Xdt+bdt+√εσ d Wwith λ>0and an initial moment-generating function nearzero.

Define K(s, t)=log E[e^(s X_t)]=Σ_(n≥1)κ_n(t)s^n/n!nearzero.

Exact OU cumulant equations

$\kappa_1' = -\lambda\kappa_1 + b$, $\kappa_2' = -2\lambda\kappa_2 + \varepsilon\sigma^2$, and $\kappa_n' = -n\,\lambda\kappa_n$forn≥3. In particular changingbonly shifts the mean and leaves λ and the variance-relaxation rate2λ unchanged.

Derivation or proof

The generator applied toe^(sx)is(−λ sx+bs+(εσ²/2)s²)e^(sx). Under the stated local exponential integrability, differentiating its expectation gives∂_t M=−λ s∂_s M+bs M+(εσ²/2)s²M. Divide by Mto obtain∂_t K=−λ s∂_s K+bs+(εσ²/2)s². Compare powers ofsusing the exponential-generating normalization:the linear term gives κ_1', the quadratic term gives κ_2', and no newhigher cumulantsare forced. Equivalently the explicit OU solutionis an affine image of X_0plus an independent Gaussian; the same equations follow from cumulant additivity and scaling.

Worked example

Take λ=σ=1andb=3.

The stationary mean isthreeand variance ε/2; all stationaryhigher cumulantsvanish. Replacingbbyfourchanges only the mean. It cannot move the drift Jacobian from−1to another eigenvalue. Any claimed first-cumulantcontrollerof a spectral escape rate therefore needs a different, explicit coupling mechanism.

The stochastic cumulantpropagationis proved for a genuine process, and the missing control assumption is isolated.

Scope

This does not identify the singular Stirling-Bernoullimatrixwith the OU generatoror Feigenbaumrenormalization. For nonlinear drifts, highermomentsgenerally couple upward and finite closure is an approximation unless proved.

Further development

For a nonlinear model, derive its exact generator moment hierarchy, then quantify a truncation error before using a finite cumulant map for rate control.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R214.

Y25-12-006 The first logistic period doublings can be derived exactly

2025-12-12 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What part of the Feigenbaum discussion can be proved directly without an unsupported operator identification?

The recovered diffusion-model appendix invoked the established Feigenbaum phenomenon and conjectured a Bernoulli-diagonal representation. Exact logistic algebra supplies a substantial baseline:the fixed-point instability and the complete stable two-cycle interval.

Definitions and assumptions

Letf_r(x)=rx(1−x)withr>0.

A periodic orbit is linearly stable if the absolute product of derivatives over one period is less thanone.

Fixed point and two-cycle stability

The nonzero fixed point1−1/ris stable for1<r<3. Forr>3, the two-cycle points arex_±=[r+1±√((r−3)(r+1))]/(2r); their multiplier is−r²+2r+4, so the cycle is stable for3<r<1+√6.

Derivation or proof

Solvingf_r(x)=xgives zero and1−1/r. At the latter, f_r′=r(1−2x)=2−r, whose modulus is less thanone exactly for1<r<3. Factoringf_r²(x)−xby the two fixed-point factors leaves the quadraticr²x²−r(r+1)x+(r+1)=0. Its discriminant isr²(r−3)(r+1), giving the displayed cycle. The sum and product of its points are(r+1)/rand(r+1)/r².
Hencef′(x_+)f′(x_-)=r²[1−2(x_++x_-)+4x_+x_-]=−r²+2r+4. This decreases fromoneatr=3to−oneatr=1+√6; the strict stability interval follows.

Worked example

Taker=16/5=3.2.

The multiplier is−256/25+32/5+4=4/25, strictly inside the unit circle. Atr=3the two points coincide and the fixed point loses stability; atr=1+√6the cycle multiplier reaches−1, marking the next doubling threshold.

The first two bifurcations are exact algebraic facts; they do not rely on a fitted universal constant.

Scope

Feigenbaum universality and renormalization hyperbolicity are deep theorems under specific analytic/unimodal hypotheses. A diffusion or transformer operator is not in that class merely because one can write S^(−1)T²S.

Further development

For a proposed new renormalization model, specify its function space and normalization, construct its fixed point, and compare finite truncations with certified operator-norm residuals.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R219.

Y25-12-013 Decimal remainders and the exact Koopman period

2025-12-12 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

When does a decimal remainder orbit havep−1modes, and how does its spectrum differ from an FFT fit?

The recovered prime dataset was interpreted through primitive roots and cyclic Koopman eigenvalues. The cycle length is the multiplicative order of10modp, which need not bep−1. This provides exact arithmetic against which any PCAor FFT interpretation must be checked.

Definitions and assumptions

For primepnot dividing10, let T(r)=10rmodp on F_p×.

Letd=ord_p(10), the least positive integer with10^d≡1modp.

Cycle length and Koopman roots of unity

Every nonzero remainder orbit has lengthd, and F_p×splits into(p−1)/dcycles. On one cycle, the Koopman operator Uf=f∘Thas eigenvalues e^(2π ik/d), k=0,..., d−1.

Derivation or proof

T^nr=rholds exactly when10^nr≡rmodp. Since r≠0it can be cancelled, giving10^n≡1; hence the orbit length isd. The disjoint cycles partitionp−1points. Label a cycle byr_j=10^jr_0, jmodd. The characterf_k(r_j)=e^(2π ikj/d)satisfies Uf_k(r_j)=f_k(r_(j+1))=e^(2π ik/d)f_k(r_j). These characters form a complete discrete Fourier basis. A digit observable can have zero coefficients in some eigendirections, so an FFTof one observable need not visibly show every Koopman eigenvalue. PCAor Isomapcoordinates further depend on the supplied dataset and metric.

Worked example

Comparep=7andp=11.

Modulo7, the orbit1,3,2,6,4,5has length6=p−1. Modulo11, the orbit1,10has length2, so there arefivecycles and no single length-tenorbit. Both primes are perfectly valid inputs, but thefull-cycleformula only applies to the first. Artin’sprimitive-root conjecture is a separate global statement about how often orderp−1occurs.

The spectral period is exactly computable from modular arithmetic and cannot be assigned from the prime’s size alone.

Scope

The historical FFT peak and entropy tables had no recovered raw digit arrays or normalization, so their numerical values and branch interpretation remain unverified.

Further development

Compute the exact orders for every recovered prime, then compare digit-observable Fourier coefficients using one declared sampling and entropy convention.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R220.

Y25-12-016 Quantization error and Layer Norm require a positive denominator

2025-12-14 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which deterministic bounds can guard the recovered quantization and normalization pipeline?

The recovered run failed numerically after a tube verification and proposed scale floors and rollback. A scale floor is a mathematical domain condition; a code label or verifier score cannot replace it. Simple bounds locate the quantization and normalization sensitivities.

Definitions and assumptions

For a nonzero scale s, uniform nearest quantization is Q_s(x)=s round(x/s).

Layer normalization on R^m is LN_ε(x)=(x−mean(x)1)/√(m^(−1)||x−mean(x)1||²+ε), with ε>0.

Quantization and global normalization bounds

Coordinatewise|Q_s(x_i)−x_i|≤s/2 without clipping. The Jacobian norm of LN_ε is at most1/√ε, so||LN_ε(x)−LN_ε(y)||≤||x−y||/√ε.

Derivation or proof

Nearest-integer rounding changes a scalar by at mostone half, proving the first bound after multiplication bys. For LN, let P=I−11^T/m, z=Px, andv=||z||²/m. Its derivative is(1/√(v+ε))P−zz^T/[m(v+ε)^(3/2)]. On the centered subspace, directions perpendicular tozhave eigenvalue1/√(v+ε); thezdirection has eigenvalue ε/(v+ε)^(3/2). The constant direction has eigenvaluezero. All are nonnegative and at most1/√ε. Integrating the derivative along the segment betweenxandygives the Lipschitz bound. If ε=0, normalization is undefined for constant inputs and no global bound exists.

Worked example

Takem=2, x=(a, a), and a quantization scale derived only frommax|x_i|.

Atx=0, a scale formula s=max|x|/(2^(b−1)−1)gives s=0, andx/scannot be evaluated. A strictly positive scale floor defines the map there. For a perturbation(δ,−δ), LN_ε returns(δ,−δ)/√(δ²+ε), showing the largest sensitivity nearzero variance.

The proposed safeguards have precise mathematical purposes and quantified effects.

Scope

Finite arithmetic overflow, clipping and mixed-precision accumulation need additional range analysis. A historical99.93%validation result is not a bound on all future runs or all numerical paths.

Further development

Propagate the quantization bound through every normalization and activation using actual local denominator lower bounds, then compare the certified error budget with the model’s decision margin.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R221.

Y25-12-015 Mahalanobis proximal steps and what a tube projector preserves

2025-12-15 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What exact optimization problem underlies a tube-aligned proximal step?

The recovered optimizer narrative combined a Mahalanobis geometry, spectral tubes, a hostileℓ^1term and delayed commitment. A finite convex update is well-defined and solvable; a low-dimensional projection can also erase the entire useful gradient. Both facts can be proved without claiming an optimizer improvement.

Definitions and assumptions

Let M be symmetric positive definite. Considerargmin_d{g^Td+(1/(2η))d^TMd+λ||d||₁}.

Let P be an orthogonal projector onto a proposed retained subspace.

Exact diagonal proximal solution and projection obstruction

If M=diag(m_i), the minimizer isd_i=−(η/m_i)sign(g_i)max(|g_i|−λ,0). If a gradient lies inker P, a projected update Pg is zero even when the objective is not stationary.

Derivation or proof

The diagonal objective separates by coordinate. On a positive or negative half-line, differentiation givesg_i+(m_i/η)d_i+λ sign(d_i)=0; checking the subgradient interval atzero yields the soft-threshold formula. Strict convexity from M>0makes the minimizer unique. For projection, if Pg=0then the projected iteration takes no descent direction althoughgmay be nonzero. For example the quadratic objectivef(x, y)=(x−1)²/2+(y−1)²/2at(1,0)has gradient(0,−1). Projecting onto thex-axis kills it, so the constrained point is stationary only within the chosen tube, not in the ambient problem.

Worked example

Take M=diag(1,4), η=1, g=(3,1), λ=2.

The update isd=(−1,0). The second coordinate is suppressed by thresholding, not by a proof that it is irrelevant. In the projection example a spectral gap or low turnover of the retained subspace cannot detect the losty-direction:the projector may remain perfectly stable while optimization stalls.

The finite update has an exact solution, and premature tube commitment has a concrete mathematical failure mode.

Scope

No theorem says that observed gap, spike, turnover or effective rank predicts a useful invariant optimization subspace. The reported negative-control failure is historical experimental evidence, not a universal result.

Further development

Prove a bound on the discarded gradient||(I−P)∇f|| and include it in the descent estimate before replacing an ambient optimizer by its projected version.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R222.

Y25-12-017 A noisy pitchfork has an exact barrier and a conditional Kramers law

2025-12-20 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which barrier and prefactor follow from the recovered quartic potential, and what remains a model of grokking?

The recovered framework proposed a pitchfork coordinate and a batch-size/learning-rate noise level. The potential’s critical points and barrier can be derived exactly. Relating them to training time requires an independently justified reduction and a small-noise asymptotic.

Definitions and assumptions

Let $U(m)=m^4/4-\alpha\, m^2/2$ with $\alpha>0$, and use $d\,M=-U'(M)dt+\sqrt{(2D)}d\,W$.

A local minimum has positive U''; the intervening saddle in one dimension is a local maximum.

Exact symmetric barrier

The minima are$m_{\pm}=\pm\sqrt{\alpha}$, the saddle is$m_s=0$, and $\Delta\, U=\alpha^2/4$. The overdamped one-dimensional Kramers prefactor gives $E\,\tau\sim(\sqrt{2\pi}/\alpha)e\hat{}(\alpha^2/(4D))$under its standard small-noise assumptions for a well-to-well transition, or an absorbing boundary strictly beyond the saddle.

Derivation or proof

$U'=m(m^2-\alpha)$, so the critical points arezero and$\pm\sqrt{\alpha}$. $U''=3m^2-\alpha$ is$-\alpha$ at zero and2α at either minimum. The minimum value is$-\alpha^2/4$and the saddle valuezero, proving the barrier. Substituting these curvatures in$2\pi/\sqrt{(|U''(m_s)|U''(m_a))}$gives$\sqrt{2\pi}/\alpha$. The exponential convention usesnoise$\sqrt{(2D)}$; with$\sqrt{D}$the rate exponent would instead be$2\Delta\, U/D$. For a saddle-node normal form $U(x)=x^3/3-\kappa\, x$, $\kappa>0$, the critical points$\pm\sqrt{\kappa}$ have barrier$4\kappa\hat{}(3/2)/3$and curvature magnitudes$2\sqrt{\kappa}$, explaining the recovered $\kappa\hat{}(3/2)$barrier and $\kappa\hat{}(-1/2)$prefactor scaling.

Worked example

Take $\alpha=2$and assume $D=c_0\eta/B$.

The exact barrier is one and the formal mean time is$(\pi/\sqrt{2})\exp(B/(c_0\eta))$. This exponential dependence is extremely sensitive to the assumed reduced-noise coefficient. Changing the neural coordinate or the fitted drift without rederiving the action can change the barrier.

The polynomial model and its scaling are mathematically coherent, while the connection to grokking remains conditional.

Scope

Kramers’formula is asymptotic, not an exact finite-noise identity. No recovered proof maps the full optimizer state to this scalar diffusion with the claimedc_0, drift and metastable wells.

Further development

Estimate the reduced drift and diffusion from independent trajectories, check the Markov approximation, and test the predicted barrier scaling over several noise levels before extrapolating training times.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R223.

Y25-12-014 Softmax Fisher geometry and a uniform fixed-point diagnostic

2025-12-22 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which diagnostics of uniform and concentrated predictions follow exactly from softmax geometry?

The recovered grokking logs reported near-uniform entropy, nearly constant Fisher weights and a vanishing update, followed by a highly concentrated run. The underlying probability geometry is exact, but it does not turn one telemetry pattern into a general grokking predictor.

Definitions and assumptions

For logitsu∈R^m, setp_i=e^(u_i)/Σ e^(u_j). The categorical Fisher matrix is F=diag(p)−pp^T.

Entropy is H(p)=−Σ p_i logp_i, and KL(q||p)=Σ q_i log(q_i/p_i).

Fisher null direction and local KL

F is positive semidefinite with F1=0. At the uniform distribution, p_i=1/m, F=(1/m)(I−11^T/m), and H=logm. For a small logit perturbationh, KL(p(u)||p(u+h))=(1/2)h^TFh+O(||h||³).

Derivation or proof

For everyv, v^TFv=Σ p_iv_i²−(Σ p_iv_i)², which is the variance ofvunderp and hence nonnegative. Constants have zero variance, proving the null direction caused by invariance under adding a common logit. Differentiating the log-partition A(u)=log Σ e^(u_i)gives∇A=p and∇²A=F. The stated KL is A(u+h)−A(u)−p(u)·h; Taylor expansion gives the quadratic term. Substituting uniformp gives the exact projector formula and entropy. Thus a natural-gradient inverse requires restricting away from the constant direction or adding a specified regularizer.

Worked example

Takem=97, as in the recovered modular-addition experiment.

Uniform entropy islog97≈4.574711, and every nonconstant Fisher eigenvalue is1/97. A diagonal-only normalized Fisher can look nearly constant without identifying the full Fisher spectrum. If an imposed schedule drives the step tozero, parameters can freeze at uniform prediction even though no useful solution was learned.

The diagnostic has exact mathematical anchors and a precise degeneracy that an optimizer must handle.

Scope

The historical validation accuracies and speedups cannot be reproduced without the programs, data and checkpoints. Low entropy alone may mean confident errors, and a tiny update does not certify convergence to a good solution.

Further development

Estimate the projected Fisher and actual KL per step, then compare prediction accuracy and held-out loss independently of the entropy or spectral diagnostic.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R224.

Y25-12-018 The recovered oscillator has minimal continuous state dimension eight

2025-12-22 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the exact eight-mode claim be proved from the recovered curve rather than declared by a certifier?

The exact curve is γ(t)=(cos t+(1/2)cos2t+(1/2)cos4t+(3/8)cos8t,2sin t−sin2t+sin4t−(3/4)sin8t). Its continuous linear realization has dimensioneight. This is a verifiable result independent of any claimed subdivision or motivic origin.

Definitions and assumptions

Write γ=(x, y), with frequencies1,2,4,8.

A continuous linear realization is γ(t)=Ce^(At)v with constant real matrices; its derivative Hankel matrix has entriesx^(i+j)(0).

Exact order-eight annihilator and minimality

Each coordinate is annihilated by P(D)=(D²+1)(D²+4)(D²+16)(D²+64)=D⁸+85D⁶+1428D⁴+5440D²+4096. The derivative-Hankel rank ofxiseight, so no lower-dimensional constant linear realization exists.

Derivation or proof

Each sinusoidal frequency solves its corresponding second-order factor, and the commuting factors annihilate their sum. Four real rotation blocks therefore construct an eight-dimensional realization. To prove minimality, write x(t)=Σ_(λ∈{±i,±2i,±4i,±8i})c_λ e^(λ t), whereeveryc_λ is nonzero. Thenx^(n)(0)=Σ c_λλ^n. The8×8Hankel matrix equals Vdiag(c_λ)V^T, where V_j λ=λ^j. Its determinant is(∏c_λ)(∏_(λ<μ)(μ−λ)²), nonzero because the eight λ are distinct. Anyd-dimensional realization factors every derivative Hankel matrix through R^d, so its rank is at mostd. The rank-eight lower bound matches the construction.

Worked example

Compute the exact derivative certificate and its sampling caveat.

The8×8determinant is331749050780925495198351360000000; the9×9determinant is zero and the displayed recurrence holds for every derivative index. Discrete samples at step Δ tretain rankeight only if the eight numbers e^(±ik Δ t), k=1,2,4,8, are distinct and no observable coefficient is lost. At Δ t=2π the entire sampled curve is constant.

The original continuous eight-state claim is proved with an exact rank certificate and explicit aliasing conditions.

Scope

It is a minimal linear realization, not a universal nonlinear latent-dimension theorem. Topological indices alone do not impose the same rank lower bound on arbitrary nonlinear models.

Further development

For a chosen nonaliased sampling step, derive the exact discrete recurrence and compare finite noisy Hankel singular values against a perturbation bound rather than a fixed rank threshold.

Source receipt: months/12/root_specifics.json record 0. Mathematical references: R225.

Y25-12-019 Exact windings of the recovered oscillator

2025-12-22 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the reported indices−4and−6be certified without trusting a sampled phase unwrap?

The recovered exact curve permits polynomial argument-principle certificates. Its tangent never vanishes, and the winding ofw=z+iz′is−6. The tangent winding itself is−2; the reported−4uses the explicitly stated doubled tangent-line convention.

Definitions and assumptions

Letz=x+iyfor the curve in the preceding entry andu=e^(it).

Usewind(f)=(1/(2π))Δ argf for a nonvanishing2π-periodic complex function.

Polynomial root counts and indices

γ′=(i/2)u^(−8)P(u), with P=−9+2u⁴−3u⁶+u⁷+3u⁹−u¹⁰+6u¹²−3u¹⁶. Also w=(1/16)u^(−8)Q(u), with Q=81−20u⁴+36u⁶−16u⁷+4u¹⁰−36u¹²+21u¹⁶. P has six zeros in the unit disk, Q has two, and neither has a unit-circle zero.

Derivation or proof

Convert every cosine and sine to u^k andu^(−k); differentiation multiplies the kth Laurent coefficient byik, giving the two displayed polynomials exactly. Their disk counts are certified by a strict integer recursion. For a real polynomialpwith leadinga, constantband reversalp#, if|a|>|b|reduce to(ap−bp#)/uand addone to the disk count; if|b|>|a|reduce tobp−ap#and addzero. At each step remove a common nonzero integer factor. A boundary zero ofpwould also be a boundary zero of the reduced polynomial. Reaching a nonzero constant therefore proves absence of boundary zeros inductively. On the circle|p|=|p#|, so Rouché gives the asserted count update. The attached exact recursion yields6and2. The argument principle then subtracts the Laurent pole order8, giving wind γ′=−2andwindw=−6. There are also exact rational Bezout certificates U P+V P#=1 and U_Q Q+V_Q Q#=1. On the unit circle |P#|=|P|, so 1≤(||U||_coeff,1+||V||_coeff,1)|P|. The verified coefficients give |P|≥6/35 and |Q|≥168643618304708106 0/2638949151339309251.

Worked example

Compare the continuous certificate to the historical sampled minima.

The Bezout bounds prove |γ′|≥3/35 and |w|>1/26, independently of the sampled minima. Uniform perturbations of the two respective complex functions by less than these margins preserve their windings: the straight-line homotopy stays away from zero by the triangle inequality. With the convention μ=Δ arg γ′/π=2wind γ′, the Maslov-style integer is−4. Using the ordinary vector winding instead gives−2, so the convention is essential.

The two historical indices are supported by exact finite algebra, with the doubling convention made explicit.

Scope

The certificate establishes no Catmull-Clark source, quantum state, or motivic quotient. These Laurent-polynomial facts are properties of the recovered curve itself.

Further development

Classify which coefficient deformations can cross the unit-root discriminant, and compare those genuine topological transitions with the independent continuous-realization rank and sampling-aliasing discriminants.

Source receipt: months/12/root_specifics.json record 0. Mathematical references: R188.

Y25-12-021 A rigorous two-point-to-gradient inequality

2025-12-24 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What precisely follows from the recovered finite-difference disorder statistic?

The recovered Navier-Stokes program contains a valid local geometric estimate. Its conversion to a weighted gradient lower bound requires an explicit intermittency factor; that factor is a hypothesis, not a consequence of unit-length directions.

Definitions and assumptions

Let $\tau:T^3\to S^2$ be smooth, $w\in L^2(T^3)$ nonnegative, and $\rho_r\in L^1(R^3)$ nonnegative with total mass one and support $|h|\le 2r$. Evaluate torus functions periodically for $h\in R^3$.

Set $A(r)=\int\int w(x)w(x+h)[1-(\tau(x)\cdot\tau(x+h))^2]\rho_r(h)dhdx$.

Exact weighted FTC-Jensen bound

$A(r)\le 4r^2\int||\nabla\tau(y)||^2J_r(y)dy$, where $J_r(y)=\operatorname{ess\,sup}_{s\in[0,1]}\int w(y-sh)w(y+(1-s)h)\rho_r(h)dh$. If $J_r\le C_{int}\,w^2$ almost everywhere for a finite $C_{int}>0$, then $\int w^2||\nabla\tau||^2\ge A(r)/(4r^2C_{int})$.

Derivation or proof

For unit vectors$u, v, 1-(u\cdot v)^2=(1-u\cdot v)(1+u\cdot v)\le 2(1-u\cdot v)=||u-v||^2$. Along the straight torus displacement$h$, the fundamental theorem gives $\tau(x+h)-\tau(x)=\int_0^1\nabla\tau(x+sh)h\,ds$. Jensen bounds its squared norm by$|h|^2\int_0^1||\nabla\tau(x+sh)||^2ds$. Insert this in A, use$|h|^2\le 4r^2$, and change variables$y=x+sh$. Tonelli applies because all terms are nonnegative. The resulting$h$-integral at fixed$s, y$is bounded by $J_r(y)$; integrating$s$gives the claimed inequality. The final lower bound follows only after inserting $J_r\le C_{int}\,w^2$. The alternative assumption $J_r\le C_{int}\,w$ instead yields a lower bound for $\int w||\nabla\tau||^2$; these are different hypotheses. If w vanishes where J_r is positive, no finite factor exists. Essential suprema avoid dependence on representatives in the interpolation parameter.

Worked example

Take$w=1$and$\text{normalize}\int\rho_r dh=1$.

Then $J_r=1$and $A(r)\le 4r^2\int||\nabla\tau||^2$. A constant τ has $A=0$andzero gradient. The inequality goes in the useful lower-bound direction only when a positive disorder $A$is independently known; it cannot create a positive lower bound from the definitions alone.

The valid recovered chain is proved with every weight and factor visible.

Scope

This is a direction-field estimate, not global Navier-Stokes regularity. It supplies neither the required disorder lower bound nor control of a nonlinear production term. Its assumptions may fail at vorticity zeros or under strong intermittency.

Further development

Derive or falsify a uniform $C_{int}$for the actual solution class, and state any nonvanishing or regularized direction convention before using the estimate in an enstrophy argument.

Source receipt: months/12/root_specifics.json record 1. Mathematical references: R226.

Y25-12-023 A conditional proof chain is not a verified closure

2025-12-24 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What does a theorem proved from named regularity axioms actually establish?

The recovered Leanfiles retainedaxioms and anadmitat the production-versus-dissipation step, while earlier output claimed compilation and completion. A precise logical model distinguishes a valid conditional theorem from evidence that its analytic hypotheses hold. The same distinction applies to a certifier that inserts its target matrix.

Definitions and assumptions

Suppose statements P, Q, Rstand for disorder, intermittency and transverse regeneration, and Gfor global regularity.

A theorem deriving Gfrom assumptions P, Q, Rproves$(P \wedge Q \wedge R) \Rightarrow G$; it does not prove the antecedent.

Countermodel test for an alleged unconditional conclusion

The implication$(P \wedge Q \wedge R) \Rightarrow G$can be true even when Gis false, because the antecedent may be false. Likewise verifying eigenvalues of a hard-coded matrix certifies that matrix, not its provenance from an unavailable source operator.

Derivation or proof

Use a Boolean model with P=false, Q=true, R=true, G=false. The conjunction is false, and material implication fromfalseis true. Thus the conditional theorem can be logically valid in a model where Gfails, proving that unconditional Gdoes not follow from that implication alone. In a proof assistant, the theorem’s dependency list records the assumed constants or axioms used; compilation checks the derivation relative to them. Replacing a missing analytic estimate by an axiom moves it into the assumptions rather than solving it. For matrix certification, the same separation is algebraic:if $M_2=[[5/8,631/25000],[0,5/8]]$is declared, its characteristic polynomial is$(t-5/8)^2$and its nilpotent residual is square-zero. Those facts do not prove $M_2$was extracted from Catmull-Clarkdata.

Worked example

Apply this audit to the recovered GC1-GC3chain and the v12 atlas certifier.

The output should list every unproved hypothesis and every independently supplied input. An exhaustive finite matrix test can be fully correct while the claimed source matrix remains unverified. A numerical#evalor a successful script run is also weaker than a theorem about an infinite operator.

The useful proof chain and finite checks can be retained with honest dependency boundaries.

Scope

This entry supplies a logic result and a finite matrix example, not an independent Leancompilation or a proof of Navier-Stokesregularity. The original full files and toolchain were not recovered.

Further development

Produce a machine-readable axiom/dependency audit and a source-to-matrix extraction certificate, then reserve “verified” for the exact statement those artifacts establish.

Source receipt: months/12/root_details.json record 1. Mathematical references: R227.

Y25-12-022 Misalignment is nonnegative but cannot universally control gradient energy

2025-12-25 · historical evidence A · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can a two-point directional-disorder integral give an unconditional coercive bound for vorticity gradients?

The recovered program isolatedmisalignmentnonnegativity and left a production-versus-dissipation closure unproved. A divergence-free field with a fixed vorticity direction shows why a universal coercive lower bound cannot follow from the misalignment polynomial alone.

Definitions and assumptions

Define $M(a, b)=||a||^2||b||^2-(a\cdot b)^2=||a\times b||^2\geq 0$.

On a2π-periodic three-torus chooseu(x, y, z)=(0, sin x,0), so ω=curlu=(0,0, cos x).

Exact directional-disorder counterexample

For everypairx, y, M(ω(x), ω(y))=0, but$\int||\nabla\omega||^2$is strictly positive. Hence no positive constantccan make a weighted two-point integral of Monly dominatec$\int||\nabla\omega||^2$for all smooth divergence-free velocity fields.

Derivation or proof

The identity for Mfollows by expanding the cross-product norm, or by Cauchy-Schwarz. In the stated shear, all vorticity vectors are collinear, whether they agree, oppose, or vanish. Thus every cross product is zero and any nonnegative weighted integral of Mvanishes. However$\partial_x\ \omega=(0,0,-\sin x)$, so its squared gradient integrates to$(\int_0\hat{}(2\pi)\sin^2x\ dx)(2\pi)^2=4\pi^3>0$. The velocity is smooth and divergence-free, so these standard kinematic assumptions do not exclude the example. This does not show a singular solution; it shows only that the proposed universal coercive inequality is false without extra hypotheses.

Worked example

Also examine the enstrophy identity for a smooth periodic Navier-Stokes solution.

$(1/2)d||\omega||_2^2/dt+\nu||\nabla\omega||_2^2=\int\omega\cdot S\ \omega$, where $S=(\nabla u+\nabla u\hat{}T)/2$. A bound on$\int\omega\cdot S\ \omega$ by $\nu||\nabla\omega||^2$requires a separate dynamical estimate. Nonnegativity of Mdoes not supply it, and a conditional “near-field dominance” assumption must be checked rather than silently inserted.

The exact shear counterexample identifies the missing coercivity mechanism while preserving the valid algebraic identity.

Scope

The counterexample is about a proposed estimate, not a resolution of the Millennium Problem. Additional disorder, alignment or curvature hypotheses can define narrower classes, but their persistence must be proved.

Further development

Formulate the strongest scale-consistent conditional estimate that excludes this shear for an explicit reason, and then prove that the solution class satisfies that reason for the required time interval.

Source receipt: months/12/root_details.json record 1. Mathematical references: R228.

Y25-12-020 A capped-addition monoid with an exact trace ideal

2025-12-31 · historical evidence U · Classical or newly chosen explicit model developed with proof; historical conjectures delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What general monoid theorem survives when the historical29×29operation table is unavailable?

The recovered source claimed exact closure, trace law, generators, idempotents and collision counts for29modes, but did not recover the table. A capped-addition construction provides a proved replacement while leaving those specific historical counts unverified.

Definitions and assumptions

Let L_k={0,..., k}withi⊕j=min(i+j, k).

For a monoid Sdefine M=S×L_kwith(s, i)·(t, j)=(st, i⊕j), andtr(s, i)=i.

Associative capped product and absorbing trace layer

M is a monoid with identity(1_S,0), andtr(xy)=min(trx+try, k). The top layer S×{k}is a two-sided ideal. If S is a group andk>0, the idempotents are exactly(1_S,0)and(1_S, k).

Derivation or proof

Capped addition is associative because both(i⊕j)⊕ℓandi⊕(j⊕ℓ)equalmin(i+j+ℓ, k). Its identity is zero. Product associativity and identity follow coordinatewise from S. If either trace isk, the product trace isk, proving the ideal statement. In a group, s^2=simpliess=1_S. The remaining equationmin(2i, k)=ihas onlyi=0ori=k:if0<i<k, either2i<kand2i≠i, or2i≥kandk≠i. This proves the idempotent characterization. More general Scontributes its own idempotents in those two layers.

Worked example

Take $S=C_2$={1, g}andk=2.

Mhas six elements, two in each trace layer. The element(g,1)has square(1,2)and cube(g,2); the trace saturates while the group coordinate keeps changing. There are exactlytwo idempotents. This illustrates why an absorbing trace ideal need not consist of absorbing individual elements.

The general construction has a complete proof and an exact small model, without falsely certifying the absent29-mode table.

Scope

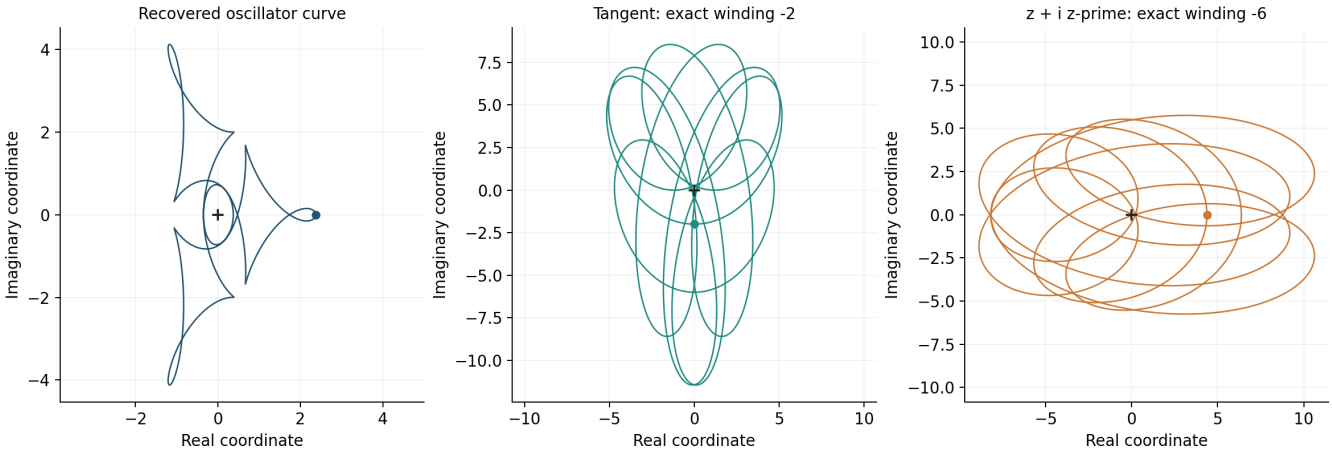
The historical counts(1,2,5,8,13), ten idempotents and generator set{1,5,6,8}cannot be deduced from this product model and remain unverified. A finite table needs exhaustive associativity checks, not only closure and commutativity.

Further development

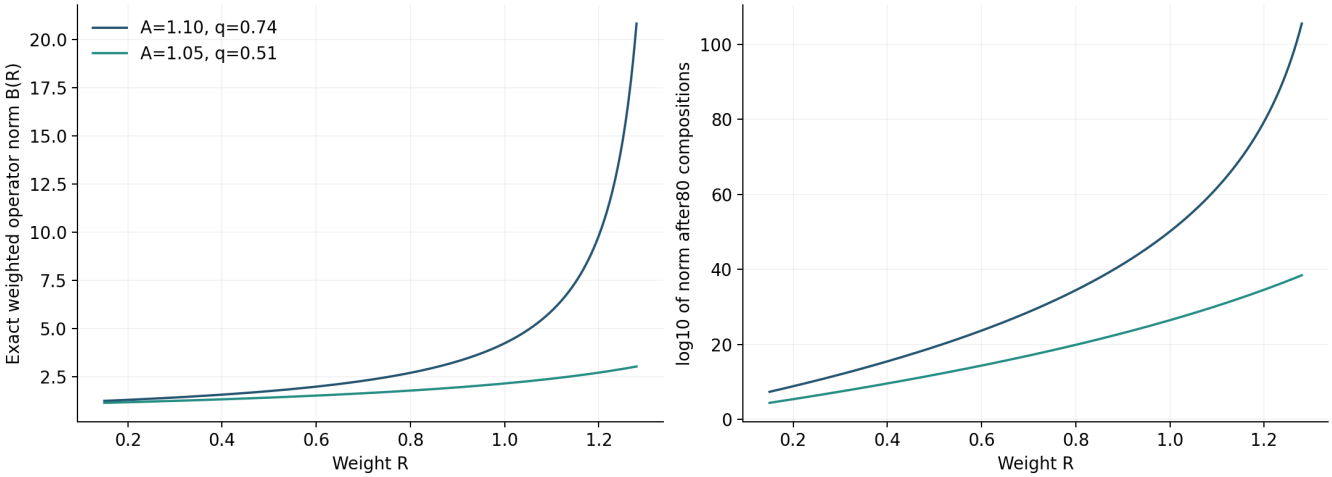
Recover the actual table, enumerate every triple for associativity, then compute ideals, generators, idempotents and trace collisions directly before formalizing the table in Lean.

Source receipt: months/12/root_receipts.json record 0. Mathematical references: R229.

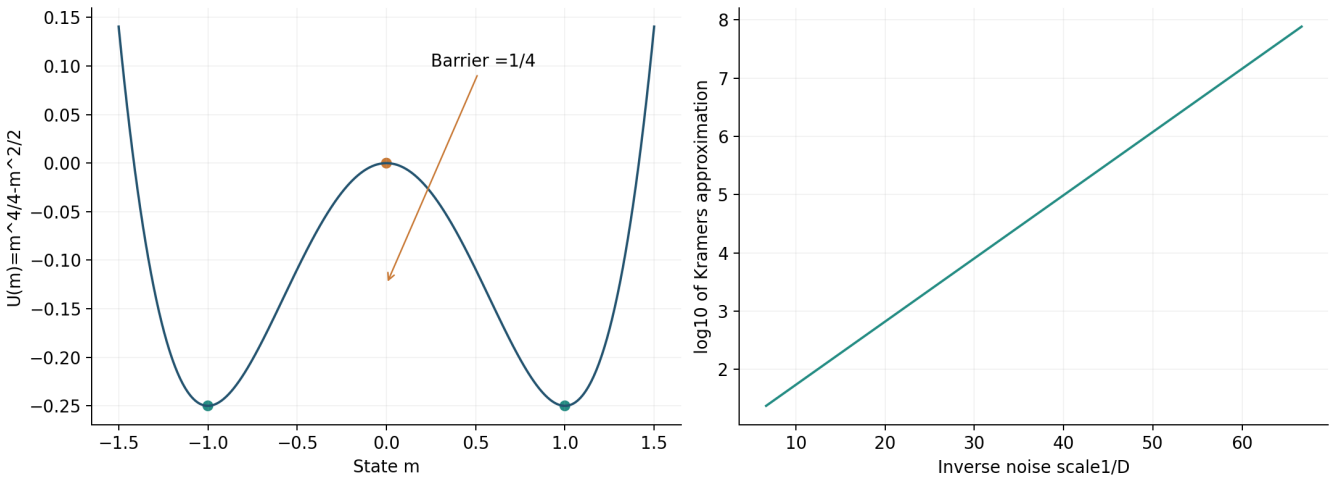
Calculated illustrations



The exact trigonometric curve recovered from22December2025, its tangent, and w=z+i z-prime. Samples draw the curves; the nonvanishing and winding claims are certified separately by integer-polynomial Rouche reduction in independent_audit.py. The scalar x coordinate already requires an8-dimensional linear realization.



Explicit positive convolution models T_ij=A q^(i-j) for i>=j on weighted ell-one, with norm sum |f_i|R^i. Their exact norms are B(R)=A/(1-qR), and their80-fold composition norms are B(R)^80. Constants define these illustrative mathematical models; the plots do not reproduce or establish a measured machine-learning advantage.



A fully specified symmetric double-well model U(m)=m^4/4-m^2/2. For dM=-U-prime(M)dt+sqrt(2D)dW, the barrier is1/4 and the small-noise Kramers approximation for transition to the other well is sqrt(2)*pi*exp(1/(4D)). The right panel displays that asymptotic formula, not simulated or measured escape times.

Verification ledger and scope

The supplied programs replay exact finite calculations and scoped numerical examples. The ledger also includes source/schema checks, which are not mathematical theorem checks. Written proofs and general literature results retain the scopes stated in their treatments. The full retained details are in the HTML edition and JSON reports.

January

Y25-01-001 · c1 · passed

Y25-01-002 · c2 · passed

Y25-01-003 · c3 · passed

Y25-01-004 · c4 · passed

Y25-01-005 · c5 · passed

Y25-01-006 · c6 · passed

Y25-01-007 · c7 · passed

Y25-01-008 · c8 · passed

Y25-01-009 · c9 · passed

Y25-01-010 · c10 · passed

Y25-01-011 · c11 · passed

Y25-01-012 · c12 · passed

Y25-01-013 · c13 · passed

Y25-01-014 · c14 · passed

Y25-01-015 · c15 · passed

Y25-01-016 · c16 · passed

Y25-01-017 · c17 · passed

Y25-01-018 · c18 · passed

Y25-01-019 · c19 · passed

Y25-01-020 · c20 · passed

Y25-01-021 · c21 · passed

Y25-01-022 · c22 · passed

Y25-01-023 · c23 · passed

Y25-01-024 · c24 · passed

Y25-01-025 · c25 · passed

February

Y25-02-001 · c1 · passed

Y25-02-002 · c2 · passed

Y25-02-003 · c3 · passed

Y25-02-004 · c4 · passed

Y25-02-005 · c5 · passed

Y25-02-006 · c6 · passed

Y25-02-007 · c7 · passed

Y25-02-008 · c8 · passed

Y25-02-009 · c9 · passed

Y25-02-010 · c10 · passed

Y25-02-011 · c11 · passed

Y25-02-012 · c12 · passed

Y25-02-013 · c13 · passed

Y25-02-014 · c14 · passed

Y25-02-015 · c15 · passed

Y25-02-016 · c16 · passed

Y25-02-017 · c17 · passed

Y25-02-018 · c18 · passed

Y25-02-019 · c19 · passed

Y25-02-020 · c20 · passed

Y25-02-021 · c21 · passed

Y25-02-022 · c22 · passed

Y25-02-023 · c23 · passed

Y25-02-024 · c24 · passed

Y25-02-025 · c25 · passed

March

Y25-03-001 · SVD error and target rank · passed

Y25-03-002 · Isometric compression residual · passed

Y25-03-003 · Triangle persistence chain matrices · passed

Y25-03-004 · QUBO exhaustive labels and QAOA state · passed

Y25-03-005 · Poisson median/count distinction · passed

Y25-03-006 · Renormalization-series truncation · passed

Y25-03-007 · Modular lattice covariance identity · passed

Y25-03-008 · Geometric scaling and spherical flow · passed

Y25-03-009 · Frobenius coproduct and handle · passed

Y25-03-010 · Connes two-point norm · passed

Y25-03-011 · Filtered cancellation · passed

Y25-03-012 · Modular matrix-unit eigenvalues · passed

Y25-03-013 · Entropy finite-change identity · passed

Y25-03-014 · Noncommuting modular cocycle · passed

Y25-03-015 · Wilson commutator loop · passed

Y25-03-016 · Gaussian RG flow · passed

Y25-03-017 · Crossing orientations · passed

Y25-03-018 · Maurer-Cartan constant curvature · passed

Y25-03-019 · Quaternion lift and semidirect input · passed

Y25-03-020 · Projective multiplier identity · passed

Y25-03-021 · Subdivision blocking homotopy · passed

Y25-03-022 · Noise bandwidth integrals · passed

Y25-03-023 · Memory kernel closed solution · passed

Y25-03-024 · Hirota heat-equation counterexample · passed

Y25-03-025 · Factorial moments · passed

Y25-03-026 · Energy-budget arithmetic · passed

Y25-03-027 · Stabilizer syndromes and logical-error polynomial · passed

Y25-03-028 · Fibonacci fusion diagonalization · passed

Y25-03-029 · Cantor probability moments · passed

Y25-03-030 · Loop lift integer endpoints · passed

Y25-03-031 · Projection-loss null direction · passed

Y25-03-032 · Four-form wedge normalization · passed

Y25-03-033 · Eta translation phase · passed

Y25-03-034 · Dual-number Hochschild resolution · passed

Y25-03-035 · Square-root spectrum count and heat bound · passed

April

Y25-04-001 · Cyclotomic units and automorphism orders · passed

Y25-04-002 · Finite determinant example · passed

Y25-04-003 · Full sine orthogonality and truncation failure · passed

Y25-04-004 · Dimension-mass contradiction and entropy inputs · passed

Y25-04-005 · Integer fusion-ring recurrence · passed

Y25-04-006 · Müger-center finite bicharacter · passed

Y25-04-007 · Z2 quantum-double labels · passed

Y25-04-008 · Lagrangian algebra multiplication certificate · passed

Y25-04-009 · Exact Brieskorn Jacobian bases · passed

Y25-04-010 · Link torsion determinant arithmetic · passed

Y25-04-011 · Top-form quotient exactness counterexample · passed

Y25-04-012 · Logarithmic-derivative example · passed

Y25-04-013 · Certified golden dilogarithm series · passed

Y25-04-014 · Hyperbolic volume exact formula · passed

Y25-04-015 · Canceled atom pairing normalization · passed

Y25-04-016 · Hyperbolic Poincaré coefficient · passed

Y25-04-017 · Harper matrix and magnetic translation · passed

Y25-04-018 · Complete Diophantine residue classification · passed

Y25-04-019 · Overlapping Cantor contraction · passed

Y25-04-020 · Golden update multipliers · passed

Y25-04-021 · Finite coherence inequality examples · passed

Y25-04-022 · Homotopy pushout cycle count · passed

Y25-04-023 · Formal-group height examples · passed

Y25-04-024 · Dimension rank and invariant test · passed

Y25-04-025 · Doubled-frequency observable · passed

Y25-04-026 · Canonical Bogoliubov coefficients · passed

Y25-04-027 · Parametric oscillator monodromy · passed

Y25-04-028 · Exact E8 root enumeration · passed

May

Y25-05-001 · Gaussian-derivative moments and Fourier normalization · passed

Y25-05-002 · Independent disk flux and circulation · passed

Y25-05-003 · Exact quartic and fifth-power counterexamples · passed

Y25-05-004 · Complete 4^3 Fourier spectrum against matrix · passed

Y25-05-005 · Exact Galerkin projection and energy example · passed

Y25-05-006 · Functional-calculus interpolation and Newton residuals · passed

Y25-05-007 · Pure GHZ and W entropy cancellation · passed

Y25-05-008 · Exact triangle Hodge matrices · passed

Y25-05-009 · Tropical necessary condition and lost unit information · passed

Y25-05-010 · Exact valuation-shell integral and tail · passed

Y25-05-011 · Dual-number free resolution certificate · passed

Y25-05-012 · Ratio denominator and duty-capacity distinction · passed

Y25-05-013 · Exact group/phase geometry · passed

Y25-05-014 · Tolerance counterexample and transitive chains · passed

Y25-05-015 · Independent product variance and repeated-variable cancellation · passed

Y25-05-016 · Finite contextual gluing and zero divisors · passed

Y25-05-017 · Exact hybrid variational optimum · passed

Y25-05-018 · Actual recovered arithmetic graph polynomial · passed

Y25-05-019 · Finite spectral-zeta values and gamma moments · passed

Y25-05-020 · Exact divisor convolution and inversion through120 · passed

Y25-05-021 · Declared PAC-Bayes numerical certificate · passed

Y25-05-022 · Thermal energy geometry and pulse enclosure · passed

Y25-05-023 · Exact series versus parallel conductivity · passed

Y25-05-024 · Compatible-layer stress and specific-mass discrepancy · passed

MONTH-2025-05 · Attribution and source boundary · passed

June

Y25-06-001 · Exact ABC source-table audit · passed

Y25-06-002 · Smoothed-radical calculus example · passed

Y25-06-003 · Power residues and factorial obstruction · passed

Y25-06-004 · Factorial gcd via Legendre valuations · passed

Y25-06-005 · Tanh positive branch and susceptibility · passed

Y25-06-006 · Point-interaction scalar equation and perturbation derivative · passed

Y25-06-007 · Regular C3 naturality certificate · passed

Y25-06-008 · Modal projector and tensor obstruction · passed

Y25-06-009 · Declared projective Chow ring and shifts · passed

Y25-06-010 · Nontrivial units witness against the false comparison · passed

Y25-06-011 · Tensor-complex signs and torsion indices · passed

Y25-06-012 · Torus unit exponents and blow-up rank shift · passed

Y25-06-013 · Projective-line finite-field and comparison test · passed

Y25-06-014 · Rational interval logarithmic identities · passed

Y25-06-015 · Tuning oscillator and energy residual · passed

Y25-06-016 · Nonlinear Jacobian and dyad composition · passed

Y25-06-017 · Exact adjoint normal-equation solution · passed

Y25-06-018 · Exact tripod overlap and sensor bitrate · passed

Y25-06-019 · Oscillator peak and regeneration denominator · passed

Y25-06-020 · Exact confounding correlation and discount sum · passed

Y25-06-021 · Unit invariance and arbitrary ranking · passed

Y25-06-022 · Fourth-power balance and complete denominators · passed

Y25-06-023 · Exhaustive B4 orbit and order certificate · passed

Y25-06-024 · Full reduced-polynomial evaluation isomorphism · passed

MONTH-2025-06 · Attribution and proof scope audit · passed

July

Y25-07-001 · Cusp triangle boundary · pass

Y25-07-002 · Finite-field count and Euler recurrence · pass

Y25-07-003 · Weierstrass discriminant · pass

Y25-07-004 · Selmer rank-bound bookkeeping · pass

Y25-07-005 · Height Gram and subgroup index · pass

Y25-07-006 · Quaternion 2-adic primitive obstruction · pass

Y25-07-007 · Primal-dual exact optimum · pass

Y25-07-008 · Vector-clock joins · pass

Y25-07-009 · Posterior moments and simulation bound · pass

Y25-07-010 · Riccati root and cost · pass

Y25-07-011 · Mean-zero conformal area variation · pass

Y25-07-012 · Heat Fourier energy identity · pass

Y25-07-013 · Dual-number extension structure · pass

Y25-07-014 · Ribbon modulus · pass

Y25-07-015 · Finite Weyl relation · pass

Y25-07-016 · E8 root count and Delta coefficients · pass

Y25-07-017 · Finite stuffle identity · pass

Y25-07-018 · Two-point distance norm · pass

Y25-07-019 · sl2 character decomposition · pass

Y25-07-020 · Ternary local obstruction · pass

Y25-07-021 · Cyclic decimal and corrected fractions · pass

Y25-07-022 · Ray-class arithmetic ingredients · pass

Y25-07-023 · Biquadratic discriminant audit · pass

Y25-07-024 · Quartic and rational ray audit · pass

Y25-07-025 · Quadratic series recurrence · pass

Y25-07-026 · Shadow weight and local differential coefficient · pass

Y25-07-027 · A2 and D4 root counts · pass

Y25-07-028 · Density residuals and interpolation ambiguity · pass

Y25-07-029 · Quartic irreducibility obstructions · pass

Y25-07-030 · Empty two-CNF complex · pass

Y25-07-031 · Inclusion can kill homology · pass

Y25-07-032 · Transport curvatures · pass

Y25-07-033 · Bounded union-closed audit · pass

Y25-07-034 · Bernoulli exact coefficient · pass

Y25-07-035 · Mackey finite character · pass

August

Y25-08-001 · Trochoid frequency period · pass

Y25-08-002 · Rotation bracket and order · pass

Y25-08-003 · TL trace Gram positivity · pass

Y25-08-004 · Perron stochastic normalization · pass

Y25-08-005 · Perron eigenvalue correction · pass

Y25-08-006 · Chebyshev degree-one example · pass

Y25-08-007 · Cubic KKT sign branch · pass

Y25-08-008 · Gaussian geometric covariance tail · pass

Y25-08-009 · Triangle cohomology and killed class · pass

Y25-08-010 · Relative-period return bound · pass

Y25-08-011 · Lp spike norm · pass

Y25-08-012 · Torus coarea factor · pass

Y25-08-013 · Atomic pairing exact integral · pass

Y25-08-014 · Landau quartic completion · pass

Y25-08-015 · Hall flag count · pass

Y25-08-016 · Quintic factorial Picard-Fuchs recurrence · pass

Y25-08-017 · Graph-phase inconsistency certificate · pass

Y25-08-018 · A-hat cross coefficient · pass

Y25-08-019 · Signed-permutation orbit size · pass

Y25-08-020 · Quadratic Armijo and BFGS secant · pass

Y25-08-021 · ISTA soft-threshold example · pass

Y25-08-022 · One-step hypergradient exact values · pass

Y25-08-023 · Fano incidence spectral certificate · pass

Y25-08-024 · Paley5 Laplacian polynomial · pass

Y25-08-025 · CRT and finite ternary entropy · pass

Y25-08-026 · Lattice symbol bounds · pass

Y25-08-027 · Riccati invariant · pass

Y25-08-028 · Complete graph audit · pass

Y25-08-029 · Rank-one inverse exact product · pass

Y25-08-030 · Residual counterexample and log-noise bound · pass

Y25-08-901 · Exact polynomial traveling-wave residual · pass

Y25-08-902 · Exhaustive Hilbert90 witness calculation over F49 · pass

Y25-08-903 · Squared-ideal certificate · pass

Y25-08-904 · Curvature integral normalization · pass

Y25-08-905 · Eisenstein coefficient normalization · pass

Y25-08-906 · Eta product, independent point counts and central-value example · pass

Y25-08-907 · p-adic nonvanishing precision example · pass

Y25-08-908 · Heegner split/ramified prime distinction · pass

Y25-08-909 · Explicit independent-test selection probability · pass

Y25-08-910 · Exact reciprocal compounding · pass

September

Y25-09-001 · Exact affine fixed point and norm · pass

Y25-09-002 · Cosine bracket and finite increment check · pass

Y25-09-003 · Cocycle and exact Lipschitz example · pass

Y25-09-004 · Cokernel of diag2,3 · pass

Y25-09-005 · Rewrite invariants · pass

Y25-09-006 · 11a1 point counts and Hecke recurrence · pass

Y25-09-007 · Symmetric-square local factor · pass

Y25-09-008 · Fourier aliasing counterexample · pass

Y25-09-009 · Two simple roots within a certified contour · pass

Y25-09-010 · Exact mu and lambda of primitive products · pass

Y25-09-011 · Six cross-ratio parameters and one j value · pass

Y25-09-012 · Quadratic residue at163 · pass

Y25-09-013 · Fractional ownership budget · pass

Y25-09-014 · Finite bisimulation certificate · pass

Y25-09-015 · Square-root smoothing and endpoint defect · pass

Y25-09-016 · Micro quantizer exact envelope · pass

Y25-09-017 · Normalized Chebyshev local coefficients · pass

Y25-09-018 · Geometric sample count and decay estimate · pass

Y25-09-019 · Rational and modular ranks · pass

Y25-09-020 · Exact finite monotonicity certificate · pass

Y25-09-021 · KdV traveling-wave profile identity · pass

October

Y25-10-001 · RG exact invariant and two-loop derivative · pass

Y25-10-002 · Exact whitening and B5 order · pass

Y25-10-003 · Compatible Hensel congruences · pass

Y25-10-004 · Cone rational acyclicity versus integer torsion · pass

Y25-10-005 · Transition cocycle and gauge invariance · pass

Y25-10-006 · Vector-space pushout certificate · pass

Y25-10-007 · Rank-two Kummer quotient sizes · pass

Y25-10-008 · Gram determinant basis and index scaling · pass

Y25-10-009 · Quadratic character cancellation and gamma ratio · pass

Y25-10-010 · Circle zeta residue normalization · pass

Y25-10-011 · Finite Hodge supertrace · pass

Y25-10-012 · Exact PF coefficient recurrence · pass

Y25-10-013 · Theta heat multiplier and elliptic shift · pass

Y25-10-014 · Fixed index and varying singular values · pass

Y25-10-015 · Tor resolution after two specializations · pass

Y25-10-016 · All E8 roots and reflection closure · pass

Y25-10-017 · Nonsplit dual-number extension witness · pass

Y25-10-018 · Exact pointed F R S and Verlinde certificate · pass

Y25-10-019 · Pareto unsupported point and redundant factor · pass

Y25-10-020 · Four-character Parseval identity · pass

Y25-10-021 · Projection location counterexample · pass

Y25-10-022 · Finite one-prime oscillatory sum · pass

Y25-10-023 · Parity midpoint quadrature · pass

Y25-10-024 · Logarithmic integral independent quadrature · pass

Y25-10-025 · Fourth-power integral transformation and closed form · pass

Y25-10-026 · Cyclotomic trace coefficient correction · pass

Y25-10-027 · Trace and Hilbert-Schmidt finite norm inequality · pass

Y25-10-028 · Conservative PDE energy-dissipation example · pass

Y25-10-029 · Fiber integration and degree mismatch · pass

Y25-10-030 · Bloch-Wigner normalization defect · pass

Y25-10-031 · Equivariant stiffness with two different ratios · pass

Y25-10-032 · Conditional oscillator damping consistency · pass

Y25-10-033 · Prime-tone addition and envelope half-life · pass

Y25-10-034 · Dirichlet fourth-moment resonance count · pass

Y25-10-035 · Exact affine recurrence and capture threshold · pass

Y25-10-036 · Product-monomial and zeta4 checks · pass

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Y25-11-001 · Quartic integral formula · pass

Y25-11-002 · Atlas integral checks · pass

Y25-11-003 · Positive-definite counterexample · pass

Y25-11-004 · Bridge variance · pass

Y25-11-005 · Beam spectral threshold · pass

Y25-11-006 · Entropy normalization · pass

Y25-11-007 · Gamma spectrum ratios · pass

Y25-11-008 · Lame n=1 residual · pass

Y25-11-009 · Humbert primitiveness · pass

Y25-11-010 · Wallis/hypergeometric recurrence · pass

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Y25-12-000 · Schema and attribution · pass

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Y25-12-020 · Capped monoid exhaustive check · pass

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Y25-12-022 · Coercivity shear counterexample · pass

Y25-12-023 · Conditional-proof dependency · pass

Y25-12-024 · Weighted rank counterexample · pass

Y25-12-025 · Portal spectral projection · pass

Y25-12-026 · Greedy deflation tails · pass

Y25-12-027 · OU cumulant hierarchy · pass

Calculated illustrations

Independent exact audit

Exact source-curve regularity and tangent winding · passed

Exact winding of z+i z-prime · passed

Independent rational Schur-Cohn matrix congruence · passed

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